



Asset Performance Management

Model Library Guide

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ABOUT MODEL LIBRARY

The Honeywell asset performance library addresses the need for wholistic monitoring of assets. The library allows for quickly deploying digital twins that turn a plant's time-series data into a host of recommended actions that improve asset productivity and reliability. The library combines three crucial asset models into a single, configurable, and asset specific template for dozens of asset types. By combining a thermodynamic model, a modern machining learning analytic model, and a mechanical health monitoring model into a unified template, the library of asset templates allows for the most in-depth view of asset performance. Combined with templated fault models and other standard metrics, asset stakeholders can get a 360-degree view of each piece of plant machinery.

The Honeywell Forge APM Asset Performance Library is a continually evolving set of templates for quickly creating digital twins for a variety of industrial machinery types. The library incorporates the latest expertise and knowledge that Honeywell machinery subject matter experts have gathered from the deployment of asset performance management (APM) solutions with clients across industries. The models within each template represent the evolution of Honeywell experience gained through decades of implementing automation systems, and modeling and simulation of numerous industrial processes.

1.1 Intended Audience

- Operations
- Maintenance
- Reliability Engineers

The purpose of this document is for the maintenance and reliability activity to fine tune downtime and for efficient operations.

1.2 How to Use this Document

A plant typically has thousands of points and control tags where data is continuously collected, yet they may have few means to process that data effectively to elicit actionable information. Likewise, model building is time consuming and has proven to be a large reason for reluctance in implementing an APM solution. Honeywell solves these issues with our asset templates.

The models within each asset template turn asset and process data into actionable insight by applying three distinct areas of monitoring:

- Performance-based Monitoring
- Analytics-based Monitoring
- Health Monitoring & Fault Models

The [Introduction to Assets](#) section describes about the basic definition and functionality of Model Library. This section helps you understand the terminologies and basic workflow. Then we have individual chapters for each assets. These chapters will include the following details:

- Introduction to the Asset
- Sub Systems
- Performance Monitoring
 - Model Assumptions
 - Variants
 - Model Input
 - KPIs
 - Performance Modeling
 - Expected KPI
 - Operating KPI
 - Model Output
- Fault Model
 - Performance Based Faults
 - Health Based Faults

1.3 Release Overview

Below are the key features in Asset Performance Management R540.1 General Availability (GA) release.

- Introduced Composite Asset Builder feature to configure composite assets using existing standard asset models
- Developed fault model for Fired Heater, Rotary Vacuum Filter, Rotary Pressure Filter
- Enabled viscosity correction factor in centrifugal pump for predicting KPI
- E - Motor
 - Digitization of characteristics curves of the motor such as Load Vs Efficiency, Load Vs Power Factor (Custom Curve). The operating point will be shown once curves are provided
 - Voltage and Current Imbalance calculations
- Shell and Tube Heat Exchanger
 - Introduced fatigue cycle calculations for the performance model
 - Introduced Expected delta pressure and modified expected heat transfer coefficient for the performance model
 - Enhanced FMEA based on performance model improvement

INTRODUCTION TO ASSETS

Asset: An Asset or Equipment is mostly a machine used in chemical and materials processing, in facilities like refineries, chemical plants, and [wastewater](#) treatment plants to do a specific process or multiple processes depending upon the facility in which it is installed. An equipment occupies the lowest level in the asset hierarchy. You must configure a plant or a unit before configuring an equipment.

An asset is an economic resource. Anything tangible or intangible that is capable of being owned or controlled to produce value and that is held to have positive economic value is considered an asset. Plant is an asset from business or organizational perspective, which is a sum of many smaller assets. The assets range from the process equipment, rotating machinery, electrical equipment, automotive equipment, to instrumentation and control equipment. Hence the industry is offered with various packages of software and hardware to manage and maintain each of the above assets.

Asset Performance Management categorizes assets by Category, Class, and Type. Asset Categories are used to categorize assets at a very high level and are the first level of asset classification. For example, all equipment in an industrial plant can be categorized into a single asset category named **Equipment** and all materials (for example, crude, gasoline) in a refinery can be categorized into single asset category named **Material**.

Rotating Equipment: Is a term generally used in the oil & gas and process industries to describe mechanical components that use kinetic energy to move fluids, gases, and other process materials. These include, but are not limited to, engines, compressors, turbines, pumps, generators, blowers, and gearboxes.

(Alternate Description: Rotary equipment is the given categorization for mechanical devices that provide kinetic energy to an operation enabling it to move. Rotating equipment is seen in various mechanisms throughout the process industry sector, and it consists of components that move solids, liquids or gases through a series of drivers, driven components and the transmission devices.)

Fixed Equipment, sometimes also known as **Static Equipment**, is a term generally used to describe non-moving equipment in the oil and gas and process industries. Some examples include things like [pressure vessels](#), [heat exchangers](#), [piping](#), [storage tanks](#), [valves](#), [pressure relieving devices](#), [boilers](#), [furnaces/heaters](#), pig launchers and structures.

Fault Symptom Models are used:

- To define rules based on the knowledge on the equipment operation and its failures
- For systematic study or brainstorming of rules using one or more methodology
- Rule logic for triggering event
- Event Actions
 - OPC A&E 'Event'
 - E-mail
 - CMMS (SAP or Maximo) work request
- Define Fault Severity
- Define Recommended Action
- Asset Fault Model can have one or more Faults defined & each fault can have one or more symptoms defined.
- Symptoms are either cause for a failure or effect of a failure

2.1 Sub Systems

Subsystems: A subsystem is a single, predefined operating environment through which the system coordinates the workflow and resource use. The system can contain several subsystems, all operating independently of each other. Subsystems manage resources in other words it can be a unit or device that is part of a larger system. Each subsystem can run unique operations. The Asset configured allows you to decide the number of subsystems and what types of work each subsystem handles.

The **Equipment Subsystems** enable us to define the subsystems of equipment units or equipment types for which we want to do calculations. We can include parts within a subsystem; they are associated with a property to identify the part or attribute. For example, in a Compressor, it could have a subsystem of a Lube Oil System. This Lube Oil System could be set to an equipment type, and therefore, apply to all assets that have that equipment type.

2.2 Performance Monitoring

The performance models calculate any variation between baseline performance and the actual performance of a machine in real-time. This provides a means to find latent inefficiency in a machine's operation and to quantify costs associated with such inefficiencies.

The performance models contained within the library are focused on recognizing two primary areas of energy loss of monitored machinery: operational loss and design loss.

- **Operational loss:** loss caused by equipment degradation and/or the machine being operated sub-optimally due to non standard process conditions or upsets.
- **Design loss:** loss caused by equipment operating off its ideal operating point during normal process operating conditions. (e.g. loss from oversized pumps and motors, undersized valves, etc.)

The results of performance monitoring can be analyzed to determine the effective performance of the machine. This indicates any maintenance need and the relative timing of the activity.

This is possible through a variety of mechanisms - the most obvious is to simply look at machine performance trends, particularly concerning previously performed maintenance to determine its effectiveness and determine the need for further activity. It is also possible to assign a cost to the degradation. When combined with an associated maintenance cost, an effective indicator of the optimal time to perform the maintenance can be easily produced. In general, it is important to recognize that machine performance is defined as much by its current operation, as by any degradation it may exhibit. To this end, the deviation from design, or expected performance, is arguably a more important indicator than the "spot value" of performance. A good performance monitoring system provides both actual performance indicators and deviation from design.

Listed below are some of the results that are derived from performance monitoring:

- Continuously reveals latent inefficiencies.
- Identifies asset performance improvement opportunities.
- Accurately quantifies energy loss.
- Provides an objective basis for maintenance planning.
- Eliminates offline, manual, and periodic calculation methods.

The performance of gas turbines, steam turbines, heat exchangers, compressors, and other types of process equipment deteriorates over time, their efficiency decreases, power consumption increases, throughput reduces, and operating costs rise. Continuous monitoring of machine vibration, thrust, and bearing temperatures as part of condition monitoring is essential for the reliable and safe operation of turbo machinery. However, this information is not enough for operators to maximize revenue and minimize maintenance expenses. The process of performance monitoring evaluates the effect of the process changes on the asset performance so that the operator can select better-operating regimes and predicts accurate maintenance schedules. This is also required to diagnose the process failures and find out better-operating conditions.

The combination of process performance monitoring (based on thermodynamic models) and mechanical condition monitoring (such as vibration) can thus allow better diagnostic capabilities for assessing the nature, severity, location, and cause of machinery performance changes. Identification of deviation between actual and design performance is vital in equipment performance monitoring for rotating and static equipment.

2.2.1 Key Performance Indicators

- **Operating KPIs** are calculated based on near-real time process inputs.
- **Expected KPIs** are calculated based on some process inputs, data sheet values, performance curves and other manufacturer reference values.
- **Losses KPIs** are calculated based on the deviations between Expected and Operating KPIs.

2.2.2 Model Inputs

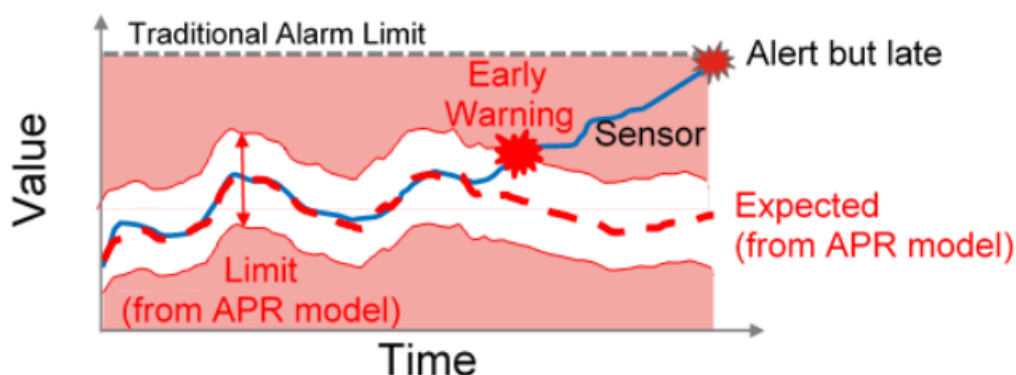
The calculation model to perform as expected, requires some inputs are to be feed into the model. The model input data consists of measured data and static data. Measured input data is supplied to the model from DCS or Historian by tag mapping. This section describes the list of input and output attributes and their properties.

2.3 Predict Model

The Predict module that builds on the Visibility module, this module is perfect for reliability teams and provides real-time monitoring of plant reliability, custom KPIs, and Overall Equipment Effectiveness. With the reliability module we look into the future to predict potential failures that could impact the health of equipment, uncover the main variables driving these deviations from normal behavior and take action before they occur. Our state-of-the-art deep learning implementation allow us to identify never before seen patterns, investigate and classify them, and ultimately ingest back into the model this new knowledge to make it smarter over time without needing to be a seasoned data scientist.

Performance models coupled with predictive machine learning data analytics is the most effective means for early recognition of asset issues.

Honeywell utilizes machine learning and advanced pattern recognition (APR) methods in its data analytics models. Each analytic model learns a machine's normal operating patterns by drawing from the plant data stream. Once trained, a model continuously compares current data patterns to the learned and expected patterns. The changes in the patterns are used to alert early on developing problems.



As with any data driven approach, the usefulness of the results depends on careful selection of both the model inputs and the training datasets. The first consideration is addressed by each embedded asset template, the second depends on the available data (see APR section of each individual asset addendum attached).

In this template First principle-based monitoring and performance-based monitoring approach is as similar as the APM template, only changed health monitoring here we are using early events detection model based on an advanced analytic algorithm (Moving Mean Principal Component Analysis (MMPCA)).

Honeywell provides an Advanced analytics capability, integrated with the Asset Performance Management solution. This approach complements the standard first principles asset templates, providing early detection of equipment and instrumentation issues and extending the capability of Asset Performance Management to monitor equipment types and sub-systems.

The Advanced analytics models use historical operating data to define a baseline of the normally expected behavior of an asset. This is captured as the normally expected correlations between the variables measured for the equipment.

The model training process requires the user to screen out abnormal / faulty operating modes of the equipment, retaining the normal operating behavior. The resulting models describe all the normally expected combinations (or patterns) of the input signals as well as model results such as health Index, Statics T2, and Square prediction error for a model. For any new set of input signals, these models can then be used to estimate the values of the measured signals (termed the cleansed, expected, predicted values). For each set of input data, along with the Model result and Root cause variable for each model.

The Asset Analytics model is used for:

- On-line development and Preprocessing data and training of the Analytics models.
- On-line execution of the models and fault detectors based on Model and Variable results Output.
- Model retraining based on changes in operating conditions. E.g. change in vibration pattern after equipment shutdown.
- Model Scheduling to assess model health and identify Root Cause Variables responsible for a degraded health condition, when a model is deployed online.

When designed and configured correctly, Asset analytics models can provide an early indication of deviations in equipment behavior which can be used to initiate more intensive monitoring, analysis, and investigation.

2.3.1 AI Model

Combined Asset and process modelling with Artificial Intelligence (AI) or Machine Learning (ML) to deliver actionable Insights.

- **AI/ML Algorithms**

- Out of the box AI/ML algorithms to predict asset health and performance degradation in advance.
 - Anomaly Detection: MMPCA, AutoEncoder, PCA
 - Fault Classification: Random Forest, LSTM
 - Prognosis: ARIMA, Regression
- Analytics workbench for engineers to create AI/ML models with minimal coding and data science knowledge
 - BYO-ML plug-in for custom python algorithms leverage existing custom models
 - Workbench to build, train, test ,deploy and run models
 - Reuse knowledge to continuously improve failure database

- **Plant-wide reliability monitoring**

- Continuously monitor plant KPIs to ensure you meet your reliability targets into the future

- **Early Identification of Anomalies**

- Predict potential issues and identify the sources of deviations before equipment degrades or fail

- **Actionable Recommendations**

- Obtain maintenance recommendations including causes, consequences, and corrective actions with quantified benefits (Cost & OEE)

2.4 Fault Model

Faults and its symptoms are used to detect anomalies with the equipment or process. It helps to monitor the health of the equipment that covers the reliability aspects. Fault symptom model explains the event rules defined to monitor health with predefined recommended actions. The parameters can be monitored by below ways.

- **Design Limits:** The parameters are compared with their applicable design high, low limits, and events are generated in case of limit violation.
- **Static Limits:** Calculated based on existing applicable Control system limits, ranges or based on equipment specifications. These are then calculated using a factor that is used to tune the degree of early warning while still having a common reference with the source of the physical limits.
- **Statistical limits:** Calculated using statistics (standard deviation, average) on historical values of expected values. These Expected values are calculated either from APM model (Expected Efficiency, Expected Head, Expected Power, etc) or APR models (expected values of measurements and KPIs determined based on empirical correlations and machine learning algorithms)
- **APR limits:** The Advanced Pattern Recognition (APR) tool can predict expected values for health monitoring parameters based on a defined correlation model and chosen calibration period used to train the model. The current values are compared with expected values and events are generated if the deviation exceeds 2 to 5 sigma.

In a process plant, FMEA is performed to determine the failure modes and its causes for an equipment. The outcome of the FMEA is converted to a Fault-Symptom model in Asset Performance Management. The failure modes from the FMEA analysis are converted to faults and all the causes from the FMEA analysis are converted to symptoms.

The Fault-symptom models from Allied Reliability is used to build fault -symptom models for process equipment that are widely used.

Other than the fault-symptom model, Asset Performance Management also supports calculations. Calculations are used for process performance calculations and KPI calculations.

The Asset Performance Management library consists of pre-built models for process equipment. The built-in process performance calculation models are available for Pumps, Compressors, Heat Exchangers, Steam Turbines and Gas Turbines.

A Performance Monitoring model is used to calculate the operating (actual) performance output, design (expected) performance output, and deviation between operating performance and design performance output. Fault models can be configured on process performance outputs like efficiency, capacity, and so on.

A vibration model is also available for monitoring vibrations of rotating equipment.

2.4.1 Fault Model Blueprint

2.4.1.1 Faults

Detail how faults are being evaluated in the engine:

Faults can be created as:

1. Simple events, with just an expression - the fault's state depends solely on the result of the expression:
 - a. If the expression is evaluated as "TRUE" then the fault will be Active.
 - b. If the expression is evaluated as "FALSE" then the fault will be Inactive.
 - c. The fault can be put in potential state only if explicitly specified in the expression. complex expressions can be used where the Fault state is attributed to the fault directly, usually using IF THEN ELSE END statements.

2. Fault based on symptoms - the fault's state depends solely on the state of the associated symptoms:
 - a. If the state of all symptoms is "Inactive" then the fault will be "Inactive" as well.
 - b. If some or all of the symptoms are "Active" then the fault will be "Potential"
 - c. Faults based on symptoms only cannot become "Active"
3. Fault with both expression and symptoms - the state of the fault depends on the expression of the fault and the state of the associated expressions:
 - a. If the expression is evaluated as "TRUE" then the fault will be Active.
 - b. If the expression is evaluated as "FALSE" then the fault will look at the symptom states
 - If the state of all symptoms is "Inactive" then the fault will be "Inactive" as well.
 - If some or all of the symptoms are "Active" then the fault will be "Potential"
 - c. Note that the fault' expression can be built using the associated symptoms as input variables, so any custom behavior can be modeled based on the states of the symptoms - complex expressions where the Fault state is attributed directly to the fault based on certain conditions.

Add section about context variables and asset states:

Apart from events and expressions, faults and symptoms also can be associated with context variables. Context variables act as enables and they take precedence over anything else. One or more context variables can be associated to an event at the same time. All context variables have to be active at the same time for the fault to be evaluated (var1 AND var2 AND ... AND varn). if the contexts variables evaluated together result in a "FALSE" result then the event won't be evaluated regardless of the other inputs. Usage example: fault generation based on running status (disable faults when assets are not running).

If a fault is active and the asset is turned off, the fault must remain locked in it's active state until the asset starts back up and the fault is being evaluated again.

Events can be enabled also by the state of the asset. when an asset is in Unmonitored or Out of Service states, faults are not being evaluated.

2.4.1.2 Symptoms

Symptom Events:

- Diagnostics - conditions that confirm the existence of the problem represented by the parent fault. these are typically effects of the problem.
- Root Causes - contributing factors that lead to the appearance of the problem represented by the parent fault.

If one or more Root Causes are present the fault is in Potential state. This is a proactive notification since the presence of contributing factors does not mean the problem is present (yet).

If at least one diagnostic (problem effect) is present then the fault is Active state. This qualifies as a reactive notification since the problem is now present and actively affecting the asset/unit performance

Each symptom that is part of a fault model is classified as Early Warning or Critical Warning based on the expressions used to evaluate the variables.

Critical Warning Symptoms shall use simple expressions to evaluate variables against direct targets.

- These symptoms will be triggered when the evaluated variable has exceeded a certain limit or has deviated a certain amount from a target.

Example Expressions:

- [Variable] > [Limit]
- [Variable] < [Limit]
- Abs([Variable] - [Target]) > [Deviation_Limit]

Early Warning Symptoms shall use expressions based on statistics to determine the behaviour of the variables they are evaluating.

- These symptoms will be triggered when the evaluated variable has exceeded a certain limit or has deviated a certain amount from a target.

Example Expressions:

- Rate (or Speed) of Change: $\text{ROC}([\text{Variable}], [\text{Duration}]) > [\text{ROC_Limit}]$
- Standard Deviation, detects sudden changes: $[\text{Variable}] - \text{AVG}([\text{Variable}], [\text{Duration}]) > [\text{STD_Limit}] * \text{STD}([\text{Variable}], [\text{Duration}])$

2.4.1.3 Expressions

Expressions are calculations used for Faults and symptoms to identify and generate anomalies as defined according to the limit values of the attributes

Fault expression is created using the associated symptoms. Please refer examples shown below

Gear Box Bearing Vib Malfunction = (LSS_RB_VIB_DE_X_H OR LSS_RB_VIB_DE_Y_H OR LSS_RB_VIB_NDE_X_H OR LSS_RB_VIB_NDE_Y_H OR HSS_RB_VIB_DE_X_H OR HSS_RB_VIB_DE_Y_H OR HSS_RB_VIB_NDE_X_H OR HSS_RB_VIB_NDE_Y_H)

RB - Radial Bearing

LSS - Low-Speed Shaft

HSS - High-Speed Shaft

VIB - Vibration

DE - Drive End

NDE - Non-Drive End

H - High

The above calculation expression generates fault if any one of the symptoms activates. The values inside the parenthesis are associated symptoms for the Gear Box Bearing Vibration Malfunction.

Symptoms associated with Advanced Pattern Recognition (APR) faults consider DCS limits and constant percentage deviation to estimate fault limits. For Example, The Below expression is used if the symptom is marked as rH:

$\text{LSS_RB_DE_VIB_X_H} = \min((\text{LSS_RB_DE_VIB_X_ExpectedValue} - (\text{LSS_RB_DE_VIB_X_Baseline_SD} - \text{LSS_RB_DE_VIB_X_DCSLowLimit}) * \text{LSS_RB_DE_VIB_X_PctDiffMeanDCS}), \text{LSS_RB_DE_VIB_X_DCSLowLimit})$

NOTE

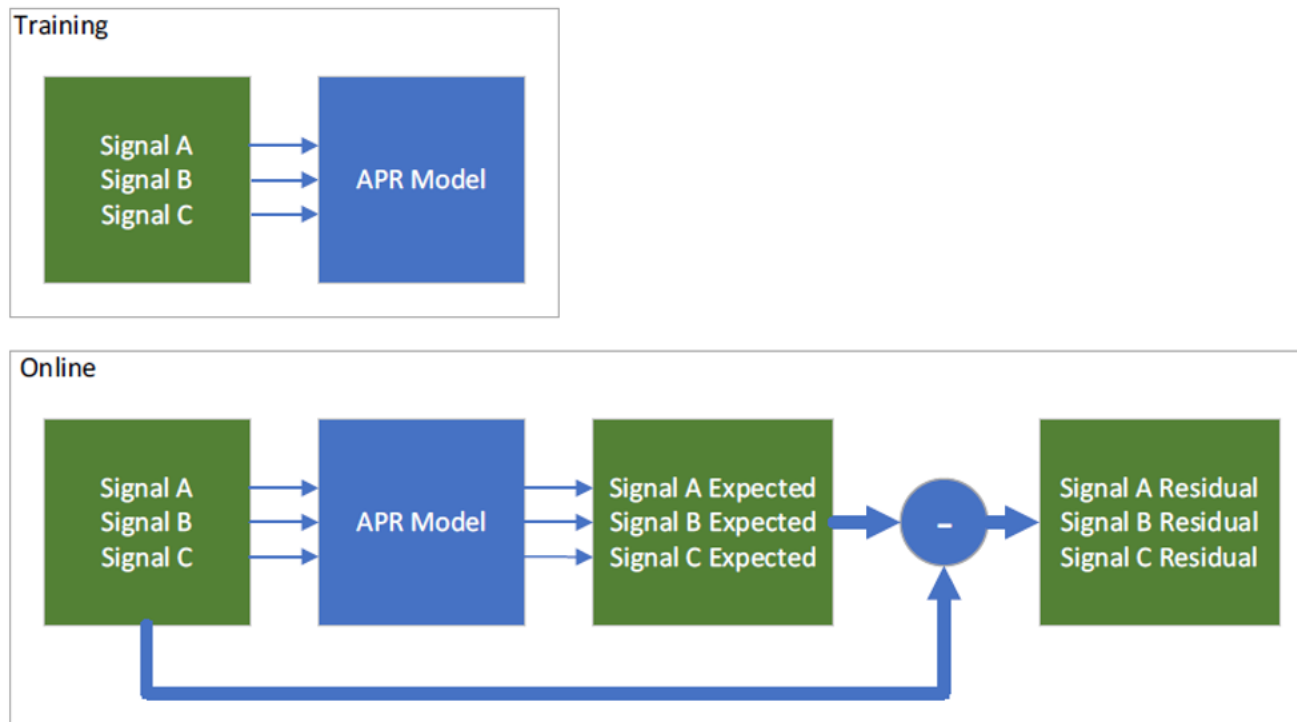
The above expression is used only if the symptom is marked as rH in the blueprint.

The above calculation expression is used to evaluate the symptom-based DCS Limits and constant percentage deviation values and the expected values calculated from the APR Models.

The APR models use historical operating data to baseline the normal expected operating pattern of an asset. This is captured as the normally expected correlations between the variables measured for the equipment.

For any new set of input signals, these models can then be used to estimate the values of the measured signals (termed the expected values). For each set of input data, a prediction error (or residual) can therefore be computed based on the delta between the expected value from the APR model and the measured value. The following figure illustrates the concept.

The model training process requires the user to screen out abnormal/faulty operating modes of the equipment, retaining the normal operating behavior. The resulting models describe all the normally expected combinations (or patterns) of the input signals.



The model is then used online with current measurements of the parameter to estimate values for the current parameter value. If the estimated values differ significantly from the measured values, this is an indication of a system malfunction.

The following figure illustrates the concept of a single variable. The red dashed line represents the time series trend of the expected measured value, and the solid blue line is the measured value. The white region around the expected value represents the alerting threshold above and below the expected value. When the measured value deviates from the estimated value by more than the threshold, then a potential fault condition is flagged. Note that the fault detectors are based on individual measured vs expected values, but at least two correlated measurements are required for an APR prediction model.

Provided the APR models are representative, this is an effective mechanism to detect equipment performance issues quickly with minimal false positives.

The below-mentioned Symptom expression is opted only if the dynamic limit is preferred.

LSS_RB_DE_VIB_X_H = (LSS_RB_DE_VIB_X_ExpectedValue - (LSS_RB_DE_VIB_X_Baseline_SD- LSS_RB_DE_VIB_X_DCSLowLimit) * LSS_RB_DE_VIB_X_PctDiffMeanDCS)

NOTE

The default values of attribute properties Baseline_SD = 5 and PctDiffMeanDCS = 20.)

The statistical high and low limits are estimated based on the statistical mean of the same attribute during the historical operating period. The Residual values are calculated as the difference of the predicted value and the actual value is used in the High and low limit calculation of APR residual faults.

The below Table shows the various symptom suffixes used to denote various symptom expressions:

Symptom suffix	Description	Symptom Name Example	Symptom Expression Examples
rL	APR residual Low	AIR_SUPPLY_PRES_POA_RL	$\text{AIR_SUPPLY_PRES.LowLimit} = (\text{Max}((\text{AIR_SUPPLY_PRES.Expected} - ((\text{AIR_SUPPLY_PRES.Mean} - \text{AIR_SUPPLY_PRES.DCSLowLimit}) * (\text{AIR_SUPPLY_PRES.MinAllowPctMeanDCSDiff} / 100))), \text{AIR_SUPPLY_PRES.DCSLowLimit}))$
rH	APR residual High	AIR_FILTER_DP_POA_RH	$\text{AIR_FILTER_DP.HighLimit} = (\text{Min}((\text{AIR_FILTER_DP.Expected} + ((\text{AIR_FILTER_DP.DCSHighLimit} - \text{AIR_FILTER_DP.Mean}) * (\text{AIR_FILTER_DP.MinAllowPctMeanDCSDiff} / 100))), \text{AIR_FILTER_DP.DCSHighLimit}))$
rL or rH	APR residual Low or High	AIR_SUPPLY_PRES_POA_RL	$\text{AIR_SUPPLY_PRES.LowLimit} = (\text{Max}((\text{AIR_SUPPLY_PRES.Expected} - ((\text{AIR_SUPPLY_PRES.Mean} - \text{AIR_SUPPLY_PRES.DCSLowLimit}) * (\text{AIR_SUPPLY_PRES.MinAllowPctMeanDCSDiff} / 100))), \text{AIR_SUPPLY_PRES.DCSLowLimit}))$
		AIR_SUPPLY_PRES_POA_RH	$\text{AIR_SUPPLY_PRES.HighLimit} = (\text{Min}((\text{AIR_SUPPLY_PRES.Expected} + ((\text{AIR_SUPPLY_PRES.DCSHighLimit} - \text{AIR_SUPPLY_PRES.Mean}) * (\text{AIR_SUPPLY_PRES.MinAllowPctMeanDCSDiff} / 100))), \text{AIR_SUPPLY_PRES.DCSHighLimit}))$
HIGH	Absolute Limit High	AIR_REGVALVE_POS_POA_HIGH	$\text{AIR_REGVALVE_POS_POA_HIGH} = (\text{AIR_REGVALVE_POS}) > (\text{AIR_REGVALVE_POS.HighLimit})$
HIGH or LOW	Absolute Limit High or Low	SUC_IN_PRES_POA_HIGH	$\text{SUC_IN_PRES_POA_HIGH} = (\text{SUC_IN_PRES}) > (\text{SUC_IN_PRES.HighLimit})$
		SUC_IN_PRES_POA_LOW	$\text{SUC_IN_PRES_POA_LOW} = (\text{SUC_IN_PRES}) < (\text{SUC_IN_PRES.LowLimit})$
LOW	Absolute Limit Low	DIS_PRES_POA_LOW	$\text{DIS_PRES_POA_LOW} = (\text{DIS_PRES}) < (\text{DIS_PRES.LowLimit})$
sL	Statistical Low	DIS_FLOW_POA_SL	$\text{DIS_FLOW.LowLimit} = (\text{Max}((\text{DIS_FLOW.Mean} - ((\text{DIS_FLOW.Mean} - \text{DIS_FLOW.DCSLowLimit}) * (\text{DIS_FLOW.MinAllowPctMeanDCSDiff} / 100))), \text{DIS_FLOW.DCSLowLimit}))$
sH	Statistical High	DIS_PRES_POA_SH	$\text{DIS_PRES.HighLimit} = (\text{Min}((\text{DIS_PRES.Mean} + ((\text{DIS_PRES.DCSHighLimit} - \text{DIS_PRES.Mean}) * (\text{DIS_PRES.MinAllowPctMeanDCSDiff} / 100))), \text{DIS_PRES.DCSHighLimit}))$
sH or sL	Statistical High or Statistical Low	FLUID_PH_POA_SH	$\text{FLUID_PH.HighLimit} = (\text{Min}((\text{FLUID_PH.Mean} + ((\text{FLUID_PH.DCSHighLimit} - \text{FLUID_PH.Mean}) * (\text{FLUID_PH.MinAllowPctMeanDCSDiff} / 100))), \text{FLUID_PH.DCSHighLimit}))$
		FLUID_PH_POA_SL	$\text{FLUID_PH.LowLimit} = (\text{Max}((\text{FLUID_PH.Mean} - ((\text{FLUID_PH.Mean} - \text{FLUID_PH.DCSLowLimit}) * (\text{FLUID_PH.MinAllowPctMeanDCSDiff} / 100))), \text{FLUID_PH.DCSLowLimit}))$

2.4.1.4 Cause, Consequences and Recommendations

Causes: Contributing factors that lead to the appearance of the problem.

Consequences: To what extent the loss will impact the process. It can run the model from easily fixed with minimal losses to catastrophic, depending on factors like repair costs, total downtime, health and safety implications.

Recommendations: This field displays any recommended actions (configured on the system) for the event.

Generally applicable Cause, Consequences and Recommendations for the fault will be configured at the fault level, while other Cause, Consequences and Recommendations specific to Root causes will be configured at the symptom level.

2.4.2 Fault Tree

The Fault Tree is a graphical display of your system assets and associated faults and symptoms with performance indicators.

You can use the Fault Tree to:

1. Locate the source of a performance issue.
2. Option to hide inactive events (event trees can be a bit overwhelming, especially for complex assets. having the option to show just the active events will simplify and speed up drill-down and root cause identification.
3. Show associated values
4. Show recommendations
5. Have links to trends
6. Support fault management actions



2.5 Configuration

This section explains the procedure to create a Fault Symptom model and a Calculation model. You can add, edit, and delete any number of models.

2.5.1 Creating a Fault Symptom Model

This section explains the procedure to create a Fault Symptom model.

A Fault Symptom model generally consists of faults and symptoms. While building a fault model, it is mandatory to have at least one fault whereas the number of symptoms can be optional. Asset type attributes are used to configure expressions for symptoms and both asset type attributes and symptoms are used to configure expressions for faults. Before creating complex fault models (fault models with fault model expressions containing one or more asset type attributes), you should create faults with empty expressions with the same names as that of the asset attribute names. During the evaluation, the symptoms are executed first and the faults are executed later.

A Fault Symptom model is configured and used in the following ways:

1. In a Fault Symptom model, if the fault expression is defined using asset type attributes instead of symptoms, then the expression is evaluated during runtime. Whenever this fault expression becomes true, the fault is set to "Active".
2. If the symptoms are configured for a fault, the fault expression is evaluated independent of the symptoms, provided there are no symptoms used in the fault expression. In this case, only asset type attributes are used in the fault expression. If any one of the symptoms is active during the runtime, the state of the fault becomes "Potential". Fault expression evaluation overrides this functionality, which means, if one symptom is active and fault expression is evaluated as "True", then the fault status is set as "Active" rather than "Potential".
3. Once the symptoms are created in the fault, they are available as attributes, which can be used in any of the expressions. The fault expression is created using these symptoms. The fault is always evaluated based on the symptom results and attribute values. During runtime, if any one of the symptoms is active, the state of the fault becomes "Potential". Fault expression evaluation overrides this functionality. That means, if one symptom is active and fault expression is evaluated as "True", then the fault status is set as "Active" rather than "Potential".
4. When no expression is created for a fault, but symptoms are configured under the fault, then the expression is evaluated during the runtime only if any one of the symptoms is active. The state of the fault becomes "Potential". If all the symptoms under the fault are active, then all the faults are set to "Potential" because no expression is defined. However, if all of the symptoms are known for a fault, then you can create a fault expression using the symptoms and set the fault as "Active" if the expression is valid. In case all of the possible symptoms for a fault are not known, then you can leave the fault expression blank after configuring the symptoms. In this case, the fault is always set as "Potential," indicating it can lead to a fault.

2.5.1.1 Adding a Fault from Calculation Configuration

1. On the Contents tree, click **Configuration > Calculation**.
2. Select the Asset.
3. By default, the Calculation Configuration Pane is displayed in the top right corner.
4. On the Calculation Configuration pane, click the + icon of fault to add a new fault to an asset.

The **Add New Fault** dialog box appears.

5. In the **Name** box, type a name for the model.
6. In the **Display Name** box, type a name in the localized language for the fault model.
7. Click **CONTINUE & ADD FAULT**.

A new fault model is created and displayed under the Associated Asset section. By default, the model is created with a fault "NewFault".

TIP

To delete the model, right-click the model and select **Delete**. You cannot rename the model.

2.5.1.2 Adding a Fault from Associated Models Section

1. From the Associated Models section, expand the model you have created and click **NewFault**.

A dialog box with several tabs is displayed on the right pane. Click each tab and enter the required information for the fault. By default, the General tab is selected and the General page is displayed.

NOTE

You can also right-click on the existing model and select **Add Fault**.

Fault Model : Test_Fault

[Fault Definition](#) / [Fault Activation](#) / [Fault Expression](#) / [Actions](#)

General	Notifications
Name* NewFault	☐ OPC Alert
Display Name NewFault	☐ Mail/Others
Description 	☐ CMMS
Severity* -- Cho	
Category* -- Cho	
Interval On Timer (min) 	
Interval Off Timer (min) 	

ADD ANOTHER FAULT **ADD SYMPTOM**

BACK **NEXT** **SAVE**

2. Provide the necessary information:

- a. In the **Name** box, type a name for the fault.

NOTE

Ensure that:

- The first character of the fault name is alphabetic.
- The fault name does not begin with C_ (this results in an error and the fault cannot be saved).
- The remaining characters in a fault name can be a combination of symbols (-,\$,%,.) and numbers.
- The maximum length of a fault name should not exceed 300 characters.
- You provide the same fault name, which is case-sensitive while defining expressions for faults.
- You cannot assign the result of the calculation/symptom/fault to itself. For example, a fault with the name "F1" cannot have the expression "F1=Variable1*Variable2".

- b. In the **Display Name** box, type a name in the localized language for the fault. On the Asset Performance Management dashboard, the name of the fault is displayed in the localized language.
- c. In the **Description** box, type a description for the fault.
- d. From the Severity drop-down list, select a severity value for the fault.

Severity

1-None
1-None
2- Minor
3-Required
4-Important
5-Serious
6-Critical

- e. From the Category drop-down list, select a category for the fault.

Category

Health
Health
Performance
Energy
Safety

- f. From the Interval drop-down list, select the time (minutes) required for the fault to be active and trigger notifications. An interval is a duration for which a condition remains persistent before it becomes a fault. If the interval is set to zero, the condition becomes a fault instantaneously. Setting up an interval avoids chattering alerts. Ensure that the interval is greater than the polling interval configured for the data source.
- g. If the fault must be reset automatically after it occurs, select the **Auto Reset** check box. When Auto-reset is set to true, the fault status is automatically updated whenever there is a change.
- When Auto-reset is set to false, the fault status is updated only once whenever the fault becomes active. The state of the fault is latched.
- h. Click **Next**.

The Notification tab is selected and the Notification page appears.

3. Select one or more **Notification** checkboxes.
 - a. OPC: a notification sent to the OPC server when a fault occurs.
 - b. Mail: an e-mail sent to the subscriber when a fault occurs.
 - c. CMMS: generates a CMMS XML file when a fault occurs. The CMMS XML file is used to create work orders.

NOTE

The CMMS XML file is saved in the default path
"C:\ProgramData\Honeywell\AssetManager\CMMSXML".

4. Click **Next**.

The Context tab is selected and the Context page appears.

The screenshot shows the 'Fault Model : Test_Fault' configuration window. At the top, there is a breadcrumb navigation bar with four tabs: 'Fault Definition', 'Fault Activation' (which is highlighted in blue), 'Fault Expression', and 'Actions'. Below the navigation bar, on the left, is a message icon with the text 'Selected variable should be True or greater than Zero'. The main area is divided into two panels. The left panel, titled 'Selected Variables', is currently empty. The right panel, titled 'Variables', contains a list of available variables: 'Patt1', 'Patt2', 'Patt3', 'Pcalc', 'Fault1', 'S1', 'S2', and 'EventMappingCalc'. Between the two panels, the text 'Drag and Drop' is displayed with arrows pointing towards both panels. At the bottom of the window, there are four buttons: 'ADD ANOTHER FAULT', 'ADD SYMPTOM', 'BACK', and 'NEXT'. The 'NEXT' button is highlighted in blue.

5. Perform the following steps:

- a. From the Available Variables list, select the variables that you want to add and click >>. The variable is displayed in the Selected Variables list.
- b. Click **Next**.

The Expression tab is selected and the Expression page is displayed. By default, this tab consists of the Expression and Attributes list. The Attribute list contains all of the attributes that you have defined for the asset type. Using these attributes, you can define expressions.

Fault Model : Test_Fault

Fault Definition / Fault Activation / **Fault Expression** / Actions

Expression

NewFault =

[Check Syntax](#)

Expression Templates

Category

DataTypes ▼

Double
double <Variable>

DoubleArray

...

<Variable>
Date Time
datetime <Variable>, <Variable>

ADD ANOTHER FAULT ADD SYMPTOM

[BACK](#) [NEXT](#) [SAVE](#)

6. On the Expression page, perform the following tasks:
- In the **Expression** box, type an equation for the fault.

OR

From the Attributes list, double-click the attribute that you want to use in the expression.

- In the Attributes list, click the **Operators** tab.

The Operators page appears.

- From the Operators drop-down list, select the operator that you want to use in the expression. For example, Conditional.

The **operator's operations** appear.

- d. Double-click the operation if you want to use them in the expression.
- e. Click **Validate** if you want to validate your expression.

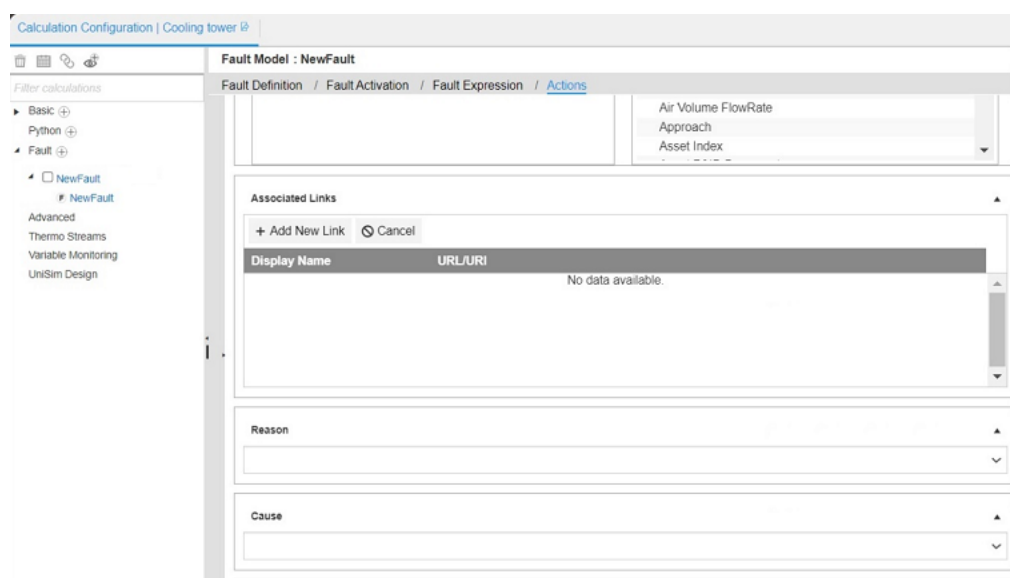
If the expression is invalid, an error message is displayed. The expression could be invalid because it had incorrect operators or attributes. Check the attributes and operators in your expression and click **Validate** again.

- f. Click **Next**.

The Actions tab is selected and the Actions page is displayed.

TIP

You can update this Actions tab from the Bulk Configuration Template sheet.



NOTE

Refer to Appendix: Calculations and Operators in Configuration Guide for more examples on how to define expressions using operators.

TIP

To delete a fault, right-click the fault and select Delete. You cannot rename the fault

2.5.1.3 Adding a Symptom

1. Click the fault you have created and select **Add Symptom**.

The General tab is selected on the right pane and the General page appears.

2. Provide the following information:
 - a. In the **Name** box, type the name of the symptom.

NOTE

Ensure that:

- The first character of the symptom name is alphabetic.
- The symptom name does not begin with C_ (this results in an error and the symptom cannot be saved).
- The remaining characters in a symptom name can be a combination of symbols (-,\$,%,.) and numbers.
- The maximum length of a symptom name does not exceed 300 characters.
- You provide the same symptom name, which is case-sensitive while defining expressions for faults.
- You cannot assign the result of the calculation/symptom/fault to itself. For example, a fault with the name "F1" cannot have the expression "F1=Variable1*Variable2".

- b. In the **Display Name** box, type a name in the localized language for the symptom. On the Asset Performance Management dashboard, the name of the symptom is displayed in the localized language.
- c. In the **Description** box, type the description of the symptom.
- d. From the Severity drop-down list, select the severity value for the symptom.
- e. From the Interval drop-down list, select the time (minutes) required for the symptom to be active and trigger notifications.

3. Click **Next**.

The Notification tab is selected and the **Notification** page is displayed.

4. Select any one or more Notification checkboxes.

- a. OPC: a notification sent to the OPC server when a symptom occurs.
- b. Mail: an e-mail sent to the subscriber when a symptom occurs.
- c. CMMS: generates a CMMS XML file when a symptom occurs. The CMMS XML file is used to create work orders.

NOTE

The CMMS XML file is saved in the default path -
"C:\ProgramData\Honeywell\AssetManager\CMMSXML".

5. Perform the following steps in the CMMS Configuration section:
 - a. From the CauseId drop-down list, select the required cause Id.
 - b. From the CauseGroup drop-down list, select the required CMMS group for the symptom.

TIP

The ProgramId, ProgramGroup, and NotificationTypes boxes are disabled for symptoms.

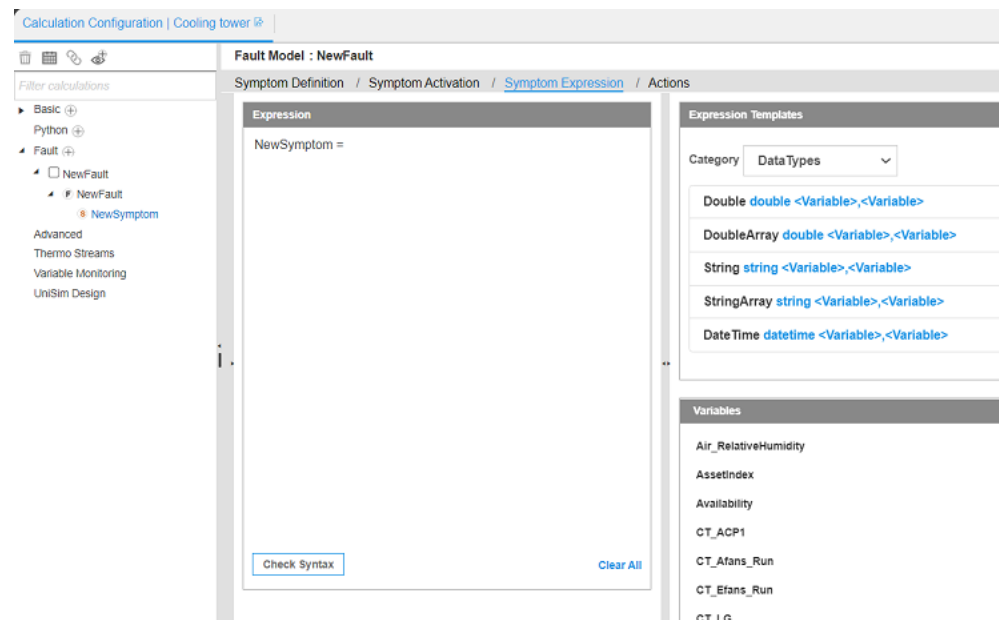
6. Click **Next**.

The Context tab is selected and the Context page is displayed.

7. On the Expression page, perform the following steps.

- a. In the Expression box, type an equation for the symptom.
- b. Click **Validate** if you want to validate your expression.

If the expression is invalid, an error message is displayed. The expression could be invalid because it had incorrect operators or attributes. Check the attributes and operators in your expression and click Validate again.



- c. Click **Next**.

The Control tab is selected on the left pane, and the Control Actions page is displayed on the right pane.

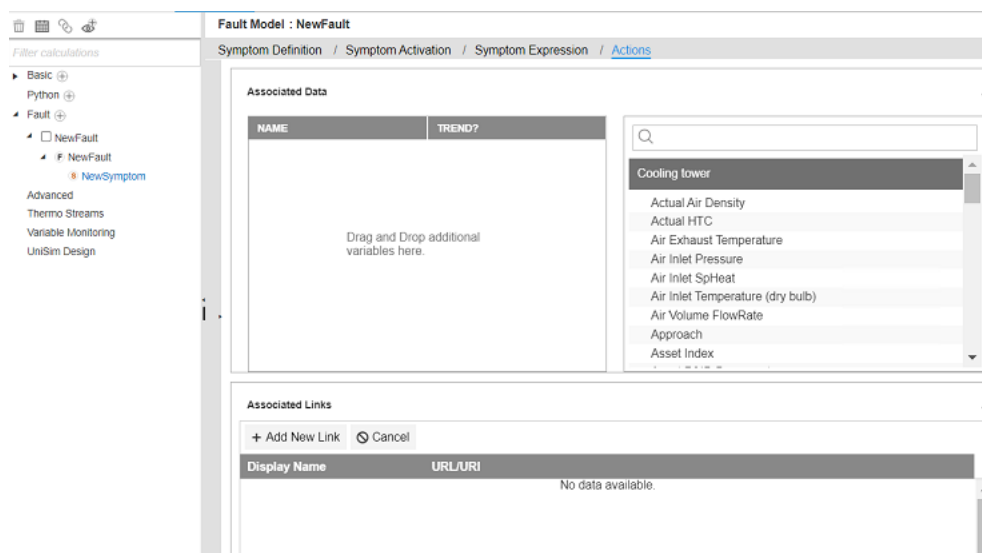
8. In the **Control Action** box, type the required control action.

- On the Models pane, click **Save** to save the model.

Similarly, you can follow the same steps to add any number of faults and symptoms and define expressions for all faults and symptoms.

TIP

To delete a symptom, right-click the symptom and select **Delete**. You cannot rename the symptom.



2.5.1.4 Making Custom Variables Available for E-mail Notifications

Custom functions on the system can contain custom variables. By default, these are not available information when sending an e-mail notification (see Notifications on page 166). To make the variable available for e-mail notifications, include a line in the fault expression for every required custom variable using the following syntax:

[CustomVariable1] = [CustomVariable1]

2.5.2 Creating a Calculation Model

Calculation models are created with real-time data and can be referred to or reused when providing expressions for faults and symptoms.

Before you create a calculation model, you must have defined the asset class, asset type, and attributes.

2.5.2.1 Adding a Calculation Model

- On the Contents tree, click **Configuration > Calculation**.

The Calculation Configuration page appears.

- On the Asset tree, expand the Asset Class and then select **Asset**.
- In the viewer, on the Calculation Configuration page, in a Basic model, click the plus icon +.

A new calculation model is created and displayed under the Models pane. A calculation "NewCalculation" is added by default to the calculation model.

2.5.2.2 Add a Calculation to a Calculation Model

1. In the Associated Models section, expand the calculation model and click NewCalculation. The General tab is selected on the right pane and the General page is displayed.
2. In the General page, perform the following steps:
 - a. In the **Name** box, type the name of the calculation.

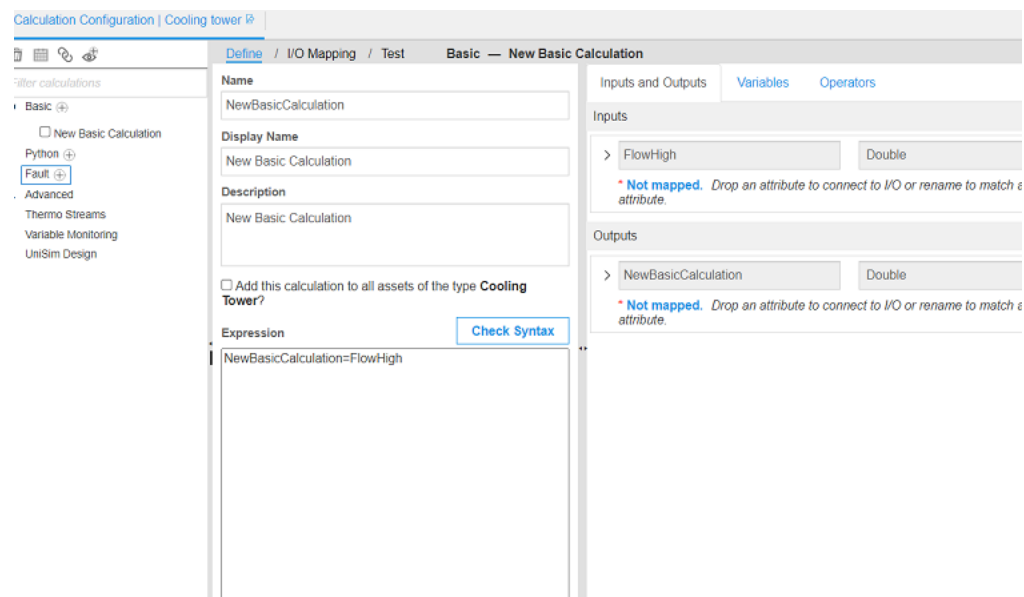
NOTE

Ensure that:

- The first character of the calculation name is alphabetic.
- The calculation name does not contain the underscore (_) character.
- The calculation name does not begin with C_ (this results in an error and the fault cannot be saved).
- The remaining characters in an attribute name can be a combination of symbols (-,\$,%,.) and numbers.
- The maximum length of the attribute name does not exceed 300 characters.

- b. In the **Display Name** box, type a name for the calculation in the localized language.
- c. In the **Description** box, type the description of the calculation.
- d. Click **Next**.

The **Expression** tab is selected and the **Expression** page is displayed. On this page, you can define expressions in the **Expression** box using the attributes listed in the Attributes box. The attributes created are listed in the **Attributes** box.



3. On the Expression page, perform the following:
 - a. In the **Expression** box, type the expression for the calculation.

NOTE

If you are creating a data-driven calculation, ensure that any variables used in the later calculations include the model name if they do not exist on the same model as the output variable. For example, to use the variable attribute2 from a different model, use ParentAsset.attribute2, instead of just attribute2.

4. Click **Check Syntax** to validate your expression.

If the expression is invalid, an error message is displayed. The expression could be invalid because it had incorrect operators or attributes. Check the attributes and operators in your expression and click **Check Syntax** again.

NOTE

You can also add/modify the details of a model. For the modified values to be available in the calculations and fault models, you must click Apply Changes on the Datasource Status page. (On the Contents tree, **Detail Views > Datasources Status**.)

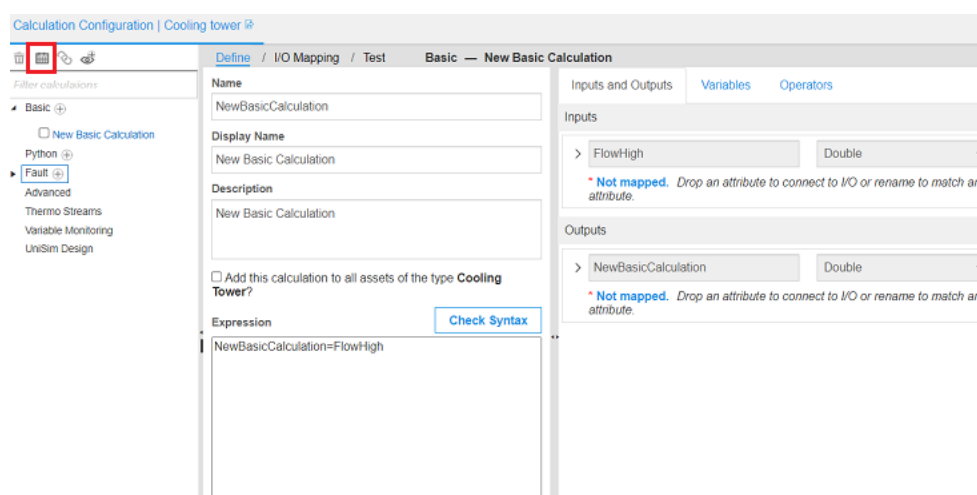
TIP

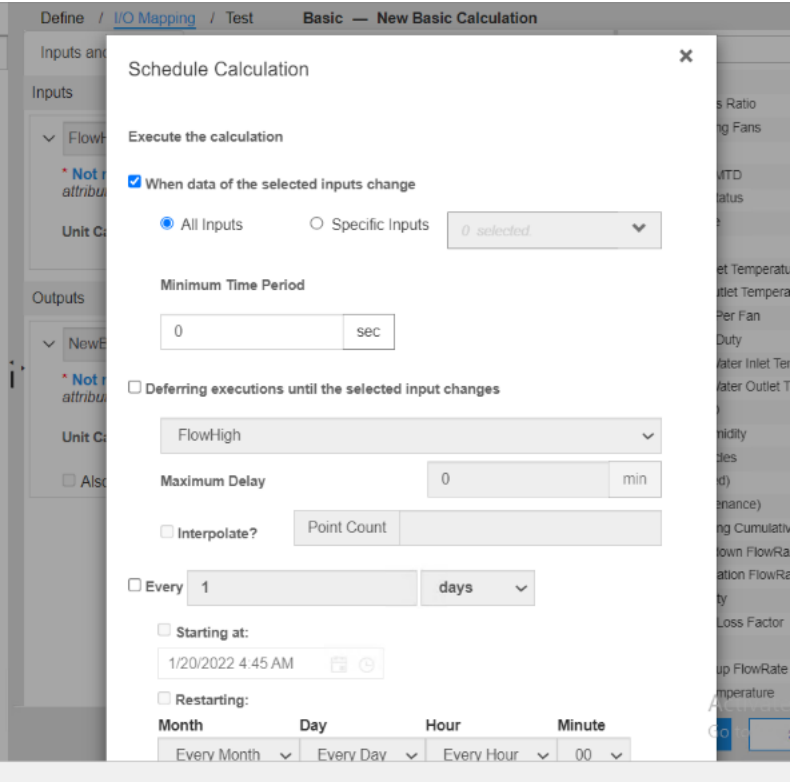
To delete a calculation, right-click the calculation and select **Delete**. You cannot rename the calculation.

To group calculations in the list, create a calculation category and add you want to group to calculations to it.

2.5.2.3 Scheduling a Calculation

1. Select the calculation name and click the **Schedule**  icon.
2. In the **schedule calculation** dialog box, do one of the following:
 - a. To schedule a data-driven calculation, select **when data changes**.
 - b. To schedule the calculation to run on an interval, select **Every**, and then enter the interval length and units.
3. Click **Save**.

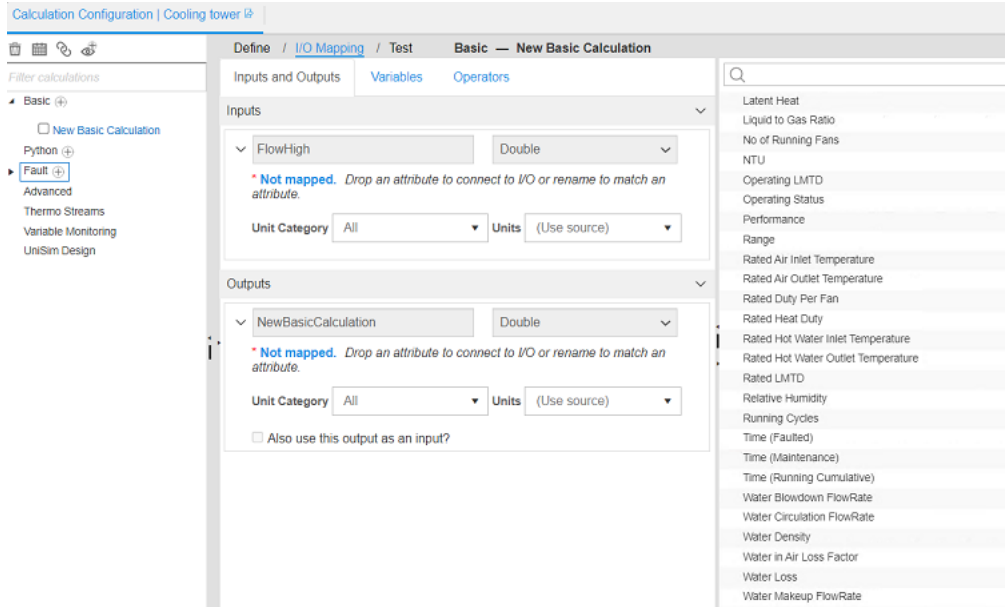




2.5.2.4 I/O Mapping

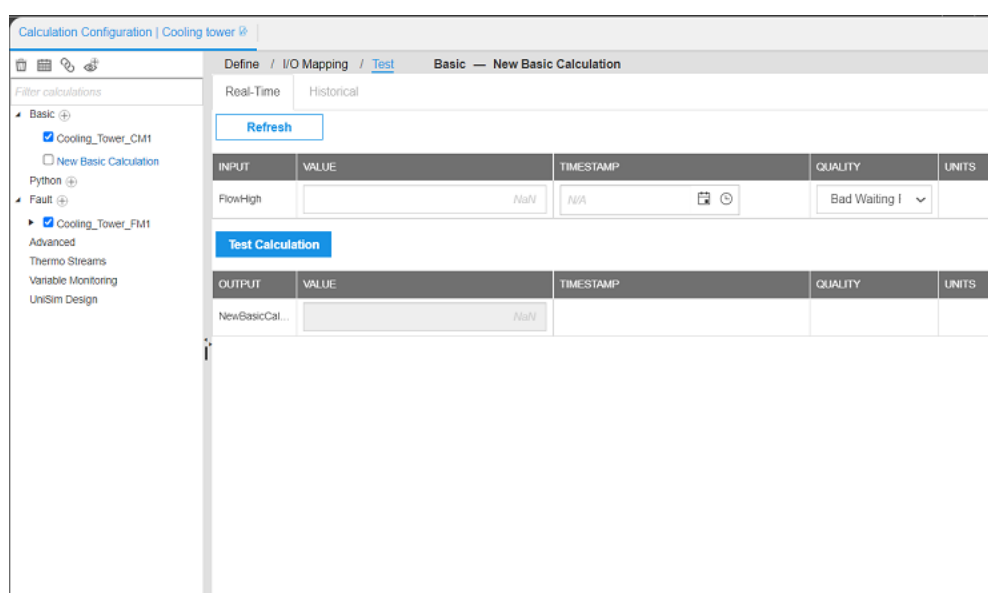
In I/O mapping, you can provide real-time data or constant value, and map input and output by dragging down or dragging variables from the same pane.

Click **I/O Mapping** in the Basic calculation.



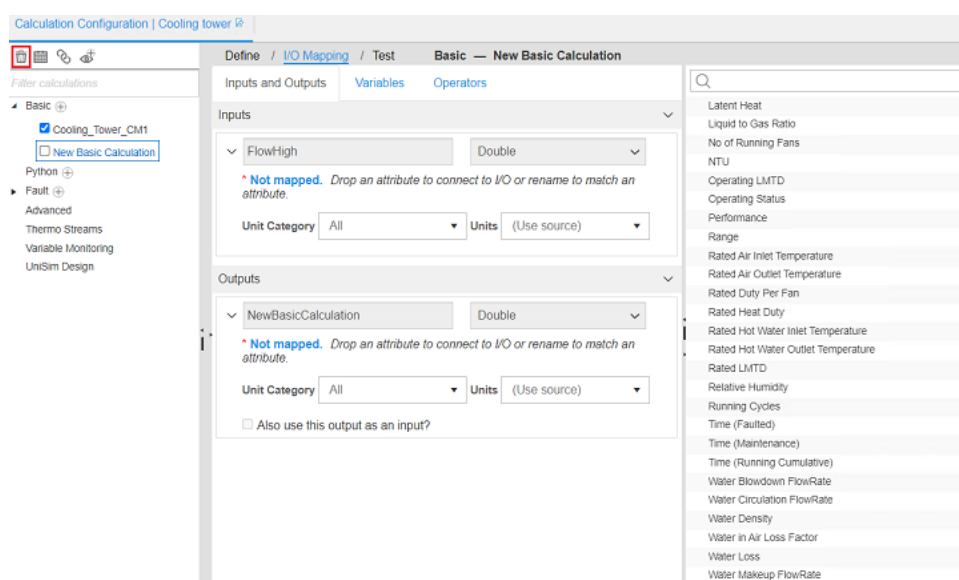
2.5.2.5 Test

Click the **Test** tab on the same pane to get into the test pane and add the input value and test calculations provided.



2.5.2.6 Calculation Delete

To delete any basic calculation, select calculation and Click  icon on the top left corner of the calculation configuration pane.



2.5.2.7 Using the Downsample Custom Function in a Calculation

Downsampling increases the interval between data samples by removing extra points. Downsampling can be useful for simplifying analysis in cases where the resolution of the data is too high. This custom function is used in cases where you have a large quantity of data and can trade off the accuracy of analyzing all the data for the increased speed of analyzing a smaller amount of downsampled data.

You can use the Downsample custom function in a calculation to define a Downsampling interval, which sets the element to keep every nth sample and eliminate the rest. An optional lag can be defined to set how many samples to skip at the beginning of the data set.

You can use the Downsample custom function in a calculation using an expression of the following format:

C_Downsample(DownsampleInterval, InputArray, StartLag), where:

DownsampleInterval is the Downsample interval (the number of interval samples to skip)

InputArray is an array attribute (for example, a Double Array)

StartLag is the number of intervals to skip at the beginning of the data set.

1. Create a calculation as described in **To add an instance calculation**.

NOTE

To use a custom function, ensure that you are using the instance calculation page referenced in the linked procedure (Configurations > Calculations) and not a calculation added on the calculation model.

2. Select the TabularArray attribute with the required data. For this example, you can use an attribute called **TestData1**.
3. In the **Expression** box, type the following expression:

```
DownsampledOutput[] = C_Downsample(5, TestData1[], 0)
```

2.5.3 Asset Model XML Import

After upgrading to R540.1, the models used in the previous versions of APM may experience some errors or the model may not execute. To avoid this issue, import the latest individual asset model xml.

Below is the list of updated Asset Types, Asset Classes and their respective xml names:

XML Name	Asset Class	Asset Type
AMExport_XML	Bearing	Brg_Bearing
ATTENTION Ensure only the listed Asset Class and Asset type is selected during import. Selecting other assets from this XML may replace the latest model data.		

XML Name	Asset Class	Asset Type
	Brakes	CAT_VIMS_ABL11_BR

XML Name	Asset Class	Asset Type
	Column	AS_DistillationColumn

XML Name	Asset Class	Asset Type
	Engine	CAT_3516B_VIMS_ABL11

XML Name	Asset Class	Asset Type
	Engine	AS_Engine_Diesel
	Fan	AS_Fan
	Filter	AS_CartridgeFilter
	Frame and Body	CAT_VIMS_ABL11_FB
	Gear	GearType
	Heat Exchanger	AS_CoolingTower
	Heaters	AS_AirPreheater
	Heaters	HRSG
	Heaters	Boiler
	Motor	AS_Motor_DC
	Power Train	CAT_VIMS_ABL11_PT
	Process Sensor	AS_GenericSensor
	Process Stream	AS_OilGas_Stream
	Process Stream	Steam_Water_Stream
	Process Stream	AS_Water_Stream
	Process Stream	AS_MEG_ProcessStream
	Process Stream	Air_Fuel_Stream
	Process Stream	AS_Ethylene_ProcessStream
	Process Stream	AS_Aromatics_Stream
	Process Stream	Air_Stream
	Process Stream	Oil_Gas_Stream
	Shaft	ShaftType
	Pump	AS_ESP
	Reactor	AS_GenericReactor
	Steam Tables	SuperheatedSteamProperties
	Steam Tables	SatT to SatP
	Steam Tables	SatP to SatT
	Steering	CAT_VIMS_ABL11_ST
	Telemetry	CAT_VIMS_ABL_TM
	Tires	CAT_VIMS_ABL_TR
	Valve	AS_Valve_Liquid
	Valve	AS_Valve_Aux
	Valve	AS_Valve_Two Phase

XML Name	Asset Class	Asset Type
AMExport_XML_BucketWheelExcavator	MiningAsset	Bucket Wheel Excavator - Belt Conveyor System
	MiningAsset	Bucket Wheel Excavator - Bucket Wheel System
	MiningAsset	Bucket Wheel Excavator - Cooling System
	MiningAsset	Bucket Wheel Excavator - Electric System
	MiningAsset	Bucket Wheel Excavator - Lower Works
	MiningAsset	Bucket Wheel Excavator - Steering System
	MiningAsset	Bucket Wheel Excavator - Transmission System
	MiningAsset	Bucket Wheel Excavator - Upper Works
AMExport_XML_CentrifugalBlower	Blower	Blower - Gearbox System
	Blower	Blower - Lube Oil System
	Blower	Blower - Unit
	Blower	Centrifugal Blower - Constant Speed
	Blower	Centrifugal Blower - Variable Guide Vane
	Blower	Centrifugal Blower - Variable Speed
AMExport_XML_CentrifugalCompressor	Compressor	Centrifugal - Constant Speed V2
	Compressor	Centrifugal - Variable Guide VaneV2
	Compressor	Centrifugal - Variable Speed V2
	Compressor	Centrifugal Compressor - Gearbox System
	Compressor	Centrifugal Compressor - Lube Oil System
	Compressor	Centrifugal Compressor - Seal Gas System
	Compressor	Centrifugal Compressor - Unit
AMExport_XML_CentrifugalCompressor_V3	Compressor	Centrifugal_ConstantSpeed_V3
	Compressor	Centrifugal_VariableGuideVane_V3
	Compressor	Centrifugal_VariableSpeed_V3

XML Name	Asset Class	Asset Type
AMExport_XML_CentrifugalPump	Bearing	Brg_Bearing
	Pump	Centrifugal
	Pump	Centrifugal
	Pump	Centrifugal
	Pump	Centrifugal
	Impeller	Pmp_Impeller
	Gear	GearType
	Shaft	ShaftType
	Pump	Centrifugal Pump - Bearing System
	Pump	Centrifugal Pump - Dry Gas Seal System
	Pump	Centrifugal Pump - Lube Oil System
	Pump	Centrifugal Pump - Mechanical Seals
AMExport_XML_Centrifuge	Centrifuge	Centrifuge - Gearbox System
	Centrifuge	Centrifuge - Bearing System
	Centrifuge	Laval Disc Centrifuge
	Centrifuge	Bird Solid Bowl Centrifuge
	Centrifuge	Centrifuge - Lube Oil System
AMExport_XML_ConveyorBelt	MiningAsset	Conveyor Belt - Belt System
	MiningAsset	Conveyor Belt - Brake System
	MiningAsset	Conveyor Belt - Drive System
	MiningAsset	Conveyor Belt - Pully System
	MiningAsset	Conveyor Belt - Transmission System
AMExport_XML_CrawlerDozer	MiningAsset	Crawler Dozer - Battery System
	MiningAsset	Crawler Dozer - Brake System
	MiningAsset	Crawler Dozer - Cooling System
	MiningAsset	Crawler Dozer - Engine System
	MiningAsset	Crawler Dozer - Fuel System
	MiningAsset	Crawler Dozer - Hydraulic System
	MiningAsset	Crawler Dozer - Steering System
	MiningAsset	Crawler Dozer - Transmissions System

XML Name	Asset Class	Asset Type
AMExport_XML_DieselEngine	MiningAsset	Hydraulic Shovel- Fuel System
	MiningAsset	Hydraulic Shovel- Engine System
	MiningAsset	Hydraulic Shovel- Cooling System
	MiningAsset	Hydraulic Shovel- Battery System
AMExport_XML_ElectricRopeShovel	MiningAsset	Electric Rope Shovel - Battery System
	MiningAsset	Electric Rope Shovel - Cooling System
	MiningAsset	Electric Rope Shovel - Electric System
	MiningAsset	Electric Rope Shovel - Lower Works
	MiningAsset	Electric Rope Shovel - Steering System
	MiningAsset	Electric Rope Shovel - Transmission System
	MiningAsset	Electric Rope Shovel - Upper Works
AMExport_XML_EMotor	Motor	EMotor
AMExport_XML_Extruder	Extruder	AT_Extruder
	Extruder	Extruder - Bearing
	Extruder	Extruder - Lube Oil System
	Extruder	Extruder - Transmission
AMExport_XML_GasTurbine	Turbine	Gas Turbine - Lube Oil System
	Gas Turbine	Gas Turbine - Multi Shaft Simple Cycle
	Gas Turbine	Gas Turbine - Single Shaft Simple Cycle
	Turbine	Gas Turbine- Air Intake
	Turbine	Gas Turbine- Bearing System
	Turbine	Gas Turbine- Exhaust System
	Turbine	Gas Turbine- FuelGas System
	Turbine	Gas Turbine- Ventilation System
AMExport_XML_HaulTruck	MovingMiningAsset	HaulTruck_Fuel_System
	MovingMiningAsset	HaulTruck_Battery_System
	MovingMiningAsset	HaulTruck_Cooling_System
	MovingMiningAsset	HaulTruck_Engine_System
	MovingMiningAsset	HaulTruck_Transmission_System
	MovingMiningAsset	HaulTruck_Brake_System
	MovingMiningAsset	HaulTruck_Suspension_System
	MovingMiningAsset	HaulTruck_Steering_System

XML Name	Asset Class	Asset Type
AMExport_XML_Heater	Heaters	Combustor
	Heaters	UltimateAnalyzer
AMExport_XML_HydraulicShovel	MiningAsset	Hydraulic Rope Shovel- Battery System
	MiningAsset	Hydraulic Rope Shovel- Brake System
	MiningAsset	Hydraulic Rope Shovel- Cooling System
	MiningAsset	Hydraulic Rope Shovel- Engine System
	MiningAsset	Hydraulic Rope Shovel- Fuel System
	MiningAsset	Hydraulic Rope Shovel- Hydraulic System
	MiningAsset	Hydraulic Rope Shovel- Steering System
	MiningAsset	Hydraulic Rope Shovel- Transmission System
AMExport_XML_LiquidRingCompressor	Compressor	Liquid Ring Compressor
	Liquid Ring	Liquid Ring Compressor - Seal Gas System
	Liquid Ring	Liquid Ring Compressor - Unit
AMExport_XML_Mixer	Mixer	AM_Mixer
AMExport_XML_Motor	Motor	AS_Motor_AC
AMExport_XML_MultiInjection_ StreamTurbine	Turbine	MIST_Bearing_System
	Turbine	MIST_LubeOil_System
	Turbine	MIST_Seal_System
AMExport_XML_ PlateAndFrameHeatExchanger	Heat Exchanger	Plate and Frame Heat Exchanger

XML Name	Asset Class	Asset Type
AMExport_XML_PositiveDisplacementPump	Pump	DiaphragmPump_AirOperated
	Pump	DiaphragmPump_HydraulicOperated
	Pump	DiaphragmPump_MotorOperated
	Pump	DiaphragmPump_Transmission
	Pump	Gear Pump
	Pump	GearPump_Bearing
	Pump	GearPump_Transmission
	Pump	Plunger Pump Bearing System
	Pump	Plunger Pump LO System
	Pump	Plunger Pump Seal System
	Pump	Reciprocating Pump
	Pump	Reciprocating Pump - Bearing System
	Pump	Reciprocating Pump - Heat Exchanger System
	Pump	Reciprocating Pump - Main LO Pump System
	Pump	Reciprocating Pump - Transmission System
	Pump	Screw Pump
	Pump	ScrewPump_Bearing
	Pump	ScrewPump_Main
	Pump	ScrewPump_Transmission
AMExport_XML_ReciprocatingCompressor	Compressor	AS_Piston_Compressor
	Reciprocating Compressor	Reciprocating Compressor - Stage 1
	Reciprocating Compressor	Reciprocating Compressor - Stage 2
	Reciprocating Compressor	Reciprocating Compressor Bearing System
	Reciprocating Compressor	Reciprocating Compressor Cooling System
	Reciprocating Compressor	Reciprocating Compressor Lube Oil System
AMExport_XML_RotaryVacuumFilter	Filter	Rotary Vacuum Filter - RVF Drum
	Filter	Rotary Vacuum Filter - Screw Conveyor and Dryer
	Filter	Rotary Vacuum Filter - Vacuum System
AMExport_XML_RPF	Filter	Rotary Pressure Filter
	Filter	Rotary Valve

XML Name	Asset Class	Asset Type
AMExport_XML_ScrewCompressor	Screw Compressor	AS_Screw_Compressor_Oil_Free
	Screw Compressor	AS_Screw_Compressor_Oil_Free_VariableSlideValve
	Screw Compressor	AS_Screw_Compressor_Oil_Free_VariableSpeed
	Screw Compressor	AS_Screw_Compressor_Oil_Injected
	Screw Compressor	AS_Screw_Compressor_Oil_Injected_VariableSlideValve
	Screw Compressor	AS_Screw_Compressor_Oil_Injected_VariableSpeed
	Screw Compressor	Screw Compressor - Bearing System
	Screw Compressor	Screw Compressor - Intercooler System
	Screw Compressor	Screw Compressor - LubeOil System
	Screw Compressor	Screw Compressor - Seal Oil System
	Screw Compressor	Screw Compressor -Dry Gas Seal System
AMExport_XML_Separator	Separator	Two-Phase Separator
AMExport_XML_ShellTubeHeatExchangerV2	Heat Exchanger	Shell and Tube Heat Exchanger V2
AMExport_XML_SteamTurbine	Heaters	Condenser
	Heaters	Desuperheater
	Turbine	Steam Turbine - Bearing System
	Turbine	Steam Turbine - Constant Speed
	Turbine	Steam Turbine - Extractive
	Turbine	Steam Turbine - Lube and Control Oil System
	Turbine	Steam Turbine - Seal System
	Turbine	Steam Turbine - Variable Speed
AMExport_XML_Stream	Process Stream	AS_FG_ProcessStream
	Process Stream	AS_LubeOilStrem

XML Name	Asset Class	Asset Type
AMExport_XML_TurboExpander	Turbine	Turbo Expander - Bearing
	Turbine	Turbo Expander - Lube Oil System
	Turbine	Turbo Expander - Seal System
	Turbine	Turbo-Expander Single Shaft Single Unit
	Turbine	Turbo-Expander Single Shaft Single Unit-Constant Speed
	Turbine	Turbo-Expander Single Shaft Single Unit-Variable Guide Vane
	Turbine	Turbo-Expander Single Shaft Single Unit-Variable Speed

ATTENTION

You must import the above models only if the model is used before upgrading to R540.1 or if any of the models listed above require reconfiguration.

Follow the below steps to import an Asset Model XML in Application Server:

1. In the installation media, navigate to the following folder.
<Installation Media>\StandardLibrary\Model Library
2. Create Model Library folder in one of the system drives.
3. Copy the XML files from the media to the created folder.
4. Navigate to the following AMImportExport folder:
<Install Location>:\Program Files(x86)\Honeywell\MES\AssetManager\Tools\AMImportExport
5. Right-click **ImportModels** and select **Run as administrator**.
6. **Asset Manager Model Tool** window appears.
7. Click **Source** drop-down.
8. Select **Asset Manager Export File**.
9. A dialog box to select the Model XML appears.
10. Navigate to the created Model Library folder.
11. Select the individual Model XML (the models given in the above table) to import.
12. Click the check box against the **Asset Class** to import.
13. Click **Copy**.
14. The copy progress is displayed at the bottom of the **Asset Manager Model Tool** window.
15. After the copy is completed the number of models successfully copied is displayed at the bottom of the **Asset Manager Model Tool** window.
16. Click **View log** to check if any errors have occurred while copying.
17. ImportModelLog Notepad appears.
18. Scroll down to check for errors during the copy.

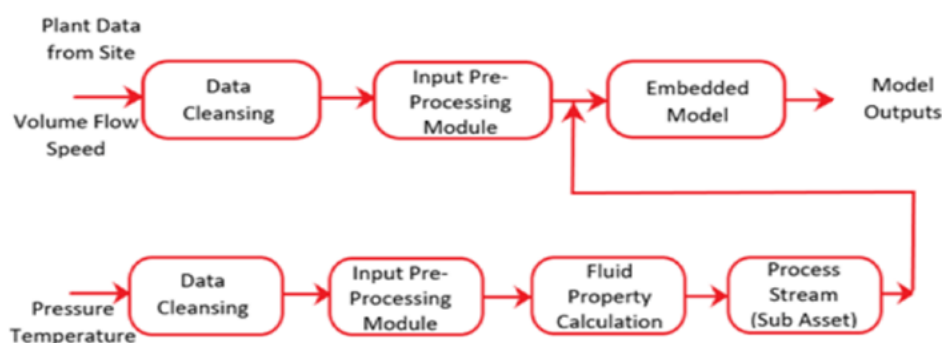
2.5.4 Input Pre-processing Module and Volumetric Flow Calculations

2.5.4.1 Introduction

This section focuses on the input pre-processing module for the centrifugal compressor and process stream asset types and volumetric flow calculations.

2.5.4.2 Centrifugal Compressor - Input Pre-processing Module

- Input preprocessing module will filter NaN, null, and bad values by specifying the operating window outside which an asset cannot operate for that particular operating case.
- You can specify those operating windows by defining the lower and upper limit for a particular variable.
- The configuration is templated at the asset level (in case of volume flow, speed and inlet guide vane positions) and at the stream level (in case of temperature and pressure). The input preprocessing module adopts dynamically based on the asset value. Hence, there is no need to configure the expressions for all assets of the same asset type.
- In the Process Stream assets, in addition to lower and higher limits, a default attribute value (cleansed value) should be defined for handling NaN, null or bad conditions.
- You can edit or update the error message for each variable.



Centrifugal Compressor Asset Type

Speed

- If compressor speed operates below its lower limit, running status will be set to 0 and no calculations are executed. An error message is displayed in the calculation status page to notify the customer.
- In case speed value is NaN, null or bad, compressor speed will be set to operate at its lower limit, and a warning is displayed in the calculation status page.
- In case of higher limit, only a warning is displayed in the calculation status page.

Volume Flow

- If volume flow of compressor is NaN, null, bad, or operating at lesser than its lower limit, then the volume flow is set to operate at its lower limit and a warning is displayed.
- In case volume flow exceeds the higher limit, compressor is set to operate at its higher limit and a warning is displayed.

Process Stream Asset Type

Pressure

- If a process stream pressure has NaN, null, bad or cleansed value, pressure for that stream is set to operate at its cleansed pressure value, and a warning is displayed.
- If a process stream pressure is operating at lesser than lower limit, then the stream pressure is set to operate at its lower limit, and a warning is displayed.
- If a process stream pressure is operating at higher than higher limit, then the stream pressure is set to operate at its higher limit, and a warning is displayed.

Temperature

- If a process stream temperature has NaN, null, bad or cleansed value, the temperature for that stream is set to operate at cleansed Temperature value, and a warning is displayed.
- If a process stream temperature is operating lesser than lower limit, the temperature value for that stream is set to operate at that lower limit, and a warning is displayed.
- If a process stream temperature is operating higher than higher limit, the particular stream temperature is set to operate at that higher limit, and a warning is displayed.

2.5.4.3 Centrifugal Compressor Asset - Input Pre-processing Module Configuration

1. Open any web browser.

Type `https:// <machine name>:<port>/MES/Default.aspx#/APM/APMLandingPage.aspx` in the address bar, where the **<machine name>** is fully qualified domain name of the APM server and **<port>** is the configured SSL port.

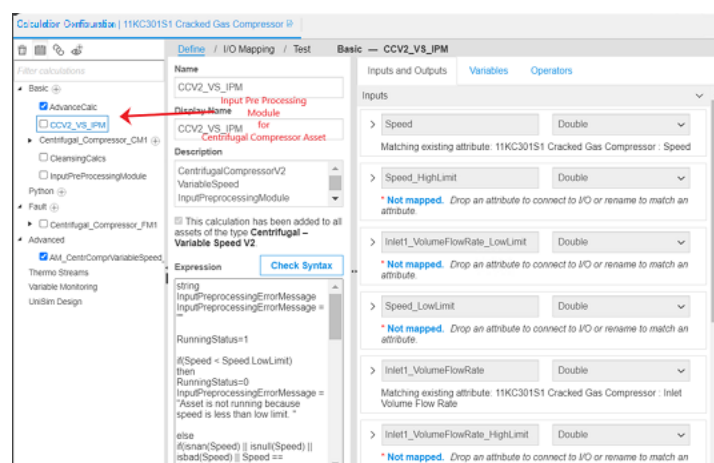
The **Honeywell Application** landing page appears.

2. In the **Asset Performance Management** area, click **Asset Performance Management**.

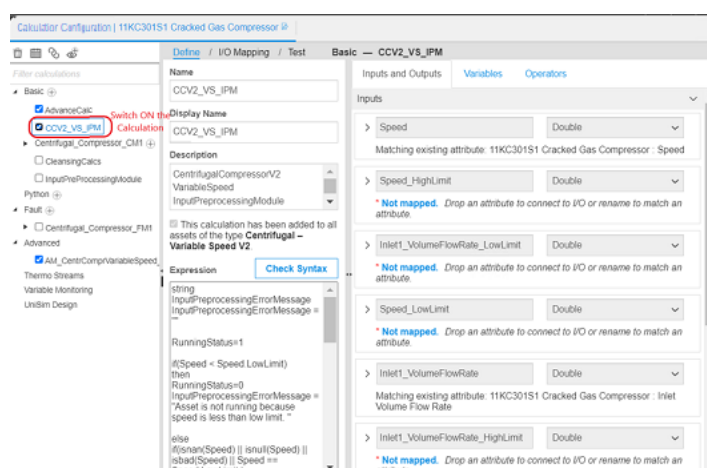
The **Asset Performance Management** page appears.

3. Select **Centrifugal Compressor Asset** from **ASSETS** pane.
4. The **CONTENTS** pane displays a list of different screen options.
5. From the **Configuration** drop-down menu, click **Calculations**.

Calculation Configuration pane for the selected asset appears on the right.



6. To enable calculation, click one of the following checkboxes under **Basic** calculations:
 - a. **CCV2_VS_IPM**: If the compressor is of Variable Speed type.
 - b. **CCV2_IGV_IPM**: If the compressor is of Inlet Guide Vane Type.
 - c. **CCV2_CS_IPM**: If the compressor is of Constant Speed type.



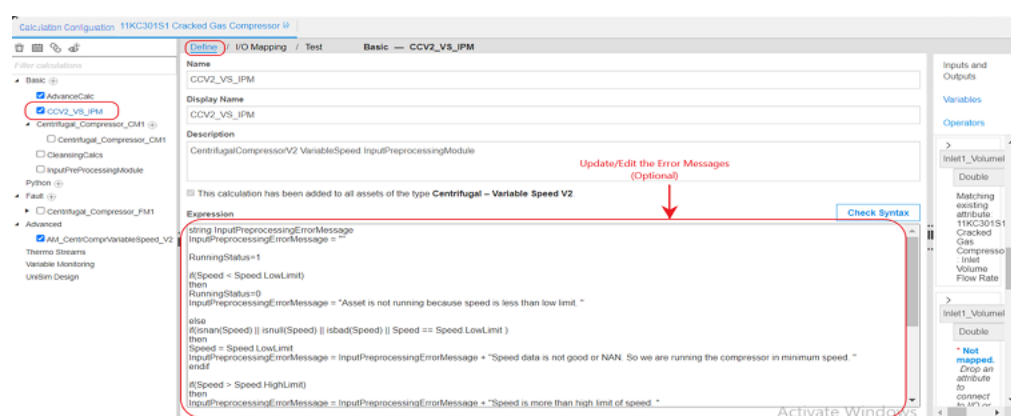
7. In **CONTENTS** pane, under **Dashboards** drop-down menu, click **Attribute Overview**.

8. The **Attribute Overview** pane for the selected asset appears on the right.

Enter the **LOW LIMIT** and **HIGH LIMIT** for Speed, Guide vane position, and Volume Flow Rate Parameters.

9. In the **Configuration** drop-down menu, click **Calculations**.

Edit the **Expression** section in **Define Tab** to customize the error message for the selected asset.



NOTE

Customization of error message is an optional procedure.

Centrifugal Compressor - Input Pre-processing Module Outputs

SI No	Parameter	Source
1	Speed	Set based on Conditions
2	Volume Flow	Set based on Conditions
3	Actual Volumetric Flow	Set based on Conditions
4	Inlet Guide Vane Position	Set based on Conditions

SI No	Parameter	Source
5	Running Status	Set based on Conditions
6	Input Pre Processing Error Message	Set based on Conditions

Centrifugal Compressor - Input Pre Processing Module Inputs

S No	Parameter	Source
1	Speed	Tag Value
2	Volume Flow	Tag Value
3	Actual Volumetric Flow	Tag Value
4	Inlet Guide Vane Position	Tag Value
5	Speed - Lower Limit	User Input
6	Speed - Higher Limit	User Input
7	Volume Flow - Lower Limit	User Input
8	Actual Volumetric Flow - Lower Limit	User Input
9	Volume Flow - Higher Limit	User Input
10	Actual Volumetric Flow - Higher Limit	User Input
11	Guide Vane Position - Lower Limit	User Input
12	Guide Vane Position - Higher Limit	User Input

2.5.4.4 Process Stream Asset - Input Pre Processing Module Configuration

1. Open any web browser.

Type `https:// <machine name>:<port>/MES/Default.aspx#/APM/APMLandingPage.aspx` in the address bar, where the **<machine name>** is fully qualified domain name of the Asset Performance Management server and **<port>** is the configured SSL port.

The **Honeywell Application** landing page appears.

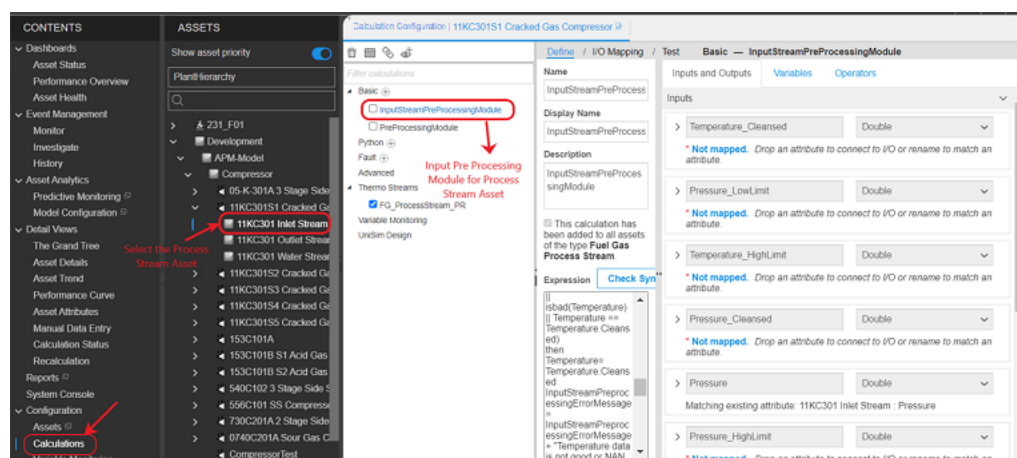
2. In the **Asset Performance Management** area, click **Asset Performance Management**.

The **Asset Performance Management** page appears.

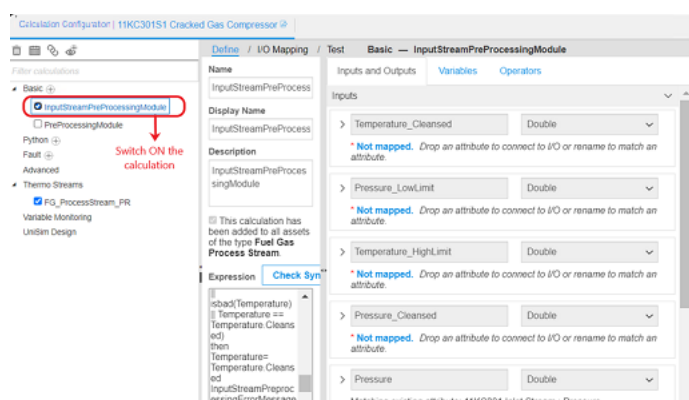
3. Select **Process Stream Asset** from **ASSETS** pane.
4. The **CONTENTS** pane displays a list of different screen options.

- From the **Configuration** drop-down menu, click **Calculations**.

Calculation Configuration pane for the selected process stream asset appears on the right.



- To enable calculation, click **InputStreamPreProcessingModule** under **Basic** calculations.

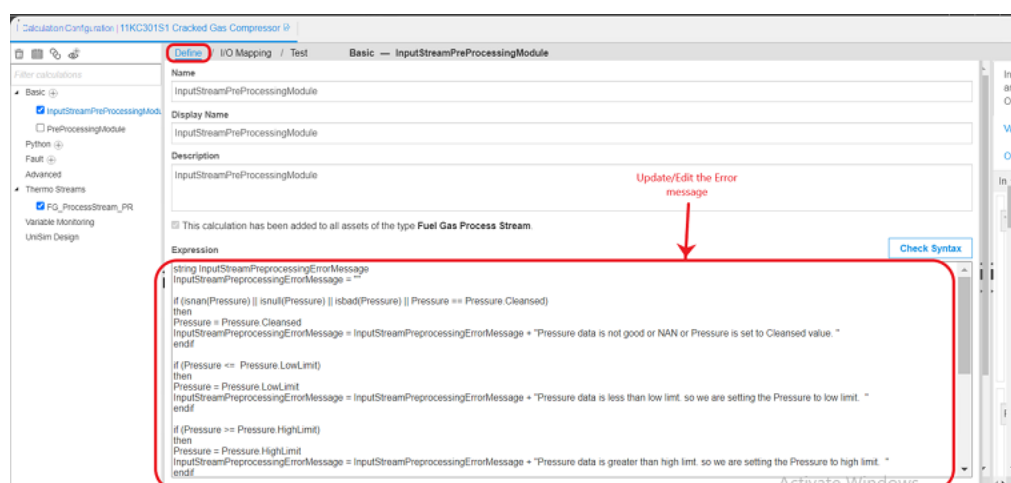


- From **CONTENTS** pane click **Attribute Overview** under **Dashboards** drop-down.
- The **Attribute Overview** pane for the selected process stream asset appears on the right.

Enter the **LOW LIMIT**, **HIGH LIMIT** and **Cleansed Value** for Temperature and Pressure Attributes.

- From the **Configuration** drop-down menu, click **Calculations**.

Edit the **Expression** section in **Define** Tab to customize the error message for the selected asset.



NOTE

Customization of error message is an optional procedure.

Process Stream - Input Pre-processing Module Inputs

SI No	Parameter	Source
1	Pressure	Tag Value
2	Temperature	Tag Value
3	Pressure - Cleansed Value	User Input
4	Pressure - Low Limit	User Input
5	Pressure - High Limit	User Input
6	Temperature - Cleansed	User Input
7	Temperature - Low Limit	User Input
8	Temperature - High Limit	User Input

Process Stream - Input Pre-processing Module Outputs

S No	Parameter	Source
1	Pressure	Set based on Conditions
2	Temperature	Set based on Conditions
3	Input Stream Pre Processing Error Message	Set based on Conditions

2.5.4.5 Flow Conversion

From any given condition (T_1, P_1), the Volumetric Flow can be calculated to the desired conditions (T_2, P_2) as follows.

$$Q(T_1, P_1) = \frac{F}{\rho(T_1, P_1)}$$

$$Q(T_2, P_2) = \frac{F}{\rho(T_2, P_2)}$$

$$\frac{Q(T_1, P_1)}{Q(T_2, P_2)} = \frac{\rho(T_2, P_2)}{\rho(T_1, P_1)}$$

$$Q(T_2, P_2) = Q(T_1, P_1) \frac{\rho(T_1, P_1)}{\rho(T_2, P_2)}$$

2.5.5 Standardizing Units of Measurement for Pressure Variable Type

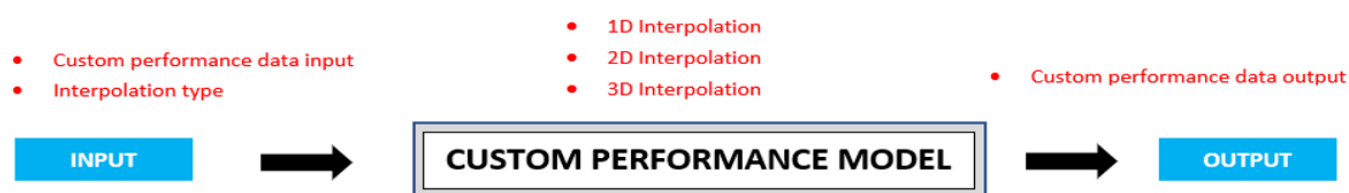
Below configuration changes are required to update type level UoM for the pressure variable:

Asset Type Name	Attribute Names	Changes Required
Centrifugal compressor - Variable speed	Inlet1_Pressure	Update type level UOM from Barg to Kg/cm ² (a)
Centrifugal compressor - Variable guide vane	Outlet1_Pressure	
Centrifugal compressor - Constant speed		

Asset Type Name	Attribute Names	Changes Required
Turbo expander - Single shaft single unit	Inlet1_Pressure	
Turbo expander - Single shaft single unit - Variable speed	Outlet1_Pressure	
Screw compressor - Oil free	Inlet1_Pressure	Update type level UOM from Bar to Kg/cm ² (a)
Screw compressor - Oil free - Variable speed	Outlet1_Pressure	
Screw compressor - Oil free - Variable slide valve	Inlet2_Pressure	
Screw compressor - Oil Injected - Variable Slide valve	Outlet2_Pressure	
Centrifugal blower - Variable guide vane	Inlet1_Pressure	• Update type level UOM from Barg to Kg/cm² (a) • Update the Outlet pressure performance curve data points from Barg to Kg/cm² (a)
Centrifugal blower - Variable speed (If any asset is present)	Outlet1_Pressure Performance Data	
Liquid ring compressor	Inlet1_Pressure	• Update type level UOM from Bar to Kg/cm² (a) • Update the Outlet pressure performance curve data points from Bar to Kg/cm² (a)
	Outlet1_Pressure Performance Data	

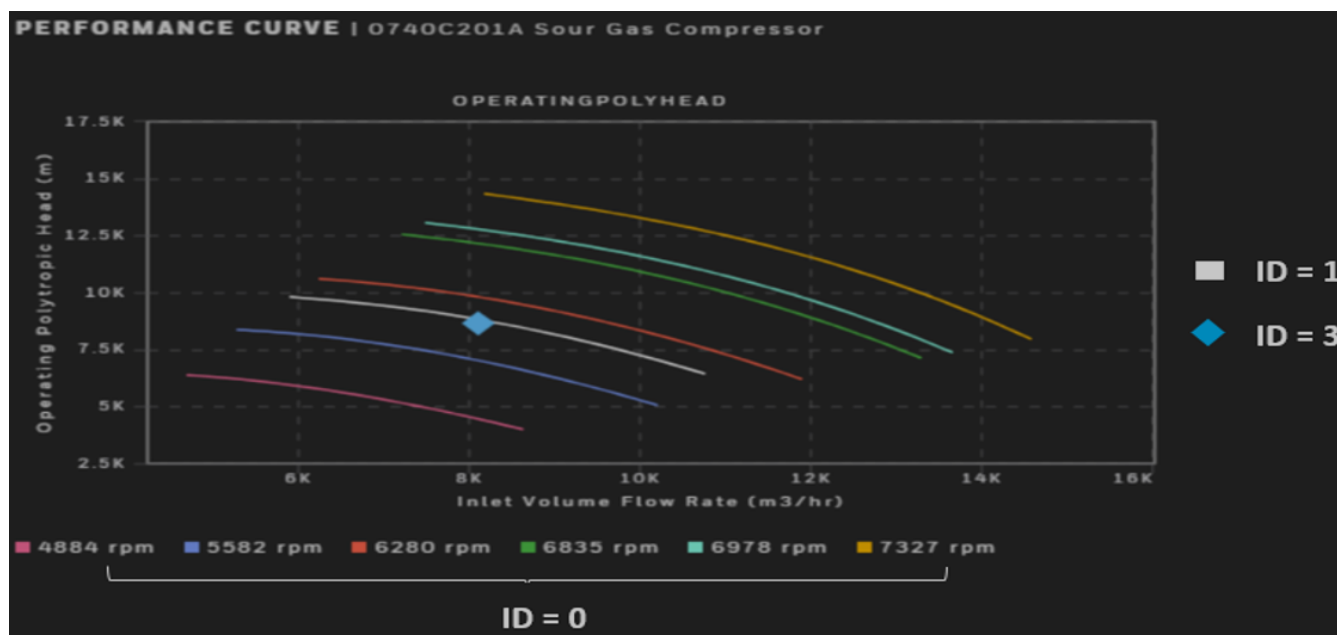
GENERIC INTERPOLATION

The Generic interpolation is an algorithm that takes input from a data table named custom performance data input and performs 1D, 2D, or 3D interpolation on user specified polynomial degrees like linear, quadratic, and cubic on input attributes named interpolation type. The model output will be a data table with a collection of data holding operating curve data points, expected data points, and current operating data points. The visualization of the final data table can be made in the performance widget in the custom curve section. This feature will be available to all asset type present in the asset model library.



3.1 Generic Interpolation Data visualization

The output data generated post interpolation in custom performance data output can be visualized or plotted on performance widget custom curve section, below is the ID 's wise representation.



3.2 Generic Interpolation Methodology

3.2.1 1D Interpolation

To perform 1D interpolation, the data points will contain (X,Y) coordinates. The custom performance data input needs to be provided in the following format. About 10 data columns of Y can be provided.

1. Custom performance data input table

COLUMN NO	1	2	3	4	5	6	7	8	9	10	11
DATA	X	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10

2. Interpolation type

User can get output table by specifying user flexible handle on interpolation type,

- 11 - 1D Linear interpolation
- 12 - 1D Quadratic interpolation
- 13 - 1D Cubic interpolation

3. Custom performance data output

Final output data table will consist of ID = 0 representing Input data, ID = 1 is operating curve points (Total 10 points), ID = 2 is expected data point from interpolation at current X, W and Z and ID = 3 is the current operating data. Same follows for all the types of interpolation

ID	1	2	3	4	5	6	7	8	9	10	11
0	X	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10
2	Current X	Expected Y data points at current X									
3		Current operating Y data points									

3.2.2 2D interpolation

To perform 2D interpolation the data points will contain only (X, Y, Z) coordinates. The custom performance data input needs to be provided in the following format.

1. Custom performance data input table

COLUMN NO	1	2	3	4	5	6	7	8	9	10	11	12
DATA	Z	X	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10

2. Interpolation type

You can get the output table by specifying the interpolation type,

- 21 - 2D Linear interpolation
- 22 - 2D Quadratic interpolation
- 23 - 2D Cubic interpolation

3. Custom performance data output

The final output data will consist of

ID	1	2	3	4	5	6	7	8	9	10	11	12
0	Z	X	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10

1	Current Z	Interpolation predicted operating curve data points (n = 10)											
1													
1													
1													
1													
1													
1													
1													
1													
1													
2	Current X	Expected Y data points at current Z and X											
3		Current operating Y data points											

3.2.3 3D interpolation

To perform 3D interpolation the data points will contain only (X, Y, Z, W) coordinates. The custom performance data input needs to be provided in the following format.

1. Custom performance data input table

COLUMN NO	1	2	3	4	5	6	7	8	9	10	11	12	13
DATA	W	Z	X	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10

2. Interpolation type

You can get the output table by specifying the interpolation type,

- 31 - 3D Linear interpolation
- 32 - 3D Quadratic interpolation
- 33 - 3D Cubic interpolation

3. Custom performance data output

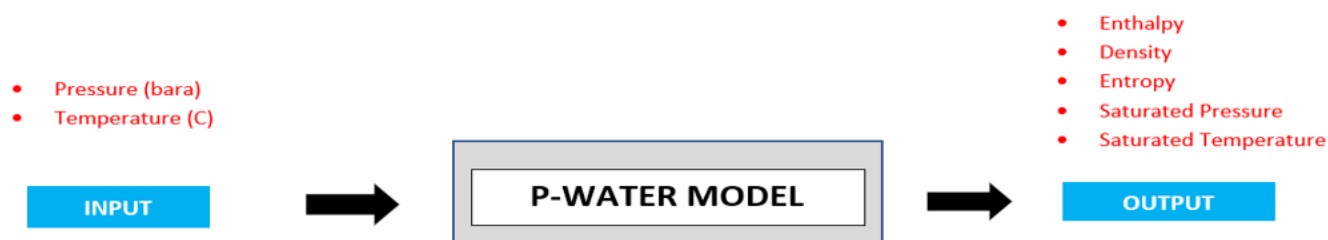
The final output data will consist of,

ID	1	2	3	4	5	6	7	8	9	10	11	12	13
0	W	Z	X	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10

1	Current W	Current Z	Interpolation predicted operating curve data points (n = 10)	
1				
1				
1				
1				
1				
1				
1				
1				
1				
2			Current X	Expected Y data points at current W, Z and X
3				Current operating Y data points

P-Water is a thermo-dynamic function used in the Asset Performance Management library to predict steam thermo-properties using a Pressure, Temperature or Pressure and Temperature flash. The p-Water function basically is a steam table, which contains thermo-properties like Enthalpy, Entropy, density, saturated pressure, and saturated temperature, etc., against steam pressure and temperature. The properties predicted from P-Water can be further used by other asset mathematical models like Turbines for predicting Isentropic and Actual KPIs.

P-Water has its predictive syntax to obtain the thermos dynamic outputs. The following syntax can also be used in the Basic calculation part of Asset Performance Management. The below image shows an overview of the model:



4.1 Methodology

The method used for predicting steam thermo-properties using the P-Water function is stated below

NOTE

Pressure needs to be passed on bara UOM whereas the temperature is °C UOM.

4.1.1 Density

$$\rho = P_{WATER} ("Density", P, T)$$

$$\rho_{liquid}^{Sat} = P_{WATER} ("SatLiqDensityTemp", T)$$

NOTE

SatLiqDensityTemp and SatLiqDensityPres predict the same thermo-properties but one uses pressure and the other uses temperature as input for prediction. The same follows for all other functions as well

$$\rho_{liquid}^{Sat} = P_{WATER} ("SatLiqDensityPres", P)$$

$$\rho_{Vapor}^{Sat} = P_{WATER} ("SatVapDensityTemp", T)$$

$$\rho_{Vapor}^{Sat} = P_{WATER} ("SatVapDensityPres", P)$$

4.1.2 Enthalpy

$$h = P_{WATER} ("Enthalpy", P, T)$$

$$h_{liquid}^{Sat} = P_{WATER} ("SatLiqEnthalpyTemp", T)$$

$$h_{liquid}^{Sat} = P_{WATER} ("SatLiqEnthalpyPres", P)$$

$$h_{Vapor}^{Sat} = P_{WATER} ("SatVapEnthalpyTemp", T)$$

$$h_{Vapor}^{Sat} = P_{WATER} ("SatVapEnthalpyPres", P)$$

4.1.3 Entropy

$$S = P_{WATER} ("Entropy", P, T)$$

$$S_{liquid}^{Sat} = P_{WATER} ("SatLiqEntropyTemp", T)$$

$$S_{liquid}^{Sat} = P_{WATER} ("SatLiqEntropyPres", P)$$

$$S_{Vapor}^{Sat} = P_{WATER} ("SatVapEntropyTemp", T)$$

$$S_{Vapor}^{Sat} = P_{WATER} ("SatVapEntropyPres", P)$$

4.1.4 Saturated Pressure

$$P_{sat} = P_{WATER} ("SatPres", T)$$

4.1.5 Saturated Temperature

$$T_{sat} = P_{WATER} ("SatTemp", P)$$

4.2 Function Usability

The basic syntax used to get the model output is given in the below example

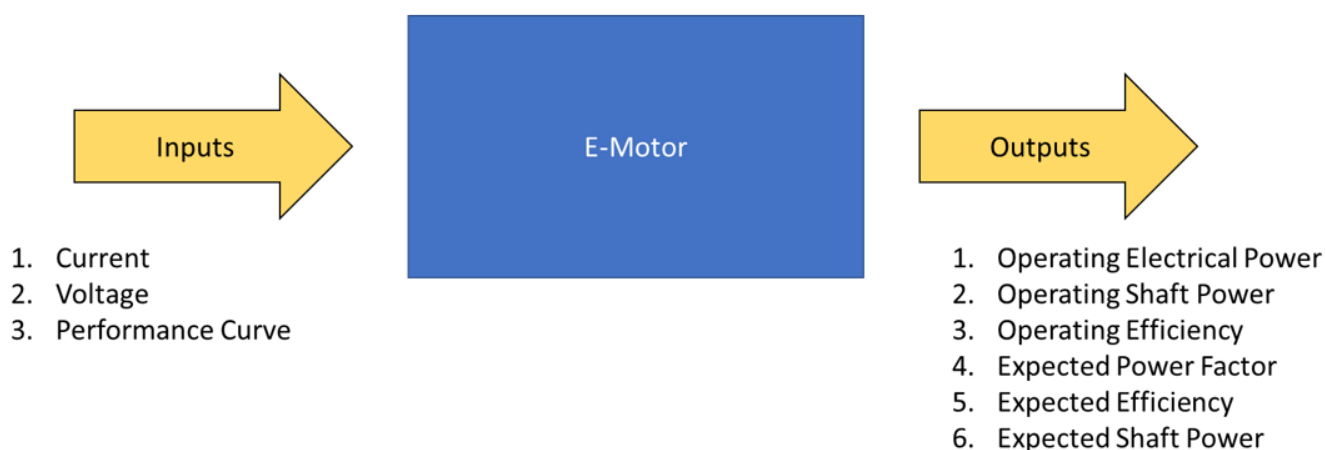
Y = P_WATER("Enthalpy",3,85)

Here, Y is the enthalpy output variable obtained at 3 bara pressure and 85C temperature.

An AC motor is an electric motor driven by an alternating current. The AC motor commonly consists of two basic parts, an outside stator having coils supplied with alternating current to produce a rotating magnetic field, and an inside rotor attached to the output shaft producing a second rotating magnetic field.

As the AC Motor used as subsystem for many industrial equipment to operate as expected, it is important to monitor the health and performance of an AC Motor to increase availability. Monitoring basic machine health data in addition to AC Motor operating conditions helps to maintain reliable operations of an AC Motor. There are three areas that helps to monitor an AC Motor performance to increase the availability.

- Process and condition monitoring
- Performance and Health monitoring



This document explains AC motor monitoring philosophy for process, performance, energy and health parameters and their importance.

5.1 AC Motor - Performance Monitoring

5.1.1 Notations

Below are the notations used in the equations of Expected KPI and Operating KPI

V	Voltage
I	Current
P	Power
P_f	Power Factor
L	Load
η	Efficiency

5.1.2 **Subscripts**

<i>Ex</i>	Expected
<i>Op</i>	Operating

5.1.3 **AC Motor - KPIs**

5.1.3.1 **Input KPI**

- Voltage Green
- Voltage Yellow
- Voltage Red
- Voltage Mean
- Current Green
- Current Yellow
- Current Red
- Current Mean
- Power Factor Constant
- Rated Power
- Rated Current
- Measured Power
- Motor Performance Data

5.1.3.2 **Output KPI**

Expected KPI

- Expected Power Factor
- Expected Shaft Power
- Expected Efficiency

Operating KPI

- Operating Electrical Power
- Operating Shaft Power
- Operating Efficiency

5.1.3.3 **AC Motor - Expected KPI**

Expected Shaft Power (kW)

$P_{Shaft} = f(L, P)$

Expected Power Factor

$P_f = f(L, P_f)$

Expected Efficiency (%)

$$\eta = f(L, \eta)$$

5.1.3.4 AC Motor - Operating KPI

NOTE

You need to either provide mean values of voltage and current or individual phase values (green, yellow, red).

$$V_{Mean} = \frac{V_{Red} + V_{Green} + V_{Yellow}}{3}$$

$$I_{Mean} = \frac{I_{Red} + I_{Green} + I_{Yellow}}{3}$$

Voltage Imbalance (%)

$$V_{Imbalance} = \frac{V_{Max} - V_{Mean}}{V_{Mean}}$$

Current Imbalance (%)

$$I_{Imbalance} = \frac{I_{Max} - I_{Mean}}{I_{Mean}}$$

Operating Electrical Power (kW)

$$P_{Electrical} = V_{Mean} * I_{Current} * P_F$$

Operating Shaft Power (kW)

$$P_{Shaft} = P_{Electrical} * \eta$$

Operating Eff (%)

$$\eta = \frac{P_{Measured}}{P_{Electrical}}$$

5.2 AC Motor - Fault Model

5.2.1 AC Motor - APM Model

Fault Name	Associated Symptoms	Recommendations
Motor degradation	Partial Discharge (PD) Fault	1. To check Partial Discharge (PD) values compared with factory test report and ROC.

Fault Name	Associated Symptoms	Recommendations
		2. With the motor stopped, check the humidity inside the motor and check the insulation level with megger and polarization index

5.2.2 AC Motor - APR Model

5.2.2.1 Motor APR Fault Model

Model Type	APR Model
Category	Motor
Equipment Class	Motor
Equipment Type	Motor AC

Fault Name	Associated Symptoms	Recommendations
DE Radial Bearing Failure RH	Radial Bearing Vibration DE X R High	1. Assess the likely cause and time to fail(trip) if intervention is not possible. 2. Intervention options; reduce speed, and restore oil properties. Parts availability. 3. Check the trends to establish if it's either both X and Y probes or one. 4. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 5. Change bearing if required.
	Radial Bearing Vibration DE Y R High	
	Radial Bearing Temp DE A R High	
	Radial Bearing Temp DE B R High	
DE Radial Bearing Failure VIB ROC	Radial Bearing Vibration DE Y R High	1. Assess the likely cause and time to fail(trip) if intervention is not possible. 2. Intervention options; reduce speed, and restore oil properties. Parts availability. 3. Check the trends to establish if it's either both X and Y probes or one. 4. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 5. Change bearing if required.
	Radial Bearing Vibration DE X R High	
	Radial Bearing Vibration DE X ROC Short-Term High	
	Radial Bearing Vibration DE Y ROC Short-Term High	
DE Radial Bearing Failure TEMP ROC	Radial Bearing Temperature DE A R High	1. Assess the likely cause and time to fail(trip) if intervention is not possible. 2. Intervention options; reduce speed, and restore oil properties. Parts availability. 3. Compare lube oil temperature/cooler temperature to determine if it is associated. 4. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 5. Change bearing if required.
	Radial Bearing Temperature DE B R High	
	Radial Bearing Temperature DE A ROC Short-Term High	
	Radial Bearing Temperature DE B ROC Short-Term High	
NDE Radial Bearing Failure RH	Radial Bearing Vibration NDE X R High	1. Assess the likely cause and time to fail(trip) if intervention is not possible.

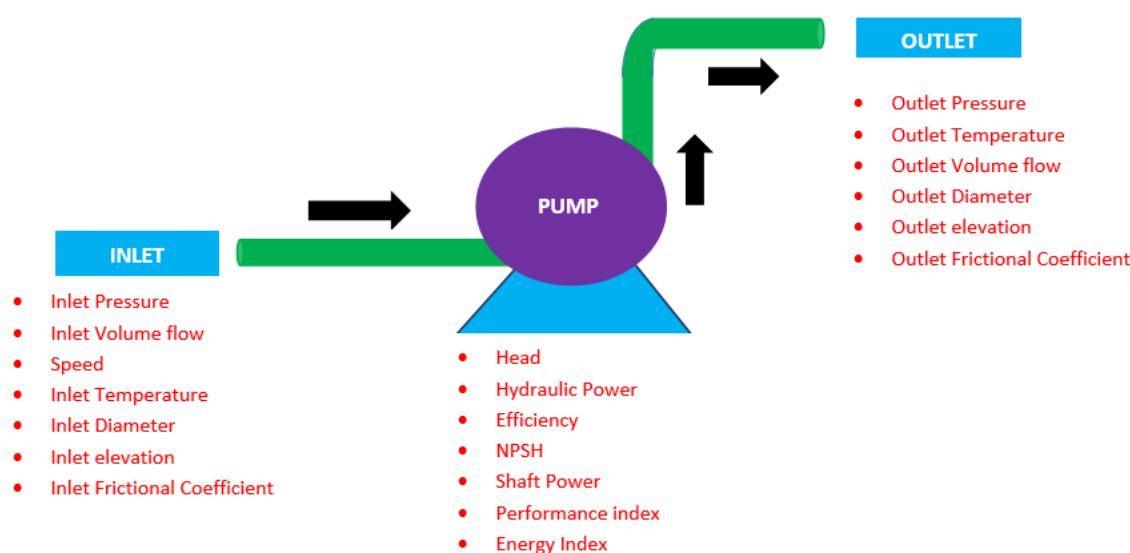
Fault Name	Associated Symptoms	Recommendations
	Radial Bearing Vibration NDE Y R High	2. Intervention options; reduce speed, and restore oil properties. Parts availability. 3. Check the trends to establish if it's either both X and Y probes or one. 4. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 5. Change bearing if required.
	Radial Bearing Temperature NDE A R High	
	Radial Bearing Temperature NDE B R High	
	Radial Bearing Temperature NDE B R High	
NDE Radial Bearing Failure VIB ROC	Radial Bearing Vibration NDE X R High	1. Assess the likely cause and time to fail(trip) if intervention is not possible. 2. Intervention options; reduce speed, and restore oil properties. Parts availability. 3. Check the trends to establish if it's either both X and Y probes or one. 4. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 5. Change bearing if required.
	Radial Bearing Vibration NDE X R High	
	Radial Bearing Vibration NDE X ROC Short-Term High	
	Radial Bearing Vibration NDE Y ROC Short-Term High	
NDE Radial Bearing Failure TEMP ROC	Radial Bearing Temperature NDE A R High	1. Assess the likely cause and time to fail(trip) if intervention is not possible. 2. Intervention options; reduce speed, and restore oil properties. 3. Ensure Parts availability. 4. Inspect bearings and internal clearances, if installed check earthing brush is working correctly, and check the bearing for arcing. 5. Check oil quality, temperature and pressure are correct, if not investigate why
	Radial Bearing Temperature NDE B R High	
	Radial Bearing Temperature NDE A ROC Short-Term High	
	Radial Bearing Temperature NDE B ROC Short-Term High	
Motor and Winding Overheating	Winding Temperature A U R High	1. Check water cooling supply (rated flow and max temperature), 2. Check airflow/blockage - filter replacing; to check motor blower correct operation. 3. Check process overload;
	Winding Temperature B U R High	
	Winding Temperature C U R High	
	Winding Temperature A V R High	
	Winding Temperature B V R High	
	Winding Temperature C V R High	
	Winding Temperature A W R High	
	Winding Temperature B W R High	

Fault Name	Associated Symptoms	Recommendations
	Winding Temperature C W R High	
	Motor Cooling Medium Outlet Temperature R High	
	Motor Power R High	
Ingress of Gas	Purge Air Pressure S High	1. Ensure gas detectors are working correctly. 2. Try to identify where the gas leak is coming from. 3. Depending if the motor is in an enclosure or not will determine the course of action

CENTRIFUGAL PUMP MONITORING

Centrifugal pump is a rotodynamic equipment that uses a rotating impeller to increase the pressure and flow rate of a fluid. It is used to transport liquids or slurries through pipes by centrifugal action. Pumps are generally used for transporting liquids in process plants, power plants, and other industries and to increase the pressure or flow. Power and process plants play a key role in the manufacturing of every product. Pumps are among the most important machines used in these plants. The centrifugal pumps are by far the most used of the pump types. Among all the installed pumps in a typical petroleum plant, almost 80-90% are centrifugal pumps.

Centrifugal pumps are widely used because of their design simplicity, high efficiency, a wide range of capacity, head, smooth flow rate, and ease of operation and maintenance. The pump operation is used to increase the pressure of an inlet liquid stream. Due to the variation in flow versus pressure and light to moderate viscous fluid handling ability, centrifugal pumps are more flexible and commonly used in process and pipeline applications. It can provide a wide range of flow rates over a certain pressure range. Hence, the pressure generated by the centrifugal pump depends on the flow rate of the pump. The operating and maintenance cost of centrifugal pumps are lower compared to Positive Displacement pumps. Around 20% of the world's electrical energy is used by pumps. If a pump fails, it can cause an entire plant to shut down and result in losses.



6.1 Centrifugal Pump - Sub Systems

6.1.1 Pump Bearing Systems

The bearing temperature, vibration, and displacement readings are monitored for the pump, motor, and gearbox. The rise in the bearing temperature and vibration indicates an improper alignment between the rotating parts. The rise in lube oil temperature increases the viscosity of the lube oil which reduces the strength of the oil film on the bearings and may lead to high vibrations.

6.1.2 Wet Seal System

The seal oil system supplies the mechanical contact and floating ring seals with an adequate flow of seal liquid always, ensuring proper function. The oil in the overhead tank is in contact with the reference gas pressure via a separate line with a static head providing the required pressure differential.

6.1.3 Lube Oil System

The main purpose of lubrication oil condition monitoring and degradation detection is to determine whether the oils have deteriorated to such a degree that they no longer fulfill their functions. The lube oil system continuously provides cool and clean oil to the bearings and gear at the proper pressure, temperature, and flow rate.

High-quality pumps require a form of lubricant to cool, seal, and lubricate key internal components such as gears, bearings, and seals. The lube oil is drawn from the reservoir by the pumps and is fed under pressure through coolers and filters to the bearings.

6.1.4 Driver System

Below are the driver systems that can be used for a Centrifugal Pump:

1. Common pump drivers:
 - a. Electric Motor
 - b. Steam Turbines
 - c. Hydraulic Turbines
 - d. Turboexpanders.
2. Uncommon pump drivers:
 - a. Diesel Engines (typically for emergency equipment)
 - b. Gas Turbine (Used for high power pumps in remote location)

6.2 Centrifugal Pump - Performance Monitoring

Base Thermodynamic Performance Model are used for:

- Process evaluation and comprehensive representation of process and asset design.
- Monitoring and analysis of the performance output.
- Determining the operating condition of the pump.

The Centrifugal Pump Performance Model calculates any variation between baseline performance and the actual performance of the machine in real-time. This difference provides a means to find the latent inefficiency in a pump's operation and quantify the cost associated with such inefficiencies. Inside the library, pump performance models are focused on recognizing two primary areas of energy loss of monitored machinery: operational loss and design loss.

1. **Operational loss:** Loss caused due to equipment degradation and/or sub-optimal machine operation because of non-standard process conditions or upsets.
2. **Design loss:** Loss caused due to equipment operating off its ideal operating point during normal process operating conditions.

Derived insight from the performance:

- Continuously reveals latent inefficiencies.
- Identifies asset performance improvement opportunities.
- Accurately quantifies energy loss.
- Provides an objective basis for maintenance planning.

- Eliminates offline, manual, and periodic calculation methods.

6.2.1 Notations

Below are the notations used in the equations of Performance Modeling, Expected KPI, and Operating KPI

F	Mass flow
Q	Volume flow
H	Head
P	Pressure
T	Temperature
η	Efficiency
W	Power
$NPSH$	Net Positive suction head
N	Speed
ρ	Density
Z	Elevation
f	Frictional Coefficients
D	Diameter
C_h	Head correction factor
C_n	Efficiency correction factor
C_q	volume flow correction factor
C_{NPSH}	NPSH correction factor

6.2.2 Subscripts

i	Inlet
$Corr$	Corrected
BEP	Best Efficiency point
o	Outlet
f	Friction
R	Reduced
vis	Viscous
w	Water

6.2.3 Centrifugal Pump - Model Assumptions

The key assumptions for developing a Performance model are:

1. Inlet frictional losses and outlet frictional losses are calculated using Darcy's equation per unit length form.

6.2.4 Centrifugal Pump - Variants

The model supports two Centrifugal Pump variants:

- Constant speed centrifugal pump
- Variable speed centrifugal pump

6.2.5 Centrifugal Pump - Model Input

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Suction Pressure	Suction Pressure (Required Input)	barg	Dynamic	Process	Field Input
2	Discharge Pressure	Discharge Pressure (Required Input)	barg	Dynamic	Process	Field Input
3	Flowing Temperature	Temperature (Input)	°C	Dynamic	Process	Field Input
4	Volume Flow Rate	Volume Flow Rate (Required Input)	m ³ /hr	Dynamic	Process	Field Input
5	Min Recirculation Flow	Min Recirculation Flow (Input)	m ³ /hr	Dynamic	Process	Field Input
6	Speed	Measured Speed (Input)	rpm	Dynamic	Process	Field Input
7	Electrical Power	Electrical Power (Required Input)	kW	Dynamic	Process	Field Input
8	Suction Diameter	Suction Diameter (Optional)	m	Static	Design	From Datasheet
9	Suction Elevation	Suction Elevation (Optional)	m	Static	Design	From Datasheet
10	Suction Frictional Losses Coefficient	Suction Frictional Losses Coefficient (Optional)		Static	Design	From Datasheet
11	Discharge Diameter	Discharge Diameter (Optional)	m	Static	Design	From Datasheet
12	Discharge Elevation	Discharge Elevation (Optional)	m	Static	Design	From Datasheet
13	Discharge Frictional Losses Coefficient	Discharge Frictional Losses Coefficient (Optional)		Static	Design	From Datasheet
14	Constant Suctional Frictional Losses	Constant Suctional Frictional Losses (Optional Input)	m	Static	Design	From Datasheet

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
15	Constant Discharge Frictional Losses	Constant Discharge Frictional Losses (Optional Input)	m	Static	Design	From Datasheet
16	Performance Data	Flow Rate, Operating Head, Operating Efficiency, Operating Shaft Power, Operating NPSH		Static	Design	Interpolated from the Performance curve
17	Property Data	Temperature, Density, Vapor Pressure		Static	Thermo properties	Constant
18	Liquid Density	Density (Overall)	kg/m ³	Static	Thermo properties	Lab Measurement / Interpolated from the Performance curve
19	Vapor Pressure	Vapor Pressure (Required Input if Property Data is not given)	bara	Static	Thermo properties	Lab Measurement / Interpolated from the Performance curve
20	Mechanical Efficiency	Mechanical Efficiency (Input)	%	Static	Design	Constant
21	Need performance data correction	Need performance data correction (1 = Yes and 0 = No) (Optional Input)		Static		User
22	Viscosity	Viscosity for performance data correction (Optional Input)	centipoise	Dynamic	Transport properties	Measured/ Data sheet
23	Impeller Suction Type	Impeller Suction Type (1 = Single and 2 = Double) (Optional Input)		Static	Design	Data sheet
24	Suction Inlet Side	Suction Inlet Geometry (1 = End Suction and 2 = Side inlet) (Optional Input)		Static	Design	Data sheet

6.2.6 Centrifugal Pump - KPIs

The following indicators help us know the overall performance of an asset.

6.2.6.1 Input KPIs:

- Inlet Pressure
- Outlet pressure
- Inlet Volume Flow
- Property data (Vapour pressure, Density against Temperature)
- Performance Curve data for Constant speed

$$Q, H, \eta, \dot{W}_{Shaft}, H_{NPSH}$$

- Power Input
- Performance Curve data for Variable speed

$$N, Q, H, \eta, \dot{W}_{Shaft}, H_{NPSH}$$

- Speed
- Inlet Diameter
- Outlet Diameter
- Inlet elevation
- Outlet elevation
- Inlet Frictional Coefficient
- Outlet Frictional Coefficient

Input data required for viscosity correction

When viscosity correction is necessary, the following inputs are required. This is only applicable if the pumping fluid has a higher viscosity than water and the OEM performance curve data is for water.

- Viscosity
- Impeller suction type (Single or double)
- Impeller suction inlet type (End suction or side inlet)

6.2.6.2 Output KPIs:

Expected KPIs

- Design Head
- Design Efficiency
- Design Efficiency at BEP
- Design Power
- Design NPSH
- Flow at BEP

Operating KPIs

- Operating head
- Operating Efficiency
- Operating Hydraulic Power
- Operating Shaft Power
- Operating NPSH

Index KPIs

- Performance Index
- Energy Index
- Flow loss

Design KPI

- Design Head
- Design Efficiency
- Design Efficiency at BEP
- Design Shaft Power
- Design NPSH
- Flow at BEP
- Design Head at BEP
- Design Power at BEP
- Corrected performance data

Reduced KPI

- Reduced Flow Rate
- Reduced Head
- Reduced Power
- Reduced Efficiency

6.2.6.3 Centrifugal Pump - Performance Modeling

The Performance modelling of centrifugal pump helps to determine the performance KPI. The following is the procedure,

Overall Mass Balance

Applying continuity equation:

$$F_i = F_o$$

Overall Force Balance

Applying energy balance equation (Bernoulli's energy theorem)

$$P_i + \frac{\rho v_i^2}{2} + \rho g Z_i - h_{f_i} = P_o + \frac{\rho v_o^2}{2} + \rho g Z_o + h_{f_o}$$

Converting to a representative head term (to meter) on dividing ρg both sides,

$$\Delta H_t = \left(\frac{P_o - P_i}{\rho g} \right) + \left(\frac{v_o^2 - v_i^2}{2g} \right) + (Z_o - Z_i) + \left(\frac{h_{f_o} + h_{f_i}}{\rho g} \right)$$

Overall Energy Balance

$$F_o h_o(P_o, T_i) = F_i h_i(P_i, T_i) + \dot{W}_{Hydraulic}$$

$$\eta_{pump} = \frac{\dot{W}_{Hydraulic}}{\dot{W}_{Electric}}$$

6.2.6.4 Centrifugal Pump - Expected KPIs

The following indicators help you to know expected overall performance of the centrifugal pump:

Centrifugal Pump Constant Speed

From curve of Head vs Volume flow, we pass operating volume flow and perform quadratic interpolation or linear interpolation based on number data points provided to get expected head.

Design head (m)

From performance curve of Head vs Volume flow, we pass operating volume flow to get expected head.

$$H_{Design} = f(Q_i)$$

Design Power (KW)

From performance curve of Power vs Volume flow, we pass operating volume flow to get expected power.

$$\dot{W}_{Shaft}^{Design} = f(Q_i)$$

Design Efficiency (%)

From performance curve of Efficiency vs Volume flow, we pass operating volume flow to get expected efficiency.

$$\eta_{Design} = f(Q_i)$$

Design Efficiency at BEP (%)

From the performance curve of Efficiency vs Volume flow, the maximum efficiency is the Best efficiency Point (BEP)

$$\eta_{BEP}^{Design} = f(Q_i)_{at\ max\ efficiency}$$

Flow at BEP (m³/hr)

From the performance curve of Efficiency vs Volume flow, the flow at BEP

$$Q_{BEP} = f(Q_i)_{\eta_{BEP}^{Design}}$$

Design NPSH (m)

From performance curve of NPSH vs Volume flow, we pass operating volume flow to get expected NPSH.

$$H_{NPSH}^{Design} = f(Q_i)$$

Design flow (m³/hr)

From curve of Head vs Volume flow, we pass operating head flow to get expected volume flow.

$$Q_{Design} = f(Q_i)$$

Centrifugal Pump Variable Speed

Reduced curve approach is taken for determining expected KPI of variable speed centrifugal pump,

Converting the performance curve data in reduced curve data:

$$Q_R = \frac{Q_i}{N}$$

$$H_R = \frac{g H}{N^2}$$

$$H_{NPSH R} = \frac{g H_{NPSH}}{N^2}$$

$$\dot{W}_R = \frac{\dot{W}_{Shaft}}{\rho N^3}$$

$$\eta_R = \frac{Q_R H_R}{\dot{W}_R}$$

At Measured Speed, on performing quadratic interpolation we get:

$$Q_R = \frac{Q_i}{N}$$

$$H_R = f(Q_R)$$

$$\dot{W}_R = f(Q_R)$$

$$\eta_{BEP} = f(Q_R) \text{ at } Q_R \text{ max efficiency}$$

$$H_{RNPSH} = f(Q_R) \text{ at } Q_R$$

$$Q_R = f(Q_R) \text{ at } H_R$$

Converting the reduced curve data to design actual values:

Design head (m)

$$H_{Design} = \frac{H_R N^2}{g}$$

Design Power (KW)

$$\dot{W}_{Shaft}^{Design} = \frac{\dot{W}_R \rho}{N^3}$$

Design Efficiency (%)

$$\eta_{Design} = \eta_R$$

Design NPSH (m)

$$H_{NPSH}^{Design} = \frac{H_{RNPSH} N^2}{g}$$

Design flow (m³/hr)

$$Q_{Design} = Q_R N$$

Design Efficiency at BEP (%)

$$\eta_{BEP}^{Design} = f(Q_R)_{at \ max \ efficiency}$$

Flow at BEP (m³/hr)

$$Q_{BEP} = f(Q_R)_{at \ \eta_{BEP}^{Design}}$$

6.2.6.5 Centrifugal Pump - Operating KPIs

The following indicators help you to know operating overall performance of the centrifugal pump:

Operating Head (m)

$$\Delta H_{Operating} = \left(\frac{P_o - P_i}{\rho g} \right) + \left(\frac{v_o^2 - v_i^2}{2g} \right) + (Z_o - Z_i) + (h_{f_o} + h_{f_i})$$

NOTE

If geometrical details of Inlet Diameter and Outer Diameter is provided then Inlet velocity and outlet velocity is calculated by below equation, else will be set to zero,

$$v_i = \frac{Q_i}{\frac{\pi}{4} D_i^2}$$

$$v_o = \frac{Q_i}{\frac{\pi}{4} D_o^2}$$

NOTE

If geometrical details of Inlet Diameter and Outer Diameter, along with Inlet frictional coefficient and Outlet frictional coefficient is provided then Inlet frictional head and outlet frictional head is calculated by below equation, else will be set to zero. On using Darcy's equation,

$$h_{f_i} = \frac{2 f_i v_i^2}{g D_i}$$

$$h_{f_o} = \frac{2 f_o v_o^2}{g D_o}$$

Operating Hydraulic Power (kW)

$$\dot{W}_{Hydraulic}^{Operating} = Q_i \rho g \Delta H_{Operating}$$

Operating Shaft Power (kW)

$$\dot{W}_{Shaftpower}^{Operating} = \dot{W}_{Electric}^{Operating} \eta_{Motor}$$

NOTE

This formula is part of basic calculations.

Operating NPSH (m)

$$H_{NPSH}^{Operating} = \frac{P_i - P_{Vapor}}{\rho g} + \frac{v_i^2}{2g} + Z_i - h_{f_i}$$

Operating Efficiency (%)

$$\eta_{operating} = \frac{\dot{W}_{Hydraulic}^{Operating}}{\dot{W}_{Electric}^{Operating}}$$

NOTE

If Driver is electric motor, then power attribute $(\dot{W}_{Electric}^{Operating})$ can be PHD tag mapped from Motor asset.

NPSH Margin

$$NPSH_{Margin} = H_{NPSH}^{Operating} - H_{NPSH}^{Design}$$

NOTE

This formula is part of basic calculations.

6.2.6.6 Centrifugal Pump - Index KPI***Performance Index***

$$Performance\ index = \frac{\eta_{Operating}}{100}$$

Energy Index

$$Energy\ index = 1 - \frac{(\dot{W}_{Electric}^{Operating} \eta_{Motor} - \dot{W}_{Design})}{\dot{W}_{Design}}$$

Flow loss

$$Q_{loss} = Q_{Design} - Q_i$$

6.2.6.7 Centrifugal Pump - Design KPI

The following indicators help us to know design overall performance of the centrifugal pump.

Centrifugal pump Constant speed***Design head (m)***

From performance curve of Head vs Volume flow, we pass operating volume flow to get design head.

$$H_{Design} = f(Q_i)$$

Design Shaft Power (kW)

From performance curve of Shaft Power vs Volume flow, we pass operating volume flow to get design power.

$$\dot{W}_{Design} = f(Q_i)$$

Design Efficiency (%)

From performance curve of Efficiency vs Volume flow, we pass operating volume flow to get design efficiency.

$$\eta_{Design} = f(Q_i)$$

Design Efficiency at BEP (%)

From the performance curve of Efficiency vs Volume flow, the maximum efficiency is the Best efficiency Point (BEP)

$$\eta_{BEP}^{Design} = f(Q_i)_{at\ max\ efficiency}$$

Flow at BEP (m^3/hr)

From the performance curve of Efficiency vs Volume flow, the flow at BEP

$$Q_{BEP} = f(Q_i)_{\eta_{BEP}^{Design}}$$

Design Head at BEP (m)

From the performance curve of Head vs Volume flow, the design head at BEP

$$H_{BEP}^{Design} = f(Q_i)_{\eta_{BEP}^{Design}}$$

Design Power at BEP (kW)

From the performance curve of Power vs Volume flow, the design power at BEP

$$\dot{W}_{Design} = f(Q_i)_{\eta_{BEP}^{Design}}$$

Design NPSH (m)

From performance curve of NPSH vs Volume flow, we pass operating volume flow to get design NPSH.

$$H_{NPSH}^{Design} = f(Q_i)$$

Design flow (m^3/hr)

From curve of Head vs Volume flow, we pass operating head flow to get design volume flow.

$$Q_{Design} = f(\Delta H_{Operating})$$

Motor Efficiency (%)

$$\eta_{Motor} = \frac{\eta_{Design}}{\eta_{BEP}^{Design}} \times 100$$

Centrifugal pump variable speed**Design head (m)**

From performance curve of Head vs Volume flow vs Speed, we pass operating volume flow and speed thereby on performing 2D interpolation we get design head.

$$H_{Design} = f(N, Q_i)$$

Design Shaft Power (kW)

From performance curve of Shaft Power vs Volume flow vs Speed, we pass operating volume flow and speed thereby on performing 2D interpolation we get design power.

$$\dot{W}_{Design} = f(N, Q_i)$$

Design Efficiency (%)

From performance curve of Efficiency vs Volume flow vs Speed, we pass operating volume flow and speed thereby on performing 2D interpolation we get design efficiency.

$$\eta_{Design} = f(N, Q_i)$$

Design NPSH (m)

From performance curve of NPSH vs Volume flow vs Speed, we pass operating volume flow and speed thereby on performing 2D interpolation we get design efficiency.

$$H_{NPSH}^{Design} = f(N, Q_i)$$

Design Flow (m³/hr)

From performance curve of head vs Volume flow vs Speed, we pass operating head and speed thereby on performing 2D interpolation we get design volume flow.

$$Q_{Design} = f(N, \Delta H_{Operating})$$

Design Efficiency at BEP (%)

From the performance curve of Efficiency vs Volume flow vs Speed, the maximum efficiency is the Best efficiency Point (BEP)

$$\eta_{BEP}^{Design} = f(N, Q_i) \text{ at max efficiency}$$

Flow at BEP (m³ /hr)

From the performance curve of Efficiency vs Volume flow vs speed, we pass operating flow and speed thereby on performing 2D interpolation we get volume flow at BEP.

$$Q_{BEP} = f(N, Q_i)_{\eta_{BEP}^{Design}}$$

Design Head at BEP (m)

From the performance curve of Head vs Volume flow vs speed, we pass operating flow and speed thereby on performing 2D interpolation we get design head at BEP.

$$H_{BEP}^{Design} = f(N, Q_i)_{\eta_{BEP}^{Design}}$$

Design Power at BEP (kW)

From the performance curve of Power vs Volume flow vs speed, we pass operating flow and speed thereby on performing 2D interpolation we get design power at BEP.

$$\dot{W}_{Design} = f(N, Q_i)_{\eta_{BEP}^{Design}}$$

6.2.6.8 Centrifugal Pump - Reduced KPI Calculation

Reduced KPI Attributes:

Reduced attributes are current reduced attribute points plotted on reduced curve, the following equations are derived from Fan laws,

Reduced Flow Rate

$$Q_R = \frac{Q_i}{N}$$

Reduced Head

$$H_R = \frac{g \Delta H_{operating}}{N^2}$$

Reduced Power

$$\dot{W}_R = \frac{\dot{W}_{Electric}^{Operating}}{\rho N^3}$$

Reduced Efficiency

$$\eta_R = \frac{Q_R H_R}{\dot{W}_R}$$

Reduced Curve:

Reduced Curve Calculation involved calculation of curve coordinate points,

$$\Delta Values = \frac{Q_{max} - Q_{min}}{\text{number of curve points (10 points)}}$$

For each number of curves points we back calculate the reduced head, power, and efficiency.

$$Q_i = (Q_{min} + \Delta Values)_{n=10}$$

$$H_{design} = f(Q_i)$$

$$\dot{W}_{Design} = f(Q_i)$$

$$Q_R = \left(\frac{Q_i}{N} \right)_{n=10}$$

$$H_R = \left(\frac{g \Delta H_{design}}{N^2} \right)_{n=10}$$

$$\dot{W}_R = \left(\frac{\dot{W}_{Design}}{\rho N^3} \right)_{n=10}$$

$$\eta_R = \left(\frac{Q_R H_R}{\dot{W}_R} \right)_{n=10}$$

For reduced curve we plot (H_R, Q_R) , $(\dot{W}_R, Q_R)$, (η_R, Q_R)

6.2.7 Centrifugal Pump - Model Output

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Design Head	Actual Head at current flow rate (Output)	m	Dynamic	Performance	Calculated
2	Design Power	Power at current flow rate from curve data (Output)	kW	Dynamic	Performance	Calculated
3	Design Efficiency	Efficiency at current flow rate from curve data (Output)	%	Dynamic	Performance	Calculated
4	Design Efficiency at BEP	Design Efficiency at BEP (Input)	%	Dynamic	Performance	Calculated
5	Volume Flow Rate at BEP	Volume Flow Rate at BEP (Output)	m ³ /hr	Dynamic	Performance	Calculated
6	Design NPSH	NPSH at current flow rate from curve data (Output)	m	Dynamic	Performance	Calculated
7	Reduced Head	Reduced Head (Output)		Dynamic	Output	Calculated
8	Reduced Power	Reduced Power (Output)		Dynamic	Output	Calculated
9	Reduced Efficiency	Reduced Efficiency (Output)		Dynamic	Output	Calculated
10	Reduced Flow Rate	Reduced Flow Rate (Output)		Dynamic	Output	Calculated
11	Operating Head	Operating Head (Output)	m	Dynamic	Performance	Calculated
12	Operating Hydraulic Power	Operating Hydraulic Power (Output)	kW	Dynamic	Performance	Calculated
13	Operating Shaft Power	Shaft Power or BHP (Output)	kW	Dynamic	Performance	Calculated
14	Operating Efficiency	Operating Efficiency (Output)	%	Dynamic	Performance	Calculated
15	Operating NPSH	NPSHA - Net Positive Suction Head Available (Output)	m	Dynamic	Performance	Calculated
16	Motor Efficiency	Motor Efficiency (Output)	%	Dynamic	Performance	Calculated
17	Deviation Head	Percent Deviation in Head between	%	Dynamic	Deviation	Calculated

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
		Operating and Design values (Output)				
18	Deviation Power	Percent Deviation in Power between Operating and Design values (Output)	%	Dynamic	Deviation	Calculated
19	Deviation Efficiency	The difference in Efficiency between Operating and Design values (Output)	%	Dynamic	Deviation	Calculated
20	Deviation NPSH	NPSH at current flow rate from curve data (Output)	%	Dynamic	Deviation	Calculated
21	BEP Loss	BEP Loss (Output)	kW	Dynamic	Energy	Calculated
22	Performance Index	Performance Index (Output)		Dynamic	Internal	Calculated
23	Energy Index	Energy Index (Output)		Dynamic	Internal	Calculated
24	Degradation Loss	Degradation Loss (Output)	kW	Dynamic	Energy	Calculated
25	Recirculation Loss	Recirculation Loss (Output)	m ³ /hr	Dynamic	Energy	Calculated
26	Health Index	Health Index (Output)		Dynamic	Internal	Calculated
27	Suction Frictional Losses	Suction Frictional Losses (Output)	m	Static	Design	Calculated
28	Discharge Frictional Losses	Discharge Frictional Losses (Output)	m	Static	Design	Calculated
29	Corrected Performance Data	Corrected Performance Data (Output)		Dynamic	Design	Calculated

6.3 Centrifugal Pump - Performance Data Correction

The performance data correction only applies when the pump is pumping a fluid with a viscosity greater than that of water, and the supplier or OEM of the pump offers performance data for water. The correction accounts for the viscous losses that occurred due to the high viscosity of fluid flowing through the pump.

Reference: - American National Standard (Guideline) for "Effect of Viscosity on Rotodynamic (Centrifugal and Vertical) Pump Performance" published by the Hydraulic Institute (Document no.- ANSI/HI 9.6.7-2010).

NOTE

The performance data correction should not be used for the pump used in service of pumping water.

6.3.1 Assumptions

1. For the pump and fluid of interest, the ANSI/HI 9.6.7-2010 technique calculates the correction factors for volumetric flow rate and efficiency and assumes that these values remain constant at all analyzed flow rates.
2. The shutoff head for the pump will be the same for both water and viscous fluid, independent of the fluid's viscosity.
3. The technique assumes that the flow and head correction factors are equal at the best efficiency point.

6.3.2 Constraints

1. Radial flow impellers
2. Newtonian Fluid
3. Kinematic viscosity from 1 to 3,000 (cSt)
4. Impeller diameters from 5.5 to 16 inches (140 to 406 mm)
5. Water BEP flow from 13 to 1800 (gpm) $\left(3 \text{ to } 410 \frac{\text{m}^3}{\text{hr}}\right)$
6. Water BEP head from 20 to 430 feet (6 to 130 m)
7. Water BEP efficiency from 28% to 86%

6.3.3 Performance correction formulae

B = Parameter used as a normalizing pump Reynolds number and to adjust the corrections.

$$B = 26.6 \times \frac{\mu_{vis}^{0.5} \times H_{BEP-w}^{0.0625}}{Q_{BEP-w}^{0.375} \times N^{0.25}}$$

For a double-suction impeller, use $Q_{BEP-w} = \frac{Q_{BEP-w}}{2}$

If $B \leq 1$, Here the viscous losses are negligible, and no need to make corrections.

Hence, $C_q = C_h = C_n = C_{NPSH} = 1$

For $1 < B < 10$,

$$C_q = 2.71^{(-0.165 \times \log_{10}(B)^{3.15})}$$

$$Q_{Corr} = C_q \times Q_w$$

$$C_h = 1 - \left((1 - C_q) \times \left(\frac{Q_w}{Q_{BEP-w}} \right)^{0.75} \right)$$

For a double-suction impeller, use $Q_w = \frac{Q_w}{2}$

$$H_{Corr} = C_h \times H_w$$

$$C_n = B^{(-0.0547B^{0.69})}$$

$$\eta_{Corr} = C_n \times \eta_w$$

$$C_{NPSH} = 1 + A \times \left(\frac{1}{C_h} - 1 \right) \times 274000 \times \left(\frac{NPSH_{BEP-w}}{Q_{BEP-w}^{0.667} \times N^{1.33}} \right)$$

For end suction pumps: $A=0.1$,



For side inlet pumps: $A=0.5$



$$NSPH_{Corr} = C_{NSH} \times NSPH_W$$

$$\dot{W}_{Corr} = \frac{Q_{Corr} \times \rho_{vis} \times H_{Corr} \times g}{\frac{\eta_{Corr}}{100}} \times \frac{3600}{1000}$$

NOTE

If you opt for the performance data correction, all design KPIs will be calculated from the corrected performance curve data instead of OEM supplied performance curve data. New corrected performance data will appear on the performance overview page. The performance data correction method is capable of handling both constant speed as well as variable speed variants.

6.4 Centrifugal Pump - Fault Model

6.4.1 Centrifugal Pump - APM Model

6.4.1.1 Pump APM Fault Model

Centrifugal Pump Performance Fault model

Model Type	APM Model
Category	Centrifugal Pump
Equipment Class	Pump
Equipment Type	Centrifugal Pump

The list of faults and symptoms configured for a Pump Performance model is below:

Fault Name	Associated Symptoms	Recommendations
Inlet Supply Anomaly	Suction flow RL (ISA)	1. Check the calibration of the sensor. 2. Assess of leaks/jams in the upstream equipment or pipeline or Suction strainers etc.
	Suction flow RH (ISA)	
	Pump Inlet Temperature RL (ISA)	
	Pump Inlet Temperature RH (ISA)	
	Pump Inlet Pressure RL (ISA)	
	Pump Inlet Pressure RH (ISA)	
Fouling / Corrosion / Erosion Risk	Efficiency deviation (Actual-Expected) ROCL LT (ER)	1. If fouling: Short term; Wash Pump if possible. 1. Long-term, investigate the application of the coating on the affected liquid path. 2. Change liquid inlet Temp. 3. Borescope inspection at the opportunity. 2. If corrosion/erosion: 1. Replace corroded parts with better material grade. 2. In case of erosion consider coating of the liquid path. 3. Borescope at the opportunity.
Insufficient Capacity or Pressure	Deviation Head LOW (ICP)	1. Check for rattling noise in the suction line or within the pump itself, in which case the pump is likely to have suction problems 2. Check the amp reading of the pump motor. 3. Check discharge pressure and flow: if lower than normal, pumped liquid, not on spec, or system resistance line problems. 4. Check vibrations with a handheld meter and if too high call engineering for further investigation.
Hydraulic Degradation	Hydraulic	1. Check the amp meter of motor

Fault Name	Associated Symptoms	Recommendations
	Efficiency LOW (HD)	- When reading low check also discharge pressure and flow. - When also these are lower than normal system resistance line is not according to design and/or liquid characteristics, not to specification. - Verify pump vibration with a hand-held meter and check for rubbing frequency if no performance deviations are observed
Low Flow	Discharge flow LOW (LF)	- Check the amp meter reading of the motor. - When too high, indication of internal problems. - Check vibrations with a hand-held meter. - If too high, call engineering to do a frequency analysis and see whether excessive piping forces are acting on the pump casing, excessive misalignment, etc. (check dominant frequencies). - In case of low amp reading and insufficient pressure and/or flow, the pump not operating at proper point.
High Flow	Discharge flow HIGH (HF)	- Check for rattling noise in the pump suction line. - If pump discharge pressure and flow are lower than the curve: suction condition problems (cavitation or vortexes). - Reconfirm by checking the pump and suction line vibrations: high readings on suction and pump signal incorrect suction conditions incorrect. - When high amp readings and lower discharge pressure than normal at high flow then the operating point of the pump is excessive to the right of the BEP.
Cavitation Risk	NPSH Margin LOW (CR)	- Check for rattling noise in the pump suction line. - If pump discharge pressure and flow are lower than the curve, and low amp reading: suction condition problems (cavitation or vortexes). - Check vibration on the suction line and pump suction: if high and the amp reading is low, then probable high internal recirculation.
Operation Outside Preferred BEP Region Insufficient NPSH Available	Pump Discharge Pressure RL (OOBEP)	- Check for rattling noise in the pump suction line. - If pump discharge pressure and flow are lower than the curve, and low amp reading: suction condition problems (cavitation or vortexes). - Check vibration on the suction line and pump suction: if high and the amp reading is low, with discharge pressure higher than normal but low flow, possible chance of high internal recirculation.
	NPSH Margin SH (OOBEP)	
	NPSH Margin SL (OOBEP)	
Centrifugal Pump Inlet Filter Anomaly	Inlet Filter DP RH (IFA)	- Check the calibration of the sensor. - Change over the filter if stand-by is available and replace the filter.

6.4.2 Centrifugal Pump - APR Model

6.4.2.1 Bearing and Vibration System

Model Type	APR Model
Category	Centrifugal Pump
Equipment Class	Pump
Equipment Type	Centrifugal Pump - Bearing System

Below table illustrates the list of faults and symptoms configured for Pump bearing system condition-based monitoring.

Fault Name	Associated Symptoms	Recommendations
Centrifugal Pump Thrust/RB Bearing Anomaly due to Lube Oil Temp	P-Radial Bearing Vibration X DE RH (LOTA)	1. Check the calibration of sensor. 2. Check cooling water supply pressure. 3. Check bypass control valve. 4. Open both coolers and stabilize temperature. 5. Check cooling water pumps for blockages, ensure strainer is clean. 6. If the cooler is an air cooler check fan and motor integrity. 7. If malfunction of one cooler is suspected, isolate and inspect it. 8. Line walk the system to check for leaks.
	P-Radial Bearing Vibration Y DE RH (LOTA)	
	P-Radial Bearing Temperature A DE RH (LOTA)	
	P-Radial Bearing Temperature B DE RH (LOTA)	
	P-Radial Bearing Vibration X NDE RH (LOTA)	
	P-Radial Bearing Vibration Y NDE RH (LOTA)	
	P-Radial Bearing Temperature A NDE RH (LOTA)	
	P-Radial Bearing Temperature B NDE RH (LOTA)	
	P-Thrust Bearing Displacement Active Side RH (LOTA)	
	P-Thrust Bearing Displacement Active Side RL (LOTA)	
	P-Thrust Bearing Temperature A Active Side RH (LOTA)	
	P-Thrust Bearing	

Fault Name	Associated Symptoms	Recommendations
	Temperature A Inactive Side RH (LOTA)	
	P-Thrust Bearing Temperature B Active Side RH (LOTA)	
	P-Thrust Bearing Temperature B Inactive Side RH (LOTA)	
	Lube Oil Header Temperature RH (LOTA)	
Centrifugal Pump Thrust/RB Bearing Anomaly due to Lube Oil Press	P-Radial Bearing Vibration X DE RH (LOPA)	1. Check the calibration of sensor. 2. Switch on both oil pumps and stabilize oil pressure. 3. Check oil level in tank. 4. Check filter DP. 5. Assess malfunction. 6. Line walk the system and check for leaks. 7. Ensure PSV's are seated, check spillback/PCV valves are set correctly and not returning oil back to the tank. 8. If maintenance has recently been performed check correct rotation of pump. 9. Ensure no hand valves have been left shut, ensure NRV's are working correctly. 10. Check functioning of the auto change over switch.
	P-Radial Bearing Vibration Y DE RH (LOPA)	
	P-Radial Bearing Temperature A DE RH (LOPA)	
	P-Radial Bearing Temperature B DE RH (LOPA)	
	P-Radial Bearing Vibration X NDE RH (LOPA)	
	P-Radial Bearing Vibration Y NDE RH (LOPA)	
	P-Radial Bearing Temperature A NDE RH (LOPA)	
	P-Radial Bearing Temperature B NDE RH (LOPA)	
	P-Thrust Bearing Displacement Active Side RH (LOPA)	
	P-Thrust Bearing Displacement Active Side RL (LOPA)	
	P-Thrust Bearing Temperature A Active Side RH (LOPA)	
	P-Thrust Bearing	

Fault Name	Associated Symptoms	Recommendations
	Temperature A Inactive Side RH (LOPA)	
	P-Thrust Bearing Temperature B Active Side RH (LOPA)	
	P-Thrust Bearing Temperature B Inactive Side RH (LOPA)	
	Lube Oil Header Pressure RL (LOPA)	
Centrifugal Pump Thrust/RB Bearing Anomaly due to Lube Oil Quality	P-Radial Bearing Vibration X DE RH (LOQA)	1. Check oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, TAN. 4. Check no breather caps or man covers have been removed and left open to allow debris into the tank. 5. Recondition or replace oil.
	P-Radial Bearing Vibration Y DE RH (LOQA)	
	P-Radial Bearing Temperature A DE RH (LOQA)	
	P-Radial Bearing Temperature B DE RH (LOQA)	
	P-Radial Bearing Vibration X NDE RH (LOQA)	
	P-Radial Bearing Vibration Y NDE RH (LOQA)	
	P-Radial Bearing Temperature A NDE RH (LOQA)	
	P-Radial Bearing Temperature B NDE RH (LOQA)	
	P-Thrust Bearing Displacement Active Side RH (LOQA)	
	P-Thrust Bearing Displacement Active Side RL (LOQA)	
	P-Thrust Bearing Temperature A Active Side RH (LOQA)	
	P-Thrust Bearing	

Fault Name	Associated Symptoms	Recommendations
	Temperature A Inactive Side RH (LOQA)	
	P-Thrust Bearing Temperature B Active Side RH (LOQA)	
	P-Thrust Bearing Temperature B Inactive Side RH (LOQA)	
	Lube Oil Viscosity SH (LOQA)	
	Lube Oil Viscosity SL (LOQA)	
	Lube Oil TAN SH (LOQA)	
Centrifugal Pump Thrust Bearing Anomaly due to Mech Issue	P-Thrust Bearing Displacement Active Side RH (TBMIA)	1. Check calibration of sensor. 2. Assess likely cause and time to trip if intervention not possible. 3. Intervention options. 4. Reduce thrust load. 5. Ensure parts availability. 6. Check axial vibration trends to establish if the machine has surged and potentially damaged/wiped the pads. 7. If required inspect the thrust pads and collar to check for damage / wiping. 8. Carry out a bump check to establish what float is in the machine, compare to previous overhaul report for measurement. 9. Check the gear coupling/pre-stretch diaphragm coupling .
	P-Thrust Bearing Displacement Active Side RL (TBMIA)	
	P-Thrust Bearing Displacement Inactive Side RH (TBMIA)	
	P-Thrust Bearing Displacement Inactive Side RL (TBMIA)	
	P-Thrust Bearing Temperature A Active Side RH (TBMIA)	
	P-Thrust Bearing Temperature A Inactive Side RH (TBMIA)	
	P-Thrust Bearing Temperature B Active Side RH (TBMIA)	
	P-Thrust Bearing Temperature B Inactive Side RH (TBMIA)	
Centrifugal Pump DE RB Anomaly due to Mech Issue	P-Radial Bearing Vibration X DE RH (DERBMIA)	1. Check calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if intervention not possible. 4. Intervention options, reduce speed, Parts availability.

Fault Name	Associated Symptoms	Recommendations
	P-Radial Bearing Vibration Y DE RH (DERBMIA)	- Check the trends to establish if its either both X and Y probes or one. - Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. - Change bearing if required.
	P-Radial Bearing Temperature A DE RH (DERBMIA)	
	P-Radial Bearing Temperature B DE RH (DERBMIA)	
Centrifugal Pump NDE RB Anomaly due to Mech Issue	P-Radial Bearing Vibration X NDE RH (NDERBMIA)	- Check calibration of sensor. - Investigate with high frequency vibration analysis. - Assess likely cause and time to fail(trip) if intervention not possible. - Intervention options; reduce speed, Parts availability. - Check the trends to establish if its either both X and Y probes or one. - Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. - Change bearing if required.
	P-Radial Bearing Vibration Y NDE RH (NDERBMIA)	
	P-Radial Bearing Temperature A NDE RH (NDERBMIA)	
	P-Radial Bearing Temperature B NDE RH (NDERBMIA)	
	Lube Oil Tank Heater Run Signal OFF (NDERBMIA)	
Centrifugal Pump Rotor Anomaly	P-Radial Bearing Vibration X DE RH (RTRA)	- Check the calibration of sensor. - Investigate with high frequency vibration analysis. - Check Carbon brush integrity. - Check magnetism of the rotor. - Check rotor unbalance, misalignment. - Looseness. - Check coupling integrity. - Check foundation bolt integrity. - Check both bearings for clearance and pad wear etc.
	P-Radial Bearing Vibration Y DE RH (RTRA)	
	P-Radial Bearing Vibration X NDE RH (RTRA)	
	P-Radial Bearing Vibration Y NDE RH (RTRA)	
	Lube Oil Tank Heater Run Signal ON (RTRA)	
Centrifugal Pump Balance Line Blockage Risk	Balance Line DP SH (BLBA)	- Check the calibration of sensor. - Check the balance line tube for any particle entrainment and clean. - Check any surge events, liquid slugs in the sample gas.
	Balance Line Flow SL (BLBA)	
Centrifugal Pump Balance	P-Thrust Bearing	Check the calibration of sensor.

Fault Name	Associated Symptoms	Recommendations
Piston Seal Wear	Displacement Active Side RH (BPSWA)	2. Monitor rotor axial position, pad temperature.
	P-Thrust Bearing Displacement Active Side RL (BPSWA)	3. Balance chamber pressure.
	P-Thrust Bearing Displacement Inactive Side RH (BPSWA)	4. Determine if failure mode is still progressing, in which case calculate time to thrust bearing displacement trip at current rate.
	P-Thrust Bearing Displacement Inactive Side RL (BPSWA)	5. Plan for repair accordingly.
	Balance Line DP SH (BPSWA)	6. If required dismantle the machine and carry out lead checks on the j strips (labyrinth seals) and determine what the clearance is.
		7. Compare this to previous overhaul reports.
		8. Excessive clearance will overload the thrust bearing beyond its design limit.
		9. The rotor will require overhaul/replacing if beyond design clearance.
		10. Clean the Balance line tube.

6.4.2.2 Mechanical Seal System

Model Type	APR Model
Category	Centrifugal Pump
Equipment Class	Pump
Equipment Type	Centrifugal Pump - Mechanical Seals

Below table illustrates the list of faults and symptoms configured for Pump Mechanical Seal System a Condition based monitoring

Fault Name	Associated Symptoms	Recommendations
Seal Malfunction	Drain Tank Level Indicator HIGH (SM)	1. Ensure spare seals cartridge is available.
	Seal Fluid Supply Temperature HIGH (SM)	2. Assess risk.
OB Seal Malfunction	Reservoir Level Indicator LOW (OB SM)	3. Monitor and replace seals at opportunity.
	Reservoir Pressure Indicator LOW (OB SM)	1. Ensure spare seals cartridge is available.
IB Seal Malfunction	Reservoir Level Indicator HIGH (IB SM)	2. Assess risk.
	Reservoir Pressure Indicator HIGH (IB SM)	3. Monitor and replace seals at first opportunity.
OB/IB Seal Pressure Malfunction	Seal Fluid Outlet Pressure LOW (SPM)	1. Ensure spare seals cartridge is available.
		2. Assess risk Monitor and replace seals at first opportunity.

6.4.2.3 Lube Oil System

Model Type	APR Model
Category	Centrifugal Pump
Equipment Class	Pump
Equipment Type	Centrifugal Pump - Lube Oil System

Below table illustrates the list of faults and symptoms configured for Pump Lube Oil Condition based monitoring.

Fault Name	Associated Symptoms	Recommendations
Centrifugal Pump Loss of Lube Oil Quality	Lube Oil Viscosity HIGH (LOQ)	1. Check oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, TAN. 4. Check no breather caps or man covers have been removed and left open to allow debris into the tank. 5. Do filter cleaning. 6. Recondition or replace oil.
	Lube Oil Viscosity LOW (LOQ)	
	Lube Oil TAN HIGH (LOQ)	
	Lube Oil Composition Analysis HIGH (LOQ)	
Centrifugal Pump Lube Oil Filter Anomaly	Lube Oil Filter DP RH (LOFA)	1. Check filter DP, changeover and clean/replace the stand by filter. 2. Check the Filter DP transmitter.
Centrifugal Pump Loss of Pressure due to Pump Anomaly	Lube Oil Header Pressure RL (LOPA)	1. Check the Discharge PT integrity. 2. Switch on both oil pumps and stabilize oil pressure. 3. Assess malfunction. 4. Line walk the system and check for leaks, ensure PSV's are seated. 5. Check spillback/PCV valves are set correctly and not returning oil back to the tank. 6. Ensure no hand valves have been left shut for the standby pump. 7. Check oil level in tank, Open tank overhead vents for release of pressure. 8. Check PCV function for Integrity. 9. Assess malfunction. 10. Check Auto Logic Function tests, check functioning of the auto change over switch. 11. If maintenance has recently been performed check correct rotation of pump. 12. Ensure NRV's are working correctly.
	Lube Oil Pump Discharge Pressure RL (LOPA)	
	Lube Oil Return Flow Rate RL (LOPA)	
	Lube Oil Main Pump Current (Run Status) ON (LOPA)	
	Lube Oil Aux Pump Current (Run Status) ON (LOPA)	
	Lube Oil PCV Position RL (LOPA)	
Centrifugal Pump Loss of Cooling due to TCV Anomaly	Lube Oil Header Temperature (Cooler Out Temp) RH	1. Check the calibration of sensor. 2. Check bypass control valve for integrity.

Fault Name	Associated Symptoms	Recommendations
	(LOCTA)	3. Open both coolers and stabilize temperature.
	LO Cooler Bypass Valve (TCV) Position RH (LOCTA)	4. Line walk the system to check for leaks. 5. Check the Isolation/bypass valve of TCV.
Centrifugal Pump Loss of Cooling due to Cooler Anomaly	Lube Oil Header Temperature (Cooler Out Temp) RH (LOCCA)	1. Check the calibration of sensor.
	LO Tank And Cooler Outlet Temperature Delta RH (LOCCA)	2. Check cooling water supply pressure. 3. Open both coolers and stabilize temperature. 4. If malfunction of one cooler is suspected, isolate and inspect it. 5. Line walk the system to check for leaks. 6. Check cooling water pumps for blockages, ensure strainer is clean.
Centrifugal Pump Loss of Heating due Lube Oil Tank Heater Off	Lube Oil Header Temperature (Cooler Out Temp) RL (LOCHOF)	1. Check the Heater ON/OFF switch integrity and Change the switch as per the requirement.
	Lube Oil Tank Temperature (Cooler Inlet Temp) RL (LOCHOF)	2. Check the heater circuit for Voltage and current.
	Lube Oil Tank Heater (Run Status) OFF (LOCHOF)	
Centrifugal Pump Loss of Cooling due Lube Oil Tank Heater On	Lube Oil Header Temperature (Cooler Out Temp) RH (LOCHO)	1. Check the Heater ON/OFF switch integrity and Change the switch as per the requirement.
	Lube Oil Tank Temperature (Cooler Inlet Temp) RH (LOCHO)	2. Offline: Check the heater circuit for Voltage and current.
	Lube Oil Tank Heater (Run Status) ON (LOCHO)	
Centrifugal Pump Loss of Cooling due to Fan Motor Issue	Lube Oil Header Temperature (Cooler Out Temp) RH (LOCFMI)	1. Offline: Check the Fan and Motor /coupling / belt Integrity and rectify.
	Lube Oil Cooler Fan 1 (Run Status) OFF (LOCFMI)	
	Lube Oil Cooler Fan 2 (Run Status) OFF (LOCFMI)	

6.4.2.4 Dry Gas Seal System

Model Type	APR Model
Category	Centrifugal Pump
Equipment Class	Pump
Equipment Type	Centrifugal Pump - Dry Gas Seal System

Below table illustrates the list of faults and symptoms configured for a Pump Dry Gas Seal Condition based monitoring.

Fault Name	Associated Symptoms	Recommendations
Centrifugal Pump Primary Seal Gas Filter Anomaly	DP over seal gas filter RH (PSGFA)	1. Check the calibration of sensor. 2. Check the filter efficiency and change to standby the filter if required. 3. Repair/replace the filter internals.
Centrifugal Pump Buffer Gas Filter Anomaly	Buffer Gas Filter DP RH (BGSFA)	1. Check the calibration of sensor. 2. Check the filter efficiency and change to standby the filter if required. 3. Repair/replace the filter internals.
Centrifugal Pump PDCV Malfunction	Reference line DP RL (PDCVA)	1. Check the calibration of sensor. 2. Check valve set point. 3. Check valve internals for any damage, and repair / replace if necessary. 4. Conduct the PDCV control loop function test .
	PDCV Valve Opening RH (PDCVA)	
Centrifugal Pump Seal Vent Control Valve Malfunction	Primary seal leak pressure DP (Leak pressure-vent pressure) SH (PSGVA)	1. Calibrate the valve and transmitters by isolation if possible. 2. Check connection loss if any. 3. Check primary vent line for any jam. 4. Check the internals of control valve of the vent line. 5. Conduct loop proof test of the control loop.
	Primary seal leak pressure DP (Leak pressure-vent pressure) SL (PSGVA)	
	Primary Seal Vent gas PCV Opening SH (PSGVA)	
	Primary Seal Vent gas PCV Opening SL (PSGVA)	
Centrifugal Pump OB Seal Malfunction	Primary vent pressure RL (OSLM)	1. Ensure upstream and downstream PSV's, check valves, transmitters are all working correctly. 2. Inspect and monitor trend sec seal gas system. 3. Ensure spare seals cartridge is available. 4. Assess risk. 5. Monitor and replace seals at opportunity. 6. Replace seal.
	Primary vent flow RL (OSLM)	
	Secondary vent pressure RH (OSLM)	
	Secondary vent flow RH (OSLM)	

Fault Name	Associated Symptoms	Recommendations
Centrifugal Pump IB Seal Malfunction	Primary vent pressure RH (IBM)	1. Ensure upstream and downstream PSV's, check valves, transmitters are all working correctly. 2. Inspect and monitor trend sec seal gas system. 3. Ensure spare seals cartridge is available. 4. Assess risk. 5. Monitor and replace seals at opportunity. 6. Replace seal.
	Primary vent flow RH (IBM)	
Centrifugal Pump Primary Seal Temp Anomaly	Primary seal gas supply temperature RL (PSGTA)	1. Check calibration of sensor. 2. Monitor seal leakage. 3. Check integrity of seal gas supply PDV. 4. Check super heater functionality if available. 5. Check separation gas. 6. Check seal supply filters for damage and are fitted correctly.
	Primary seal gas supply temperature RH (PSGTA)	
Centrifugal Pump No Lift Off or Back-Flow (No Seal Gas) At Low-Speed Transients	Primary vent pressure RH (NSLA)	
	Primary vent flow RH (NSLA)	
	Secondary vent pressure RH (NSLA)	
	Secondary vent flow RH (NSLA)	
	Speed LOW (NSLA)	
	Driver Speed LOW (NSLA)	
Centrifugal Pump Primary Seal Gas Supply Anomaly	Primary seal gas supply pressure RL (PSGSA)	1. Check the calibration of sensor. 2. Ensure upstream and downstream PSV's, check valves, transmitters are all working correctly. 3. Monitor trend of seal gas system. 4. Inspect system for leakage. 5. Inspect PCV and check functionality. 6. Replace filters.
	Primary seal gas supply pressure RH (PSGSA)	
	Primary seal gas supply flow RL (PSGSA)	
	Primary seal gas supply flow RH (PSGSA)	
Centrifugal Pump Buffer Gas Supply Anomaly	Buffer gas supply pressure RL (BGSA)	

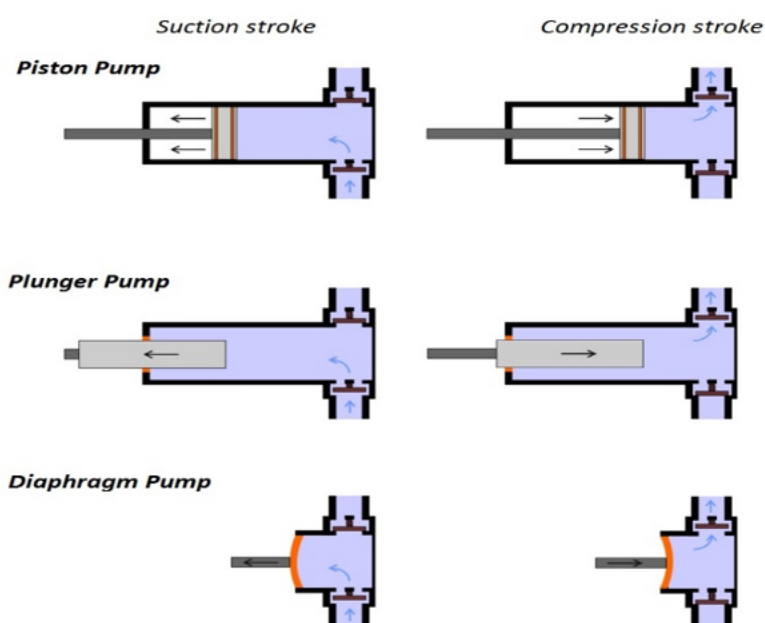
Fault Name	Associated Symptoms	Recommendations
		all working correctly.
	Buffer gas supply pressure RH (BGSA)	4. Inspect PCV and check functionality.
	Buffer gas supply flow RL (BGSA)	5. Replace Filters.
	Buffer gas supply flow RH (BGSA)	
	Secondary supply PCV Opening SH (BGSA)	
	Buffer Gas PCV Opening SH (BGSA)	
Centrifugal Pump Separation Gas Supply Anomaly	Separation gas pressure RL (SGSA)	1. Check the calibration of sensor.
	Separation gas pressure RH (SGSA)	2. Check N2 supply system.
	Separation gas flow RL (SGSA)	3. Increased risk of outboard seal contamination.
	Separation gas flow RH (SGSA)	4. Increased risk of bearing oil ingress into seal.
	Separation gas PCV Opening SH (SGSA)	5. Inspect supply line upstream of seal.
	Buffer Gas PCV Opening SH (SGSA)	6. Check labyrinth seal for blockages.
Centrifugal Pump Seal Gas Booster Pump Malfunction	Reference line DP RL (SGBPM)	Check Seal Gas Booster Pump.
	Primary seal gas supply pressure RL (SGBPM)	
	Primary seal gas supply pressure RH (SGBPM)	
	Primary seal gas supply flow RL (SGBPM)	
	Primary seal gas supply flow RH (SGBPM)	
Centrifugal Pump Secondary Supply to Primary Vent DP Anomaly	Dp secondary supply to primary vent RL (SSTPVA)	1. Check calibration of sensor.
	Dp secondary supply to primary vent RH (SSTPVA)	2. Check and control N2 supply.
		3. Check for any line leaks.
		4. Check for seal functionality and replace if necessary.

RECIPROCATING PUMP

A reciprocating pump is a class of positive-displacement pumps, which is often used where a relatively small quantity of liquid is to be handled and where delivery pressure is quite large. It works on the principle of reversing piston motion within a cylinder drawing in liquid during the forward stroke and delivering it under pressure during the return or backward stroke. The reciprocating pump can deliver the required flow rate very precisely and no priming is needed.

The three classes of reciprocating pumps are

- **Piston Type:** The pump in which fluid is moved by the repeated back-and-forth movement (strokes) of a piston. Generally, are used where medium to high delivery pressures are required, such as for high-pressure flushing of vessel interiors and tanks. These can be obtained in multicylinder form and can be single or double acting.
- **Plunger Type:** The pump in which the high-pressure seal is stationary and fluid is moved as smooth cylindrical plunger slides through the seal.
- **Diaphragm Type:** The pump in which a combination of the reciprocating action of a rubber, thermoplastic, or Teflon diaphragm is used to pump a fluid. These are invariably air driven and very compact, also no seals or packing is exposed to the liquid being pumped, which makes them ideal for handling hazardous or toxic liquids.



Based on mechanism reciprocating pump are classified as

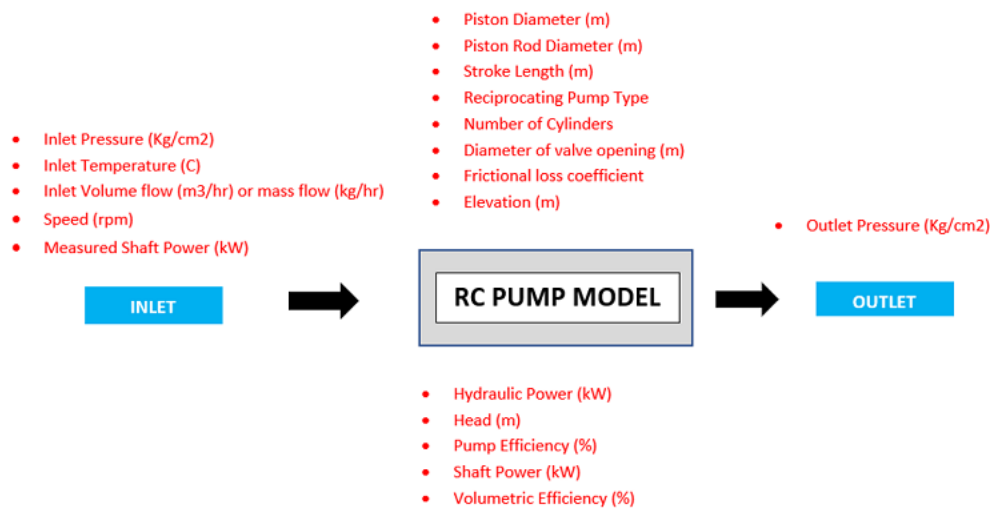
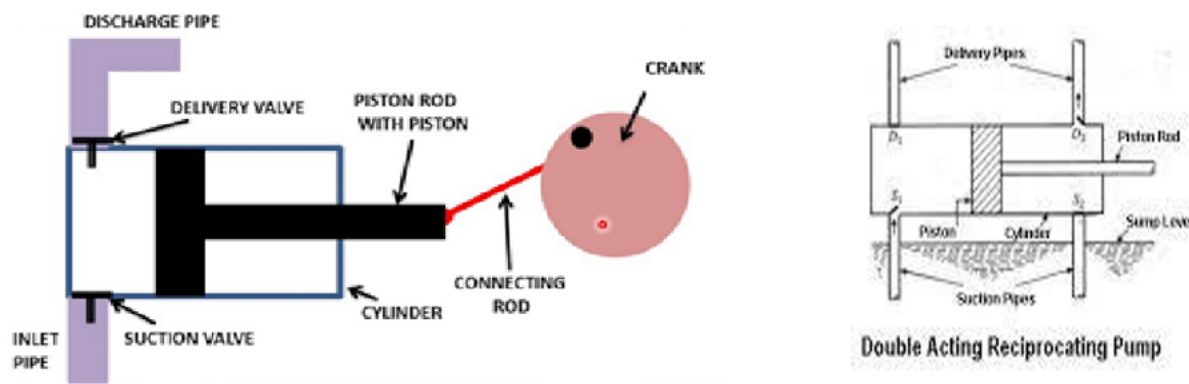
- **Single-acting reciprocating pump:** It consists of a piston of which only one side engages the fluid being displaced.
- **Double-acting reciprocating pump:** engage with both sides of the piston, each stroke of the piston carries out both suction and expulsion at the same time. Thus, it requires two inflow pipes and two outflow pipes.

Based on number of cylinders reciprocating pump are classified as

- Single cylinder - consists of a single cylinder connected to a shaft.
- Double cylinder - consists of two cylinders connected to a shaft.
- Triple cylinder - consists of three cylinders connected to a shaft.

Application of Reciprocating pumps, are as follows:

- Vessel, pipe, tank, tube, condensate pipe, heat exchanger etc. cleaning.
- Oil drilling, refineries, production, disposal, injections.
- Pneumatic pressure applications.
- Vehicle cleaning.
- Sewer line cleaning.
- Wet sandblasting.
- Boiler feeding.
- High-pressure pumps for the RO system (Reverse osmosis).
- Hydro testing of tanks, vessels, etc.
- Firefighting system.
- Wastewater treatment system.



7.1 Reciprocating Pump - Performance Monitoring

7.1.1 Notations

Below are the notations used in the equations of Performance Modeling, Expected KPI, and Operating KPI

F	Mass Flow
h	Head
Q	Volume flow
P	Pressure
η	Efficiency
D_p	Piston Diameter
d_p	Piston Rod Diameter
L	Stroke Length
n	Number of Cylinders
N_t	Reciprocating pump type
W	Power
N	Speed
d	Diameter of valve opening
f	Frictional loss coefficient
v	Velocity
h_f	Frictional head loss

7.1.2 Subscripts

i	Inlet
o	Outlet

7.1.3 Reciprocating Pump - Model Assumptions

The key assumptions taken in developing a Performance model are:

1. The density of process liquid is assumed to be constant as its incompressible.

NOTE

1. If the composition of process liquid is available, there is a provision for creating an inlet thermo stream that can be port mapped to the main asset to get thermo properties like density and vapor pressure.
2. If composition is unavailable, whereas density V/S temperature and vapor pressure V/S temperature data are available we can map these in the Property data (Tabular array) attribute.
3. If only one data point of density V/S temperature and vapor pressure V/S temperature is available, then place the data point value in vapor pressure attribute and density attribute declared at main asset instead of property data

7.1.4 Reciprocating Pump - Model Inputs

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Inlet1_Pressure	Inlet Stream Pressure (Required Input)	kg/cm ² (a)	Dynamic	Measurement	Field Input
2	Outlet1_Pressure	Outlet Stream Pressure (Required Input)	kg/cm ² (a)	Dynamic	Measurement	Field Input
3	Inlet1_Temperature	Inlet Stream Temperature (Optional if mass flow is available)	°C	Dynamic	Measurement	Field Input
4	Inlet1_VolumeFlow	Inlet Stream Volume Flow Rate (Required Input)	m ³ /hr	Dynamic	Measurement	Field Input
5	Inlet1_Actual Volumetric Flow	Inlet Stream Actual Volume Flow (Input - Port mapped from suction stream)	m ³ /hr	Dynamic	Internal	Field Input
6	Inlet1_MassFlow	Inlet Stream Mass Flow Rate (Input)	kg/hr	Dynamic	Measurement	Field Input
7	Speed	Speed (Required Input)	rpm	Dynamic	Measurement	Field Input
8	Measured Shaft Power	Measured Shaft Power (Required Input)	kW	Dynamic	Measurement	Field Input
9	Performance Data	Curve Data between Shaft Power, Volumetric Efficiency, Mechanical Efficiency VS Differential Pressure VS Speed (Input)		Static	Performance Map	Interpolated from the performance curve
10	Property Data	Curve Data between Liquid Density, Vapor Pressure VS Temperature (Required Input if composition data is not available)		Static	Internal (Fluid Property Data)	Interpolated from Property data curve
11	Liquid Density	Liquid Density (Optional Input)	kg/m ³	Static	Internal	Constant
12	Vapor Pressure	Vapor Pressure (Optional Input)	kg/cm ² (a)	Static	Internal	Constant
13	NPSH Required	NPSH Required (Input)	m	Static	Design	Constant
14	Reciprocating Pump Type	Reciprocating Pump Type (Input)		Static	Design	Constant
15	Number Of Cylinders	Number Of Cylinders (Input)		Static	Design	Constant
16	Piston Diameter	Piston Diameter (Input)	m	Static	Design	Constant
17	Piston Rod Diameter	Piston Rod Diameter (Input)	m	Static	Design	Constant
18	Stroke Length	Stroke Length (Input)	m	Static	Design	Constant
19	Gear Box Efficiency	Gear Box Efficiency (Optional Input)	%	Static	Design	Constant

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
20	Power Required	Power Required (Required Input if no performance curve is available)	kW	Static	Design	Constant
21	CT_Pump Efficiency	Constant Pump Efficiency (Required Input if no performance curve is available)	%	Static	Design	Constant
22	CT_Volumetric Efficiency	Constant Volumetric Efficiency (Required Input if no performance curve is available)	%	Static	Design	Constant
23	Suction Diameter	Suction Diameter (Input)	m	Static	Design	Constant
24	Suction Elevation	Suction Elevation (Input)	m	Static	Design	Constant
25	Suction Frictional Losses	Suction Frictional Losses (Input)	m	Static	Design	Constant
26	Suction Frictional Losses Coefficient	Suction Frictional Losses Coefficient (Input)	m	Static	Design	Constant
27	Discharge Diameter	Discharge Diameter (Input)	m	Static	Design	Constant
28	Discharge Elevation	Discharge Elevation (Input)	m	Static	Design	Constant
29	Discharge Frictional Losses	Discharge Frictional Losses (Input)	m	Static	Design	Constant
30	Discharge Frictional Losses Coefficient	Discharge Frictional Losses Coefficient (Input)	m	Static	Design	Constant

7.1.5 Reciprocating Pump - KPIs

The following indicators help us to know the overall performance of the asset:

7.1.5.1 Model Inputs

- Inlet Pressure
- Outlet Pressure
- Inlet Temperature
- Inlet Volume Flow
- Speed
- Stroke Length
- Piston Diameter

- Piston Rod Diameter
- Property Data - (ρ, T) and (P_{vapour}, T)
- Number of Cylinders
- Compressor Type
- Inlet Diameter
- Outlet Diameter
- Inlet elevation
- Outlet elevation
- Inlet Frictional Coefficient
- Outlet Frictional Coefficient
- Performance Data

$$(\dot{W}_{BHP}, \Delta_{Pressure}), (\eta_V, \Delta_{Pressure}) \text{ and } (\eta_{Mech}, \Delta_{Pressure})$$

- Measured Shaft Power

7.1.5.2 Model Output KPI's

Expected KPI

- Expected Volumetric efficiency
- Expected Volume Flow
- Expected Head
- Expected Hydraulic Power
- Expected Isothermal efficiency
- Expected Pump efficiency
- Expected Shaft Power

Operating KPI

- Operating Volumetric Efficiency
- Operating Head
- Operating Hydraulic Power
- Operating Isothermal efficiency
- Operating Pump efficiency
- Operating Shaft Power
- NPSH available

7.1.5.3 Reciprocating Pump - Performance Modeling

The Performance modeling of the reciprocating pump helps to determine the performance KPI. The following is the procedure,

Overall Mass Balance

$$F_i = F_o$$

Overall Force Balance

Applying the force balance equation,

$$P_i + \frac{\rho v_i^2}{2} + \rho g Z_i = P_o + \frac{\rho v_o^2}{2} + \rho g Z_o + \rho g h_{f_o} + \rho g h_{f_i}$$

Converting to a representative head term (to meter) on dividing ρg both sides,

$$\Delta h_t = \left(\frac{P_o - P_i}{\rho g} \right) + \left(\frac{v_o^2 - v_i^2}{2g} \right) + (Z_o - Z_i) + (h_{f_o} + h_{f_i})$$

Overall Energy Balance

$$F_o h_o (P_o, T_o) = F_i h_i (P_i, T_i) + \dot{W}_{Hydraulic\ Actual}$$

$$F_o h_o (P_o, T_i) = F_i h_i (P_i, T_i) + \dot{W}_{Hydraulic\ Isothermal}$$

$$\eta_{Isothermal} = \frac{\dot{W}_{Hydraulic\ Isothermal}}{\dot{W}_{Hydraulic\ Actual}}$$

$$\eta_{Isothermal} = \frac{\dot{W}_{Hydraulic\ Isothermal}}{\dot{W}_{Shaft} \eta_{Mech} \eta_{GearBox}}$$

$$\eta_{Pump} = \frac{\dot{W}_{Hydraulic}}{\dot{W}_{Shaft}}$$

NOTE

1. As a governing balance, the principle and equations are same for piston type, plunger type, and diaphragm. Therefore, the same model can be used for the above three variants of reciprocating pumps.
2. The pump efficiency calculation is finalized post data sheet analysis and literature survey.

7.1.5.4 Reciprocating Pump - Expected KPIs

The following indicators help us to know the expected overall performance of the Reciprocating Pump. The following will be the approach if a performance curve is available for the model.

If the Performance Curve is Available***Expected Volumetric Efficiency (%) and Expected Shaft Power (kW)***

From the performance curve of Volumetric efficiency vs pressure difference with speed in Z coordinate, we calculate expected volumetric efficiency by passing measured differential pressure and speed.

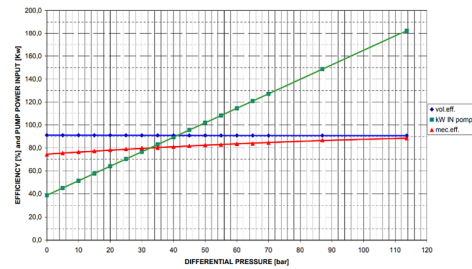
The expected volumetric efficiency is calculated using the performance curve:

$$\eta_V^{Expected} = f(\Delta_{Pressure})$$

The expected shaft Power is calculated using the performance curve:

$$\dot{W}_{Shaft}^{Expected} = f(\Delta_{Pressure})$$

The typical performance curve is represented as shown in the right cell. The curve has differential pressure in the independent axis whereas volumetric efficiency and shaft power are in the primary axis. The above equation would predict the expected volumetric efficiency and expected shaft power.



NOTE

If curve data of volumetric efficiency or mechanical efficiency is not available or is a constant value, you can map the data value to the volumetric efficiency or mechanical efficiency attribute created in the main asset.

Expected Volume Flow (m^3/hr)

$$Q_i^{Expected} = \eta_V^{Expected} Q_{Swept}$$

$$Q_{Swept} = n A_{eff} L_{Stroke} N$$

$$A_{eff} = \frac{\pi}{4} D_{eff}^2$$

$$D_{eff} = D_p \quad (\text{for Single acting})$$

$$D_{eff} = (D_p + D_p) - d_p \quad (\text{for double acting})$$

Expected Head (m)

From the performance curve of Brake HP vs differential, we pass measured power and calculate the expected differential pressure

$$\Delta h_t^{Expected} = \frac{(P_o - P_i)_{Expected}}{\rho g} + \left(\frac{v_o^2 - v_i^2}{2g} \right) + (Z_o - Z_i) + (h_{f_o} + h_{f_i})$$

$$(P_o - P_i)_{Expected} = f(\dot{W}_{Shaft}, N)$$

NOTE

If geometrical details of inlet diameter and outer diameter are provided then inlet velocity and outlet velocity are calculated by the below equation, else will be set to zero.

$$v_i^{expected} = \frac{Q_i^{Expected}}{\frac{\pi}{4} d_i^2}$$

$$v_o^{expected} = \frac{Q_i^{Expected}}{\frac{\pi}{4} d_o^2}$$

NOTE

If geometrical details of inlet diameter and outer diameter, along with inlet frictional coefficient and outlet frictional coefficient, are provided, then inlet frictional head losses and outlet frictional head losses are calculated by the below equation else will be set to zero.

On using Darcy's equation,

$$h_{f_i} = \frac{2 f_i v_i^2}{g d_i}$$

$$h_{f_o} = \frac{2 f_o v_o^2}{g d_o}$$

Expected Hydraulic Power (m³/hr)

$$\dot{W}_{Hydraulic}^{Expected} = Q_i^{Expected} (P_o - P_i)_{Expected}$$

Expected Pump Efficiency (%)

$$\eta_{Pump}^{Expected} = \frac{\dot{W}_{Hydraulic}^{Expected}}{\dot{W}_{Shaft}^{Expected}}$$

If Performance Curve is Not Available**Expected Volumetric Efficiency (%)**

If curve data of volumetric efficiency is not available or is a constant value, we can map the data value to volumetric efficiency created in the main asset.

$$\eta_v^{Expected} = \eta_v$$

Expected Shaft Power (kW)

If curve data of shaft power is not available or is a constant value, we can map the data value power required attribute created in the main asset.

$$\dot{W}_{Shaft}^{Expected} = \dot{W}_{Shaft}^{Required}$$

Expected Volume Flow (m³/hr)

The calculation of the expected volume flow remains the same as derived in the above section.

Expected Pump Efficiency (%)

The rated efficiency as provided in the datasheet needs to be provided on the rated pump efficiency attribute created at the main asset of the pump.

$$\eta_{Pump}^{Expected} = \eta_{Pump}^{Rated}$$

Expected Hydraulic Power (kW)

$$\dot{W}_{Hydraulic}^{Expected} = \dot{W}_{Shaft}^{Expected} \eta_{Pump}^{Expected}$$

Expected Head (m)

$$\Delta h_t^{Expected} = \frac{\dot{W}_{Hydraulic}^{Expected}}{Q_i^{Expected} \rho g}$$

7.1.5.5 Reciprocating Pump - Operating KPIs

The following indicators help us to know the operating overall performance of the Reciprocating Pump.

Operating Volumetric Efficiency (%)

$$\eta_V^{Operating} = \frac{Q_i}{Q_{swept}}$$

Operating Head (m)

$$\Delta h_t^{operating} = \left(\frac{P_o - P_i}{\rho g} \right) + \left(\frac{v_o^2 - v_i^2}{2g} \right) + (Z_o - Z_i) + (h_{f_o} + h_{f_i})$$

Operating Hydraulic Power (kW)

$$\dot{W}_{Hydraulic}^{Operating} = Q_i (P_o - P_i)$$

Operating Shaft Power (kW)

If Input power is available, we can PHD tag map the power to a measured shaft power attribute declared to the model.

$$\dot{W}_{Shaft}^{Operating} = (Measured\ Power)_{Tag\ value}$$

NOTE

Suppose the driver is an electric motor and the current tag value is available. In that case, the measured power attribute will be PHD tag mapped to electric power attribute of the motor asset.

Operating Pump Efficiency (%)

$$\eta_{Pump}^{Operating} = \frac{\dot{W}_{Hydraulic}^{Operating}}{\dot{W}_{Shaft}^{Operating}}$$

NPSH Available (m)

$$\Delta h_{NPSH} = \frac{P_i - P_{vapour}}{\rho g}$$

7.1.6 Reciprocating Pump - Model Outputs

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Expected Head	Expected Head (Output)	m	Dynamic	Performance	Calculated
2	Expected Hydraulic Power	Expected Hydraulic Power (Output)	kW	Dynamic	Performance	Calculated
3	Expected Pump Efficiency	Expected Pump Efficiency (Output)	%	Dynamic	Performance	Calculated
4	Expected Shaft Power	Expected Shaft Power (Output)	kW	Dynamic	Performance	Calculated
5	Expected Volume Flow	Expected Volume Flow (Output)	m ³ /hr	Dynamic	Performance	Calculated
6	Expected Volumetric Efficiency	Expected Volumetric Efficiency (Output)	%	Dynamic	Performance	Calculated
7	Differential Pressure	Differential Pressure (Output)	kg/cm ²	Dynamic	Performance	Calculated
8	Operating Head	Operating Head (Output)	m	Dynamic	Performance	Calculated
9	Operating Hydraulic Power	Operating Hydraulic Power (Output)	kW	Dynamic	Performance	Calculated
10	Operating Pump Efficiency	Operating Pump Efficiency (Output)	%	Dynamic	Performance	Calculated
11	Operating Shaft	Operating Shaft Power (Output)	kW	Dynamic	Performance	Calculated

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
	Power					
12	Operating Volumetric Efficiency	Operating Volumetric Efficiency (Output)	%	Dynamic	Performance	Calculated
13	NPSH Available	NPSH Available (Output)	m	Dynamic	Performance	Calculated
14	Deviation Head	Percent deviation in Head between Operating and Expected values (Output)	%	Dynamic	Deviation	Calculated
15	Deviation Hydraulic Power	Percent deviation in Hydraulic Power between Operating and Expected values (Output)	%	Dynamic	Deviation	Calculated
16	Deviation Pump Efficiency	Percent deviation in Pump Efficiency between Operating and Expected values (Output)	%	Dynamic	Deviation	Calculated
17	Deviation Shaft Power	Percent deviation in Shaft Power between Operating and Expected values (Output)	%	Dynamic	Deviation	Calculated
18	Deviation Volume Flow	Percent deviation in Volume Flow between Operating and Expected values (Output)	%	Dynamic	Deviation	Calculated
19	Deviation Volumetric Efficiency	Percent deviation in Volumetric Efficiency between Operating and Expected values (Output)	%	Dynamic	Deviation	Calculated
20	NPSH Margin	NPSH Margin (Output)	%	Dynamic	Deviation	Calculated

7.2 Reciprocating Pump - Fault Model

7.2.1 Reciprocating Pump - APM Model

7.2.1.1 Piston Pump APM Fault Model

Item	Type
Category	Equipment
Equipment Class	Pump
Equipment Type	Reciprocating Pump

The below table illustrates the list of faults and symptoms configured for a Reciprocating Pump for Performance

Fault Name	Associated Symptoms	Recommendations
Performance Parameter Deviation High	Volumetric Efficiency Deviation High HIGH (RCPPPDH)	1. Check Pumping Liquid Quality 2. Check for Inlet Filter for any debris Perform functional 3. Check the Suction and Discharge Valve (including the opening sequence) 4. Check the stiffness of springs in valves, Pulley and V-belt 5. Verify the Pressure after the control stage.

7.2.1.2 Plunger Pump APM Fault Model

Item	Type
Category	Equipment
Equipment Class	Pump
Equipment Type	Reciprocating Pump

The below table illustrates the list of faults and symptoms configured for a Plunger Pump for Performance.

Fault Name	Associated Symptoms	Recommendations
Performance Parameter Deviation High	Volumetric Efficiency Deviation High HIGH (RCPPPDH)	1. Check Pumping Liquid Quality, 2. Check for Inlet Filter for any debris Perform a functional check of the Suction and Discharge Valve (including the opening sequence), 3. Check the stiffness of springs in valves, Pulley and V-belt Verify the Pressure after the control stage.
	Head Deviation High HIGH (RCPPPDH)	
	Shaft Power Deviation High HIGH (RCPPPDH)	
	Discharge Flow Deviation High HIGH (RCPPPDH)	

7.2.1.3 Diaphragm Pump APM Fault Model

Being a positive displacement pump, APM of the diaphragm pump uses the piston pump model for its efficiency calculation.

7.2.2 Reciprocating Pump - APR Model

7.2.2.1 Piston Pump Bearing Fault Model

Item	Type
Category	Equipment
Equipment Class	Pump
Equipment Type	Reciprocating Pump - Bearing System

The below table illustrates the list of faults and symptoms configured for a Reciprocating Pump - Bearing System

Fault Name	Associated Symptoms	Recommendations
Main Bearing Failure /Malfunctions	Main Bearing Temperature RH (RCPBMBF)	1. Assess time to failure (trip). 2. Intervention options: 1. reduce speed. 2. reduce oil temperature to a minimum. 3. If temperature and vibrations do not stabilize plan for bearing replacement.

7.2.2.2 Piston Pump Heat Exchanger System

Item	Type
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Category	Equipment
Equipment Class	Pump
Equipment Type	Reciprocating Pump - Heat Exchanger System

The below table illustrates the list of faults and symptoms configured for a Reciprocating Pump - Heat Exchanger System.

Fault Name	Associated Symptoms	Recommendations
Heat Exchanger Loss in Efficiency	LO Inlet temperature RH (RCPHELOE)	1. To check the In process condition of the equipment to identify deviations in the process. 2. Cleaning of tube bundle & shell.
	LO outlet Temperature RH (RCPHELOE)	
	LO Outlet pressure RL (RCPHELOE)	
	Cooling Water Inlet Pressure RL (RCPHELOE)	
	Cooling Water Outlet Pressure RL (RCPHELOE)	
	Cooling Water Inlet Temp RH (RCPHELOE)	
	Cooling Water Outlet temp RH (RCPHELOE)	

7.2.2.3 Piston Pump - Transmission System Fault Model

Item	Type
Category	Equipment
Equipment Class	Pump
Equipment Type	Reciprocating Pump - Transmission System

The below table illustrates the list of faults and symptoms configured for a Reciprocating Pump - Transmission System.

Fault Name	Associated Symptoms	Recommendations
Gearbox Radial Bearings malfunction /Failure	Gearbox IP Shaft Bearing Vibration DE X RH (RCPGBRBM)	1. Assess likely cause and time to trip if intervention is not possible. 2. Intervention options; reduce load, oil temperature, and restore oil properties. 3. Ensure parts availability. 4. Take an oil sample and have it tested. 5. At first opportunity: Inspect journal and thrust bearing, and check clearances. 6. If temperature and vibrations do not stabilize plan for bearing replacement.
	Gearbox IP Shaft Bearing Vibration DE Y RH (RCPGBRBM)	
	Gearbox IP Shaft Bearing Vibration NDE X RH (RCPGBRBM)	
	Gearbox IP Shaft Bearing Vibration NDE Y RH (RCPGBRBM)	

Fault Name	Associated Symptoms	Recommendations
	Gearbox OP Shaft Bearing Vibration DE X RH (RCPGBRBM)	
	Gearbox OP Shaft Bearing Vibration DE Y RH (RCPGBRBM)	
	Gearbox OP Shaft Bearing Vibration NDE X RH (RCPGBRBM)	
	Gearbox OP Shaft Bearing Vibration NDE Y RH (RCPGBRBM)	
	Gearbox Oil TAN HIGH (RCPGBRBM)	
	Gearbox Oil TAN LOW (RCPGBRBM)	
	Gearbox Oil Viscosity HIGH (RCPGBRBM)	
	Gearbox Oil Viscosity LOW (RCPGBRBM)	
Gearbox loss of cooling	Cooling Water Inlet Pressure LOW (RCPGBLOC)	1. Check cooling water supply pressure. 2. Check the bypass control valve. 3. Open both coolers and stabilize the temperature. 4. Isolate and inspect it. 5. Line walk the system to check for leaks, Check cooling water pumps for blockages, and ensure the strainer is clean.
	Cooling Water Outlet Pressure LOW (RCPGBLOC)	
	Oil Inlet Pressure RL (RCPGBLOC)	
	Oil Outlet Pressure RL (RCPGBLOC)	

7.2.2.4 Piston Pump - Main LO Pump System

Item	Type
Category	Equipment
Equipment Class	Pump
Equipment Type	Reciprocating Pump - Main LO Pump System

The below table illustrates the list of faults and symptoms configured for a Reciprocating Pump - Main LO Pump System.

Fault Name	Associated Symptoms	Recommendations
Main Pump Pin Bearing	Pin Bearing LO Supply pressure RL	1. Recondition or replace oil. 2. Check the oil tank for free water by draining 1L into a sample bottle.

Fault Name	Associated Symptoms	Recommendations
failure	(RCPMPBRGF)	3. Switch cooler if water contamination. 4. Take an oil sample and check water content, viscosity, and TAN. 5. Check no breather caps or man covers have been removed and left open to allow debris into the tank.
	Pin Bearing LO PSV Position/return flow HIGH (RCPMPBRGF)	
Loss of oil cooling-Main pump	Lube Oil Return Temp SH (RCPMPLOC)	1. Check cooling water supply pressure. 2. Check the bypass control valve. 3. Open both coolers and stabilize the temperature. 4. If a malfunction of one cooler is suspected, isolate and inspect it. 5. Line walk the system to check for leaks. 6. Check cooling water pumps for blockages, and ensure the strainer is clean.
	Lube Oil Supply Temp SH (RCPMPLOC)	
	Heat Exchanger Cooling Water Inlet RL (RCPMPLOC)	
	Heat Exchanger Cooling Water Outlet RL (RCPMPLOC)	
Loss of oil pressure-Main pump	Duplux Oil filter DP RH (RCPMPLOP)	1. Start the stand-by pump and stabilize the oil pressure. 2. Assess malfunction. 3. Line walk the system and check for leaks, check filter DP, and ensure PSVs are seated. 4. Check spillback valves are set correctly and not returning oil back to the tank, if maintenance has recently been performed check the correct rotation of the pump. 5. Ensure no hand valves have been left shut, ensure NRVs are working correctly. 6. Check the oil level in the tank. 7. Switch the filter lever to the spare cartridge and clean/replace it.
	LO Pressure Transmitter RL (RCPMPLOP)	
	LO PSV Position/return flow HIGH (RCPMPLOP)	
	Main Lube oil Srew pump current (run signal) OFF (RCPMPLOP)	
	LO Differential Pressure RH (RCPMPLOP)	
	Lube oil level in Main pump Sump LOW (RCPMPLOP)	
Loss of lube oil quality-Main pump	LO Viscosity HIGH (RCPMPLOQ)	1. Recondition or replace oil. 2. Check the oil tank for free water by draining 1L into a sample bottle. 3. Switch the cooler if the water is contaminated. 4. Take an oil sample and check water content, viscosity, and TAN. 5. Check no breather caps or man covers have been removed and left open to allow debris into the tank.
	LO Viscosity LOW (RCPMPLOQ)	
	LO TAN HIGH (RCPMPLOQ)	
	LO TAN LOW (RCPMPLOQ)	

7.2.2.5 Plunger Pump Bearing Fault Model

Item	Type
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Category	Equipment
Equipment Class	Pump
Equipment Type	Plunger Pump Bearing System

The below table illustrates the list of faults and symptoms configured for a Plunger Pump Bearing System

Fault Name	Associated Symptoms	Recommendations
RD Bearing Fail Due to Lube Oil Pressure	Radial Bearing Vibration DE X RH (PPRBLOP)	1. Switch on both oil pumps (if possible) and stabilize the oil pressure. 2. Assess malfunction. 3. Line walk oil pipework if accessible and try to identify leaks in flanges/pipework. 4. Inspect filters and strainers for blockages. 5. Ensure the LO pressure transmitter/gauge is reading correctly. 6. Inspect/borescope oil drains for oil coking and blockages.
	Radial Bearing Vibration DE Y RH (PPRBLOP)	
	Radial Bearing Vibration NDE X RH (PPRBLOP)	
	Radial Bearing Vibration NDE Y RH (PPRBLOP)	
	Radial Bearing Temperature DE A RH (PPRBLOP)	
	Radial Bearing Temperature DE B RH (PPRBLOP)	
	Radial Bearing Temperature NDE A RH (PPRBLOP)	
	Radial Bearing Temperature NDE B RH (PPRBLOP)	
	LO header pressure SL (PPRBLOP)	
RD Bearing Fail Due to Lube Oil Temperature	Radial Bearing Vibration DE X RH (PPRBLOT)	1. Check cooling water supply pressure. 2. Check the bypass control valve. 3. Open both coolers and stabilize the temperature. 4. Check oil heater is not on (even if instruments indicate it is off). 5. Line walk the LO cooling pipework to check for leaks. 6. Inspect strainers on pumps. 7. Check instrumentation is working correctly. 8. If a malfunction of one cooler is suspected, isolate and inspect it.
	Radial Bearing Vibration DE Y RH (PPRBLOT)	
	Radial Bearing Vibration NDE X RH (PPRBLOT)	
	Radial Bearing Vibration NDE Y RH (PPRBLOT)	
	Radial Bearing Temperature DE A RH (PPRBLOT)	
	Radial Bearing Temperature DE B RH (PPRBLOT)	
	Radial Bearing Temperature NDE A RH (PPRBLOT)	
	Radial Bearing Temperature NDE B RH (PPRBLOT)	
	LO header temperature SH (PPRBLOT)	

Fault Name	Associated Symptoms	Recommendations
DE Radial Bearing Failure/Malfunction	Radial Bearing Vibration DE X RH (PPDERBM)	1. Assess time to failure (trip). 2. Intervention options; reduce speed, and reduce oil temperature to a minimum. 3. If temperature and vibrations do not stabilize plan for bearing replacement.
	Radial Bearing Vibration DE Y RH (PPDERBM)	
	Radial Bearing Temperature DE A RH (PPDERBM)	
	Radial Bearing Temperature DE B RH (PPDERBM)	
NDE Radial Bearing Failure/Malfunction	Radial Bearing Vibration NDE X RH (PPNDERBM)	1. Assess time to failure (trip). 2. Intervention options; reduce speed, and reduce oil temperature to a minimum. 3. If temperature and vibrations do not stabilize plan for bearing replacement.
	Radial Bearing Vibration NDE Y RH (PPNDERBM)	
	Radial Bearing Temperature NDE A RH (PPNDERBM)	
	Radial Bearing Temperature NDE B RH (PPNDERBM)	

7.2.2.6 Plunger Pump Lube Oil System Fault Model

Item	Type
Category	Equipment
Equipment Class	Pump
Equipment Type	Plunger Pump LO System

The below table illustrates the list of faults and symptoms configured for a Plunger Pump LO System.

Fault Name	Associated Symptoms	Recommendations
Loss of lube oil quality	LO Viscosity HIGH (PPLOQ)	1. Recondition or replace oil. 2. Check the oil tank for free water by draining 1L into a sample bottle. 3. Switch cooler if water contamination. 4. Take an oil sample and check water content, viscosity, and TAN. 5. Ensure all man covers on tanks are closed.
	LO TAN HIGH (PPLOQ)	
	LO TAN LOW (PPLOQ)	
Loss of pressure	LO header pressure RL (PPLOP)	1. Switch on both oil pumps (if possible) and stabilize the oil pressure. 2. Assess malfunction. 3. Line walk oil pipework if accessible and try to identify leaks in flanges/pipework. 4. Inspect filters and strainers for blockages. 5. Inspect/borescope oil drains for oil coking and blockages. 6. Ensure the LO pressure transmitter/gauge is reading

Fault Name	Associated Symptoms	Recommendations
	Oil tank level RL (PPLOP)	correctly.
Loss of cooling	LO header temperature RH (PPLOC)	1. Check cooling water supply pressure. 2. Check the bypass control valve. 3. Open both coolers and stabilize the temperature. 4. Check oil heater is not on (even if instruments indicate it is off). 5. If a malfunction of one cooler is suspected, isolate and inspect it. 6. Line walk the LO cooling pipework to check for leaks. 7. Inspect strainers on pumps. 8. Check instrumentation is working correctly.
	Oil tank temperature RH (PPLOC)	
Lube Oil Filter DP High	Oil filter DP RH (PPLOFDP)	1. Change the filter. 2. Replace the filter. 3. Check the calibration of the transmitter.

7.2.2.7 Plunger Pump Seal System Fault Model

Item	Type
Category	Equipment
Equipment Class	Pump
Equipment Type	Plunger Pump Seal System

The below table illustrates the list of faults and symptoms configured for a Plunger Pump Seal System.

Fault Name	Associated Symptoms	Recommendations
Shaft Seal failure	Outlet Pressure RL (PPSSF)	1. Increase cooling of seal faces. 2. Increase flush flow rate. 3. Check for improper materials or seals used for application. 4. Prevent abrasives from accumulating at seal faces. 5. Use an abrasive separator or filter. 6. Check for obstructed flow in cooling lines. 7. Check for rubbing of the seal on the shaft. 8. Improve cooling flush lines.
	Outlet Flow RL (PPSSF)	
	Seal pressure DE RL (PPSSF)	
	Seal pressure NDE RL (PPSSF)	

7.2.2.8 Diaphragm Pump - Air Operated Fault Model

Item	Type
Category	Equipment
Equipment Class	Pump
Equipment Type	DiaphragmPump_AirOperated

The below table illustrates the list of faults and symptoms configured for a Diaphragm Pump Air operated.

Fault Name	Associated Symptoms	Control Action
DP HighPH Anomaly	AODP FLUID PH LOW (HPA)	1. Run The Pump Slowly, It is suggested to run Abrasive Fluids on Bigger Pumps at a Slower Speed.
DP PumpCycle Anomaly	AODP AIR SUPPLY PRES RL (PCA)	1. Set the lubricator on the lowest possible setting or remove it. Units are designed for lube-free operation.
	AODP AIR REGVALVE POS RL (PCA)	2. Check the Hydraulic line size and length, and compressor capacity (HP VS CFM required).
	AODP DIS FLOW RL (PCA)	3. Disassemble and inspect the main Hydraulic distribution valve, pilot valve, and pilot valve actuators.
	AODP DIS PRES RL (PCA)	4. Check for inadvertently closed discharge line valves.
DP ExhaustLeaking Anomaly	AODP AIR SUPPLY PRES RH (ELA)	5. Clean discharge manifolds/piping.
	AODP DIS FLOW RL (ELA)	6. Increase the inlet Hydraulic pressure to the pump. The pump is designed for a 1:1 pressure ratio at zero flow (Does not apply to high-pressure 2:1 units).
		7. Remove the muffler screen, clean or de-ice, and re-install.
DP StallingFlow Anomaly	AODP AIR SUPPLY PRES RL (SFA)	8. Disassemble pump chambers.
		9. Inspect for diaphragm rupture or loose diaphragm plate assembly.
		10. Disassemble and inspect wetted chambers.
		11. Remove or flush any obstructions

Fault Name	Associated Symptoms	Control Action
		12. Remove or flush obstruction. Check and clear all suction screens or strainers.
	AODP SUC IN PRES RL (SFA)	13. Disassemble pump chambers. Inspect for diaphragm rupture or loose diaphragm plate assembly.
	AODP DIS FLOW RL (SFA)	14. Disassemble the wet end of the pump and manually dislodge the obstruction in the check valve pocket.
	AODP DIS PRES RL (SFA)	15. Inspect and check valves and seats for wear and proper setting. Replace if necessary. 16. Purge chambers through tapped chamber vent plugs. 17. Purging the chambers of Hydraulic can be dangerous.
DP Premature Diaphragm Failure	AODP AIR SUPPLY PRES RH (PDF)	1. Enlarge pipe diameter on the suction side of the pump. 2. Move the pump closer to the product. 3. Raise the pump/place the pump on top of the tank to reduce inlet pressure. 4. Install Back pressure device. 5. Add an accumulation tank or pulsation dampener. 6. Consult Chemical Resistance Chart for compatibility with products, cleaners, temperature limitations, and lubrication. 7. Check Operating Manual to check for the correct part and installation. Ensure outer plates have not been worn to a sharp edge.
	AODP AIR SUPPLY TEMP RH (PDF)	
	AODP DIS PRES RH (PDF)	
	AODP DIS TEMP RH (PDF)	
DP Pump Operation Anomaly	AODP AIR SUPPLY PRES RL (POA)	1. Alter the inlet Hydraulic pressure to the design Hydraulic pressure required. The pump is designed for a 1:1 pressure ratio at zero flow (Does not apply to high-pressure 2:1 units). 2. Install gaskets with holes properly aligned. 3. Remove the pilot valve and inspect the actuator plungers
	AODP AIR SUPPLY PRES RH (POA)	
	AODP AIR FILTER DP RH (POA)	
	AODP AIR REGVALVE POS RL (POA)	
	AODP AIR REGVALVE POS RH (POA)	

7.2.2.9 Diaphragm Pump - Hydraulic Operated Fault Model

Item	Type
Category	Equipment
Equipment Class	Pump

Equipment Type	DiaphragmPump_HydraulicOperated
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The below table illustrates the list of faults and symptoms configured for a Diaphragm Pump Hydraulic operated model System.

Fault Name	Associated Symptoms	Control Action
HYD DP Pump Operation Anomaly	HYD IN PRESS RL (HPOA)	1. Alter the inlet Hydraulic pressure to the design Hydraulic pressure required. The pump is designed for a 1:1 pressure ratio at zero flow (Does not apply to high-pressure 2:1 units). 2. Install gaskets with holes properly aligned. 3. Remove the pilot valve and inspect the actuator plungers.
HYD DP Discharge Anomaly	HYD DISC PRES RL (HDA)	The discharge valve part of the pump has to be removed and cleaned and reinstalled.
	HYD HBV FLOW RH (HDA)	
	HYD DISC FLOW RL (HDA)	
HYD DP PumpCycle Anomaly	HYD IN PRESS RL (HPCA)	1. Set the lubricator on the lowest possible setting or remove it. Units are designed for lube-free operation. 2. Check the Hydraulic line size and length, and compressor capacity (HP VS CFM required). 3. Disassemble and inspect the main Hydraulic distribution valve, pilot valve, and pilot valve actuators. 4. Check for inadvertently closed discharge line valves. 5. Clean discharge manifolds/piping.
	HYD DISC PRES RL (HPCA)	6. Increase the inlet Hydraulic pressure to the pump. The pump is designed for a 1:1 pressure ratio at zero flow (Does not apply to high-pressure 2:1 units). 7. Remove the muffler screen, clean or de-ice, and re-install. 8. Disassemble pump chambers. 9. Inspect for diaphragm rupture or loose diaphragm plate assembly. 10. Disassemble and inspect wetted chambers. 11. Remove or flush any obstructions.
HYD DP NoPriming Anomaly	HYD SUC IN PRESS RL (HNPA)	1. Check suction condition (move pump closer to product). 2. Disassemble the wet end of the pump and manually dislodge the obstruction in the check valve pocket. 3. Clean out around the valve ball cage and valve seat area. 4. Replace the valve ball or valve seat if damaged. 5. Use heavier valve ball material. 6. Worn valve ball or valve seat. 7. Worn fingers in valve ball cage (replace part). 8. Check Chemical Resistance Guide for compatibility.

Fault Name	Associated Symptoms	Control Action
		9. Inspect and check valves and seats for wear and proper setting. Replace if necessary. 10. Remove or flush obstruction. 11. Check and clear all suction screens or strainers. 12. Visually inspect all suction-side gaskets and pipe connections. 13. Disassemble pump chambers. 14. Inspect for diaphragm rupture or loose diaphragm plate assembly.
HYD DP StallingFlow Anomaly	HYD DISC PRES RL (HSFA)	1. Set the lubricator on the lowest possible setting or remove it. Units are designed for lube-free operation. 2. Remove the muffler screen, de-ice, and re-install. 3. Install a point-of-use Hydraulic drier. 4. Clean manifolds to allow proper Hydraulic flow. 5. Increase the inlet Hydraulic pressure to the pump. 6. The pump is designed for a 1:1 pressure ratio at zero flow (Does not apply to high-pressure 2:1 units). 7. Check suction (move pump closer to product). 8. Check the Hydraulic line size, length, and compressor capacity. 9. Decrease inlet Hydraulic (pressure and volume) to the pump. 10. The pump is cavitating the fluid by fast cycling. 11. Meet or exceed pump connections. 12. Install a larger Hydraulic line and connection. 13. Visually inspect all suction-side gaskets and pipe connections. 14. Remove or flush obstruction. 15. Check and clear all suction screens or strainers. 16. Disassemble pump chambers. 17. Inspect for diaphragm rupture or loose diaphragm plate assembly. 18. Disassemble the wet end of the pump and manually dislodge the obstruction in the check valve pocket. 19. Inspect and check valves and seats for wear and proper setting. Replace if necessary. 20. Purge chambers through tapped chamber vent plugs. 21. Purging the chambers of Hydraulic can be dangerous.
HYD DP PrematureDiaphragm Failure	HYD DIAPHRAGM STATUS ON (HPDF)	1. Enlarge pipe diameter on the suction side of the pump. 2. Move the pump closer to the product. 3. Raise the pump/place pump on top of the tank to reduce inlet pressure. 4. Install Back pressure device. 5. Add an accumulation tank or pulsation dampener. 6. Consult Chemical Resistance Chart for compatibility with products, cleaners, temperature limitations, and lubrication.

Fault Name	Associated Symptoms	Control Action
	HYD DISC FLOW RL (HPDF)	7. Check Operating Manual to check for the correct part and installation. 8. Ensure outer plates have not been worn to a sharp edge.

7.2.2.10 Diaphragm Pump Motor Operated Fault Model

Item	Type
Category	Equipment
Equipment Class	Pump
Equipment Type	DiaphragmPump_MotorOperated

The Below table illustrates the list of faults and symptoms configured for a Diaphragm Pump motor-operated fault model system.

Fault Name	Associated Symptoms	Control Action
MO DP Pump Operation Anomaly	Suction Inlet Pressure RL (MPOA)	1. Alter the inlet air pressure to the design air pressure required. 2. Install gaskets with holes properly aligned. 3. Remove pilot valve and inspect actuator plungers.
MO DP PumpCycle Anomaly	Discharge Pressure RL (MPCA)	1. Set lubricator on lowest possible setting or remove. 2. Units are designed for lube free operation. 3. Check the air line size and length, compressor capacity (HP VS cfm required). 4. Disassemble and inspect main air distribution valve, pilot valve and pilot valve actuators. 5. Check for inadvertently closed discharge line valves. 6. Clean discharge manifolds/piping. 7. Increase the inlet air pressure to the pump. 8. Remove muffler screen, clean or de-ice, and re-install. 9. Disassemble pump chambers. 10. Inspect for diaphragm rupture or loose diaphragm plate assembly. 11. Disassemble and inspect wetted chambers. 12. Remove or flush any obstructions.
MO DP NoPriming Anomaly	Discharge Flow RL (MNPA)	1. Check suction condition (move pump closer to product). 2. Disassemble the wet end of the pump and manually dislodge

Fault Name	Associated Symptoms	Control Action
		obstruction in the check valve pocket. 3. Clean out around valve ball cage and valve seat area. 4. Replace valve ball or valve seat if damaged. 5. Use heavier valve ball material. 6. Worn valve ball or valve seat. 7. Worn fingers in valve ball cage (replace part). 8. Check Chemical Resistance Guide for compatibility. 9. Check Chemical Resistance Guide for compatibility. 10. Inspect check valves and seats for wear and proper setting. 11. Replace if necessary. 12. Remove or flush obstruction. 13. Check and clear all suction screens or strainers. 14. Visually inspect all suction-side gaskets and pipe connections. 15. Disassemble pump chambers. 16. Inspect for diaphragm rupture or loose diaphragm plate assembly.
MODP StallingFlow Anomaly	Suction Inlet Pressure RL (MSFA)	1. Set lubricator on lowest possible setting or remove. 2. Units are designed for lube free operation. 3. Remove muffler screen, de-ice, and re-install. 4. Install a point of use air drier. 5. Clean manifolds to allow proper air flow. 6. Increase the inlet air pressure to the pump. 7. Pump is designed for 1:1 pressure ratio at zero flow. 8. Check suction (move pump closer to product). 9. Check the air line size, length, compressor capacity. 10. Decrease inlet air (pressure and volume) to the pump. 11. Pump is cavitating the fluid by fast cycling. 12. Meet or exceed pump connections. 13. Install a larger air line and connection.

Fault Name	Associated Symptoms	Control Action
		14. Visually inspect all suction-side gaskets and pipe connections. 15. Remove or flush obstruction. 16. Check and clear all suction screens or strainers. 17. Disassemble pump chambers.
	Discharge Pressure RL (MSFA)	18. Inspect for diaphragm rupture or loose diaphragm plate assembly. 19. Disassemble the wet end of the pump and manually dislodge obstruction in the check valve pocket. 20. Inspect check valves and seats for wear and proper setting. 21. Replace if necessary. 22. Purge chambers through tapped chamber vent plugs. 23. Purging the chambers of air can be dangerous.
MODP Premature Diaphragm Failure	Discharge Pressure RH (MPDF)	1. Enlarge pipe diameter on suction side of pump. 2. Move pump closer to product. 3. Raise pump/place pump on top of tank to reduce inlet pressure. 4. Install Back pressure device.
	Discharge Temperature RH (MPDF)	5. Add accumulation tank or pulsation dampener. 6. Consult Chemical Resistance Chart for compatibility with products, cleaners, temperature limitations and lubrication. 7. Check Operating Manual to check for correct part and installation. 8. Ensure outer plates have not been worn to a sharp edge.
MODP DryRun	Suction Inlet Pressure RL (MDR)	1. Monitoring the inlet fluid level will ensure that the pump will have continues supply of fluid resulting in a run dry condition.
	Suction inlet filter DP RL (MDR)	
MODP LowPH Anomaly	PH of Process Fluid RL (MHPA)	1. Run The Pump Slowly, It is suggested to run Abraasive Fluids on Bigger Pumps with Slower Speed.
MODP Diaphragm Anomaly	Diaphragm Rupture Status ON (MDRA)	1. Check the integrity of the diaphragm rupture status sensor. 2. Change the diaphragm with a suitable diaphragm.

Diaphragm Pump- Transmission System Fault Model

Item	Type
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Category	Equipment
Equipment Class	Pump
Equipment Type	DiaphragmPump_Transmission

Below table illustrates the list of faults and symptoms configured for a Diaphragm Pump Transmission system.

Fault Name	Associated Symptoms	Control Action
DP GB RB MALF	Gearbox IPShaftBearingVibration DE X RH (DPRB)	1. Assess likely cause and time to trip if intervention not possible. 2. Intervention options; reduce load, oil temperature, restore oil properties. 3. Ensure parts availability. 4. Take an oil sample and have it tested. 5. At first opportunity: Inspect journal and thrust bearing, check clearances. 6. If temperature and vibrations do not stabilize plan for bearing replacement.
	Gearbox IPShaftBearingVibration DE Y RH (DPRB)	
	Gearbox IPShaftBearingVibration NDE X RH (DPRB)	
	Gearbox IPShaftBearingVibration NDE Y RH (DPRB)	
	Gearbox OPShaftBearingVibration DE X RH (DPRB)	
	Gearbox OPShaftBearingVibration DE Y RH (DPRB)	
	Gearbox OPShaftBearingVibration NDE X RH (DPRB)	
	Gearbox OPShaftBearingVibration NDE Y RH (DPRB)	
	Gearbox Oil TAN HIGH (DPRB)	
	Gearbox Oil TAN LOW (DPRB)	
	Gearbox Oil Viscosity HIGH (DPRB)	
	Gearbox Oil Viscosity LOW (DPRB)	
DP GB LOC	Cooling Water Inlet Pressure LOW (DPLOC)	1. Check cooling water supply pressure. 2. Check bypass control valve. 3. Open both coolers and stabilise temperature. 4. isolate and inspect it. 5. Line walk the system to check for leaks, Check cooling water pumps for blockages, ensure strainer is clean, .
	Cooling Water Outlet Pressure LOW (DPLOC)	
	Oil Inlet Pressure RL (DPLOC)	
	Oil Outlet Pressure RL (DPLOC)	

SCREW PUMP

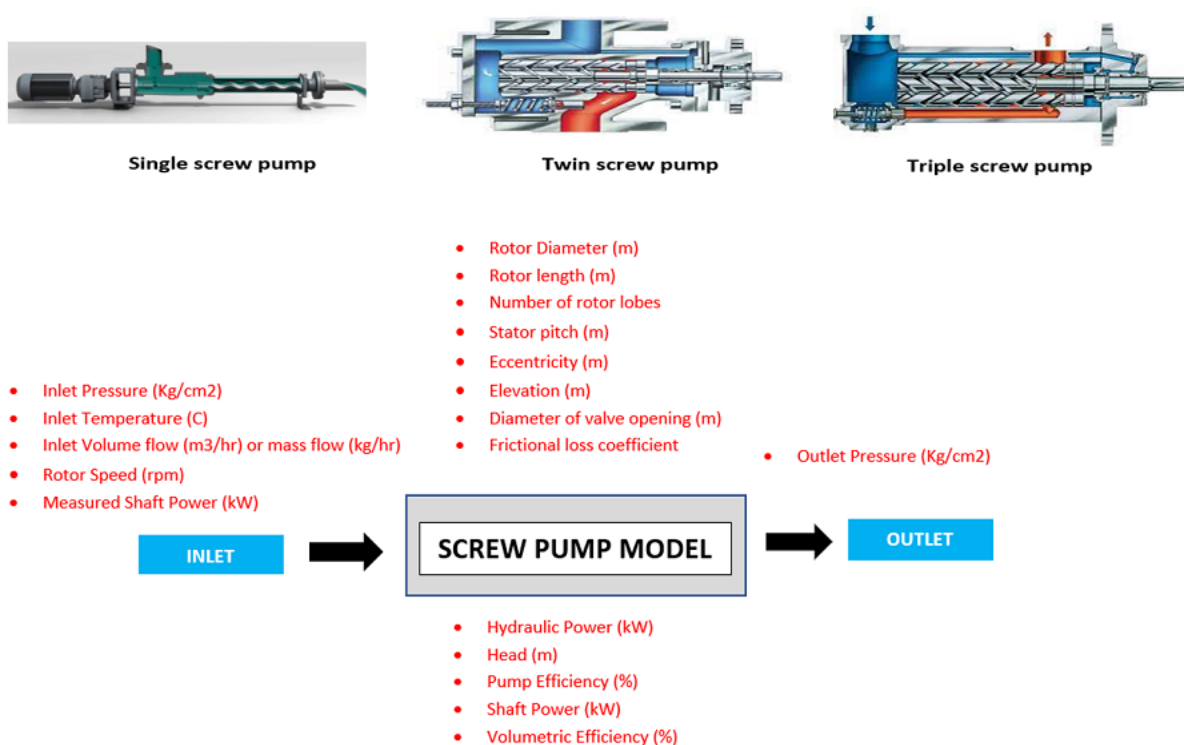
Is a Positive displacement pump that uses one or several screws to move fluid solids or liquids along the screw(s) axis. In its simplest form (the Archimedes' screw pump), a single screw rotates in a cylindrical cavity, moving the material along the screw's spindle.

Some screw pump advantages include versatility with different flow rates, pressures, liquid types, and viscosities. There are many uses for screw pumps in various industries, including Manufacturing, Mining, and Oil & Gas. Most of these applications deal with high-viscosity fluid, such as oil or asphalt, or multi-phase fluid, which means there is liquid, and vapor mixed in the fluid stream.

The screw pump is classified as,

- **Single Screw:** A progressive cavity pump that includes one shaft with somewhat twisted within the form of a screw & is enclosed within a pumping hall that is generally rubber lined.
- **Twin Screw:** These pumps are also named the double screw pump, and it is the most general type used for high-power applications like weighty oil tube transfer. The screws in the pumps are driven from the motor, & timing gears are generally included to turn the next screw.
- **Triple Screw:** These pumps are usually used in small applications like lubrication systems. The screws in the pump are determined from the motor to rotate the remaining two screws which are around it without using timing gears.

Even the screw pump is classified as multiple-screw pump (More than three screws).



8.1 Screw Pump - Performance Monitoring

8.1.1 Notations

F	Mass Flow
h	Head
H	Enthalpy
Q	Volume flow
P	Pressure
η	Efficiency
D	Rotor diameter
l	Rotor length
L_r	Number of lobes of the rotor
W	Power
N	Rotor Speed
e	Eccentricity
T_s	Pitch of stator
T_r	Pitch of rotor
d	Diameter of valve opening
f	Frictional loss coefficient
v	Velocity
h_f	Frictional head loss

8.1.2 Subscripts

i	Inlet
o	Outlet

8.1.3 Screw Pump - Model Assumptions

The key assumptions taken in developing a Performance model are:

1. The density of process liquid is assumed to be constant as its incompressible.

NOTE

1. If the composition of process liquid is available, there is the provision for creating an inlet thermo-stream that can be port mapped to the main asset to get thermo-properties like density and vapor pressure.
2. If the composition is unavailable, whereas density V/S temperature and vapor pressure V/S temperature data is available we can map these in the Property data (Tabular array) attribute.
3. If only one data point of density V/S temperature and vapor pressure V/S temperature is available, then place the data point value in the vapor pressure attribute and density attribute declared at main asset instead of property data

8.1.4 Screw Pump - Model Input

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Inlet1_Pressure	Inlet Stream Pressure (Required Input)	kg/cm ² (a)	Dynamic	Measurement	Field Input
2	Outlet1_Pressure	Outlet Stream Pressure (Required Input)	kg/cm ² (a)	Dynamic	Measurement	Field Input
3	Inlet1_Temperature	Inlet Stream Temperature (Required Input)	°C	Dynamic	Measurement	Field Input
4	Inlet1_Actual VolumetricFlow	Inlet Stream Actual Volume Flow (Input - Port mapped from suction stream)	m ³ /hr	Dynamic	Internal	Field Input
5	Inlet1_MassFlow	Inlet Stream Mass Flow Rate (Input)	kg/hr	Dynamic	Measurement	Field Input
6	Speed	Speed (Required Input)	rpm	Dynamic	Measurement	Field Input
7	Measured Shaft Power	Measured Shaft Power (Required Input)	kW	Dynamic	Measurement	Field Input
8	Performance Data	Curve Data between Shaft Power, Efficiency VS Differential Pressure VS Speed (Input)		Static	Performance Map	Interpolated from the performance curve
9	Property Data	Curve Data between Liquid Density, Vapour Pressure VS Temperature (Required Input if composition data is not available)		Static	Internal (Fluid Property Data)	Interpolated from Property data curve
10	Liquid Density	Liquid Density (Optional Input)	kg/m ³	Static	Internal	Constant
11	Vapour Pressure	Vapour Pressure (Optional Input)	kg/cm ² (a)	Static	Internal	Constant
12	NPSH Required	NPSH Required (Input)	m	Static	Design	Constant
13	Number of Rotor Lobes	Number of Rotor Lobes (Input)		Static	Design	Constant
14	Rotor Diameter	Rotor Diameter (Input)	m	Static	Design	Constant

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
15	Rotor Length	Rotor Length (Input)	m	Static	Design	Constant
16	Rotor Pitch	Rotor Pitch (Input)	m	Static	Design	Constant
17	Rotor Profile Constant	Rotor Profile Constant (Input)		Static	Design	Constant
18	Stator Pitch	Stator Pitch (Input)	m	Static	Design	Constant
19	Eccentricity	Eccentricity (Input)	m	Static	Design	Constant
20	Power Required	Power Required (Required Input if no performance curve is available)	kW	Static	Design	Constant
21	CT_Pump Efficiency	Constant Pump Efficiency (Required Input if no performance curve is available)	%	Static	Design	Constant
22	CT_Volumetric Efficiency	Constant Volumetric Efficiency (Required Input if no performance curve is available)	%	Static	Design	Constant
23	Suction Diameter	Suction Diameter (Input)	m	Static	Design	Constant
24	Suction Elevation	Suction Elevation (Input)	m	Static	Design	Constant
25	Suction Frictional Losses	Suction Frictional Losses (Input)	m	Static	Design	Constant
26	Suction Frictional Losses Coefficient	Suction Frictional Losses Coefficient (Input)	m	Static	Design	Constant
27	Discharge Diameter	Discharge Diameter (Input)	m	Static	Design	Constant
28	Discharge Elevation	Discharge Elevation (Input)	m	Static	Design	Constant
29	Discharge Frictional Losses	Discharge Frictional Losses (Input)	m	Static	Design	Constant
30	Discharge Frictional Losses Coefficient	Discharge Frictional Losses Coefficient (Input)	m	Static	Design	Constant

8.1.5 Screw Pump - KPIs

The following indicators help us to know the overall performance of the asset.

8.1.5.1 Model Inputs

- Inlet Pressure
- Outlet Pressure
- Inlet Temperature
- Inlet Volume Flow
- Rotor Speed
- Rotor Length
- Rotor Diameter
- Pitch of Stator
- Eccentricity
- Inlet Diameter
- Outlet Diameter
- Inlet elevation
- Outlet elevation
- Inlet Frictional Coefficient
- Outlet Frictional Coefficient
- Property Data
 (ρ, T) and P_{vapour}, T
- Performance Data
 $(\dot{W}_{BHP}, \Delta P_{Pressure}, N)$
- Measured Shaft Power

8.1.5.2 Model Output

Expected KPI

- Expected Volumetric efficiency
- Expected Volume Flow
- Expected Head
- Expected Hydraulic Power
- Expected Pump efficiency
- Expected Shaft Power

Operating KPI

- Operating Volumetric Efficiency
- Operating Head

- Operating Hydraulic Power
- Operating Pump efficiency
- Operating Shaft Power
- NPSH available

8.1.5.3 Screw Pump - Performance Modelling

The Performance modeling of the screw pump helps to determine the performance KPI. The following is the procedure:

Overall Mass Balance

Applying the continuity equation,

$$F_i = F_o$$

Overall Force Balance

Applying the force balance equation,

$$P_i + \frac{\rho v_i^2}{2} + \rho g Z_i = P_o + \frac{\rho v_o^2}{2} + \rho g Z_o + \rho g h_{f_o} + \rho g h_{f_i}$$

Converting to a representative head term, by dividing both sides with ρg

$$\Delta h_t = \left(\frac{P_o - P_i}{\rho g} \right) + \left(\frac{v_o^2 - v_i^2}{2g} \right) + (Z_o - Z_i) + (h_{f_o} + h_{f_i})$$

Overall Energy Balance

$$F_o H_o(P_o, T_o) = F_i H_i(P_i, T_i) + \dot{W}_{Hydraulic\ Actual}$$

$$F_o H_o(P_o, T_i) = F_i H_i(P_i, T_i) + \dot{W}_{Hydraulic\ Isothermal}$$

$$\eta_{Isothermal} = \frac{\dot{W}_{Hydraulic\ Isothermal}}{\dot{W}_{Hydraulic\ Actual}}$$

$$\eta_{Isothermal} = \frac{\dot{W}_{Hydraulic\ Isothermal}}{\dot{W}_{Shaft} \eta_{Mech} \eta_{GearBox}}$$

$$\eta_{Pump} = \frac{\dot{W}_{Hydraulic}}{\dot{W}_{Shaft}}$$

NOTE

The pump efficiency calculation is finalized post data sheet analysis and literature survey.

8.1.5.4 Screw Pump - Expected KPI

The following indicators help us to know the expected overall performance of the Screw Pump. The following will be the approach if a performance curve is available for the model.

If the Performance Curve is Available

Expected Volumetric Efficiency (%)

The Volumetric efficiency is from the data sheet value that needs to be mapped to the volumetric efficiency attribute present in the main asset.

The expected shaft Power is calculated using a performance curve supplied by the user,

$$\eta_V^{Expected} = \eta_V$$

Expected Shaft Power (kW)

From the performance curve of Shaft power vs pressure difference with speed in Z coordinate, we calculate expected shaft power by passing measured differential pressure and speed.

The expected shaft Power is calculated using a performance curve supplied by the user,

$$W_{Shaft}^{Expected} = f(\Delta_{Pressure}, N)$$

Expected Volume Flow (m³/hr)

The theoretical swept volume flow is given by

Alternative 1:

$$Q_{Swept} = 0.0007928717 e D T_s N$$

$$T_s = \frac{L_r}{L_r + 1} T_r$$

Where,

e = Distance between stator centerline and rotor centerline. Eccentricity can also be referred to as the difference between the radius of stator and the radius of the rotor.

Reference: Boyun Guo, William C. Lyons, Ali Ghalambor., Petroleum Production Engineering - A computer assisted approach.,2007. (Page Number: 14/214)

Alternative 2:

$$Q_{Swept} = \frac{D^3 \left(\frac{l}{D}\right)}{c} N$$

Where,

c = 2.231 circular profile and 2.055 if asymmetric profile.

$$Q_i^{Expected} = Q_{Swept} \eta_V^{Expected}$$

Expected Head (m)

From the performance curve of Brake HP vs differential pressure with speed in Z coordinate, we pass measured power and speed to calculate the expected differential pressure

$$\Delta h_t^{Expected} = \frac{(P_o - P_i)_{Expected}}{\rho g} + \left(\frac{v_o^2 - v_i^2}{2g} \right) + (Z_o - Z_i) + (h_{f_o} + h_{f_i})$$

$$(P_o - P_i)_{Expected} = f(\dot{W}_{Shaft}, N)$$

NOTE

If geometrical details of inlet diameter and outer diameter are provided then inlet velocity and outlet velocity are calculated by the below equation, else will be set to zero,

$$v_i^{expected} = \frac{Q_i^{Expected}}{\frac{\pi}{4} d_i^2}$$

$$v_o^{expected} = \frac{Q_i^{Expected}}{\frac{\pi}{4} d_o^2}$$

NOTE

If geometrical details of inlet diameter and outer diameter, along with inlet frictional coefficient and outlet frictional coefficient are provided then inlet frictional head losses and outlet frictional head losses are calculated by the below equation, else will be set to zero.

$$h_{f_i} = \frac{2 f_i v_i^2}{g d_i}$$

$$h_{f_o} = \frac{2 f_o v_o^2}{g d_o}$$

Expected Hydraulic Power (kW)

$$\dot{W}_{Hydraulic}^{Expected} = Q_i^{Expected} (P_o - P_i)_{Expected}$$

Expected Pump Efficiency (%)

$$\eta_{Pump}^{Expected} = \frac{\dot{W}_{Hydraulic}^{Expected}}{\dot{W}_{Shaft}^{Expected}}$$

If the Performance Curve is Not Available***Expected Volumetric Efficiency (%)***

If curve data of volumetric efficiency is not available or is a constant value, we can map the data value to volumetric efficiency created in the main asset.

$$\eta_V^{Expected} = \eta_V$$

Expected Shaft Power (kW)

$$\dot{W}_{Shaft}^{Expected} = \dot{W}_{Shaft}^{Required}$$

Expected Volume Flow (m³/hr)

The calculation of the expected volume flow remains the same as derived in the above section.

Expected Pump Efficiency (%)

The rated efficiency as provided in the datasheet needs to be provided on the rated pump efficiency attribute created at the main asset of the pump.

$$\eta_{Pump}^{Expected} = \eta_{Pump}^{Rated}$$

Expected Hydraulic Power (kW)

$$\dot{W}_{Hydraulic}^{Expected} = \dot{W}_{Shaft}^{Expected} \eta_{Pump}^{Expected}$$

Expected Head (m)

$$\Delta h_t^{Expected} = \frac{\dot{W}_{Hydraulic}^{Expected}}{Q_i^{Expected} \rho g}$$

8.1.5.5 Screw Pump - Operating KPI

The following indicators help us to know the operating overall performance of the Screw Pump.

Operating Volumetric Efficiency (%)

$$\eta_V^{Operating} = \frac{Q_i}{Q_{swept}}$$

Operating Head (m)

$$\Delta h_t^{Operating} = \left(\frac{P_o - P_i}{\rho g} \right) + \left(\frac{v_o^2 - v_i^2}{2g} \right) + (Z_o - Z_i) + (h_{f_o} + h_{f_i})$$

NOTE

If geometrical details of inlet diameter and outer Diameter are provided then inlet velocity and outlet velocity are calculated by the below equation, else will be set to zero,

$$v_i = \frac{Q_i}{\frac{\pi}{4} d_i^2}$$

$$v_o = \frac{Q_i}{\frac{\pi}{4} d_o^2}$$

NOTE

If geometrical details of inlet diameter and outer diameter, along with Inlet frictional coefficient and outlet frictional coefficient are provided then Inlet frictional head losses and outlet frictional head losses are calculated by the below equation, else will be set to zero.

On using Darcy's equation,

$$h_{fi} = \frac{2 f_i v_i^2}{g d_i}$$

$$h_{fo} = \frac{2 f_o v_o^2}{g d_o}$$

Operating Hydraulic Power (kW)

$$\dot{W}_{Hydraulic}^{Operating} = Q_i (P_o - P_i)$$

Operating Shaft Power (kW)

If the Input power is available, we can PHD tag map the power to a measured shaft power attribute declared to the model.

$$\dot{W}_{Shaft}^{Operating} = (Measured\ Shaft\ Power)_{Tag\ value}$$

NOTE

If the driver is an electric motor and the current tag value is available, the measured shaft power attribute will be PHD tag mapped to the electric power attribute of a motor asset with the motor efficiency factor taken.

Operating Pump Efficiency (%)

$$\eta_{Pump}^{Operating} = \frac{\dot{W}_{Hydraulic}^{Operating}}{\dot{W}_{Shaft}^{Operating}}$$

NPSH Available (m)

$$\Delta h_{NPSH} = \frac{P_i - P_{vapour}}{\rho g} + \frac{v_i^2}{2g} + Z_i - h_{f_i}$$

8.1.6 Screw Pump - Model Output

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Expected Head	Expected Head (Output)	m	Dynamic	Performance	Calculated
2	Expected Hydraulic Power	Expected Hydraulic Power (Output)	kW	Dynamic	Performance	Calculated
3	Expected Pump Efficiency	Expected Pump Efficiency (Output)	%	Dynamic	Performance	Calculated
4	Expected Shaft Power	Expected Shaft Power (Output)	kW	Dynamic	Performance	Calculated
5	Expected VolumeFlow	Expected Volume Flow (Output)	m ³ /hr	Dynamic	Performance	Calculated
6	Expected Volumetric Efficiency	Expected Volumetric Efficiency (Output)	%	Dynamic	Performance	Calculated
7	Differential Pressure	Differential Pressure (Output)	kg/cm ²	Dynamic	Performance	Calculated
8	Operating Head	Operating Head (Output)	m	Dynamic	Performance	Calculated
9	Operating Hydraulic Power	Operating Hydraulic Power (Output)	kW	Dynamic	Performance	Calculated
10	Operating Pump Efficiency	Operating Pump Efficiency (Output)	%	Dynamic	Performance	Calculated
11	Operating Shaft Power	Operating Shaft Power (Output)	kW	Dynamic	Performance	Calculated
12	Operating Volumetric Efficiency	Operating Volumetric Efficiency (Output)	%	Dynamic	Performance	Calculated
13	NPSH Available	NPSH Available (Output)	m	Dynamic	Performance	Calculated
14	Deviation Head	Percent deviation in Head between Operating and Expected values (Output)	%	Dynamic	Deviation	Calculated
15	Deviation Hydraulic Power	Percent deviation in Hydraulic Power between Operating and Expected values (Output)	%	Dynamic	Deviation	Calculated
16	Deviation Pump Efficiency	Percent deviation in Pump Efficiency between Operating and Expected values (Output)	%	Dynamic	Deviation	Calculated
17	Deviation Shaft Power	Percent deviation in Shaft Power between Operating and Expected values (Output)	%	Dynamic	Deviation	Calculated

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
18	Deviation VolumeFlow	Percent deviation in Volume Flow between Operating and Expected values (Output)	%	Dynamic	Deviation	Calculated
19	Deviation Volumetric Efficiency	Percent deviation in Volumetric Efficiency between Operating and Expected values (Output)	%	Dynamic	Deviation	Calculated
20	NPSH Margin	NPSH Margin (Output)	%	Dynamic	Deviation	Calculated

8.2 Screw Pump - Fault Model

8.2.1 Screw Pump - APM Fault Model

Item	Type
Category	Equipment
Equipment Class	Pump
Equipment Type	Screw Pump

Below table illustrates the list of faults and symptoms configured for a Screw Pump for Performance

Fault Name	Associated Symptoms	Recommendations
Low production Feed	Inlet Pressure RL (SCPLPF)	1. Take repair action for the pump and Casing. 2. Piping cleaning.
	Inlet Flow RL (SCPLPF)	
Performance Parameter Deviation	Deviation Head HIGH (SCPPPD)	1. Check Pumping Liquid Quality. 2. Check for Inlet Filter for any debris. 3. Perform functional check of the Suction and Discharge Valve (including the opening sequence). 4. Check the stiffness of springs in valves. 5. Casing repair. 6. Piping cleaning.
	Deviation Shaft Power HIGH (SCPPPD)	
	Deviation Pump efficiency HIGH (SCPPPD)	
	Deviation Volumetric efficiency HIGH (SCPPPD)	

8.2.2 Screw Pump - APR Fault Model

8.2.2.1 Screw Pump Bearing Fault Model

Item	Type
Category	Equipment
Equipment Class	Pump
Equipment Type	ScrewPump_Bearing

Below table illustrates the list of faults and symptoms configured for a Screw Pump Bearing

Fault Name	Associated Symptoms	Recommendations
DE Radial Bearing	Radial Bearing Vibration DE X RH	1. Assess time to failure (trip).

Fault Name	Associated Symptoms	Recommendations
Failure/Malfunction	(SCPDE)	2. Intervention options 1. Reduce speed 2. Reduce oil temperature to minimum 3. If temperature and vibrations do not stabilize plan for bearing replacement.
	Radial Bearing Vibration DE Y RH (SCPDE)	
	Radial Bearing Temperature DE A RH (SCPDE)	
	Radial Bearing Temperature DE B RH (SCPDE)	
NDE Radial Bearing Failure/Malfunction	Radial Bearing Vibration NDE X RH (SCPDE)	1. Assess time to failure (trip). 2. Intervention options 1. Reduce speed 2. Reduce oil temperature to minimum 3. If temperature and vibrations do not stabilize plan for bearing replacement.
	Radial Bearing Vibration NDE Y RH (SCPDE)	
	Radial Bearing Temperature NDE A RH (SCPDE)	
	Radial Bearing Temperature NDE B RH (SCPDE)	

8.2.2.2 Screw Pump Main Fault Model

Item	Type
Category	Equipment
Equipment Class	Pump
Equipment Type	Screw Pump_Main

Below table illustrates the list of faults and symptoms configured for a Screw Pump Main Systems.

Fault Name	Associated Symptoms	Recommendations
Abrasive Wear	Casing Vibration RH (SCPABRW)	1. Adding additional seal lines will reduce the pressure per sealed cavity thereby reducing the effect of abrasives on the rotor. 2. Running a pump with a larger displacement per RPM will lower the required pump speed and thereby reduce the particle velocity within the pump.
Acid Attack & Corrosion	Casing Vibration RH (SCPAAC)	1. If the fluid is to be acidized, pull the rotor before conducting the acid job. 2. If the fluid normal pH is normally acidic, a stainless steel chrome plated rotor should be used. 3. By selecting proper material of rotor as per fluid which is going to use in pump.
	pH of fluid LOW (SCPAAC)	
Run Dry	Casing Vibration RH (SCPRD)	1. Monitoring the inlet fluid level will ensure that the pump will have continues supply of fluid resulting in a run dry condition.
	Inlet Pressure RL (SCPRD)	

8.2.2.3 Screw Pump Emotor Fault Model

Item	Type
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Category	Equipment
Equipment Class	Emotor
Equipment Type	ScrewPump_Emotor

Below table illustrates the list of faults and symptoms configured for a Screw Pump Emotor.

Fault Name	Associated Symptoms	Recommendations
DE Radial Bearing Failure/Malfunction	Radial Bearing Vibration DE X RH (EMDEM)	1. Assess likely cause and time to fail(trip) if intervention not possible. 2. Reduce speed. 3. Inspect bearings and internal clearances, if installed check earthing brush is working correctly and check bearing for arcing.
	Radial Bearing Vibration DE Y RH (EMDEM)	
	Radial Bearing Temperature DE A RH (EMDEM)	
	Radial Bearing Temperature DE B RH (EMDEM)	
NDE Radial Bearing Failure/Malfunction	Radial Bearing Vibration NDE X RH (EMNDEM)	1. Assess likely cause and time to fail(trip) if intervention not possible. 2. Reduce speed. 3. Inspect bearings and internal clearances, if installed check earthing brush is working correctly and check bearing for arcing.
	Radial Bearing Vibration NDE Y RH (EMNDEM)	
	Radial Bearing Temperature NDE A RH (EMNDEM)	
	Radial Bearing Temperature NDE B RH (EMNDEM)	

8.2.2.4 Screw Pump- Transmission System Fault Model

Item	Type
Category	Equipment
Equipment Class	Pump
Equipment Type	ScrewPump_Transmission

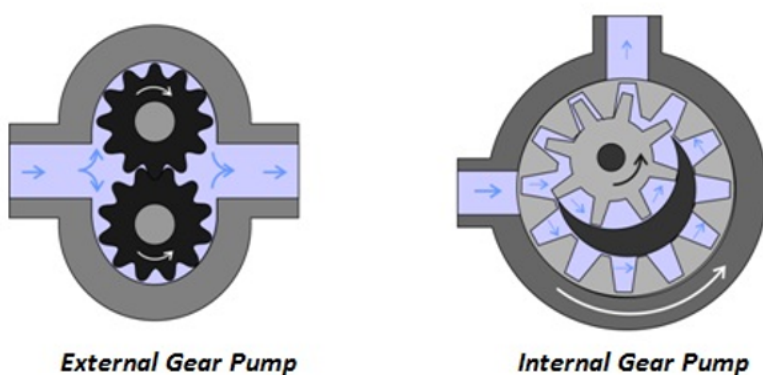
Below table illustrates the list of faults and symptoms configured for a Screw Pump Transmission system.

Fault Name	Associated Symptoms	Recommendations
Gearbox Radial Bearings malfunction /Failure	Gearbox IP Shaft Bearing Vibration DE X RH (SCPRB)	1. Assess likely cause and time to trip if intervention not possible. 2. Intervention options; reduce load, oil temperature, restore oil properties. 3. Ensure parts availability. 4. Take an oil sample and have it tested. 5. At first opportunity: Inspect journal and thrust

Fault Name	Associated Symptoms	Recommendations
	Gearbox IP Shaft Bearing Vibration DE Y RH (SCPRB)	bearing, check clearances. 6. If temperature and vibrations do not stabilize plan for bearing replacement.
	Gearbox IP Shaft Bearing Vibration NDE X RH (SCPRB)	
	Gearbox IP Shaft Bearing Vibration NDE Y RH (SCPRB)	
	Gearbox OP Shaft Bearing Vibration DE X RH (SCPRB)	
	Gearbox OP Shaft Bearing Vibration DE Y RH (SCPRB)	
	Gearbox OP Shaft Bearing Vibration NDE X RH (SCPRB)	
	Gearbox OP Shaft Bearing Vibration NDE Y RH (SCPRB)	
	Gearbox oil Analysis RH (SCPRB)	
Gearbox loss of cooling	Cooling Water Inlet Pressure LOW (SCPLOC)	1. Check cooling water supply pressure. 2. Check bypass control valve. 3. Open both coolers and stabilize temperature. 4. Isolate and inspect it. 5. Line walk the system to check for leaks, Check cooling water pumps for blockages, ensure strainer is clean.
	Cooling Water Outlet Pressure LOW (SCPLOC)	
	Oil Inlet Pressure RL (SCPLOC)	
	Oil Outlet Pressure RL (SCPLOC)	

GEAR PUMP

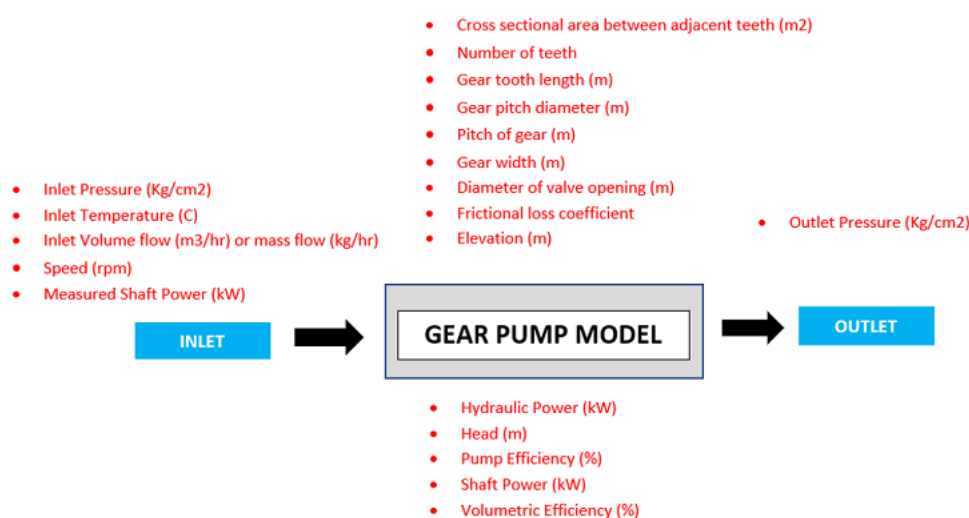
A gear pump uses the meshing of gears to pump fluid by displacement. They are one of the most common types of pumps for hydraulic fluid power applications.



Gear pumps are also widely used in chemical installations to pump high-viscosity fluids.

The gear pump is classified as,

- **External gear pump:** It utilizes two identical gears meshed side by side, where one gear (driving) is driven by a motor, and it - in turn - drives the other gear (driven) driving gear driving the idle (driven) gear. Each gear is supported by a shaft with bearings on both sides of the gear.
- **Internal gear pump:** The internal gear pump is non-pulsing, self-priming, and can run dry for short periods. They're also bi-rotational, meaning that the same pump can be used to load and unload vessels. Because internal gear pumps have only two moving parts, they are reliable, simple to operate, and easy to maintain.



9.1 Gear Pump - Performance Monitoring

9.1.1 Notations

Below are the different notations used:

F	Mass Flow
h	Head
Q	Volume flow
P	Pressure
η	Efficiency
a	The cross-sectional area between adjacent gear teeth
n	Number of gear teeth
W	Power
N	Speed
L	Gear tooth length
D	Gear tip diameter
c	Gear centers when meshed
T_d	Gear pitch diameter
T	Pitch of gear
w	Width of gear
d	Diameter of valve opening
f	Frictional loss coefficient
v	Velocity
h_f	Frictional head loss

9.1.2 Subscripts

i	Inlet
o	Outlet

9.1.3 Gear Pump - Model Assumptions

The key assumption taken in developing a Performance model is:

1. The density of process liquid is assumed to be constant as its incompressible.

NOTE

1. If the composition of process liquid is available, there is a provision for creating an inlet thermos-stream that can be port mapped to the main asset to get thermal properties like density and vapor pressure.
2. If composition is unavailable, whereas density V/S temperature and vapor pressure V/S temperature data are available you can map these in the Property data (Tabular array) attribute.
3. If only one data point of density V/S temperature and vapor pressure V/S temperature is available, then place the data point value in vapor pressure attribute and density attribute declared at the main asset instead of property data.

9.1.4 Gear Pump - Model Input

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Inlet1_Pressure	Inlet Stream Pressure (Required Input)	kg/cm ² (a)	Dynamic	Measurement	Field Input
2	Outlet1_Pressure	Outlet Stream Pressure (Required Input)	kg/cm ² (a)	Dynamic	Measurement	Field Input
3	Inlet1_Temperature	Inlet Stream Temperature (Required Input)	°C	Dynamic	Measurement	Field Input
4	Inlet1_Actual Volumetric Flow	Inlet Stream Actual Volume Flow (Input - Port mapped from suction stream)	m ³ /hr	Dynamic	Internal	Field Input
5	Inlet1_MassFlow	Inlet Stream Mass Flow Rate (Input)	kg/hr	Dynamic	Measurement	Field Input
6	Speed	Speed (Required Input)	rpm	Dynamic	Measurement	Field Input
7	Measured Shaft Power	Measured Shaft Power (Required Input)	kW	Dynamic	Measurement	Field Input
8	Performance Data	Curve Data between Shaft Power, Efficiency VS Differential Pressure VS Speed (Input)		Static	Performance Map	Interpolated from the performance curve
9	Property Data	Curve Data between Liquid Density, Vapour Pressure VS Temperature (Required Input if composition data is not available)		Static	Internal (Fluid Property Data)	Interpolated from Property data curve
10	Liquid Density	Liquid Density (Optional Input)	kg/m ³	Static	Internal	Constant
11	Vapour Pressure	Vapour Pressure (Optional Input)	kg/cm ² (a)	Static	Internal	Constant
12	NPSH Required	NPSH Required (Input)	m	Static	Design	Constant
13	Number Of Gear Teeth	Number Of Gear Teeth (Input)		Static	Design	Constant
14	Gear Centers Width	Gear Centers Width (Input)	m	Static	Design	Constant

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
15	Gear Pitch	Gear Pitch (Input)	m	Static	Design	Constant
16	Gear Pitch Diameter	Gear Pitch Diameter (Input)	m	Static	Design	Constant
17	Gear Tip Diameter	Gear Tip Diameter (Input)	m	Static	Design	Constant
18	Gear Tooth Length	Gear Tooth Length (Input)	m	Static	Design	Constant
19	Gear Width	Gear Width (Input)	m	Static	Design	Constant
20	Cavity Area	Cavity Area (Input)	m ²	Static	Design	Constant
21	Theoretical Capacity	Theoretical Capacity (Input)	m ³	Static	Design	Constant
22	Power Required	Power Required (Required Input if no performance curve is available)	kW	Static	Design	Constant
23	CT_Pump Efficiency	Constant Pump Efficiency (Required Input if no performance curve is available)	%	Static	Design	Constant
24	CT_Volumetric Efficiency	Constant Volumetric Efficiency (Required Input if no performance curve is available)	%	Static	Design	Constant
25	Suction Diameter	Suction Diameter (Input)	m	Static	Design	Constant
26	Suction Elevation	Suction Elevation (Input)	m	Static	Design	Constant
27	Suction Frictional Losses	Suction Frictional Losses (Input)	m	Static	Design	Constant
28	Suction Frictional Losses Coefficient	Suction Frictional Losses Coefficient (Input)	m	Static	Design	Constant
29	Discharge Diameter	Discharge Diameter (Input)	m	Static	Design	Constant
30	Discharge Elevation	Discharge Elevation (Input)	m	Static	Design	Constant
31	Discharge Frictional Losses	Discharge Frictional Losses (Input)	m	Static	Design	Constant
32	Discharge Frictional Losses Coefficient	Discharge Frictional Losses Coefficient (Input)	m	Static	Design	Constant

9.1.5 Gear Pump - KPIs

9.1.5.1 Model Inputs

- Inlet Pressure
- Outlet Pressure
- Inlet Temperature
- Inlet Volume Flow
- Speed
- Cross sectional area of space between adjacent gear teeth
- Number of gear teeth
- Gear tooth length
- Gear pitch diameter
- Pitch of the gear
- Gear width
- Inlet diameter
- Outlet diameter
- Inlet elevation
- Outlet elevation
- Inlet frictional coefficient
- Outlet frictional coefficient
- Property Data

$$\rho, T \text{ and } P_{vapour}, T$$

- Performance Data

$$\dot{W}_{BHP}, \Delta_{Pressure}, N$$

- Measured shaft Power

9.1.5.2 Model Output KPI's

Expected KPI

- Expected Volumetric efficiency
- Expected Volume Flow
- Expected Head
- Expected Hydraulic Power
- Expected Pump efficiency
- Expected Shaft Power

Operating KPI

- Operating Volumetric Efficiency
- Operating Head
- Operating Hydraulic Power
- Operating Pump efficiency
- Operating Shaft Power
- NPSH available

9.1.5.3 Gear Pump - Performance Modeling

The Performance modeling of the Gear pump helps to determine the performance KPI.

Overall Mass Balance

Applying the continuity equation,

$$F_i = F_o$$

Overall Force Balance

Applying the force balance equation,

$$P_i + \frac{\rho v_i^2}{2} + \rho g Z_i = P_o + \frac{\rho v_o^2}{2} + \rho g Z_o + \rho g h_{f_o} + \rho g h_{f_i}$$

Converting to a representative head term, by dividing both side with ρg

$$\Delta h_t = \left(\frac{P_o - P_i}{\rho g} \right) + \left(\frac{v_o^2 - v_i^2}{2g} \right) + (Z_o - Z_i) + (h_{f_o} + h_{f_i})$$

Overall Energy Balance

$$F_o H_o(P_o, T_o) = F_i H_i(P_i, T_i) + \dot{W}_{Hydraulic\ Actual}$$

$$F_o H_o(P_o, T_i) = F_i H_i(P_i, T_i) + \dot{W}_{Hydraulic\ Isothermal}$$

$$\eta_{Isothermal} = \frac{\dot{W}_{Hydraulic\ Isothermal}}{\dot{W}_{Hydraulic\ Actual}}$$

$$\eta_{Isothermal} = \frac{\dot{W}_{Hydraulic\ Isothermal}}{\dot{W}_{Shaft} \eta_{Mech} \eta_{GearBox}}$$

$$\eta_{Pump} = \frac{\dot{W}_{Hydraulic}}{\dot{W}_{Shaft}}$$

NOTE

The pump efficiency calculation is finalized post data sheet analysis and literature survey.

9.1.5.4 Gear Pump - Expected KPIs

The following indicators help us to know the expected overall performance of the Gear Pump. Depending on the approach the model can be classified into two:

- Performance curve available
- Performance curve not available

Performance Curve Available

Expected Volumetric Efficiency (%)

The Volumetric efficiency is from the data sheet value that needs to be mapped to the volumetric efficiency attribute present in main asset.

$$\eta_V^{Expected} = \eta_V$$

Expected Shaft Power (kW)

From the performance curve of Shaft power vs pressure difference with speed in Z coordinate, we calculate expected shaft power by passing measured differential pressure and speed.

The expected shaft Power is calculated using a performance curve supplied by the user,

$$W_{Shaft}^{Expected} = f(\Delta_{Pressure}, N)$$

Expected Volume Flow (m3/hr)

$$Q_{Swept} = 2 \pi a n L N$$

There is also an alternative formula to calculate swept volume flow,

Alternative 1

$$Q_{Swept} = 2 \pi T_d T w N$$

Reference: ence-pumps.ru

Alternative 2

$$Q_{Swept} = \frac{\pi}{2} w (D^2 - c^2)$$

Reference: unwinhydraulic.co.uk

$$Q_i^{Expected} = Q_{Swept} \eta_V^{Expected}$$

Expected Head (m)

From performance curve of Brake HP vs differential pressure with speed in Z coordinate, we pass measured power and speed to calculate expected differential pressure.

$$\Delta h_t^{Expected} = \frac{(P_o - P_i)_{Expected}}{\rho g} + \left(\frac{v_o^2 - v_i^2}{2g} \right) + (Z_o - Z_i) + (h_{f_o} + h_{f_i})$$

$$(P_o - P_i)_{Expected} = f(\dot{W}_{Shaft}, N)$$

NOTE

If geometrical details of inlet diameter and outer diameter is provided then inlet velocity and outlet velocity are calculated by the below equation, else will be set to zero.

$$v_i^{expected} = \frac{Q_i^{Expected}}{\frac{\pi}{4} d_i^2}$$

$$v_o^{expected} = \frac{Q_i^{Expected}}{\frac{\pi}{4} d_o^2}$$

NOTE

If geometrical details of inlet diameter and outer diameter, along with inlet frictional coefficient and outlet frictional coefficient are provided then inlet frictional head losses and outlet frictional head losses are calculated by the below equation, else will be set to zero.

On using Darcy's equation,

$$h_{f_i} = \frac{2 f_i v_i^2}{g d_i}$$

$$h_{f_o} = \frac{2 f_o v_o^2}{g d_o}$$

Performance Curve Not Available**Expected Volumetric Efficiency (%)**

If curve data of volumetric efficiency is not available or is a constant value, we can map the data value to volumetric efficiency created in the main asset.

$$\eta_v^{Expected} = \eta_v$$

Expected Shaft Power (kW)

If curve data of shaft power is not available or is a constant value, we can map the data value power required attribute created in the main asset.

$$W_{Shaft}^{Expected} = W_{Shaft}^{Required}$$

Expected Volume Flow (m3/hr)

The calculation of expected volume flow remains the same as derived in the above section.

Expected Pump Efficiency (%)

The rated efficiency as provided in datasheet needs to be provided on the rated pump efficiency attribute created at the main asset of the pump.

$$\eta_{Pump}^{Expected} = \eta_{Pump}^{Rated}$$

Expected Hydraulic Power (kW)

$$W_{Hydraulic}^{Expected} = W_{Shaft}^{Expected} \eta_{Pump}^{Expected}$$

Expected Head (m)

$$W_{Hydraulic}^{Expected} = W_{Shaft}^{Expected} \eta_{Pump}^{Expected}$$

9.1.5.5 Gear Pump - Operating KPIs

The following indicators help us to know the overall operating performance of the Gear Pump.

Operating Volumetric Efficiency (%)

$$\eta_v^{Operating} = \frac{Q_i}{Q_{swept}}$$

Operating Head (m)

$$\Delta h_t^{Operating} = \left(\frac{P_o - P_i}{\rho g} \right) + \left(\frac{v_o^2 - v_i^2}{2g} \right) + (Z_o - Z_i) + (h_{f_o} + h_{f_i})$$

NOTE

If geometrical details of inlet diameter and outer diameter are provided then inlet velocity and outlet velocity are calculated by the below equation, else will be set to zero,

$$v_i = \frac{Q_i}{\frac{\pi}{4} d_i^2}$$

$$v_o = \frac{Q_i}{\frac{\pi}{4} d_o^2}$$

NOTE

If geometrical details of inlet diameter and outer diameter, along with Inlet frictional coefficient and outlet frictional coefficient, are provided, then Inlet frictional head losses and outlet frictional head losses are calculated by the below equation else will be set to zero.

On using Darcy's equation,

$$h_{f_i} = \frac{2 f_i v_i^2}{g d_i}$$

$$h_{f_o} = \frac{2 f_o v_o^2}{g d_o}$$

Operating Hydraulic Power (kW)

$$\dot{W}_{Hydraulic}^{Operating} = Q_i (P_o - P_i)$$

Operating Shaft Power (kW)

If the Input power is available, we can PHD tag map the power to a measured shaft power attribute declared to the model.

$$\dot{W}_{Shaft}^{Operating} = (Measured\ Shaft\ Power)_{Tag\ value}$$

NOTE

If driver is electric motor and current tag value is available, the measured shaft power attribute will be PHD tag mapped to electric power attribute of motor asset with motor efficiency factor taken.

Operating Pump Efficiency (%)

$$\eta_{Pump}^{Operating} = \frac{\dot{W}_{Hydraulic}^{Operating}}{\dot{W}_{Shaft}^{Operating}}$$

NPSH Available (m)

$$\Delta h_{NPSH} = \frac{P_i - P_{vapour}}{\rho g} + \frac{v_i^2}{2g} + Z_i - h_{f_i}$$

9.1.6 Gear Pump - Model Output

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Expected Head	Expected Head (Output)	m	Dynamic	Performance	Calculated
2	Expected Hydraulic Power	Expected Hydraulic Power (Output)	kW	Dynamic	Performance	Calculated
3	Expected Pump Efficiency	Expected Pump Efficiency (Output)	%	Dynamic	Performance	Calculated
4	Expected Shaft Power	Expected Shaft Power (Output)	kW	Dynamic	Performance	Calculated
5	Expected VolumeFlow	Expected Volume Flow (Output)	m ³ /hr	Dynamic	Performance	Calculated
6	Expected Volumetric Efficiency	Expected Volumetric Efficiency (Output)	%	Dynamic	Performance	Calculated
7	Differential Pressure	Differential Pressure (Output)	kg/cm ²	Dynamic	Performance	Calculated
8	Operating Head	Operating Head (Output)	m	Dynamic	Performance	Calculated
9	Operating Hydraulic Power	Operating Hydraulic Power (Output)	kW	Dynamic	Performance	Calculated
10	Operating Pump Efficiency	Operating Pump Efficiency (Output)	%	Dynamic	Performance	Calculated
11	Operating Shaft Power	Operating Shaft Power (Output)	kW	Dynamic	Performance	Calculated
12	Operating Volumetric Efficiency	Operating Volumetric Efficiency (Output)	%	Dynamic	Performance	Calculated
13	NPSH Available	NPSH Available (Output)	m	Dynamic	Performance	Calculated
14	Deviation Head	Percent deviation in Head between Operating and Expected values (Output)	%	Dynamic	Deviation	Calculated
15	Deviation Hydraulic Power	Percent deviation in Hydraulic Power between Operating and Expected values (Output)	%	Dynamic	Deviation	Calculated
16	Deviation Pump Efficiency	Percent deviation in Pump Efficiency between Operating and Expected values (Output)	%	Dynamic	Deviation	Calculated
17	Deviation Shaft Power	Percent deviation in Shaft Power between Operating and Expected values (Output)	%	Dynamic	Deviation	Calculated
18	Deviation VolumeFlow	Percent deviation in Volume Flow between Operating and Expected values (Output)	%	Dynamic	Deviation	Calculated
19	Deviation Volumetric Efficiency	Percent deviation in Volumetric Efficiency between Operating and Expected values (Output)	%	Dynamic	Deviation	Calculated
20	NPSH Margin	NPSH Margin (Output)	%	Dynamic	Deviation	Calculated

9.2 Gear Pump - Fault Model

9.2.1 Gear Pump - APM Fault Model

Item	Type
Category	Equipment
Equipment Class	Pump
Equipment Type	Gear Pump

Below table illustrates the list of faults and symptoms configured for a Gear Pump for Performance

Fault Name	Associated Symptoms	Recommendations
Dry Run/No Flow	Speed RL (NF)	1. Monitoring the inlet fluid level will ensure that the pump will have a continuous supply of fluid resulting in a run-dry condition.
	NPSH Available RL (NF)	
Low Flow/Capacity	Suction flow RL (LNF)	1. Clean the Clogged strainer/filter. 2. Check the Suction valve and Discharge valve Position. 3. Check the Adjustment Position of the Pressure Relief valve. 4. Check the Speed and direction of rotation. 5. Check for Leakages In Suction. 6. Inspect pump gaskets and seal for leakages. 7. Check out the Suction line pressure Gauge.
	Suction Temperature RH (LNF)	
	Speed RH (LNF)	
	Speed RL (LNF)	
	NPSH Available RL (LNF)	
Performance Parameter Deviation	Deviation Head HIGH (GPPPD)	1. Check Pumping Liquid Quality. 2. Check for Inlet Filter for any debris. 3. Perform a functional check of the Suction and Discharge Valve (including the opening sequence). 4. Check the stiffness of springs in valves. 5. Casing repair. 6. Piping cleaning.
	Deviation Pump efficiency HIGH (GPPPD)	
	Deviation Volumetric efficiency HIGH (GPPPD)	
	Deviation Shaft Power HIGH (GPPPD)	
	Deviation Volume Flow HIGH (GPPPD)	
	Deviation Hydraulic Power HIGH (GPPPD)	

9.2.2 Gear Pump - APR Fault Model

9.2.2.1 Gear Pump Bearing Fault Model

Item	Type
Category	Equipment
Equipment Class	Pump
Equipment Type	GearPump_Bearing

Below table illustrates the list of faults and symptoms configured for a Gear Pump Bearing

Fault Name	Associated Symptoms	Recommendations
DE Radial Bearing Failure/Malfunction	Radial Bearing Vibration DE X RH (GPDE)	1. Assess time to failure (trip). 2. Intervention options: 1. Reduce speed. 2. Reduce oil temperature to minimum. 3. If temperature and vibrations do not stabilize plan for bearing replacement.
	Radial Bearing Vibration DE Y RH (GPDE)	
	Radial Bearing Temperature DE A RH (GPDE)	
	Radial Bearing Temperature DE B RH (GPDE)	
NDE Radial Bearing Failure/Malfunction	Radial Bearing Vibration NDE X RH (GPNDE)	1. Assess time to failure (trip). 2. Intervention options: 1. Reduce speed, 2. Reduce oil temperature to minimum 3. If temperature and vibrations do not stabilize plan for bearing replacement.
	Radial Bearing Vibration NDE Y RH (GPNDE)	
	Radial Bearing Temperature NDE A RH (GPNDE)	
	Radial Bearing Temperature NDE B RH (GPNDE)	
Pump runs noisy / excessive vibration	Radial Bearing Vibration DE X RH (PN)	1. Check fluid viscosity. 2. Clean filters. 3. Reduce pump speed. 4. Check gears and Bearings. 5. Replace if necessary. 6. Realign Pump. 7. Check coupling and rubber. 8. Increase pressure setting or replace. 9. Check Strainer. 10. Check the fittings and tighten it.
	Radial Bearing Vibration DE Y RH (PN)	
	Radial Bearing Vibration NDE X RH (PN)	
	Radial Bearing Vibration NDE Y RH (PN)	
	Speed RH (PN)	
	Inlet filter DP RH (PN)	

9.2.2.2 Gear Pump E-motor Fault Model

Item	Type
Category	Equipment
Equipment Class	E-motor
Equipment Type	GearPump_Emotor

Below table illustrates the list of faults and symptoms configured for a Gear Pump E-motor.

Fault Name	Associated Symptoms	Recommendations
DE Radial Bearing Failure/Malfunction	Radial Bearing Vibration DE X RH (EMDEM)	1. Assess the likely cause and time to fail(trip) if intervention is not possible. 2. Reduce speed. 3. Inspect bearings and internal clearances.

Fault Name	Associated Symptoms	Recommendations
	Radial Bearing Vibration DE Y RH (EMDEM)	4. Installed check earthing brush is working correctly and check bearing for arcing.
	Radial Bearing Temperature DE A RH (EMDEM)	
	Radial Bearing Temperature DE B RH (EMDEM)	
NDE Radial Bearing Failure/Malfunction	Radial Bearing Vibration NDE X RH (EMNDEM)	1. Assess the likely cause and time to fail(trip) if intervention is not possible. 2. Reduce speed. 3. Inspect bearings and internal clearances. 4. Installed check earthing brush is working correctly and check bearing for arcing.
	Radial Bearing Vibration NDE Y RH (EMNDEM)	
	Radial Bearing Temperature NDE A RH (EMNDEM)	
	Radial Bearing Temperature NDE B RH (EMNDEM)	

9.2.2.3 Gear Pump- Transmission System Fault Model

Item	Type
Category	Equipment
Equipment Class	Pump
Equipment Type	GearPump_Transmission

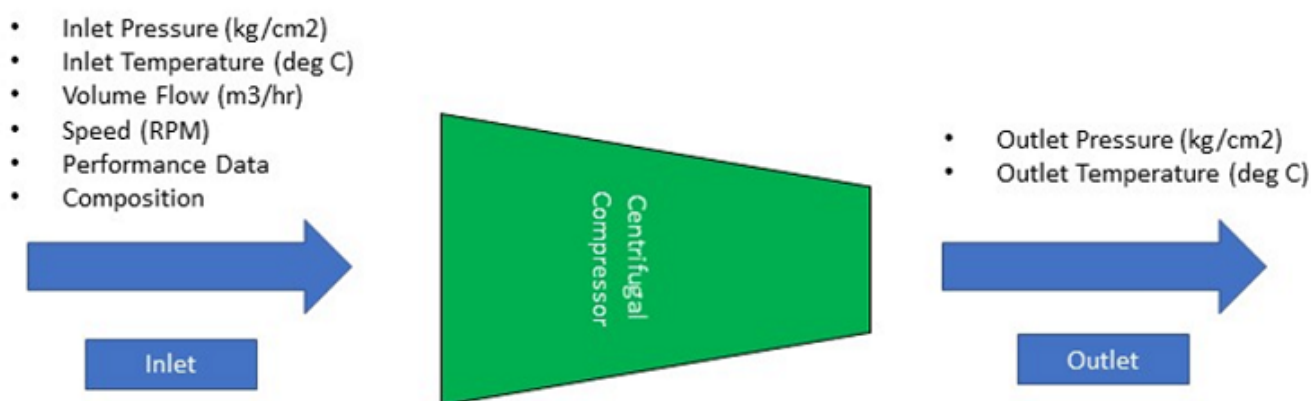
Below table illustrates the list of faults and symptoms configured for a Gear Pump Transmission system.

Fault Name	Associated Symptoms	Recommendations
Gearbox Radial Bearings malfunction /Failure	GP Gearbox IP Shaft Bearing Vibration X RH (GPRB)	1. Assess likely cause and time to trip if intervention is not possible. 2. Intervention options: reduce load, oil temperature, and restore oil properties. 3. Ensure parts availability. 4. Take an oil sample and have it tested. 5. At first opportunity: Inspect journal and thrust bearing, check clearances. 6. If temperature and vibrations do not stabilize plan for bearing replacement.
	GP Gearbox IP Shaft Bearing Vibration Y RH (GPRB)	
	GP Gearbox OP Shaft Bearing Vibration X RH (GPRB)	
	GP Gearbox OP Shaft Bearing Vibration Y RH (GPRB)	
	GP Gearbox Oil Temperature RH (GPRB)	
	GP Gearbox Oil Analysis SH (GPRB)	
	GP Gearbox Oil Analysis SL (GPRB)	

CENTRIFUGAL COMPRESSOR

Compressors are used to transport gases and to increase the pressure of gases in process plants, power plants, and other industries. Compressor performance has a significant impact on the overall plant performance in terms of energy usage, efficiency, and throughput.

Centrifugal compressors, also known as turbo-compressors, belong to the roto-dynamic type of compressors. In these compressors, the required pressure rise takes place due to the continuous conversion of angular momentum imparted to the gas vapor by a high-speed impeller into static pressure. As shown in the figure below, low-pressure gas enters the compressor through the eye of the impeller. The impeller consists of several blades, which form flow passages for gas. From the eye, the gas enters the flow passages formed by the impeller blades, which rotate at a very high speed. As the gas flows through the blade passages towards the tip of the impeller, it gains momentum and its static pressure also increases. From the tip of the impeller, the gas flows into a stationary diffuser. In the diffuser, the gas is decelerated and, as a result, the dynamic pressure drop is converted into static pressure rise, thus increasing the static pressure further. The vapor from the diffuser enters the volute casing where further conversion of velocity into static pressure takes place due to the divergent shape of the volute. Finally, the pressurized gas leaves the compressor from the volute casing.



A centrifugal compressor performance monitoring model is used to:

1. Monitor and analyze the performance output.
2. Determine the condition of a centrifugal compressor.

Using the outputs of a compressor performance monitoring model, you can:

1. Configure the fault/symptom notification to identify the compressor faults such as surge, choking, excessive discharge temperature, fouling, and so on.
2. Calculate the performance of the compressor based on the capacity loss.

10.1 Centrifugal Compressor Sub System

10.1.1 Compressor Bearing Temperature and Vibrations

Monitoring compressor vibrations is by far the predominant and most widely used method to determine the mechanical conditions of the compressor. Many different failure modes can cause an increase in Compressor vibrations.

Condition monitoring is used for the early detection of noise and vibration signature and bearing temperature of centrifugal compressor. Rise in bearing vibration and temperature is due to inappropriate fitting, overgreasing or cage fracture. Surface distortion is due to poor lubricant and overload. Normally, bearings display symptoms of distress before they fail. Symptoms of a failing bearing include higher vibration levels and rising bearing or oil sump temperatures. Rise in lube oil temperature increases the viscosity of the lube oil, which reduces the strength of the oil film on the bearings and may lead to high vibrations.

10.1.2 Seal Gas System

A dry gas seal system is designed to provide reliable shaft sealing on rotating equipment to reduce the amount of process gas escaping from the machine's primary flow path to an acceptable level.

Loss of buffer gas will, in most cases, reverse the differential pressure across the seal faces, which will damage the seal in short order. The seals will operate at a zero differential pressure level.

Dry Gas Seal Degradation: Contamination of the seal by foreign objects leads to seal failures. The running gap between the primary and mating gas seal rings is typically around three microns. Injection of any type of solids or liquids into this very narrow seal running gap can cause degradation of seal performance. This would create excessive gas leakage to the vent and eventual failure of the seal.

The major areas from which gas seals contamination occurs are:

1. Process gas leakages from the inboard or high-pressure side of the seal.
2. Bearing lubrication oil from the outboard or low-pressure side of the seal.
3. The seal gas injected into the seal being contaminated upstream of the injection.

Key parameters to monitor for a dry gas seal are as follows:

1. Primary seal gas differential pressure—monitor and trend control valve positions.
2. Primary vent flow or pressure—high for primary seal leakage, low for secondary seal leakage.
3. Intermediate N₂ DP (If available).
4. Primary vent control valve position (if available)—if opened - the primary seal leakage has increased; if closed - the secondary seal leakage has increased.
5. Separation N₂ pressure or differential pressure (if equipped with DP control).

10.1.3 Lube Oil System

High-quality pumps require a form of lubricant to either cool, seal, or lubricate key internal components such as gears, bearings, and seals. The lube oil system supplies oil to the pump and driver bearings and the gears and couplings. The lube oil is drawn from the reservoir by the pumps and is fed under pressure through coolers and filters to the bearings.

Tank level: Loss in lube oil tank level can lead to loss of lubrication and pump trip scenario. The reservoir may be pressurized or vented.

Supply pressure and temperature: Temperatures and pressures are measured at all important locations in the system, including temperatures from oil sumps, return lines from bearings, gears, and other mechanical components. Temperatures and pressures are often recorded on the suction and discharge sides of each compression stage to offer the operator a sense of the health of the system. The readings can be taken locally or transmitted to a monitoring station. The flow of oil to each bearing is regulated individually by orifices, particularly important for lubrication points requiring different pressures.

Return temperature: Heat generated by friction in the bearings is transferred to the cooling medium in the oil coolers. Air-cooled oil coolers may be employed as an alternative to water-cooled oil coolers.

Differential pressure across Filters: Filters clean the lube oil before it reaches the lubrication points and a differential pressure gauge monitors the degree of fouling (flow restriction) of the filters.

10.1.4 Compressor Motor

Motor used as subsystem for many industrial equipment to operate as expected, it is important to monitor the health and performance of an AC Motor to increase availability. In addition to tracking AC motor operating conditions, monitoring basic machine health data helps maintain reliable AC motor operations.

10.2 Centrifugal Compressor - Performance Monitoring

The following measured parameters are monitored as they can negatively impact the performance and health of the compressor if not maintained in the optimum operating range.

Suction Temperature: Increase in suction temperature reduces feed density which increases volumetric flow rate to be compressed for delivery of the same amount of discharge mass. At the same time, the lower suction temperature may lead to condensation of vapor feed which can lead to an increase in vibration and trip scenario.

Suction Pressure: Decrease in suction pressure indicates the low feed to the compressor which can lead to a critical asset health condition like surging, a lower suction pressure may also increase the pressure ratio across the compressor if the discharge pressure is controlled, which can lead to surging.

Discharge Temperature: Higher discharge temperature indicates upset in the overall operations of the equipment and may lead to a trip scenario. For example, it may be because of low coolant flow, high coolant temperature, high bearing temperatures, or drop in efficiency, etc.

Discharge Pressure: Higher discharge pressure indicates upset in the overall operations of equipment and may lead to a trip scenario. It may be due to different reasons such as obstacles in downstream operations, fouled aftercooler, etc.

Speed: It is a direct indication of the load on the compressor.

Anti-surge Valve Opening and flow: Surge is an unstable and undesirable operating condition of the compressor that occurs when the flow through it is less than surge flow. This causes a momentary flow reversal, reducing line pressure and causing erratic behavior. For ensuring the safety of equipment, an anti-surge controller maintains flow near the control line. Additional circulation beyond the control line is operational loss due to excess compression.

10.2.1 Notations

F	Flow
h	Specific enthalpy
Q	Volume flow
P	Pressure
T	Temperature
η	Efficiency
W	Power
N	Speed
H	Overall Enthalpy

10.2.2 Subscripts

In	Inlet
Out	Outlet

10.2.3 Superscripts

<i>Isen</i>	Isentropic
<i>Poly</i>	Polytropic

10.2.4 Centrifugal Compressor - Variants

The centrifugal compressor variants are:

- Variable speed
- Variable Guide Vane
- Constant Speed
- Water Injection
- Side Stream

10.2.5 Centrifugal Compressor - Model Input

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Diameter	Impeller Diameter	m	Static	Design	Data Sheet
2	Inlet1 Pressure	Inlet Stream Pressure (Required Input)	kg/cm ² (a)	Dynamic	Measurement	Field Input
3	Outlet1 Pressure	Inlet Stream Pressure (Required Input)	kg/cm ² (a)	Dynamic	Measurement	Field Input
4	Inlet1 Temperature	Inlet Stream Temperature (Required Input)	°C	Dynamic	Measurement	Field Input
5	Outlet1 Temperature	Outlet Stream Temperature (Required Input)	°C	Dynamic	Measurement	Field Input
6	Inlet2 Pressure	Inlet Stream Pressure (Required Input) (For Side Stream Compressor)	kg/cm ² (a)	Dynamic	Measurement	Field Input
7	Inlet3 Pressure	Inlet Stream Pressure (Required Input) (For Side Stream Compressor)	kg/cm ² (a)	Dynamic	Measurement	Field Input
8	Inlet1 Volume Flow Rate	Inlet Stream Volume Flow Rate (Input)	m ³ /hr	Dynamic	Measurement	Field Input
9	Inlet1 Actual Volumetric Flow	Inlet Stream Actual Volume Flow (Input - Port mapped from suction stream) NOTE Inlet actual volumetric flow can be used while configuring a new model.	m ³ /hr	Dynamic	Internal	Field Input
10	Inlet2 Volume Flow Rate	Inlet Stream Volume Flow Rate (Input) (For Side Stream Compressor)	m ³ /hr	Dynamic	Measurement	Field Input
11	Inlet2 Actual Volumetric	Inlet Stream Actual Volume Flow (Input -	m ³ /hr	Dynamic	Internal	Field

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
	Flow	Port mapped from side stream 1)				Input
12	Inlet3 Volume Flow Rate	Inlet Stream Volume Flow Rate (Input) (For Side Stream Compressor)	m ³ /hr	Dynamic	Measurement	Field Input
13	Inlet3 Actual Volumetric Flow	Inlet Stream Actual Volume Flow (Input - Port mapped from side stream 2)	m ³ /hr	Dynamic	Internal	Field Input
14	Separator Volume Flow	Separator Volume Flow (Input)	m ³ /hr	Dynamic	Measurement	Field Input
15	Seperator Side Stream 1 Volume Flow Rate	Separator Volume Flow (Input) (For Side Stream `Compressor)	m ³ /hr	Dynamic	Measurement	Field Input
16	Seperator Side Stream 2 Volume Flow Rate	Separator Volume Flow (Input) (For Side Stream Compressor)	m ³ /hr	Dynamic	Measurement	Field Input

10.2.6 Centrifugal Compressor - KPIs

The following indicators help us know the overall asset performance:

10.2.6.1 Input KPIs

- Inlet Pressure
- Outlet Pressure
- Inlet Temperature
- Outlet Temperature
- Inlet Volume Flow Rate
- Speed / Inlet Guide Vane Position
- Performance Data (Reduced Curve, OEM Curve data, Stage Wise Data for Side Stream Compressor)
- Mechanical Efficiency

10.2.6.2 Output KPIs

Expected KPIs

- Expected Gas Power
- Expected Shaft Power
- Expected Polytropic Efficiency
- Expected Head

Operating KPIs

- Operating Gas Power
- Operating Shaft Power
- Operating Polytropic Efficiency
- Operating Isentropic Efficiency
- Operating Head

10.2.6.3 Centrifugal Compressor - Performance Modeling***Overall Mass Balance***

Applying continuity equation:

$$F_{In} = F_{Out}$$

$$\rho_{In}Q_{In} = \rho_{Out}Q_{Out}$$

For water injection:

$$F_{In} + F_{Water} = F_{Out}$$

$$\rho_{In}Q_{In} + \rho_{Water}Q_{Water} = \rho_{Out}Q_{Out}$$

Overall Force Balance

$$P_{Out} = P_{In} + \Delta P$$

Overall Energy Balance

$$H_{In} + W_{Gas} = H_{Out}$$

For Water Injection:

$$H_{In} + H_{Water} + \dot{W}_{Gas} = H_{Out}$$

10.2.6.4 Centrifugal Compressor - Expected KPIs***Expected Shaft Power (kW)***

$$\dot{W}_{ShaftPower} = f(W, F) = \text{From flow vs power curve}$$

Expected Gas Power (kW)

$$\dot{W}_{GasPower} = \dot{W}_{ShaftPower}\eta_{MechEff}$$

Expected Polytropic Efficiency (%)

$$\eta_{Poly} = f(\eta, F) = \text{From flow vs Eff curve}$$

Expected Head (m)

$$H = f(H, F) = \text{From flow vs Head curve}$$

10.2.6.5 Centrifugal Compressor - Operating KPIs

Operating Gas Power (kW)

$$W_{Gas} = H_{Out} - H_{In}$$

For Water Injection,

$$\dot{W}_{Gas} = H_{Out} - (H_{In} + H_{Water})$$

Operating Shaft Power (kW)

$$\dot{W}_{ShaftPower} = \frac{\dot{W}_{GasPower}}{\eta_{MechEff}}$$

Operating Isentropic Efficiency (%)

$$\eta_{Isen} = \frac{\dot{W}_{IsenPower}}{\dot{W}_{GasPower}}$$

$$\dot{W}_{IsenPower} = H_{Out}^{Isen} - H_{In}$$

$$H_{Out}^{Isen} = F_{Out} h_{Out}(P_{Out}, S_{In})$$

Operating Polytropic Efficiency (%)

$$\eta_{Isen} = \frac{\dot{W}_{PolyPower}}{\dot{W}_{GasPower}}$$

$$\dot{W}_{PolyPower} = H_{Poly} F_{Out}$$

$$H_{Poly} = \frac{n_{Coeff} T_{In} Z_{In}}{MW} \left(\left(\frac{P_{Out}}{P_{In}} \right)^{\frac{1}{n_{Coeff}}} - 1 \right)$$

$$n_{Coeff} = \frac{\text{Log} \left(\frac{P_{Out}}{P_{In}} \right)}{\text{Log} \left(\left(\frac{T_{Out} Z_{Out}^{Isen}}{T_{In} Z_{In}} \right) \right)}$$

10.2.7 Centrifugal Compressor - Model Output

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Capacity Loss	Capacity or Flow Loss (Output)	%	Dynamic	Performance	Calculated
2	Degradation Loss	Degradation Loss(Output)	kW	Dynamic	Performance	Calculated

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
3	Deviation GasPower	Deviation GasPower (Output)	%	Dynamic	Performance	Calculated
4	Deviation GasPower Stage1	Deviation GasPower Stage1 (Output)	%	Dynamic	Performance	Calculated
5	Deviation GasPower Stage2	Deviation GasPower Stage2 (Output)	%	Dynamic	Performance	Calculated
6	Deviation GasPower Stage3	Deviation GasPower Stage3 (Output)	%	Dynamic	Performance	Calculated
7	Deviation Head	Difference between Expected and Operating Polytropic Head(Output)	%	Dynamic	Performance	Calculated
8	Deviation Efficiency	Difference between Expected and Operating Efficiency(Output)	%	Dynamic	Performance	Calculated
9	Deviation PolyEfficiency	Deviation PolyEfficiency (Output)	%	Dynamic	Performance	Calculated
10	Deviation PolyEfficiency Stage1	Deviation PolyEfficiency Stage1 (Output)	%	Dynamic	Performance	Calculated
11	Deviation PolyEfficiency Stage2	Deviation PolyEfficiency Stage2 (Output)	%	Dynamic	Performance	Calculated
12	Deviation PolyEfficiency Stage3	Deviation PolyEfficiency Stage3 (Output)	%	Dynamic	Performance	Calculated
13	Deviation PolyHead Stage1	Deviation PolyHead Stage1 (Output)	%	Dynamic	Performance	Calculated
14	Deviation PolyHead Stage2	Deviation PolyHead Stage2 (Output)	%	Dynamic	Performance	Calculated
15	Deviation PolyHead Stage3	Deviation PolyHead Stage3 (Output)	%	Dynamic	Performance	Calculated
16	Deviation Power	Difference between Expected and Operating Power(Output)	%	Dynamic	Performance	Calculated
17	Deviation ShaftPower	Deviation ShaftPower (Output)	%	Dynamic	Performance	Calculated
18	Deviation ShaftPower Stage1	Deviation ShaftPower Stage1 (Output)	%	Dynamic	Performance	Calculated
19	Deviation ShaftPower Stage2	Deviation ShaftPower Stage2 (Output)	%	Dynamic	Performance	Calculated
20	Deviation ShaftPower Stage3	Deviation ShaftPower Stage3 (Output)	%	Dynamic	Performance	Calculated
21	Energy Index	Energy Index (Output)		Dynamic	Performance	Calculated
22	Expected Gas Power	Expected Gas Power at current flow rate (Output)	kW	Dynamic	Performance	Calculated
23	Expected GasPower Stage1	Expected GasPower Stage1 (Output)	kW	Dynamic	Performance	Calculated
24	Expected GasPower	Expected GasPower Stage2	kW	Dynamic	Performance	Calculated

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
	Stage2	(Output)				
25	Expected GasPower Stage3	Expected GasPower Stage3 (Output)	kW	Dynamic	Performance	Calculated
26	Expected Head	Expected Head at current flow rate (Output)	m	Dynamic	Performance	Calculated
27	Expected Poly Efficiency	Polytropic Efficiency from curve data (Output)	%	Dynamic	Performance	Calculated
28	Expected PolyEfficiency Stage1	Expected PolyEfficiency Stage1 (Output)	%	Dynamic	Performance	Calculated
29	Expected PolyEfficiency Stage2	Expected PolyEfficiency Stage2 (Output)	%	Dynamic	Performance	Calculated
30	Expected PolyEfficiency Stage3	Expected PolyEfficiency Stage3 (Output)	%	Dynamic	Performance	Calculated
31	Expected PolyHead Stage1	Expected PolyHead Stage1 (Output)	m	Dynamic	Performance	Calculated
32	Expected PolyHead Stage2	Expected PolyHead Stage2 (Output)	m	Dynamic	Performance	Calculated
33	Expected PolyHead Stage3	Expected PolyHead Stage3 (Output)	m	Dynamic	Performance	Calculated
34	Expected Shaft Power	Expected Shaft Power (Output) at current flow rate (Output)	kW	Dynamic	Performance	Calculated
35	Expected ShaftPower Stage1	Expected ShaftPower Stage1 (Output)	kW	Dynamic	Performance	Calculated
36	Expected ShaftPower Stage2	Expected ShaftPower Stage2 (Output)	kW	Dynamic	Performance	Calculated
37	Expected ShaftPower Stage3	Expected ShaftPower Stage3 (Output)	kW	Dynamic	Performance	Calculated
38	Expected Speed	Expected Speed for current flow (Output)	rpm	Dynamic	Performance	Calculated
39	Expected Volume Flow	Expected Flow from Performance Curve (Output)	m ³ /hr	Dynamic	Performance	Calculated
40	Flow Factor	Flow Factor (Output)		Dynamic	Performance	Calculated
41	Head Factor	Head Factor (Output)		Dynamic	Performance	Calculated
42	Health Index	Health Index (Output)		Dynamic	Performance	Calculated
43	Opeartional Loss	Opeartional Loss(Output)	kW	Dynamic	Performance	Calculated
44	Operating Gas Power	Operating Gas Power (Output)	kW	Dynamic	Performance	Calculated
45	Operating GasPower Stage1	Operating GasPower Stage1 (Output)	kW	Dynamic	Performance	Calculated
46	Operating GasPower	Operating GasPower Stage2	kW	Dynamic	Performance	Calculated

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
	Stage2	(Output)				
47	Operating GasPower Stage3	Operating GasPower Stage3 (Output)	kW	Dynamic	Performance	Calculated
48	Operating Head	Operating Head (Output)	m	Dynamic	Performance	Calculated
49	OperatingHighFlow	Operating High Volume Flow Rate (Output)	m ³ /hr	Dynamic	Performance	Calculated
50	Operating Isen Eff	Operating Isentropic Efficiency (Output)	%	Dynamic	Performance	Calculated
51	Operating Isen Exponent	Operating Isentropic Exponent (Output)		Dynamic	Performance	Calculated
52	Operating Isen Head	Operating Isentropic Head (Output)	m	Dynamic	Performance	Calculated
53	OperatingLowFlow	Operating Low Volume Flow Rate (Output)	m ³ /hr	Dynamic	Performance	Calculated
54	Operating Poly Efficiency	Operating Polytropic Efficiency (Output)	%	Dynamic	Performance	Calculated
55	Operating PolyEfficiency Stage1	Operating PolyEfficiency Stage1 (Output)	%	Dynamic	Performance	Calculated
56	Operating PolyEfficiency Stage2	Operating PolyEfficiency Stage2 (Output)	%	Dynamic	Performance	Calculated
57	Operating PolyEfficiency Stage3	Operating PolyEfficiency Stage3 (Output)	%	Dynamic	Performance	Calculated
58	Operating Poly Exponent	Operating Polytropic Exponent (Output)		Dynamic	Performance	Calculated
59	Operating Polytropic Head	Operating Polytropic Head (Output)	m	Dynamic	Performance	Calculated
60	Operating PolyHead Stage1	Operating PolyHead Stage1 (Output)	m	Dynamic	Performance	Calculated
61	Operating PolyHead Stage2	Operating PolyHead Stage2 (Output)	m	Dynamic	Performance	Calculated
62	Operating PolyHead Stage3	Operating PolyHead Stage3 (Output)	m	Dynamic	Performance	Calculated
63	Operating Shaft Power	Operating Shaft Power (Output)	kW	Dynamic	Performance	Calculated
64	Operating ShaftPower Stage1	Operating ShaftPower Stage1 (Output)	kW	Dynamic	Performance	Calculated
65	Operating ShaftPower Stage2	Operating ShaftPower Stage2 (Output)	kW	Dynamic	Performance	Calculated
66	Operating ShaftPower Stage3	Operating ShaftPower Stage3 (Output)	kW	Dynamic	Performance	Calculated
67	Performance Index	Performance Index (Output)		Dynamic	Performance	Calculated

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
68	Power Factor	Power Factor (Output)		Static	Performance	Calculated
69	Recycle Trip and Surge Control Tabular Data	Recycle Trip and Surge Control Tabular Data (Output)		Static	Performance	Calculated
70	Recycle Trip Efficiency	Recycle Trip Efficiency (Output)	%	Static	Performance	Calculated
71	Recycle Trip Flow	Recycle Trip Flow (Output)	m ³ /hr	Static	Performance	Calculated
72	Recycle Trip Head	Recycle Trip Head (Output)	m	Static	Performance	Calculated
73	Recycle Trip Power	Recycle Trip Power (Output)	kW	Static	Performance	Calculated
74	Reduced Head	Reduced Head (Output)		Dynamic	Performance	Calculated
75	Reduced Head Flow Rate	Reduced Head Flow Rate (Output)		Dynamic	Performance	Calculated
76	Reduced Power	Reduced Power (Output)		Dynamic	Performance	Calculated
77	Reduced Power Flow Rate	Reduced Power Flow Rate (Output)		Dynamic	Performance	Calculated
78	Predicted Outlet Temperature	Predicted Outlet Temperature (Output)	°C	Dynamic	Performance	Calculated
79	Speed Factor	Speed Factor (Output)		Dynamic	Performance	Calculated
80	Stage1 MolecularWeight	Stage1 MolecularWeight (Output)		Dynamic	Performance	Calculated
81	Stage1 Outlet Temperature	Stage1 Outlet Temperature (Output)	°C	Dynamic	Performance	Calculated
82	Stage1 StoneWall Margin	Stage1 StoneWall Margin (Output)	%	Dynamic	Performance	Calculated
83	Stage1 Surge Margin	Stage1 Surge Margin (Output)	%	Dynamic	Performance	Calculated
84	Stage2 MolecularWeight	Stage2 MolecularWeight (Output)		Dynamic	Performance	Calculated
85	Stage2 Outlet Temperature	Stage2 Outlet Temperature (Output)	°C	Dynamic	Performance	Calculated
86	Stage2 StoneWall Margin	Stage2 StoneWall Margin (Output)	%	Dynamic	Performance	Calculated
87	Stage2 Surge Margin	Stage2 Surge Margin (Output)	%	Dynamic	Performance	Calculated
88	Stage3 MolecularWeight	Stage3 MolecularWeight (Output)		Dynamic	Performance	Calculated
89	Stage3 Outlet Temperature	Stage3 Outlet Temperature (Output)	°C	Dynamic	Performance	Calculated
90	Stage3 StoneWall Margin	Stage3 StoneWall Margin (Output)	%	Dynamic	Performance	Calculated
91	Stage3 Surge Margin	Stage3 Surge Margin (Output)	%	Dynamic	Performance	Calculated
92	Stone Wall Margin	Stone Wall Margin (Output)	%	Dynamic	Performance	Calculated
93	Surge Control Efficiency	Surge Control Efficiency (Output)	%	Static	Performance	Calculated

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
94	Surge Control Flow	Surge Control Flow (Output)	m ³ /hr	Static	Performance	Calculated
95	Surge Control Head	Surge Control Head (Output)	m	Static	Performance	Calculated
96	Surge Control Power	Surge Control Power (Output)	kW	Static	Performance	Calculated
97	Surge Margin	Surge Margin (Output)	%	Dynamic	Performance	Calculated

10.3 Centrifugal Compressor - Fault Model

10.3.1 Centrifugal Compressor - APM Model

Model Type	APM Model
Category	Centrifugal Compressor
Equipment Class	Compressor
Equipment Type	Centrifugal - Constant Speed V2

Below table illustrates the list of faults and symptoms configured for Compressor Performance-based monitoring.

Fault Name	Associated Symptoms	Recommendations
Anti-Surge Valve Opening Anomaly	Inlet Volume Flow Rate	1. Check the operating requirement if not then 2. Check calibration of ASV or its flow & pressure transmitter. 3. Inspect the ASV for any internal damage, and rectify if necessary
	Anti-surge valve Opening %	
	Minimum Surge Flow	
Change of Gas Properties (MW)	Efficiency deviation (Actual-Expected) HIGH (CGP)	1. Verify if process conditions have changed, verify if there is any correlation with environmental conditions (sun exposure, ambient temp, rain, etc. 2. Take a sample of the gas if possible.
	(Drive Input Power - Calculated Gas Power) deviation HIGH (CGP)	
	(Drive Input Power - Calculated Gas Power) deviation LOW (CGP)	
Compressor Inlet Supply Anomaly	Suction Volume Flow Rate RL (CISA)	1. Check calibration of sensor. 2. Asses of leaks/jam in the upstream equipment or pipe line or Suction strainers etc.
	Suction Volume Flow Rate RH (CISA)	
	Compressor Inlet Temperature RL (CISA)	
	Compressor Inlet Temperature RH (CISA)	
	Compressor Inlet Pressure RL (CISA)	
	Compressor Inlet Pressure	

Fault Name	Associated Symptoms	Recommendations
	RH (CISA)	
Compressor Performance degradation	Efficiency deviation (Actual-Expected) LOW (CPD)	1. For non-air, non-closed-loop applications: Perform a gas analysis or reconfirm molecular weight. 2. Check instruments accuracy (flow meter, suction and discharge temperature and pressure transmitters). 3. If all of the above are ok, then performance deviation is genuine.
	Compressor Discharge Temperature RH (CPD)	4. At this point: Confirm IGV operability. 5. Check for casing leaks. 6. Confirm seal integrity. 7. If all of the above is ok the reason for degradation could be internal clearances have increased and/or flow path has fouled up or has been damaged.
	Compressor Discharge Pressure RL (CPD)	8. In this case, at first opportunity: Carry out borescope inspection to verify if there is internal damage. 9. If fouling is identified, wash the machine if possible. 10. Monitor efficiency to determine speed of degradation and plan rotor replacement when it becomes unsustainable.
Fouling / Corrosion / Erosion Risk	Deviation Polytrophic Efficiency ROC Long Term Low (FCE)	1. If fouling: Short term; Wash compressor if possible. 2. Long term, investigate application of coating on affected gas path. 3. Change gas inlet Temp. 4. Borescope inspection at opportunity. 5. If corrosion / erosion: Replace corroded parts with better material grade. 6. In case of erosion consider coating of gas path. 7. Borescope at opportunity.
Inefficient Operation	Operating Efficiency	1. Verify if operating point can be moved closer to BEP.
	Best Efficiency	2. Check if control strategy can be modified to a more efficient method (i.e. suction throttling, speed control, etc.)

Model Type	APM Model
Category	Centrifugal Compressor
Equipment Class	Compressor
Equipment Type	Centrifugal - Variable Guide VaneV2

Below table illustrates the list of faults and symptoms configured for a Compressor Performance based monitoring.

Fault Name	Associated Symptoms	Recommendations
Anti-Surge Valve Opening Anomaly	Inlet Volume Flow Rate	1. Check the operating requirement if not then 2. Check calibration of ASV or its flow & pressure transmitter. 3. Inspect the ASV for any internal damage, and rectify if necessary
	Anti-surge valve Opening %	
	Minimum Surge Flow	
Change of Gas Properties (MW)	Efficiency deviation (Actual-Expected) HIGH (CGP)	1. Verify if process conditions have changed 2. Verify if there is any correlation with environmental conditions (sun exposure, ambient temp, rain, etc.) 3. Take a sample of the gas if possible.
	(Drive Input Power - Calculated Gas Power) deviation HIGH (CGP)	
	(Drive Input Power - Calculated Gas Power) deviation LOW (CGP)	
Compressor Inlet Supply Anomaly	Suction Volume Flow Rate RL (CISA)	1. Check calibration of sensor. 2. Asses of leaks/jam in the upstream equipment or pipe line or Suction strainers etc.
	Suction Volume Flow Rate RH (CISA)	
	Compressor Inlet Temperature RL (CISA)	
	Compressor Inlet Temperature RH (CISA)	
	Compressor Inlet Pressure RL (CISA)	
	Compressor Inlet Pressure RH (CISA)	
Compressor Performance degradation	Efficiency deviation (Actual-Expected) LOW (CPD)	1. For non-air, non-closed-loop applications: Perform a gas analysis or reconfirm molecular weight. 2. Check instruments accuracy (flow meter, suction and discharge temperature and pressure transmitters). 3. If all of the above are ok, then performance deviation is genuine. 4. At this point: Confirm IGV operability. 5. Check for casing leaks. 6. Confirm seal integrity. 7. If all of the above is ok the reason for degradation could be internal clearances have increased and/or flow path has fouled up or has been damaged. 8. In this case, at first opportunity: Carry out borescope inspection to verify if there is internal damage. 9. If fouling is identified, wash the machine if possible. 10. Monitor efficiency to determine speed of degradation and plan rotor replacement when it becomes unsustainable.
	Compressor Discharge Temperature RH (CPD)	
	Compressor Discharge Pressure RL (CPD)	
Fouling /	Deviation Polytrophic	1. If fouling: Short term; Wash compressor if possible.

Fault Name	Associated Symptoms	Recommendations
Corrosion / Erosion Risk	Efficiency ROC Long Term Low (FCE)	2. Long term, investigate application of coating on affected gas path. 3. Change gas inlet Temp. 4. Borescope inspection at opportunity. 5. If corrosion / erosion: Replace corroded parts with better material grade. 6. In case of erosion consider coating of gas path. 7. Borescope at opportunity.
Inefficient Operation	Operating Efficiency	1. Verify if operating point can be moved closer to BEP. 2. Check if control strategy can be modified to a more efficient method (i.e. suction throttling, speed control, etc.)
	Best Efficiency	
IGV/Suction Throttle Valve Malfunction	IGV Angle / Actuator Position RL (ISTVMF)	1. Check IGV/Suction throttle integrity and operability. 2. Check calibration of the positioner. 3. Check IGV/Suction throttle actuator and feed. 4. Check all linkage connections. 5. Carry out borescope examination to identify presence of internal damage
	IGV Angle / Actuator Position RH (ISTVMF)	
	Suction Throttling Valve % RL (ISTVMF)	
	Suction Throttling Valve % RH (ISTVMF)	

Model Type	APM Model
Category	Centrifugal Compressor
Equipment Class	Compressor
Equipment Type	Centrifugal - Variable Speed V2

Below table illustrates the list of faults and symptoms configured for a Compressor Performance based monitoring.

Fault Name	Associated Symptoms	Recommendations
Anti-Surge Valve Opening Anomaly	Inlet Volume Flow Rate	1. Check the operating requirement if not then 2. Check calibration of ASV or its flow & pressure transmitter. 3. Inspect the ASV for any internal damage, and rectify if necessary
	Anti-surge valve Opening %	
	Minimum Surge Flow	
Change of Gas Properties (MW)	Efficiency deviation (Actual-Expected) HIGH (CGP)	1. Verify if process conditions have changed, verify if there is any correlation with environmental conditions (sun exposure, ambient temp, rain, etc.) 2. Take a sample of the gas if possible.
	(Drive Input Power - Calculated Gas Power) deviation HIGH (CGP)	
	(Drive Input Power - Calculated Gas Power) deviation LOW (CGP)	
Compressor Inlet Supply	Suction Volume Flow Rate RL (CISA)	1. Check calibration of sensor.

Fault Name	Associated Symptoms	Recommendations
Anomaly		2. Asses of leaks/jam in the upstream equipment or pipe line or Suction strainers etc.
	Suction Volume Flow Rate RH (CISA)	
	Compressor Inlet Temperature RL (CISA)	
	Compressor Inlet Temperature RH (CISA)	
	Compressor Inlet Pressure RL (CISA)	
	Compressor Inlet Pressure RH (CISA)	
Compressor Performance degradation	Efficiency deviation (Actual-Expected) LOW (CPD)	1. For non-air, non-closed-loop applications: Perform a gas analysis or reconfirm molecular weight. 2. Check instruments accuracy (flow meter, suction and discharge temperature and pressure transmitters). 3. If all of the above are ok, then performance deviation is genuine. 4. At this point: Confirm IGV operability. 5. Check for casing leaks. 6. Confirm seal integrity. 7. If all of the above is ok the reason for degradation could be internal clearances have increased and/or flow path has fouled up or has been damaged. 8. In this case, at first opportunity: Carry out borescope inspection to verify if there is internal damage. 9. If fouling is identified, wash the machine if possible. 10. Monitor efficiency to determine speed of degradation and plan rotor replacement when it becomes unsustainable.
	Compressor Discharge Temperature RH (CPD)	
	Compressor Discharge Pressure RL (CPD)	
Fouling / Corrosion / Erosion Risk	Deviation Polytrophic Efficiency ROC Long Term Low (FCE)	1. If fouling: Short term; Wash compressor if possible. 2. Long term, investigate application of coating on affected gas path. 3. Change gas inlet Temp. 4. Borescope inspection at opportunity. 5. If corrosion / erosion: Replace corroded parts with better material grade. 6. In case of erosion consider coating of gas path. 7. Borescope at opportunity.
Inefficient Operation	Operating Efficiency	1. Verify if operating point can be moved closer to BEP. 2. Check if control strategy can be modified to a more efficient method (i.e. suction throttling, speed control, etc.
	Best Efficiency	

10.3.2 Centrifugal Compressor - APR Model

10.3.2.1 Centrifugal Compressor - Constant speed V2

Centrifugal Compressor Unit Performance and Health Fault Model - Constant Speed V2

Model Type	APR Model
Category	Centrifugal Compressor
Equipment Class	Compressor
Equipment Type	Centrifugal Compressor - Unit

Below table illustrates the list of faults and symptoms configured for a Compressor Unit.

Fault Name	Associated Symptoms	Recommendations
Bearing Anomaly due to Lube Oil Temperature	Thrust Bearing Axial shaft Displacement Active Side R High	1. Check the calibration of sensor. 2. Check cooling water supply pressure. 3. Check bypass control valve. 4. Open both coolers and stabilize temperature. 5. Check cooling water pumps for blockages, ensure strainer is clean. 6. If the cooler is an air cooler check fan and motor integrity. 7. If malfunction of one cooler is suspected, isolate and inspect it. 8. Line walk the system to check for leaks.
	Thrust Bearing Axial shaft Displacement Active Side R Low	
	Radial Bearing Vibration X DE R High	
	Radial Bearing Vibration Y DE R High	
	Radial Bearing Vibration X NDE R High	
	Radial Bearing Vibration Y NDE R High	
	Radial Bearing Temperature A DE R High	
	Radial Bearing Temperature B DE R High	
	Radial Bearing Temperature A NDE R high	
	Radial Bearing Temperature B NDE R high	
	Thrust Bearing Temperature A Active Side High	
	Thrust Bearing Temperature B Active Side High	
	Thrust Bearing	

Fault Name	Associated Symptoms	Recommendations
	Temperature C Active Side High	
	Thrust Bearing Temperature D Active Side High	
	Thrust Bearing Temperature A Inactive Side High	
	Thrust Bearing Temperature B Inactive Side High	
	Thrust Bearing Temperature C Inactive Side High	
	Thrust Bearing Temperature D Inactive Side High	
	Lube Oil Header Temperature R High	
Bearing Anomaly due to Lube Oil Pressure	Thrust Bearing Axial shaft Displacement Active Side R High	1. Check the calibration of sensor. 2. Switch on both oil pumps and stabilize oil pressure. 3. Check oil level in tank. 4. Check filter DP. 5. Assess malfunction. 6. Line walk the system and check for leaks. 7. Ensure PSV's are seated, check spillback/PCV valves are set correctly and not returning oil back to the tank. 8. If maintenance has recently been performed check correct rotation of pump. 9. Ensure no hand valves have been left shut, ensure NRV's are working correctly. 10. Check functioning of the auto change over switch.
	Thrust Bearing Axial shaft Displacement Active Side R Low	
	Radial Bearing Vibration X DE R High	
	Radial Bearing Vibration Y DE R High	
	Radial Bearing Vibration X NDE R High	
	Radial Bearing Vibration Y NDE R High	
	Radial Bearing Temperature A DE R High	
	Radial Bearing Temperature B DE R High	
	Radial Bearing Temperature A NDE R high	
	Radial Bearing Temperature B NDE R high	

Fault Name	Associated Symptoms	Recommendations
	Thrust Bearing Temperature A Active Side High	
	Thrust Bearing Temperature B Active Side High	
	Thrust Bearing Temperature C Active Side High	
	Thrust Bearing Temperature D Active High	
	Thrust Bearing Temperature A Inactive Side High	
	Thrust Bearing Temperature B Inactive Side High	
	Thrust Bearing Temperature C Inactive Side High	
	Thrust Bearing Temperature D Inactive High	
	Lube Oil Header Pressure R Low	
Bearing Anomaly due to Lube Oil Quality	Thrust Bearing Axial shaft Displacement Active Side R High	1. Check oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, TAN. 4. Check no breather caps or man covers have been removed and left open to allow debris into the tank. 5. Recondition or replace oil.
	Thrust Bearing Axial shaft Displacement Active Side R Low	
	Radial Bearing Vibration X DE R High	
	Radial Bearing Vibration Y DE R High	
	Radial Bearing Vibration X NDE R High	
	Radial Bearing Vibration Y NDE R High	
	Radial Bearing Temperature A DE R High	
	Radial Bearing	

Fault Name	Associated Symptoms	Recommendations
	Temperature B DE R High	
	Radial Bearing Temperature A NDE R high	
	Radial Bearing Temperature B NDE R high	
	Thrust Bearing Temperature A Active Side High	
	Thrust Bearing Temperature B Active Side High	
	Thrust Bearing Temperature C Active Side High	
	Thrust Bearing Temperature D Active High	
	Thrust Bearing Temperature A Inactive Side High	
	Thrust Bearing Temperature B Inactive Side High	
	Thrust Bearing Temperature C Inactive Side High	
	Thrust Bearing Temperature D Inactive Side High	
	Lube Oil Viscosity High	
	Lube Oil Viscosity Low	
	Lube Oil TAN High	
DE Radial Bearing Failure due to Mechanical Issue	Radial Bearing Vibration X DE R High	1. Check calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if intervention not possible. 4. Intervention options, reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots.

Fault Name	Associated Symptoms	Recommendations
	Radial Bearing Vibration Y DE R High	7. Change bearing if required.
	Radial Bearing Temperature A DE R High	
	Radial Bearing Temperature B DE R High	
NDE Radial Bearing Failure due to Mechanical Issue	Radial Bearing Vibration X NDE R High	1. Check calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if intervention not possible. 4. Intervention options; reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 7. Change bearing if required.
	Radial Bearing Vibration Y NDE R High	
	Radial Bearing Temperature A NDE R High	
	Radial Bearing Temperature B NDE R High	
Rotor Anomaly	Radial Bearing Vibration X DE R High	1. Check the calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Check Carbon brush integrity. 4. Check magnetism of the rotor. 5. Check rotor unbalance, misalignment. 6. Looseness. 7. Check coupling integrity. 8. Check foundation bolt integrity. 9. Check both bearings for clearance and pad wear etc.
	Radial Bearing Vibration Y DE R High	
	Radial Bearing Vibration X NDE R High	
	Radial Bearing Vibration Y NDE R High	
Thrust Bearing Failure due to Mechanical Issue	Thrust Bearing Axial Shaft Displacement Active Side R High	1. Check calibration of sensor. 2. Assess likely cause and time to trip if intervention not possible. 3. Intervention options. 4. Reduce thrust load. 5. Ensure parts availability. 6. Check axial vibration trends to establish if the machine has surged and potentially damaged/wiped the pads. 7. If required inspect the thrust pads and collar to check for damage / wiping. 8. Carry out a bump check to establish what float is in the machine, compare to previous overhaul report for measurement. 9. Check the gear coupling/pre-stretch diaphragm coupling.
	Thrust Bearing Axial Shaft Displacement Inactive Side R High	
	Thrust Bearing Axial Shaft Displacement Active Side R Low	
	Thrust Bearing Axial Shaft Displacement Inactive Side R Low	
	Thrust Bearing Temperature A Active Side High	
	Thrust Bearing	

Fault Name	Associated Symptoms	Recommendations
	Temperature B Active Side High	
	Thrust Bearing Temperature C Active Side High	
	Thrust Bearing Temperature D Active Side High	
	Thrust Bearing Temperature A Inactive Side High	
	Thrust Bearing Temperature B Inactive Side High	
	Thrust Bearing Temperature C Inactive Side High	
	Thrust Bearing Temperature D Inactive Side High	
Balance Piston Seal Failure	Thrust Bearing Axial Shaft Displacement Active Side R High	1. Check the calibration of sensor. 2. Monitor rotor axial position, pad temperature. 3. Balance chamber pressure. 4. Determine if failure mode is still progressing, in which case calculate time to thrust bearing displacement trip at current rate. 5. Plan for repair accordingly. 6. If required dismantle the machine and carry out lead checks on the j strips (labyrinth seals) and determine what the clearance is. 7. Compare this to previous overhaul reports. 8. Excessive clearance will overload the thrust bearing beyond its design limit. 9. The rotor will require overhaul/replacing if beyond design clearance. 10. Clean the Balance line tube.
	Thrust Bearing Axial Shaft Displacement Inactive Side R High	
	Thrust Bearing Axial Shaft Displacement Active Side R Low	
	Thrust Bearing Axial Shaft Displacement Inactive Side R Low	
	Balance line DP S High	
Balance Line Blockage Risk	Balance line Blockage S High	1. Check the calibration of sensor. 2. Check the balance line tube for any particle entrainment and clean. 3. Check any surge events, liquid slugs in the sample gas.
	Balance line Blockage S Low	
Inlet Filter Anomaly	Inlet Filter DP RH (IFDA)	1. Check the calibration of sensor. 2. Change over the filter if stand by is available and replace the filter.

Fault Name	Associated Symptoms	Recommendations
Liquid Slugs	Radial Bearing Vibration X DE RH (LS)	- Short term: reduce flow. - Correct problem at process level (reduce scrubber level or act on process), monitor seals leakages, axial shaft position, thrust bearing temperature.
	Radial Bearing Vibration Y DE RH (LS)	
	Radial Bearing Vibration X NDE RH (LS)	
	Radial Bearing Vibration Y NDE RH (LS)	
	Suction Scrubber Level RH (LS)	
Inter stage Cooler Anomaly	Interstage Cooler Gas Outlet Temperature SH (ISCA)	- Check the interstage cooler's integrity. - Check the temp control valve for any anomaly. - Conduct loop proof test for temp control valve. - Check cooling water supply line for any leakage / blockage/ supply issue.
	Interstage Cooler Temp Control Valve Opening % SH (ISCA)	
	Interstage Cooler Gas Outlet Temperature SL (ISCA)	
	Interstage Cooler Temp Control Valve Opening % SL (ISCA)	
Interstage Separator Anomaly	Interstage Separator Level SH (ISSA)	- Check Separator / scrubber's internals. - Check drain line of the separator. - Check the transmitter.

Centrifugal Compressor Seal Gas System Fault Model - Constant Speed V2

Model Type	APR Model
Category	Centrifugal Compressor
Equipment Class	Compressor
Equipment Type	Centrifugal Compressor - Seal Gas System

Below table illustrates the list of faults and symptoms configured for a Centrifugal Compressor Seal Gas System.

Fault Name	Associated Symptoms	Recommendations
Primary Seal Gas Filter Anomaly	Primary Seal Gas Filter DP RH (PSGFA)	- Check the calibration of sensor. - Check the filter efficiency and change to standby the filter if required. - Repair/replace the filter internals.

Fault Name	Associated Symptoms	Recommendations
Buffer Gas Filter Anomaly	Buffer Seal Gas Filter DP RH (BGSFA)	1. Check the calibration of sensor. 2. Check the filter efficiency and change to standby the filter if required. 3. Repair/replace the filter internals.
PDCV Malfunction	Reference Line DP RL (PDCVA)	1. Check the calibration of sensor. 2. Check valve set point. 3. Check valve internals for any damage, and repair / replace if necessary. 4. Conduct the PDCV control loop function test .
	PDCV Valve Opening RH (PDCVA)	
Primary Seal Gas Vent PCV Anomaly	Primary Seal Leak Pressure DP SH (PSGVA)	1. Calibrate the valve and transmitters by isolation if possible. 2. Check connection loss if any. 3. Check primary vent line for any jam. 4. Check the internals of control valve of the vent line. 5. Conduct loop proof test of the control loop.
	Primary Seal Leak Pressure DP SL (PSGVA)	
	Primary Seal Vent Gas PCV SH (PSGVA)	
	Primary Seal Vent Gas PCV SL (PSGVA)	
OB Seal Laby Malfunction	Primary Vent Gas Pressure RL (OSLM)	1. Ensure upstream and downstream PSV's, check valves, transmitters are all working correctly. 2. Inspect and monitor trend sec seal gas system. 3. Ensure spare seals cartridge is available. 4. Assess risk. 5. Monitor and replace seals at opportunity. 6. Replace seal.
	Primary Vent Gas Flow RL (OSLM)	
	Secondary Vent Gas Pressure RH (OSLM)	
	Secondary Vent Gas Flow RH (OSLM)	
OB Seal Malfunction	Primary Vent Gas Pressure RL (OSM)	1. Ensure upstream and downstream PSV's, check valves, transmitters are all working correctly. 2. Inspect and monitor trend sec seal gas system. 3. Ensure spare seals cartridge is available. 4. Assess risk. 5. Monitor and replace seals at opportunity. 6. Replace seal.
	Primary Vent Gas Flow RL (OSM)	
	Secondary Vent Gas Pressure RH (OSM)	
	Secondary Vent Gas Flow RH (OSM)	
IB Seal Malfunction	Primary Vent Gas Pressure RH (IBM)	1. Ensure upstream and downstream PSV's, check valves, transmitters are all working correctly. 2. Inspect and monitor trend sec seal gas system. 3. Ensure spare seals cartridge is available. 4. Assess risk. 5. Monitor and replace seals at opportunity.

Fault Name	Associated Symptoms	Recommendations
	Primary Vent Gas Flow RH (IBM)	6. Replace seal.
Primary Seal Supply Gas Temperature Anomaly	Primary Supply Gas Temperature RL (PSGTA)	1. Check calibration of sensor.
	Primary Supply Gas Temperature RH (PSGTA)	2. Monitor seal leakage.
		3. Check integrity of seal gas supply PDV.
		4. Check super heater functionality if available.
		5. Check separation gas.
		6. Check seal supply filters for damage and are fitted correctly.
No Seal Lift off Anomaly	Primary Vent Gas Pressure RH (NSLA)	1. Seal faces: overheating, cracking, scoring.
	Primary Vent Gas Flow RH (NSLA)	2. Inspect seal and replace.
	Secondary Vent Gas Pressure RH (NSLA)	
	Secondary Vent Gas Flow RH (NSLA)	
	Compressor Speed LOW (NSLA)	
Primary Seal Gas Supply Anomaly	Primary Seal Gas Supply Pressure RL (PSGSA)	1. Check the calibration of sensor.
	Primary Seal Gas Supply Pressure RH (PSGSA)	2. Ensure upstream and downstream PSV's, check valves, transmitters are all working correctly.
	Primary Seal Gas Supply Flow RL (PSGSA)	3. Monitor trend of seal gas system.
	Primary Seal Gas Supply Flow RH (PSGSA)	4. Inspect system for leakage.
		5. Inspect PCV and check functionality.
		6. Replace filters.
Buffer Gas Supply Anomaly	Buffer Gas Supply Pressure RL (BGSA)	1. Check the calibration of sensor.
	Buffer Gas Supply Pressure RH (BGSA)	2. Check N2 supply system for any leaks or damage.
	Buffer Gas Supply Flow RL (BGSA)	3. Ensure upstream and downstream PSV's, check valves, transmitters are all working correctly.
	Buffer Gas Supply Flow RH (BGSA)	4. Inspect PCV and check functionality.
	Secondary Supply PCV Opening SH (BGSA)	5. Replace Filters.
	Buffer Gas PCV Position SH (BGSA)	
Separation Gas Supply	Separation Supply Gas	1. Check the calibration of sensor.

Fault Name	Associated Symptoms	Recommendations
Anomaly	Pressure RL (SGSA)	2. Check N2 supply system. 3. Increased risk of outboard seal contamination. 4. Increased risk of bearing oil ingress into seal. 5. Inspect supply line upstream of seal. 6. Check labyrinth seal for blockages.
	Separation Supply Gas Pressure RH (SGSA)	
	Separation Supply Gas Flow RL (SGSA)	
	Separation Supply Gas Flow RH (SGSA)	
	Separation Gas PCV Opening SH (SGSA)	
	Buffer Gas PCV Position SH (SGSA)	
Seal Gas Booster Compressor Malfunction	Reference Line DP RL (SGBCM)	Check Seal Gas Booster Compressor.
	Primary Seal Gas Supply Pressure RL (SGBCM)	
	Primary Seal Gas Supply Pressure RH (SGBCM)	
	Primary Seal Gas Supply Flow RL (SGBCM)	
	Primary Seal Gas Supply Flow RH (SGBCM)	
Secondary Supply to Primary Vent Anomaly	Secondary Supply to Primary Vent DP RL (SSTPVA)	1. Check calibration of sensor. 2. Check and control N2 supply. 3. Check for any line leaks. 4. Check for seal functionality and replace if necessary.
	Secondary Supply to Primary Vent DP RH (SSTPVA)	

Centrifugal Compressor Gearbox Fault Model - Constant Speed V2

Model Type	APR Model
Category	Centrifugal Compressor
Equipment Class	Compressor
Equipment Type	Centrifugal Compressor - Gearbox System

Below table illustrates the list of faults and symptoms configured for a Centrifugal Compressor Gearbox System.

Fault Name	Associated Symptoms	Recommendations
Gearbox LSS Radial Bearing DE Anomaly	LSS GB Radial Bearing Vibration X DE RH (GLRDA)	1. Check the calibration of sensor. 2. Check if the lube oil header temp, pressure and quality has not altered(if lube oil system is available otherwise

Fault Name	Associated Symptoms	Recommendations
	LSS GB Radial Bearing Vibration Y DE RH (GLRDA)	check greasing or clog). 3. Investigate with high frequency vibration analysis. 4. Assess likely cause and time to fail(trip) if intervention not possible. 5. Intervention options; reduce speed, Parts availability. 6. Check the trends to establish if its either both X and Y probes or one. 7. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 8. Change bearing if required.
	LSS GB Radial Bearing Temperature A DE RH (GLRDA)	
	LSS GB Radial Bearing Temperature B DE RH (GLRDA)	
	LSS GB Gearbox Casing Vibration DE RH (GLRDA)	
Gearbox LSS Radial Bearing NDE Anomaly	LSS GB Radial Bearing Vibration X NDE RH (GLRNDA)	1. Check the calibration of sensor. 2. Check if the lube oil header temp, pressure and quality has not altered(if lube oil system is available otherwise check greasing or clog). 3. Investigate with high frequency vibration analysis. 4. Assess likely cause and time to fail(trip) if intervention not possible. 5. Intervention options; reduce speed, Parts availability. 6. Check the trends to establish if its either both X and Y probes or one. 7. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 8. Change bearing if required.
	LSS GB Radial Bearing Vibration Y NDE RH (GLRNDA)	
	LSS GB Radial Bearing Temperature A NDE RH (GLRNDA)	
	LSS GB Radial Bearing Temperature B NDE RH (GLRNDA)	
	LSS GB Gearbox Casing Vibration NDE RH (GLRNDA)	
Gearbox HSS Radial Bearing DE Anomaly	HSS GB Radial Bearing Vibration X DE RH (GHRDA)	1. Check the calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if intervention not possible. 4. Intervention options; reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 7. Change bearing if required.
	HSS GB Radial Bearing Vibration Y DE RH (GHRDA)	
	HSS-DE Radial bearing temperature RH (GHRDA)	
	HSS-DE Radial bearing temperature RH (GHRDA)	
	HSS GB Gearbox Casing Vibration DE RH (GHRDA)	
Gearbox HSS Radial Bearing NDE Anomaly	HSS GB Radial Bearing Vibration X NDE RH (GHRNDA)	1. Check the calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if intervention not possible. 4. Intervention options; reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal

Fault Name	Associated Symptoms	Recommendations
	HSS GB Radial Bearing Vibration Y NDE RH (GHRNDA)	clearances, oil coking / hot spots. 7. Change bearing if required.
	HSS-NDE Radial bearing temperature RH (GHRNDA)	
	HSS-NDE Radial bearing temperature RH (GHRNDA)	
	HSS GB Gearbox Casing Vibration NDE RH (GHRNDA)	
Gearbox LSS Shaft Anomaly	LSS GB Radial Bearing Vibration X DE RH (GLSA)	1. Check the calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Check Carbon brush integrity. 4. Check magnetism of the rotor. 5. Check rotor unbalance, misalignment. 6. Looseness. 7. Check coupling integrity. 8. Check foundation bolt integrity. 9. Check both bearings for clearance and pad wear etc. 10. Check gear integrity.
	LSS GB Radial Bearing Vibration Y DE RH (GLSA)	
	LSS GB Radial Bearing Vibration X NDE RH (GLSA)	
	LSS GB Radial Bearing Vibration Y NDE RH (GLSA)	
	LSS GB Gearbox Casing Vibration DE RH (GLSA)	
	LSS GB Gearbox Casing Vibration NDE RH (GLSA)	
Gearbox HSS Shaft Anomaly	HSS GB Radial Bearing Vibration X DE RH (GHSA)	1. Check calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Check Carbon brush integrity. 4. Check magnetism of the rotor Check rotor unbalance, misalignment. 5. Looseness. 6. Check coupling integrity. 7. Check foundation bolt integrity. 8. Check both bearings for clearance and pad wear etc. 9. Check gear integrity.
	HSS GB Radial Bearing Vibration Y DE RH (GHSA)	
	HSS GB Radial Bearing Vibration X NDE RH (GHSA)	
	HSS GB Radial Bearing Vibration Y NDE RH (GHSA)	
	HSS GB Gearbox Casing Vibration DE RH (GHSA)	
	HSS GB Gearbox Casing Vibration NDE RH (GHSA)	
Gearbox LSS Thrust Bearing Anomaly	LSS GB Thrust Bearing Displacement Active Side RH (GLTBA)	1. Check if the lube oil header temp, pressure and quality has not altered(if lube oil system is available). 2. Assess likely cause and time to trip if intervention not possible. 3. Check calibration of sensor. 4. Intervention options; reduce thrust load, Ensure parts availability. 5. Check axial vibration trends to establish if the driver or

Fault Name	Associated Symptoms	Recommendations
	LSS GB Thrust Bearing Displacement Active Side RL (GLTBA)	driven has change in axial thrust and potentially damaged/wiped the pads. - If required inspect the thrust pads and collar to check for damage / wiping. - Carry out a bump check to establish what float is in the machine, compare to previous overhaul report for measurement. - Check the gear coupling/pre-stretch diaphragm coupling.
	LSS GB Thrust Bearing Displacement Inactive Side RH (GLTBA)	
	LSS GB Thrust Bearing Displacement Inactive Side RL (GLTBA)	
	LSS GB Thrust Bearing Temperature A Active Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature B Active Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature C Active Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature A Inactive Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature B Inactive Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature C Inactive Side RH (GLTBA)	
Gearbox HSS Thrust Bearing Anomaly	HSS GB Thrust Bearing Displacement Active Side RH (GHTBA)	- Check if the lube oil header temp, pressure and quality has not altered(if lube oil system is available). - Assess likely cause and time to trip if intervention not possible. - Check calibration of sensor. - Intervention options; reduce thrust load, Ensure parts availability. - Check axial vibration trends to establish if the driver or driven has change in axial thrust and potentially damaged/wiped the pads. - If required inspect the thrust pads and collar to check for damage / wiping. - Carry out a bump check to establish what float is in the machine, compare to previous overhaul report for measurement. - Check the gear coupling/pre-stretch diaphragm coupling.
	HSS GB Thrust Bearing Displacement Active Side RL (GHTBA)	
	HSS GB Thrust Bearing Displacement Inactive Side RH (GHTBA)	
	HSS GB Thrust Bearing Displacement Inactive Side RL (GHTBA)	
	HSS GB Thrust Bearing Temperature A Active Side RH (GHTBA)	
	HSS GB Thrust Bearing Temperature B Active Side RH	

Fault Name	Associated Symptoms	Recommendations
	(GHTBA)	
	HSS GB Thrust Bearing Temperature C Active Side RH (GHTBA)	
	HSS GB Thrust Bearing Temperature A Inactive Side RH (GHTBA)	
	HSS GB Thrust Bearing Temperature B Inactive Side RH (GHTBA)	
	HSS GB Thrust Bearing Temperature C Inactive Side RH (GHTBA)	

Centrifugal Compressor Lube Oil Fault Model - Constant Speed V2

Model Type	APR Model
Category	Centrifugal Compressor
Equipment Class	Compressor
Equipment Type	Centrifugal Compressor - Lube Oil System

Below table illustrates the list of faults and symptoms configured for a Centrifugal Compressor Lube Oil System.

Fault Name	Associated Symptoms	Recommendations
Loss of Lube Oil Quality	Lube Oil Viscosity HIGH (LOLQ)	1. Check oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, TAN. 4. Check no breather caps or man covers have been removed and left open to allow debris into the tank. 5. Do filter cleaning. 6. Recondition or replace oil.
	Lube Oil Viscosity LOW (LOLQ)	
	Lube Oil TAN HIGH (LOLQ)	
	Lube Oil Composition Analysis HIGH (LOLQ)	
Lube Oil Filter Anomaly	Lube Oil Filter DP RH (LFDF)	1. Check filter DP, changeover and clean/replace the stand by filter. 2. Check the Filter DP transmitter.
Loss of Pressure due to Pump Anomaly	Lube Oil Header Pressure RL (LOPPF)	1. Check the Discharge PT integrity. 2. Switch on both oil pumps and stabilize oil pressure. 3. Assess malfunction. 4. Line walk the system and check for leaks, ensure

Fault Name	Associated Symptoms	Recommendations
		PSV's are seated.
	Lube Oil Pump Discharge Pressure RL (LOPPF)	5. Check spillback/PCV valves are set correctly and not returning oil back to the tank.
	Lube Oil PCV Position RL (LOPPF)	6. Ensure no hand valves have been left shut for the standby pump.
	Lube Oil Return Flow Rate RL (LOPPF)	7. Check oil level in tank, Open tank overhead vents for release of pressure.
	Lube Oil Main Pump Current (Run Status) ON (LOPPF)	8. Check PCV function for Integrity.
	Lube Oil Aux Pump Current (Run Status) ON (LOPPF)	9. Assess malfunction.
Loss of Cooling due to TCV Anomaly	Lube Oil Header Temperature (Cooler Out Temp) RH (LOCTF)	10. Check Auto Logic Function tests, check functioning of the auto change over switch.
	LO Cooler Bypass Valve (TCV) Position RH (LOCTF)	11. If maintenance has recently been performed check correct rotation of pump.
Loss of Cooling due to Cooler Anomaly	Lube Oil Header Temperature (Cooler Out Temp) RH (LOCCF)	12. Ensure NRV's are working correctly.
	LO Tank And Cooler Outlet Temperature Delta RH (LOCCF)	
Loss of Heating Due to Lube Oil Tank Heater Off	Lube Oil Header Temperature (Cooler Out Temp) RL (LOCHOFF)	1. Check the calibration of sensor.
	Lube Oil Tank Temperature (Cooler Inlet Temp) RL (LOCHOFF)	2. Check bypass control valve for integrity.
	Lube Oil Tank Heater (Run Status) OFF (LOCHOFF)	3. Open both coolers and stabilize temperature.
Loss of Cooling due to Lube Oil Tank Heater On	Lube Oil Header Temperature (Cooler Out Temp) RH (LOCHON)	4. Line walk the system to check for leaks.
	Lube Oil Tank Temperature (Cooler Inlet Temp) RH (LOCHON)	5. Check the Isolation/bypass valve of TCV.
	Lube Oil Tank Heater (Run Status) ON (LOCHON)	

Fault Name	Associated Symptoms	Recommendations
	Status) ON (LOCHON)	
Loss of Cooling due to Fan Motor Issue	Lube Oil Header Temperature (Cooler Out Temp) RH (LOCFMI)	Offline: Check the Fan and Motor /coupling / belt Integrity and rectify.
	Lube Oil Cooler Fan 1 (Run Status) OFF (LOCFMI)	
	Lube Oil Cooler Fan 2 (Run Status) OFF (LOCFMI)	

10.3.2.2 Centrifugal Compressor - Variable Guide Vane V2

Centrifugal Compressor Unit Performance and Health Fault Model - Variable Guide Vane V2

Model Type	APR Model
Category	Centrifugal Compressor
Equipment Class	Compressor
Equipment Type	Centrifugal Compressor - Unit

Below table illustrates the list of faults and symptoms configured for a Compressor Unit.

Fault Name	Associated Symptoms	Recommendations
Bearing Anomaly due to Lube Oil Temperature	Thrust Bearing Axial shaft Displacement Active Side R High	1. Check the calibration of sensor. 2. Check cooling water supply pressure. 3. Check bypass control valve. 4. Open both coolers and stabilize temperature. 5. Check cooling water pumps for blockages, ensure strainer is clean. 6. If the cooler is an air cooler check fan and motor integrity. 7. If malfunction of one cooler is suspected, isolate and inspect it. 8. Line walk the system to check for leaks.
	Thrust Bearing Axial shaft Displacement Active Side R Low	
	Radial Bearing Vibration X DE R High	
	Radial Bearing Vibration Y DE R High	
	Radial Bearing Vibration X NDE R High	
	Radial Bearing Vibration Y NDE R High	
	Radial Bearing Temperature A DE R High	
	Radial Bearing Temperature B DE R High	
	Radial Bearing	

Fault Name	Associated Symptoms	Recommendations
	Temperature A NDE R high	
	Radial Bearing Temperature B NDE R high	
	Thrust Bearing Temperature A Active Side High	
	Thrust Bearing Temperature B Active Side High	
	Thrust Bearing Temperature C Active Side High	
	Thrust Bearing Temperature D Active Side High	
	Thrust Bearing Temperature A Inactive Side High	
	Thrust Bearing Temperature B Inactive Side High	
	Thrust Bearing Temperature C Inactive Side High	
	Thrust Bearing Temperature D Inactive Side High	
	Lube Oil Header Temperature R High	
Bearing Anomaly due to Lube Oil Pressure	Thrust Bearing Axial shaft Displacement Active Side R High	1. Check the calibration of sensor. 2. Switch on both oil pumps and stabilize oil pressure. 3. Check oil level in tank. 4. Check filter DP. 5. Assess malfunction. 6. Line walk the system and check for leaks. 7. Ensure PSV's are seated, check spillback/PCV valves are set correctly and not returning oil back to the tank. 8. If maintenance has recently been performed check correct rotation of pump. 9. Ensure no hand valves have been left shut, ensure

Fault Name	Associated Symptoms	Recommendations
		NRV's are working correctly. 10. Check functioning of the auto change over switch.

Fault Name	Associated Symptoms	Recommendations
	Thrust Bearing Axial shaft Displacement Active Side R Low	
	Radial Bearing Vibration X DE R High	
	Radial Bearing Vibration Y DE R High	
	Radial Bearing Vibration X NDE R High	
	Radial Bearing Vibration Y NDE R High	
	Radial Bearing Temperature A DE R High	
	Radial Bearing Temperature B DE R High	
	Radial Bearing Temperature A NDE R high	
	Radial Bearing Temperature B NDE R high	
	Thrust Bearing Temperature A Active Side High	
	Thrust Bearing Temperature B Active Side High	
	Thrust Bearing Temperature C Active Side High	
	Thrust Bearing Temperature D Active High	
	Thrust Bearing Temperature A Inactive Side High	
	Thrust Bearing Temperature B Inactive Side High	
	Thrust Bearing	

Fault Name	Associated Symptoms	Recommendations
	Temperature C Inactive Side High	
	Thrust Bearing Temperature D Inactive High	
	Lube Oil Header Pressure R Low	
Bearing Anomaly due to Lube Oil Quality	Thrust Bearing Axial shaft Displacement Active Side R High	1. Check oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, TAN. 4. Check no breather caps or man covers have been removed and left open to allow debris into the tank. 5. Recondition or replace oil.
	Thrust Bearing Axial shaft Displacement Active Side R Low	
	Radial Bearing Vibration X DE R High	
	Radial Bearing Vibration Y DE R High	
	Radial Bearing Vibration X NDE R High	
	Radial Bearing Vibration Y NDE R High	
	Radial Bearing Temperature A DE R High	
	Radial Bearing Temperature B DE R High	
	Radial Bearing Temperature A NDE R high	
	Radial Bearing Temperature B NDE R high	
	Thrust Bearing Temperature A Active Side High	
	Thrust Bearing Temperature B Active Side High	
	Thrust Bearing Temperature C Active Side High	

Fault Name	Associated Symptoms	Recommendations
	Thrust Bearing Temperature D Active High	
	Thrust Bearing Temperature A Inactive Side High	
	Thrust Bearing Temperature B Inactive Side High	
	Thrust Bearing Temperature C Inactive Side High	
	Thrust Bearing Temperature D Inactive Side High	
	Lube Oil Viscosity High	
	Lube Oil Viscosity Low	
	Lube Oil TAN High	
DE Radial Bearing Failure due to Mechanical Issue	Radial Bearing Vibration X DE R High	1. Check calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if intervention not possible. 4. Intervention options, reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 7. Change bearing if required.
	Radial Bearing Vibration Y DE R High	
	Radial Bearing Temperature A DE R High	
	Radial Bearing Temperature B DE R High	
NDE Radial Bearing Failure due to Mechanical Issue	Radial Bearing Vibration X NDE R High	1. Check calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if intervention not possible. 4. Intervention options; reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 7. Change bearing if required.
	Radial Bearing Vibration Y NDE R High	
	Radial Bearing Temperature A NDE R High	
	Radial Bearing Temperature B NDE R High	
Rotor Anomaly	Radial Bearing Vibration X DE R High	1. Check the calibration of sensor. 2. Investigate with high frequency vibration analysis.

Fault Name	Associated Symptoms	Recommendations
	Radial Bearing Vibration Y DE R High	3. Check Carbon brush integrity. 4. Check magnetism of the rotor. 5. Check rotor unbalance, misalignment. 6. Looseness. 7. Check coupling integrity. 8. Check foundation bolt integrity. 9. Check both bearings for clearance and pad wear etc.
	Radial Bearing Vibration X NDE R High	
	Radial Bearing Vibration Y NDE R High	
Thrust Bearing Failure due to Mechanical Issue	Thrust Bearing Axial Shaft Displacement Active Side R High	1. Check calibration of sensor. 2. Assess likely cause and time to trip if intervention not possible. 3. Intervention options. 4. Reduce thrust load. 5. Ensure parts availability. 6. Check axial vibration trends to establish if the machine has surged and potentially damaged/wiped the pads. 7. If required inspect the thrust pads and collar to check for damage / wiping. 8. Carry out a bump check to establish what float is in the machine, compare to previous overhaul report for measurement. 9. Check the gear coupling/pre-stretch diaphragm coupling.
	Thrust Bearing Axial Shaft Displacement Inactive Side R High	
	Thrust Bearing Axial Shaft Displacement Active Side R Low	
	Thrust Bearing Axial Shaft Displacement Inactive Side R Low	
	Thrust Bearing Temperature A Active Side High	
	Thrust Bearing Temperature B Active Side High	
	Thrust Bearing Temperature C Active Side High	
	Thrust Bearing Temperature D Active Side High	
	Thrust Bearing Temperature A Inactive Side High	
	Thrust Bearing Temperature B Inactive Side High	
	Thrust Bearing Temperature C Inactive Side High	
	Thrust Bearing Temperature D Inactive Side High	

Fault Name	Associated Symptoms	Recommendations
Balance Piston Seal Failure	Thrust Bearing Axial Shaft Displacement Active Side R High	1. Check the calibration of sensor. 2. Monitor rotor axial position, pad temperature. 3. Balance chamber pressure. 4. Determine if failure mode is still progressing, in which case calculate time to thrust bearing displacement trip at current rate. 5. Plan for repair accordingly. 6. If required dismantle the machine and carry out lead checks on the j strips (labyrinth seals) and determine what the clearance is. 7. Compare this to previous overhaul reports. 8. Excessive clearance will overload the thrust bearing beyond its design limit. 9. The rotor will require overhaul/replacing if beyond design clearance. 10. Clean the Balance line tube.
	Thrust Bearing Axial Shaft Displacement Inactive Side R High	
	Thrust Bearing Axial Shaft Displacement Active Side R Low	
	Thrust Bearing Axial Shaft Displacement Inactive Side R Low	
	Balance line DP S High	
Balance Line Blockage Risk	Balance line Blockage S High	1. Check the calibration of sensor. 2. Check the balance line tube for any particle entrainment and clean. 3. Check any surge events, liquid slugs in the sample gas.
	Balance line Blockage S Low	
Inlet Filter Anomaly	Inlet Filter DP RH (IFDA)	1. Check the calibration of sensor. 2. Change over the filter if stand by is available and replace the filter.
Liquid Slugs	Radial Bearing Vibration X DE RH (LS)	1. Short term: reduce flow. 2. Correct problem at process level (reduce scrubber level or act on process), monitor seals leakages, axial shaft position, thrust bearing temperature.
	Radial Bearing Vibration Y DE RH (LS)	
	Radial Bearing Vibration X NDE RH (LS)	
	Radial Bearing Vibration Y NDE RH (LS)	
	Suction Scrubber Level RH (LS)	
Inter stage Cooler Anomaly	Interstage Cooler Gas Outlet Temperature SH (ISCA)	1. Check the interstage cooler's integrity. 2. Check the temp control valve for any anomaly. 3. Conduct loop proof test for temp control valve. 4. Check cooling water supply line for any leakage / blockage/ supply issue.

Fault Name	Associated Symptoms	Recommendations
	Interstage Cooler Temp Control Valve Opening % SH (ISCA)	
	Interstage Cooler Gas Outlet Temperature SL (ISCA)	
	Interstage Cooler Temp Control Valve Opening % SL (ISCA)	
Interstage Separator Anomaly	Interstage Separator Level SH (ISSA)	1. Check Separator / scrubber's internals. 2. Check drain line of the separator. 3. Check the transmitter.

Centrifugal Compressor Seal Gas System Fault Model - Variable Guide Vane V2

Model Type	APR Model
Category	Centrifugal Compressor
Equipment Class	Compressor
Equipment Type	Centrifugal Compressor - Seal Gas System

Below table illustrates the list of faults and symptoms configured for a Centrifugal Compressor Seal Gas System.

Fault Name	Associated Symptoms	Recommendations
Primary Seal Gas Filter Anomaly	Primary Seal Gas Filter DP RH (PSGFA)	1. Check the calibration of sensor. 2. Check the filter efficiency and change to standby the filter if required. 3. Repair/replace the filter internals.
Buffer Gas Filter Anomaly	Buffer Seal Gas Filter DP RH (BGSFA)	1. Check the calibration of sensor. 2. Check the filter efficiency and change to standby the filter if required. 3. Repair/replace the filter internals.
PDCV Malfunction	Reference Line DP RL (PDCVA)	1. Check the calibration of sensor. 2. Check valve set point. 3. Check valve internals for any damage, and repair / replace if necessary. 4. Conduct the PDCV control loop function test.
	PDCV Valve Opening RH (PDCVA)	
Primary Seal Gas Vent PCV Anomaly	Primary Seal Leak Pressure DP SH (PSGVA)	1. Calibrate the valve and transmitters by isolation if possible. 2. Check connection loss if any. 3. Check primary vent line for any jam.

Fault Name	Associated Symptoms	Recommendations
	Primary Seal Leak Pressure DP SL (PSGVA)	4. Check the internals of control valve of the vent line. 5. Conduct loop proof test of the control loop.
	Primary Seal Vent Gas PCV SH (PSGVA)	
	Primary Seal Vent Gas PCV SL (PSGVA)	
OB Seal Laby Malfunction	Primary Vent Gas Pressure RL (OSLM)	1. Ensure upstream and downstream PSV's, check valves, transmitters are all working correctly. 2. Inspect and monitor trend sec seal gas system. 3. Ensure spare seals cartridge is available. 4. Assess risk. 5. Monitor and replace seals at opportunity. 6. Replace seal.
	Primary Vent Gas Flow RL (OSLM)	
	Secondary Vent Gas Pressure RH (OSLM)	
	Secondary Vent Gas Flow RH (OSLM)	
OB Seal Malfunction	Primary Vent Gas Pressure RL (OSM)	1. Ensure upstream and downstream PSV's, check valves, transmitters are all working correctly. 2. Inspect and monitor trend sec seal gas system. 3. Ensure spare seals cartridge is available. 4. Assess risk. 5. Monitor and replace seals at opportunity. 6. Replace seal.
	Primary Vent Gas Flow RL (OSM)	
	Secondary Vent Gas Pressure RH (OSM)	
	Secondary Vent Gas Flow RH (OSM)	
IB Seal Malfunction	Primary Vent Gas Pressure RH (IBM)	1. Ensure upstream and downstream PSV's, check valves, transmitters are all working correctly. 2. Inspect and monitor trend sec seal gas system. 3. Ensure spare seals cartridge is available. 4. Assess risk. 5. Monitor and replace seals at opportunity. 6. Replace seal.
	Primary Vent Gas Flow RH (IBM)	
Primary Seal Supply Gas Temperature Anomaly	Primary Supply Gas Temperature RL (PSGTA)	1. Check calibration of sensor. 2. Monitor seal leakage. 3. Check integrity of seal gas supply PDV. 4. Check super heater functionality if available. 5. Check separation gas. 6. Check seal supply filters for damage and are fitted correctly.
	Primary Supply Gas Temperature RH (PSGTA)	
No Seal Lift off Anomaly	Primary Vent Gas Pressure RH (NSLA)	1. Seal faces: overheating, cracking, scoring. 2. Inspect seal and replace.
	Primary Vent Gas Flow RH (NSLA)	

Fault Name	Associated Symptoms	Recommendations
	Secondary Vent Gas Pressure RH (NSLA)	
	Secondary Vent Gas Flow RH (NSLA)	
	Compressor Speed LOW (NSLA)	
Primary Seal Gas Supply Anomaly	Primary Seal Gas Supply Pressure RL (PSGSA)	1. Check the calibration of sensor. 2. Ensure upstream and downstream PSV's, check valves, transmitters are all working correctly. 3. Monitor trend of seal gas system. 4. Inspect system for leakage. 5. Inspect PCV and check functionality. 6. Replace filters.
	Primary Seal Gas Supply Pressure RH (PSGSA)	
	Primary Seal Gas Supply Flow RL (PSGSA)	
	Primary Seal Gas Supply Flow RH (PSGSA)	
Buffer Gas Supply Anomaly	Buffer Gas Supply Pressure RL (BGSA)	1. Check the calibration of sensor. 2. Check N2 supply system for any leaks or damage. 3. Ensure upstream and downstream PSV's, check valves, transmitters are all working correctly. 4. Inspect PCV and check functionality. 5. Replace Filters.
	Buffer Gas Supply Pressure RH (BGSA)	
	Buffer Gas Supply Flow RL (BGSA)	
	Buffer Gas Supply Flow RH (BGSA)	
	Secondary Supply PCV Opening SH (BGSA)	
	Buffer Gas PCV Position SH (BGSA)	
Separation Gas Supply Anomaly	Separation Supply Gas Pressure RL (SGSA)	1. Check the calibration of sensor. 2. Check N2 supply system. 3. Increased risk of outboard seal contamination. 4. Increased risk of bearing oil ingress into seal. 5. Inspect supply line upstream of seal. 6. Check labyrinth seal for blockages.
	Separation Supply Gas Pressure RH (SGSA)	
	Separation Supply Gas Flow RL (SGSA)	
	Separation Supply Gas Flow RH (SGSA)	
	Separation Gas PCV Opening SH (SGSA)	
	Buffer Gas PCV Position SH (SGSA)	
Seal Gas Booster Compressor Malfunction	Reference Line DP RL (SGBCM)	Check Seal Gas Booster Compressor.
	Primary Seal Gas Supply	

Fault Name	Associated Symptoms	Recommendations
	Pressure RL (SGBCM)	
	Primary Seal Gas Supply Pressure RH (SGBCM)	
	Primary Seal Gas Supply Flow RL (SGBCM)	
	Primary Seal Gas Supply Flow RH (SGBCM)	
Secondary Supply to Primary Vent Anomaly	Secondary Supply to Primary Vent DP RL (SSTPVA)	1. Check calibration of sensor. 2. Check and control N2 supply. 3. Check for any line leaks. 4. Check for seal functionality and replace if necessary.
	Secondary Supply to Primary Vent DP RH (SSTPVA)	

Centrifugal Compressor Gearbox Fault Model - Variable Guide Vane V2

Model Type	APR Model
Category	Centrifugal Compressor
Equipment Class	Compressor
Equipment Type	Centrifugal Compressor - Gearbox System

Below table illustrates the list of faults and symptoms configured for a Centrifugal Compressor Gearbox System.

Fault Name	Associated Symptoms	Recommendations
Gearbox LSS Radial Bearing DE Anomaly	LSS GB Radial Bearing Vibration X DE RH (GLRDA)	1. Check the calibration of sensor. 2. Check if the lube oil header temp, pressure and quality has not altered(if lube oil system is available otherwise check greasing or clog). 3. Investigate with high frequency vibration analysis. 4. Assess likely cause and time to fail(trip) if intervention not possible. 5. Intervention options; reduce speed, Parts availability. 6. Check the trends to establish if its either both X and Y probes or one. 7. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 8. Change bearing if required.
	LSS GB Radial Bearing Vibration Y DE RH (GLRDA)	
	LSS GB Radial Bearing Temperature A DE RH (GLRDA)	
	LSS GB Radial Bearing Temperature B DE RH (GLRDA)	
	LSS GB Gearbox Casing Vibration DE RH (GLRDA)	
Gearbox LSS Radial Bearing NDE Anomaly	LSS GB Radial Bearing Vibration X NDE RH (GLRNDA)	1. Check the calibration of sensor. 2. Check if the lube oil header temp, pressure and quality has not altered(if lube oil system is available otherwise check greasing or clog). 3. Investigate with high frequency vibration analysis. 4. Assess likely cause and time to fail(trip) if intervention not

Fault Name	Associated Symptoms	Recommendations
	LSS GB Radial Bearing Vibration Y NDE RH (GLRNDA)	possible. - Intervention options; reduce speed, Parts availability. - Check the trends to establish if its either both X and Y probes or one. - Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. - Change bearing if required.
	LSS GB Radial Bearing Temperature A NDE RH (GLRNDA)	
	LSS GB Radial Bearing Temperature B NDE RH (GLRNDA)	
	LSS GB Gearbox Casing Vibration NDE RH (GLRNDA)	
Gearbox HSS Radial Bearing DE Anomaly	HSS GB Radial Bearing Vibration X DE RH (GHRDA)	- Check the calibration of sensor. - Investigate with high frequency vibration analysis. - Assess likely cause and time to fail(trip) if intervention not possible. - Intervention options; reduce speed, Parts availability. - Check the trends to establish if its either both X and Y probes or one. - Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. - Change bearing if required.
	HSS GB Radial Bearing Vibration Y DE RH (GHRDA)	
	HSS-DE Radial bearing temperature RH (GHRDA)	
	HSS-DE Radial bearing temperature RH (GHRDA)	
	HSS GB Gearbox Casing Vibration DE RH (GHRDA)	
Gearbox HSS Radial Bearing NDE Anomaly	HSS GB Radial Bearing Vibration X NDE RH (GHRNDA)	- Check the calibration of sensor. - Investigate with high frequency vibration analysis. - Assess likely cause and time to fail(trip) if intervention not possible. - Intervention options; reduce speed, Parts availability. - Check the trends to establish if its either both X and Y probes or one. - Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. - Change bearing if required.
	HSS GB Radial Bearing Vibration Y NDE RH (GHRNDA)	
	HSS-NDE Radial bearing temperature RH (GHRNDA)	
	HSS-NDE Radial bearing temperature RH (GHRNDA)	
	HSS GB Gearbox Casing Vibration NDE RH (GHRNDA)	
Gearbox LSS Shaft Anomaly	LSS GB Radial Bearing Vibration X DE RH (GLSA)	- Check the calibration of sensor. - Investigate with high frequency vibration analysis. - Check Carbon brush integrity. - Check magnetism of the rotor. - Check rotor unbalance, misalignment. - Looseness. - Check coupling integrity. - Check foundation bolt integrity.

Fault Name	Associated Symptoms	Recommendations
	LSS GB Radial Bearing Vibration Y DE RH (GLSA)	9. Check both bearings for clearance and pad wear etc. 10. Check gear integrity.
	LSS GB Radial Bearing Vibration X NDE RH (GLSA)	
	LSS GB Radial Bearing Vibration Y NDE RH (GLSA)	
	LSS GB Gearbox Casing Vibration DE RH (GLSA)	
	LSS GB Gearbox Casing Vibration NDE RH (GLSA)	
Gearbox HSS Shaft Anomaly	HSS GB Radial Bearing Vibration X DE RH (GHSA)	1. Check calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Check Carbon brush integrity. 4. Check magnetism of the rotor Check rotor unbalance, misalignment. 5. Looseness. 6. Check coupling integrity. 7. Check foundation bolt integrity. 8. Check both bearings for clearance and pad wear etc. 9. Check gear integrity.
	HSS GB Radial Bearing Vibration Y DE RH (GHSA)	
	HSS GB Radial Bearing Vibration X NDE RH (GHSA)	
	HSS GB Radial Bearing Vibration Y NDE RH (GHSA)	
	HSS GB Gearbox Casing Vibration DE RH (GHSA)	
	HSS GB Gearbox Casing Vibration NDE RH (GHSA)	
Gearbox LSS Thrust Bearing Anomaly	LSS GB Thrust Bearing Displacement Active Side RH (GLTBA)	1. Check if the lube oil header temp, pressure and quality has not altered(if lube oil system is available). 2. Assess likely cause and time to trip if intervention not possible. 3. Check calibration of sensor. 4. Intervention options; reduce thrust load, Ensure parts availability. 5. Check axial vibration trends to establish if the driver or driven has change in axial thrust and potentially damaged/wiped the pads. 6. If required inspect the thrust pads and collar to check for damage / wiping. 7. Carry out a bump check to establish what float is in the machine, compare to previous overhaul report for measurement. 8. Check the gear coupling/pre-stretch diaphragm coupling.
	LSS GB Thrust Bearing Displacement Active Side RL (GLTBA)	
	LSS GB Thrust Bearing Displacement Inactive Side RH (GLTBA)	
	LSS GB Thrust Bearing Displacement Inactive Side RL (GLTBA)	
	LSS GB Thrust Bearing Temperature A Active Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature B Active Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature C Active Side RH	

Fault Name	Associated Symptoms	Recommendations
	(GLTBA)	
	LSS GB Thrust Bearing Temperature A Inactive Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature B Inactive Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature C Inactive Side RH (GLTBA)	
Gearbox HSS Thrust Bearing Anomaly	HSS GB Thrust Bearing Displacement Active Side RH (GHTBA)	1. Check if the lube oil header temp, pressure and quality has not altered(if lube oil system is available). 2. Assess likely cause and time to trip if intervention not possible. 3. Check calibration of sensor. 4. Intervention options; reduce thrust load, Ensure parts availability. 5. Check axial vibration trends to establish if the driver or driven has change in axial thrust and potentially damaged/wiped the pads. 6. If required inspect the thrust pads and collar to check for damage / wiping. 7. Carry out a bump check to establish what float is in the machine, compare to previous overhaul report for measurement. 8. Check the gear coupling/pre-stretch diaphragm coupling.
	HSS GB Thrust Bearing Displacement Active Side RL (GHTBA)	
	HSS GB Thrust Bearing Displacement Inactive Side RH (GHTBA)	
	HSS GB Thrust Bearing Displacement Inactive Side RL (GHTBA)	
	HSS GB Thrust Bearing Temperature A Active Side RH (GHTBA)	
	HSS GB Thrust Bearing Temperature B Active Side RH (GHTBA)	
	HSS GB Thrust Bearing Temperature C Active Side RH (GHTBA)	
	HSS GB Thrust Bearing Temperature A Inactive Side RH (GHTBA)	
	HSS GB Thrust Bearing Temperature B Inactive Side RH (GHTBA)	
	HSS GB Thrust Bearing Temperature C Inactive Side RH (GHTBA)	

Centrifugal Compressor Lube Oil Fault Model - Variable Guide Vane V2

Model Type	APR Model
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Category	Centrifugal Compressor
Equipment Class	Compressor
Equipment Type	Centrifugal Compressor - Lube Oil System

Below table illustrates the list of faults and symptoms configured for a Centrifugal Compressor Lube Oil System.

Fault Name	Associated Symptoms	Recommendations
Loss of Lube Oil Quality	Lube Oil Viscosity HIGH (LOLQ)	1. Check oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, TAN. 4. Check no breather caps or man covers have been removed and left open to allow debris into the tank. 5. Do filter cleaning. 6. Recondition or replace oil.
	Lube Oil Viscosity LOW (LOLQ)	
	Lube Oil TAN HIGH (LOLQ)	
	Lube Oil Composition Analysis HIGH (LOLQ)	
Lube Oil Filter Anomaly	Lube Oil Filter DP RH (LFDF)	1. Check filter DP, changeover and clean/replace the stand by filter. 2. Check the Filter DP transmitter.
Loss of Pressure due to Pump Anomaly	Lube Oil Header Pressure RL (LOPPF)	1. Check the Discharge PT integrity. 2. Switch on both oil pumps and stabilize oil pressure. 3. Assess malfunction. 4. Line walk the system and check for leaks, ensure PSV's are seated. 5. Check spillback/PCV valves are set correctly and not returning oil back to the tank. 6. Ensure no hand valves have been left shut for the standby pump. 7. Check oil level in tank, Open tank overhead vents for release of pressure. 8. Check PCV function for Integrity. 9. Assess malfunction. 10. Check Auto Logic Function tests, check functioning of the auto change over switch. 11. If maintenance has recently been performed check correct rotation of pump. 12. Ensure NRV's are working correctly.
	Lube Oil Pump Discharge Pressure RL (LOPPF)	
	Lube Oil PCV Position RL (LOPPF)	
	Lube Oil Return Flow Rate RL (LOPPF)	
	Lube Oil Main Pump Current (Run Status) ON (LOPPF)	
	Lube Oil Aux Pump Current (Run Status) ON (LOPPF)	
Loss of Cooling due to TCV Anomaly	Lube Oil Header Temperature (Cooler Out Temp) RH (LOCTF)	1. Check the calibration of sensor. 2. Check bypass control valve for integrity. 3. Open both coolers and stabilize temperature.

Fault Name	Associated Symptoms	Recommendations
	LO Cooler Bypass Valve (TCV) Position RH (LOCTF)	4. Line walk the system to check for leaks. 5. Check the Isolation/bypass valve of TCV.
Loss of Cooling due to Cooler Anomaly	Lube Oil Header Temperature (Cooler Out Temp) RH (LOCCF)	1. Check the calibration of sensor. 2. Check cooling water supply pressure. 3. Open both coolers and stabilize temperature. 4. If malfunction of one cooler is suspected, isolate and inspect it.
	LO Tank And Cooler Outlet Temperature Delta RH (LOCCF)	5. Line walk the system to check for leaks. 6. Check cooling water pumps for blockages, ensure strainer is clean.
Loss of Heating Due to Lube Oil Tank Heater Off	Lube Oil Header Temperature (Cooler Out Temp) RL (LOCHOFF)	1. Check the Heater ON/OFF switch integrity and Change the switch as per the requirement.
	Lube Oil Tank Temperature (Cooler Inlet Temp) RL (LOCHOFF)	2. Check the heater circuit for Voltage and current.
	Lube Oil Tank Heater (Run Status) OFF (LOCHOFF)	
Loss of Cooling due Lube Oil Tank Heater On	Lube Oil Header Temperature (Cooler Out Temp) RH (LOCHON)	1. Check the Heater ON/OFF switch integrity and Change the switch as per the requirement.
	Lube Oil Tank Temperature (Cooler Inlet Temp) RH (LOCHON)	2. Offline 1.
	Lube Oil Tank Heater (Run Status) ON (LOCHON)	3. Check the heater circuit for Voltage and current.
Loss of Cooling due to Fan Motor Issue	Lube Oil Header Temperature (Cooler Out Temp) RH (LOCFMI)	Offline: Check the Fan and Motor /coupling / belt Integrity and rectify.
	Lube Oil Cooler Fan 1 (Run Status) OFF (LOCFMI)	
	Lube Oil Cooler Fan 2 (Run Status) OFF (LOCFMI)	

10.3.2.3 Centrifugal Compressor - Variable speed V2

Centrifugal Compressor Unit Performance and Health Fault Model - Variable Speed V2

Model Type	APR Model
Category	Centrifugal Compressor
Equipment Class	Compressor
Equipment Type	Centrifugal Compressor - Unit

Below table illustrates the list of faults and symptoms configured for a Compressor Unit.

Fault Name	Associated Symptoms	Recommendations
Bearing Anomaly due	Thrust Bearing Axial shaft	1. Check the calibration of sensor.

Fault Name	Associated Symptoms	Recommendations
to Lube Oil Temperature	Displacement Active Side R High	2. Check cooling water supply pressure. 3. Check bypass control valve. 4. Open both coolers and stabilize temperature. 5. Check cooling water pumps for blockages, ensure strainer is clean. 6. If the cooler is an air cooler check fan and motor integrity. 7. If malfunction of one cooler is suspected, isolate and inspect it. 8. Line walk the system to check for leaks.
	Thrust Bearing Axial shaft Displacement Active Side R Low	
	Radial Bearing Vibration X DE R High	
	Radial Bearing Vibration Y DE R High	
	Radial Bearing Vibration X NDE R High	
	Radial Bearing Vibration Y NDE R High	
	Radial Bearing Temperature A DE R High	
	Radial Bearing Temperature B DE R High	
	Radial Bearing Temperature A NDE R high	
	Radial Bearing Temperature B NDE R high	
	Thrust Bearing Temperature A Active Side High	
	Thrust Bearing Temperature B Active Side High	
	Thrust Bearing Temperature C Active Side High	
	Thrust Bearing Temperature D Active Side High	
	Thrust Bearing Temperature A Inactive Side High	
	Thrust Bearing Temperature B Inactive Side High	
	Thrust Bearing Temperature C Inactive	

Fault Name	Associated Symptoms	Recommendations
	Side High	
	Thrust Bearing Temperature D Inactive Side High	
	Lube Oil Header Temperature R High	
Bearing Anomaly due to Lube Oil Pressure	Thrust Bearing Axial shaft Displacement Active Side R High	1. Check the calibration of sensor. 2. Switch on both oil pumps and stabilize oil pressure. 3. Check oil level in tank. 4. Check filter DP. 5. Assess malfunction. 6. Line walk the system and check for leaks. 7. Ensure PSV's are seated, check spillback/PCV valves are set correctly and not returning oil back to the tank. 8. If maintenance has recently been performed check correct rotation of pump. 9. Ensure no hand valves have been left shut, ensure NRV's are working correctly. 10. Check functioning of the auto change over switch.
	Thrust Bearing Axial shaft Displacement Active Side R Low	
	Radial Bearing Vibration X DE R High	
	Radial Bearing Vibration Y DE R High	
	Radial Bearing Vibration X NDE R High	
	Radial Bearing Vibration Y NDE R High	
	Radial Bearing Temperature A DE R High	
	Radial Bearing Temperature B DE R High	
	Radial Bearing Temperature A NDE R high	
	Radial Bearing Temperature B NDE R high	
	Thrust Bearing Temperature A Active Side High	
	Thrust Bearing Temperature B Active Side High	
	Thrust Bearing Temperature C Active Side High	
	Thrust Bearing Temperature D Active High	
	Thrust Bearing	

Fault Name	Associated Symptoms	Recommendations
	Temperature A Inactive Side High	
	Thrust Bearing Temperature B Inactive Side High	
	Thrust Bearing Temperature C Inactive Side High	
	Thrust Bearing Temperature D Inactive High	
	Lube Oil Header Pressure R Low	
Bearing Anomaly due to Lube Oil Quality	Thrust Bearing Axial shaft Displacement Active Side R High	1. Check oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, TAN. 4. Check no breather caps or man covers have been removed and left open to allow debris into the tank. 5. Recondition or replace oil.
	Thrust Bearing Axial shaft Displacement Active Side R Low	
	Radial Bearing Vibration X DE R High	
	Radial Bearing Vibration Y DE R High	
	Radial Bearing Vibration X NDE R High	
	Radial Bearing Vibration Y NDE R High	
	Radial Bearing Temperature A DE R High	
	Radial Bearing Temperature B DE R High	
	Radial Bearing Temperature A NDE R high	
	Radial Bearing Temperature B NDE R high	
	Thrust Bearing Temperature A Active Side High	
	Thrust Bearing Temperature B Active Side High	

Fault Name	Associated Symptoms	Recommendations
	Thrust Bearing Temperature C Active Side High	
	Thrust Bearing Temperature D Active High	
	Thrust Bearing Temperature A Inactive Side High	
	Thrust Bearing Temperature B Inactive Side High	
	Thrust Bearing Temperature C Inactive Side High	
	Thrust Bearing Temperature D Inactive Side High	
	Lube Oil Viscosity High	
	Lube Oil Viscosity Low	
	Lube Oil TAN High	
DE Radial Bearing Failure due to Mechanical Issue	Radial Bearing Vibration X DE R High	1. Check calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if intervention not possible. 4. Intervention options, reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 7. Change bearing if required.
	Radial Bearing Vibration Y DE R High	
	Radial Bearing Temperature A DE R High	
	Radial Bearing Temperature B DE R High	
NDE Radial Bearing Failure due to Mechanical Issue	Radial Bearing Vibration X NDE R High	1. Check calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if intervention not possible. 4. Intervention options; reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 7. Change bearing if required.
	Radial Bearing Vibration Y NDE R High	
	Radial Bearing Temperature A NDE R High	
	Radial Bearing Temperature B NDE R High	
Rotor Anomaly	Radial Bearing Vibration X	1. Check the calibration of sensor.

Fault Name	Associated Symptoms	Recommendations
	DE R High	- Investigate with high frequency vibration analysis. - Check Carbon brush integrity. - Check magnetism of the rotor. - Check rotor unbalance, misalignment. - Looseness. - Check coupling integrity. - Check foundation bolt integrity. - Check both bearings for clearance and pad wear etc.
	Radial Bearing Vibration Y DE R High	
	Radial Bearing Vibration X NDE R High	
	Radial Bearing Vibration Y NDE R High	
Thrust Bearing Failure due to Mechanical Issue	Thrust Bearing Axial Shaft Displacement Active Side R High	- Check calibration of sensor. - Assess likely cause and time to trip if intervention not possible. - Intervention options. - Reduce thrust load. - Ensure parts availability. - Check axial vibration trends to establish if the machine has surged and potentially damaged/wiped the pads. - If required inspect the thrust pads and collar to check for damage / wiping. - Carry out a bump check to establish what float is in the machine, compare to previous overhaul report for measurement. - Check the gear coupling/pre-stretch diaphragm coupling.
	Thrust Bearing Axial Shaft Displacement Inactive Side R High	
	Thrust Bearing Axial Shaft Displacement Active Side R Low	
	Thrust Bearing Axial Shaft Displacement Inactive Side R Low	
	Thrust Bearing Temperature A Active Side High	
	Thrust Bearing Temperature B Active Side High	
	Thrust Bearing Temperature C Active Side High	
	Thrust Bearing Temperature D Active Side High	
	Thrust Bearing Temperature A Inactive Side High	
	Thrust Bearing Temperature B Inactive Side High	
	Thrust Bearing Temperature C Inactive Side High	
	Thrust Bearing Temperature D Inactive	

Fault Name	Associated Symptoms	Recommendations
	Side High	
Balance Piston Seal Failure	Thrust Bearing Axial Shaft Displacement Active Side R High	1. Check the calibration of sensor. 2. Monitor rotor axial position, pad temperature. 3. Balance chamber pressure.
	Thrust Bearing Axial Shaft Displacement Inactive Side R High	4. Determine if failure mode is still progressing, in which case calculate time to thrust bearing displacement trip at current rate. 5. Plan for repair accordingly.
	Thrust Bearing Axial Shaft Displacement Active Side R Low	6. If required dismantle the machine and carry out lead checks on the j strips (labyrinth seals) and determine what the clearance is. 7. Compare this to previous overhaul reports.
	Thrust Bearing Axial Shaft Displacement Inactive Side R Low	8. Excessive clearance will overload the thrust bearing beyond its design limit. 9. The rotor will require overhaul/replacing if beyond design clearance.
	Balance line DP S High	10. Clean the Balance line tube.
Balance Line Blockage Risk	Balance line Blockage S High	1. Check the calibration of sensor. 2. Check the balance line tube for any particle entrainment and clean.
	Balance line Blockage S Low	3. Check any surge events, liquid slugs in the sample gas.
Inlet Filter Anomaly	Inlet Filter DP RH (IFDA)	1. Check the calibration of sensor. 2. Change over the filter if stand by is available and replace the filter.
Liquid Slugs	Radial Bearing Vibration X DE RH (LS)	1. Short term: reduce flow. 2. Correct problem at process level (reduce scrubber level or act on process), monitor seals leakages, axial shaft position, thrust bearing temperature.
	Radial Bearing Vibration Y DE RH (LS)	
	Radial Bearing Vibration X NDE RH (LS)	
	Radial Bearing Vibration Y NDE RH (LS)	
	Suction Scrubber Level RH (LS)	
Inter stage Cooler Anomaly	Interstage Cooler Gas Outlet Temperature SH (ISCA)	1. Check the interstage cooler's integrity. 2. Check the temp control valve for any anomaly. 3. Conduct loop proof test for temp control valve. 4. Check cooling water supply line for any leakage / blockage/ supply issue.

Fault Name	Associated Symptoms	Recommendations
	Interstage Cooler Temp Control Valve Opening % SH (ISCA)	
	Interstage Cooler Gas Outlet Temperature SL (ISCA)	
	Interstage Cooler Temp Control Valve Opening % SL (ISCA)	
Interstage Separator Anomaly	Interstage Separator Level SH (ISSA)	1. Check Separator / scrubber's internals. 2. Check drain line of the separator. 3. Check the transmitter.

Centrifugal Compressor Seal Gas System Fault Model - Variable Speed V2

Model Type	APR Model
Category	Centrifugal Compressor
Equipment Class	Compressor
Equipment Type	Centrifugal Compressor - Seal Gas System

Below table illustrates the list of faults and symptoms configured for a Centrifugal Compressor Seal Gas System.

Fault Name	Associated Symptoms	Recommendations
Primary Seal Gas Filter Anomaly	Primary Seal Gas Filter DP RH (PSGFA)	1. Check the calibration of sensor. 2. Check the filter efficiency and change to standby the filter if required. 3. Repair/replace the filter internals.
Buffer Gas Filter Anomaly	Buffer Seal Gas Filter DP RH (BGSFA)	1. Check the calibration of sensor. 2. Check the filter efficiency and change to standby the filter if required. 3. Repair/replace the filter internals.
PDCV Malfunction	Reference Line DP RL (PDCVA)	1. Check the calibration of sensor. 2. Check valve set point. 3. Check valve internals for any damage, and repair / replace if necessary. 4. Conduct the PDCV control loop function test .
	PDCV Valve Opening RH (PDCVA)	
Primary Seal Gas Vent PCV Anomaly	Primary Seal Leak Pressure DP SH (PSGVA)	1. Calibrate the valve and transmitters by isolation if possible. 2. Check connection loss if any. 3. Check primary vent line for any jam.

Fault Name	Associated Symptoms	Recommendations
	Primary Seal Leak Pressure DP SL (PSGVA)	4. Check the internals of control valve of the vent line. 5. Conduct loop proof test of the control loop.
	Primary Seal Vent Gas PCV SH (PSGVA)	
	Primary Seal Vent Gas PCV SL (PSGVA)	
OB Seal Laby Malfunction	Primary Vent Gas Pressure RL (OSLM)	1. Ensure upstream and downstream PSV's, check valves, transmitters are all working correctly. 2. Inspect and monitor trend sec seal gas system. 3. Ensure spare seals cartridge is available. 4. Assess risk. 5. Monitor and replace seals at opportunity. 6. Replace seal.
	Primary Vent Gas Flow RL (OSLM)	
	Secondary Vent Gas Pressure RH (OSLM)	
	Secondary Vent Gas Flow RH (OSLM)	
OB Seal Malfunction	Primary Vent Gas Pressure RL (OSM)	1. Ensure upstream and downstream PSV's, check valves, transmitters are all working correctly. 2. Inspect and monitor trend sec seal gas system. 3. Ensure spare seals cartridge is available. 4. Assess risk. 5. Monitor and replace seals at opportunity. 6. Replace seal.
	Primary Vent Gas Flow RL (OSM)	
	Secondary Vent Gas Pressure RH (OSM)	
	Secondary Vent Gas Flow RH (OSM)	
IB Seal Malfunction	Primary Vent Gas Pressure RH (IBM)	1. Ensure upstream and downstream PSV's, check valves, transmitters are all working correctly. 2. Inspect and monitor trend sec seal gas system. 3. Ensure spare seals cartridge is available. 4. Assess risk. 5. Monitor and replace seals at opportunity. 6. Replace seal.
	Primary Vent Gas Flow RH (IBM)	
Primary Seal Supply Gas Temperature Anomaly	Primary Supply Gas Temperature RL (PSGTA)	1. Check calibration of sensor. 2. Monitor seal leakage. 3. Check integrity of seal gas supply PDV. 4. Check super heater functionality if available. 5. Check separation gas. 6. Check seal supply filters for damage and are fitted correctly.
	Primary Supply Gas Temperature RH (PSGTA)	
No Seal Lift off Anomaly	Primary Vent Gas Pressure RH (NSLA)	1. Seal faces: overheating, cracking, scoring. 2. Inspect seal and replace.
	Primary Vent Gas Flow RH (NSLA)	

Fault Name	Associated Symptoms	Recommendations
	Secondary Vent Gas Pressure RH (NSLA)	
	Secondary Vent Gas Flow RH (NSLA)	
	Compressor Speed LOW (NSLA)	
Primary Seal Gas Supply Anomaly	Primary Seal Gas Supply Pressure RL (PSGSA)	1. Check the calibration of sensor. 2. Ensure upstream and downstream PSV's, check valves, transmitters are all working correctly. 3. Monitor trend of seal gas system. 4. Inspect system for leakage. 5. Inspect PCV and check functionality. 6. Replace filters.
	Primary Seal Gas Supply Pressure RH (PSGSA)	
	Primary Seal Gas Supply Flow RL (PSGSA)	
	Primary Seal Gas Supply Flow RH (PSGSA)	
Buffer Gas Supply Anomaly	Buffer Gas Supply Pressure RL (BGSA)	1. Check the calibration of sensor. 2. Check N2 supply system for any leaks or damage. 3. Ensure upstream and downstream PSV's, check valves, transmitters are all working correctly. 4. Inspect PCV and check functionality. 5. Replace Filters.
	Buffer Gas Supply Pressure RH (BGSA)	
	Buffer Gas Supply Flow RL (BGSA)	
	Buffer Gas Supply Flow RH (BGSA)	
	Secondary Supply PCV Opening SH (BGSA)	
	Buffer Gas PCV Position SH (BGSA)	
Separation Gas Supply Anomaly	Separation Supply Gas Pressure RL (SGSA)	1. Check the calibration of sensor. 2. Check N2 supply system. 3. Increased risk of outboard seal contamination. 4. Increased risk of bearing oil ingress into seal. 5. Inspect supply line upstream of seal. 6. Check labyrinth seal for blockages.
	Separation Supply Gas Pressure RH (SGSA)	
	Separation Supply Gas Flow RL (SGSA)	
	Separation Supply Gas Flow RH (SGSA)	
	Separation Gas PCV Opening SH (SGSA)	
	Buffer Gas PCV Position SH (SGSA)	
Seal Gas Booster Compressor Malfunction	Reference Line DP RL (SGBCM)	Check Seal Gas Booster Compressor.
	Primary Seal Gas Supply	

Fault Name	Associated Symptoms	Recommendations
	Pressure RL (SGBCM)	
	Primary Seal Gas Supply Pressure RH (SGBCM)	
	Primary Seal Gas Supply Flow RL (SGBCM)	
	Primary Seal Gas Supply Flow RH (SGBCM)	
Secondary Supply to Primary Vent Anomaly	Secondary Supply to Primary Vent DP RL (SSTPVA)	1. Check calibration of sensor. 2. Check and control N2 supply. 3. Check for any line leaks. 4. Check for seal functionality and replace if necessary.
	Secondary Supply to Primary Vent DP RH (SSTPVA)	

Centrifugal Compressor Gearbox Fault Model - Variable Speed V2

Model Type	APR Model
Category	Centrifugal Compressor
Equipment Class	Compressor
Equipment Type	Centrifugal Compressor - Gearbox System

Below table illustrates the list of faults and symptoms configured for a Centrifugal Compressor Gearbox System.

Fault Name	Associated Symptoms	Recommendations
Gearbox LSS Radial Bearing DE Anomaly	LSS GB Radial Bearing Vibration X DE RH (GLRDA)	1. Check the calibration of sensor. 2. Check if the lube oil header temp, pressure and quality has not altered(if lube oil system is available otherwise check greasing or clog). 3. Investigate with high frequency vibration analysis. 4. Assess likely cause and time to fail(trip) if intervention not possible. 5. Intervention options; reduce speed, Parts availability. 6. Check the trends to establish if its either both X and Y probes or one. 7. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 8. Change bearing if required.
	LSS GB Radial Bearing Vibration Y DE RH (GLRDA)	
	LSS GB Radial Bearing Temperature A DE RH (GLRDA)	
	LSS GB Radial Bearing Temperature B DE RH (GLRDA)	
	LSS GB Gearbox Casing Vibration DE RH (GLRDA)	
Gearbox LSS Radial Bearing NDE Anomaly	LSS GB Radial Bearing Vibration X NDE RH (GLRNDA)	1. Check the calibration of sensor. 2. Check if the lube oil header temp, pressure and quality has not altered(if lube oil system is available otherwise check greasing or clog). 3. Investigate with high frequency vibration analysis. 4. Assess likely cause and time to fail(trip) if intervention not

Fault Name	Associated Symptoms	Recommendations
	LSS GB Radial Bearing Vibration Y NDE RH (GLRNDA)	possible. - Intervention options; reduce speed, Parts availability. - Check the trends to establish if its either both X and Y probes or one. - Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. - Change bearing if required.
	LSS GB Radial Bearing Temperature A NDE RH (GLRNDA)	
	LSS GB Radial Bearing Temperature B NDE RH (GLRNDA)	
	LSS GB Gearbox Casing Vibration NDE RH (GLRNDA)	
Gearbox HSS Radial Bearing DE Anomaly	HSS GB Radial Bearing Vibration X DE RH (GHRDA)	- Check the calibration of sensor. - Investigate with high frequency vibration analysis. - Assess likely cause and time to fail(trip) if intervention not possible. - Intervention options; reduce speed, Parts availability. - Check the trends to establish if its either both X and Y probes or one. - Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. - Change bearing if required.
	HSS GB Radial Bearing Vibration Y DE RH (GHRDA)	
	HSS-DE Radial bearing temperature RH (GHRDA)	
	HSS-DE Radial bearing temperature RH (GHRDA)	
	HSS GB Gearbox Casing Vibration DE RH (GHRDA)	
Gearbox HSS Radial Bearing NDE Anomaly	HSS GB Radial Bearing Vibration X NDE RH (GHRNDA)	- Check the calibration of sensor. - Investigate with high frequency vibration analysis. - Assess likely cause and time to fail(trip) if intervention not possible. - Intervention options; reduce speed, Parts availability. - Check the trends to establish if its either both X and Y probes or one. - Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. - Change bearing if required.
	HSS GB Radial Bearing Vibration Y NDE RH (GHRNDA)	
	HSS-NDE Radial bearing temperature RH (GHRNDA)	
	HSS-NDE Radial bearing temperature RH (GHRNDA)	
	HSS GB Gearbox Casing Vibration NDE RH (GHRNDA)	
Gearbox LSS Shaft Anomaly	LSS GB Radial Bearing Vibration X DE RH (GLSA)	- Check the calibration of sensor. - Investigate with high frequency vibration analysis. - Check Carbon brush integrity. - Check magnetism of the rotor. - Check rotor unbalance, misalignment. - Looseness. - Check coupling integrity. - Check foundation bolt integrity.

Fault Name	Associated Symptoms	Recommendations
	LSS GB Radial Bearing Vibration Y DE RH (GLSA)	9. Check both bearings for clearance and pad wear etc. 10. Check gear integrity.
	LSS GB Radial Bearing Vibration X NDE RH (GLSA)	
	LSS GB Radial Bearing Vibration Y NDE RH (GLSA)	
	LSS GB Gearbox Casing Vibration DE RH (GLSA)	
	LSS GB Gearbox Casing Vibration NDE RH (GLSA)	
Gearbox HSS Shaft Anomaly	HSS GB Radial Bearing Vibration X DE RH (GHSA)	1. Check calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Check Carbon brush integrity. 4. Check magnetism of the rotor Check rotor unbalance, misalignment. 5. Looseness. 6. Check coupling integrity. 7. Check foundation bolt integrity. 8. Check both bearings for clearance and pad wear etc. 9. Check gear integrity.
	HSS GB Radial Bearing Vibration Y DE RH (GHSA)	
	HSS GB Radial Bearing Vibration X NDE RH (GHSA)	
	HSS GB Radial Bearing Vibration Y NDE RH (GHSA)	
	HSS GB Gearbox Casing Vibration DE RH (GHSA)	
	HSS GB Gearbox Casing Vibration NDE RH (GHSA)	
Gearbox LSS Thrust Bearing Anomaly	LSS GB Thrust Bearing Displacement Active Side RH (GLTBA)	1. Check if the lube oil header temp, pressure and quality has not altered(if lube oil system is available). 2. Assess likely cause and time to trip if intervention not possible. 3. Check calibration of sensor. 4. Intervention options; reduce thrust load, Ensure parts availability. 5. Check axial vibration trends to establish if the driver or driven has change in axial thrust and potentially damaged/wiped the pads. 6. If required inspect the thrust pads and collar to check for damage / wiping. 7. Carry out a bump check to establish what float is in the machine, compare to previous overhaul report for measurement. 8. Check the gear coupling/pre-stretch diaphragm coupling.
	LSS GB Thrust Bearing Displacement Active Side RL (GLTBA)	
	LSS GB Thrust Bearing Displacement Inactive Side RH (GLTBA)	
	LSS GB Thrust Bearing Displacement Inactive Side RL (GLTBA)	
	LSS GB Thrust Bearing Temperature A Active Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature B Active Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature C Active Side RH	

Fault Name	Associated Symptoms	Recommendations
	(GLTBA)	
	LSS GB Thrust Bearing Temperature A Inactive Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature B Inactive Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature C Inactive Side RH (GLTBA)	
Gearbox HSS Thrust Bearing Anomaly	HSS GB Thrust Bearing Displacement Active Side RH (GHTBA)	1. Check if the lube oil header temp, pressure and quality has not altered(if lube oil system is available). 2. Assess likely cause and time to trip if intervention not possible. 3. Check calibration of sensor. 4. Intervention options; reduce thrust load, Ensure parts availability. 5. Check axial vibration trends to establish if the driver or driven has change in axial thrust and potentially damaged/wiped the pads. 6. If required inspect the thrust pads and collar to check for damage / wiping. 7. Carry out a bump check to establish what float is in the machine, compare to previous overhaul report for measurement. 8. Check the gear coupling/pre-stretch diaphragm coupling.
	HSS GB Thrust Bearing Displacement Active Side RL (GHTBA)	
	HSS GB Thrust Bearing Displacement Inactive Side RH (GHTBA)	
	HSS GB Thrust Bearing Displacement Inactive Side RL (GHTBA)	
	HSS GB Thrust Bearing Temperature A Active Side RH (GHTBA)	
	HSS GB Thrust Bearing Temperature B Active Side RH (GHTBA)	
	HSS GB Thrust Bearing Temperature C Active Side RH (GHTBA)	
	HSS GB Thrust Bearing Temperature A Inactive Side RH (GHTBA)	
	HSS GB Thrust Bearing Temperature B Inactive Side RH (GHTBA)	
	HSS GB Thrust Bearing Temperature C Inactive Side RH (GHTBA)	

Centrifugal Compressor Lube Oil Fault Model - Variable Speed V2

Model Type	APR Model
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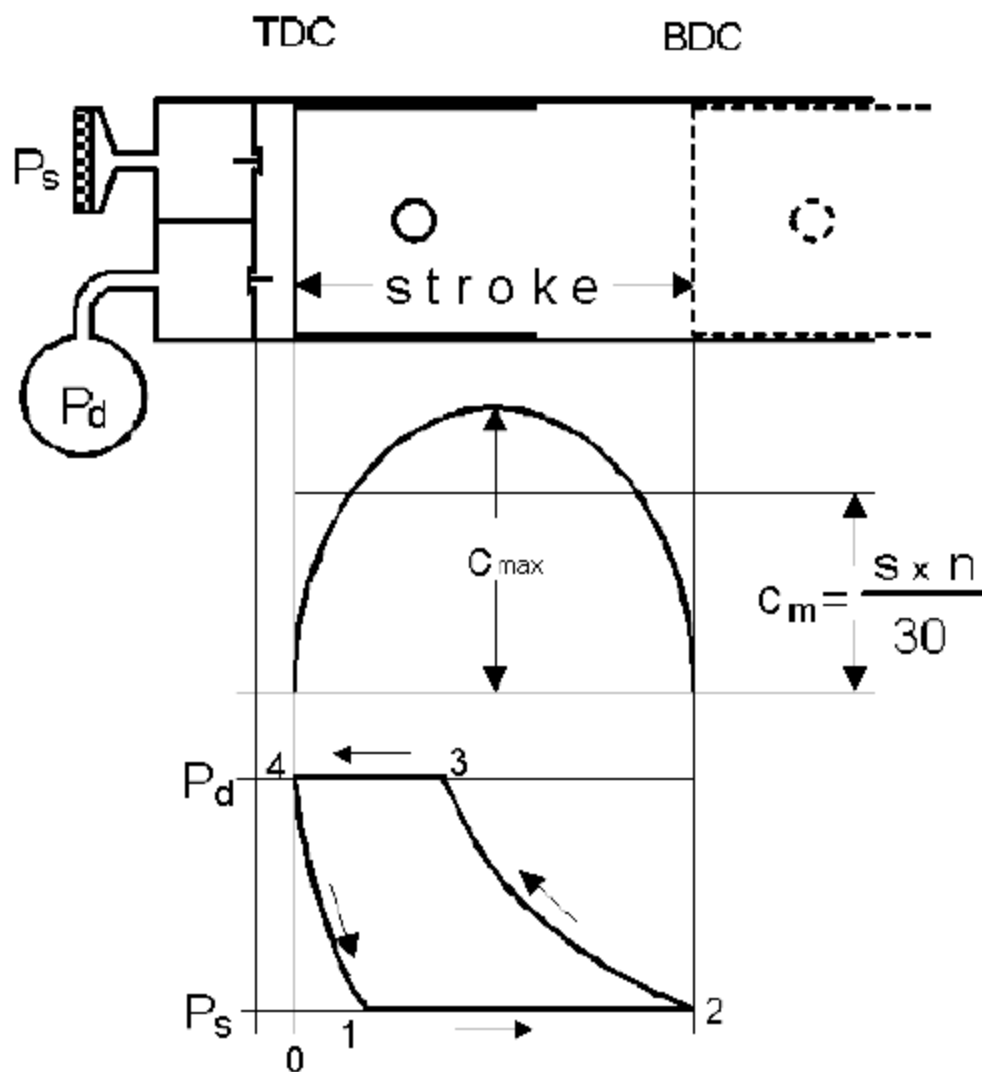
Category	Centrifugal Compressor
Equipment Class	Compressor
Equipment Type	Centrifugal Compressor - Lube Oil System

Below table illustrates the list of faults and symptoms configured for a Centrifugal Compressor Lube Oil System.

Fault Name	Associated Symptoms	Recommendations
Loss of Lube Oil Quality	Lube Oil Viscosity HIGH (LOLQ)	1. Check oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, TAN. 4. Check no breather caps or man covers have been removed and left open to allow debris into the tank. 5. Do filter cleaning. 6. Recondition or replace oil.
	Lube Oil Viscosity LOW (LOLQ)	
	Lube Oil TAN HIGH (LOLQ)	
	Lube Oil Composition Analysis HIGH (LOLQ)	
Lube Oil Filter Anomaly	Lube Oil Filter DP RH (LFDF)	1. Check filter DP, changeover and clean/replace the stand by filter. 2. Check the Filter DP transmitter.
Loss of Pressure due to Pump Anomaly	Lube Oil Header Pressure RL (LOPPF)	1. Check the Discharge PT integrity. 2. Switch on both oil pumps and stabilize oil pressure. 3. Assess malfunction. 4. Line walk the system and check for leaks, ensure PSV's are seated. 5. Check spillback/PCV valves are set correctly and not returning oil back to the tank. 6. Ensure no hand valves have been left shut for the standby pump. 7. Check oil level in tank, Open tank overhead vents for release of pressure. 8. Check PCV function for Integrity. 9. Assess malfunction. 10. Check Auto Logic Function tests, check functioning of the auto change over switch. 11. If maintenance has recently been performed check correct rotation of pump. 12. Ensure NRV's are working correctly.
	Lube Oil Pump Discharge Pressure RL (LOPPF)	
	Lube Oil PCV Position RL (LOPPF)	
	Lube Oil Return Flow Rate RL (LOPPF)	
	Lube Oil Main Pump Current (Run Status) ON (LOPPF)	
	Lube Oil Aux Pump Current (Run Status) ON (LOPPF)	
Loss of Cooling due to TCV Anomaly	Lube Oil Header Temperature (Cooler Out Temp) RH (LOCTF)	1. Check the calibration of sensor. 2. Check bypass control valve for integrity. 3. Open both coolers and stabilize temperature. 4. Line walk the system to check for leaks.

Fault Name	Associated Symptoms	Recommendations
	LO Cooler Bypass Valve (TCV) Position RH (LOCTF)	5. Check the Isolation/bypass valve of TCV.
Loss of Cooling due to Cooler Anomaly	Lube Oil Header Temperature (Cooler Out Temp) RH (LOCCF)	1. Check the calibration of sensor. 2. Check cooling water supply pressure. 3. Open both coolers and stabilize temperature. 4. If malfunction of one cooler is suspected, isolate and inspect it.
	LO Tank And Cooler Outlet Temperature Delta RH (LOCCF)	5. Line walk the system to check for leaks. 6. Check cooling water pumps for blockages, ensure strainer is clean.
Loss of Heating Due to Lube Oil Tank Heater Off	Lube Oil Header Temperature (Cooler Out Temp) RL (LOCHOFF)	1. Check the Heater ON/OFF switch integrity and Change the switch as per the requirement.
	Lube Oil Tank Temperature (Cooler Inlet Temp) RL (LOCHOFF)	2. Check the heater circuit for Voltage and current.
	Lube Oil Tank Heater (Run Status) OFF (LOCHOFF)	
Loss of Cooling due Lube Oil Tank Heater On	Lube Oil Header Temperature (Cooler Out Temp) RH (LOCHON)	1. Check the Heater ON/OFF switch integrity and Change the switch as per the requirement.
	Lube Oil Tank Temperature (Cooler Inlet Temp) RH (LOCHON)	2. Offline 1.
	Lube Oil Tank Heater (Run Status) ON (LOCHON)	3. Check the heater circuit for Voltage and current.
Loss of Cooling due to Fan Motor Issue	Lube Oil Header Temperature (Cooler Out Temp) RH (LOCFMI)	Offline: Check the Fan and Motor /coupling / belt Integrity and rectify.
	Lube Oil Cooler Fan 1 (Run Status) OFF (LOCFMI)	
	Lube Oil Cooler Fan 2 (Run Status) OFF (LOCFMI)	

RECIPROCATING COMPRESSOR

**Introduction**

This document will specify the variables and equations required to model a generic reciprocating compressor stage and its implementation.

Background

Reciprocating compressors are used to compress gas in a number of common processes in an industry. Good prediction and control of the compressor interstage temperatures and pressures are of great concern when operating near their limits. These limits can be encountered when the unit production rate is being maximized and cooling limitations are reached. In addition to operating limits, there are a number of mechanical constraints in the compressor. Prediction of these constraints is critical for the safe operation of the process units.

In addition to modeling the compression effects of the compressor, there is a need to predict compressor rod loading. The loading on the compressor rods come from the loading imposed by the gas and the loading from inertial forces. These loading conditions are desirable to predict as a function of the compressor operating variables. The compressor model described in this document will implement both the equations for the prediction of compressor outlet conditions and the total rod loading as a function of compressor operating variables.

Basis

This model requires thermodynamic calculations at various conditions associated with the compression cycle. A diagram of the compressor stage is shown below. In this diagram, the Y-Axis is pressure, and the X-axis is volume. Pressure P represents inlet (suction) pressure and P_d represents outlet (discharge) pressure. P-V diagram shows the relationship of pressure and volume of the gas within one end of a cylinder of reciprocating compressor to the displacement of the piston. It is assumed that the compression is adiabatic. Referring to the P-V diagram at the bottom, a typical reciprocating compressor cycle can be described as follows:

1. Suction valve opens and gas is drawn into the cylinder (1 - 2).
2. Suction valve closes and gas compression begins (2 - 3).
3. Discharge valve opens and compressed gas is discharged from the cylinder (3 - 4).
4. Discharge valve closes. In this diagram, note that a gap is shown between zero volume and the volume at position 4. This represents the clearance volume in the cylinder. As the piston begins its return stroke, the gas which remains in this space re-expands (4 - 1).

Gas is trapped in the clearance volume after each stroke. This volume of gas must be re-expanded before gas at suction conditions is admitted into the cylinder to be compressed.

The model requires calculation of the Base Volume, which is the volume of the cylinder displaced by the piston. The compressor stroke length, piston diameter and number of cylinders are inputs to the model. The cylindrical volume displaced by the piston is equal to the length of the piston times the cross-sectional area of the piston. The model also requires Clearance Volume Fraction, which is the fractional volume in the cylinder (relative to the base volume) when the piston is at full stroke. The model will be developed for a single stage compressor. Multiple instances of the stage model can be used to construct multi-stage compressor models.

11.1 Reciprocating Compressor Sub System

11.1.1 Compressor Bearing Temperature and Vibrations

Monitoring compressor vibrations is by far the predominant and most widely used method to determine the mechanical conditions of the compressor. Many different failure modes can cause an increase in compressor vibrations.

Condition monitoring is used to early detect noise and vibration signature and bearing temperature of centrifugal compressor. Rise in bearing vibration and temperature is due to inappropriate fitting, overgreasing, or cage fracture. And surface distortion is due to poor lubricant and overload. Normally, bearings display symptoms of distress before they fail. Symptoms of a failing bearing include higher vibration levels and rising bearing or oil sump temperatures. The rise in Lube oil temperature increases the viscosity of the lube oil, which reduces the strength of the oil film on the bearings and may lead to high vibrations.

11.2 Reciprocating Compressor - Performance Monitoring

The following measured parameters are monitored as they can negatively impact the performance and health of reciprocating compressor if not maintained in the optimum operating range.

Suction Temperature: Inlet temperature changes produce large changes in performance. Lower temperature increases the capacity (weight flow) at a given discharge pressure and hence increases power consumption. Whereas higher temperature decreases the capacity (weight flow) at a given discharge pressure and reduces power consumption.

Suction Pressure: Lower inlet pressure decreases the capacity (weight flow) and power consumption due to reduced weight flow.

Discharge Temperature: If the compressor discharge temperature is very high, then it may result in the breakdown of lubricating oil, causing excessive wear and reduced life of the compressor valves (mainly the discharge valve). When the temperature is high, undesirable chemical reactions may take place inside the compressor, especially in the presence of water. This may ultimately damage the compressor, increasing the power consumption eventually.

Discharge Pressure: Changes in discharge pressure have effect on cylinder capacity and compression ratio. Volumetric efficiency varies slightly with compression ratio, and the required power is directly proportional to the change in compression ratio.

Volumetric Flow Rate: Volumetric flow is a key parameter for calculating a compressor's volumetric efficiency. Reduced volumetric flow hampers a screw compressor's volumetric efficiency and vice versa. In screw compressor, there is no clearance volume expansion, as in reciprocation compressor. Volumetric efficiency is a function of rotor slip. This is the internal leakage from the higher pressure to lower pressure side through blowhole area, interlobe area, and discharge clearance. This leakage reduces compressor's potential volume capacity and adiabatic efficiency.

Notations

F	Flow
h	Specific enthalpy
$\dot{F}$	Mass flow
P	Pressure
T	Temperature
η	Efficiency
$\dot{W}$	Power
N	Speed

Subscripts

1	Suction Valve Opens
2	Suction Valve Closes
3	Discharge Valve Opens
4	Discharge Valve Closes

Superscripts

$Exp.$	Expected
$Oper.$	Operating

11.2.1 Reciprocating Compressor - Model Assumptions

1. The diameter in one compressor stage is identical.
2. There is no temperature gain or drop when suction valve opens and mixes with residual gas in the compressor.

11.2.2 Reciprocating Compressor - Model Input

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Piston Bore Diameter	Piston Bore Diameter (Input)	m	Dynamic	Design	Constant
2	Clearance	Clearance (Input)	%	Dynamic	Design	Constant
3	Stage Cylinder Number	Stage Cylinder Number (Input)		Dynamic	Design	Constant
4	Suction Specific Heat	Suction Specific Heat (Input)	kJ/kg-C	Dynamic	Thermo Properties	Field Input
5	Discharge Specific Heat	Discharge Specific Heat (Input)	kJ/kg-C	Dynamic	Thermo Properties	Field Input
6	Mechanical Efficiency	Mechanical Efficiency (Input)	%	Static	Design	Field Input
7	Gear Ratio	Gear Ratio (Input)		Static	Design	Constant
8	Adiabatic Exponent K1	Suction Adiabatic Exponent K1 (Input)		Dynamic	Thermo Properties	Constant
9	Piston Rod Diameter	Piston Rod Diameter (Input)	m	Dynamic	Design	Constant
10	Piston Stroke Length	Piston Stroke Length (Input)	m	Dynamic	Design	Constant
11	Z1 (compressibility inlet)	Z1 (compressibility inlet)		Dynamic	Thermo Properties	Constant
12	Suction Pressure	Suction Pressure (Input)	barg	Dynamic	Process	Field Input
13	Discharge Pressure	Discharge Pressure (Input)	barg	Dynamic	Process	Field Input
14	Compressor Volumetric Flow	Compressor Volumetric Flow (Input)	m ³ /hr	Dynamic	Process	Field Input
15	Inlet1 Actual Volumetric Flow	Inlet Stream Actual Volume Flow (Input - Port mapped from suction stream)	m ³ /hr	Dynamic	Internal	Field Input
		NOTE Inlet actual volumetric flow can be used while configuring a new model.				
16	Shaft Rotation Speed	Shaft Rotation Speed (Input)	rpm	Dynamic	Process	Field Input
17	Piston Stroke %	Piston Stroke % (Input)	%	Dynamic	Design	Field Input
18	Suction Temperature	Suction Temperature (Input)	°C	Dynamic	Process	Field Input

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
19	Discharge Temperature	Discharge Temperature (Input)	°C	Dynamic	Process	Field Input

11.2.3 Reciprocating Compressor - KPIs

The following indicators help us know the overall performance of the asset.

11.2.3.1 Expected KPIs

- Expected Gas Power
- Expected Shaft Power
- Expected Compressor Power
- Expected Inboard Inertial Load
- Expected Inboard Gas Load
- Expected Inboard Total Load
- Base Volume
- Stroke Volume
- Swept Volume

11.2.3.2 Operating KPIs

- Operating Gas Power
- Operating Adiabatic Efficiency
- Operating Shaft Power
- Operating Compressor Power
- Operating Inboard Inertial Load
- Operating Inboard Gas Load
- Operating Inboard Total Load

11.2.3.3 Reciprocating Compressor - Performance Modeling

Suction Valve Opens

Incoming feed mixes with the residual gas content

Overall Mass Balance

$$F_1 = F_I + F_{Residual}$$

Force Balance

The feed pressure and residual gas are both at feed pressure.

$$P_1 = P_I$$

Suction Valve Closes

Adiabatic compression cycle expected behavior is pressure and temperature rise.

Overall Mass Balance

$$F_2 = F_1$$

Force Balance

The feed pressure and residual gas are both at feed pressure.

$$P_2 = P_1 * P_{Ratio}$$

$$P_{Ratio} = \frac{P_d}{P_I}$$

Energy Balance

$$T_2 = T_1 * (P_{Ratio})^{\frac{(\gamma-1)}{\gamma}}$$

$$H_2(P_2, T_2) = F_2 h(P_2, T_2)$$

Discharge Valve Opens

Discharge valve opens and compressed gas is discharged from the cylinder.

Overall Mass Balance

$$F_3 = F_2 - F_R$$

Force Balance

The delivery is at discharge pressure.

$$P_3 = P_d$$

Energy Balance

$$T_2 = T_1 * (P_{Ratio})^{\frac{(\gamma-1)}{\gamma}}$$

$$H_2(P_2, T_2) = F_2 h(P_2, T_2)$$

Discharge Valve Closes

Residual gas is expanded to inlet conditions.

Overall Mass Balance

$$F_4 = F_{Reidual}$$

Force Balance

$$P_4 = P_1$$

Energy Balance

$$T_4 = T_1$$

$$H_4(P_4, T_4) = F_4 h(P_4, T_4)$$

11.2.3.4 Reciprocating Compressor - Expected KPIs**Expected Gas Power (kW)**

It is the gas power absorbed by the process stream.

$$\dot{W}_{Gas}^{Expected} = \frac{F_1^{Exp.} (h_4(P_4, S_1) - h_1(P_1, T_1))}{\eta_{Adiabatic}}$$

Expected Shaft Power (kW)

It is the power delivered to the compressor shaft.

$$\dot{W}_{Shaft}^{Expected} = \frac{\dot{W}_{Gas}^{Expected}}{\eta_{Mechanical}}$$

Expected Compressor Power (kW)

$$\dot{W}_{Compressor}^{Expected} = f \frac{\dot{W}_{Shaft}^{Expected}}{\eta_{Compressor}}$$

Expected Inboard/Outboard Inertial Load (kN)

$$\Phi = w_m \alpha$$

$$\alpha = \frac{\omega L_S}{2} \left(\cos\theta - \frac{\lambda \cos 2\theta}{\sqrt{1 - \lambda^2 \sin^2 \theta}} - \frac{\lambda^3 \sin^2 2\theta}{4 \left(\sqrt{1 - \lambda^2 \sin^2 \theta} \right)^3} \right)$$

$$\omega = \frac{2\pi\Gamma}{60}$$

$$\lambda = \frac{R}{L_R}$$

$$R = \frac{L_S}{2}$$

Φ	Inertial Load
w_m	Weight of Machinery
α	Acceleration
ω	Angular Velocity
λ	Ratio of crank radius to rod length
R	Crank Radius
L_R	Rod length
L_S	Stroke Length
θ	inboard/outboard crank angle

Expected Inboard/Outboard Gas Load (kN)

$$\Psi = (P_1 + 2\Delta P - P_4) \left(\frac{\pi D_P^2}{4} \right)$$

$$\Delta P = \frac{1}{2} \rho_4^{Exp} \nu_G^2$$

$$\nu_G = \nu_P \left(\frac{\frac{\pi D_P^2}{4}}{\sigma_{Open} \sigma_{Lift} \cos \delta} \right)$$

$$\nu_P = \omega R \left(\sin \theta - \frac{\lambda \sin 2\theta}{2 \cos \beta} \right)$$

$$\beta = \arccos \left(\sqrt{1 - \lambda^2 \sin^2 \theta} \right)$$

Ψ	Gas Load
D_P	Piston Diameter
ν_G	Gas Velocity
ν_P	Piston Velocity
R	Crank Radius
θ	Inboard/Outboard Crank Angle
β	Rod Angle
σ_{Open}	Valve opening length

σ_{Lift}	Valve lift length
δ	Seat angle

Expected Inboard/Outboard Total Load (kN)

$$T = \frac{\Phi + \Psi}{\sqrt{1 - \lambda^2 \sin^2 \theta}}$$

Φ	Inertial Load
Ψ	Gas Load
λ	Ratio of crank radius to rod length
θ	Inboard/Outboard Crank Angle
T	Total Load

11.2.3.5 Reciprocating Compressor - Operating KPIs

Operating Gas Power (kW)

$$\dot{W}_{Gas}^{Operating} = F_1 (h_4 (P_4, T_4) - h_1 (P_1, T_1))$$

Operating Adiabatic Efficiency (%)

$$\eta_{Adiabatic} = \frac{\dot{W}_{Gas}^{Operating Isen}}{\dot{W}_{Gas}^{Operating}}$$

$$\dot{W}_{Gas}^{Operating Isen} = F_1 (h (P_4, S_1) - h (P_1, T_1))$$

Operating Shaft Power (kW)

$$\dot{W}_{Shaft}^{Operating} = \frac{\dot{W}_{Gas}^{Operating}}{\eta_{Mechanical}}$$

Operating Compressor Power (kW)

$$\dot{W}_{Compressor}^{Operating} = \frac{\dot{W}_{Shaft}^{Operating}}{\eta_{Compressor}}$$

Operating Inboard/Outboard Inertial Load (kN)

$$\Phi = w_m \alpha$$

$$\alpha = \frac{\omega L_S}{2} \left(\cos \theta - \frac{\lambda \cos 2\theta}{\sqrt{1 - \lambda^2 \sin^2 \theta}} - \frac{\lambda^3 \sin^2 2\theta}{4 \left(\sqrt{1 - \lambda^2 \sin^2 \theta} \right)^3} \right)$$

$$\omega = \frac{2\pi\Gamma}{60}$$

$$\lambda = \frac{R}{L_R}$$

$$R = \frac{L_S}{2}$$

Φ	Inertial Load
w_m	Weight of Machinery
α	Acceleration
ω	Angular Velocity
λ	Ratio of crank radius to rod length
R	Crank Radius
L_R	Rod length
L_S	Stroke Length
θ	inboard/outboard crank angle

Operating Inboard/Outboard Gas Load (kN)

$$\Psi = (P_1 + 2\Delta P - P_4) \left(\frac{\pi D_P^2}{4} \right)$$

$$\Delta P = \frac{1}{2} \rho_4^{oper.} \nu_G^2$$

$$\nu_G = \nu_P \left(\frac{\frac{\pi D_P^2}{4}}{\sigma_{Open} \sigma_{Lift} \cos \delta} \right)$$

$$\nu_P = \omega R \left(\sin \theta - \frac{\lambda \sin 2\theta}{2 \cos \beta} \right)$$

$$\beta = \arccos \left(\sqrt{1 - \lambda^2 \sin^2 \theta} \right)$$

Ψ	Gas Load
D_P	Piston Diameter
ν_G	Gas Velocity
ν_P	Piston Velocity

R	Crank Radius
θ	Inboard/Outboard Crank Angle
β	Rod Angle
σ_{Open}	Valve opening length
σ_{Lift}	Valve lift length
δ	Seat angle

Operating Inboard/Outboard Total Load (kN)

$$T = \frac{\Phi + \Psi}{\sqrt{1 - \lambda^2 \sin^2 \theta}}$$

Φ	Inertial Load
Ψ	Gas Load
λ	Ratio of crank radius to rod length
θ	Inboard/Outboard Crank Angle
T	Total Load

11.2.4 Reciprocating Compressor - Model Output

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Isentropic Efficiency	Isentropic Efficiency (Output)	%	Dynamic	Performance	Calculated
2	Compressor Load	Compressor Load (Output)	%	Dynamic	Performance	Calculated
3	Shaft Efficiency	Shaft Efficiency (Output)	%	Dynamic	Performance	Calculated
4	Operating Shaft Power	Operating Shaft Power (Output)	kW	Dynamic	Performance	Calculated
5	Volumetric Efficiency	Volumetric Efficiency (Output)	%	Dynamic	Performance	Calculated

11.3 Reciprocating Compressor - Fault Model

11.3.1 Reciprocating Compressor - APM Model

11.3.1.1 Compressor Stage One Performance Fault Model

Model Type	APM Model
Category	Reciprocating
Equipment Class	ReciprocatingCompressor
Equipment Type	Reciprocating Compressor- Single Stage

Below table illustrates the list of faults and symptoms configured for Compressor Performance based monitoring.

Fault Name	Associated Symptoms	Recommendations
S1 Liquid carryover	Stage 1 Separator inlet pressure RL (LCO)	1. Inspect precooler integrity. 2. Inspect stage 1 separator integrity. 3. Check mesh thickness/integrity. 4. Check stage 1 separator drain LCV valve integrity and drain line for any jam. 5. Check precooler motor / fan issue. 6. Inspect suction bottle LCV and drains/ inspect piston packing and seal rings.
	Stage 1 Separator inlet pressure RH (LCO)	
	Stage 1 Separator outlet pressure RH (LCO)	
	Stage 1 Separator pressure differential RL (LCO)	
	Stage 1 Separator pressure differential RH (LCO)	
	Stage 1 Separator level RH (LCO)	
	Stage 1 Separator level valve position RL (LCO)	
	Stage 1 suction volume bottle liquid level RH (LCO)	
	Precooler outlet temperature RH (LCO)	
	Precooler Operating Duty RL (LCO)	
S1 Recip Suction CE pressure loss	Stage 1 Suction pressure RL (CE SUC PRES)	1. Inspect the stage 1 suction bottle for its integrity and its line for any jam. 2. Check the stage 1 CE suction valve lift and spring stiffness.
	Stage 1 Discharge Pressure RL (CE SUC PRES)	
	Stage 1 Recip Differential Pressure RH (CE SUC PRES)	
	Stage 1 Instantaneous pressure (Crank Side) RL (CE SUC PRES)	
	Stage 1 suction volume bottle pressure RL (CE SUC PRES)	
S1 Recip CE Discharge pressure loss	Stage 1 Suction pressure RH (CE DIS PRES)	1. Inspect the stage 1 discharge bottle for its integrity and its line for any jam. 2. Check the stage 1 CE discharge valve lift and spring stiffness. 3. CE suction valve looseness/spring stiffness loss.
	Stage 1 Discharge Pressure RL (CE DIS PRES)	
	Stage 1 CE suction valve loader position RH (CE DIS PRES)	
	Stage 1 discharge volume bottle pressure RL (CE DIS PRES)	
S1 Recip Suction HE pressure loss	Stage 1 Suction pressure RL (HE SUC PRES)	1. Check the stage 1 HE suction valve lift and spring stiffness. 2. Check stage 1 inlet and outlet pressure.

Fault Name	Associated Symptoms	Recommendations
	Stage 1 Discharge Pressure RL (HE SUC PRES)	3. Check separator PSV.
	Stage 1 Recip Differential Pressure RH (HE SUC PRES)	
	Stage 1 Instantaneous pressure (Head Side) RL (HE SUC PRES)	
	Stage 1 HE suctions valve loader position RL (HE SUC PRES)	
S1 Recip HE Discharge pressure loss	Stage 1 Suction pressure RH (HE DIS PRES)	1. Check the stage 1 HE discharges valve lift and spring stiffness. 2. Check for cylinder liner damage. 3. Check stage 1 inlet and outlet pressure/check separator PSV.
	Stage 1 Discharge Pressure RL (HE DIS PRES)	
	Stage 1 HE suctions valve loader position RH (HE DIS PRES)	
S1 Piston ring/Ryder ring/cylinder liner clearance increase	Stage 1 Discharge Pressure RL (CCA)	1. Check integrity of Stage 1 rider ring, piston ring, cylinder liner, and packings. 2. Check crank case lube oil gauge/lube oil pump and filters/cylinder loader valves integrity.
	Stage 1 Piston rod drop displacement RH (CCA)	
	Lube Oil Header Pressure LOW (CCA)	
	Crankcase Lube oil pressure RL (CCA)	
S1 Recip Performance degradation loss	Stage 1 Operating volumetric efficiency RL (PDA)	1. Check integrity of Stage 1 valves, valve unloaders. 2. Check gaskets. 3. Check packings. 4. Check cooling system.
	Stage 1 Operating isentropic efficiency RL (PDA)	
	Stage 1 Operating shaft efficiency RL (PDA)	
	Stage 1 Operating shaft efficiency RH (PDA)	
	Stage 1 Shaft power RL (PDA)	
	Stage 1 Shaft power RH (PDA)	
S1 Crosshead Vibration High	Stage 1 Piston rod drop displacement RH (CHVA)	1. Inspect the lube oil system and check any obstruction in its path. 2. check the alignment of distance piece guide. 3. Check crosshead wear and tear. 4. Check crosshead pin and its components' integrity. 5. Check packing damage.

Fault Name	Associated Symptoms	Recommendations
	Comp DE Radial vibration overall (X) RH (CHVA)	6. Check cylinder volume clearance.
	Comp DE Radial vibration overall (Y) RH (CHVA)	
	Comp NDE Radial vibration overall (X) RH (CHVA)	
	Comp NDE Radial vibration overall (Y) RH (CHVA)	
S1 suction pressure pulsation high	Stage 1 suction volume bottle pressure RL (SPPA)	1. Inspect the stage 1 suction pulsation bottle and its pipeline. 2. Inspect the stage 1 recycle valve.
	Stage 1 recycle valve position RH (SPPA)	
S1 discharge pulsation high	Stage 1 discharge volume bottle pressure RL (DPPA)	Inspect the Stage 1 discharge pulsation bottle and its pipeline.

Compressor Stage Two Performance Fault Model

Model Type	APM Model
Category	Reciprocating
Equipment Class	Reciprocating Compressor
Equipment Type	Reciprocating Compressor- Second Stage

Below table illustrates the list of faults and symptoms configured for a Reciprocating Compressor Second Stage Performance based monitoring.

Fault Name	Associated Symptoms	Recommendations
S2 Liquid carryover	Stage 2 Separator pressure differential RL (LCO)	1. Inspect intercooler integrity. 2. Inspect stage 2 separator integrity. 3. Check mesh thickness/integrity. 4. Check stage 2 separator drain valve integrity and drain line for any jam. 5. Check precoolers motor, and fan issue. 6. Check discharge bottle LCV and drain valves, inspect piston packing, and seal rings.
	Stage 2 Separator pressure differential RH (LCO)	
	Stage 2 Separator level RH (LCO)	
	Stage 2 Separator level valve position RL (LCO)	
	Stage 2 discharge volume bottle level RH (LCO)	
	Intercooler outlet temperature RH (LCO)	
	Intercooler Operating Duty RL (LCO)	
S2 Recip Suction CE pressure loss	Stage 2 Suction pressure RL (CE SUC PRES)	1. Inspect the stage 2 suction bottle for its integrity, and its line for any jam. 2. Check the stage 2 CE suction valve lift

Fault Name	Associated Symptoms	Recommendations
	Stage 2 Discharge Pressure RL (CE SUC PRES) Stage 2 Recip Differential Pressure RH (CE SUC PRES) Stage 2 Instantaneous pressure (Crank Side) RL (CE SUC PRES) Stage 2 discharge volume bottle pressure RL (CE SUC PRES)	and spring stiffness.
S2 Recip CE Discharge pressure loss	Stage 2 Suction pressure RH (CE DIS PRES) Stage 2 Discharge Pressure RL (CE DIS PRES) Stage 2 CE suction valve loader position RH (CE DIS PRES) Stage 2 discharge volume bottle pressure RL (CE DIS PRES)	1. Inspect stage 2 discharge bottle for its integrity, and its line for any jam. 2. Check the stage 2 CE discharge valve lift and spring stiffness. 3. CE suction valve looseness/spring stiffness loss.
S2 Recip Suction HE pressure loss	Stage 2 Separator outlet pressure RL (HE SUC PRES) Stage 2 Suction pressure RL (HE SUC PRES) Stage 2 Discharge Pressure RL (HE SUC PRES) Stage 2 Recip Differential Pressure RH (HE SUC PRES) Stage 2 Instantaneous pressure (Head Side) RL (HE SUC PRES) Stage 2 HE suction valve loader position RL (HE SUC PRES) Stage 2 discharge volume bottle pressure RL (HE SUC PRES) Stage 1 Separator outlet pressure RL (HE SUC PRES)	1. Check the stage 2 HE suction valve lift and spring stiffness. 2. Check stage 2 separator parameters (pressure, level, high). 3. Check stage1 compressor inlet parameters.
S2 Recip HE Discharge pressure loss	Stage 2 Suction pressure RH (HE DIS PRES) Stage 2 Discharge Pressure RL (HE DIS PRES) Stage 2 HE suction valve loader position RH (HE DIS PRES) Stage 2 discharge volume bottle pressure RL (HE DIS PRES)	1. Check the stage 2 HE discharge valve lift and spring stiffness.

Fault Name	Associated Symptoms	Recommendations
S2 Piston ring/Rider ring/cylinder liner clearance increase	Stage 2 Discharge Pressure RL (CCA)	1. Check the integrity of Stage 2 rider ring, piston ring, cylinder liner, and packings. 2. Check crank case lube oil gauge/lube oil pump and filters/cylinder loader valve integrity.
	Stage 2 Piston rod drop displacement RH (CCA)	
	Lube Oil Header Pressure LOW (CCA)	
	Crankcase Lube oil pressure RL (CCA)	
S2 Crosshead Vibration High	Stage 2 Piston rod drop displacement RH (CHVA)	1. Inspect the stage 2 lube oil system and check any obstruction in its path. 2. Check the alignment of distance piece guide. 3. Check croshead wear and tear. 4. Check croshead pin and its components' integrity. 5. Check packing damage/ check cylinder volume clearance.
	Comp DE Radial vibration overall (X) RH (CHVA)	
	Comp DE Radial vibration overall (Y) RH (CHVA)	
	Comp NDE Radial vibration overall (X) RH (CHVA)	
	Comp NDE Radial vibration overall (Y) RH (CHVA)	
S2 suction pressure pulsation high	Stage 2 suction volume bottle pressure RL (SPPA)	1. Inspect the 2nd stage suction pulsation bottle and its pipeline. 2. Inspect the 2nd stage recycle valve.
	Stage 2 recycle valve position RH (SPPA)	
S2 discharge pulsation high	Stage 2 discharge volume bottle pressure RH (DPPA)	1. Inspect the 2nd stage-discharge pulsation bottle and its pipeline.

11.3.2 Reciprocating Compressor - APR Model

11.3.2.1 Compressor Bearing Vibration and Temperature Fault Model.

Model Type	APR Model
Category	Reciprocating
Equipment Class	ReciprocatingCompressor
Equipment Type	Reciprocating Compressor Bearing System

Below table illustrates the list of faults and symptoms configured for a Compressor Bearing Vibration and Temperature Fault Model.

Fault Name	Associated Symptoms	Recommendations
Crank Shaft Unbalance Malfunction	Comp DE Radial vibration overall (X) RH (CSA)	1. Check the crankshaft for scaling, clean if necessary. 2. Check the integrity of the balancing of the crank shaft.

Fault Name	Associated Symptoms	Recommendations
	Comp DE Radial vibration overall (Y) RH (CSA)	3. Check bush, bolt, pin conditions of the connecting rod.
	Comp NDE Radial vibration overall (X) RH (CSA)	
	Comp NDE Radial vibration overall (Y) RH (CSA)	
	Casing Vibration RH (CSA)	
Stage 1 Piston Rod Drop Displacement Anomaly	Stage 1 Piston Rod Drop Displacement RL (S1PDA)	1. Check the integrity of the Displacement sensors. 2. Check the System for displacement, Adjust the displacement if required. 3. Check for internal damage or wear of internal components and replace or correct as required.
	Stage 1 Piston Rod Drop Displacement RH (S1PDA)	
DE Radial Bearing Mechanical Anomaly rH	Comp DE Radial vibration overall (X) RH (DE RB MA)	1. Assess the likely cause and time to fail(trip) if intervention is not possible. 2. Intervention options; reduce speed, and restore oil properties. 3. Parts availability. 4. Check the trends to establish if it is either both X and Y probes or one. 5. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 6. Change bearing if required.
	Comp DE Radial vibration overall (Y) RH (DE RB MA)	
	DE Radial bearing temperature (A) RH (DE RB MA)	
	DE Radial bearing temperature (B) RH (DE RB MA)	
Stage 2 Piston Rod Drop Displacement Anomaly	Stage 2 Piston Rod Drop Displacement RL (S2PDA)	1. Check the integrity of the Displacement sensors. 2. Check the System for displacement, Adjust the displacement if required. 3. Check for internal damage or wear of internal components and replace or correct as required.
	Stage 2 Piston Rod Drop Displacement RH (S2PDA)	
Rotor Anomaly	Casing Vibration RH (RTRA)	1. Check the rotor alignment, balancing, offset, and runout.
	Radial vibration DE overall (X)/NDE overall (X) HIGH (RTRA)	
	Radial vibration DE overall (X)/NDE overall (X) LOW (RTRA)	
	Radial vibration DE overall (Y)/NDE overall (Y) HIGH (RTRA)	
	Radial vibration DE overall (Y)/NDE overall (Y) LOW (RTRA)	
Thrust/Radial Bearing Anomaly due to Lube Oil	Comp DE Radial vibration overall (X) RH (LOQA)	1. Recondition or replace oil. 2. Check the oil tank for free water by draining 1L

Fault Name	Associated Symptoms	Recommendations
Quality	Comp DE Radial vibration overall (Y) RH (LOQA)	into a sample bottle. - Switch cooler if water contamination. - Take an oil sample and check water content viscosity, TAN. - Check no breather caps or man covers have been removed and left open to allow debris into the tank.
	DE Radial bearing temperature (A) RH (LOQA)	
	DE Radial bearing temperature (B) RH (LOQA)	
	Comp NDE Radial vibration overall (X) RH (LOQA)	
	Comp NDE Radial vibration overall (Y) RH (LOQA)	
	NDE Radial bearing temperature (A) RH (LOQA)	
	NDE Radial bearing temperature (B) RH (LOQA)	
	Lube Oil Viscosity SH (LOQA)	
	Lube Oil Viscosity SL (LOQA)	
	Lube Oil TAN SH (LOQA)	
	Lube Oil TAN SL (LOQA)	
Thrust/Radial Bearing Anomaly due to Lube Oil Press	Comp DE Radial vibration overall (X) RH (LOPA)	- Switch on both oil pumps and stabilize the oil pressure. - Assess malfunction. - Line walk the system and check for leaks. - Check filter DP. - Ensure PSVs are seated. - Check spillback/PCV valves are set correctly and not returning oil back to the tank. - If maintenance has recently been performed check the correct rotation of pump. - Ensure no hand valves have been left shut. - Ensure NRVs are working correctly. - Check the oil level in tank. - Check functioning of the auto change-over switch.
	Comp DE Radial vibration overall (Y) RH (LOPA)	
	Comp NDE Radial vibration overall (X) RH (LOPA)	
	Comp NDE Radial vibration overall (Y) RH (LOPA)	
	Lube Oil Header Pressure RL (LOPA)	
	Lube Oil Header Pressure RH (LOPA)	
	DE Radial bearing temperature (A) RH (LOPA)	
	DE Radial bearing temperature (B) RH (LOPA)	
	NDE Radial bearing temperature (A) RH (LOPA)	
	NDE Radial bearing temperature (B) RH (LOPA)	
Thrust Bearing Active Side Mechanical Anomaly rH/rL	Thrust Bearing Temp - Active A RL (TB AS MA)	Inspect the thrust bearing and its pads, thrust collar etc.

Fault Name	Associated Symptoms	Recommendations
	Thrust Bearing Temp - Active A RH (TB AS MA)	2. Check lube oil system. 3. Check coupling DBSE.
	Thrust Bearing Temp - Active B RL (TB AS MA)	
	Thrust Bearing Temp - Active B RH (TB AS MA)	
	Thrust Bearing Temp - Active C RL (TB AS MA)	
	Thrust Bearing Temp - Active C RH (TB AS MA)	
	Thrust Bearing Temp - Active D RL (TB AS MA)	
	Thrust Bearing Temp - Active D RH (TB AS MA)	
Thrust Bearing Inactive Side Mechanical Anomaly rH/rL	Thrust Bearing Temp - Inactive A RL (TB IS MA)	1. Inspect the thrust bearing and its pads, thrust collar etc. 2. Check lube oil system. 3. Check coupling DBSE.
	Thrust Bearing Temp - Inactive A RH (TB IS MA)	
	Thrust Bearing Temp - Inactive B RL (TB IS MA)	
	Thrust Bearing Temp - Inactive B RH (TB IS MA)	
	Thrust Bearing Temp - Inactive C RL (TB IS MA)	
	Thrust Bearing Temp - Inactive C RH (TB IS MA)	
	Thrust Bearing Temp - Inactive D RL (TB IS MA)	
	Thrust Bearing Temp - Inactive D RH (TB IS MA)	
NDE Radial Bearing Mechanical Anomaly rH	Comp NDE Radial vibration overall (X) RH (NDE RB MA)	1. Assess likely cause and time to fail(trip) if intervention not possible. 2. Intervention options; reduce speed, restore oil properties. 3. Parts availability 4. Check the trends to establish if its either both X and Y probes or one. 5. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 6. Change bearing if required.
	Comp NDE Radial vibration overall (Y) RH (NDE RB MA)	
	NDE Radial bearing temperature (A) RH (NDE RB MA)	
	NDE Radial bearing temperature (B) RH (NDE RB MA)	

Fault Name	Associated Symptoms	Recommendations
Thrust/Radial Bearing Anomaly due to Lube Oil Temp	DE Radial bearing temperature (A) RH (LOTA)	1. Check cooling water supply pressure. 2. Check bypass control valve. 3. Open both coolers and stabilise temperature. 4. If malfunction of one cooler is suspected. 5. Isolate and inspect it. 6. Line walk the system to check for leaks. 7. Check cooling water pumps for blockages. 8. Ensure strainer is clean. 9. If the cooler is an air cooler check fan and motor integrity.
	DE Radial bearing temperature (B) RH (LOTA)	
	NDE Radial bearing temperature (A) RH (LOTA)	
	NDE Radial bearing temperature (B) RH (LOTA)	
	Lube Oil Header Temperature (Cooler Out Temp) RH (LOTA)	
	Comp DE Radial vibration overall (X) RH (LOTA)	
	Comp DE Radial vibration overall (Y) RH (LOTA)	
	Comp NDE Radial vibration overall (X) RH (LOTA)	

11.3.2.2 Compressor Cooler System Fault Model

Model Type	APR Model
Category	Reciprocating
Equipment Class	ReciprocatingCompressor
Equipment Type	Reciprocating Compressor Cooler System

Below table illustrates the list of faults and symptoms configured for a Reciprocating Compressor Cooler System.

Fault Name	Associated Symptoms	Recommendations
Precooler Tube Fouling anomaly	Precooler pressure differential RH (PTF)	Inspect Precooler integrity
	Precooler Operating Heat Transfer Coefficient RH (PTF)	
Precooler Ineffectiveness anomaly	Precooler outlet temperature RH (PIA)	1. Check the precooler fan or motor condition and repair if necessary. 2. Check V-belt and Pulley condition and replace them if necessary. 3. Clean the external tube surface./ Periodic. 4. Cleaning of cooler fins with water sprinkler system for effective heat transfer.
	Precooler fan-1 motor running status OFF (PIA)	
	Precooler fan-2 motor running status OFF (PIA)	
	Precooler Expected Heat Transfer Coefficient RL (PIA)	
	Precooler Operating Duty RL (PIA)	
Intercooler Tube Fouling anomaly	Intercooler pressure differential RH (ITF)	Inspect intercooler integrity

Fault Name	Associated Symptoms	Recommendations
	Intercooler Fouling Percentage RH (ITF)	
Intercooler Ineffectiveness anomaly	Intercooler outlet temperature RH (IIA)	1. Check the intercooler fan or motor condition and repair if necessary. 2. Check V-belt and Pulley condition and replace them if necessary. 3. Clean the external tube surface./ Periodic. 4. Cleaning of cooler fins with water sprinkler system for effective heat transfer
	Intercooler Operating Heat Transfer Coefficient RL (IIA)	
	Intercooler Operating Duty RL (IIA)	
	Intercooler fan-1 motor running status OFF (IIA)	
	Intercooler fan-2 motor running status OFF (IIA)	
S2 Recip CE Inlet Temp Anomaly	Intercooler outlet temperature RH (OTA)	1. Inspect the stage 2 discharge bottle for its integrity and its line for any jam. 2. Check the stage 2 CE discharge valve lift and spring stiffness. 3. CE suction valve looseness / spring stiffness loss.
COOLING WATER SUPPLY ANOMALY	Outlet cooling water temp RL (WSA)	1. Check the temperature probe for the accurate measurement. 2. Check the line for blockages and fouling debris. 3. Perform line walkthrough for leakages. 4. Check all vents(if any).
	Outlet cooling water temp RH (WSA)	
	Outlet cooling water pressure RL (WSA)	
	Outlet cooling water pressure RH (WSA)	

11.3.2.3 Compressor Lube Oil System Fault Model

Model Type	APR Model
Category	Reciprocating
Equipment Class	ReciprocatingCompressor
Equipment Type	Reciprocating Compressor Lube Oil System

Below table illustrates the list of faults and symptoms configured for a Reciprocating Compressor Lube Oil System

Fault Name	Associated Symptoms	Recommendations
Lube Oil Filter Anomaly	Lube Oil Filter DP RH (LFDF)	1. Check filter DP, changeover and clean/replace the stand-by filter. 2. Check the Filter DP transmitter.
Loss Of Cooling due to Cooler Anomaly	Lube Oil Header Temperature (Cooler Out Temp) RH (LOCCF)	1. Check the calibration of the sensor. 2. Check cooling water supply pressure. 3. Open both coolers and stabilize the temperature.

Fault Name	Associated Symptoms	Recommendations
		4. If a malfunction of one cooler is suspected, isolate and inspect it.
	LO Tank And Cooler Outlet Temperature Delta RH (LOCCF)	5. Line walk the system to check for leaks. 6. Check cooling water pumps for blockages, and ensure the strainer is clean.
Loss Of Cooling due to Fan Motor Issue	Lube Oil Header Temperature (Cooler Out Temp) RH (LOCFMI)	1. Offline: Check the Fan and Motor /coupling/belt Integrity and rectify.
	Lube Oil Cooler Fan 1 (Run Status) OFF (LOCFMI)	
	Lube Oil Cooler Fan 2 (Run Status) OFF (LOCFMI)	
Loss Of Cooling due Lube Oil Tank Heater On	Lube Oil Header Temperature (Cooler Out Temp) RH (LOCHON)	1. Check the Heater ON/OFF switch integrity and Change the switch as per the requirement. 2. Offline 1: 1. Check the heater circuit for Voltage and current.
	Lube Oil Tank Temperature (Cooler Inlet Temp) RH (LOCHON)	
	Lube Oil Tank Heater (Run Status) ON (LOCHON)	
Loss Of Lube Oil Quality	Lube Oil Viscosity HIGH (LOLQ)	1. Check the oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, TAN. 4. Check no breather caps or man covers have been removed and left open to allow debris into the tank. 5. Do filter cleaning. 6. Recondition or replace oil.
	Lube Oil Viscosity LOW (LOLQ)	
	Lube Oil TAN HIGH (LOLQ)	
Loss Of Pressure due to Pump Anomaly	Lube Oil Header Pressure RL (LOPPF)	1. Check the Discharge PT integrity. 2. Switch on both oil pumps and stabilize oil pressure. 3. Assess malfunction. 4. Line walk the system and check for leaks, ensure PSV's are seated. 5. Check spillback/PCV valves are set correctly and not returning oil back to the tank. 6. Ensure no hand valves have been left shut for the standby pump. 7. Check the oil level in tank, Open tank overhead vents for release of pressure. 8. Check the PCV function for Integrity. 9. Assess malfunction. 10. Check Auto Logic Function tests, and check

Fault Name	Associated Symptoms	Recommendations
	Lube Oil Pump Discharge Pressure RL (LOPPF)	functioning of the auto change-over switch. 11. If maintenance has recently been performed check the correct rotation of pump. 12. Ensure NRVs are working correctly.
	Lube Oil PCV Position RL (LOPPF)	
	Lube Oil Return Flow Rate RL (LOPPF)	
	Lube Oil Main Pump Current (Run Status) ON (LOPPF)	
	Lube Oil Aux Pump Current (Run Status) ON (LOPPF)	
Loss Of Heating due Lube Oil Tank Heater Off	Lube Oil Header Temperature (Cooler Out Temp) RL (LOCHOFF)	1. Check the Heater ON/OFF switch integrity and Change the switch as per the requirement. 2. Check the heater circuit for Voltage and current.
	Lube Oil Tank Temperature (Cooler Inlet Temp) RL (LOCHOFF)	
	Lube Oil Tank Heater (Run Status) OFF (LOCHOFF)	
Loss Of Cooling due to TCV Anomaly	Lube Oil Header Temperature (Cooler Out Temp) RH (LOCTF)	1. Check the calibration of the sensor. 2. Check the bypass control valve for integrity. 3. Open both coolers and stabilize the temperature. 4. Line walk the system to check for leaks. 5. Check the Isolation/bypass valve of TCV.
	LO Cooler Bypass Valve (TCV) Position RH (LOCTF)	

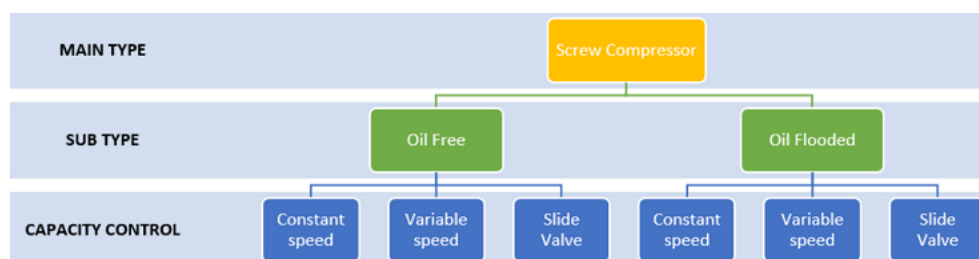
SCREW COMPRESSOR MONITORING

Screw compressor is a positive-displacement compressor that uses intermeshing pair of male and female lobe that is driven by a timing gear mounted on shaft to deliver gases at high pressure. Power is applied to the male rotor and as a lobe of the male rotor starts to move out of mesh with the female rotor a void is created, and the gas is taken in at the inlet port. The continuously turning rotor leads to an increase in the intermesh space and to continuous flow of gas into the compressor until the entire inter lobe space is filled. A continued rotation brings male lobe into the inter lobe space compressing and moving in the direction of the discharge port. Compressed gas starts to flow out of the compressor. Reciprocating compressors are typically used where high compression ratios (ratio of discharge to suction pressures) are required per stage.

Screw machines are equally suited to compress process gases such as ammonia, argon, ethylene, acetylene, butadiene, chlorine, hydrochloric gas, natural and synthetic pipeline gases, flare gas mixtures, blast furnace gas, swamp and biomass gases, coke oven or coal gas, carbon monoxide, town gas, methane, propane, propylene, flue gas, crude or raw gas, sulfur dioxide, nitrous oxide, vinyl chloride, styrene, and hydrogen.

The Screw compressor are classified as:

- Oil Free screw compressor (Dry)
- Oil Injected screw compressor (Oil flooded)



12.1 Screw Compressor - Sub Systems

Screw Compressor Subsystems	Oil Free Screw Compressor	Oil Injected Screw Compressor
Screw Compressor - Unit	Applicable	Applicable
Screw Compressor - Gearbox System	Applicable	Applicable
Screw Compressor - Inter Stage Cooler	Applicable	Applicable
Screw Compressor - Seal Gas System	Not Applicable	Applicable

12.1.1 Screw Compressor Bearing Temperature and Vibrations

Condition monitoring is used for the early detection of noise, vibration signature, and bearing temperature of screw compressor. Rise of bearing vibration and temperature due to inappropriate fitting or over greasing or cage fracture and surface distortion due to poor lubricant and overload. Bearings display symptoms of distress before they fail. These symptoms include higher vibration levels and rising bearing or oil sump temperatures. The rise in Lube oil temperature leads to an increase in the viscosity of the lube oil, reducing the strength of the oil film on the bearings that might lead to high vibrations.

12.1.2 Seal Gas System

Dry gas seal system is designed to provide reliable shaft sealing on rotating equipment to reduce the amount of process gas escaping from the machine's primary flow path to an acceptable level.

Loss of buffer gas will, in most cases, reverse the differential pressure across the seal faces, damaging the seal in short order. The seals will operate at a zero differential pressure level.

Dry Gas Seal Degradation: Contamination of the seal by foreign objects leads to seal failures. The running gap between the primary and mating gas seal rings is typically around three microns. Injection of any type of solids or liquids into this very narrow seal running gap can cause degradation of seal performance. This would lead to an excessive gas leakage to the vent and failure of the seal eventually.

The prominent sources of gas seal contamination include:

- Process gas leakages from the inboard or high-pressure side of the seal.
- Bearing lubrication oil from the outboard or low-pressure side of the seal.
- Seal gas injected into the seal being contaminated upstream of the injection.

The key parameters to monitor for a dry gas seal are as follows:

- Primary seal gas differential pressure—monitor and trend control valve positions.
- Primary vent flow or pressure—high for primary seal leakage; low for secondary seal leakage.
- Intermediate N2 DP (If available).
- Primary vent control valve position (if available):

If opening, the primary seal leakage has increased

If closing, the secondary seal leakage has increased.

- Separation N2 pressure or differential pressure (if equipped with DP control)

12.1.3 Lube Oil System

The main purpose of lubrication oil condition monitoring and degradation detection is to determine whether the oils have deteriorated to such a degree that they no longer fulfill their functions. The lube oil system continuously provides cool and clean oil to the bearings and gear at the proper pressure, temperature, and flow rate.

High-quality pumps require a form of lubricant to cool, seal, and lubricate key internal components such as gears, bearings, and seals. The lube oil is drawn from the reservoir by the pumps and is fed under pressure through coolers and filters to the bearings.

12.2 Screw Compressor - Oil Free

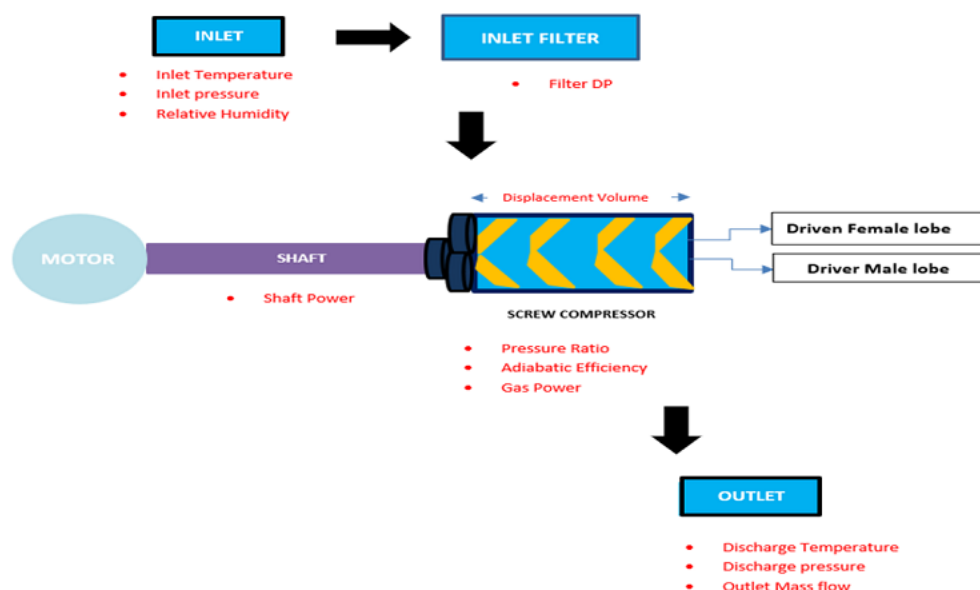
Compression is achieved by the intermeshing of the male and female rotors. Power is applied to the male rotor and as a lobe of the male rotor starts to move out of mesh with the female rotor a void is created, and the gas is taken in at the inlet port. The continuously turning rotor leads to an increase in the intermesh space and the continuous flow of gas into the compressor until the entire interlobe space is filled. Continued rotation brings male lobe into the interlobe space compressing

and moving in the direction of the discharge port. Compressed gas starts to flow out of the compressor.

Unlike reciprocation compressor, screw compressor has no clearance volume expansion. The volumetric efficiency is a function of rotor slip. **Internal** leakage from the higher pressure side to the lower pressure side happens through the following:

- Blowhole area
- Interlobe area
- Discharge clearance

Such internal leakage reduces a compressor's potential volume **capacity** and adiabatic efficiency.



12.2.1 Screw Compressor - Oil Free Performance Monitoring

The following measured parameters are monitored for if they are not maintained in the optimal operating range they bear a negative impact on performance and health of the reciprocating compressor.

Suction Temperature: Changes in the inlet temperature leads to a significant variation in performance. Lower temperature increases the capacity (weight flow) at a given discharge pressure thereby increasing power consumption. Whereas higher temperature decreases the capacity (weight flow) at a given discharge pressure and power consumption.

Suction Pressure: Lower inlet pressure decreases the capacity (weight flow), leading to reduced power consumption.

Discharge Temperature: If the compressor discharge temperature is very high then it may result in the breakdown of the lubricating oil, causing excessive wear and reduced life of the compressor valves (mainly, the discharge valve). On the contrary, if the temperature is high, undesirable chemical reactions may take place inside the compressor, especially in the presence of water. This may ultimately damage the compressor, which, in turn, increases power consumption.

Discharge Pressure: Changes in discharge pressure have effect on cylinder capacity and compression ratio. Volumetric efficiency varies slightly with compression ratio, and the required power is directly proportional to the change in compression ratio.

Volumetric Flow Rate: Volumetric flow is a key parameter for calculating volumetric efficiency of compressor. A reduced volumetric flow results in a decreased volumetric efficiency of screw compressor and vice versa. Unlike reciprocation compressor, screw compressor has no clearance volume expansion. Internal leakage from the higher pressure side to the lower pressure side happens through the following:

- Blowhole area
- Interlobe area
- Discharge clearance

Such internal leakage reduces a compressor's potential volume capacity and adiabatic efficiency.

Process Performance Overview: The process performance monitoring view in APM provides a list of the asset attributes and their details including the current value. This list can be used to review and update asset attribute information and to open an Asset Trend for viewing additional details. It also highlights (in red) the attributes if deviates from the high or low limits.

12.2.1.1 **Notations**

Below are the notations used in the equations for Performance Modeling, Expected KPI, and Operating KPI of the Oil Free model.

F	Flow
h	Specific enthalpy
d	Diameter
L	Length
Q	Volume flow
P	Pressure
T	Temperature
η	Efficiency
W	Power
γ	Adiabatic exponent
V_d	Displacement Volume
C	Rotor Profile constant
N	Rotor Speed

12.2.1.2 **Subscripts**

i	Inlet
o	Outlet

12.2.1.3 **Screw Compressor - Oil Free Model Assumptions**

The key assumptions made for developing a performance model are:

1. The model is oil free.
2. The compression process is Adiabatic.

NOTE

After analyzing various data sheets, it was observed that performance curve was of Season type and multiple molecular weight type. Hence, we have declared 6 tabular array attributes to handle curves as per variants.

NOTE

Separator calculation is included internally to model, so that the model can handle liquid fraction at inlet stream of compressor.

NOTE

If there is no performance curve data available, a separate performance model is created for oil free screw compressor. In this model, expected shaft power would be an input attribute.

12.2.1.4 Screw Compressor - Oil Free Variants

The model supports two Oil free Screw compressor variants:

- Variable speed
- Variable slide valve

12.2.1.5 Screw Compressor - Oil Free Model Input

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Inlet1 Pressure	Inlet Pressure (Required Input)	Kg/cm ² (a)	Dynamic	Measurement	Field Input
2	Outlet1 Pressure	Outlet Pressure (Required Input)	Kg/cm ² (a)	Dynamic	Measurement	Field Input
3	Inlet1 Temperature	Inlet Temperature (Required Input)	°C	Dynamic	Measurement	Field Input
4	Outlet1 Temperature	Outlet Temperature (Required Input)	°C	Dynamic	Measurement	Field Input
5	Inlet1 Volume Flow	Inlet Volume Flow (Required Input)	m ³ /hr	Dynamic	Measurement	Field Input
6	Inlet1 Actual Volumetric Flow	Inlet Stream Actual Volume Flow (Input - Port mapped from suction stream)	m ³ /hr	Dynamic	Internal	Field Input
		NOTE Inlet1 actual volumetric flow can be used while configuring a new model.				
7	Separator	Separator Volume Flow (Input)	m ³ /hr	Dynamic	Measurement	Field Input

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
	Volume Flow					
8	Rotor Speed / Slide Valve	Rotor Speed if Variable speed variant / Slide Valve position if variable slide valve variant (Required Input)	rpm	Dynamic	Measurement	Field Input
9	Measured Shaft Power	Measured Shaft Power (Required Input)	kW	Dynamic	Measurement	Field Input
10	Rotor Diameter	Rotor Diameter (Required Input)	m	Static	Design	Constant
11	Rotor Length	Rotor Length (Required Input)	m	Static	Design	Constant
12	Rotor Profile Constant	Rotor Profile Constant (Required Input)		Static	Design	Constant
13	Performance Data	Curve Data of Rotor Speed Vs Operating Shaft Power for Variable speed / Slide valve position Vs Inlet Volume flow , Operating Shaft Power for Variable slide valve		Static	Performance Map	Interpolated from performance curve
14	Efficiency Data	Volumetric Efficiency and Mechanical efficiency data		Static	Design	Constant
15	Volumetric Efficiency	Volumetric Efficiency (Input)	%	Static	Design	Constant
16	Eff mech	Mechanical Efficiency (Input)	%	Static	Design	Constant

12.2.1.6 Screw Compressor - Oil Free KPIs

The following indicators help us know the overall performance of the asset.

Input KPIs

- Inlet Pressure
- Inlet Temperature
- Outlet Pressure
- Outlet Temperature
- Inlet Volume Flow/Outlet Volume Flow
- Rotor Speed/Slide Valve (%)
- Rotor Diameter
- Rotor Length
- Rotor Profile Constant
- Volumetric Efficiency
- Performance Data

- Variable Speed

$$(N, \dot{W}_{Shaft})$$

- Variable Slide Valve

$$(\text{Slide valve position}, Q_i) \text{ and } (\text{Slide valve position}, \dot{W}_{Shaft})$$

Output KPIs

Expected KPIs

- Expected Adiabatic Efficiency
- Expected Gas Power
- Expected Shaft Power
- Expected Volume Flow

Operating KPIs

- Operating Adiabatic Efficiency
- Operating Gas Power
- Operating Shaft Power
- Operating Volumetric Efficiency

Screw Compressor - Oil Free Performance Modeling

Performance modelling of screw compressor helps determine the performance KPI. The procedure is as follows:

Overall Mass Balance

Applying continuity equation

$$F_i = F_o + F_{leakage}$$

Overall Force Balance

$$P_o = P_i + \Delta_{Pressure}$$

Overall Energy Balance

$$F_o h_o(P_o, T_o) = F_i h_i(P_i, T_i) + \dot{W}_{Gas}$$

$$F_o h_o^{Adia}(P_o, T_o^{Adia}, S_i) = F_i h_i(P_i, T_i, S_i) + \dot{W}_{Adiabatic}$$

$$\eta_{Adiabatic} = \frac{\dot{W}_{Adiabatic}}{\dot{W}_{Gas}}$$

Screw Compressor - Oil Free Expected KPIs

The following indicators help you to know the expected overall performance of the oil free screw compressor.

Expected Volume Flow (m³/hr)

If model is Oil free Screw compressor variable speed, expected volume is calculated by,

$$V_d = \frac{d^3 \left(\frac{L}{d} \right)}{C}$$

$$Q_i^{Expected} = V_d N \eta_{Volumetric}$$

If model is Oil free Screw compressor variable slide valve, expected volume is calculated by,

$$Q_i^{Expected} = f(\text{Slide Valve } \%)$$

Expected Shaft Power (kW)

The expected shaft Power is calculated using performance curve supplied by the user,

If model is Oil free Screw compressor variable speed, expected shaft power is calculated by,

$$\dot{W}_{Shaft}^{Expected} = f(N)$$

If model is Oil free Screw compressor variable slide valve, expected shaft power is calculated by,

$$\dot{W}_{Shaft}^{Expected} = f(\text{Slide Valve } \%)$$

Expected Gas Power (kW)

$$\dot{W}_{Gas}^{Expected} = \dot{W}_{Shaft}^{Expected} \eta_{Mech.}^{Expected}$$

Expected Adiabatic Efficiency (%)

$$\eta_{Adiabatic}^{Expected} = \frac{\left(\frac{\gamma}{\gamma-1} Q_i^{Expected} P_i c f_s \left(\left(\frac{P_o}{P_i} \right)^{\frac{(\gamma-1)}{\gamma}} - 1 \right) \right)}{\dot{W}_{Gas}^{Expected}}$$

Screw Compressor - Oil Free Operating KPIs

The following indicators help you to know operating overall performance of the Screw compressor:

Operating Gas Power (kW)

$$\dot{W}_{Gas}^{Operating} = F_o h_o (P_o, T_o) - F_1 h_1 (P_1, T_1)$$

Operating Shaft Power (kW)

Mechanical efficiency is a user defined input.

$$\dot{W}_{Shaft}^{Operating} = \frac{\dot{W}_{Gas}^{Operating}}{\eta_{Mech.}^{Operating}}$$

Operating Volumetric Efficiency (%)

If model is Oil free screw compressor variable speed.

$$\eta_{Volumetric}^{Operating} = \frac{Q_i}{V_d N}$$

Operating Adiabatic Efficiency (%)

$$\eta_{Adiabatic}^{Operating} = \frac{\left(\frac{\gamma}{\gamma-1} Q_i P_i c f_s \left(\left(\frac{P_o}{P_i} \right)^{\frac{(\gamma-1)}{\gamma}} - 1 \right) \right)}{\dot{W}_{Gas}^{Operating}}$$

12.2.1.7 Screw Compressor - Model Output

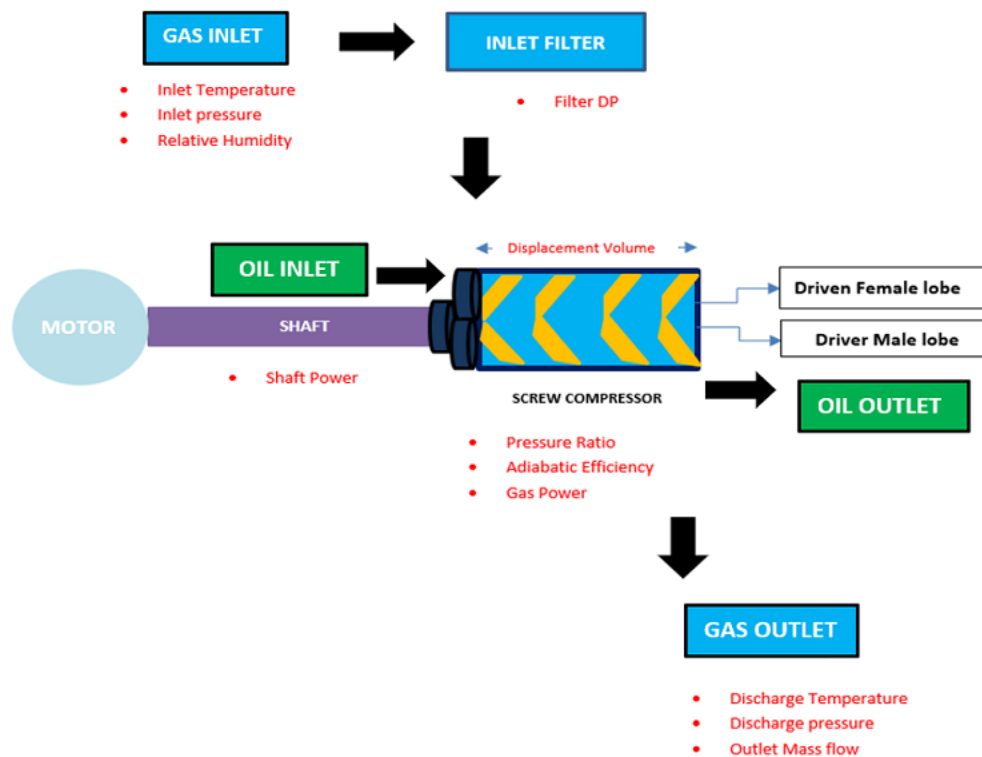
No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Expected Adiabatic Efficiency	Expected Adiabatic Efficiency (Output)	%	Dynamic	Performance	Calculated
2	Expected Gas Power	Expected Gas Power (Output)	kW	Dynamic	Performance	Calculated
3	Expected Shaft Power	Expected Shaft Power (Output)	kW	Dynamic	Performance	Calculated
4	Expected Volume Flow	Expected Volume Flow (Output)	m ³ /hr	Dynamic	Performance	Calculated
5	Expected Volumetric Efficiency	Expected Volumetric Efficiency (Output)	%	Dynamic	Performance	Calculated
6	Operating Adiabatic Efficiency	Operating Adiabatic Efficiency (Output)	%	Dynamic	Performance	Calculated
7	Operating Gas Power	Operating Gas Power (Output)	kW	Dynamic	Performance	Calculated
8	Operating Shaft Power	Operating Shaft Power (Output)	kW	Dynamic	Performance	Calculated
9	Operating Volumetric Efficiency	Operating Volumetric Efficiency (Output)	%	Dynamic	Performance	Calculated
10	Deviation Adiabatic Efficiency	Deviation Adiabatic Efficiency (Output)	%	Dynamic	Deviation	Calculated
11	Deviation Gas Power	Deviation Gas Power (Output)	%	Dynamic	Deviation	Calculated
12	Deviation Shaft Power	Deviation Shaft Power (Output)	%	Dynamic	Deviation	Calculated

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
13	Deviation Volumetric Efficiency	Deviation Volumetric Efficiency (Output)	%	Dynamic	Deviation	Calculated
14	Deviation Volume Flow	Deviation Volume Flow (Output)	%	Dynamic	Deviation	Calculated

12.3 Screw Compressor - Oil Injected

Oil-injected rotary screw compressors are generally supplied without timing gears. In oil-injected machines, the lubricant provides a layer separating the two screw profiles even as one screw drives the other. All liquid-injected machines offer the following advantages:

- The oil-injected machines provide internal cooling. Certain gases are thus kept from polymerizing or operating in an explosion-prone temperature range.
- Compared to their dry counterparts, these units achieve considerably higher compression ratios. This capability is, in part, attributable to the fact that in many services, liquid-injected machines do not require seals between the rotor chamber and the bearings. This reduces bearing span and therefore rotor deflection.



12.3.1 Screw Compressor - Oil Injected Performance Monitoring

The following measured parameters are monitored as they have impact on the performance and health of reciprocating compressor if not maintained in the operating range.

Gas Suction Temperature: Inlet temperature variations produce significant changes in performance. Lower temperature increases the capacity (weight flow) at a given discharge pressure and hence increases power consumption. Whereas Higher temperature decreases the capacity (weight flow) at a given discharge pressure and decreases power consumption.

Gas Suction Pressure: Lower inlet pressure decreases the capacity (weight flow) and power consumption due to reduced weight flow.

Gas Discharge Temperature: If the compressor discharge temperature is very high then it may result in the breakdown of the lubricating oil, causing excessive wear and reduced life of the compressor valves (mainly the discharge valve). When the temperature is high, undesirable chemical reactions may take place inside the compressor, especially in the presence of water. This may ultimately damage the compressor, increasing the power consumption eventually.

Gas Discharge Pressure: Changes in discharge pressure have effect on cylinder capacity and compression ratio. Volumetric efficiency varies slightly with compression ratio and the required power is directly proportional to the change in compression ratio.

Gas Volumetric Flow Rate: Volumetric flow is a key parameter for calculating volumetric efficiency of compressor. As volumetric flow decreases, volumetric efficiency of screw compressor, too, reduces and vice versa. Unlike reciprocation compressor, screw compressor has no clearance volume expansion. The volumetric efficiency is a function of rotor slip. This is the internal leakage from the higher pressure side to the lower pressure side through

- Blowhole area
- Inter lobe area
- Discharge clearance,

Thereby reducing potential volume capacity, and adiabatic efficiency of compressor

Oil Suction Temperature: Inlet temperature variations produce significant changes in performance. Lower temperature increases the capacity (weight flow) at a given discharge pressure and hence increases power consumption. Whereas higher temperature decreases the capacity (weight flow) at a given discharge pressure and reduces power consumption.

Oil Suction Pressure: Lower inlet pressure decreases the capacity (weight flow) and power consumption due to reduced weight flow.

Oil Discharge Temperature: If the compressor discharge temperature is very high then it may result in breakdown of the lubricating oil, causing excessive wear and reduced life of the compressor valves (mainly the discharge valve). When the temperature is high, undesirable chemical reactions may take place inside the compressor, especially in the presence of water. This may ultimately damage the compressor, increasing the power consumption eventually.

Oil Discharge Pressure: Changes in the discharge pressure have an effect on cylinder capacity and compression ratio. Volumetric efficiency varies slightly with compression ratio and the required power is directly proportional to the change in compression ratio.

Process Performance Overview: Process performance monitoring view in APM provides a list of the asset attributes and their details including the current value. This list can be used to review and update asset attribute information and open an Asset Trend to view more details. It also highlights (in red color) the attributes if deviates from the high or low limits.

12.3.1.1 Notations

F	Flow
h	Specific enthalpy
d	Diameter
L	Length
Q	Volume flow
P	Pressure
T	Temperature
η	Efficiency
W	Power

γ	Adiabatic exponent
V_d	Displacement Volume
C	Rotor Profile constant
N	Rotor Speed

12.3.1.2 Subscripts

G_i	Inlet
G_o	Outlet
O_i	Oil inlet
O_o	Oil outlet

12.3.1.3 Screw Compressor - Oil Injected Model Assumptions

The key assumptions made for developing a performance model are:

1. The model is Oil Injected.
2. The compression process is Adiabatic.

NOTE

After analyzing various data sheet it was observed that performance curve was of Season type and multiple molecular weight type. Hence, we have declared 6 tabular array attributes to handle curves as per variants.

NOTE

Separator calculation is included internally to model, so that model can handle liquid fraction at gas inlet stream of compressor.

NOTE

There is separate performance model created for oil injected screw compressor if there is no performance curve data available. For this model expected shaft power would be an input attribute.

12.3.1.4 Screw Compressor - Oil Injected Variants

The model supports two Oil Injected Screw compressor variants:

- Variable speed
- Variable slide valve

12.3.1.5 Screw Compressor - Oil Injected Model Input

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Inlet1 Pressure	Process gas inlet Pressure (Required Input)	Kg/cm ² (a)	Dynamic	Measurement	Field Input

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
2	Outlet1 Pressure	Process gas outlet Pressure (Required Input)	Kg/cm ² (a)	Dynamic	Measurement	Field Input
3	Inlet2 Pressure	Oil Inlet Pressure (Required Input)	Kg/cm ² (a)	Dynamic	Measurement	Field Input
4	Outlet2 Pressure	Oil Outlet Pressure (Required Input)	Kg/cm ² (a)	Dynamic	Measurement	Field Input
5	Inlet1 Temperature	Process gas Inlet Temperature (Required Input)	°C	Dynamic	Measurement	Field Input
6	Outlet1 Temperature	Process gas Outlet temperature (Required Input)	°C	Dynamic	Measurement	Field Input
7	Inlet2 Temperature	Oil Inlet Temperature (Required Input)	°C	Dynamic	Measurement	Field Input
8	Outlet2 Temperature	Oil Outlet temperature (Required Input)	°C	Dynamic	Measurement	Field Input
9	Inlet1 Volume Flow	Process gas Inlet Volume Flow (Required Input)	m ³ /hr	Dynamic	Measurement	Field Input
10	Inlet2 Volume Flow	Oil Inlet flow or Circulation rate (Required Input)	m ³ /hr	Dynamic	Measurement	Field Input
11	Inlet1 Actual Volumetric Flow	Process gas Actual Volume Flow (Input - Port mapped from process gas suction stream) NOTE Inlet1 actual volumetric flow can be used while configuring a new model.	m ³ /hr	Dynamic	Internal	Field Input
12	Inlet2 Actual Volumetric Flow	Oil Actual Volume Flow (Input - Port mapped from oil suction stream) NOTE Inlet2 actual volumetric flow can be used while configuring a new model.	m ³ /hr	Dynamic	Internal	Field Input
13	Separator Volume Flow	Separator Volume Flow (Input)	m ³ /hr	Dynamic	Measurement	Field Input
14	Rotor Speed / Slide Valve	Rotor Speed (Required Input) if Variable speed variant / Slide Valve position if variable slide valve variant	rpm	Dynamic	Measurement	Field Input
15	Measured	Measured Shaft Power	kW	Dynamic	Measurement	Field Input

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
	Shaft Power	(Required Input)				
16	Rotor Diameter	Rotor Diameter (Required Input)	m	Static	Design	Constant
17	Rotor Length	Rotor Length (Required Input)	m	Static	Design	Constant
18	Rotor Profile Constant	Rotor Profile Constant (Required Input)		Static	Design	Constant
19	Performance Data	Curve Data of Rotor Speed Vs Operating Shaft Power for Variable speed / Slide valve position Vs Inlet Volume flow , Operating Shaft Power for Variable slide valve		Static	Performance Map	Interpolated from performance curve
20	Efficiency Data	Volumetric Efficiency and Mechanical efficiency		Static	Design	Constant
21	Volumetric Efficiency	Volumetric Efficiency (Input)	%	Static	Design	Constant
22	Eff mech	Mechanical Efficiency (Input)	%	Static	Design	Constant

12.3.1.6 Screw Compressor - Oil Injected KPIs

The following indicators help know the overall performance of the asset.

Input KPIs

- Inlet Pressure
- Inlet Temperature
- Outlet Pressure
- Outlet Temperature
- Oil Inlet Pressure
- Oil Inlet Temperature
- Oil Outlet Pressure (If not available can be set equal to Gas Outlet pressure)
- Oil Outlet Temperature (If not available can be set equal to Gas Outlet temperature)
- Gas Inlet Volume Flow / Outlet Volume Flow
- Oil Inlet Volume Flow / Outlet Volume Flow
- Rotor Speed / Slide Valve (%)
- Rotor Diameter
- Rotor length
- Rotor profile constant
- Volumetric efficiency
- Performance Data

- Variable Speed

$$(N, \dot{W}_{Shaft})$$

- Variable Slide Valve

$$(Slide\ valve\ position, Q_i) \text{ and } (Slide\ valve\ position, \dot{W}_{Shaft})$$

Output KPIs

Expected KPIs

- Expected Adiabatic Efficiency
- Expected Gas Power
- Expected Shaft Power
- Expected Volume Flow

Operating KPIs

- Operating Adiabatic Efficiency
- Operating Gas Power
- Operating Shaft Power
- Operating Volumetric Efficiency

Screw Compressor - Oil Injected Performance Modeling

Performance modelling of screw compressor helps determine the performance KPI. The procedure is as follows:

Overall Mass Balance

Applying continuity equation:

$$F_{G_o} = F_{G_i} + F_{leakage}$$

$$F_{O_o} = F_{O_i}$$

Overall Force Balance

$$P_{G_o} = P_{G_i} + \Delta_{Pressure}$$

$$P_{O_o} = P_{G_o}$$

Overall Energy Balance

$$F_{G_o} h_{G_o}(P_{G_o}, T_{G_o}) = F_{G_i} h_{G_i}(P_{G_i}, T_{G_i}) + F_{O_o} h_{O_o}(P_{O_o}, T_{O_o}) - F_{O_i} h_{O_i}(P_{O_i}, T_{O_i}) + \dot{W}_{Gas}$$

$$\eta_{Adiabatic} = \frac{\dot{W}_{Adiabatic}}{\dot{W}_{Gas}}$$

Screw Compressor - Oil Injected Expected KPIs

The following indicators help you know the expected overall performance of the oil injected screw compressor.

Expected Volume Flow (m³/hr)

If model is Oil Injected Screw compressor variable speed, expected volume is calculated by the following equation:

$$V_d = \frac{d^3 \left(\frac{L}{d} \right)}{C}$$

$$Q_{G_i}^{Expected} = V_d N \eta_{Volumetric}$$

If model is Oil Injected Screw compressor variable slide valve, expected volume is calculated by the following equation:

$$Q_{G_i}^{Expected} = f(\text{Slide Valve } \%)$$

Expected Shaft Power (kW)

The expected shaft Power is calculated using performance curve supplied by the user.

If model is Oil Injected Screw compressor variable speed, expected shaft power is calculated by the following equation:

$$\dot{W}_{Shaft}^{Expected} = f(N)$$

If model is Oil Injected Screw compressor variable slide valve, expected shaft power is calculated by the following equation:

$$\dot{W}_{Shaft}^{Expected} = f(\text{Slide Valve } \%)$$

Expected Gas Power (kW)

$$\dot{W}_{Gas}^{Expected} = \dot{W}_{Shaft}^{Expected} \eta_{Mech.}^{Expected}$$

Expected Adiabatic Efficiency (%)

$$\eta_{Adiabatic}^{Expected} = \frac{\left(\frac{\gamma}{\gamma-1} Q_{G_i}^{Expected} P_{G_i} c f_s \left(\left(\frac{P_{G_o}}{P_{G_i}} \right)^{\frac{(\gamma-1)}{\gamma}} - 1 \right) \right)}{\dot{W}_{Gas}^{Expected}}$$

Screw Compressor - Oil Injected Operating KPIs

The following indicators help us know the overall operating performance of the screw compressor.

Operating Gas Power (kW)

$$\dot{W}_{Gas}^{Operating} = (F_{G_o} h_{G_o}(P_{G_o}, T_{G_o}) - F_{G_i} h_{G_i}(P_{G_i}, T_{G_i})) + (F_{O_o} h_{O_o}(P_{O_o}, T_{O_o}) - F_{O_i} h_{O_i}(P_{O_i}, T_{O_i}))$$

Operating Shaft Power (kW)

Mechanical efficiency is a user-defined input:

$$\dot{W}_{Shaft}^{Operating} = \frac{\dot{W}_{Gas}^{Operating}}{\eta_{Mech.}^{Operating}}$$

Operating Volumetric Efficiency (%)

If the model is of the oil free screw compressor variable speed type:

$$\eta_{Volumetric}^{Operating} = \frac{Q_{Gi}}{V_d N}$$

Operating Adiabatic Efficiency (%)

$$\eta_{Adiabatic}^{Operating} = \frac{\left(\frac{\gamma}{\gamma-1} Q_{Gi} P_{Gi} c_{fs} \left(\left(\frac{P_{Ge}}{P_{Gi}} \right)^{\frac{(\gamma-1)}{\gamma}} - 1 \right) \right)}{\dot{W}_{Gas}^{Operating}}$$

12.3.1.7 Screw Compressor - Oil Injected Model Output

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Expected Adiabatic Efficiency	Expected Adiabatic Efficiency (Output)	%	Dynamic	Performance	Calculated
2	Expected Gas Power	Expected Gas Power (Output)	kW	Dynamic	Performance	Calculated
3	Expected Shaft Power	Expected Shaft Power (Output)	kW	Dynamic	Performance	Calculated
4	Expected Volume Flow	Expected Volume Flow (Output)	m ³ /hr	Dynamic	Performance	Calculated
5	Expected Volumetric Efficiency	Expected Volumetric Efficiency (Output)	%	Dynamic	Performance	Calculated
6	Operating Adiabatic Efficiency	Operating Adiabatic Efficiency (Output)	%	Dynamic	Performance	Calculated
7	Operating Gas Power	Operating Gas Power (Output)	kW	Dynamic	Performance	Calculated
8	Operating Shaft Power	Operating Shaft Power (Output)	kW	Dynamic	Performance	Calculated
9	Operating Volumetric Efficiency	Operating Volumetric Efficiency (Output)	%	Dynamic	Performance	Calculated
10	Deviation Adiabatic Efficiency	Deviation Adiabatic Efficiency (Output)	%	Dynamic	Deviation	Calculated
11	Deviation Gas Power	Deviation Gas Power (Output)	%	Dynamic	Deviation	Calculated
12	Deviation Shaft Power	Deviation Shaft Power (Output)	%	Dynamic	Deviation	Calculated

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
13	Deviation Volumetric Efficiency	Deviation Volumetric Efficiency (Output)	%	Dynamic	Deviation	Calculated
14	Deviation Volume Flow	Deviation Volume Flow (Output)	%	Dynamic	Deviation	Calculated

12.4 Screw Compressor - Fault Model

12.4.1 Screw Compressor - APM Model

Item	Type
Category	Equipment
Equipment Class	Screw Compressor
Equipment Type	Oil Free Variable Slide Valve Screw Compressor,
	Oil Free Variable Speed Screw Compressor,
	Oil Injected Variable Slide Valve Screw Compressor,
	Oil Injected Variable Speed Screw Compressor.

Below table illustrates the list of faults and symptoms configured for a Screw Compressor APM - System for Performance

Fault Name	Associated Symptoms	Recommendations
Compressor performance degradation	Pressure deviation RH (CPD)	- For non-air, non-closed-loop applications: Perform a gas analysis or reconfirm molecular weight. - Check instrument accuracy (flow meter suction and discharge temperature and pressure transmitters). - If all of the above is ok, then performance deviation is genuine. - At this point: - Confirm IGV operability. - Check for casing leaks. - Confirm seal integrity. - If all of the above is ok the reason for degradation could be internal clearances have increased and/or the flow path has fouled up or has been damaged. - In this case at the first opportunity: - Carry out a borescope inspection to verify if there is internal damage. - If fouling is identified. - Wash the machine if possible. - Monitor efficiency to determine the speed of degradation and plan rotor replacement when it becomes unsustainable. - If fouling: - Short term: Wash compressor if possible.

Fault Name	Associated Symptoms	Recommendations
	Compressor Volumetric efficiency Deviation RH (CPD)	- Long term: investigate the application of coating on the affected gas path. - Change gas inlet Temp. - Borescope inspection at the opportunity. - If corrosion/erosion: Replace corroded parts with better material grade. - In case of erosion consider coating of gas path. - Borescope at the opportunity.
Fouling / Corrosion / Erosion Risk	Compressor Volumetric efficiency Deviation RH (FCER)	- Verify if process conditions have changed. - Verify if there is any correlation with environmental conditions (sun exposure, ambient temp, rain, etc.) - Take a sample of the gas if possible.
Change of gas properties (MW)	Compressor Volumetric efficiency Deviation RH (CGPA)	- For non-air, non-closed-loop applications: Perform a gas analysis or reconfirm molecular weight. - Check instruments accuracy (flow meter suction and discharge temperature and pressure transmitters). - If all of the above is ok then performance deviation is genuine. - At this point: - Confirm IGV operability. - Check for casing leaks. - Confirm seal integrity.
	Gas/air Composition Analysis deviation RH (CGPA)	- If all of the above is ok the reason for degradation could be internal clearances have increased and/or the flow path has fouled up or has been damaged. - In this case at the first opportunity: - Carry out borescope inspection to verify if there is internal damage. - If fouling is identified. - wash the machine if possible. - Monitor efficiency to determine the speed of degradation and plan rotor replacement when it becomes unsustainable.

12.4.2 Screw Compressor - APR Model

12.4.2.1 Screw Compressor Bearing Fault Model

Item	Type
Category	Equipment
Equipment Class	Screw Compressor
Equipment Type	Screw Compressor - Bearing System

The below table illustrates the list of faults and symptoms configured for the Screw Compressor Bearing Fault Model.

Fault Name	Associated Symptoms	Recommendations
SC BRG BEARING OIL QUAL ANOMALY	M RTR RB VIB X DE RH (BLOQ)	1. Recondition or replace oil. 2. Check the oil tank for free water by draining 1L into a sample bottle. 3. Switch cooler if water contamination. 4. Take an oil sample and check water content, Viscosity, and TAN. 5. Check no breather caps or man covers have been removed and left open to allow debris into the tank.
	M RTR RB VIB Y DE RH (BLOQ)	
	M RTR RB TEMP A DE RH (BLOQ)	
	M RTR RB TEMP B DE RH (BLOQ)	
	M RTR RB VIB X NDE RH (BLOQ)	
	M RTR RB VIB Y NDE RH (BLOQ)	
	M RTR RB TEMP A NDE RH (BLOQ)	
	M RTR RB TEMP B NDE RH (BLOQ)	
	FM RTR RB VIB X DE RH (BLOQ)	
	FM RTR RB VIB Y DE RH (BLOQ)	
	FM RTR RB TEMP A DE RH (BLOQ)	
	FM RTR RB TEMP B DE RH (BLOQ)	
	FM RTR RB VIB X NDE RH (BLOQ)	
	FM RTR RB VIB Y NDE RH (BLOQ)	
	FM RTR RB TEMP A NDE RH (BLOQ)	
SC BER DE ML TB MECH ANOMALY	FM RTR RB TEMP B NDE RH (BLOQ)	1. Assess the likely cause and time to trip if intervention is not possible. 2. Intervention options; reduce thrust load. 3. Ensure parts availability 4. Check axial vibration trends to establish if the machine has surged and potentially damaged/wiped the pads. 5. If required inspect the thrust pads and collar to check for damage/wiping. 6. Carry out a bump check to establish what float is in the machine. 7. Compare to the previous overhaul report for measurement.

Fault Name	Associated Symptoms	Recommendations
		8. Check the gear coupling/pre-stretch diaphragm coupling.
SC BER DE FL TB MECH ANOMALY	SC LO VISCOSITY SH (BLOQ)	1. Assess the likely cause and time to trip if intervention is not possible. 2. Intervention options; reduce thrust load. 3. Ensure parts availability 4. Check axial vibration trends to establish if the machine has surged and potentially damaged/wiped the pads. 5. If required inspect the thrust pads and collar to check for damage/wiping. 6. Carry out a bump check to establish what float is in the machine. 7. Compare to the previous overhaul report for measurement. 8. Check the gear coupling/pre-stretch diaphragm coupling.
SC BER DE ML RB MECH ANOMALY	SC LO VISCOSITY SL (BLOQ)	1. Investigate with high-frequency vibration analysis. 2. Assess the likely cause and time to fail(trip) if intervention is not possible. 3. Intervention options; reduce speed. 4. Parts availability. 5. Check the trends to establish if it's either both X and Y probes or one. 6. Inspect the bearing for the following. 1. wiping. 2. Internal clearances. 3. Oil coking/hot spots. 7. Check for correctness in alignment. 8. Change bearing if required.
SC BER NDE ML RB MECH ANOMALY	SC LO TAN SH (BLOQ)	1. Investigate with high-frequency vibration analysis. 2. Assess the likely cause and time to fail(trip) if intervention is not possible. 3. Intervention options; reduce speed. 4. Parts availability. 5. Check the trends to establish if it's either both X and Y probes or one. 6. Inspect the bearing for the following. 1. Wiping. 2. Internal clearances. 3. Oil coking/hot spots. 7. Check for correctness in alignment.

Fault Name	Associated Symptoms	Recommendations
		8. Change bearing if required.
SC BER DE FL RB MECH ANOMALY	FM RTR RB VIB X DE RH (DE FL RB MA)	1. Investigate with high-frequency vibration analysis. 2. Assess the likely cause and time to fail(trip) if intervention is not possible. 3. Intervention options; reduce speed. 4. Parts availability. 5. Check the trends to establish if it is either both X and Y probes or one. 6. Inspect bearing for the following. 1. Wiping. 2. Internal clearances. 3. Oil coking/hot spots. 7. Check for correctness in alignment. 8. Change bearing if required.
	FM RTR RB VIB Y DE RH (DE FL RB MA)	
	FM RTR RB TEMP A DE RH (DE FL RB MA)	
	FM RTR RB TEMP B DE RH (DE FL RB MA)	
SC BER NDE FL RB MECH ANOMALY	FM RTR RB VIB X NDE RH (NDE FL RB MA)	1. Investigate with high-frequency vibration analysis. 2. Assess the likely cause and time to fail(trip) if intervention is not possible. 3. Intervention options; reduce speed. 4. Parts availability. 5. Check the trends to establish if it is either both X and Y probes or one. 6. Inspect bearing for the following. 1. Wiping. 2. Internal clearances. 3. Oil coking/hot spots. 7. Check for correctness in alignment. 8. Change bearing if required.
	FM RTR RB VIB Y NDE RH (NDE FL RB MA)	
	FM RTR RB TEMP A NDE RH (NDE FL RB MA)	
	FM RTR RB TEMP B NDE RH (NDE FL RB MA)	
SC BER ROTOR ANOMALY	M RTR RB VIB X DE RH (RTRA)	1. Investigate with high-frequency vibration analysis. 2. Check Carbon brush integrity. 3. Check the magnetism of the rotor. 4. Check rotor imbalance, misalignment, and Looseness. 5. Check coupling integrity. 6. Check foundation bolt integrity. 7. Check both bearings for clearance and pad wear etc.
	M RTR RB VIB Y DE RH (RTRA)	
	M RTR RB VIB X NDE RH (RTRA)	
	M RTR RB VIB Y NDE RH (RTRA)	
	FM RTR RB VIB X DE RH (RTRA)	
	FM RTR RB VIB Y DE RH (RTRA)	
	FM RTR RB VIB X NDE	

Fault Name	Associated Symptoms	Recommendations
	RH (RTRA)	
	FM RTR RB VIB Y NDE RH (RTRA)	
	SC MTR SHAFT VIB A RH (RTRA)	
SC BRG TIMING GEAR ANOMALY	SC MTR SHAFT VIB B RH (RTRA)	1. Maintain proper lubrication. 2. Alarm high-frequency vibrations. 3. Inspect timing gear.
SC BRG BEARING OIL TEMP ANOMALY	SC CASING VIB RH (RTRA)	1. Check cooling water supply pressure. 2. Check the bypass control valve. 3. Open both coolers and stabilize the temperature. 4. If a malfunction of one cooler is suspected isolate and inspect it. 5. Line walk the system to check for leaks. 6. Check cooling water pumps for blockages. 7. Ensure the strainer is clean. 8. If the cooler is an air cooler check fan and motor integrity.
SC BRG BEARING OIL PRES ANOMALY	SC CASING VIB 1 RH (RTRA)	1. Switch on both oil pumps and stabilize the oil pressure. 2. Assess malfunction. 3. Line walk the system and check for leaks. 4. Check filter DP. 5. Ensure PSV's are seated. 6. Check spillback/PCV valves are set correctly and not returning oil back to the tank. 7. If maintenance has recently been performed. 8. Check the correct rotation of the pump. 9. Ensure no hand valves have been left shut. 10. Ensure NRV's are working correctly. 11. Check the oil level in tank. 12. Check to function of the auto change-over switch.
SC PERF INLET FIL ANOMALY	SC IN FILTER DP RH (IFA)	1. Check the calibration of the sensor. 2. Change over the filter if stand-by is available and replace the filter.
SC PERF LIQ SLUG	SC SCR LIQ LEVEL RH (LS)	1. Short term: reduce flow. 2. The correct problem at the process level (reduce scrubber level or act on the process). 3. Monitor seals leakages. 4. Axial shaft position.

Fault Name	Associated Symptoms	Recommendations
		5. Thrust bearing temperature.
SC PERF SUC THROTTLE VALVE ANOMALY	SC SUC THROT VAL RL (IGVA)	1. Check IGV/Suction throttle integrity and operability. 2. Check the calibration of the positioner. 3. Check IGV/Suction throttle actuator and feed. 4. Check all linkage connections. 5. Carry out a borescope examination to identify the presence of internal damage.
	SC SUC THROT VAL RH (IGVA)	
SC PERF IS COOLER ANOMALY	SC INT STG CLR OUT TEMP SH (ISCA)	1. Check the interstage cooler's integrity. 2. Check the temp control valve for any anomaly. 3. Conduct loop proof test for the temperature control valve. 4. Check the cooling water supply line for any leakage/blockage/ supply issue.
	SC INT STG CLR OUT TEMP SL (ISCA)	
	SC INT STG CLR TCV POS SH (ISCA)	
	SC INT STG CLR TCV POS SL (ISCA)	
SC PERF IS SEP ANOMALY	SC INT STG SEP LVL SH (ISSA)	1. Check the Separator/scrubber's internals. 2. Check the drain line of the separator. 3. Check the transmitter.

12.4.2.2 Screw Compressor - Intercooler System Fault Model

Item	Type
Category	Equipment
Equipment Class	Screw Compressor
Equipment Type	Screw Compressor - Intercooler System

The below table illustrates the list of faults and symptoms configured for a Screw Compressor - Intercooler System.

Fault Name	Associated Symptoms	Recommendations
SC IC LOSS AIR COOLING	Intercooler gas outlet temperature RH (LOCA)	1. Check cooling water supply pressure and temperature. 2. Check CW flow control valve. 3. Check cooling water pumps for blockages. 4. Ensure the strainer is clean. 5. If all of the above are ok, then intercooler fouling could be present.
	Interstage Cooler fan speed RL (LOCA)	
	Interstage Cooler fan speed RH (LOCA)	
SC IC LOSS WATER COOLING	Intercooler gas outlet temperature RH (LOCA)	1. Check cooling water supply pressure and temperature. 2. Check CW flow control valve. 3. Check cooling water pumps for blockages. 4. Ensure the strainer is clean.

Fault Name	Associated Symptoms	Recommendations
	Interstage Cooler fan speed RL (LOCA)	5. If all the above are ok, then intercooler fouling could be present.
	Interstage Cooler fan speed RH (LOCA)	
	Intercooler gas outlet temperature RH (LOCA)	
	Interstage Cooler fan speed RL (LOCA)	

12.4.2.3 Screw Compressor - Lube Oil System

Item	Type
Category	Equipment
Equipment Class	Screw Compressor
Equipment Type	Screw Compressor - LubeOil System

The below table illustrates the list of faults and symptoms configured for a Screw Compressor - Lube Oil System.

Fault Name	Associated Symptoms	Recommendations
Loss of lube oil quality	SC LO VISCOSITY HIGH (LOLQ)	1. Check the oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, and TAN. 4. Check no breather caps or man covers have been removed and left open to allow debris into the tank. 5. Do filter cleaning. 6. Recondition or replace oil.
	SC LO VISCOSITY LOW (LOLQ)	
	SC LO TAN HIGH (LOLQ)	
Lube Oil Filter Anomaly	SC LO FLTR DP RH (LOFA)	1. Check filter DP. 2. Changeover and clean/replace the stand-by filter. 3. Check the Filter DP transmitter.
Loss of pressure due to Pump anomaly	SC LO HEADER PRES RL (LOPA)	1. Check the Discharge PT integrity. 2. Switch on both oil pumps and stabilize the oil pressure. 3. Assess malfunction. 4. Line walk the system and check for leaks. 5. Ensure PSVs are seated. 6. Check spillback/PCV valves are set correctly and not returning oil back to the tank. 7. Ensure no hand valves have been left shut for the standby pump. 8. Check the oil level in tank.

Fault Name	Associated Symptoms	Recommendations
	SC LO PUMP DISC PRES RL (LOPA)	9. Open tank overhead vents for release of pressure. 10. Check PCV function for Integrity. 11. Assess malfunction. 12. Check Auto Logic Function tests. 13. Check the functioning of the auto change-over switch. 14. If maintenance has recently been performed check the correct rotation of the pump. 15. Ensure NRV's are working correctly.
	SC LO PCV POSITION RL (LOPA)	
	SC LO M PUMP RUN STATUS ON (LOPA)	
	SC LO A PUMP RUN STATUS ON (LOPA)	
	SC LO PSL STATUS ON (LOPA)	
Loss of cooling due to TCV Anomaly	SC LO HEADER TEMP RH (TCVA)	1. Check the calibration of the sensor. 2. Check the bypass control valve for integrity. 3. Open both coolers and stabilize the temperature. 4. Line walk the system to check for leaks. 5. Check the Isolation/bypass valve of TCV.
	SC LO CO BYPASS TCV POSITION RH (TCVA)	
	SC LO COOLER OUT TEMP RH (TCVA)	
Loss of cooling due to Cooler Anomaly	SC LO HEADER TEMP RH (LOCA)	1. Check the calibration of the sensor. 2. Check the cooling water supply pressure. 3. Open both coolers and stabilize the temperature. 4. If a malfunction of one cooler is suspected, isolate and inspect it. 5. Line walk the system to check for leaks. 6. Check cooling water pumps for blockages. 7. Ensure the strainer is clean.
	SC LO TANK CO DEL TEMP RH (LOCA)	
Loss of heating due lube oil tank heater OFF	SC LO HEADER TEMP RL (THOA)	1. Check the Heater ON/OFF switch integrity and Change the switch as per the requirement. 2. Check the heater circuit for Voltage and current.
	SC LO TANK TEMP RL (THOA)	
	SC LO HTR STATUS OFF (THOA)	
Loss of cooling due lube oil tank heater ON	SC LO HEADER TEMP RH (THO)	1. Check the Heater ON/OFF switch integrity and Change the switch as per the requirement. 2. Offline 1. 3. Check the heater circuit for Voltage and current.
	SC LO TANK TEMP RH (THO)	
	SC LO HTR STATUS ON (THO)	
Loss of cooling Due to Fan-Motor Issue	SC LO HEADER TEMP RH (LCMI)	1. Offline: Check the Fan and Motor /coupling/belt Integrity and rectify.
	SC LO FAN1 STATUS OFF (LCMI)	

Fault Name	Associated Symptoms	Recommendations
	SC LO FAN2 STATUS OFF (LCMI)	
	SC LO COOLER OUT TEMP RH (LCMI)	
Low Reservoir Oil Level	SC LO TANK LEVEL LOW (ROLL)	1. Check The tank for Sensor related faults. 2. Check the line for leakages, and Ensure that the Lube oil is refilled above the min operating level.
High Reservoir Oil Level	SC LO TANK LEVEL HIGH (ROLH)	1. Check The tank for Sensor related faults. 2. Check the vent Position, and Ensure that the Lube oil is maintained below the maximum operating level.
Blockage in Cooler Rerun Line	SC LO COOLER OUT FLOW LOW (RFL)	1. Check the line for blockages and ensure that the flow throughput is at the required level.

12.4.2.4 Screw Compressor - Seal Oil System

Item	Type
Category	Equipment
Equipment Class	Screw Compressor
Equipment Type	Screw Compressor - Seal Oil System

The below table illustrates the list of faults and symptoms configured for a Screw Compressor - Seal Oil System.

Fault Name	Associated Symptoms	Recommendations
SC SO SEAL FIL DP ANOMALY	Seal Oil Filter DP 1 Main Pump RH (SFDPA)	1. Check the calibration of the transmitter 2. Change the filter 3. Replace the affected filter.
	Seal Oil Filter DP 2 Auxiliary Pump RH (SFDPA)	
SC SO OIL PRES LOW PCV ANOMALY	Seal Oil Header Pressure (1st casing) RL (PCVA)	1. Check calibration of pressure transmitters, PCVs, etc. 2. Check calibration/integrity of PCVs etc.
	Seal Oil Header Pressure (2nd casing) RL (PCVA)	
	Seal oil PCV Opening RL (PCVA)	
	Seal oil return PCV Opening RH (PCVA)	
SC SO OIL PRES ANOMALY	Seal Oil Header Pressure (1st casing) RL (PLA)	1. Check the calibration of pressure transmitters. 2. Check if the leak is appearing in RV/PSV. 1. if yes, replace RV/ repair or replace PSV. 3. Check if the leak is happening in the seal line.
	Seal Oil Header Pressure (2nd casing) RL (PLA)	
SC SO OIL TEMP HIGH	Seal Oil Header Temp (1st casing) RH (THA)	1. Check the calibration of the temperature transmitter. 2. Assess the reason for higher temp and seal damage if

Fault Name	Associated Symptoms	Recommendations
	Seal Oil Header Temp (2nd casing) RH (THA)	any.
	Seal Tank Temp (Casing 1) RH (THA)	
	Seal Tank Temp (Casing 2) RH (THA)	
SC SO OIL PUMP ANOMALY	Seal Oil Pump -1 delivery pressure RL (PA)	1. Check the calibration of pressure transmitters. 2. Check the integrity of the motor/ pump/ coupling. 3. Check if the suction line is choked/leaked.
	Seal Oil Pump -2 delivery pressure RL (PA)	
SC SO OIL SEP LEVEL LOW	Overhead seal oil tank level RL (LSOL)	1. Check The tank for Sensor related faults. 2. Check the line for leakages, and Ensure that the Lube oil is refilled above the min operating level.
	Separator LSL Switch ON (LSOL)	
SC SO OIL SEP LEVEL HIGH	Overhead seal oil tank level RH (HSOL)	1. Check The tank for Sensor related faults. 2. Check the vent Position, and Ensure that the Lube oil is maintained below the maximum operating level.
SC SO OIL SEP LEVEL LOW PERMISSIBLE	Overhead seal oil tank level ROCL LT (SOBPL)	1. Check The tank for Sensor related faults. 2. Check the line for leakages, and Ensure that the Lube oil is refilled above the min operating level.
	Separator LSL Switch ON (SOBPL)	

12.4.2.5 Screw Compressor -Dry Gas Seal System

Item	Type
Category	Equipment
Equipment Class	Screw Compressor
Equipment Type	Screw Compressor -Dry Gas Seal System

The below table illustrates the list of faults and symptoms configured for a Screw Pump Transmission system.

Fault Name	Associated Symptoms	Recommendations
Seal Gas Filter Anomaly	Supply Seal Gas Filter DP RH (SGFA)	1. Check the calibration of the sensor. 2. Check the filter efficiency and change to standby filter if required. 3. Repair/replace the filter internals.
DE Primary PDCV Malfunction	DE Primary-Suction DP RL (DE PDCVA)	1. Check the calibration of the sensor. 2. Check valve set point. 3. Check valve internals for any damage and repair / replace if necessary.

Fault Name	Associated Symptoms	Recommendations
	DE PDCV Valve Opening RH (DE PDCVA)	4. Conduct the PDCV control loop function test.
NDE Primary PDCV Malfunction	NDE Primary-Suction DP RL (NDE PDCVA)	1. Check the calibration of the sensor. 2. Check valve set point. 3. Check valve internals for any damage and repair / replace if necessary.
	NDE PDCV Valve Opening RH (NDE PDCVA)	4. Conduct the PDCV control loop function test.
No Lift off or Back-Flow (No Seal Gas) At Low-Speed Transients	Vent Pressure RH (BFA)	1. Seal faces: overheating. 2. Cracking.
	Vent Flow RL (BFA)	3. Scoring. 4. Inspect seal and replace.
DE Male Primary Seal Gas Supply Anomaly	DE Male Primary Secondary DP RL (DE ML PSGSA)	1. Check the calibration of the sensor. 2. Check valves (if any). 3. The transmitters are all working correctly. 4. Monitor the trend of seal gas system. 5. Inspect the system for leakage. 6. Inspect Line and check functionality. 7. Replace filters (if any) if necessary.
DE Female Primary Seal Gas Supply Anomaly	DE Female Primary and Secondary DP RL (DE FL PSGSA)	1. Check the calibration of the sensor. 2. Check valves (if any). 3. The transmitters are all working correctly. 4. Monitor the trend of the seal gas system. 5. Inspect the system for leakage. 6. Inspect Line and check functionality. 7. Replace filters (if any) if necessary.
NDE Male Primary Seal Gas Supply Anomaly	NDE Male Primary and Secondary DP RL (NDE ML PSGSA)	1. Check the calibration of the sensor. 2. Check valves (if any). 3. The transmitters are all working correctly. 4. Monitor the trend of seal gas system. 5. Inspect the system for leakage. 6. Inspect Line and check

Fault Name	Associated Symptoms	Recommendations
		functionality. 7. Replace filters (if any) if necessary.
NDE Female Primary Seal Gas Supply Anomaly	NDE Female Primary and Secondary DP RL (NDE FL PSGSA)	1. Check the calibration of the sensor. 2. Check valves (if any). 3. The transmitters are all working correctly. 4. Monitor the trend of seal gas system. 5. Inspect the system for leakage. 6. Inspect Line and check functionality. 7. Replace filters (if any) if necessary.
Secondary Gas Supply Pressure Anomaly	Secondary Gas Supply Pressure RL (SGSPA)	1. Check the calibration of the sensor. 2. Check valves (if any). 3. The transmitters are all working correctly. 4. Monitor the trend of seal gas system. 5. Inspect the system for leakage. 6. Inspect Line and check functionality. 7. Replace filters (if any) if necessary.
	Secondary Gas Supply Pressure RH (SGSPA)	
DE Male Secondary Seal Gas Supply Anomaly	DE Male Secondary Flow RL (DE ML SSGSA)	1. Check the calibration of the sensor. 2. Check valves (if any). 3. The transmitters are all working correctly. 4. Monitor the trend of seal gas system. 5. Inspect the system for leakage. 6. Inspect Line and check functionality. 7. Replace filters (if any) if necessary.
	DE Male Secondary Flow RH (DE ML SSGSA)	
DE Female Secondary Seal Gas Supply Anomaly	DE Female Secondary Flow RL (DE FL SSGSA)	1. Check the calibration of the sensor. 2. Check valves (if any). 3. The transmitters are all working correctly. 4. Monitor the trend of seal gas system. 5. Inspect the system for leakage. 6. Inspect Line and check functionality.

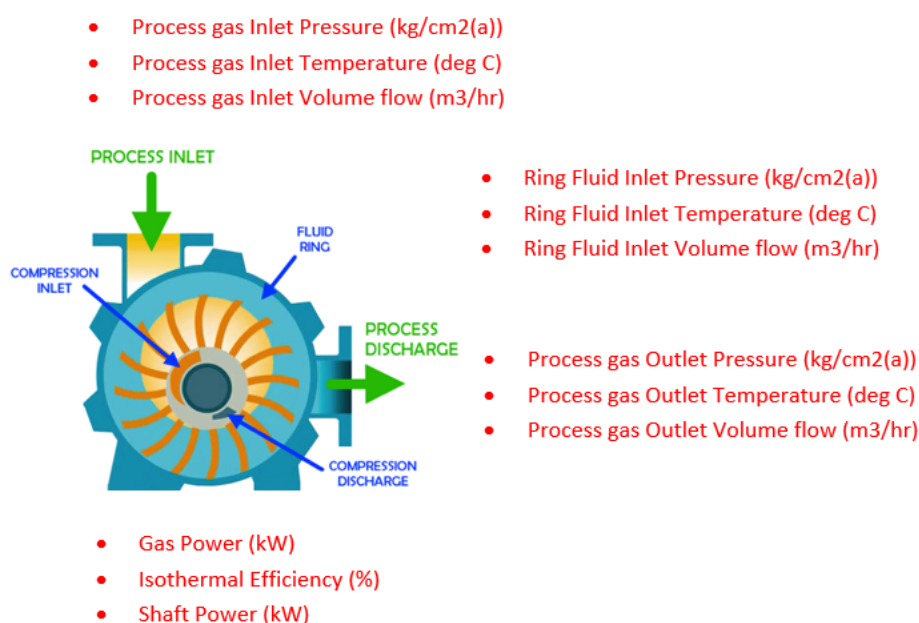
Fault Name	Associated Symptoms	Recommendations
	DE Female Secondary Flow RH (DE FL SSGSA)	7. Replace filters (if any) if necessary.
NDE Male Secondary Seal Gas Supply Anomaly	NDE Male Secondary Flow RL (NDE ML SSGSA)	1. Check the calibration of the sensor. 2. Check valves (if any). 3. The transmitters are all working correctly. 4. Monitor the trend of seal gas system. 5. Inspect the system for leakage. 6. Inspect Line and check functionality. 7. Replace filters (if any) if necessary.
	NDE Male Secondary Flow RH (NDE ML SSGSA)	
NDE Female Secondary Seal Gas Supply Anomaly	NDE Female Secondary Flow RL (NDE FL SSGSA)	1. Check the calibration of the sensor. 2. Check valves (if any). 3. The transmitters are all working correctly. 4. Monitor the trend of seal gas system. 5. Inspect the system for leakage. 6. Inspect Line and check functionality. 7. Replace filters (if any) if necessary.
	NDE Female Secondary Flow RH (NDE FL SSGSA)	
Separator Gas Supply Anomaly.	Separation Gas Pressure RL (SGSA)	1. Check the calibration of the sensor. 2. Check valves (if any). 3. The transmitters are all working correctly. 4. Monitor the trend of seal gas system. 5. Inspect the system for leakage. 6. Inspect Line and check functionality. 7. Replace filters (if any) if necessary.
	Separation Gas Pressure RH (SGSA)	
	Separation Gas Flow RL (SGSA)	
	Separation Gas Flow RH (SGSA)	

LIQUID RING COMPRESSOR MONITORING

The liquid-ring compressor, compresses gas by rotating a vaned impeller located eccentrically within a cylindrical casing. Ring liquid (usually water) is fed into the compressor and, by centrifugal acceleration, forms a moving cylindrical ring against the inside of the casing. This liquid ring creates a series of seals in the space between the impeller vanes, which form compression chambers. The eccentricity between the impeller's axis of rotation and the casing geometric axis results in a cyclic variation of the volume enclosed by the vanes and the ring.

Gas, often air, is drawn into the compressor through an inlet port in the end of the casing. The gas is trapped in the compression chambers formed by the impeller vanes and the liquid ring. The reduction in volume caused by the impeller rotation compresses the gas, which reports to the discharge port in the end of the casing. Compressed gas on discharge of compressor contains certain amount of working liquid which is usually removed in vapor-liquid separator.

Liquid-ring compressor can use any liquid compatible with the process, provided it has appropriate vapor pressure properties as the sealant liquid. Although the most common sealant is water, almost any liquid can be used. The second most common is oil as it has a very low vapor pressure and oil-sealed liquid-ring compressor is typically air-cooled. For dry chlorine gas applications, concentrated sulfuric acid is used.



13.1 Liquid Ring Compressor - Sub Systems

13.1.1 Seal Gas System

A Dry gas seal system is designed to provide reliable shaft sealing on rotating equipment to reduce the amount of process gas escaping from the machine's primary flow path to an acceptable level.

Loss of buffer gas will in most cases reverse the differential pressure across the seal faces, which will damage the seal in short order. The seals will operate at a zero differential pressure level.

Dry Gas Seal Degradation: Contamination of the seal by foreign objects leads to seal failures. The running gap between the primary and mating gas seal rings is typically around three microns. Injection of any type of solids or liquids into this very narrow seal running gap can cause degradation of seal performance. This would create excessive gas leakage to the vent and eventual failure of the seal.

The major areas from which gas seals contamination occurs are:

1. Process gas leakages from the inboard or high-pressure side of the seal.
2. Bearing lubrication oil from the outboard or low-pressure side of the seal.
3. The seal gas injected into the seal being contaminated upstream of the injection.

Key parameters to monitor for a dry gas seal are as follows.

1. Primary seal gas differential pressure—monitor and trend control valve positions.
2. Primary vent flow or pressure—high for primary seal leakage, low for secondary seal leakage.
3. Intermediate N2 DP (If available).
4. Primary vent control valve position (if available)—if opening, the primary seal leakage has increased, if closing the secondary seal leakage has increased.
5. Separation N2 pressure or differential pressure (if equipped with DP control).

13.2 Liquid Ring Compressor - Performance Monitoring

13.2.1 Notations

F	Flow
h	Specific enthalpy
Q	Volume flow
P	Pressure
T	Temperature
η	Efficiency
W	Power
S	Entropy
ρ	Mass Density

13.2.2 Subscripts

G_i	Gas Inlet
G_o	Gas Outlet
R_i	Ring fluid Inlet
R_o	Ring fluid Outlet

13.2.3 Liquid Ring Compressor - Model Assumptions

The key assumptions made for developing a performance model are:

1. There is no internal leakage (No mass transfer between liquid ring and process gas; liquid ring always encompasses the impeller so that the process gas cannot move from one slot to the adjacent slot.)

NOTE

After analyzing various data sheets, it was observed that performance curve of the asset was based on **Seasontype** (e.g., Summer and Winter) or **Composition type** (e.g., Hydrocarbons and Acid gas). Hence there are two tabular array data attributes declared in performance model to support two performance curves in one asset.

13.2.4 Liquid Ring Compressor - Model Input

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Inlet1 Pressure	Process gas Inlet Pressure (Required Input)	Kg/cm ² (a)	Dynamic	Process	Field Input
2	Inlet2 Pressure	Ring Fluid Inlet Pressure (Required Input)	Kg/cm ² (a)	Dynamic	Process	Field Input
3	Outlet1 Pressure	Process gas Outlet Pressure (Required Input)	Kg/cm ² (a)	Dynamic	Process	Field Input
4	Outlet2 Pressure	Ring Fluid Outlet Pressure (Required Input)	Kg/cm ² (a)	Dynamic	Process	Field Input
5	Inlet1 Temperature	Process gas Inlet Temperature (Required Input)	°C	Dynamic	Process	Field Input
6	Inlet2 Temperature	Ring Fluid Inlet Temperature (Required Input)	°C	Dynamic	Process	Field Input
7	Outlet1 Temperature	Process gas Outlet Temperature (Required Input)	°C	Dynamic	Process	Field Input
8	Outlet2 Temperature	Ring Fluid Outlet Temperature (Required Input)	°C	Dynamic	Process	Field Input
9	Inlet1 Volume Flow Rate	Process gas Inlet Volume Flow Rate (Required Input)	m ³ /hr	Dynamic	Process	Field Input
10	Inlet2 Volume Flow Rate	Ring Fluid Inlet Volume Flow Rate (Required Input)	m ³ /hr	Dynamic	Process	Field Input
11	Inlet1 Actual Volumetric Flow	Process gas Actual Volume Flow (Input - Port mapped from process gas suction stream)	m ³ /hr	Dynamic	Internal	Field Input
		NOTE Inlet actual volumetric flow can be used while configuring a new model.				
12	Inlet2 Actual Volumetric Flow	Ring Fluid Actual Volume Flow (Input - Port mapped from ring fluid suction stream)	m ³ /hr	Dynamic	Internal	Field Input

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
		NOTE Inlet actual volumetric flow can be used while configuring a new model.				
13	Separator Volume Flow	Separator Volume Flow (Input)	m ³ /hr	Dynamic	Process	Field Input
14	Measured Shaft Power	Measured Shaft Power (Required Input)	kW	Dynamic	Measurement	Field Input
15	Expected Mechanical Efficiency	Expected Mechanical efficiency (Input)	%	Static	Design	Constant
16	Operating Mechanical Efficiency	Operating Mechanical efficiency (Input)	%	Static	Design	Constant
17	Performance Data	Curve data for Process gas outlet pressure (kg/cm ² (a)) V/S Process gas inlet volume flow and Power		Static	Performance	Interpolated from performance curve

13.2.5 Liquid Ring Compressor - KPIs

The following indicators help us know the overall performance of the asset:

13.2.5.1 Input KPIs

- Inlet process gas pressure
- Outlet process gas pressure
- Inlet process gas temperature
- Outlet process gas temperature
- Inlet ring fluid pressure
- Outlet ring fluid pressure
- Inlet ring fluid temperature
- Outlet ring fluid temperature
- Inlet process gas volume flow rate
- Inlet ring fluid volume flow rate
- Performance data

$$Q \text{ VS } P_{G_o} \quad \dot{W}_{Shaft} \text{ VS } P_{G_o}$$

13.2.5.2 Output KPIs

Expected KPIs

- Expected Isothermal Efficiency
- Expected Gas Power
- Expected Shaft Power
- Expected Volume Flow

Operating KPIs

- Operating Isothermal Efficiency
- Operating Gas Power
- Operating Shaft Power

13.2.5.3 Liquid Ring Compressor - Performance Modeling

The Performance modelling of liquid ring compressor helps to determine the expected performance KPI. The following is the procedure:

Overall Mass Balance

There are two independent hydraulic system. One is process gas which is getting compressed, and another is open system for liquid ring to maintain its temperature.

Applying continuity equation for both hydraulic systems,

$$F_{G_i} = F_{G_o}$$

$$F_{R_i} = F_{R_o}$$

Overall Force Balance

$$P_{G_o} = P_{G_i} + \Delta P_{Pressure}$$

$$P_{R_o} = P_{R_i}$$

Overall Energy Balance

Unlike flow, which is independent in their flow path, the energy is open to exchange between compressed process gas and liquid ring. Hence the overall energy balance is written as

$$F_{G_i} h_{G_i}(P_{G_i}, T_{G_i}) + F_{R_i} h_{R_i}(P_{R_i}, T_{R_i}) + \dot{W}_{Gas} = F_{G_o} h_{G_o}(P_{G_o}, T_{G_o}) + F_{R_o} h_{R_o}(P_{R_o}, T_{R_o})$$

In terms of shaft power or driver power, we can represent gas power as:

$$\dot{W}_{Gas} = \eta_{Mech} \dot{W}_{Shaft}$$

$$F_{G_i} h_{G_i}(P_{G_i}, T_{G_i}) + F_{R_i} h_{R_i}(P_{R_i}, T_{R_i}) + \eta_{Mech} \dot{W}_{Shaft} = F_{G_o} h_{G_o}(P_{G_o}, T_{G_o}) + F_{R_o} h_{R_o}(P_{R_o}, T_{R_o})$$

13.2.5.4 Liquid Ring Compressor - Expected KPIs

The following indicators help us to know expected overall performance of the liquid ring compressor:

Expected Volume Flow (m³/hr)

Expected volume flow is calculated using performance curve supplied by the user.

$$Q_{G_i}^{Expected} = f(P_{G_o})$$

Expected Shaft Power (kW)

Expected shaft power is calculated using performance curve supplied by the user.

$$Q_{G_i}^{Expected} = f(P_{G_o})$$

Expected Gas Power (kW)

$$\dot{W}_{Gas}^{Expected} = \eta_{Mech}^{Expected} \dot{W}_{Shaft}^{Expected}$$

Expected Isothermal Efficiency (%)

$$\eta_{Isothermal}^{Expected} = \frac{\text{Isothermal power received by the gas}}{\text{Actual Gas Power to the system}} = \frac{\text{Isothermal power received by the gas}}{\eta_{Mech} \times \text{Actual Shaft Power to the system}}$$

$$\eta_{Isothermal}^{Expected} = \frac{F \left(h_o^{Iso}(P_{G_o}, T_{G_i}) - h_i(P_{G_i}, T_{G_i}) - T \left(s_{G_o}^{Iso} - s_{G_i} \right) \right)}{\dot{W}_{Gas}^{Expected}}$$

$$\eta_{Isothermal}^{Expected} = \frac{\left(Q_{G_i}^{Expected} \rho_{G_i} \right) \left(h_o^{Iso}(P_{G_o}, T_{G_i}) - h_i(P_{G_i}, T_{G_i}) - T \left(s_{G_o}^{Iso} - s_{G_i} \right) \right)}{\dot{W}_{Gas}^{Expected}}$$

13.2.5.5 Liquid Ring Compressor - Operating KPIs

The following indicators help us know the overall operating performance of the Liquid Ring compressor:

Operating Gas Power (kW)

$$\dot{W}_{Gas}^{Operating} = \left(F_{G_o} h_{G_o}(P_{G_o}, T_{G_o}) + F_{R_0} h_{R_0}(P_{R_0}, T_{R_0}) \right) - \left(F_{G_i} h_{G_i}(P_{G_i}, T_{G_i}) + F_{R_i} h_{R_i}(P_{R_i}, T_{R_i}) \right)$$

Operating Shaft Power (kW)

The mechanical efficiency is a user defined input,

$$\dot{W}_{Shaft}^{Operating} = \frac{\dot{W}_{Gas}^{Operating}}{\eta_{Mech.}}$$

Operating Isothermal Efficiency (%)

$$\eta_{Isothermal}^{Operating} = \frac{(Q_{G_i} \rho_{G_i}) \left(h_o^{Iso}(P_{G_o}, T_{G_i}) - h_i(P_{G_i}, T_{G_i}) - T_{G_i} (s_{G_o}^{Iso} - s_{G_i}) \right)}{\dot{W}_{Gas}^{Expected}}$$

13.2.6 Liquid Ring Compressor - Model Output

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Expected Gas Power	Expected Gas Power (Output)	kW	Dynamic	Performance	Calculated
2	Expected Isothermal Efficiency	Expected Isothermal Efficiency (Output)	%	Dynamic	Performance	Calculated
3	Expected Shaft Power	Expected Shaft Power (Output)	kW	Dynamic	Performance	Calculated
4	Expected Volume Flow Rate	Expected Volume Flow Rate (Output)	m ³ /hr	Dynamic	Performance	Calculated
5	Operating Gas Power	Operating Gas Power (Output)	kW	Dynamic	Performance	Calculated
6	Operating Isothermal Efficiency	Operating Isothermal Efficiency (Output)	%	Dynamic	Performance	Calculated
7	Operating Shaft Power	Operating Shaft Power (Output)	kW	Dynamic	Performance	Calculated
8	Deviation Gas Power	Deviation Gas Power (Output)	%	Dynamic	Deviation	Calculated
9	Deviation Isothermal Efficiency	Deviation Isothermal Efficiency (Output)	%	Dynamic	Deviation	Calculated
10	Deviation Shaft Power	Deviation Shaft Power (Output)	%	Dynamic	Deviation	Calculated
11	Deviation Volume Flow Rate	Deviation Volume Flow Rate (Output)	%	Dynamic	Deviation	Calculated

13.3 Liquid Ring Compressor - Fault Model**13.3.1 Liquid Ring Compressor - APM Model**

Model Type	APM Model
Category	Liquid Ring Compressor
Equipment Class	Compressor
Equipment Type	Liquid Ring Compressor

Below table illustrates the list of faults and symptoms configured for a Liquid ring compressor to identify possible anomalies that affect the performance or health of a Gas Turbine

Fault Name	Associated Symptoms	Recommendations
Cavitation Risks	LRC Suction Pressure-Gas RL (CR)	1. Check suction / Recycle pressure valve 2. Clean suction line / strainer.
	LRC Discharge Temperature RH (CR)	
	Pressure difference between suction pressure and vapour pressure LOW (CR)	
Efficiency Degradation Anomaly	Deviation Isothermal Efficiency RL (EDA)	1. Check the impeller vane condition. 2. Clean the impeller. 3. Check the amount of eccentricity.
Gas Discharge Temperature High Anomaly	LRC Suction Temperature-Gas RH (GDTHA)	1. Check the condition of the recycle valve.
	LRC Discharge Temperature RH (GDTHA)	
	Inlet Pressure Control Valve Position RH (GDTHA)	
LRC Compressor Gas supply Anomaly	LRC Suction Pressure-Gas RL (LRCGSA)	1. Check the recycle valve.
	LRC Discharge Pressure RL (LRCGSA)	
	Suction Pressure-Ring Liquid RL (LRCGSA)	
	Inlet Pressure Control Valve Position OFF (LRCGSA)	

13.3.2 Liquid Ring Compressor - APR Model

13.3.2.1 Liquid Ring Compressor Unit System Fault Model

Model Type	APR Model
Category	Liquid Ring Compressor
Equipment Class	Liquid Ring
Equipment Type	Liquid Ring Compressor Unit

Below table illustrates the list of faults and symptoms configured for a Liquid Ring Compressor Unit

Fault Name	Associated Symptoms	Recommendations
LRC Large Liquid Ring inside Compressor Anomaly	LRC - Casing Vibration X DE RH (LLRICA)	Check the liquid level inside the compressor.
	LRC - Casing Vibration Y DE RH (LLRICA)	
	LRC - Casing Vibration X NDE RH (LLRICA)	
	LRC - Casing Vibration Y NDE RH (LLRICA)	
	LRC Suction Flow-Gas RL (LLRICA)	

Fault Name	Associated Symptoms	Recommendations
	LRC Motor Current RH (LLRICA)	
	LRC Motor Power RH (LLRICA)	
	LRC Discharge Pressure RH (LLRICA)	
LRC Thrust Bearing Malfunction due to Mechanical Failure	LRC - Thrust Bearing Displacement Active Side RH (TBMF)	1. Check calibration of sensor. 2. Assess likely cause and time to trip if intervention not possible. 3. Intervention options. 4. Reduce thrust load. 5. Ensure parts availability. 6. Check axial vibration trends to establish if the machine has surged and potentially damaged/wiped the pads. 7. If required inspect the thrust pads and collar to check for damage / wiping. 8. Carry out a bump check to establish what float is in the machine, compare to previous overhaul report for measurement. 9. Check the gear coupling/pre-stretch diaphragm coupling.
	LRC - Thrust Bearing Displacement Active Side RL (TBMF)	
	LRC - Thrust Bearing Displacement Inactive Side RH (TBMF)	
	LRC - Thrust Bearing Displacement Inactive Side RL (TBMF)	
	LRC - Thrust Bearing Temperature A Active Side RH (TBMF)	
	LRC - Thrust Bearing Temperature B Active Side RH (TBMF)	
	LRC - Thrust Bearing Temperature C Active Side RH (TBMF)	
	LRC - Thrust Bearing Temperature D Active Side RH (TBMF)	
	LRC - Thrust Bearing Temperature A Inactive Side RH (TBMF)	
	LRC - Thrust Bearing Temperature B Inactive Side RH (TBMF)	
	LRC - Thrust Bearing Temperature C Inactive Side RH (TBMF)	
	LRC - Thrust Bearing Temperature D Inactive Side RH (TBMF)	
LRC DE Radial Bearing Malfunction due to	LRC - Radial Bearing Vibration X DE RH	1. Check calibration of sensor. 2. Investigate with high frequency vibration analysis.

Fault Name	Associated Symptoms	Recommendations
Mechanical Failure	(RBDEMF)	3. Assess likely cause and time to fail(trip) if intervention not possible. 4. Intervention options, reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 7. Change bearing if required.
	LRC - Radial Bearing Vibration Y DE RH (RBDEMF)	
	LRC - Radial Bearing Temperature A DE RH (RBDEMF)	
	LRC - Radial Bearing Temperature B DE RH (RBDEMF)	
LRC NDE Radial Bearing Malfunction due to Mechanical Failure	LRC - Radial Bearing Vibration X NDE RH (RBNDEMF)	1. Check calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if intervention not possible. 4. Intervention options; reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 7. Change bearing if required.
	LRC - Radial Bearing Vibration Y NDE RH (RBNDEMF)	
	LRC - Radial Bearing Temperature A NDE RH (RBNDEMF)	
	LRC - Radial Bearing Temperature B NDE RH (RBNDEMF)	
LRC Bearing Vibration due to Cavitation Malfunction	LRC - Casing Vibration X DE RH (BVMF)	1. Check suction / Recycle pressure valve. 2. Clean suction line / strainer. 3. Check the sensors for calibration.
	LRC - Casing Vibration Y DE RH (BVMF)	
	LRC - Casing Vibration X NDE RH (BVMF)	
	LRC - Casing Vibration Y NDE RH (BVMF)	
LRC Rotor Anomaly	LRC - Radial Bearing Vibration X DE RH (RTR)	1. Check the calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Check Carbon brush integrity. 4. Check magnetism of the rotor. 5. Check rotor unbalance, misalignment. 6. Looseness. 7. Check coupling integrity. 8. Check foundation bolt integrity. 9. Check both bearings for clearance and pad wear etc. 10. Check the efficiency to know if there is scaling or corrosion of impellers.

Fault Name	Associated Symptoms	Recommendations
	LRC - Radial Bearing Vibration Y DE RH (RTR)	11. Check the impeller condition.
	LRC - Radial Bearing Vibration X NDE RH (RTR)	
	LRC - Radial Bearing Vibration Y NDE RH (RTR)	
Ring Liquid suction Temp anomaly	Suction Temperature-Ring Liquid RH (LRSTA)	1. Inspect the cooler for corrosion, particle entrainment. 2. Check the fan/ motor issues. 3. Check make up water temperature.
	Exchanger Discharge Temp RH (LRSTA)	
	Exchanger Fan Motor-1 Status OFF (LRSTA)	
	Exchanger Fan Motor-2 Status OFF (LRSTA)	
	Exchanger Fan Motor-3 Status OFF (LRSTA)	
	Exchanger Fan Motor-4 Status OFF (LRSTA)	
Ring Liquid Suction Anomaly	LRC Discharge Pressure RL (LRSA)	1. Check the ring water pump and motor / separator level. 2. Check and clean the suction strainer, line. 3. Check make up /recirculation line water flow sufficiency.
	Suction Flow-Ring Liquid RL (LRSA)	
	Separator liquid level RL (LRSA)	
	Ring Water Pump Motor Status OFF (LRSA)	
	Ring Water Pump Motor Ampere RH (LRSA)	
LRC Compressor Dry Run	LRC Discharge Temperature RH (LRCDR)	1. Carry out boroscope inspection. 2. Check Lube Oil System for internal oil leaks. 3. Prepare for machine overhaul.
	Rated Suction Capacity LOW (LRCDR)	
	LRC Motor Ampere RH (LRCDR)	
	LRC Motor Power RH (LRCDR)	

13.3.2.2 Liquid Ring Compressor Seal Gas System Fault Model

Model Type	APR Model
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Category	Liquid Ring Compressor
Equipment Class	Liquid Ring
Equipment Type	Liquid Ring Compressor Seal Gas System

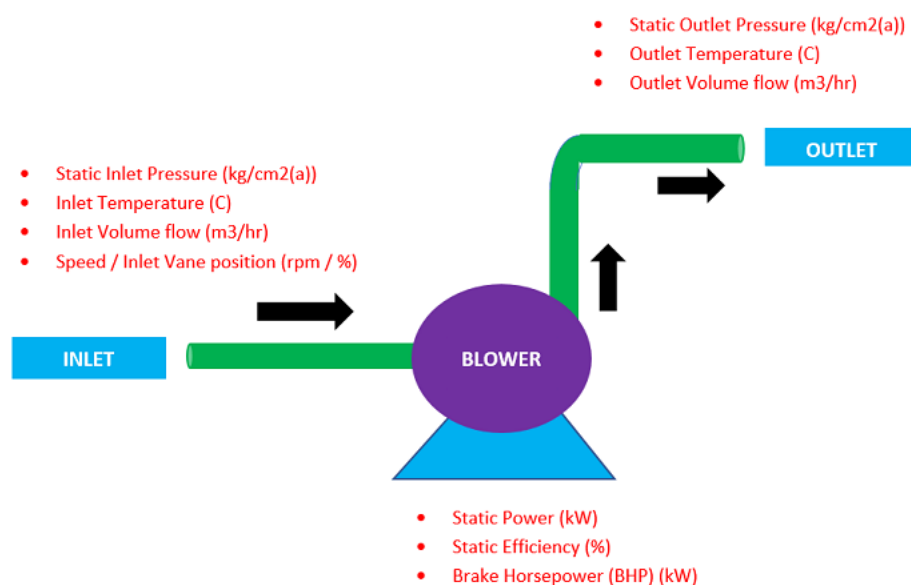
Below table illustrates the list of faults and symptoms configured for a Liquid Ring Compressor Seal Gas System

Fault Name	Associated Symptoms	Recommendations
DE Seal Anomaly	DE Seal flush liquid Pressure RL (DSA)	1. Inspect any blockage in the seal system, 2. Inspect the drum integrity 3. Inspect the seal integrity.
	DE Seal flush liquid Flow RL (DSA)	
	DE Seal drum level RL (DSA)	
NDE Seal Anomaly	NDE Seal flush liquid Pressure RL (NDSA)	1. Inspect any blockage in the seal system 2. Inspect the drum integrity 3. Inspect the seal integrity.
	NDE Seal flush liquid Flow RL (NDSA)	
	NDE Seal drum level RL (NDSA)	

CENTRIFUGAL BLOWER

Industrial blower is a mechanical device for moving air or gases in a direction at an angle to the incoming fluid for enhanced and efficient air flow in a workspace. Industrial blowers are mainly used in many cooling and drying applications. Blowers are routinely used for applications such as dust control, combustion air supplies, on cooling, drying systems, for fluid bed aerators with air conveyor systems etc. These highly technical devices are designed to produce higher pressure than a fan at a ratio of 1.11 to 1.2, which places them somewhere between a fan and a compressor. Industrial blowers can be either centrifugal, axial, or positive displacement.

Centrifugal industrial blowers are the most frequently used type of industrial blowers implemented in ventilation systems. These blowers can continuously transfer gas and air and can block air pollutants. The air stream goes through a set of rotating impellers where the speed and volume of the air is steadily increased. The fan wheel shifts direction and accelerates as the amount of air passing through increases. As more air enters, the kinetic energy of the blower increases.



14.1 Centrifugal Blower - Performance Monitoring

Notations

F	Flow
h	Head
Q	Volume flow
P	Pressure
η	Efficiency
W	Power
N	Speed

ρ	Mass Density
--------	--------------

Subscript

i	Inlet
o	outlet

14.1.1 Centrifugal Blower - Model Assumptions

The key assumptions made for developing a performance model are:

1. Potential head and velocity head are negligible.
2. No internal leakage.

NOTE

The model is supported with separator calculation internally to handle any liquid fraction at the inlet stream.

14.1.2 Centrifugal Blower - Variants

The centrifugal blower model supports handling of the below three variants. Those are:

- Centrifugal Blower - Variable speed
- Centrifugal Blower - Variable Guide Vane
- Centrifugal Blower - Constant Speed

14.1.3 Centrifugal Blower - Model Input

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Inlet1 Pressure	Inlet Stream Pressure (Required Input)	Kg/cm ² (a)	Dynamic	Measurement	Field Input
2	Outlet1 Pressure	Outlet Stream Pressure (Required Input)	Kg/cm ² (a)	Dynamic	Measurement	Field Input
3	Inlet1 Temperature	Inlet Stream Temperature (Required Input)	°C	Dynamic	Measurement	Field Input
4	Outlet1 Temperature	Outlet Stream Temperature (Required Input)	°C	Dynamic	Measurement	Field Input
5	Inlet1 Volume Flow Rate	Inlet Stream Volume Flow Rate (Input)	m ³ /hr	Dynamic	Measurement	Field Input
6	Inlet1 Actual Volumetric Flow	Inlet Stream Actual Volume Flow (Input - Port mapped from suction stream)	m ³ /hr	Dynamic	Internal	Field Input
		NOTE Inlet actual volumetric flow can be used while configuring a new model.				

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
7	Separator Volume Flow	Separator Volume Flow (Input)	m ³ /hr	Dynamic	Measurement	Field Input
8	Speed / Inlet vane position	Speed (Required Input) - variable speed and constant speed variant / Inlet vane position (Required Input) if variable guide vane variant	rpm / %	Dynamic	Measurement	Field Input
9	Measured Power	Measured Power (Required Input)	kW	Dynamic	Measurement	Field Input
10	Performance Data	Speed / Inlet vane position, Inlet Mass Flow, Outlet pressure, Operating BHP		Static	Performance Map	Interpolated from performance curve

14.1.4 Centrifugal Blower - KPIs

The following indicators help us know the overall asset performance:

14.1.4.1 Input KPIs

- Inlet Pressure
- Outlet Pressure
- Inlet Temperature
- Outlet Temperature
- Inlet Volume Flow Rate
- Speed / Inlet Vane Position
- Voltage
- Current
- Performance Data

- Variable speed:

$$(\dot{W}_{BHP}, F_i, N) \text{ and } (P_o^{Static}, F_i, N)$$

- Variable guide vane:

$$(\dot{W}_{BHP}, F_i, \text{vane position}) \text{ and } (P_o^{Static}, F_i, \text{vane position})$$

- Constant speed:

$$(\dot{W}_{BHP}, (P_o^{Static}, F_i))$$

14.1.4.2 Output KPIs

Expected KPIs

- Expected Static Power
- Expected Static Efficiency
- Expected Volume flow
- Expected BHP

Operating KPIs

- Operating Static Power
- Operating Static Efficiency
- Operating BHP

14.1.4.3 Centrifugal Blower - Performance Modeling

The Performance modelling of centrifugal blower helps to determine the performance KPI. The following is the procedure:

Overall Mass Balance

Applying continuity equation,

$$F_i = F_o$$

Overall Force Balance

Applying force balance equation (Bernoulli's energy theorem) and assuming negligible potential and Kinetic contribution.

$$P_i^{Static} + \frac{\rho v_i^2}{2} + \rho g Z_i = P_o^{Static} + \frac{\rho v_o^2}{2} + \rho g Z_o$$

$$(\Delta Z = 0, \frac{\Delta v^2}{2} = 0)$$

Converting to a representative head,

$$\rho g h_t = P_o^{Static} - P_i^{Static}$$

$$h_t = \frac{P_o^{Static} - P_i^{Static}}{\rho g}$$

Overall Energy Balance

$$F_o h_o (P_o^{Static}, T_i) = F_i h_i (P_i^{Static}, T_i) + \dot{W}_{Static}$$

$$\eta = \frac{\dot{W}_{Static}}{\dot{W}_{BHP}}$$

14.1.4.4 Centrifugal Blower - Expected KPIs

The following indicators help us to know expected overall performance of the Centrifugal Blower.

Expected Static Power (kW)

It is static power conveyed to process stream.

$$\dot{W}_{Static}^{Expected} = Q_i (P_o^{Static} - P_i^{Static}) 98.0665$$

$$\Delta P_{Static\ rise} = P_o^{Static} - P_i^{Static}$$

From performance curve of Static outlet pressure vs Mass flow, we get expected static outlet pressure.

For variable speed,

$$P_o^{Static} = f(F_i, N)$$

For variable guide vane,

$$P_o^{Static} = f(F_i, \text{vane position})$$

For constant speed,

$$P_o^{Static} = f(F_i)$$

NOTE

98.0665 is conversion factor to keep power terms in kW.

$$\text{Bar} \rightarrow \text{N/m}^2 = 98066.5 \text{ --- (1)}$$

$$\text{W} \rightarrow \text{kW} = 0.001 \text{ -----(2)}$$

Therefore, on multiplication of conversion (1) and (2) we get conversion factor of 100.

Expected Static Efficiency (%)

It is the ratio of static power conveyed to process stream to the power delivered by motor to the blower.

$$\eta_{Static}^{Expected} = \frac{\dot{W}_{Static}^{Expected}}{\dot{W}_{BHP}^{Expected}}$$

Expected BHP (kW)

From performance curve of BHP vs Mass flow, we perform 2D interpolation to get Expected BHP for measured volume flow.

For variable speed variant,

$$\dot{W}_{BHP}^{Expected} = f(F_i, N)$$

For variable guide vane variant,

$$\dot{W}_{BHP}^{Expected} = f (F_i , vane\ position)$$

For constant speed,

$$\dot{W}_{BHP}^{Expected} = f (F_i)$$

Expected Volume Flow

Using curve for Static outlet pressure vs Mass flow, we can calculate volume flow for the measured static outlet pressure.

For variable speed variant,

$$Q_i^{expected} = f(P_o^{Static}, N, \rho_i)$$

For variable guide vane variant,

$$Q_i^{expected} = f(P_o^{Static}, vane\ position, \rho_i)$$

For constant speed,

$$Q_i^{expected} = f(P_o^{Static}, \rho_i)$$

14.1.4.5 Centrifugal Blower - Operating KPIs

The following indicators help us to know operating overall performance of the Blower.

Operating Static Power (kW)

It is static power conveyed to process stream. Please refer the note under the section **Centrifugal Blower - Expected KPIs** for conversion factor of 100 used in below equation.

$$\dot{W}_{Static}^{Operating} = Q_i (P_o^{Static} - P_i^{Static}) 98.0665$$

Operating Static Efficiency (%)

It is the ratio of static power conveyed to process stream to the power delivered to the blower.

$$\eta_{Static}^{Operating} = \frac{\dot{W}_{Static}^{Operating}}{\dot{W}_{BHP}^{Operating}}$$

Operating BHP (kW)

If Input power is available, we can PHD tag map the power to a measured power attribute declared to the model.

$$\dot{W}_{BHP}^{Operating} = (Measured\ Power)_{Tag\ value}$$

NOTE

If driver is electric motor and current tag value is available, then we can calculate BHP from Motor Asset model and Map it to Measured power attribute.

14.1.5 Centrifugal Blower - Model Output

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Expected BHP	Expected BHP from curve data (Output)	kW	Dynamic	Performance	Calculated
2	Expected Static Efficiency	Expected Static Efficiency (Output)	%	Dynamic	Performance	Calculated
3	Expected Static Power	Expected Static Power (Output)	kW	Dynamic	Performance	Calculated
4	Expected Volume Flow	Expected Volume Flow from curve data (Output)	m ³ /hr	Dynamic	Performance	Calculated
5	Operating BHP	Operating BHP (Output)	kW	Dynamic	Performance	Calculated
6	Operating Static Efficiency	Operating Static Efficiency (Output)	%	Dynamic	Performance	Calculated
7	Operating Static Power	Operating Static Power (Output)	kW	Dynamic	Performance	Calculated
8	Deviation BHP	Percent deviation in BHP between Operating and Expected values (Output)	%	Dynamic	Performance	Calculated
9	Deviation Static Efficiency	Percent deviation in Static Efficiency between Operating and Expected values (Output)	%	Dynamic	Performance	Calculated
10	Deviation Static Power	Percent deviation in Static Power between Operating and Expected values (Output)	%	Dynamic	Performance	Calculated
11	Deviation Volume Flow	Percent deviation in Volume Flow between Operating and Expected values (Output)	%	Dynamic	Performance	Calculated

14.2 Centrifugal Blower - Fault Model

14.2.1 Centrifugal Blower - APM Model

Model Type	APM Model
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Category	Centrifugal Blower
Equipment Class	Blower
Equipment Type	Centrifugal Blower - Constant Speed

Below table illustrates the list of faults and symptoms configured for a Blower Performance-based monitoring.

Fault Name	Associated Symptoms	Recommendations
Blower Performance (Air/Gas)	Outlet1 Temperature RH (BP)	- For non-air, non-closed-loop applications: Perform a gas analysis or reconfirm molecular weight. - Check instruments accuracy (flow meter, suction and discharge temperature and pressure transmitters) If all of the above are ok, then performance deviation is genuine at this point: - Confirm IGV integrity and operability - Check for casing leaks - Confirm seal integrity - If all of the above is ok the reason for degradation could be internal clearances have increased and/or flow path has fouled up or has been damaged. - In this case, at first opportunity: Carry out borescope inspection to verify if there is internal damage. - If fouling is identified, wash the blower if possible. - Monitor efficiency to determine speed of degradation and plan rotor replacement when it becomes unsustainable.
	Deviation Static Efficiency HIGH (BP)	
	Deviation Static Efficiency LOW (BP)	
Change of gas properties (MW) (Not Applicable for Air Blowers)	Deviation Static Efficiency HIGH (CGP)	- Verify if process conditions have changed - Verify if there is any correlation with environmental conditions (sun exposure, ambient temp, rain, etc. - Take a sample of the gas if possible.
	Deviation Static Power SH (CGP)	
	Deviation Static Power SL (CGP)	
Fouling / Corrosion / Erosion (Air/Gas)	Deviation Static Efficiency ROC Long Term Low Limit(FCE)	- If fouling: Short term; Wash Blower if possible. - Long term, investigate application of coating on affected gas path. - Change gas inlet T. - Borescope inspection at opportunity. - If corrosion / erosion: No short-term corrective action is possible. - Replace corroded parts with better material grade. - In case of erosion consider coating of gas path. - Borescope at opportunity.

Model Type	APM Model
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Category	Centrifugal Blower
Equipment Class	Blower
Equipment Type	Centrifugal Blower - Variable Guide Vane

Fault Name	Associated Symptoms	Recommendations
Blower Performance (Air/Gas)	Outlet1 Temperature RH (BP)	- For non-air, non-closed-loop applications: Perform a gas analysis or reconfirm molecular weight. - Check instruments accuracy (flow meter, suction and discharge temperature and pressure transmitters) If all of the above are ok, then performance deviation is genuine at this point: - Confirm IGV integrity and operability - Check for casing leaks - Confirm seal integrity - If all of the above is ok the reason for degradation could be internal clearances have increased and/or flow path has fouled up or has been damaged. - In this case, at first opportunity: Carry out borescope inspection to verify if there is internal damage. - If fouling is identified, wash the blower if possible. - Monitor efficiency to determine speed of degradation and plan rotor replacement when it becomes unsustainable.
	Deviation Static Efficiency HIGH (BP)	
	Deviation Static Efficiency LOW (BP)	
Change of gas properties (MW) (Not Applicable for Air Blowers)	Deviation Static Efficiency HIGH (CGP)	- Verify if process conditions have changed, verify if there is any correlation with environmental conditions (sun exposure, ambient temp, rain, etc). - Take a sample of the gas if possible.
	Deviation Static Power SH (CGP)	
	Deviation Static Power SL (CGP)	
Fouling / Corrosion / Erosion (Air/Gas)	Deviation Static Efficiency ROC Long Term Low Limit(FCE)	- If fouling: Short term; Wash Blower if possible. - Long term, investigate application of coating on affected gas path. - Change gas inlet T. - Borescope inspection at opportunity. - If corrosion / erosion: No short-term corrective action is possible. - Replace corroded parts with better material grade. - In case of erosion consider coating of gas path. - Borescope at opportunity.
IGV malfunction (Air/Gas)	Inlet Vane Position RH (IGVM)	Check IGV actuator and feed Check all linkage connections Carry out borescope examination to identify presence of internal damage.
	Inlet Vane Position RL (IGVM)	

Model Type	APM Model
Category	Centrifugal Blower
Equipment Class	Blower
Equipment Type	Centrifugal Blower - Variable Speed

Fault Name	Associated Symptoms	Recommendations
Blower Performance (Air/Gas)	Outlet1 Temperature RH (BP)	- For non-air, non-closed-loop applications: Perform a gas analysis or reconfirm molecular weight. - Check instruments accuracy (flow meter, suction and discharge temperature and pressure transmitters) If all of the above are ok, then performance deviation is genuine at this point: - Confirm IGV integrity and operability - Check for casing leaks - Confirm seal integrity - If all of the above is ok the reason for degradation could be internal clearances have increased and/or flow path has fouled up or has been damaged. - In this case, at first opportunity: Carry out borescope inspection to verify if there is internal damage. - If fouling is identified, wash the blower if possible. - Monitor efficiency to determine speed of degradation and plan rotor replacement when it becomes unsustainable.
	Deviation Static Efficiency HIGH (BP)	
	Deviation Static Efficiency LOW (BP)	
Change of gas properties (MW) (Not Applicable for Air Blowers)	Deviation Static Efficiency HIGH (CGP)	- Verify if process conditions have changed - Verify if there is any correlation with environmental conditions (sun exposure, ambient temp, rain, etc. - Take a sample of the gas if possible.
	Deviation Static Power SH (CGP)	
	Deviation Static Power SL (CGP)	
Fouling / Corrosion / Erosion (Air/Gas)	Deviation Static Efficiency ROC Long Term Low Limit(FCE)	- If fouling: Short term; Wash Blower if possible. - Long term, investigate application of coating on affected gas path. - Change gas inlet T. - Borescope inspection at opportunity. - If corrosion / erosion: No short-term corrective action is possible. - Replace corroded parts with better material grade. - In case of erosion consider coating of gas path. - Borescope at opportunity.

14.2.2 Centrifugal Blower - APR Model

14.2.2.1 Centrifugal Blower - Constant Speed

Centrifugal Blower Unit Performance and Health Fault Model (Constant Speed)

Model Type	APR Model
Category	Centrifugal Blower
Equipment Class	Blower
Equipment Type	Blower - Unit

Below table illustrates the list of faults and symptoms configured for a Blower Unit.

Fault Name	Associated Symptoms	Recommendations
Bearing Anomaly due to Lube Oil Temperature	B-Radial Bearing Vibration X DE RH (LOT)	1. Check the calibration of sensor. 2. Check cooling water supply pressure. 3. Check bypass control valve. 4. Open both coolers and stabilize temperature. 5. Check cooling water pumps for blockages, ensure strainer is clean. 6. If the cooler is an air cooler check fan and motor integrity. 7. If malfunction of one cooler is suspected, isolate and inspect it. 8. Line walk the system to check for leaks.
	B-Radial Bearing Vibration Y DE RH (LOT)	
	B-Radial Bearing Temperature A DE RH (LOT)	
	B-Radial Bearing Temperature B DE RH (LOT)	
	B-Radial Bearing Vibration X NDE RH (LOT)	
	B-Radial Bearing Vibration Y NDE RH (LOT)	
	B-NDE Radial bearing temperature A NDE RH (LOT)	
	B-NDE Radial bearing temperature B NDE RH (LOT)	
	B-Thrust Bearing Displacement Active Side RH (LOT)	
	B-Thrust Bearing Displacement Active Side RL (LOT)	
	B-Thrust Bearing Temperature A Active Side RH (LOT)	
	B-Thrust Bearing Temperature B Active Side RH (LOT)	

Fault Name	Associated Symptoms	Recommendations
	B-Thrust Bearing Temperature C Active Side RH (LOT)	
	B-Thrust Bearing Temperature D Active Side RH (LOT)	
	B-Thrust Bearing Temperature A Inactive Side RH (LOT)	
	B-Thrust Bearing Temperature B Inactive Side RH (LOT)	
	B-Thrust Bearing Temperature C Inactive Side RH (LOT)	
	B-Thrust Bearing Temperature D Inactive Side RH (LOT)	
	LO header temperature RH (LOT)	
Bearing Anomaly due to Lube Oil Pressure	B-Radial Bearing Vibration X DE RH (LOP)	1. Check the calibration of sensor. 2. Switch on both oil pumps and stabilize oil pressure. 3. Check oil level in tank. 4. Check filter DP. 5. Assess malfunction. 6. Line walk the system and check for leaks. 7. Ensure PSV's are seated, check spillback/PCV valves are set correctly and not returning oil back to the tank. 8. If maintenance has recently been performed check correct rotation of pump. 9. Ensure no hand valves have been left shut, ensure NRV's are working correctly. 10. Check functioning of the auto change over switch.
	B-Radial Bearing Vibration Y DE RH (LOP)	
	B-Radial Bearing Temperature A DE RH (LOP)	
	B-Radial Bearing Temperature B DE RH (LOP)	
	B-Radial Bearing Vibration X NDE RH (LOP)	
	B-Radial Bearing Vibration Y NDE RH (LOP)	
	B-NDE Radial bearing temperature A NDE RH (LOP)	
	B-NDE Radial bearing temperature B NDE RH (LOP)	
	B-Thrust Bearing Displacement Active Side RH (LOP)	

Fault Name	Associated Symptoms	Recommendations
	B-Thrust Bearing Displacement Active Side RL (LOP)	
	B-Thrust Bearing Temperature A Active Side RH (LOP)	
	B-Thrust Bearing Temperature B Active Side RH (LOP)	
	B-Thrust Bearing Temperature C Active Side RH (LOP)	
	B-Thrust Bearing Temperature D Active Side RH (LOP)	
	B-Thrust Bearing Temperature A Inactive Side RH (LOP)	
	B-Thrust Bearing Temperature B Inactive Side RH (LOP)	
	B-Thrust Bearing Temperature C Inactive Side RH (LOP)	
	B-Thrust Bearing Temperature D Inactive Side RH (LOP)	
	LO header pressure RL (LOP)	
Bearing Anomaly due to Lube Oil Quality	B-Radial Bearing Vibration X DE RH (LOQ)	1. Check oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, TAN. 4. Check no breather caps or man covers have been removed and left open to allow debris into the tank. 5. Recondition or replace oil.
	B-Radial Bearing Vibration Y DE RH (LOQ)	
	B-Radial Bearing Temperature A DE RH (LOQ)	
	B-Radial Bearing Temperature B DE RH (LOQ)	
	B-Radial Bearing Vibration X NDE RH (LOQ)	
	B-Radial Bearing Vibration Y NDE RH (LOQ)	

Fault Name	Associated Symptoms	Recommendations
	B-NDE Radial bearing temperature A NDE RH (LOQ)	
	B-NDE Radial bearing temperature B NDE RH (LOQ)	
	B-Thrust Bearing Displacement Active Side RH (LOQ)	
	B-Thrust Bearing Displacement Active Side RL (LOQ)	
	B-Thrust Bearing Temperature A Active Side RH (LOQ)	
	B-Thrust Bearing Temperature B Active Side RH (LOQ)	
	B-Thrust Bearing Temperature C Active Side RH (LOQ)	
	B-Thrust Bearing Temperature D Active Side RH (LOQ)	
	B-Thrust Bearing Temperature A Inactive Side RH (LOQ)	
	B-Thrust Bearing Temperature B Inactive Side RH (LOQ)	
	B-Thrust Bearing Temperature C Inactive Side RH (LOQ)	
	B-Thrust Bearing Temperature D Inactive Side RH (LOQ)	
	LO Viscosity SH (LOQ)	
	LO Viscosity SL (LOQ)	
	LO TAN SH (LOQ)	
Thrust Bearing Failure due to Mechanical Issue	B-Thrust Bearing Displacement Active Side RH (TBM)	1. Check calibration of sensor. 2. Assess likely cause and time to trip if intervention not possible. 3. Intervention options.

Fault Name	Associated Symptoms	Recommendations
	B-Thrust Bearing Displacement Active Side RL (TBM)	4. Reduce thrust load. 5. Ensure parts availability. 6. Check axial vibration trends to establish if the machine has surged and potentially damaged/wiped the pads. 7. If required inspect the thrust pads and collar to check for damage / wiping. 8. Carry out a bump check to establish what float is in the machine, compare to previous overhaul report for measurement. 9. Check the gear coupling/pre-stretch diaphragm coupling .
	B-Thrust Bearing Displacement Inactive Side RH (TBM)	
	B-Thrust Bearing Displacement Inactive Side RL (TBM)	
	B-Thrust Bearing Temperature A Active Side RH (TBM)	
	B-Thrust Bearing Temperature B Active Side RH (TBM)	
	B-Thrust Bearing Temperature C Active Side RH (TBM)	
	B-Thrust Bearing Temperature D Active Side RH (TBM)	
	B-Thrust Bearing Temperature A Inactive Side RH (TBM)	
	B-Thrust Bearing Temperature B Inactive Side RH (TBM)	
	B-Thrust Bearing Temperature C Inactive Side RH (TBM)	
	B-Thrust Bearing Temperature D Inactive Side RH (TBM)	
DE Radial Bearing Failure due to Mechanical Issue	B-Radial Bearing Vibration X DE RH (RBDM)	1. Check calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if intervention not possible. 4. Intervention options, reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 7. Change bearing if required.
	B-Radial Bearing Vibration Y DE RH (RBDM)	
	B-Radial Bearing Temperature A DE RH (RBDM)	
	B-Radial Bearing Temperature B DE RH (RBDM)	

Fault Name	Associated Symptoms	Recommendations
NDE Radial Bearing Failure due to Mechanical Issue	B-Radial Bearing Vibration X NDE RH (RBNDM)	1. Check calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if intervention not possible. 4. Intervention options; reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 7. Change bearing if required.
	B-Radial Bearing Vibration Y NDE RH (RBNDM)	
	B-NDE Radial bearing temperature A NDE RH (RBNDM)	
	B-NDE Radial bearing temperature B NDE RH (RBNDM)	
Rotor Anomaly	B-Radial Bearing Vibration X DE RH (RTR)	1. Check the calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Check Carbon brush integrity. 4. Check magnetism of the rotor. 5. Check rotor unbalance, misalignment. 6. Looseness. 7. Check coupling integrity. 8. Check foundation bolt integrity. 9. Check both bearings for clearance and pad wear etc.
	B-Radial Bearing Vibration Y DE RH (RTR)	
	B-Radial Bearing Vibration X NDE RH (RTR)	
	B-Radial Bearing Vibration Y NDE RH (RTR)	

Centrifugal Blower Gearbox Fault Model (Constant Speed)

Model Type	APR Model
Category	Centrifugal Blower
Equipment Class	Blower
Equipment Type	Blower - Gearbox System

Below table illustrates the list of faults and symptoms configured for a Centrifugal Blower Gearbox System.

Fault Name	Associated Symptoms	Recommendations
Blower Gearbox LSS Radial Bearing DE Anomaly	LSS GB Radial Bearing Vibration X DE RH (GLRDA)	1. Check the calibration of sensor. 2. Check if the lube oil header temperature, pressure and quality has not altered(if lube oil system is available otherwise check greasing or clo). 3. Investigate with high frequency vibration analysis. 4. Assess likely cause and time to fail(trip) if intervention not possible. 5. Intervention options; reduce speed, Parts availability. 6. Check the trends to establish if its either both X and Y probes or one. 7. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots.

Fault Name	Associated Symptoms	Recommendations
	LSS GB Radial Bearing Vibration Y DE RH (GLRDA) LSS GB Radial Bearing Temperature A DE RH (GLRDA) LSS GB Radial Bearing Temperature B DE RH (GLRDA) LSS GB Gearbox Casing Vibration DE RH (GLRDA)	8. Change bearing if required.
Blower Gearbox LSS Radial Bearing NDE Anomaly	LSS GB Radial Bearing Vibration X NDE RH (GLRNDA) LSS GB Radial Bearing Vibration Y NDE RH (GLRNDA) LSS GB Radial Bearing Temperature A NDE RH (GLRNDA) LSS GB Radial Bearing Temperature B NDE RH (GLRNDA) LSS GB Gearbox Casing Vibration NDE RH (GLRNDA)	1. Check the calibration of sensor. 2. Check if the lube oil header temperature, pressure and quality has not altered (if lube oil system is available otherwise check greasing or clo). 3. Investigate with high frequency vibration analysis. 4. Assess likely cause and time to fail (trip) if intervention not possible. 5. Intervention options; reduce speed, Parts availability. 6. Check the trends to establish if its either both X and Y probes or one. 7. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 8. Change bearing if required.
Blower Gearbox HSS Radial Bearing DE Anomaly	HSS GB Radial Bearing Vibration X DE RH (GHRDA) HSS GB Radial Bearing Vibration Y DE RH (GHRDA) HSS-DE Radial bearing temperature A RH (GHRDA) HSS-DE Radial bearing temperature B RH (GHRDA) HSS GB Gearbox Casing Vibration DE RH (GHRDA)	1. Check the calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Assess likely cause and time to fail (trip) if intervention not possible. 4. Intervention options; reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 7. Change bearing if required.
Blower Gearbox HSS Radial Bearing NDE Anomaly	HSS GB Radial Bearing Vibration X NDE RH (GHRNDA)	1. Check the calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Assess likely cause and time to fail (trip) if intervention not possible. 4. Intervention options; reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one.

Fault Name	Associated Symptoms	Recommendations
	HSS GB Radial Bearing Vibration Y NDE RH (GHRNDA)	- Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. - Change bearing if required.
	HSS-NDE Radial bearing temperature A RH (GHRNDA)	
	HSS-NDE Radial bearing temperature B RH (GHRNDA)	
	HSS GB Gearbox Casing Vibration NDE RH (GHRNDA)	
Blower Gearbox LSS Shaft Anomaly	LSS GB Radial Bearing Vibration X DE RH (GLSA)	- Check the calibration of sensor. - Investigate with high frequency vibration analysis. - Check Carbon brush integrity. - Check magnetism of the rotor. - Check rotor unbalance, misalignment. - Looseness. - Check coupling integrity. - Check foundation bolt integrity. - Check both bearings for clearance and pad wear etc. - Check gear integrity.
	LSS GB Radial Bearing Vibration Y DE RH (GLSA)	
	LSS GB Radial Bearing Vibration X NDE RH (GLSA)	
	LSS GB Radial Bearing Vibration Y NDE RH (GLSA)	
	LSS GB Gearbox Casing Vibration DE RH (GLSA)	
	LSS GB Gearbox Casing Vibration NDE RH (GLSA)	
Blower Gearbox HSS Shaft Anomaly	HSS GB Radial Bearing Vibration X DE RH (GHSA)	- Check calibration of sensor. - Investigate with high frequency vibration analysis. - Check Carbon brush integrity. - Check magnetism of the rotor Check rotor unbalance, misalignment. - Looseness. - Check coupling integrity. - Check foundation bolt integrity. - Check both bearings for clearance and pad wear etc. - Check gear integrity.
	HSS GB Radial Bearing Vibration Y DE RH (GHSA)	
	HSS GB Radial Bearing Vibration X NDE RH (GHSA)	
	HSS GB Radial Bearing Vibration Y NDE RH (GHSA)	
	HSS GB Gearbox Casing Vibration DE RH (GHSA)	
	HSS GB Gearbox Casing Vibration NDE RH (GHSA)	
Blower Gearbox LSS Thrust Bearing Anomaly	LSS GB Thrust Bearing Displacement Active Side RH (GLTBA)	- Check if the lube oil header temperature, pressure and quality has not altered(if lube oil system is available). - Assess likely cause and time to trip if intervention not possible. - Check calibration of sensor.

Fault Name	Associated Symptoms	Recommendations
	LSS GB Thrust Bearing Displacement Active Side RL (GLTBA)	- Intervention options; reduce thrust load, Ensure parts availability. - Check axial vibration trends to establish if the driver or driven has change in axial thrust and potentially damaged/wiped the pads. - If required inspect the thrust pads and collar to check for damage / wiping. - Carry out a bump check to establish what float is in the machine, compare to previous overhaul report for measurement. - Check the gear coupling/pre-stretch diaphragm coupling.
	LSS GB Thrust Bearing Displacement Inactive Side RH (GLTBA)	
	LSS GB Thrust Bearing Displacement Inactive Side RL (GLTBA)	
	LSS GB Thrust Bearing Temperature A Active Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature B Active Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature C Active Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature A Inactive Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature B Inactive Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature C Inactive Side RH (GLTBA)	
Blower Gearbox HSS Thrust Bearing Anomaly	HSS GB Thrust Bearing Displacement Active Side RH (GHTBA)	- Check if the lube oil header temperature, pressure and quality has not altered(if lube oil system is available). - Assess likely cause and time to trip if intervention not possible. - Check calibration of sensor. - Intervention options; reduce thrust load, Ensure parts availability. - Check axial vibration trends to establish if the driver or driven has change in axial thrust and potentially damaged/wiped the pads. - If required inspect the thrust pads and collar to check for damage / wiping. - Carry out a bump check to establish what float is in the machine, compare to previous overhaul report for measurement. - Check the gear coupling/pre-stretch diaphragm coupling.

Fault Name	Associated Symptoms	Recommendations
	HSS GB Thrust Bearing Displacement Active Side RL (GHTBA)	
	HSS GB Thrust Bearing Displacement Inactive Side RH (GHTBA)	
	HSS GB Thrust Bearing Displacement Inactive Side RL (GHTBA)	
	HSS GB Thrust Bearing Temperature A Active Side RH (GHTBA)	
	HSS GB Thrust Bearing Temperature B Active Side RH (GHTBA)	
	HSS GB Thrust Bearing Temperature C Active Side RH (GHTBA)	
	HSS GB Thrust Bearing Temperature A Inactive Side RH (GHTBA)	
	HSS GB Thrust Bearing Temperature B Inactive Side RH (GHTBA)	
	HSS GB Thrust Bearing Temperature C Inactive Side RH (GHTBA)	

Centrifugal Blower Lube Oil (Constant Speed)

Model Type	APR Model
Category	Centrifugal Blower
Equipment Class	Blower
Equipment Type	Blower - Lube Oil System

Below table illustrates the list of faults and symptoms configured for a Centrifugal Blower Lube Oil System.

Fault Name	Associated Symptoms	Recommendations
Loss of Lube Oil Quality	Lube Oil Viscosity HIGH (LOLQ)	1. Check oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, TAN. 4. Check no breather caps or man covers have been removed and left open to allow debris into the tank. 5. Do filter cleaning. 6. Recondition or replace oil.
	Lube Oil Viscosity LOW (LOLQ)	
	Lube Oil TAN HIGH (LOLQ)	
	Lube Oil Composition Analysis HIGH (LOLQ)	
Lube Oil Filter Anomaly	Lube Oil Filter DP RH (LFDF)	1. Check filter DP, changeover and clean/replace the stand by filter. 2. Check the Filter DP transmitter.
Loss of Pressure due to Pump Anomaly	Lube Oil Header Pressure RL (LOPPF)	1. Check the Discharge PT integrity. 2. Switch on both oil pumps and stabilize oil pressure. 3. Assess malfunction. 4. Line walk the system and check for leaks, ensure PSV's are seated. 5. Check spillback/PCV valves are set correctly and not returning oil back to the tank. 6. Ensure no hand valves have been left shut for the standby pump. 7. Check oil level in tank, Open tank overhead vents for release of pressure. 8. Check PCV function for Integrity. 9. Assess malfunction. 10. Check Auto Logic Function tests, check functioning of the auto change over switch. 11. If maintenance has recently been performed check correct rotation of pump. 12. Ensure NRV's are working correctly.
	Lube Oil Pump Discharge Pressure RL (LOPPF)	
	Lube Oil PCV Position RL (LOPPF)	
	Lube Oil Return Flow Rate RL (LOPPF)	
	Lube Oil Main Pump Current (Run Status) ON (LOPPF)	
	Lube Oil Aux Pump Current (Run Status) ON (LOPPF)	
Loss of Cooling due to TCV Anomaly	Lube Oil Header Temperature (Cooler Out temperature) RH (LOCTF)	1. Check the calibration of sensor. 2. Check bypass control valve for integrity. 3. Open both coolers and stabilize temperature. 4. Line walk the system to check for leaks. 5. Check the Isolation/bypass valve of TCV.
	LO Cooler Bypass Valve (TCV) Position RH (LOCTF)	
Loss of Cooling due to Cooler Anomaly	Lube Oil Header Temperature (Cooler Out temperature) RH (LOCCF)	1. Check the calibration of sensor. 2. Check cooling water supply pressure. 3. Open both coolers and stabilize temperature. 4. If malfunction of one cooler is suspected, isolate and inspect it.

Fault Name	Associated Symptoms	Recommendations
		5. Line walk the system to check for leaks.
	LO Tank And Cooler Outlet Temperature Delta RH (LOCCF)	6. Check cooling water pumps for blockages 7. Ensure strainer is clean.
Loss of Heating due Lube Oil Tank Heater Off	Lube Oil Header Temperature (Cooler Out temperature) RL (LOCHOFF)	1. Check the Heater ON/OFF switch integrity and Change the switch as per the requirement. 2. Check the heater circuit for Voltage and current.
	Lube Oil Tank Temperature (Cooler Inlet temperature) RL (LOCHOFF)	
	Lube Oil Tank Heater (Run Status) OFF (LOCHOFF)	
Loss of Cooling due Lube Oil Tank Heater On	Lube Oil Header Temperature (Cooler Out temperature) RH (LOCHON)	1. Check the Heater ON/OFF switch integrity and Change the switch as per the requirement. 2. Offline 1. 3. Check the heater circuit for Voltage and current.
	Lube Oil Tank Temperature (Cooler Inlet temperature) RH (LOCHON)	
	Lube Oil Tank Heater (Run Status) ON (LOCHON)	
Loss of Cooling due to Fan Motor Issue	Lube Oil Header Temperature (Cooler Out temperature) RH (LOCFMI)	Offline: Check the Fan and Motor /coupling / belt Integrity and rectify.
	Lube Oil Cooler Fan 1 (Run Status) OFF (LOCFMI)	
	Lube Oil Cooler Fan 2 (Run Status) OFF (LOCFMI)	

14.2.2.2 Centrifugal Blower - Variable Guide Vane

Centrifugal Blower Unit Performance and Health Fault Model (Variable Guide Vane)

Model Type	APR Model
Category	Centrifugal Blower
Equipment Class	Blower
Equipment Type	Blower - Unit

Below table illustrates the list of faults and symptoms configured for a Blower Unit.

Fault Name	Associated Symptoms	Recommendations
Bearing Anomaly due to Lube Oil Temperature	B-Radial Bearing Vibration X DE RH (LOT)	1. Check the calibration of sensor. 2. Check cooling water supply pressure. 3. Check bypass control valve. 4. Open both coolers and stabilize temperature. 5. Check cooling water pumps for blockages, ensure strainer is clean. 6. If the cooler is an air cooler check fan and motor integrity.

Fault Name	Associated Symptoms	Recommendations
		7. If malfunction of one cooler is suspected, isolate and inspect it. 8. Line walk the system to check for leaks.

Fault Name	Associated Symptoms	Recommendations
	B-Radial Bearing Vibration Y DE RH (LOT)	
	B-Radial Bearing Temperature A DE RH (LOT)	
	B-Radial Bearing Temperature B DE RH (LOT)	
	B-Radial Bearing Vibration X NDE RH (LOT)	
	B-Radial Bearing Vibration Y NDE RH (LOT)	
	B-NDE Radial bearing temperature A NDE RH (LOT)	
	B-NDE Radial bearing temperature B NDE RH (LOT)	
	B-Thrust Bearing Displacement Active Side RH (LOT)	
	B-Thrust Bearing Displacement Active Side RL (LOT)	
	B-Thrust Bearing Temperature A Active Side RH (LOT)	
	B-Thrust Bearing Temperature B Active Side RH (LOT)	
	B-Thrust Bearing Temperature C Active Side RH (LOT)	
	B-Thrust Bearing Temperature D Active Side RH (LOT)	
	B-Thrust Bearing Temperature A Inactive Side RH (LOT)	
	B-Thrust Bearing Temperature B Inactive Side RH (LOT)	
	B-Thrust Bearing Temperature C Inactive Side	

Fault Name	Associated Symptoms	Recommendations
	RH (LOT)	
	B-Thrust Bearing Temperature D Inactive Side RH (LOT)	
	LO header temperature RH (LOT)	
Bearing Anomaly due to Lube Oil Pressure	B-Radial Bearing Vibration X DE RH (LOP)	1. Check the calibration of sensor. 2. Switch on both oil pumps and stabilize oil pressure. 3. Check oil level in tank. 4. Check filter DP. 5. Assess malfunction. 6. Line walk the system and check for leaks. 7. Ensure PSV's are seated, check spillback/PCV valves are set correctly and not returning oil back to the tank. 8. If maintenance has recently been performed check correct rotation of pump. 9. Ensure no hand valves have been left shut, ensure NRV's are working correctly. 10. Check functioning of the auto change over switch.
	B-Radial Bearing Vibration Y DE RH (LOP)	
	B-Radial Bearing Temperature A DE RH (LOP)	
	B-Radial Bearing Temperature B DE RH (LOP)	
	B-Radial Bearing Vibration X NDE RH (LOP)	
	B-Radial Bearing Vibration Y NDE RH (LOP)	
	B-NDE Radial bearing temperature A NDE RH (LOP)	
	B-NDE Radial bearing temperature B NDE RH (LOP)	
	B-Thrust Bearing Displacement Active Side RH (LOP)	
	B-Thrust Bearing Displacement Active Side RL (LOP)	
	B-Thrust Bearing Temperature A Active Side RH (LOP)	
	B-Thrust Bearing Temperature B Active Side RH (LOP)	
	B-Thrust Bearing Temperature C Active Side RH (LOP)	
	B-Thrust Bearing Temperature D Active Side	

Fault Name	Associated Symptoms	Recommendations
	RH (LOP)	
	B-Thrust Bearing Temperature A Inactive Side RH (LOP)	
	B-Thrust Bearing Temperature B Inactive Side RH (LOP)	
	B-Thrust Bearing Temperature C Inactive Side RH (LOP)	
	B-Thrust Bearing Temperature D Inactive Side RH (LOP)	
	LO header pressure RL (LOP)	
Bearing Anomaly due to Lube Oil Quality	B-Radial Bearing Vibration X DE RH (LOQ)	1. Check oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, TAN. 4. Check no breather caps or man covers have been removed and left open to allow debris into the tank. 5. Recondition or replace oil.
	B-Radial Bearing Vibration Y DE RH (LOQ)	
	B-Radial Bearing Temperature A DE RH (LOQ)	
	B-Radial Bearing Temperature B DE RH (LOQ)	
	B-Radial Bearing Vibration X NDE RH (LOQ)	
	B-Radial Bearing Vibration Y NDE RH (LOQ)	
	B-NDE Radial bearing temperature A NDE RH (LOQ)	
	B-NDE Radial bearing temperature B NDE RH (LOQ)	
	B-Thrust Bearing Displacement Active Side RH (LOQ)	
	B-Thrust Bearing Displacement Active Side RL (LOQ)	
	B-Thrust Bearing Temperature A Active Side	

Fault Name	Associated Symptoms	Recommendations
	RH (LOQ)	
	B-Thrust Bearing Temperature B Active Side RH (LOQ)	
	B-Thrust Bearing Temperature C Active Side RH (LOQ)	
	B-Thrust Bearing Temperature D Active Side RH (LOQ)	
	B-Thrust Bearing Temperature A Inactive Side RH (LOQ)	
	B-Thrust Bearing Temperature B Inactive Side RH (LOQ)	
	B-Thrust Bearing Temperature C Inactive Side RH (LOQ)	
	B-Thrust Bearing Temperature D Inactive Side RH (LOQ)	
	LO Viscosity SH (LOQ)	
	LO Viscosity SL (LOQ)	
	LO TAN SH (LOQ)	
Thrust Bearing Failure due to Mechanical Issue	B-Thrust Bearing Displacement Active Side RH (TBM)	1. Check calibration of sensor. 2. Assess likely cause and time to trip if intervention not possible. 3. Intervention options. 4. Reduce thrust load. 5. Ensure parts availability. 6. Check axial vibration trends to establish if the machine has surged and potentially damaged/wiped the pads. 7. If required inspect the thrust pads and collar to check for damage / wiping. 8. Carry out a bump check to establish what float is in the machine, compare to previous overhaul report for measurement. 9. Check the gear coupling/pre-stretch diaphragm coupling .
	B-Thrust Bearing Displacement Active Side RL (TBM)	
	B-Thrust Bearing Displacement Inactive Side RH (TBM)	
	B-Thrust Bearing Displacement Inactive Side RL (TBM)	
	B-Thrust Bearing Temperature A Active Side RH (TBM)	
	B-Thrust Bearing Temperature B Active Side RH (TBM)	

Fault Name	Associated Symptoms	Recommendations
	B-Thrust Bearing Temperature C Active Side RH (TBM)	
	B-Thrust Bearing Temperature D Active Side RH (TBM)	
	B-Thrust Bearing Temperature A Inactive Side RH (TBM)	
	B-Thrust Bearing Temperature B Inactive Side RH (TBM)	
	B-Thrust Bearing Temperature C Inactive Side RH (TBM)	
	B-Thrust Bearing Temperature D Inactive Side RH (TBM)	
DE Radial Bearing Failure due to Mechanical Issue	B-Radial Bearing Vibration X DE RH (RBDM)	1. Check calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if intervention not possible. 4. Intervention options, reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 7. Change bearing if required.
	B-Radial Bearing Vibration Y DE RH (RBDM)	
	B-Radial Bearing Temperature A DE RH (RBDM)	
	B-Radial Bearing Temperature B DE RH (RBDM)	
NDE Radial Bearing Failure due to Mechanical Issue	B-Radial Bearing Vibration X NDE RH (RBNDM)	1. Check calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if intervention not possible. 4. Intervention options; reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 7. Change bearing if required.
	B-Radial Bearing Vibration Y NDE RH (RBNDM)	
	B-NDE Radial bearing temperature A NDE RH (RBNDM)	
	B-NDE Radial bearing temperature B NDE RH (RBNDM)	
Rotor Anomaly	B-Radial Bearing Vibration X DE RH (RTR)	1. Check the calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Check Carbon brush integrity. 4. Check magnetism of the rotor.

Fault Name	Associated Symptoms	Recommendations
	B-Radial Bearing Vibration Y DE RH (RTR)	5. Check rotor unbalance, misalignment. 6. Looseness.
	B-Radial Bearing Vibration X NDE RH (RTR)	7. Check coupling integrity. 8. Check foundation bolt integrity.
	B-Radial Bearing Vibration Y NDE RH (RTR)	9. Check both bearings for clearance and pad wear etc.

Centrifugal Blower Gearbox Fault Model (Variable Guide Vane)

Model Type	APR Model
Category	Centrifugal Blower
Equipment Class	Blower
Equipment Type	Blower - Gearbox System

Below table illustrates the list of faults and symptoms configured for a Centrifugal Blower Gearbox System.

Fault Name	Associated Symptoms	Recommendations
Blower Gearbox LSS Radial Bearing DE Anomaly	LSS GB Radial Bearing Vibration X DE RH (GLRDA)	1. Check the calibration of sensor. 2. Check if the lube oil header temperature, pressure and quality has not altered(if lube oil system is available otherwise check greasing or clo). 3. Investigate with high frequency vibration analysis. 4. Assess likely cause and time to fail(trip) if intervention not possible. 5. Intervention options; reduce speed, Parts availability. 6. Check the trends to establish if its either both X and Y probes or one. 7. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 8. Change bearing if required.
	LSS GB Radial Bearing Vibration Y DE RH (GLRDA)	
	LSS GB Radial Bearing Temperature A DE RH (GLRDA)	
	LSS GB Radial Bearing Temperature B DE RH (GLRDA)	
	LSS GB Gearbox Casing Vibration DE RH (GLRDA)	
Blower Gearbox LSS Radial Bearing NDE Anomaly	LSS GB Radial Bearing Vibration X NDE RH (GLRNDA)	1. Check the calibration of sensor. 2. Check if the lube oil header temperature, pressure and quality has not altered(if lube oil system is available otherwise check greasing or clo). 3. Investigate with high frequency vibration analysis. 4. Assess likely cause and time to fail(trip) if intervention not possible. 5. Intervention options; reduce speed, Parts availability. 6. Check the trends to establish if its either both X and Y probes or one. 7. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 8. Change bearing if required.
	LSS GB Radial Bearing Vibration Y NDE RH (GLRNDA)	
	LSS GB Radial Bearing Temperature A NDE RH (GLRNDA)	
	LSS GB Radial Bearing Temperature B NDE RH (GLRNDA)	
	LSS GB Gearbox Casing	

Fault Name	Associated Symptoms	Recommendations
	Vibration NDE RH (GLRNDA)	
Blower Gearbox HSS Radial Bearing DE Anomaly	HSS GB Radial Bearing Vibration X DE RH (GHRDA)	1. Check the calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if intervention not possible. 4. Intervention options; reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 7. Change bearing if required.
	HSS GB Radial Bearing Vibration Y DE RH (GHRDA)	
	HSS-DE Radial bearing temperature A RH (GHRDA)	
	HSS-DE Radial bearing temperature B RH (GHRDA)	
	HSS GB Gearbox Casing Vibration DE RH (GHRDA)	
Blower Gearbox HSS Radial Bearing NDE Anomaly	HSS GB Radial Bearing Vibration X NDE RH (GHRNDA)	1. Check the calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if intervention not possible. 4. Intervention options; reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 7. Change bearing if required.
	HSS GB Radial Bearing Vibration Y NDE RH (GHRNDA)	
	HSS-NDE Radial bearing temperature A RH (GHRNDA)	
	HSS-NDE Radial bearing temperature B RH (GHRNDA)	
	HSS GB Gearbox Casing Vibration NDE RH (GHRNDA)	
Blower Gearbox LSS Shaft Anomaly	LSS GB Radial Bearing Vibration X DE RH (GLSA)	1. Check the calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Check Carbon brush integrity. 4. Check magnetism of the rotor. 5. Check rotor unbalance, misalignment. 6. Looseness. 7. Check coupling integrity. 8. Check foundation bolt integrity. 9. Check both bearings for clearance and pad wear etc. 10. Check gear integrity.
	LSS GB Radial Bearing Vibration Y DE RH (GLSA)	
	LSS GB Radial Bearing Vibration X NDE RH (GLSA)	
	LSS GB Radial Bearing Vibration Y NDE RH (GLSA)	
	LSS GB Gearbox Casing Vibration DE RH (GLSA)	
	LSS GB Gearbox Casing Vibration NDE RH (GLSA)	
Blower Gearbox HSS Shaft Anomaly	HSS GB Radial Bearing Vibration X DE RH (GHSA)	1. Check calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Check Carbon brush integrity.

Fault Name	Associated Symptoms	Recommendations
	HSS GB Radial Bearing Vibration Y DE RH (GHSA)	4. Check magnetism of the rotor Check rotor unbalance, misalignment. 5. Looseness. 6. Check coupling integrity. 7. Check foundation bolt integrity. 8. Check both bearings for clearance and pad wear etc. 9. Check gear integrity.
	HSS GB Radial Bearing Vibration X NDE RH (GHSA)	
	HSS GB Radial Bearing Vibration Y NDE RH (GHSA)	
	HSS GB Gearbox Casing Vibration DE RH (GHSA)	
	HSS GB Gearbox Casing Vibration NDE RH (GHSA)	
Blower Gearbox LSS Thrust Bearing Anomaly	LSS GB Thrust Bearing Displacement Active Side RH (GLTBA)	1. Check if the lube oil header temperature, pressure and quality has not altered(if lube oil system is available). 2. Assess likely cause and time to trip if intervention not possible. 3. Check calibration of sensor. 4. Intervention options; reduce thrust load, Ensure parts availability. 5. Check axial vibration trends to establish if the driver or driven has change in axial thrust and potentially damaged/wiped the pads. 6. If required inspect the thrust pads and collar to check for damage / wiping. 7. Carry out a bump check to establish what float is in the machine, compare to previous overhaul report for measurement. 8. Check the gear coupling/pre-stretch diaphragm coupling.
	LSS GB Thrust Bearing Displacement Active Side RL (GLTBA)	
	LSS GB Thrust Bearing Displacement Inactive Side RH (GLTBA)	
	LSS GB Thrust Bearing Displacement Inactive Side RL (GLTBA)	
	LSS GB Thrust Bearing Temperature A Active Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature B Active Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature C Active Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature A Inactive Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature B Inactive Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature C Inactive Side RH (GLTBA)	
Blower Gearbox HSS Thrust Bearing Anomaly	HSS GB Thrust Bearing Displacement Active Side RH (GHTBA)	1. Check if the lube oil header temperature, pressure and quality has not altered(if lube oil system is available). 2. Assess likely cause and time to trip if intervention not

Fault Name	Associated Symptoms	Recommendations
		possible.
	HSS GB Thrust Bearing Displacement Active Side RL (GHTBA)	3. Check calibration of sensor.
	HSS GB Thrust Bearing Displacement Inactive Side RH (GHTBA)	4. Intervention options; reduce thrust load, Ensure parts availability.
	HSS GB Thrust Bearing Displacement Inactive Side RL (GHTBA)	5. Check axial vibration trends to establish if the driver or driven has change in axial thrust and potentially damaged/wiped the pads.
	HSS GB Thrust Bearing Temperature A Active Side RH (GHTBA)	6. If required inspect the thrust pads and collar to check for damage / wiping.
	HSS GB Thrust Bearing Temperature B Active Side RH (GHTBA)	7. Carry out a bump check to establish what float is in the machine, compare to previous overhaul report for measurement.
	HSS GB Thrust Bearing Temperature C Active Side RH (GHTBA)	8. Check the gear coupling/pre-stretch diaphragm coupling.
	HSS GB Thrust Bearing Temperature A Inactive Side RH (GHTBA)	
	HSS GB Thrust Bearing Temperature B Inactive Side RH (GHTBA)	
	HSS GB Thrust Bearing Temperature C Inactive Side RH (GHTBA)	

Centrifugal Blower Lube Oil (Variable Guide Vane)

Model Type	APR Model
Category	Centrifugal Blower
Equipment Class	Blower
Equipment Type	Blower - Lube Oil System

Below table illustrates the list of faults and symptoms configured for a Centrifugal Blower Lube Oil System.

Fault Name	Associated Symptoms	Recommendations
Loss of Lube Oil Quality	Lube Oil Viscosity HIGH (LOLQ)	1. Check oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, TAN. 4. Check no breather caps or man covers have been removed and left open to allow debris into the tank. 5. Do filter cleaning. 6. Recondition or replace oil.
	Lube Oil Viscosity LOW (LOLQ)	
	Lube Oil TAN HIGH (LOLQ)	
	Lube Oil Composition Analysis HIGH (LOLQ)	
Lube Oil Filter Anomaly	Lube Oil Filter DP RH (LFDF)	1. Check filter DP, changeover and clean/replace the stand by filter. 2. Check the Filter DP transmitter.
Loss of Pressure due to Pump Anomaly	Lube Oil Header Pressure RL (LOPPF)	1. Check the Discharge PT integrity. 2. Switch on both oil pumps and stabilize oil pressure. 3. Assess malfunction. 4. Line walk the system and check for leaks, ensure PSV's are seated. 5. Check spillback/PCV valves are set correctly and not returning oil back to the tank. 6. Ensure no hand valves have been left shut for the standby pump. 7. Check oil level in tank, Open tank overhead vents for release of pressure. 8. Check PCV function for Integrity. 9. Assess malfunction. 10. Check Auto Logic Function tests, check functioning of the auto change over switch. 11. If maintenance has recently been performed check correct rotation of pump. 12. Ensure NRV's are working correctly.
	Lube Oil Pump Discharge Pressure RL (LOPPF)	
	Lube Oil PCV Position RL (LOPPF)	
	Lube Oil Return Flow Rate RL (LOPPF)	
	Lube Oil Main Pump Current (Run Status) ON (LOPPF)	
	Lube Oil Aux Pump Current (Run Status) ON (LOPPF)	
Loss of Cooling due to TCV Anomaly	Lube Oil Header Temperature (Cooler Out temperature) RH (LOCTF)	1. Check the calibration of sensor. 2. Check bypass control valve for integrity. 3. Open both coolers and stabilize temperature. 4. Line walk the system to check for leaks. 5. Check the Isolation/bypass valve of TCV.
	LO Cooler Bypass Valve (TCV) Position RH (LOCTF)	
Loss of Cooling due to Cooler Anomaly	Lube Oil Header Temperature (Cooler Out temperature) RH (LOCCF)	1. Check the calibration of sensor. 2. Check cooling water supply pressure. 3. Open both coolers and stabilize temperature. 4. If malfunction of one cooler is suspected, isolate and inspect it.

Fault Name	Associated Symptoms	Recommendations
Loss of Heating due Lube Oil Tank Heater Off	LO Tank And Cooler Outlet Temperature Delta RH (LOCCF)	5. Line walk the system to check for leaks. 6. Check cooling water pumps for blockages, ensure strainer is clean.
	Lube Oil Header Temperature (Cooler Out temperature) RL (LOCHOFF)	1. Check the Heater ON/OFF switch integrity and Change the switch as per the requirement. 2. Check the heater circuit for Voltage and current.
	Lube Oil Tank Temperature (Cooler Inlet temperature) RL (LOCHOFF)	
Loss of Cooling due Lube Oil Tank Heater On	Lube Oil Tank Heater (Run Status) OFF (LOCHOFF)	1. Check the Heater ON/OFF switch integrity and Change the switch as per the requirement. 2. Offline 1. 3. Check the heater circuit for Voltage and current.
	Lube Oil Header Temperature (Cooler Out temperature) RH (LOCHON)	
	Lube Oil Tank Temperature (Cooler Inlet temperature) RH (LOCHON)	
Loss of Cooling due to Fan Motor Issue	Lube Oil Tank Heater (Run Status) ON (LOCHON)	Offline: Check the Fan and Motor /coupling / belt Integrity and rectify.
	Lube Oil Header Temperature (Cooler Out temperature) RH (LOCFMI)	
	Lube Oil Cooler Fan 1 (Run Status) OFF (LOCFMI)	
	Lube Oil Cooler Fan 2 (Run Status) OFF (LOCFMI)	

14.2.2.3 Centrifugal Blower - Variable Speed

Centrifugal Blower Unit Performance and Health Fault Model (Variable Speed)

Model Type	APR Model
Category	Centrifugal Blower
Equipment Class	Blower
Equipment Type	Blower - Unit

Below table illustrates the list of faults and symptoms configured for a Blower Unit.

Fault Name	Associated Symptoms	Recommendations
Bearing Anomaly due to Lube Oil Temperature	B-Radial Bearing Vibration X DE RH (LOT)	1. Check the calibration of sensor. 2. Check cooling water supply pressure. 3. Check bypass control valve. 4. Open both coolers and stabilize temperature. 5. Check cooling water pumps for blockages, ensure strainer is clean. 6. If the cooler is an air cooler check fan and motor integrity. 7. If malfunction of one cooler is suspected, isolate and

Fault Name	Associated Symptoms	Recommendations
		inspect it. 8. Line walk the system to check for leaks.

Fault Name	Associated Symptoms	Recommendations
	B-Radial Bearing Vibration Y DE RH (LOT)	
	B-Radial Bearing Temperature A DE RH (LOT)	
	B-Radial Bearing Temperature B DE RH (LOT)	
	B-Radial Bearing Vibration X NDE RH (LOT)	
	B-Radial Bearing Vibration Y NDE RH (LOT)	
	B-NDE Radial bearing temperature A NDE RH (LOT)	
	B-NDE Radial bearing temperature B NDE RH (LOT)	
	B-Thrust Bearing Displacement Active Side RH (LOT)	
	B-Thrust Bearing Displacement Active Side RL (LOT)	
	B-Thrust Bearing Temperature A Active Side RH (LOT)	
	B-Thrust Bearing Temperature B Active Side RH (LOT)	
	B-Thrust Bearing Temperature C Active Side RH (LOT)	
	B-Thrust Bearing Temperature D Active Side RH (LOT)	
	B-Thrust Bearing Temperature A Inactive Side RH (LOT)	
	B-Thrust Bearing Temperature B Inactive Side RH (LOT)	
	B-Thrust Bearing Temperature C Inactive Side	

Fault Name	Associated Symptoms	Recommendations
	RH (LOT)	
	B-Thrust Bearing Temperature D Inactive Side RH (LOT)	
	LO header temperature RH (LOT)	
Bearing Anomaly due to Lube Oil Pressure	B-Radial Bearing Vibration X DE RH (LOP)	1. Check the calibration of sensor. 2. Switch on both oil pumps and stabilize oil pressure. 3. Check oil level in tank. 4. Check filter DP. 5. Assess malfunction. 6. Line walk the system and check for leaks. 7. Ensure PSV's are seated, check spillback/PCV valves are set correctly and not returning oil back to the tank. 8. If maintenance has recently been performed check correct rotation of pump. 9. Ensure no hand valves have been left shut, ensure NRV's are working correctly. 10. Check functioning of the auto change over switch.
	B-Radial Bearing Vibration Y DE RH (LOP)	
	B-Radial Bearing Temperature A DE RH (LOP)	
	B-Radial Bearing Temperature B DE RH (LOP)	
	B-Radial Bearing Vibration X NDE RH (LOP)	
	B-Radial Bearing Vibration Y NDE RH (LOP)	
	B-NDE Radial bearing temperature A NDE RH (LOP)	
	B-NDE Radial bearing temperature B NDE RH (LOP)	
	B-Thrust Bearing Displacement Active Side RH (LOP)	
	B-Thrust Bearing Displacement Active Side RL (LOP)	
	B-Thrust Bearing Temperature A Active Side RH (LOP)	
	B-Thrust Bearing Temperature B Active Side RH (LOP)	
	B-Thrust Bearing Temperature C Active Side RH (LOP)	
	B-Thrust Bearing Temperature D Active Side	

Fault Name	Associated Symptoms	Recommendations
	RH (LOP)	
	B-Thrust Bearing Temperature A Inactive Side RH (LOP)	
	B-Thrust Bearing Temperature B Inactive Side RH (LOP)	
	B-Thrust Bearing Temperature C Inactive Side RH (LOP)	
	B-Thrust Bearing Temperature D Inactive Side RH (LOP)	
	LO header pressure RL (LOP)	
Bearing Anomaly due to Lube Oil Quality	B-Radial Bearing Vibration X DE RH (LOQ)	1. Check oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, TAN. 4. Check no breather caps or man covers have been removed and left open to allow debris into the tank. 5. Recondition or replace oil.
	B-Radial Bearing Vibration Y DE RH (LOQ)	
	B-Radial Bearing Temperature A DE RH (LOQ)	
	B-Radial Bearing Temperature B DE RH (LOQ)	
	B-Radial Bearing Vibration X NDE RH (LOQ)	
	B-Radial Bearing Vibration Y NDE RH (LOQ)	
	B-NDE Radial bearing temperature A NDE RH (LOQ)	
	B-NDE Radial bearing temperature B NDE RH (LOQ)	
	B-Thrust Bearing Displacement Active Side RH (LOQ)	
	B-Thrust Bearing Displacement Active Side RL (LOQ)	
	B-Thrust Bearing Temperature A Active Side	

Fault Name	Associated Symptoms	Recommendations
	RH (LOQ)	
	B-Thrust Bearing Temperature B Active Side RH (LOQ)	
	B-Thrust Bearing Temperature C Active Side RH (LOQ)	
	B-Thrust Bearing Temperature D Active Side RH (LOQ)	
	B-Thrust Bearing Temperature A Inactive Side RH (LOQ)	
	B-Thrust Bearing Temperature B Inactive Side RH (LOQ)	
	B-Thrust Bearing Temperature C Inactive Side RH (LOQ)	
	B-Thrust Bearing Temperature D Inactive Side RH (LOQ)	
	LO Viscosity SH (LOQ)	
	LO Viscosity SL (LOQ)	
	LO TAN SH (LOQ)	
Thrust Bearing Failure due to Mechanical Issue	B-Thrust Bearing Displacement Active Side RH (TBM)	1. Check calibration of sensor. 2. Assess likely cause and time to trip if intervention not possible. 3. Intervention options. 4. Reduce thrust load. 5. Ensure parts availability. 6. Check axial vibration trends to establish if the machine has surged and potentially damaged/wiped the pads. 7. If required inspect the thrust pads and collar to check for damage / wiping. 8. Carry out a bump check to establish what float is in the machine, compare to previous overhaul report for measurement. 9. Check the gear coupling/pre-stretch diaphragm coupling .
	B-Thrust Bearing Displacement Active Side RL (TBM)	
	B-Thrust Bearing Displacement Inactive Side RH (TBM)	
	B-Thrust Bearing Displacement Inactive Side RL (TBM)	
	B-Thrust Bearing Temperature A Active Side RH (TBM)	
	B-Thrust Bearing Temperature B Active Side RH (TBM)	

Fault Name	Associated Symptoms	Recommendations
	B-Thrust Bearing Temperature C Active Side RH (TBM)	
	B-Thrust Bearing Temperature D Active Side RH (TBM)	
	B-Thrust Bearing Temperature A Inactive Side RH (TBM)	
	B-Thrust Bearing Temperature B Inactive Side RH (TBM)	
	B-Thrust Bearing Temperature C Inactive Side RH (TBM)	
	B-Thrust Bearing Temperature D Inactive Side RH (TBM)	
DE Radial Bearing Failure due to Mechanical Issue	B-Radial Bearing Vibration X DE RH (RBDM)	1. Check calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if intervention not possible. 4. Intervention options, reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 7. Change bearing if required.
	B-Radial Bearing Vibration Y DE RH (RBDM)	
	B-Radial Bearing Temperature A DE RH (RBDM)	
	B-Radial Bearing Temperature B DE RH (RBDM)	
NDE Radial Bearing Failure due to Mechanical Issue	B-Radial Bearing Vibration X NDE RH (RBNDM)	1. Check calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if intervention not possible. 4. Intervention options; reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 7. Change bearing if required.
	B-Radial Bearing Vibration Y NDE RH (RBNDM)	
	B-NDE Radial bearing temperature A NDE RH (RBNDM)	
	B-NDE Radial bearing temperature B NDE RH (RBNDM)	
Rotor Anomaly	B-Radial Bearing Vibration X DE RH (RTR)	1. Check the calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Check Carbon brush integrity. 4. Check magnetism of the rotor.

Fault Name	Associated Symptoms	Recommendations
	B-Radial Bearing Vibration Y DE RH (RTR)	5. Check rotor unbalance, misalignment. 6. Looseness.
	B-Radial Bearing Vibration X NDE RH (RTR)	7. Check coupling integrity. 8. Check foundation bolt integrity.
	B-Radial Bearing Vibration Y NDE RH (RTR)	9. Check both bearings for clearance and pad wear etc.

Centrifugal Blower Gearbox Fault Model (Variable Speed)

Model Type	APR Model
Category	Centrifugal Blower
Equipment Class	Blower
Equipment Type	Blower - Gearbox System

Below table illustrates the list of faults and symptoms configured for a Centrifugal Blower Gearbox System.

Fault Name	Associated Symptoms	Recommendations
Blower Gearbox LSS Radial Bearing DE Anomaly	LSS GB Radial Bearing Vibration X DE RH (GLRDA)	1. Check the calibration of sensor. 2. Check if the lube oil header temperature, pressure and quality has not altered(if lube oil system is available otherwise check greasing or clo). 3. Investigate with high frequency vibration analysis. 4. Assess likely cause and time to fail(trip) if intervention not possible. 5. Intervention options; reduce speed, Parts availability. 6. Check the trends to establish if its either both X and Y probes or one. 7. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 8. Change bearing if required.
	LSS GB Radial Bearing Vibration Y DE RH (GLRDA)	
	LSS GB Radial Bearing Temperature A DE RH (GLRDA)	
	LSS GB Radial Bearing Temperature B DE RH (GLRDA)	
	LSS GB Gearbox Casing Vibration DE RH (GLRDA)	
Blower Gearbox LSS Radial Bearing NDE Anomaly	LSS GB Radial Bearing Vibration X NDE RH (GLRNDA)	1. Check the calibration of sensor. 2. Check if the lube oil header temperature, pressure and quality has not altered(if lube oil system is available otherwise check greasing or clo). 3. Investigate with high frequency vibration analysis. 4. Assess likely cause and time to fail(trip) if intervention not possible. 5. Intervention options; reduce speed, Parts availability. 6. Check the trends to establish if its either both X and Y probes or one. 7. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 8. Change bearing if required.
	LSS GB Radial Bearing Vibration Y NDE RH (GLRNDA)	
	LSS GB Radial Bearing Temperature A NDE RH (GLRNDA)	
	LSS GB Radial Bearing Temperature B NDE RH (GLRNDA)	
	LSS GB Gearbox Casing	

Fault Name	Associated Symptoms	Recommendations
	Vibration NDE RH (GLRNDA)	
Blower Gearbox HSS Radial Bearing DE Anomaly	HSS GB Radial Bearing Vibration X DE RH (GHRDA)	1. Check the calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if intervention not possible. 4. Intervention options; reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 7. Change bearing if required.
	HSS GB Radial Bearing Vibration Y DE RH (GHRDA)	
	HSS-DE Radial bearing temperature A RH (GHRDA)	
	HSS-DE Radial bearing temperature B RH (GHRDA)	
	HSS GB Gearbox Casing Vibration DE RH (GHRDA)	
Blower Gearbox HSS Radial Bearing NDE Anomaly	HSS GB Radial Bearing Vibration X NDE RH (GHRNDA)	1. Check the calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if intervention not possible. 4. Intervention options; reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 7. Change bearing if required.
	HSS GB Radial Bearing Vibration Y NDE RH (GHRNDA)	
	HSS-NDE Radial bearing temperature A RH (GHRNDA)	
	HSS-NDE Radial bearing temperature B RH (GHRNDA)	
	HSS GB Gearbox Casing Vibration NDE RH (GHRNDA)	
Blower Gearbox LSS Shaft Anomaly	LSS GB Radial Bearing Vibration X DE RH (GLSA)	1. Check the calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Check Carbon brush integrity. 4. Check magnetism of the rotor. 5. Check rotor unbalance, misalignment. 6. Looseness. 7. Check coupling integrity. 8. Check foundation bolt integrity. 9. Check both bearings for clearance and pad wear etc. 10. Check gear integrity.
	LSS GB Radial Bearing Vibration Y DE RH (GLSA)	
	LSS GB Radial Bearing Vibration X NDE RH (GLSA)	
	LSS GB Radial Bearing Vibration Y NDE RH (GLSA)	
	LSS GB Gearbox Casing Vibration DE RH (GLSA)	
	LSS GB Gearbox Casing Vibration NDE RH (GLSA)	
Blower Gearbox HSS Shaft Anomaly	HSS GB Radial Bearing Vibration X DE RH (GHSA)	1. Check calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Check Carbon brush integrity.

Fault Name	Associated Symptoms	Recommendations
	HSS GB Radial Bearing Vibration Y DE RH (GHSA)	4. Check magnetism of the rotor Check rotor unbalance, misalignment, Looseness. 5. Check coupling integrity. 6. Check foundation bolt integrity. 7. Check both bearings for clearance and pad wear etc. 8. Check gear integrity.
	HSS GB Radial Bearing Vibration X NDE RH (GHSA)	
	HSS GB Radial Bearing Vibration Y NDE RH (GHSA)	
	HSS GB Gearbox Casing Vibration DE RH (GHSA)	
	HSS GB Gearbox Casing Vibration NDE RH (GHSA)	
Blower Gearbox LSS Thrust Bearing Anomaly	LSS GB Thrust Bearing Displacement Active Side RH (GLTBA)	1. Check if the lube oil header temperature, pressure and quality has not altered(if lube oil system is available). 2. Assess likely cause and time to trip if intervention not possible. 3. Check calibration of sensor. 4. Intervention options; reduce thrust load, Ensure parts availability. 5. Check axial vibration trends to establish if the driver or driven has change in axial thrust and potentially damaged/wiped the pads. 6. If required inspect the thrust pads and collar to check for damage / wiping. 7. Carry out a bump check to establish what float is in the machine, compare to previous overhaul report for measurement. 8. Check the gear coupling/pre-stretch diaphragm coupling.
	LSS GB Thrust Bearing Displacement Active Side RL (GLTBA)	
	LSS GB Thrust Bearing Displacement Inactive Side RH (GLTBA)	
	LSS GB Thrust Bearing Displacement Inactive Side RL (GLTBA)	
	LSS GB Thrust Bearing Temperature A Active Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature B Active Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature C Active Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature A Inactive Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature B Inactive Side RH (GLTBA)	
	LSS GB Thrust Bearing Temperature C Inactive Side RH (GLTBA)	
Blower Gearbox HSS Thrust Bearing Anomaly	HSS GB Thrust Bearing Displacement Active Side RH (GHTBA)	1. Check if the lube oil header temperature, pressure and quality has not altered(if lube oil system is available). 2. Assess likely cause and time to trip if intervention not

Fault Name	Associated Symptoms	Recommendations
		possible.
	HSS GB Thrust Bearing Displacement Active Side RL (GHTBA)	3. Check calibration of sensor.
	HSS GB Thrust Bearing Displacement Inactive Side RH (GHTBA)	4. Intervention options; reduce thrust load, Ensure parts availability.
	HSS GB Thrust Bearing Displacement Inactive Side RL (GHTBA)	5. Check axial vibration trends to establish if the driver or driven has change in axial thrust and potentially damaged/wiped the pads.
	HSS GB Thrust Bearing Temperature A Active Side RH (GHTBA)	6. If required inspect the thrust pads and collar to check for damage / wiping.
	HSS GB Thrust Bearing Temperature B Active Side RH (GHTBA)	7. Carry out a bump check to establish what float is in the machine, compare to previous overhaul report for measurement.
	HSS GB Thrust Bearing Temperature C Active Side RH (GHTBA)	8. Check the gear coupling/pre-stretch diaphragm coupling.
	HSS GB Thrust Bearing Temperature A Inactive Side RH (GHTBA)	
	HSS GB Thrust Bearing Temperature B Inactive Side RH (GHTBA)	
	HSS GB Thrust Bearing Temperature C Inactive Side RH (GHTBA)	

Centrifugal Blower Lube Oil (Variable Speed)

Model Type	APR Model
Category	Centrifugal Blower
Equipment Class	Blower
Equipment Type	Blower - Lube Oil System

Below table illustrates the list of faults and symptoms configured for a Centrifugal Blower Lube Oil System.

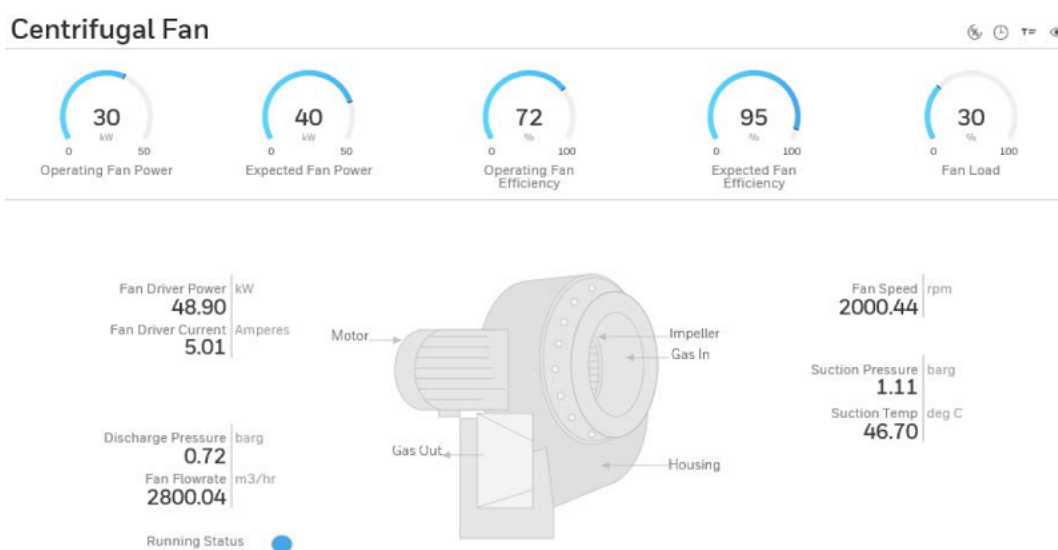
Fault Name	Associated Symptoms	Recommendations
Loss of Lube Oil Quality	Lube Oil Viscosity HIGH (LOLQ)	1. Check oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, TAN. 4. Check no breather caps or man covers have been removed and left open to allow debris into the tank. 5. Do filter cleaning. 6. Recondition or replace oil.
	Lube Oil Viscosity LOW (LOLQ)	
	Lube Oil TAN HIGH (LOLQ)	
	Lube Oil Composition Analysis HIGH (LOLQ)	
Lube Oil Filter Anomaly	Lube Oil Filter DP RH (LFDF)	1. Check filter DP, changeover and clean/replace the stand by filter. 2. Check the Filter DP transmitter.
Loss of Pressure due to Pump Anomaly	Lube Oil Header Pressure RL (LOPPF)	1. Check the Discharge PT integrity. 2. Switch on both oil pumps and stabilize oil pressure. 3. Assess malfunction. 4. Line walk the system and check for leaks, ensure PSV's are seated. 5. Check spillback/PCV valves are set correctly and not returning oil back to the tank. 6. Ensure no hand valves have been left shut for the standby pump. 7. Check oil level in tank, Open tank overhead vents for release of pressure. 8. Check PCV function for Integrity. 9. Assess malfunction. 10. Check Auto Logic Function tests, check functioning of the auto change over switch. 11. If maintenance has recently been performed check correct rotation of pump. 12. Ensure NRV's are working correctly.
	Lube Oil Pump Discharge Pressure RL (LOPPF)	
	Lube Oil PCV Position RL (LOPPF)	
	Lube Oil Return Flow Rate RL (LOPPF)	
	Lube Oil Main Pump Current (Run Status) ON (LOPPF)	
	Lube Oil Aux Pump Current (Run Status) ON (LOPPF)	
Loss of Cooling due to TCV Anomaly	Lube Oil Header Temperature (Cooler Out temperature) RH (LOCTF)	1. Check the calibration of sensor. 2. Check bypass control valve for integrity. 3. Open both coolers and stabilize temperature. 4. Line walk the system to check for leaks. 5. Check the Isolation/bypass valve of TCV.
	LO Cooler Bypass Valve (TCV) Position RH (LOCTF)	
Loss of Cooling due to Cooler Anomaly	Lube Oil Header Temperature (Cooler Out temperature) RH (LOCCF)	1. Check the calibration of sensor. 2. Check cooling water supply pressure. 3. Open both coolers and stabilize temperature. 4. If malfunction of one cooler is suspected, isolate and inspect it.

Fault Name	Associated Symptoms	Recommendations
		5. Line walk the system to check for leaks.
	LO Tank And Cooler Outlet Temperature Delta RH (LOCCF)	6. Check cooling water pumps for blockages, ensure strainer is clean.
Loss of Heating due Lube Oil Tank Heater Off	Lube Oil Header Temperature (Cooler Out temperature) RL (LOCHOFF)	1. Check the Heater ON/OFF switch integrity and Change the switch as per the requirement. 2. Check the heater circuit for Voltage and current.
	Lube Oil Tank Temperature (Cooler Inlet temperature) RL (LOCHOFF)	
	Lube Oil Tank Heater (Run Status) OFF (LOCHOFF)	
Loss of Cooling due Lube Oil Tank Heater On	Lube Oil Header Temperature (Cooler Out temperature) RH (LOCHON)	1. Check the Heater ON/OFF switch integrity and Change the switch as per the requirement. 2. Offline 1. 3. Check the heater circuit for Voltage and current.
	Lube Oil Tank Temperature (Cooler Inlet temperature) RH (LOCHON)	
	Lube Oil Tank Heater (Run Status) ON (LOCHON)	
Loss of Cooling due to Fan Motor Issue	Lube Oil Header Temperature (Cooler Out temperature) RH (LOCFMI)	Offline: Check the Fan and Motor /coupling / belt Integrity and rectify.
	Lube Oil Cooler Fan 1 (Run Status) OFF (LOCFMI)	
	Lube Oil Cooler Fan 2 (Run Status) OFF (LOCFMI)	

CENTRIFUGAL FAN MONITORING

Centrifugal fans use the kinetic energy of the impellers to increase the volume of the air stream, which in turn moves against the resistance caused by ducts, dampers and other components. Centrifugal fans displace air radially, changing the direction of the airflow. The main parts that is responsible for the conversion of energy is Impeller.

Impellers: An impeller is a rotating component in a centrifugal Fan which is equipped with vanes or blades that rotate and moves the fluid in a Fan. These vanes or blades are coupled to a shaft. When the impeller rotates, it converts the energy derived from a source i.e. motor to induce flow of the fluid. Impellers form an integral part of a Fan since it creates velocity to the fluid through its blades/vanes.



Bearings: A bearing is a component that helps to reduce friction on moving parts in a Fan. It supports the shaft to rotate smoothly. Sleeve-ring oil bearings are used extensively in fans.

Fan dampers and vanes: Fan dampers are used to control gas flow into and out of the centrifugal fan. They either installed on the inlet side or on the outlet side of the fan, or both. Dampers on the outlet side impose a flow resistance that is used to control gas flow. Dampers on the inlet side (inlet vanes) are designed to control gas flow by changing the amount of gas or air admitted to the fan inlet.

The centrifugal fan is one of the most widely used fans in Process Industry. They are used in transporting gas or materials and in ventilation systems for buildings and vehicles. They are also well- suited for industrial processes and air pollution control systems. Monitoring Basic machine health data in addition to fan operating conditions helps to maintain reliable fan operations. There are two areas that helps to monitor fan performance to increase the availability and decrease the energy costs.

1. Process and condition monitoring
2. Performance and Health monitoring

This document explains centrifugal Fan monitoring philosophy for process, performance, energy and health parameters and their importance.

15.1 Centrifugal Fan Monitoring - Performance Monitoring

The following measured parameters are monitored as it has following impact on performance and health of the centrifugal fan, if not operated in operating range.

Suction flow rate: Increasing suction flow due to increased density at lower intake temperatures will result in a higher free air delivery and also higher power consumption of the fan.

Suction Pressure: A decrease in suction pressure will reduce the density of the air at the fan intake. As with higher temperatures, it will result in lower free air delivery and power. Changes in inlet pressure can be caused by fouled inlet filters or changing barometric pressure.

Suction and Discharge temperature: The significant increase in discharge temperature over the suction temperature indicates degradation in fan efficiency. It may occur due to improper operating conditions or health issue of the fan.

Discharge Pressure: A lower discharge pressure indicates than the desired value indicates either high flow or the inability of fan to fulfill the head requirement.

Process Performance Overview: Process Performance monitoring view in APM provides a list of the asset attributes and their details, including the current value. This list can be used to review and update asset attribute information, as well as to open an Asset Trend to view more details. It also highlights (in red color) the attributes if deviates from the high or low limits.

15.1.1 Centrifugal Fan Monitoring - Model Inputs

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Running Status	Operational Status of Asset (Input)	Dynamic		Process	Field Input
2	Fan Driver power	Fan Driver power(Input)	Dynamic	kW	Process	Field Input
3	Current	Current (Required Input if Electrical Power is not given)	Dynamic	Amperes	Process	Field Input
4	Speed	Measured Speed(Input)	Dynamic	rpm	Process	Field Input
5	Suction Pressure	Suction Pressure (Required Input)	Dynamic	barg	Process	Field Input
6	Discharge Pressure	Discharge Pressure(Input)	Dynamic	barg	Process	Field Input
7	Flow Rate	Volume Flow Rate(Required Input)	Dynamic	m ³ /hr	Process	Field Input
8	Suction Temperature	Suction Temperature (Required Input)	Dynamic	°C	Process	Field Input
9	Radial Bearing 1 Vibration X	Radial Bearing 1 Vibration X (Input)	Dynamic	millimetres/second	Process	Field Input
10	Radial Bearing 1 Vibration Y	Radial Bearing 1 Vibration Y (Input)	Dynamic	millimetres/second	Process	Field Input
11	Radial Bearing 2 Vibration X	Radial Bearing 2 Vibration X (Input)	Dynamic	millimetres/second	Process	Field Input
12	Radial Bearing 2 Vibration Y	Radial Bearing 2 Vibration Y (Input)	Dynamic	millimetres/second	Process	Field Input

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
13	Efficiency Performance ct.A0	Efficiency Performance ct.A0 (Required Input)	Dynamic		Design	Constant
14	Efficiency Performance ct.A1	Efficiency Performance ct.A1 (Required Input)	Dynamic		Design	Constant
15	Efficiency Performance ct.A10	Efficiency Performance ct.A10(Required Input)	Dynamic		Design	Constant
16	Efficiency Performance ct.A11	Efficiency Performance ct.A11(Required Input)	Dynamic		Design	Constant
17	Efficiency Performance ct.A2	Efficiency Performance ct.A2 (Required Input)	Dynamic		Design	Constant
18	Efficiency Performance ct.A3	Efficiency Performance ct.A3 (Required Input)	Dynamic		Design	Constant
19	Efficiency Performance ct.A4	Efficiency Performance ct.A4 (Required Input)	Dynamic		Design	Constant
20	Efficiency Performance ct.A5	Efficiency Performance ct.A5 (Required Input)	Dynamic		Design	Constant
21	Efficiency Performance ct.A6	Efficiency Performance ct.A6 (Required Input)	Dynamic		Design	Constant
22	Efficiency Performance ct.A7	Efficiency Performance ct.A7 (Required Input)	Dynamic		Design	Constant
23	Efficiency Performance ct.A8	Efficiency Performance ct.A8 (Required Input)	Dynamic		Design	Constant
24	Efficiency Performance ct.A9	Efficiency Performance ct.A9 (Required Input)	Dynamic		Design	Constant
25	Blade Natural Frequency	Blade Natural Frequency (Required Input)	Dynamic		Design	Constant
26	Blade Pitch Angle (Reference)	Blade Pitch Angle (Reference)	Dynamic		Design	Constant
27	Blade Clearence	Blade Clearence (Required Input)	Dynamic		Design	Constant
28	Rotor Diameter	Rotor Diameter (Required Input)	Dynamic		Design	Constant
29	Driver Rated Power	Driver Rated Power[optional]	Dynamic	kW	Design	Constant
30	Mech Efficiency	Mechanical Efficiency (Required Input)	Dynamic	%	Design	Constant
31	Gear Ratio	Gear Ratio (Required Input)	Dynamic		Design	Constant
32	Motor Power Factor	Motor Power Factor[optional]	Dynamic		Design	Constant
33	Voltage	Voltage (Required Input if Electrical Power is not given)	Dynamic	V	Design	Constant

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
34	Number of Blades	Number of Blades(Required Input)	Dynamic		Design	Constant
35	Number of Obstacles	Number of Obstacles (Required Input)	Dynamic		Design	Constant
36	Rated Speed	Rated Fan Speed(Required Input)	Dynamic	rpm	Design	Constant
37	Rated Power	Rated Power (Required Input)	Dynamic	kW	Design	Constant
38	Rated Fan Static Pressure	Rated Fan Static Pressure (Required Input)	Dynamic		Design	Constant
39	Noise Level Calculation Type	Noise Level Calculation Type (Required Input)	Dynamic		Design	Constant
40	Blade Rotational Displacement Volume	Blade Rotational Displacement Volume (Required Input)	Dynamic		Design	Constant
41	Fan Blade Control Pitch Angle	Fan Blade Control Pitch Angle (Required Input)	Dynamic		Design	Constant
42	Fan Inlet Guide Vane Angle	Fan Inlet Guide Vane Angle (Required Input)	Dynamic	%	Design	Constant

15.1.2 Centrifugal Fan Monitoring - KPIs

The following indicators help to know the overall performance of the centrifugal fan.

Operating Efficiency

Fan efficiency is defined as the ratio of actual horsepower output from the fan to the shaft horsepower input for the fan.

Operating Efficiency = (Operating Power/ (Mechanical efficiency*Electric Power))

Operating Power

Operating power to drive fan would be lift volumetric flowrate from one elevation to another.

Operating Power (KW)= (Discharge Pressure - Suction Pressure) * Volumetric Flow Rate

Expected Efficiency

Expected efficiency is the actual design efficiency which is derived from the fan performance curve at current volumetric flow rate.

Expected Power

Expected power is the actual design power which is derived from the pump performance curve at current volumetric flow rate.

Fan Load

It is ration of Actual power to Expected power.

Fan Load = 100* Actual power / Expected power

15.1.3 Centrifugal Fan Monitoring - Model Outputs

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Operating Efficiency	Fan Actual Efficiency (Output)	Dynamic	%	Performance	Calculated
2	Operating Power	Actual Fan Power (Output)	Dynamic	kW	Performance	Calculated
3	Expected Power	Expected Fan Power (Output)	Dynamic	kW	Performance	Calculated
4	Expected Efficiency	Fan Expected Efficiency (Output)	Dynamic	%	Performance	Calculated
5	Fan Load	Fan Load(Output)	Dynamic	%	Performance	Calculated
6	Sound Power Level	Sound Power Level (Output)	Dynamic	decibel	Performance	Calculated

15.2 Centrifugal Fan Monitoring - Fault Model

In Asset Performance Management, you can configure faults, symptoms, and fault models on Fan performance output to diagnose the problems early. The fault management can inform operators of any violations of the nominal fan operating range and deviations from the expected characteristic and make this data available for further processing through the historian. The fault symptoms or models are used to:

1. Warn against potential damage to fan under unfavorable operating conditions.
2. Provide early warning of damages to the fan.

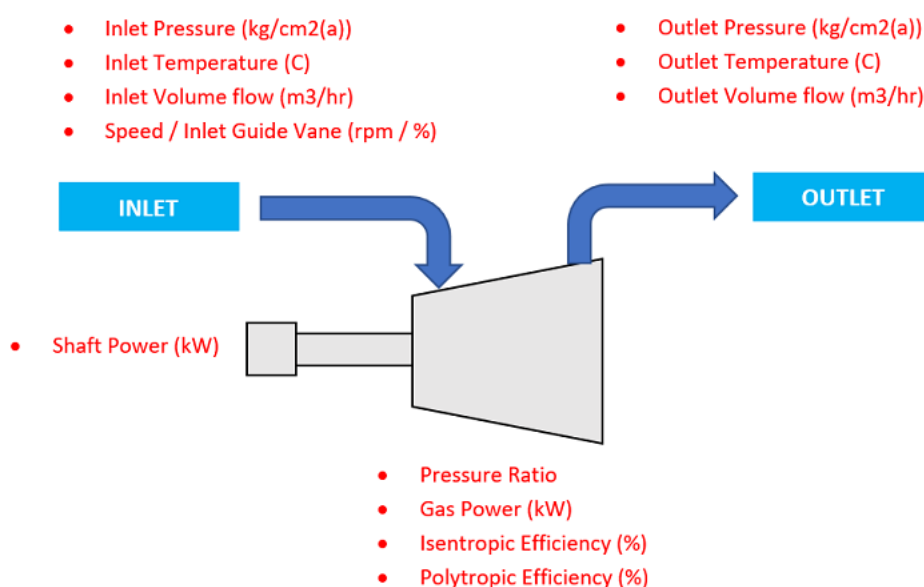
The following messages can be generated for process alarms or notifications to warn operating personnel of unfavorable operating conditions

1. **Low fan efficiency:** It is determined by means of the deviation of the current fan efficiency from the efficiency characteristic.
2. **High solid content:** It is determined by means of the reduction in delivery height or head.
3. **Dry running:** It is determined on the basis of a (second, lower) limit value for an electrical power being undershot.
4. **Incorrect direction of rotation (the motor was connected incorrectly and rotates in the wrong direction):** It is determined when the delivery head falls significantly ($> 40\%$) but with only a slight deviation ($< 20\%$) of the power characteristic.
5. **Limit value violation for power values (e.g. electrical overload).**
6. **Deviation of the operating point from the flow characteristic (i.e. reduction in delivery height or head:** indication of high gas content, blockage, or dry running). This is configured by flow loss or capacity loss of fan.
7. **Deviation of the operating point from the power characteristic:** It is configured by percent deviation in power between operating and design values.

Fault Name	Associated Symptoms
Performance Degradation	Efficiency Low
	Load High
	Load Low
	Efficiency Deviation High

Fault Name	Associated Symptoms
Health Degradation	Fan Noise Level High

TURBO-EXPANDER MONITORING



A turbo-expander (also referred to as expansion turbine) is a centrifugal or axial flow turbine in which a high-pressure gas or vapor expands to produce useful work, generally driving rotating equipment or machinery (like a secondary generator or axial compressor). The device often provides an attractive option for recovering energy when the pressure of a gas stream needs reducing and so finds use in a wide variety of plants. Because the work comes from the expanding high-pressure gas, the expansion is approximated by an isentropic (nearly constant entropy) process; the reduced pressure exhaust gas from the turbo-expander is at a lower temperature than that of the inlet gas.

For many years, turbo expanders have been used in cryogenic processing plants to provide low-temperature refrigeration. Power recovery has been of secondary importance. The main market for turbo expanders has been in low-pressure air separation plants, expanding down from 5 bar, and in hydrocarbon processing plants, expanding natural gas from as high as 200 bar. Several processes, such as dew-pointing, refrigeration and natural gas liquefaction can benefit from the additional cooling capability that a turbo-expander provides over the simple Joule-Thomson valve.

The industrial applications of Turbo Expander are as follows:

- Production of liquefied petroleum gas (LPG)
- Production of natural gas liquids (NLG) from natural gas (cryogenic expanders)
- Energy recovery in power generation by driving secondary generators
- Cryogenic services for cooling and refrigeration.
- Application in chillers and air conditioning.
- Power recovery in fluid catalytic cracker
- Liquefaction of gases like (Oxygen, Nitrogen, Argon, Helium, krypton)

Most of these industrial applications involve a combination of Turbo-Expander solid shaft coupled axial compressor except in power generation, TE is coupled with secondary generator, for cryogenic application work output is considered as by-product, cooling to around -150 and even lower.

16.1 Turbo-Expander - Sub Systems

16.1.1 Turbo-Expander Bearing Temperature and Vibrations

Condition monitoring is used for the early detection of noise, vibration signature, and bearing temperature of turbo-expander. Normally, bearings display symptoms of distress before they fail. Symptoms of a failing bearing include higher vibration levels and rising bearing or oil sump temperatures. The rise in lube oil temperature increases the viscosity of the lube oil, reducing the strength of the oil film on the bearings, which, in turn, may lead to high vibrations.

Rotor: A turbo-expander rotor is typically composed of a shaft with an expander wheel mounted on one end and a compressor wheel mounted on the other.

Bearings: Turbo-expanders can be equipped with either oil bearings or magnetic bearings. In addition to supporting the weight of the rotor, bearings must overcome the axial forces generated in the wheels (due to the difference in pressure between the front and back of each wheel.)

Casings: Turbo-expander casings are usually radial split and grouped in three distinct parts. The bearing housing is usually carbon steel and contains almost the entire machine, from expander wheel to compressor wheel, and everything in between, including bearings and seals.

16.1.2 Seal Gas System

The dry gas seal system is designed to provide a reliable shaft sealing on the rotating equipment to reduce the amount of process gas escaping from the machine's primary flow path to an acceptable level.

Loss of buffer gas will, in most cases, reverse the differential pressure across seal faces, which will damage the seal in short order. The seals will operate at a zero differential pressure level.

Dry Gas Seal Degradation: Contamination of the seal by foreign objects leads to seal failures. The running gap between the primary and mating gas seal rings is typically around three microns. Injection of any type of solids or liquids into this very narrow seal running gap can cause degradation of seal performance. This would create excessive gas leakage to the vent and would lead to seal failure eventually.

Gas seal contamination occurs mainly due to:

- Process gas leakages from the inboard or high-pressure side of the seal.
- Bearing lubrication oil from the outboard or low-pressure side of the seal.
- The seal gas injected into the seal being contaminated upstream of the injection.

The key parameters to monitor for a dry gas seal are as follows.

- Primary seal gas differential pressure—monitor and trend control valve positions.
- Primary vent flow or pressure—high for primary seal leakage, low for secondary seal leakage.
- Intermediate N₂ DP (If available).
- Primary vent control valve position (if available):

If opening, the primary seal leakage has increased

If closing, the secondary seal leakage has increased.

- Separation N₂ pressure or differential pressure (if equipped with DP control)

16.1.3 Lube Oil System

High-quality pumps require a form of lubricant to either cool, seal, or lubricate key internal components such as gears, bearings, and seals. The lube oil system supplies oil to the pump and driver bearings gears and couplings. The lube oil is drawn from the reservoir by the pumps and is fed under pressure through coolers and filters to the bearings.

Tank level: Loss in lube oil tank level can lead to loss of lubrication and pump trip scenario. The reservoir may be pressurized or vented.

Supply pressure and temperature: Temperatures and pressures are measured at all important locations in the system, including temperatures from oil sumps, return lines from bearings, gears, and other mechanical components. Temperatures and pressures are often recorded on the suction and discharge sides of each compression stage to offer the operator a sense of the system health of the system. The readings can be taken locally or transmitted to a monitoring station. The flow of oil to each bearing is regulated individually by orifices, particularly important for lubrication points requiring different pressures.

Return temperature: Heat generated by friction in the bearings is transferred to the cooling medium in the oil coolers. Air-cooled oil coolers may be employed as an alternative to water-cooled oil coolers.

Differential pressure across Filters: Filters clean the lube oil before it reaches the lubrication points and a differential pressure gauge monitors the degree of fouling (flow restriction) of the filters.

16.2 Turbo-Expander - Performance Monitoring

This section consists of Turbo Expander Single Shaft Single Unit monitoring philosophy for process, performance and health parameters and their importance.

Monitoring should include the usual parameters worthy of surveillance in high-speed turbomachinery, the temperature of journal bearings, vibration and axial position of the pinions, inlet, and discharge temperatures combined with discharge pressure.

16.2.1 Notations

F	Flow
h	Specific enthalpy
H	Heat flow
Q	Volume flow
P	Pressure
T	Temperature
η	Efficiency
W	Power
n	Polytropic exponent
γ	Isentropic exponent
P_r	Pressure Ratio
N	Speed

16.2.2 Subscripts

i	Inlet
o	Outlet

16.2.3 Turbo-Expander - Model Assumptions

The key assumptions made for developing performance model are:

- No internal leakage is assumed as tag values for both inlet and outlet are unavailable. So leakage determination would be difficult.

NOTE

If there is no performance curve data available, a separate performance model is created for turbo-expander. For this model, expected polytropic efficiency and Isentropic efficiency would be input variables whereas subsequent equations remain the same.

NOTE

For a given asset, either Isentropic performance curve data or polytropic performance curve data should be given at a time.

NOTE

After various data sheet analysis, it was observed that the performance curve was of multiple molecular weight type. Hence 3 tabular array attributes are declared to support three molecular weight curves at a time.

NOTE

The model is supported with separator calculation internally to handle the liquid fraction at the inlet stream of Turbo expander.

16.2.4 Turbo-Expander - Variants

The model supports three turbo-expander variants:

- Variable Speed
- Variable Guide Vane
- Constant speed

NOTE

If there is a turbo expander re-compressor type of variant then the operating shaft power of expander will be PHD tag mapped to the compressor's operating shaft power.

16.2.5 Turbo-Expander - Model Input

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Inlet1 Pressure	Inlet Stream 1 Pressure (Required Input)	Kg/cm ² (a)	Dynamic	Measurement	Field Input
2	Outlet1 Pressure	Outlet Stream 1 Pressure (Required Input)	Kg/cm ² (a)	Dynamic	Measurement	Field Input
3	Inlet1 Temperature	Inlet Stream 1 Temperature (Required Input)	°C	Dynamic	Measurement	Field Input
4	Outlet1 Temperature	Outlet Stream 1 Temperature (Required Input)	°C	Dynamic	Measurement	Field Input

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
5	Inlet1 Volume Flow	Inlet Stream 1 Volume Flow (Required Input)	m ³ /hr	Dynamic	Measurement	Field Input
6	Inlet1 Actual Volumetric Flow	Inlet Stream Actual Volume Flow (Input - Port mapped from suction stream) NOTE Inlet1 actual volumetric flow can be used while configuring a new model.	m ³ /hr	Dynamic	Internal	Field Input
7	Separator Volume Flow	Separator Volume Flow (Input)	m ³ /hr	Dynamic	Measurement	Field Input
8	Speed / Inlet guide vane position	Measured Speed / guide vane position (Required Input)	rpm / %	Dynamic	Measurement	Field Input
9	Measured Shaft Power	Measured Shaft Power (Input)	kW	Dynamic	Measurement	Field Input
10	Isen Performance Data	Performance curve data of Speed or Guide vane position, Operating Isentropic efficiency, Inlet Mass Flow, Operating Shaft Power		Static	Performance Map	Interpolated from performance curve
11	Poly Performance Data	Performance curve data of Speed or Guide vane position Operating Polytropic efficiency, Inlet Mass Flow, Operating Shaft Power		Static	Performance Map	Interpolated from performance curve
12	Expected Mechanical Efficiency	Expected Mechanical Efficiency (Input)	%	Static	Design	Constant
13	Operating Mechanical Efficiency	Operating Mechanical Efficiency (Input)	%	Static	Design	Constant

16.2.6 Turbo-Expander - KPIs

The following indicators help us know the overall asset performance:

16.2.6.1 Input KPIs

- Inlet Pressure
- Inlet Temperature
- Outlet pressure
- Outlet Temperature
- Inlet Volume flow rate
- Outlet Volume flow rate
- Speed / IGV position

Isentropic performance data/Polytropic performance data

- Variable speed

$$(\eta, F_i, N) \text{ and } (\dot{W}_{Shaft}, F_i, N)$$

- Variable guide vane

$$(\eta, F_i, \text{guide vane position}) \text{ and } (\dot{W}_{Shaft}, F_i, \text{guide vane position})$$

- Constant Speed

$$(\eta, F_i) \text{ and } (\dot{W}_{Shaft}, F_i)$$

16.2.6.2 Output KPIs

Expected KPIs

- Expected Isentropic Efficiency
- Expected Polytropic Efficiency
- Expected Gas Power
- Expected Shaft Power
- Expected Pressure Ratio

Operating KPIs

- Operating Isentropic Efficiency
- Operating Polytropic Efficiency
- Operating Gas Power
- Operating Shaft Power
- Operating Pressure Ratio

16.2.6.3 Turbo-Expander - Performance Modeling

Overall Mass Balance

Applying continuity equation:

$$F_i = F_o$$

Overall Force Balance

$$P_o = P_i + \Delta_{Pressure}$$

Overall Energy Balance

$$F_i h_i(P_i, T_i) = F_o h_o(P_o, T_o) + \dot{W}_{Gas}$$

$$F_i h_i(P_i, T_i, S_i) = F_o h_o^{Isen}(P_o, T_o^{Isen}, S_i) + \dot{W}_{Isentropic}$$

$$\eta_{Isentropic} = \frac{\dot{W}_{Gas}}{\dot{W}_{Isentropic}}$$

16.2.6.4 Turbo-Expander - Expected KPIs

The following indicators help us know the expected overall performance of the turbo-expander:

Expected Shaft Power (kW)

The expected shaft Power is calculated using performance curve supplied by the user (Shaft Power Vs Mass Flow).

If variable speed variant:

$$\dot{W}_{Shaft}^{Expected} = f(F_i, N)$$

If variable guide vane variant:

$$\dot{W}_{Shaft}^{Expected} = f(F_i, \text{guide vane position})$$

If constant speed variant:

$$\dot{W}_{Shaft}^{Expected} = f(F_i)$$

Expected Gas Power (kW)

$$\dot{W}_{Gas}^{Expected} = \frac{\dot{W}_{Shaft}^{Expected}}{\eta_{Mech.}^{Expected}}$$

Expected Pressure Ratio

If performance curve data of Isentropic efficiency vs mass flow is given then pressure ratio is calculated using the below formula:

$$P_r^{Expected} = \left(1 - \frac{\dot{W}_{Gas}^{Expected}}{\eta_{Isentropic}^{Expected} c f_s \left(\frac{\gamma}{\gamma-1} \right) Q_i P_i} \right)^{\frac{\gamma}{\gamma-1}}$$

If performance curve data of Polytropic efficiency vs mass flow is given:

$$P_r^{Expected} = \left(1 - \frac{\dot{W}_{Gas}^{Expected}}{\eta_{Polytropic}^{Expected} c f_n \left(\frac{n}{n-1} \right) Q_1 P_1} \right)^{\frac{n}{n-1}}$$

Expected Isentropic Efficiency (%)

If performance curve has Isentropic Efficiency Vs Mass Flow then the expected Isentropic efficiency is calculated using performance curve.

If variable speed variant:

$$\eta_{Isentropic}^{Expected} = f(F_i, N)$$

If variable guide vane variant:

$$\eta_{Isentropic}^{Expected} = f(F_i, \text{guide vane position})$$

If constant speed variant:

$$\eta_{Isentropic}^{Expected} = f(F_i)$$

Using the expected isentropic efficiency, calculate polytropic efficiency as per the below formula:

$$\eta_{Polytropic}^{Expected} = \eta_{Isentropic}^{Expected} \frac{\frac{\gamma}{\gamma-1} \left(1 - \left(P_r^{Expected} \right)^{\frac{\gamma-1}{\gamma}} \right)}{\frac{n}{n-1} \left(1 - \left(P_r^{Expected} \right)^{\frac{n-1}{n}} \right)}$$

Expected Polytropic Efficiency (%)

If performance curve has Polytropic Efficiency Vs Mass Flow then the expected polytropic efficiency is calculated using performance curve.

If variable speed variant:

$$\eta_{Polytropic}^{Expected} = f(F_i, N)$$

If variable guide vane variant:

$$\eta_{Polytropic}^{Expected} = f(F_i, \text{guide vane position})$$

If constant speed variant:

$$\eta_{Polytropic}^{Expected} = f(F_i)$$

Using the expected polytropic efficiency, calculate isentropic efficiency as per the below formula:

$$\eta_{Isentropic}^{Expected} = \eta_{Polytropic}^{Expected} \frac{\frac{n}{n-1} \left(1 - \left(P_r^{Expected} \right)^{\frac{n-1}{n}} \right)}{\frac{\gamma}{\gamma-1} \left(1 - \left(P_r^{Expected} \right)^{\frac{\gamma-1}{\gamma}} \right)}$$

16.2.6.5 Turbo Expander - Operating KPIs

The following indicators help us to know operating overall performance of the Turbo expander.

Operating Pressure Ratio

$$P_r^{Operating} = \frac{P_o}{P_i}$$

Operating Gas Power (kW)

$$F_o h_o (P_o, T_o) + \dot{W}_{Gas}^{Operating} = F_i h_i (P_i, T_i)$$

Operating Shaft Power (kW)

The mechanical efficiency is a user defined input,

$$\dot{W}_{Shaft}^{Operating} = \dot{W}_{Gas}^{Operating} \eta_{Mech.}^{Operating}$$

Operating Polytropic Efficiency (%)

$$\eta_{Polytropic}^{Operating} = \frac{\dot{W}_{Gas}^{Operating}}{\left(\frac{n}{n-1} Q_i P_i c f_n \left(1 - \left(P_r^{Operating} \right)^{\frac{(n-1)}{n}} \right) \right)}$$

Operating Isentropic Efficiency (%)

$$\eta_{Isentropic}^{Operating} = \frac{\dot{W}_{Gas}^{Operating}}{\left(\frac{\gamma}{\gamma-1} Q_i P_i c f_s \left(1 - \left(P_r^{Operating} \right)^{\frac{(\gamma-1)}{\gamma}} \right) \right)}$$

16.2.7 Turbo-Expander - Model Output

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Expected Gas Power	Expected Gas Power (Output)	kW	Dynamic	Performance	Calculated
2	Expected Isentropic Efficiency	Expected Isentropic Efficiency (Output)	%	Dynamic	Performance	Calculated
3	Expected Polytropic Efficiency	Expected Polytropic Efficiency (Output)	%	Dynamic	Performance	Calculated
4	Expected Pressure Ratio	Expected Pressure Ratio (Output)		Dynamic	Performance	Calculated
5	Expected Shaft Power	Expected Shaft Power (Output)	kW	Dynamic	Performance	Calculated
6	Operating Gas Power	Operating Gas Power (Output)	kW	Dynamic	Performance	Calculated
7	Operating Isentropic Efficiency	Operating Isentropic Efficiency (Output)	%	Dynamic	Performance	Calculated
8	Operating Polytropic Efficiency	Operating Polytropic Efficiency (Output)	%	Dynamic	Performance	Calculated
9	Operating Pressure Ratio	Operating Pressure Ratio (Output)		Dynamic	Performance	Calculated
10	Operating Shaft Power	Operating Shaft Power	kW	Dynamic	Performance	Calculated

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
		(Output)				
11	Deviation Gas Power	Deviation Gas Power (Output)	kW	Dynamic	Deviation	Calculated
12	Deviation Isentropic Efficiency	Deviation Isentropic Efficiency (Output)	%	Dynamic	Deviation	Calculated
13	Deviation Polytropic Efficiency	Deviation Polytropic Efficiency (Output)	%	Dynamic	Deviation	Calculated
14	Deviation Pressure Ratio	Deviation Pressure Ratio (Output)	%	Dynamic	Deviation	Calculated
15	Deviation Shaft Power	Deviation Shaft Power (Output)	%	Dynamic	Deviation	Calculated

16.3 Turbo-Expander - Fault Model

16.3.1 Turbo-Expander - APM Model

16.3.1.1 Turbo-Expander and Re-Compressor Bearing Temperature and Vibrations

Condition monitoring is used to early detect noise and vibration signature and bearing temperature of Screw compressor. Rise of bearing vibration and temperature due to inappropriate fitting or over greasing or cage fracture and surface distortion due to poor lubricant and overload. Normally, bearings display symptoms of distress before they fail. Symptoms of a failing bearing include higher vibration levels and rising bearing or oil sump temperatures. The rise in Lube oil temperature increases the viscosity of the lube oil which reduces the strength of the oil film on the bearings and may lead to high vibrations.

16.3.1.2 Turbo-Expander Single Shaft and Combined Shaft APM Fault Model

Category	Equipment
Equipment Class	Turbine
Equipment Type	Turbo-Expander Single Shaft Single Unit-Variable Speed/Constant Speed

Below table illustrates the list of faults and symptoms configured for a Turbo-Expander Single Shaft Single Unit-Variable Speed/Constant Speed for Performance

Fault Name	Associated Symptoms	Recommendations
Expander Inlet Anomaly	TE Inlet Pressure RL (VSEIA)	1. Check the operating context for any change 2. Check the suction header supply for TE.
	TE Inlet Pressure RH (VSEIA)	
	TE Inlet VolumeFlowRate RL (VSEIA)	
	TE Inlet VolumeFlowRate RH (VSEIA)	
TE Suction strainer Choke	TE InletStrainerDP RH (VSTESSC)	1. Clean the TE suction strainer 2. Check integrity of the mesh.

Fault Name	Associated Symptoms	Recommendations
Low Flow & Underspeed	TE Inlet VolumeFlowRate RL (VSLFU)	1. Check recycle line for any leak 2. Inspect JT valve internals and replace parts as necessary 3. Check the IGV integrity.
	TE ExpanderSpeed SL (VSLFU)	
	TE JT Valve Position RH (VSLFU)	
Overspeed	TE Inlet VolumeFlowRate RH (VSOVS)	1. Check recycle line for any jam 2. Inspect J-T Valve for integrity 3. Inspect the Compressor Impeller for its integrity.
	TE ExpanderSpeed SH (VSOVS)	
	TE JT Valve Position RL (VSOVS)	
TE Impeller Anomaly	TE PressureRatio RH (VSTEIA)	1. Inspect the Impeller Integrity of TE.
	TE Inlet VolumeFlowRate RL (VSTEIA)	
	TE ExpanderSpeed SL (VSTEIA)	
	TE Polytropic Efficiency Deviation LOW (VSTEIA)	
Performance Degradation	TE Outlet Temperature RH (VSPERDRG)	1. Check if hydrates could be forming. 2. Perform a gas analysis or reconfirm molecular weight. 3. If molecular weight is correct, performance deviation is genuine. 4. Verify DPI on process strainers. 5. Check correct IGV position. 6. Confirm J-T and recycle valve isn't passing when in the closed position. 7. Check for leaks.
	TE ExpanderSpeed RL (VSPERDRG)	
	TE ExpanderSpeed RH (VSPERDRG)	
	TE Polytropic Efficiency Deviation LOW (VSPERDRG)	
		If the above is ok the reason for degradation could be internal clearances have increased and also internals surface finish has corroded Boroscope inspection.

16.3.1.3 Turbo-Expander Single Shaft Single Unit-Variable Guide vane APM Fault Model

Category	Equipment
Equipment Class	Turbine
Equipment Type	Turbo-Expander Single Shaft Single Unit-Variable Guide vane

Below table illustrates the list of faults and symptoms configured for a Turbo-Expander Single Shaft Single Unit-Variable Guide vane for Performance

Fault Name	Associated Symptoms	Recommendations
Expander	TE Inlet Pressure RL	1. Check the operating context for any change

Fault Name	Associated Symptoms	Recommendations
Inlet Anomaly	(VGVEIA)	2. Check the suction header supply for TE.
	TE Inlet Pressure RH (VGVEIA)	
	TE Inlet VolumeFlowRate RL (VGVEIA)	
	TE Inlet VolumeFlowRate RH (VGVEIA)	
TE Suction strainer Choke	TE InletStrainerDP RH (VGVTSSC)	1. Clean the TE suction strainer 2. Check integrity of the mesh.
Low Flow & Underspeed	TE Inlet VolumeFlowRate RL (VGVLFU)	1. Check recycle line for any leak 2. Inspect JT valve internals and replace parts as necessary 3. Check the IGV integrity.
	TE ExpanderSpeed SL (VGVLFU)	
	TE InletGuideVanePos LOW (VGVLFU)	
	TE JT Valve Position RH (VGVLFU)	
Overspeed	TE Inlet VolumeFlowRate RH (VGVOVS)	1. Check recycle line for any jam 2. Inspect J-T Valve for integrity 3. Inspect the Compressor Impeller for its integrity.
	TE ExpanderSpeed SH (VGVOVS)	
	TE JT Valve Position RL (VGVOVS)	
TE Impeller Anomaly	TE PressureRatio RH (VGVTEIA)	1. Inspect the Impeller Integrity of TE.
	TE Inlet VolumeFlowRate RL (VGVTEIA)	
	TE ExpanderSpeed SL (VGVTEIA)	
	TE Polytropic Efficiency Deviation LOW (VGVTEIA)	
Performance Degradation	TE Outlet Temperature RH (VGVPERDRG)	1. Check if hydrates could be forming. 2. Perform a gas analysis or reconfirm molecular weight. 3. If molecular weight is correct, performance deviation is genuine. 4. Verify DPI on process strainers. 5. Check correct IGV position. 6. Confirm J-T and recycle valve isn't passing when in the closed position.

Fault Name	Associated Symptoms	Recommendations
		7. Check for leaks
	TE ExpanderSpeed RL (VGUPERDRG)	If the above is ok the reason for degradation could be internal clearances have increased and also internals surface finish has corroded Boroscope inspection.
	TE ExpanderSpeed RH (VGUPERDRG)	
	TE Polytropic Efficiency Deviation LOW (VGUPERDRG)	

16.3.2 Turbo-Expander - APR Model

16.3.2.1 Turbo Expander - Bearing APR Fault Model

Category	Equipment
Equipment Class	Turbine
Equipment Type	Turbo Expander - Bearing

Below table illustrates the list of faults and symptoms configured for a Turbo Expander Bearing Systems.

Fault Name	Associated Symptoms	Recommendations
Thrust/RB brg Anomaly due to Lube Oil Temp	TE-ES DE Radial vibration overall X RH (JBLOT)	1. Check the calibration of sensor. 2. Check cooling water supply pressure. 3. Check bypass control valve. 4. Open both coolers and stabilize temperature. 5. Check cooling water pumps for blockages, ensure strainer is clean. 6. If the cooler is an air cooler check fan and motor integrity. 7. If malfunction of one cooler is suspected, isolate and inspect it. 8. Line walk the system to check for leaks.
	TE-ES DE Radial vibration overall Y RH (JBLOT)	
	TE-ES DE Radial bearing temperature A RH (JBLOT)	
	TE-ES DE Radial bearing temperature B RH (JBLOT)	
	TE-ES NDE Radial vibration overall X RH (JBLOT)	
	TE-ES NDE Radial vibration overall Y RH (JBLOT)	
	TE-ES NDE Radial bearing temperature A RH (JBLOT)	
	TE-ES NDE Radial bearing temperature B RH (JBLOT)	
	TE-ES Thrust bearing active side gap A RH (JBLOT)	
	TE-ES Thrust bearing active side gap A RL (JBLOT)	
	TE-ES Thrust bearing inactive side gap A RH (JBLOT)	
	TE-ES Thrust bearing inactive side gap A RL (JBLOT)	

Fault Name	Associated Symptoms	Recommendations
	TE-ES Thrust bearing active side gap B RH (JBLOT)	
	TE-ES Thrust bearing active side gap B RL (JBLOT)	
	TE-ES Thrust bearing inactive side gap B RH (JBLOT)	
	TE-ES Thrust bearing inactive side gap B RL (JBLOT)	
	TE-ES Thrust bearing active side gap C RH (JBLOT)	
	TE-ES Thrust bearing active side gap C RL (JBLOT)	
	TE-ES Thrust bearing inactive side gap C RH (JBLOT)	
	TE-ES Thrust bearing inactive side gap C RL (JBLOT)	
	TE-ES Thrust bearing temperature active A RH (JBLOT)	
	TE-ES Thrust bearing temperature Inactive A RH (JBLOT)	
	TE-ES Thrust bearing temperature active B RH (JBLOT)	
	TE-ES Thrust bearing temperature Inactive B RH (JBLOT)	
	TE-ES Thrust bearing temperature active C RH (JBLOT)	
	TE-ES Thrust bearing temperature Inactive C RH (JBLOT)	
	LO header temperature RH (JBLOT)	
Thrust/RB brg Anomaly due to Lube Oil Press	TE-ES DE Radial vibration overall X RH (JBLOP)	1. Check the calibration of sensor. 2. Switch on both oil pumps and stabilize oil pressure. 3. Check oil level in tank. 4. Check filter DP. 5. Assess malfunction. 6. Line walk the system and check for leaks.

Fault Name	Associated Symptoms	Recommendations
		7. Ensure PSV's are seated, check spillback/PCV valves are set correctly and not returning oil back to the tank. 8. If maintenance has recently been performed check correct rotation of pump. 9. Ensure no hand valves have been left shut, ensure NRV's are working correctly. 10. Check functioning of the auto change over switch.

Fault Name	Associated Symptoms	Recommendations
	TE-ES DE Radial vibration overall Y RH (JBLOP)	
	TE-ES DE Radial bearing temperature A RH (JBLOP)	
	TE-ES DE Radial bearing temperature B RH (JBLOP)	
	TE-ES NDE Radial vibration overall X RH (JBLOP)	
	TE-ES NDE Radial vibration overall Y RH (JBLOP)	
	TE-ES NDE Radial bearing temperature A RH (JBLOP)	
	TE-ES NDE Radial bearing temperature B RH (JBLOP)	
	TE-ES Thrust bearing active side gap A RH (JBLOP)	
	TE-ES Thrust bearing active side gap A RL (JBLOP)	
	TE-ES Thrust bearing inactive side gap A RH (JBLOP)	
	TE-ES Thrust bearing inactive side gap A RL (JBLOP)	
	TE-ES Thrust bearing active side gap B RH (JBLOP)	
	TE-ES Thrust bearing active side gap B RL (JBLOP)	
	TE-ES Thrust bearing inactive side gap B RH (JBLOP)	
	TE-ES Thrust bearing inactive side gap B RL (JBLOP)	
	TE-ES Thrust bearing active side gap C RH (JBLOP)	
	TE-ES Thrust bearing active side gap C RL (JBLOP)	
	TE-ES Thrust bearing inactive side gap C RH (JBLOP)	
	TE-ES Thrust bearing inactive side gap C RL (JBLOP)	
	TE-ES Thrust bearing temperature active A RH (JBLOP)	

Fault Name	Associated Symptoms	Recommendations
	TE-ES Thrust bearing temperature Inactive A RH (JBLOP)	
	TE-ES Thrust bearing temperature active B RH (JBLOP)	
	TE-ES Thrust bearing temperature Inactive B RH (JBLOP)	
	TE-ES Thrust bearing temperature active C RH (JBLOP)	
	TE-ES Thrust bearing temperature Inactive C RH (JBLOP)	
	LO header pressure RL (JBLOP)	
Thrust/RB brg Anomaly due to Lube Oil Quality	TE-ES DE Radial vibration overall X RH (JBLOQ)	1. Check oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, TAN. 4. Check no breather caps or man covers have been removed and left open to allow debris into the tank. 5. Recondition or replace oil.
	TE-ES DE Radial vibration overall Y RH (JBLOQ)	
	TE-ES DE Radial bearing temperature A RH (JBLOQ)	
	TE-ES DE Radial bearing temperature B RH (JBLOQ)	
	TE-ES NDE Radial vibration overall X RH (JBLOQ)	
	TE-ES NDE Radial vibration overall Y RH (JBLOQ)	
	TE-ES NDE Radial bearing temperature A RH (JBLOQ)	
	TE-ES NDE Radial bearing temperature B RH (JBLOQ)	
	TE-ES Thrust bearing active side gap A RH (JBLOQ)	
	TE-ES Thrust bearing active side gap A RL (JBLOQ)	
	TE-ES Thrust bearing inactive side gap A RH (JBLOQ)	
	TE-ES Thrust bearing inactive side gap A RL (JBLOQ)	
	TE-ES Thrust bearing active	

Fault Name	Associated Symptoms	Recommendations
	side gap B RH (JBLOQ)	
	TE-ES Thrust bearing active side gap B RL (JBLOQ)	
	TE-ES Thrust bearing inactive side gap B RH (JBLOQ)	
	TE-ES Thrust bearing inactive side gap B RL (JBLOQ)	
	TE-ES Thrust bearing active side gap C RH (JBLOQ)	
	TE-ES Thrust bearing active side gap C RL (JBLOQ)	
	TE-ES Thrust bearing inactive side gap C RH (JBLOQ)	
	TE-ES Thrust bearing inactive side gap C RL (JBLOQ)	
	TE-ES Thrust bearing temperature active A RH (JBLOQ)	
	TE-ES Thrust bearing temperature Inactive A RH (JBLOQ)	
	TE-ES Thrust bearing temperature active B RH (JBLOQ)	
	TE-ES Thrust bearing temperature Inactive B RH (JBLOQ)	
	TE-ES Thrust bearing temperature active C RH (JBLOQ)	
	TE-ES Thrust bearing temperature Inactive C RH (JBLOQ)	
	LO Viscosity SH (JBLOQ)	
	LO Viscosity SL (JBLOQ)	
	LO TAN SH (JBLOQ)	
Thrust Brg Anomaly due to Mech Issue	TE-ES Thrust bearing active side gap A RH (JBLOQ)	1. Check calibration of sensor. 2. Assess likely cause and time to trip if intervention not possible. 3. Intervention options. 4. Reduce thrust load.

Fault Name	Associated Symptoms	Recommendations
	TE-ES Thrust bearing active side gap A RL (JBTMI)	5. Ensure parts availability. 6. Check axial vibration trends to establish if the machine has surged and potentially damaged/wiped the pads. 7. If required inspect the thrust pads and collar to check for damage / wiping. 8. Carry out a bump check to establish what float is in the machine, compare to previous overhaul report for measurement. 9. Check the gear coupling/pre-strech diaphragm coupling.
	TE-ES Thrust bearing inactive side gap A RH (JBTMI)	
	TE-ES Thrust bearing inactive side gap A RL (JBTMI)	
	TE-ES Thrust bearing active side gap B RH (JBTMI)	
	TE-ES Thrust bearing active side gap B RL (JBTMI)	
	TE-ES Thrust bearing inactive side gap B RH (JBTMI)	
	TE-ES Thrust bearing inactive side gap B RL (JBTMI)	
	TE-ES Thrust bearing active side gap C RH (JBTMI)	
	TE-ES Thrust bearing active side gap C RL (JBTMI)	
	TE-ES Thrust bearing inactive side gap C RH (JBTMI)	
	TE-ES Thrust bearing inactive side gap C RL (JBTMI)	
	TE-ES Thrust bearing temperature active A RH (JBTMI)	
	TE-ES Thrust bearing temperature Inactive A RH (JBTMI)	
	TE-ES Thrust bearing temperature active B RH (JBTMI)	
	TE-ES Thrust bearing temperature Inactive B RH (JBTMI)	
	TE-ES Thrust bearing temperature active C RH (JBTMI)	
	TE-ES Thrust bearing temperature Inactive C RH (JBTMI)	
DE RB Anomaly due to Mech Issue	TE-ES DE Radial vibration overall X RH (JBDEMI)	1. Check calibration of sensor. 2. Investigate with high frequency vibration analysis.

Fault Name	Associated Symptoms	Recommendations
	TE-ES DE Radial vibration overall Y RH (JBDEMI)	3. Assess likely cause and time to fail(trip) if intervention not possible. 4. Intervention options, reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 7. Change bearing if required.
	TE-ES DE Radial bearing temperature A RH (JBDEMI)	
	TE-ES DE Radial bearing temperature B RH (JBDEMI)	
	TE-ES DE Radial bearing temperature B RH (JBDEMI)	
NDE RB Anomaly due to Mech Issue	TE-ES NDE Radial vibration overall X RH (JBNDEMI)	1. Check calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if intervention not possible. 4. Intervention options; reduce speed, Parts availability. 5. Check the trends to establish if its either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 7. Change bearing if required.
	TE-ES NDE Radial vibration overall Y RH (JBNDEMI)	
	TE-ES NDE Radial bearing temperature A RH (JBNDEMI)	
	TE-ES NDE Radial bearing temperature B RH (JBNDEMI)	
Rotor Anomaly	TE-ES DE Radial vibration overall X RH (JBRTR)	1. Check the calibration of sensor. 2. Investigate with high frequency vibration analysis. 3. Check Carbon brush integrity. 4. Check magnetism of the rotor. 5. Check rotor unbalance, misalignment. 6. Looseness. 7. Check coupling integrity. 8. Check foundation bolt integrity. 9. Check both bearings for clearance and pad wear etc.
	TE-ES DE Radial vibration overall Y RH (JBRTR)	
	TE-ES NDE Radial vibration overall X RH (JBRTR)	
	TE-ES NDE Radial vibration overall Y RH (JBRTR)	

16.3.2.2 Turbo Expander - Seal System Fault Model

Category	Equipment
Equipment Class	Turbine
Equipment Type	Turbo Expander - Seal System

Below table illustrates the list of faults and symptoms configured for a Turbo Expander Seal System.

Fault Name	Associated Symptoms	Recommendations
Seal gas filter anomaly	DP over seal gas filter RH (SGFA)	1. Check the calibration of sensor. 2. Check the filter efficiency and change to standby the filter if required. 3. Repair/replace the filter internals.

Fault Name	Associated Symptoms	Recommendations
PDCV Malfunction	PDCV Valve Opening RL (PDCVM)	1. Check the calibration of sensor. 2. Check valve set point. 3. Check valve internals for any damage, and repair / replace if necessary. 4. Conduct the PDCV control loop function test.
	PDCV Valve Opening RH (PDCVM)	
	Seal gas circular Dp RL (PDCVM)	
	Seal gas circular Dp RH (PDCVM)	
Primary seal temp anomaly	Primary seal gas supply temperature RL (PSTA)	1. Check calibration of sensor. 2. Monitor seal leakage. 3. Check integrity of seal gas supply PDV. 4. Check super heater functionality if available. 5. Check separation gas. 6. Check seal supply filters for damage and are fitted correctly.
	Primary seal gas supply temperature RH (PSTA)	
Seal chamber pressure anomaly	Seal Chamber Temperature RH (SCPA)	1. Check calibration of sensor. 2. Check the PDCV alerts if activated. 3. Check drain lines and valves integrity.
	Seal Chamber Pressure RL (SCPA)	
	Seal Chamber Pressure RH (SCPA)	

16.3.2.3 Turbo Expander - Lube Oil System Fault Model

Category	Equipment
Equipment Class	Turbine
Equipment Type	Turbo Expander - Lube Oil System

Below table illustrates the list of faults and symptoms configured for a Lube Oil system.

Fault Name	Associated Symptoms	Recommendations
Loss of lube oil quality	LO Viscosity HIGH (LLOQ)	1. Check oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, TAN. 4. Check no breather caps or man covers have been removed and left open to allow debris into the tank. 5. Do filter cleaning. 6. Recondition or replace oil.
	LO Viscosity LOW (LLOQ)	
	LO TAN HIGH (LLOQ)	
	LO Composition Analysis HIGH (LLOQ)	
Lube Oil Filter Anomaly	Oil filter DP RH (LOFA)	1. Check filter DP, changeover and clean/replace the stand by filter. 2. Check the Filter DP transmitter.

Fault Name	Associated Symptoms	Recommendations
Loss of pressure due to Pump anomaly	LO header pressure RL (LOPPA)	1. Check the Discharge PT integrity. 2. Switch on both oil pumps and stabilize oil pressure. 3. Assess malfunction. 4. Line walk the system and check for leaks, ensure PSV's are seated. 5. Check spillback/PCV valves are set correctly and not returning oil back to the tank. 6. Ensure no hand valves have been left shut for the standby pump. 7. Check oil level in tank, Open tank overhead vents for release of pressure. 8. Check PCV function for Integrity. 9. Assess malfunction. 10. Check Auto Logic Function tests, check functioning of the auto change over switch. 11. If maintenance has recently been performed check correct rotation of pump. 12. Ensure NRV's are working correctly.
	LO Pumps discharge pressure RL (LOPPA)	
	LO PCV position RL (LOPPA)	
	LO PCV return flow RL (LOPPA)	
	Main Lube oil pump current (run signal) ON (LOPPA)	
	Aux Lube oil pump current (run signal) ON (LOPPA)	
Loss of cooling due to TCV Anomaly	LO header temperature RH (LOCTCVA)	1. Check the calibration of sensor. 2. Check bypass control valve for integrity. 3. Open both coolers and stabilize temperature. 4. Line walk the system to check for leaks. 5. Check the Isolation/bypass valve of TCV.
	LO Cooler bypass valve (TCV Position) RH (LOCTCVA)	
Loss of cooling due to Cooler Anomaly	LO header temperature RH (LOCCA)	1. Check the calibration of sensor. 2. Check cooling water supply pressure. 3. Open both coolers and stabilize temperature. 4. If malfunction of one cooler is suspected, isolate and inspect it. 5. Line walk the system to check for leaks. 6. Check cooling water pumps for blockages, ensure strainer is clean.
	Delta Temperature LO tank vs. LO cooler discharge RH (LOCCA)	
Loss of heating due lube oil tank heater OFF	LO header temperature RL (LOHOTH)	1. Check the Heater ON/OFF switch integrity and Change the switch as per the requirement. 2. Check the heater circuit for Voltage and current.
	Oil tank temperature RL (LOHOTH)	
	Lube oil tank heater run signal OFF (LOHOTH)	
Loss of cooling due lube oil tank heater ON	LO header temperature RH (LOCOTH)	1. Check the Heater ON/OFF switch integrity and Change the switch as per the requirement. 2. Check the heater circuit for Voltage and current.
	Oil tank temperature RH (LOCOTH)	
	Lube oil tank heater run signal ON	

Fault Name	Associated Symptoms	Recommendations
	(LOCOTH)	
Loss of cooling Due to Fan-Motor Issue	LO header temperature RH (LOCFMI)	1. Check the Fan and Motor /coupling / belt Integrity and rectify.
	Air Cooler Fan-1 (Run Status) OFF (LOCFMI)	
	Air Cooler Fan-2 (Run Status) OFF (LOCFMI)	

STEAM TURBINE MONITORING

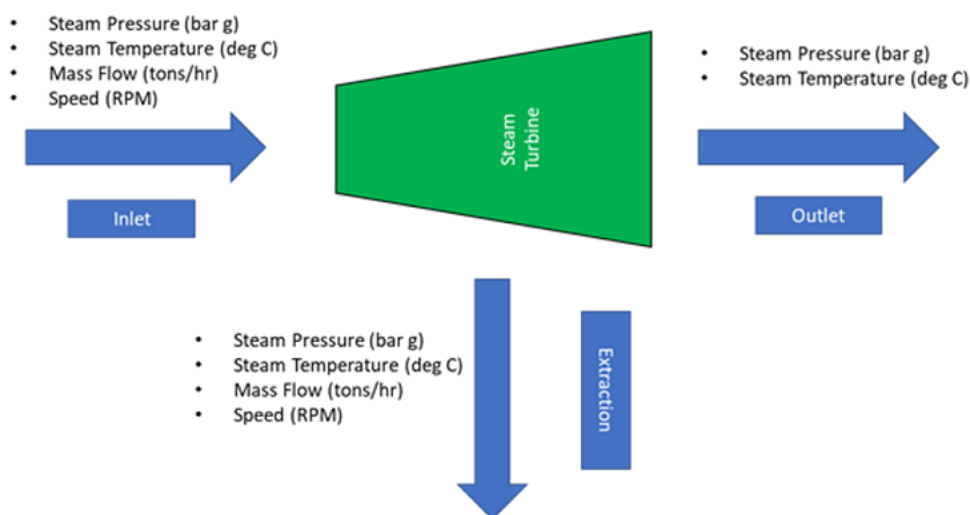
Thermal Power unit has rotating components in which the enthalpy drop of steam is converted into mechanical energy in one or several stages. The steam turbine is used in connection with industrial processes where there is a need for low- or medium-pressure steam. Turbine stages consist of a stationary stator row (guide vanes or nozzle ring) and a rotating rotor row. In guide vanes, high-pressure, high-temperature steam is expanded, resulting in high velocity. The guide vanes direct the flow to the rotor blades at an appropriate angle. In rotor, the flow direction is changed, and the kinetic energy of the working fluid is absorbed by the rotor shaft, producing mechanical energy. This mechanical energy is used to run an electric generator or mechanical equipment such as pumps, fans, compressors, etc.

As steam turbines are energy-intensive equipment, using performance monitoring tools can enable steam turbine to operate in desired conditions, saving the energy eventually. Therefore, the availability of steam turbines is an extremely important factor. A special monitoring system is required for improving turbine availability and decreasing energy cost.

This chapter explains the steam turbine monitoring philosophy related to process, performance, energy, and health parameters and their importance.

In addition to tracking turbine operating conditions, monitoring basic machine health data helps maintain reliable steam turbine operation. Monitoring of steam turbine performance leads to increased turbine availability and decreased energy cost. The three areas of monitoring in focus are:

1. Process and condition monitoring
2. Energy monitoring
3. Performance and health monitoring



The steam turbine model supports one extraction stream as well. These extraction streams are typically used as utility streams.

17.1 Steam Turbine - Sub Systems

17.1.1 Steam Turbine Bearing System

Monitoring of bearing vibrations is very important for two reasons:

It is necessary to protect turbine generator from excessive vibration. If allowed to persist, the vibration could cause damage due to rubbing, fatigue, or loosening of components.

Unusual bearing vibrations, even if they are below the alarm range, are often the first indication that something is wrong. Monitoring bearing vibration allows for the early detection of some potentially operational problems such as water induction, rubbing, or cracks in the rotor.

17.1.2 Steam Turbine Shaft Seal System

In Asset Performance Management, you can configure faults and symptoms models, steam turbine inputs, and performance output to diagnose the problems early. Fault management can inform operators of any violations in terms of the nominal turbine operating range and deviations from the expected characteristics. The fault symptoms or models are used to:

1. Warn against any potential damage to the steam turbine under unfavorable operating conditions.
2. Provide early warning of damages to the steam turbine.
3. The Fault Tree provides graphical display of assets and associated faults and symptoms.
4. The Fault Tree is used to locate the source of a fault in a single page. Hence, analyzing the fault becomes easier with navigation views.

17.2 Steam Turbine - Performance Monitoring

Thermal Power unit with rotating components in which the enthalpy drop of steam is converted into mechanical energy in one or several stages. The steam turbine is used in connection with industrial processes where there is a need for low- or medium-pressure steam. Turbine stages consist of a stationary stator row (guide vanes or nozzle ring) and a rotating rotor row. In the guide vanes, high-pressure, high-temperature steam is expanded, resulting in high velocity. The guide vanes direct the flow to the rotor blades at an appropriate angle. In the rotor, the flow direction is changed, and the kinetic energy of the working fluid is absorbed by the rotor shaft, producing mechanical energy. Mechanical energy is used to run an electric generator or mechanical equipment such as pumps, fans, and compressors, among others.

Steam turbines are energy-intensive equipment. By using performance monitoring tools, however, the steam turbine can be operated in the desired operating range, saving energy eventually.

Therefore, the availability of steam turbines is an extremely important factor. Three areas help monitor steam turbine performance for increased turbine availability and decreased energy cost.

This chapter explains the steam turbine monitoring philosophy related to process, performance, energy, and health parameters and their importance.

Three areas help monitor steam turbine performance for increased turbine availability and decreased energy cost.

In addition to turbine operating conditions, monitoring of the basic machine health data helps maintain reliable steam turbine operations.

Three areas help monitor steam turbine performance for increased turbine availability and decreased energy cost.

1. Process and condition monitoring
2. Energy monitoring
3. Performance and Health monitoring

Category	Equipment
Equipment Class	Turbine
Equipment Type	Steam Turbine (Constant Speed/Variable Speed)

17.2.1 Notations

F	Mass Flow
h	Specific enthalpy
Q	Volume flow
P	Pressure
T	Temperature
S	Entropy
n	Efficiency
W	Power
N	Speed

17.2.2 Subscripts

1	Inlet
2	Outlet
$Isen$	At Isentropic conditions
Ex	Extractive

17.2.3 Steam Turbine - Model Assumptions

The key assumptions made for developing performance model are:

1. The exhaust for back pressure steam turbine is superheated.
2. The exhaust for condensing steam turbine is saturated or below saturation point.

17.2.4 Steam Turbine - Variants

There are three steam turbine variants:

- Constant Speed
- Variable Speed
- Extractive

17.2.5 Steam Turbine - Model Input

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Barometric Pressure	Barometric Pressure (Input)	bara	Dynamic	Measurement	Field Input
2	Condenser Duty	Condenser Duty (Optional Input)	kW	Dynamic	Measurement	Field Input
3	Default Exhaust Quality	Default Exhaust Quality of Steam (Input between 0.0 and 1.0)		Static	Design	Data Sheet
4	Desuperheater Duty	Desuperheater Duty (Optional Input)	kW	Dynamic	Measurement	Field Input
5	Inlet Steam Mass Flow Rate	Inlet Steam Mass Flow Rate (Required Input)	tons/hr	Dynamic	Measurement	Field Input
6	Inlet Steam Pressure	Inlet Steam Pressure (Required Input)	barg	Dynamic	Measurement	Field Input
7	Inlet Steam Temperature	Inlet Steam Temperature (Required Input)	°C	Dynamic	Measurement	Field Input
8	Measured Shaft Power	Measured Shaft Power (Input)	kW	Dynamic	Measurement	Field Input
9	Mechanical Efficiency	Mechanical Efficiency (Input)	%	Static	Design	Data Sheet
10	Outlet Steam Pressure	Outlet Steam Pressure (Required Input)	barg	Dynamic	Measurement	Field Input
11	Outlet Steam Temperature	Outlet Steam Temperature (Required Input, if back-pressure exhaust option is selected)	°C	Dynamic	Measurement	Field Input
12	Rated Inlet Pressure	Rated Inlet Pressure (Input)	bara	Static	Design	Data Sheet
13	Rated Inlet Temperature	Rated Inlet Temperature (Input)	°C	Static	Design	Data Sheet
14	Rated Power	Rated Power (Input)	kW	Static	Design	Data Sheet
15	Rated Speed	Rated Speed (Input)	rpm	Static	Design	Data Sheet
16	Steam Exhaust Type	Steam Exhaust Type (Input as 1-for condensate exhaust, 2- for back-pressure exhaust)		Static	Measurement	Field Input
17	Steam Turbine Speed	Measured Steam Turbine Speed (Required Input)	rpm	Dynamic	Measurement	Field Input

17.2.6 Steam Turbine - KPIs

The following indicators help us know the overall performance of the asset:

17.2.6.1 Input KPIs

- Inlet Pressure
- Outlet Pressure
- Inlet Temperature
- Outlet Temperature
- Inlet Mass Flow Rate
- Speed
- Performance Data (Power, Flow)

17.2.6.2 Output KPIs

Design KPIs

- Design Shaft Power
- Design Actual Steam Rate
- Design Isentropic Efficiency
- Design Mechanical Efficiency
- Design Exhaust Quality
- Design Exhaust Temp

Operating KPIs

- Operating Shaft Power
- Operating Actual Steam Rate
- Operating Isentropic Efficiency
- Operating Mechanical Efficiency
- Operating Exhaust Quality
- Operating Exhaust Temp

17.2.6.3 Steam Turbine - Performance Modeling

Overall Mass Balance

Applying continuity equation,

$$F_1 = F_2 + F_{Ex}$$

Overall Force Balance

$$P_2 = P_1 - \Delta P$$

Overall Energy Balance

$$F_1 h_1(T_1, P_1) = F_{Ex} h_{Ex}(T_{Ex}, P_{Ex}) + F_2 h_2(T_2, P_2) + \dot{W}_{ShaftPower}$$

In certain cases, it might be possible that we don't have pressure and temperature tags just at the exhaust of steam turbine but after a condenser or desuperheater. Then we will have to correct the power calculation by accounting Desuperheater/Condenser duty.

17.2.6.4 Steam Turbine - Design KPIs

Design Shaft Power (kW)

$$\dot{W}_{ShaftPower} = f(W, F) = \text{From flow vs power curve}$$

Design Isentropic Efficiency (%)

$$\eta_{Isen} = \frac{W_{ShaftPower}}{W_{Isen}}$$

$$W_{Isen} = F_1 h_1(T_1, P_1) - F_{Ex} h_{Ex}(S_1, P_{Ex}) - F_2 h_2(S_1, P_2)$$

Design Actual Steam Rate

$$W_{SteamRate} = \frac{F_1}{W_{ShaftPower}}$$

Design Exhaust Quality

It is the amount of liquid present in the exhaust. Exhaust quality of 1 denotes pure vapor, whereas 0 denotes completely liquid and anything in between is of two phases.

$$\alpha = a(P_2, h_2)$$

The phase fraction at the exhaust is calculated by using Pressure-Enthalpy flash.

$$h_2(P_2, T_2) = \frac{(F_1 h_1(T_1, P_1) - F_{Ex} h_{Ex}(T_{Ex}, P_{Ex}) - \dot{W}_{ShaftPower})}{F_2}$$

17.2.6.5 Steam Turbine - Operating KPIs

Operating Shaft Power (kW)

$$\dot{W}_{ShaftPower} = F_1 h_1(T_1, P_1) - F_{Ex} h_{Ex}(T_{Ex}, P_{Ex}) - F_2 h_2(T_2, P_2)$$

If Desuperheater or Condenser Present

$$\dot{W}_{ShaftPower} = F_1 h_1(T_1, P_1) - F_{Ex} h_{Ex}(T_{Ex}, P_{Ex}) - F_2 h_2(T_2, P_2) - \dot{W}_{Cond./Desup}$$

Operating Isentropic Efficiency (%)

$$\eta_{Isen} = \frac{W_{ShaftPower}}{W_{Isen}}$$

$$W_{Isen} = F_1 h_1(T_1, P_1) - F_{Ex} h_{Ex}(S_1, P_{Ex}) - F_2 h_2(S_1, P_2)$$

Operating Actual Steam Rate

$$W_{SteamRate} = \frac{F_1}{W_{ShaftPower}}$$

Operating Exhaust Quality

It is the amount of liquid present in the exhaust. Exhaust quality of 1 denotes pure vapor, whereas 0 denotes completely liquid and anything in between is of two phases.

$$\alpha = a(P_2, h_2)$$

The phase fraction at the exhaust is calculated by using Pressure-Enthalpy flash.

$$h_2 = h_2(T_2, P_2)$$

17.2.7 Steam Turbine - Model Output

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Corrected Condensing Temperature	Corrected Condensing Temperature (Output)	°C	Dynamic	Performance	Calculated
2	Corrected Superheated Temperature	Corrected Superheated Temperature (Output)	°C	Dynamic	Performance	Calculated
3	Design SteamRate	Design Actual Steam Rate (Output)	kg/hr/kW	Dynamic	Performance	Calculated
4	Design Efficiency	Design Efficiency (Output)	%	Dynamic	Performance	Calculated
5	Design Exhaust Quality	Design Exhaust Quality of Steam (Output)		Dynamic	Performance	Calculated
6	Design Exhaust Temperature	Design Exhaust Temperature (Output)	°C	Dynamic	Performance	Calculated
7	Design Mech Efficiency	Design Mechanical Efficiency (Output)	%	Dynamic	Performance	Calculated
8	Design Shaft Power	Design Shaft Power (Output)	kW	Dynamic	Performance	Calculated
9	Design Transpose SteamRate	Design Transpose Steam Rate (Output)	kg/hr/kW	Dynamic	Performance	Calculated
10	Efficiency Deviation	Deviation in Efficiency between Operating and Design values (Output)	%	Dynamic	Performance	Calculated
11	Exhaust Quality Deviation	Deviation in Exhaust Quality of Steam between Operating and Design values (Output)		Dynamic	Performance	Calculated
12	Exhaust	Deviation in Exhaust Temperature	°C	Dynamic	Performance	Calculated

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
	Temperature Deviation	between Operating and Design values (Output)				
13	EnergyIndex	Energy Index (Output)		Dynamic	Performance	Calculated
14	FoulingIndex	Fouling Index (Output)		Dynamic	Performance	Calculated
15	HealthIndex	Health Index (Output)		Dynamic	Performance	Calculated
16	Ideal Steam Rate	Ideal Steam Rate (Output)	kg/hr/kW	Dynamic	Performance	Calculated
17	Operating Actual Steam Rate	Operating Actual Steam Rate (Output)	kg/hr/kW	Dynamic	Performance	Calculated
18	Operating Efficiency	Operating Efficiency (Output)	%	Dynamic	Performance	Calculated
19	Operating Exhaust Quality	Operating Exhaust Quality of Steam (Output)		Dynamic	Performance	Calculated
20	Operating Exhaust Temperature	Operating Exhaust Temperature (Output)	°C	Dynamic	Performance	Calculated
21	Operating Mech Efficiency	Operating Mechanical Efficiency (Output)	%	Dynamic	Performance	Calculated
22	Operating Shaft Power	Operating Shaft Power (Output)	kW	Dynamic	Performance	Calculated
23	Operating Transpose Steam Rate	Operating Transpose Steam Rate (Output)	kg/hr/kW	Dynamic	Performance	Calculated
24	Actual Steam Rate % Deviation	Percent Deviation in Actual Steam Rate between Operating and Design values (Output)	%	Dynamic	Performance	Calculated
25	Shaft Power % Deviation	Percent Deviation in Shaft Power between Operating and Design values (Output)	%	Dynamic	Performance	Calculated
26	Transpose Steam Rate % Deviation	Percent Deviation in Transpose Steam Rate between Operating and Design values (Output)	%	Dynamic	Performance	Calculated
27	PerformanceIndex	Performance Index (Output)		Dynamic	Performance	Calculated

17.3 Steam Turbine - Fault Model

17.3.1 Steam Turbine - APM Model

Model Type	APM Model
Category	Steam Turbine
Equipment Class	Turbine
Equipment Type	Steam Turbine - Variable Speed/Constant Speed

The following messages can be generated for process alarms or notifications to warn operating personnel of unfavorable operating conditions:

1. **Operating Efficiency Low:** Occurs due to variations in temperatures at the inlet and outlet. It can be avoided by operating the steam turbine at optimal conditions.
2. **Steam Exhaust Quality Low:** High-quality steam ensures steam is in a gaseous state. Whereas low-quality steam comprises a mixture of liquid and vapor parts. The liquid part of steam may erode the turbine blades due to the turbine exit pressure being lower than the vapor pressure of the liquid part, causing the Liquid to form bubbles, and repeated hitting of these bubbles on turbine the. blades lead to pitting. So, to avoid cavitation on turbine blades, and to have a smooth flow, it is made sure that steam exiting the turbine is of high quality.
3. **Steam Exhaust Pressure High:** High pressure of exhaust steam causes the exiting steam to be hotter and denser than it would be at lower backpressure, which puts stress on the turbine blades at the exit. If the pressure increases the more the alarm limit, the turbine will normally trip off-line.
4. **Inlet Steam Temperature Low:** The steam Temperature should be high to increase the high-quality exhaust steam and the power output of the turbine. Hence the temperature should be monitored against a limit to optimize the performance of the steam turbine.
5. **Inlet Steam Flowrate Deviations:** It is required to optimize the flow conditions to meet ideal operating conditions and to achieve high efficiency. Hence monitoring inlet steam within limits is important for the efficient operation of a steam turbine.

Fault Name	Associated Symptoms	Recommendations
ST Efficiency Degradation	Measured Shaft Power RL (ED)	1. Check Sensor Calibration, Check Steam Quantity & Quality. 2. Verify the Performance Of The Machine Versus The Enthalpy-Entropy Steam Diagram, Verify The Pressure After The Control Stage. 3. Perform a Functional Check of The Control Valve (Including Opening Sequence), Possibly, Carry Out Borescope Examination (Control Valve, Nozzles, Blades, And Labyrinths). 4. Check If Any Other Fault Has Caused These Fault Alert And Not To Include In Condensing Fault. 5. Check For Faults Related To Extraction Steam Anomaly, Exhaust Steam Anomaly, And Driven Motor Faults.
	Measured Shaft Power RH (ED)	
	Efficiency Deviation LOW (ED)	
	DP 1st Stage Wheel Chamber And Extraction Steam Pressure RL (ED)	
	DP 1st Stage Wheel Chamber And Extraction Steam Pressure RH (ED)	
	DP 2nd Stage Wheel Chamber And Exhaust Steam Pressure RL (ED)	
	DP 2nd Stage Wheel Chamber and Exhaust Steam Pressure RH (ED)	
Train Component Underperformance	Efficiency Deviation LOW (TCU)	1. Check Sensor Calibration, Identify Which Are the Components Responsible For The Underperformance. 2. Perform the Actions Described In The Dedicated Component Underperformance Fault.
	Driven Equipment Speed LOW (TCU)	
	Driven Equipment Pressure LOW (TCU)	
	Driven Equipment Power Deviation LOW (TCU)	
High Condensing Temperature and/or Pressure	Absolute Condensing Pressure (Condensing Turbine) RL (HCTP)	1. Check Sensor Calibration, Check Condenser Cooling Water Function (Flow and Temperature). 2. Verify If Vacuum Ejector Performance Is Sufficient (For

Fault Name	Associated Symptoms	Recommendations
	Absolute Condensing Pressure (Condensing Turbine) RH (HCTP)	Instance by Connecting A Large Start-Up Ejector To See Whether Condenser Pressure Can Be Lowered To Normal Levels). 3. Check the Vacuum Breaker (If Present) For Leaks. 4. Check Condensate Pump Performance. 5. If All of The Above Is Ok, Verify If Fouling If Heavy Present At The First Opportunity. 6. If Discharge Temperature Increases Significantly and Normal Temperature Cannot Be Quickly Restored, 7. Check the Compatibility Of Casing And Labyrinth Materials With Such Temperatures.
	Condensing Temperature (Condensing Turbine) RL (HCTP)	
	Condensing Temperature (Condensing Turbine) RH (HCTP)	
	Condensing Vacuum Pressure (Condensing Turbine) RL (HCTP)	
	Condensing Vacuum Pressure (Condensing Turbine) RH (HCTP)	
	Condenser Level Measurement (Condensing Turbine) RL (HCTP)	
	Condenser Level Measurement (Condensing Turbine) RH (HCTP)	
Inlet Steam Quality / Flow/Pressure/ Temp Anomaly	Inlet Steam Mass Flow Rate RL (ISQF)	1. Check Sensor Calibration, Check Inlet Steam Quantity, Pressure, Temp, and Quality. 2. Check The Condition Of The Header And The Valves In Between.
	Inlet Steam Mass Flow Rate RH (ISQF)	
	Inlet Steam Header Pressure RL (ISQF)	
	Inlet Steam Header Pressure RH (ISQF)	
	Inlet Steam Header Temperature RL (ISQF)	
	Inlet Steam Header Temperature RH (ISQF)	
GV/TV/Nozzle Anomaly	Difference In Temp of Inlet Steam Header And Inlet Steam After Throttle Valve RL (GVM)	1. Check Sensor Calibration. 2. Check GV Integrity. 3. Check TV Integrity. 4. Check Actuator Calibration. 5. Conduct Loop Proof Test of All Controllers. 6. Check Nozzle Integrity.
	Difference In Temp of Inlet Steam Header And Inlet Steam After Throttle Valve RH (GVM)	
	1st Stage Wheel Chamber Pressure RL (GVM)	
	1st Stage Wheel Chamber Pressure RH (GVM)	

Fault Name	Associated Symptoms	Recommendations
	Governor Valve Position RL (GVM)	
	Governor Valve Position RH (GVM)	
	Inlet Steam After Motor Valve And 1st Wheel Chamber Pressure Difference RL (GVM)	
	Inlet Steam After Motor Valve And 1st Wheel Chamber Pressure Difference RH (GVM)	
Motor Valve Issue	DP Of Inlet Steam Header And Inlet Steam After Motor Valve RL (MVI)	1. Check Sensor Calibration, Check Proper Supply To the Motor. 2. Check MOV Integrity. 3. Check Actuator Calibration.
	DP Of Inlet Steam Header And Inlet Steam After Motor Valve RH (MVI)	
Exhaust Steam Anamoly	Exhaust Steam Valve Position (Turbine) RL (EXSA)	1. Check Sensor Calibration. 2. Check SV/Extraction Valve Integrity. 3. Conduct Loop Proof Test for Valve Controllers. 4. Inspect Steam Line And Its Insulation.
	Exhaust Steam Valve Position (Turbine) RH (EXSA)	
	Exhaust Steam Pressure (Back Pressure turbine) RL (EXSA)	
	Exhaust Steam Pressure (Back Pressure turbine) RH (EXSA)	
	Exhaust Steam Flow (Back Pressure turbine) RL (EXSA)	
	Exhaust Steam Flow (Back Pressure turbine) RH (EXSA)	
	Exhaust Steam Temperature (Back Pressure turbine) RL (EXSA)	
	Exhaust Steam Temperature (Back Pressure turbine) RH (EXSA)	
Re-admission Steam Anomaly	Second Stage Re-Admission Pressure (Extraction & Back Pressure Turbine) RL (RASA)	1. Check Sensor Calibration, Check Nozzle, and Clean. 2. Check For Steam Line And Clean.
	Second Stage Re-Admission Pressure (Extraction & Back Pressure Turbine) RH (RASA)	
	DP Of Extraction And 2nd Stage Chamber RL (RASA)	
	DP Of Extraction And 2nd Stage	

Fault Name	Associated Symptoms	Recommendations
	Chamber RH (RASA)	
Exhaust Desuperheater Valve Anomaly	DP Of Exhaust Pressure And Exhaust Pressure After BFW Injection RH (EXDAVL)	1. Check Sensor Calibration, Check For Inlet Steam Flow. 2. Check The Requirement Of DSV Injection. 3. Check The Calibration /Leakage/Damage/Worn Off TV.
	Difference Of Exhaust Temperature And Exhaust Temperature After BFW Injection RL (EXDAVL)	
	Difference Of Exhaust Temperature And Exhaust Temperature After BFW Injection RH (EXDAVL)	
	BFW Injection Valve Opening Of Exhaust Steam RH (EXDAVL)	

Model Type	APM Model
Category	Steam Turbine
Equipment Class	Turbine
Equipment Type	Steam Turbine - Extractive

The below table illustrates the list of faults and symptoms configured for a Steam Turbine Extractive system:

Fault Name	Associated Symptoms	Recommendations
ST Efficiency Degradation	Measured Shaft Power RL (ED)	1. Check Sensor Calibration, Check Steam Quantity & Quality. 2. Verify The Performance Of The Machine Versus The Enthalpy-Entropy Steam Diagram, Verify The Pressure After The Control Stage. 3. Perform a Functional Check Of The Control Valve (Including Opening Sequence), If Possible, Carry Out Borescope Examination (Control Valve, Nozzles, Blades, And Labyrinths). 4. Check If Any other Fault Has Caused These Fault Alert And Not To Include In Condensing Fault. 5. Check For Faults Related To Extraction Steam Anomaly, Exhaust Steam Anomaly, And Driven Motor Faults.
	Measured Shaft Power RH (ED)	
	Efficiency Deviation LOW (ED)	
	DP 1st Stage Wheel Chamber And Extraction Steam Pressure RL (ED)	
	DP 1st Stage Wheel Chamber And Extraction Steam Pressure RH (ED)	
	DP 2nd Stage Wheel Chamber And Exhaust Steam Pressure RL (ED)	
	DP 2nd Stage Wheel Chamber And Exhaust Steam Pressure RH (ED)	

Fault Name	Associated Symptoms	Recommendations
Train Component Underperformance	Efficiency Deviation LOW (TCU)	1. Check Sensor Calibration, Identity Which Are The Components Responsible Of The Underperformance. 2. Perform The Actions Described In The Dedicated Component Underperformance Fault.
	Driven Equipment Speed LOW (TCU)	
	Driven Equipment Pressure LOW (TCU)	
	Driven Equipment Power Deviation LOW (TCU)	
Inlet Steam Quality / Flow/Pressure/ Temp Anomaly	Inlet Steam Mass Flow Rate RL (ISQF)	1. Check Sensor Calibration, Check Inlet Steam Quantity, Pressure, Temp, And Quality. 2. Check The Condition Of the Header And The Valves In Between.
	Inlet Steam Mass Flow Rate RH (ISQF)	
	Inlet Steam Header Pressure RL (ISQF)	
	Inlet Steam Header Pressure RH (ISQF)	
	Inlet Steam Header Temperature RL (ISQF)	
	Inlet Steam Header Temperature RH (ISQF)	
GV/TV/Nozzle Anomaly	Difference In Temp of Inlet Steam Header And Inlet Steam After Throttle Valve RL (GVM)	1. Check Sensor Calibration. 2. Check GV Integrity. 3. Check TV Integrity. 4. Check Actuator Calibration. 5. Conduct Loop Proof Test of All Controllers. 6. Check Nozzle Integrity.
	Difference In Temp of Inlet Steam Header And Inlet Steam After Throttle Valve RH (GVM)	
	1st Stage Wheel Chamber Pressure RL (GVM)	
	1st Stage Wheel Chamber Pressure RH (GVM)	
	Governor Valve Position RL (GVM)	
	Governor Valve Position RH (GVM)	
	Inlet Steam After Motor Valve And 1st Wheel Chamber Pressure Difference RL (GVM)	
	Inlet Steam After Motor Valve And 1st Wheel Chamber Pressure Difference RH (GVM)	
Motor Valve Issue	DP Of Inlet Steam Header And Inlet Steam After Motor Valve RL (MVI)	1. Check Sensor Calibration. 2. Check Proper Supply To Motor. 3. Check MOV Integrity.

Fault Name	Associated Symptoms	Recommendations
	DP Of Inlet Steam Header And Inlet Steam After Motor Valve RH (MVI)	4. Check Actuator Calibration.
Extraction Steam Anomaly	Extraction Steam Valve Position (Extraction Turbine) RL (ESA)	1. Check Sensor Calibration. 2. Check SV/Extraction Valve/On/Off Valve Integrity. 3. Conduct Loop Proof Test for Valve Controllers. 4. Inspect Steam Line and Its Insulation.
	Extraction Steam Valve Position (Extraction Turbine) RH (ESA)	
	Extraction Steam Flow (Extraction Turbine) RL (ESA)	
	Extraction Steam Flow (Extraction Turbine) RH (ESA)	
	Extraction Steam Pressure (Extraction Turbine) RL (ESA)	
	Extraction Steam Pressure (Extraction Turbine) RH (ESA)	
	Extraction Steam Temperature (Extraction Turbine) RL (ESA)	
	Extraction Steam Temperature (Extraction Turbine) RH (ESA)	
Re-admission Steam Anomaly	Second Stage Re-Admission Pressure (Extraction & Back Pressure Turbine) RL (RASA)	1. Check Sensor Calibration. 2. Check Nozzle And Clean. 3. Check For Steam Line And Clean.
	Second Stage Re-Admission Pressure (Extraction & Back Pressure Turbine) RH (RASA)	
	DP Of Extraction And 2nd Stage Chamber RL (RASA)	
	DP Of Extraction And 2nd Stage Chamber RH (RASA)	
Extraction Desuperheater Valve Anomaly	DP Of Extraction Pressure And Extraction Pressure After BFW Injection RH (EDSVL)	1. Check Sensor Calibration, Check For Inlet Steam Flow. 2. Check The Requirement Of DSV Injection. 3. Check The Calibration /Leakage/Damage/Worn of the TV.
	Difference Of Extraction Temperature And Extraction Temperature After BFW Injection RL (EDSVL)	
	Difference Of Extraction Temperature And Extraction Temperature After BFW Injection RH (EDSVL)	
	BFW Injection Valve Opening Of Extraction Steam RH (EDSVL)	

17.3.2 Steam Turbine - APR Model

17.3.2.1 Steam Turbine Bearing System

Model Type	APR Model
Category	Steam Turbine
Equipment Class	Turbine
Equipment Type	Steam Turbine - Bearing System

The below table illustrates the list of faults and symptoms configured for a Steam Turbine Bearing system:

Fault Name	Associated Symptoms	Recommendations
Thrust/Radial Bearing Anomaly due to Lube Oil Temp	ST-DE Radial vibration overall (X) RH (LOT)	1. Check cooling water supply pressure. 2. Check the bypass control valve. 3. Open both coolers and stabilize the temperature. 4. Check oil heater is not on (even if instruments indicate it is off). 5. Line walk the LO cooling pipework to check for leaks. 6. Inspect strainers on pumps. 7. Check instrumentation is working correctly. 8. If a malfunction of one cooler is suspected, isolate and inspect it.
	ST-DE Radial vibration overall (Y) RH (LOT)	
	ST-DE Radial bearing A temperature RH (LOT)	
	ST-DE Radial bearing B temperature RH (LOT)	
	ST-NDE Radial vibration overall (X) RH (LOT)	
	ST-NDE Radial vibration overall (Y) RH (LOT)	
	ST-NDE Radial bearing A temperature RH (LOT)	
	ST-NDE Radial bearing B temperature RH (LOT)	
	ST-Thrust bearing active side gap Z1 RH (LOT)	
	ST-Thrust bearing active side gap Z2 RH (LOT)	
	ST-OB Thrust Bearing A Temperature RH (LOT)	
	ST-OB Thrust Bearing B Temperature RH (LOT)	
	ST-IB Thrust Bearing A Temperature RH (LOT)	
	ST-IB Thrust Bearing B Temperature RH (LOT)	
	LO header temperature RH (LOT)	
Thrust/Radial Bearing Anomaly	ST-DE Radial vibration overall (X) RH (LOP)	1. Switch on both oil pumps (if possible) and stabilize the oil pressure.

Fault Name	Associated Symptoms	Recommendations
due to Lube Oil Press	ST-DE Radial vibration overall (Y) RH (LOP)	2. Assess malfunction. Line walk oil pipework if accessible and try to identify leaks in flanges/pipework. 3. Inspect filters and strainers for blockages. 4. Ensure the LO pressure transmitter/gauge is reading correctly. 5. Inspect/borescope oil drains for oil coking and blockages.
	ST-DE Radial bearing A temperature RH (LOP)	
	ST-DE Radial bearing B temperature RH (LOP)	
	ST-NDE Radial vibration overall (X) RH (LOP)	
	ST-NDE Radial vibration overall (Y) RH (LOP)	
	ST-NDE Radial bearing A temperature RH (LOP)	
	ST-NDE Radial bearing B temperature RH (LOP)	
	ST-Thrust bearing active side gap Z1 RH (LOP)	
	ST-Thrust bearing active side gap Z2 RH (LOP)	
	ST-OB Thrust Bearing A Temperature RH (LOP)	
	ST-OB Thrust Bearing B Temperature RH (LOP)	
	ST-IB Thrust Bearing A Temperature RH (LOP)	
	ST-IB Thrust Bearing B Temperature RH (LOP)	
	LO header pressure RL (LOP)	
Thrust/Radial Bearing Anomaly due to Lube Oil Quality	ST-DE Radial vibration overall (X) RH (LOQ)	1. Check the oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, and TAN. 4. Ensure all man covers on tanks are closed. 5. Recondition or replace oil.
	ST-DE Radial vibration overall (Y) RH (LOQ)	
	ST-DE Radial bearing A temperature RH (LOQ)	
	ST-DE Radial bearing B temperature RH (LOQ)	
	ST-NDE Radial vibration overall (X) RH (LOQ)	
	ST-NDE Radial vibration	

Fault Name	Associated Symptoms	Recommendations
	overall (Y) RH (LOQ)	
	ST-NDE Radial bearing A temperature RH (LOQ)	
	ST-NDE Radial bearing B temperature RH (LOQ)	
	ST-Thrust bearing active side gap Z1 RH (LOQ)	
	ST-Thrust bearing active side gap Z2 RH (LOQ)	
	ST-OB Thrust Bearing A Temperature RH (LOQ)	
	ST-OB Thrust Bearing B Temperature RH (LOQ)	
	ST-IB Thrust Bearing A Temperature RH (LOQ)	
	ST-IB Thrust Bearing B Temperature RH (LOQ)	
	LO Viscosity SH (LOQ)	
	LO Viscosity SL (LOQ)	
	LO TAN SH (LOQ)	
	LO TAN SL (LOQ)	
Balance Drum Seal Damage	ST-OB Thrust Bearing A Temperature RL (BDSD)	1. Verify turbine thermal expansion gauges. 2. Verify the Pressure Differential over the balance drum. 3. Carry out borescope examination. 4. If maintenance was recently done check current thrust bearing temperatures with pre-maintenance ones to determine if the rotor axial adjustment was incorrect. 5. If rotor axial adjustment is incorrect check for alignment and bearing clearance.
	ST-OB Thrust Bearing A Temperature RH (BDSD)	
	ST-OB Thrust Bearing B Temperature RL (BDSD)	
	ST-OB Thrust Bearing B Temperature RH (BDSD)	
	ST-IB Thrust Bearing A Temperature RL (BDSD)	
	ST-IB Thrust Bearing A Temperature RH (BDSD)	

Fault Name	Associated Symptoms	Recommendations
	ST-IB Thrust Bearing B Temperature RL (BDSD)	
	ST-IB Thrust Bearing B Temperature RH (BDSD)	
	DP 1st stage wheel chamber and extraction steam pressure RH (BDSD)	
Thrust Bearing Anomaly due to Mechanical Issue	ST-Thrust bearing active side gap Z1 RH (MI)	1. Inspect vibration instrumentation to check integrity. 2. If the unit has tripped borescope the internals of the turbine to identify any potential damage and inspect the thrust bearing (if this is accessible). 3. If damage has been seen the turbine will require overhauling. 4. If the unit hasn't tripped yet and has passed the alarm but approaching the trip shut the unit down and carry out a borescope inspection and thrust bearing inspection (if the bearing is accessible).
	ST-Thrust bearing active side gap Z2 RH (MI)	
	ST-OB Thrust Bearing A Temperature RH (MI)	
	ST-OB Thrust Bearing B Temperature RH (MI)	
	ST-IB Thrust Bearing A Temperature RH (MI)	
	ST-IB Thrust Bearing B Temperature RH (MI)	
DE RB Anomaly due to Mechanical Issue	ST-DE Radial vibration overall (X) RH (DERBMI)	1. Inspect vibration instrumentation to check integrity. 2. If the unit has tripped borescope the internals of the turbine to identify any potential damage and inspect the thrust bearing (if this is accessible), 3. If damage has been seen the turbine will require overhauling. 4. If the unit hasn't tripped yet and has passed the alarm but approaching the trip shut the unit down and carry out a borescope inspection and thrust bearing inspection (if the bearing is accessible). 5. Check the gland seal performance.
	ST-DE Radial vibration overall (Y) RH (DERBMI)	
	ST-DE Radial bearing A temperature RH (DERBMI)	
	ST-DE Radial bearing B temperature RH (DERBMI)	
NDE RB Anomaly due to Mechanical Issue	ST-NDE Radial vibration overall (X) RH (NDERBMI)	1. Inspect vibration instrumentation to check integrity. 2. If the unit has tripped borescope the internals of the turbine to identify any potential damage and inspect the thrust bearing (if this is accessible), if damage has been seen the turbine will require overhauling. 3. If the unit hasn't tripped yet and has passed the alarm but approaching the trip shut the unit down and carry out a borescope inspection and thrust bearing inspection (if the bearing is accessible). 4. Check the gland seal performance.
	ST-NDE Radial vibration overall (Y) RH (NDERBMI)	
	ST-NDE Radial bearing A temperature RH (NDERBMI)	

Fault Name	Associated Symptoms	Recommendations
	ST-NDE Radial bearing B temperature RH (NDERBMI)	
Rotor Anomaly	ST-DE Radial vibration overall (X) RH (RTR)	1. Investigate with high-frequency vibration analysis. 2. Check Carbon brush integrity. 3. Check the magnetism of the rotor. 4. Check rotor unbalance misalignment looseness, 5. Check coupling integrity. 6. Check foundation bolt integrity. 7. Check both bearings for clearance. 8. Check the gland seal performance.
	ST-DE Radial vibration overall (Y) RH (RTR)	
	ST-NDE Radial vibration overall (X) RH (RTR)	
	ST-NDE Radial vibration overall (Y) RH (RTR)	
Casing Temperature Anomaly	Casing Temperature Top RL (CTA)	1. Check Sensor Calibration, Check Insulation, 2. Check If Any Thermal Expansion Is Happening by Monitoring LVDT Sensor, bearing vibration, and axial displacement.
	Casing Temperature Top RH (CTA)	
	Casing Temperature Center RL (CTA)	
	Casing Temperature Center RH (CTA)	
	Casing Temperature Bottom RL (CTA)	
	Casing Temperature Bottom RH (CTA)	

17.3.2.2 Steam Turbine Seal System

Model Type	APR Model
Category	Steam Turbine
Equipment Class	Turbine
Equipment Type	Steam Turbine - Seal System

The below table illustrates the list of faults and symptoms configured for a Steam Turbine Seal System

Fault Name	Associated Symptoms	Recommendations
Gland Steam Condenser Malfunction	Gland Steam Condenser Outlet Pressure RH (GSCH)	1. Check the cooling water flow. 2. Check the ejector function. 3. Check the vacuum slot of the condenser on liquid content. 4. Check condenser temperature vs. pressure. 5. If all the above are ok, either line from the shaft seal to the condenser is restricted /blocked or excessive leakage through worn turbine shaft labyrinths may cause the malfunction of the system.
	Gland Steam Condenser Outlet Temperature RH (GSCH)	

Fault Name	Associated Symptoms	Recommendations
Loss of Seal Steam	Seal Steam Header Pressure RL (LSS)	1. Verify the seal steam supply.
	Seal Steam Pressure DE RL (LSS)	2. The PCV and the connection to the steam header.
	Seal Steam Pressure NDE RL (LSS)	3. Ensure the seal steam valve is free and open.
Shaft Seal Failure	Seal Steam Flow RH (SSF)	4. Inspect valve carbon rings.
		5. Check the performance /condition of the gland steam condenser.
		6. As last resort, use lead strips to ensure correct laby seal clearance.
Shaft Seal Failure	Seal Steam Flow RH (SSF)	1. Verify steam quality.
		2. Check turbine thermal expansion gauges vs manual.
		3. Ensure steam seals are fitted correctly with the correct clearance.

17.3.2.3 Steam Turbine Lube and Control Oil System APR Fault Model

Model Type	APR Model
Category	Steam Turbine
Equipment Class	Turbine
Equipment Type	Steam Turbine - Lube and Control Oil System

Below table illustrates the list of faults and symptoms configured for a Steam Turbine lube & Control Oil system

Fault Name	Associated Symptoms	Recommendations
Loss of lube oil quality	LO Viscosity HIGH (LLOQ)	1. Check the oil tank for free water by draining 1L into a sample bottle.
	LO Viscosity LOW (LLOQ)	2. Switch cooler if water contamination.
	LO TAN HIGH (LLOQ)	3. Take an oil sample and check water content, viscosity, and TAN.
	LO TAN LOW (LLOQ)	4. Check no breather caps or man covers have been removed and left open to allow debris into the tank.
Oil circuit leak	Oil tank level RH (OCL)	5. Do filter cleaning Recondition or replace oil.
	LO PCV position/return flow RL (OCL)	1. Line walk oil pipework if accessible and try to identify leaks in flanges /pipework.
Lube Oil Filter Anomaly	Oil filter DP RH (LOFA)	2. Inspect filters and strainers for leaks ensure.
		3. Check Oil pressure transmitter/gauge is reading correctly.
		4. Switch the oil cooler.
Lube Oil Filter Anomaly	Oil filter DP RH (LOFA)	5. If leakage stops inspect the offline cooler and plan its repair.
		1. Check filter DP.
Lube Oil Filter Anomaly	Oil filter DP RH (LOFA)	2. Changeover and clean /replace the standby filter.
		3. Check the Filter DP transmitter.

Fault Name	Associated Symptoms	Recommendations
Loss of pressure due to Pump anomaly	LO header pressure RL (LPPA)	1. Check the Discharge PT integrity. 2. Switch on both oil pumps and stabilize the oil pressure. 3. Assess malfunction. 4. Line walk the system and check for leaks. 5. Ensure PSV's are seated. 6. Check spillback/PCV valves are set correctly and not returning oil back to the tank. 7. Ensure no hand valves have been left shut for the standby pump. 8. Check the oil level in the tank. Open the tank overhead vents for the release of pressure. 9. Check the PCV function for Integrity. 10. Assess malfunction. 11. Check Auto Logic Function tests, and check the functioning of the auto change-over switch. 12. If maintenance has recently been performed check the correct rotation of the pump. 13. Ensure NRV's are working correctly.
	LO Pumps discharge pressure RL (LPPA)	
	LO PCV position/return flow RL (LPPA)	
	Main Lube oil pump current (run signal) ON (LPPA)	
	Aux Lube oil pump current (run signal) ON (LPPA)	
Loss of cooling due to TCV Anomaly	LO header temperature RH (TCVA)	1. Check the bypass control valve for integrity. 2. Open both coolers and stabilize the temperature. 3. Line walk the system to check for leaks. 4. Check the Isolation/ bypass valve of TCV.
	LO Cooler bypass valve (TCV Position) RH (TCVA)	
Loss of cooling due to Cooler Anomaly	LO header temperature RH (LCCA)	1. Check cooling water supply pressure. 2. Open both coolers and stabilize the temperature. 3. If a malfunction of one cooler is suspected, isolate and inspect it. 4. Line walk the system to check for leaks. 5. Check cooling water pumps for blockages, and ensure the strainer is clean.
	Delta Temperature LO tank vs. LO cooler discharge RH (LCCA)	
Loss of heating due lube oil tank heater OFF	LO header temperature RL (LHOT)	1. Check the Heater ON/OFF. 2. Switch integrity and Change the switch as per the requirement. 3. Check the heater circuit for Voltage and current.
	Oil tank temperature RL (LHOT)	
	Lube oil tank heater run signal OFF (LHOT)	
Loss of cooling due lube oil tank heater ON	LO header temperature RH (LCOT)	1. Check the Heater ON/OFF. 2. Switch integrity and Change the switch as per the requirement. 3. Offline Check the heater circuit for Voltage and current.
	Oil tank temperature RH (LCOT)	
	Lube oil tank heater run signal ON (LCOT)	
Loss of cooling Due to	LO header temperature RH	1. Check the Fan and Motor /coupling/belt Integrity

Fault Name	Associated Symptoms	Recommendations
Fan-Motor Issue	(LCFMI)	and rectify.
	Air Cooler Fan-1 (Run Status) OFF (LCFMI)	
	Air Cooler Fan-2 (Run Status) OFF (LCFMI)	
Loss of Control Oil pressure	Control Oil header pressure RL (LCOP)	1. Switch on both oil pumps (if possible) and stabilize the oil pressure. 2. Assess malfunction. 3. Line walk oil pipework if accessible and try to identify leaks in flanges /pipework. 4. Inspect filters and strainers for blockages. 5. Ensure the Control Oil pressure transmitter /gauge is reading correctly.
Control Oil Filter Malfunction	Control oil Filter DP RH (COFM)	1. Check the control oil filter and clean it.
LO overpressure failure	Tank over pressure RH (LOOPF)	1. Check the tank relief valve.

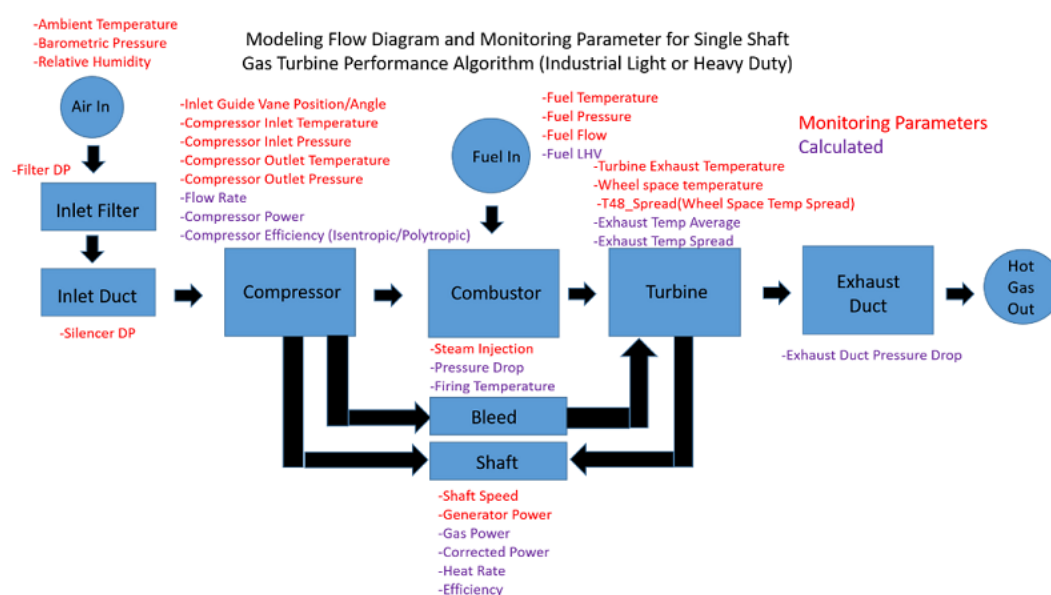
GAS TURBINE MONITORING

Gas turbines generate power using gas. This gas, in turn, is produced by burning fuel in combustor in the presence of compressed air. The fuels considered are natural gas, diesel oil, and so on. Gas turbines convert fuel energy into mechanical or electric power (by connecting to electric generators). Most of the gas turbines are internal combustion machines. Other machines are fired externally. Especially for high power applications the gas turbines are widely used compared to conventional reciprocating engines due to its high-power density. Gas turbines have high reliability as they have no reciprocating and rubbing members inside them, unlike in reciprocating engine.

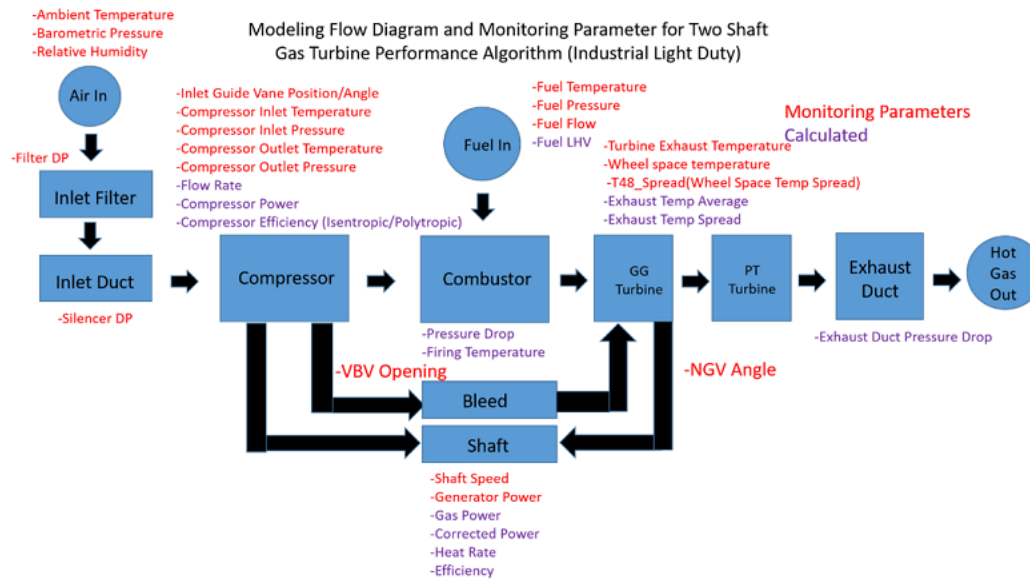
Entering the intake at constant pressure, the air is compressed as it passes through the compressor (area A in the figure 1.2). Combustion occurs at the intersection of area B and C, increasing the volume. Passing through area D, the gases expand because of the reducing pressure and increasing volume. The gases are discharged to the atmosphere through the exhaust duct at constant pressure.

Gas turbine works on the ideal Brayton cycle. The underlying thermodynamic cycle involves the compression of a gaseous medium (typically air) and the addition of fuel energy through combustion or heat exchange and expansion of the hot compressed gas through a turbine to convert thermal energy into shaft power. In contrast to ideal thermodynamic cycle, a gas turbine cannot perform these processes without friction or losses. The profitability of the gas turbine process is related to the efficiency (shaft power generated per unit fuel consumption). The importance of maximizing efficiency is thus evident. Gas turbine (GT) is a complex installation. It requires high reliability and availability for a safe output delivery. Therefore, there is a need for the improvement in engine reliability. Maintenance cost reduction can be achieved through effective and continual condition and health monitoring of the engine.

In case of single-shaft, gas turbine is designed with roughly equal pressure ratios across all the expansion stages and are mechanically coupled to the gas compressor and generator to operate at the generator speed. Only one turbine is present which is used for driving the compressor and transferring shaft work.



In case of multi-shaft, the compressor is mechanically driven by a set of expansion stages sized to produce the amount of mechanical work that is required by the compressor. Consequently, the shaft can rotate at different speeds, without it being connected to the electrical generator. The air produced from this gas generator is heated and then directed to a turbo generator. As the final expansion stage on a separate power turbine shaft, turbo generator rotates at the optimal generator speed. There are two shafts. The high-pressure turbine (gas generator) is used for driving the compressor and the low-pressure turbine (power turbine) is used to transfer the generated shaft work.



In addition to tracking turbine operating conditions, monitoring basic machine health data helps maintain reliable gas turbine operations. Three areas help monitor gas turbine performance (for increased availability and decreased energy costs):

1. Process and condition monitoring
2. Energy monitoring
3. Performance and Health monitoring

18.1 Gas Turbine - Sub Systems

18.1.1 Gas Turbine Bearing System APM Template

Monitoring of bearing vibrations is very important for two reasons: - It is necessary to protect the turbine generator from excessive vibration. If allowed to persist the vibration could cause damage due to rubbing, fatigue or loosening of components.

Unusual bearing vibrations (even if still below the alarm range) are very often the first indication of something wrong. Thus, their monitoring allows early detection of some potentially operational problems such as water induction, rubbing, or cracks in the rotor.

18.2 Gas Turbine - Performance Monitoring

The following measured parameters are monitored as they can hamper the health and performance of gas turbine if not maintained in an optimal operating range.

Turbine Inlet Pressure and Temperature: The power output of a gas turbine is a function of turbine inlet temperature (TIT). TIT plays an important role in the performance of a combined cycle. The maximum value of TIT is fixed due to the metallurgical problem of turbine blade cooling. TIT should be kept on the higher side, because at lower values, the exergy destruction is higher. High pressure and high temperature at the turbine inlet reduces the durability of the gas path

component, decreasing turbine efficiency. However, to achieve the thermal efficiencies delivered by modern gas turbines, they must work at process conditions that push the mechanical and thermal stress of the materials used in the machine's gas path components to their limits. High temperature and low pressure at the compressor inlet reduces the mass flow that can be generated through the turbine.

Humidity: Higher specific humidity increases the specific volume of the inlet air flow, so that the mass flow through the turbine is reduced resulting in less power output and increased heat rate.

Turbine Outlet Pressure: At the turbine outlet, high pressure (either exhausting into a stack, or a heat recovery steam generator in combined cycle plants) reduces the energy that can be converted to shaft power.

Load: If the gas turbine is not operated at its optimal load level, the flow triangles in the compressor and turbine expander stages will differ from design assumptions, resulting in the dissipation of more energy.

Compressor Pressure Ratio: The temperature of air leaving the compressor and entering in combustion chamber depends upon the compressor pressure ratio. As the compression ratio increases, the air exiting the compressors is hotter. As a result, lesser fuel is required (lowering the air fuel ratio) to reach the desired turbine inlet temperature for a fixed gas flow. The energy required in the compressor and the power output of the gas turbine steadily increases with compression ratio, decreasing the exhaust gas temperature. This lower gas temperature causes less steam to be produced in the HRSG, lowering the outputs of the steam cycle.

Fuel heating value: It depends on the energy per mass unit of fuel injected along with steam or water into the combustors, which helps boost power output and control NOx emission level.

18.2.1 Notations

F	Mass Flow
h	Specific enthalpy
Q	Volume flow
P	Pressure
T	Temperature
η	Efficiency
W	Power
N	Rotor Speed

18.2.2 Subscripts

I	Inlet
O	Outlet
$Isen$	At Isentropic conditions
$Poly$	At Polytropic conditions

18.2.3 Gas Turbine - Variants

The model supports to handle three variants those are

- Gas Turbine - Single Shaft
- Gas Turbine - Multi Shaft

18.2.4 Gas Turbine - Model Input

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	AirBellMouthStaticPressure	Air Bell Mouth Static Pressure or Pressure Drop (Required Input, if air inlet volume flow rate and GT measured shaft power are not given)	barg	Dynamic	Measurement	Field Input
2	AirInletVolumeFlowRate	AirInletVolumeFlowRate (Required Input, if air Bell Mouth static pressure and GT measured shaft power are not given)	m ³ /hr	Dynamic	Measurement	Field Input
3	AmbientRelativeHumidity	Ambient Relative Humidity (Input)	%	Dynamic	Measurement	Field Input
4	AmbientTemperature	Ambient Temperature (Input)	°C	Dynamic	Measurement	Field Input
5	BarometricPressure	Barometric Pressure (Input)	bara	Dynamic	Measurement	Field Input
6	CombustorJacketHeatLoss	Combustor Jacket Heat Loss (Input)	kJ/hr	Static	Design	Data Sheet
7	CombustorPresLossFactor	Combustor Pressure Loss Factor (Input)		Static	Design	Data Sheet
8	CombustorRatedEff	Combustor Rated Efficiency (Input)	%	Static	Design	Data Sheet
9	CompressorDischargePres	Compressor Discharge Pressure (Required Input)	barg	Dynamic	Measurement	Field Input
10	CompressorDischargeTemp	Compressor Discharge Temperature (Required Input)	°C	Dynamic	Measurement	Field Input
11	CompressorDisPresCorrFactor	Compressor Discharge Pressure Correction Factor (Input)		Static	Design	Data Sheet
12	CompressorGuideVanePos	Compressor Guide Vane Position (Required Input for variable guide vane option)	Degrees	Dynamic	Measurement	Field Input
13	CompressorInletArea	Compressor Inlet Area at IGV (Required Input, if air inlet volume flow rate and GT measured shaft power are not given)	m ²	Dynamic	Measurement	Field Input
14	CompressorInletPressure	Compressor Inlet Pressure (Input)	barg	Dynamic	Measurement	Field Input
15	CompressorInletTemperature	Compressor Inlet	°C	Dynamic	Measurement	Field

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
		Temperature (Input)				Input
16	CompressorMechEff	Compressor Mechanical Efficiency (Input)	%	Dynamic	Measurement	Field Input
17	CompressorOption	Compressor Type (Input as 1-Constant Speed and Constant Guide Vane Pos (default), 2- Constant Speed Variable Guide Vane Pos, 3-Variable Speed)		Static	Design	Data Sheet
18	CompressorRecycleBleedFraction	Compressor Recycle Bleed Fraction for Inlet Heating (Input)		Dynamic	Measurement	Field Input
19	DiffusorHeatLoss	Diffusor Heat Loss (Input)	kJ/hr	Static	Design	Data Sheet
20	DiffusorPresLossFactor	Diffusor Pressure Loss Factor (Input)		Static	Design	Data Sheet
21	Fuel_Argon	Argon Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
22	Fuel_Ash	Ash Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
23	Fuel_Butane	Butane Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
24	Fuel_Butene	Butene Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
25	Fuel_CarbonDioxide	Carbon Dioxide Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
26	Fuel_CarbonMonoxide	Carbon Monoxide Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
27	Fuel_Decane	Decane Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
28	Fuel_Ethane	Ethane Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
29	Fuel_Ethylene	Ethylene Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
30	Fuel_Helium	Helium Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
31	Fuel_Heptane	Heptane Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
32	Fuel_Hexane	Hexane Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
33	Fuel_Hydrogen	Hydrogen Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
34	Fuel_HydrogenSulfide	Hydrogen Sulfide Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
35	Fuel_IsoButane	IsoButane Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
36	Fuel_IsoPentane	IsoPentane Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
37	Fuel_Methane	Methane Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
38	Fuel_Nitrogen	Nitrogen Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
39	Fuel_Nonane	Nonane Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
40	Fuel_Octane	Octane Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
41	Fuel_Oxygen	Oxygen Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
42	Fuel_Pentane	Pentane Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
43	Fuel_Propane	Propane Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
44	Fuel_Propylene	Propylene Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
45	Fuel_Sulfur	Sulfur Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
46	Fuel_SulfurDioxide	Sulfur Dioxid Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
47	Fuel_Water	Water Mole Percent in Fuel (Input)	%	Static	Design	Data Sheet
48	FuelInletMassFlowRate	Fuel Inlet Mass Flow Rate (Required Input)	kg/hr	Dynamic	Measurement	Field Input
49	FuelInletPressure	Fuel Inlet Pressure (Input)	barg	Dynamic	Measurement	Field Input
50	FuelInletTemperature	Fuel Inlet Temperature (Input)	°C	Dynamic	Measurement	Field Input
51	FuelLHV	Fuel Lower Heating Value (Required Input)	kJ/kg	Dynamic	Measurement	Field Input
52	Gas Generator Turbine Outlet Pressure	Gas Generator Turbine Outlet Pressure (Required Input)	barg	Dynamic	Measurement	Field Input

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
53	Gas Generator Turbine Outlet Temperature	Gas Generator Turbine Outlet Temperature (Required Input)	°C	Dynamic	Measurement	Field Input
54	Gas generator turbine to compressor mechanical Efficiency	Gas generator turbine to compressor mechanical Efficiency (Input)	%	Dynamic	Measurement	Field Input
55	GT Mechanical Efficiency	Gas Turbine Mechanical Efficiency (Input)	%	Dynamic	Measurement	Field Input
56	GT Nozzle Angle	Gas Turbine Nozzle Angle (Input)	Degrees	Dynamic	Measurement	Field Input
57	GT Pitch Radius	Gas Turbine Pitch Radius (Input)	m	Dynamic	Measurement	Field Input
58	GTInletAirFlowCorrFactor	Gas Turbine Inlet Air Flow Correction Factor (Input)		Dynamic	Measurement	Field Input
59	GTMeasuredShaftPower	Gas Turbine Measured Shaft Power or Power Produced by Gas Turbine (Required Input, if air inlet volume flow rate and air Bell Mouth static pressure are not given)	kW	Dynamic	Measurement	Field Input
60	GTOutletPressure	Gas Turbine Outlet Pressure (Required Input)	barg	Dynamic	Measurement	Field Input
61	GTOutletPressure in millibar	Gas Turbine Outlet Pressure in millibar (Required Input)	millibar	Dynamic	Measurement	Field Input
62	GTOutletTemperature	Gas Turbine Outlet Temperature (Required Input)	°C	Dynamic	Measurement	Field Input
63	GTRatedPressure	Gas Turbine Rated Pressure or Combustor Rated Pressure (Input)	bara	Static	Design	Data Sheet
64	GTRatedShaftPower	Gas Turbine Rated Shaft Power (Input)	kW	Static	Design	Data Sheet
65	GTRatedTemperature	Gas Turbine Rated Temperature or Combustor Rated Temperature (Input)	°C	Static	Design	Data Sheet
66	GTReferenceMolecularWeight	Gas Turbine Reference Molecular Weight (Input)		Static	Design	Data Sheet
67	GTReferencePressure	Gas Turbine Reference Pressure (Input)	bara	Static	Design	Data Sheet
68	GTReferenceTemperature	Gas Turbine Reference Temperature (Input)	°C	Static	Design	Data Sheet

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
69	Nozzle Leakage Air Fraction	Nozzle Leakage Air Fraction (Input)		Static	Design	Data Sheet
70	NozzleCoolingAirFraction	Nozzle Cooling Air Fraction (Input)		Static	Design	Data Sheet
71	RatedAirMassFlowRate	Rated Air Inlet Mass Flow Rate (Required Input if air BellMouth static pressure and air inlet volume flow rate are not given)	kg/hr	Static	Design	Data Sheet
72	Turbine Leakage Air Fraction	Turbine Leakage Air Fraction (Input)		Static	Design	Data Sheet
73	TurbineCoolingAirFraction	Turbine Cooling Air Fraction (Input)		Static	Design	Data Sheet
74	TurbineMechEff	Turbine Mechanical Efficiency (Input)	%	Static	Design	Data Sheet
75	TurbineNozzleAngle	Turbine Nozzle Angle (Required Input for Variable Turbine Nozzle option)	Degrees	Dynamic	Measurement	Field Input
76	TurbineOption	Turbine Type (Input as 1-Fixed Nozzle (default), 2-Variable Nozzle)		Static	Design	Data Sheet
77	TurbinePowerLossFactor	Turbine Power Loss Factor (Input)		Dynamic	Measurement	Field Input
78	TurbineRatedSpeed	Turbine Rated Speed (Input)	rpm	Static	Design	Data Sheet
79	TurbineRotorPitchRadius	TurbineRotorPitchRadius (Required Input for design calculations)	m	Dynamic	Measurement	Field Input
80	TurbineSpeed	Turbine Speed (Required Input)	rpm	Dynamic	Measurement	Field Input

18.2.5 Gas Turbine Single Shaft - KPI

18.2.5.1 Operating KPI

- Operating GT Shaft Power

18.2.5.2 Gas Turbine Single Shaft - Performance Modeling

Compressor

Overall Mass Balance

Inlet mass flow rate, and bleed and turbine cooling loss fractions are user specified.

$$F_O = F_I - F_{Bleed} - F_{Turb.Cooling}$$

Force Balance

The inlet pressure of compressor and pressure ratio are input parameters.

$$P_O = P_I + \Delta_{Pressure}$$

Energy Balance

Isentropic outlet temperature is calculated using isentropic relations.

$$\frac{T_{O_{Isen.}}}{T_I} = \left(\frac{P_O}{P_I}\right)^{\frac{(\gamma-1)}{\gamma}}$$

The expected compressor efficiency is calculated using performance curve supplied by the user.

$$H_O(P_O^{Comp.}, T_O^{Comp.}) = H_I(P_I^{Comp.}, T_I^{Comp.}) + \frac{H_{O_{Isen.}}(P_O^{Comp.}, T_{O_{Isen.}}) - H_I(P_I^{Comp.}, T_I^{Comp.})}{\eta_{Expected}^{Comp.}}$$

The expected compressor outlet is calculated from the calculated outlet enthalpy by the means of an appropriate thermo package.

$$T_O = f(H_O)$$

Combustor

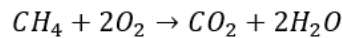
Overall Mass Balance

Inlet mass flow rate as well as inlet fuel rate is user specified.

$$F_O + F_{Recycle} = F_O^{Comp.} + F_{Fuel}$$

Component Balance

The composition will change as the feed undergoes combustion reactions. Ex: reaction is as follows,



$$F_I^j - F_O^j - F_{Extent}^j = 0$$

$$F_I^{CH_4} - F_O^{CH_4} - F_{Extent}^{CH_4} = 0$$

$$F_I^{O_2} - F_O^{O_2} - F_E^{O_2} = 0$$

$$F_I^{CO_2} - F_O^{CO_2} + F_{Extent}^{CO_2} = 0$$

$$F_I^{H_2O} - F_O^{H_2O} + F_{Extent}^{H_2O} = 0$$

$$F_{Extent}^{CH_4} = X_{CH_4} F_I^{CH_4} = k_0 e^{-\frac{E_a}{RT}} C_{CH_4} * Volume$$

$$F_{Extent}^{O_2} = 2F_{Extent}^{CH_4}$$

$$F_{Extent}^{CO_2} = F_{Extent}^{CH_4}$$

$$F_{Extent}^{H_2O} = 2F_{Extent}^{CH_4}$$

Force Balance

The inlet pressure of compressor and pressure loss factor are input parameters.

$$P_O = P_I + \Delta_{Pressure}$$

Energy Balance

The combustor inlet temperature is compressor expected outlet temperature. The heat loss (Q_{Loss}) is primarily due to cooling jacket.

$$T_I = T_O^{Comp.}$$

$$H_O = H_I + \dot{Q}_{HeatGen.} + H_{Fuel} - \dot{Q}_{JacketLoss}$$

$$T_O^{Combustor} = f(H_O)$$

Gas Turbine

Overall Mass Balance

$$F_I = F_O^{Combustor} + F_{Bleed} - F_{BleedLeak}$$

$$F_O = F_I + F_{TurbineCooling} - F_{TurbineCoolingLeak}$$

Force Balance

The turbine inlet pressure is same as combustor outlet pressure and the delta pressure is user specified.

$$P_O = P_O^{Combustor} + \Delta_{Pressure}$$

Energy Balance

$$T_I = T_O^{Combustor}$$

$$\frac{T_{OIsen}}{T_I} = \left(\frac{P_O}{P_I} \right)^{\frac{(\gamma-1)}{\gamma}}$$

The expected turbine efficiency is calculated using performance curve supplied by the user.

$$H_O = H_I - (H_I - H_{O_{Isen}}) * \eta_{Expected}$$

$$T_{Exhaust} = f(H_O)$$

$$\dot{W}_{TurbineGasPower} = H_I - H_O$$

$$\dot{W}_{ShaftPower} = \dot{W}_{TurbineGasPower} * \eta_{Mech.}$$

$$\eta_{Mech.} = 98\%$$

$$\dot{W}_{Comp.Power} = H_O^{Comp.}(P_O^{Comp.}, T_O^{Comp.}) - H_I^{Comp.}(P_I^{Comp.}, T_I^{Comp.})$$

$$\dot{W}_{ShaftComp.Power} = \frac{\dot{W}_{Comp.Power}}{\eta_{Mech.}^{Comp.}}$$

$$\eta_{Mech.}^{Comp.} = 98\%$$

$$\dot{W}_{GTShaftPower} = \dot{W}_{ShaftPower} - \dot{W}_{ShaftComp.Power}$$

18.2.5.3 Gas Turbine Single Shaft - Operating KPIs

Operating GT Shaft Power

$$\dot{W}_{TurbineGasPower} = H_I - H_O$$

$$\dot{W}_{ShaftPower} = \dot{W}_{TurbineGasPower} * \eta_{Mech.}$$

$$\dot{W}_{GTShaftPower} = \dot{W}_{ShaftPower} - \dot{W}_{ShaftComp.Power}$$

18.2.6 Gas Turbine Multi Shaft - KPIs

18.2.6.1 Operating KPIs

- Operating GT Shaft Power
- Operating Compressor Poly Efficiency
- Operating Compressor Isen Efficiency
- Operating Compressor Shaft Power
- Operating GT Heat Rate
- Operating GT Thermal Efficiency

18.2.6.2 Gas Turbine Multi Shaft - Performance Modeling**Compressor****Overall Mass Balance**

Inlet mass flow rate, nozzle and gas turbine cooling loss fractions are user specified.

$$F_I = F_O + F_{NozzleCooling} + F_{TurbineCooling}$$

18.2.6.3 Force Balance

The inlet pressure of compressor and pressure ratio are input parameters.

$$P_O = P_I + \Delta_{Pressure}$$

18.2.6.4 Energy Balance

Isentropic outlet temperature is calculated using isentropic relations.

$$\frac{T_{O_{Isen.}}}{T_I} = \left(\frac{P_O}{P_I}\right)^{\frac{(\gamma-1)}{\gamma}}$$

The expected compressor efficiency is calculated using performance curve supplied by the user.

$$F_I h_I(P_I^{Comp.}, T_I^{Comp.}) + W_{Shaft} = (F_O + F_{NozzleCooling} + F_{TurbineCooling}) h_O(P_O^{Comp.}, T_O^{Comp.})$$

$$F_I h_I(P_I^{Comp.}, T_I^{Comp.}) + W_{Shaft}^{Isentropic} = (F_O + F_{NozzleCooling} + F_{TurbineCooling}) h_O(P_O^{Comp.}, S_I^{Comp.})$$

$$\eta_{Expected}^{Comp.} = \frac{W_{Shaft}^{Isentropic}}{W_{Shaft}}$$

The expected compressor outlet is calculated from the calculated outlet enthalpy by the means of a appropriate thermo package.

$$T_O = f(H_O)$$

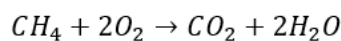
Combustor**18.2.6.5 Overall Mass Balance**

Inlet mass flow rate as well as inlet fuel rate is user specified.

$$F_O + F_{Recycle} = F_O^{Comp.} + F_{Fuel}$$

18.2.6.6 Component Balance

The composition will change as the feed undergoes combustion reactions. Eg: reaction is as follows:



$$F_I^j - F_O^j - F_{Extent}^j = 0$$

$$F_I^{CH_4} - F_O^{CH_4} - F_{Extent}^{CH_4} = 0$$

$$F_I^{O_2} - F_O^{O_2} - F_E^{O_2} = 0$$

$$F_I^{CO_2} - F_O^{CO_2} + F_{Extent}^{CO_2} = 0$$

$$F_I^{H_2O} - F_O^{H_2O} + F_{Extent}^{H_2O} = 0$$

$$F_{Extent}^{CH_4} = X_{CH_4} F_I^{CH_4} = k_0 e^{-\frac{E_a}{RT}} C_{CH_4} * Volume$$

$$F_{Extent}^{O_2} = 2F_{Extent}^{CH_4}$$

$$F_{Extent}^{CO_2} = F_{Extent}^{CH_4}$$

$$F_{Extent}^{H_2O} = 2F_{Extent}^{CH_4}$$

18.2.6.7 Force Balance

The inlet pressure of compressor and pressure loss factor are input parameters.

$$P_O = P_I * (1 - \Delta P_{Loss Factor})$$

18.2.6.8 Energy Balance

The combustor inlet temperature is compressor expected outlet temperature. The heat loss (Q_{Loss}) is primarily to due to cooling jacket.

$$T_I = T_O^{Comp.}$$

$$H_O = H_I + \dot{Q}_{HeatGen.} + H_{Fuel} - \dot{Q}_{JacketLoss}$$

$$T_O^{Combustor} = f(H_O)$$

Gas Generator Turbine

Overall Mass Balance

$$F_I = F_O^{Comb.}$$

$$F_I + F_{NozzleCooling} - F_{NozzleLeak} = F_O$$

18.2.6.9 Force Balance

The turbine inlet pressure is same as combustor outlet pressure and the delta pressure is user specified.

$$P_O = P_I^{Combustor} - \Delta_{Pressure}$$

18.2.6.10 Energy Balance

$$T_I = T_O^{Combustor}$$

$$\frac{T_{OIsen}}{T_I} = \left(\frac{P_O}{P_I}\right)^{\frac{(\gamma-1)}{\gamma}}$$

$$F_I h_I(P_O^{Comb.}, T_O^{Comb.}) + (F_{Nozzle\ Cooling} + F_{Nozzle\ Leak.}) h_O(P_O^{Comp.}, T_O^{Comp.}) - W_{Shaft} = F_O h_O(P_O^{GG}, T_O^{GG})$$

$$F_I h_I(P_O^{Comb.}, T_O^{Comb.}) + (F_{Nozzle\ Cooling} + F_{Nozzle\ Leak.}) h_O(P_O^{Comp.}, T_O^{Comp.}) - W_{Shaft_{Isentropic}}$$

$$= F_O h_O(P_O^{GG}, T_O^{Isentropic\ GG})$$

$$\eta_{Operating}^{GG} = \frac{W_{Shaft}}{W_{Shaft}^{Isentropic}}$$

There are two ways to calculate $\dot{W}_{ShaftComp.Power}$, either by enthalpies of the inlet and outlet of the compressor or using the shaft power generated by the GG turbine

Power Turbine

Overall Mass Balance

$$F_I = F_O^{GG}.$$

$$F_I + F_{TurbineCooling} - F_{TurbineLeak} = F_O$$

18.2.6.11 Force Balance

The power turbine inlet pressure is same as gas generator outlet pressure and the delta pressure is user specified.

$$P_O = P_O^{GG} - \Delta_{Pressure}$$

18.2.6.12 Energy Balance

$$T_I = T_O^{GG}$$

$$\frac{T_{OIsen}}{T_I} = \left(\frac{P_O}{P_I}\right)^{\frac{(\gamma-1)}{\gamma}}$$

$$F_I h_I(P_O^{GG}, T_O^{GG}) + (F_{Nozzle\ Cooling} + F_{Nozzle\ Leak.}) h_O(P_O^{Comp.}, T_O^{Comp.}) - W_{Shaft} = F_O h_O(P_O^{PT}, T_O^{PT})$$

$$F_I h_I(P_O^{GG}, T_O^{Comb.}) + (F_{Nozzle\ Cooling} + F_{Nozzle\ Leak.}) h_O(P_O^{Comp.}, T_O^{Comp.}) - W_{Shaft\ Isentropic}$$

$$= F_O h_O(P_O^{GG}, T_O^{Isentropic\ PT})$$

$$\eta_{Operating}^{PT} = \frac{W_{Shaft}}{W_{Shaft}^{Isentropic}}$$

18.2.6.13 Gas Turbine Multi Shaft - Operating KPIs

Operating GT Shaft Power

$$W_{GTShaftPower} = F_O h_O(P_O^{PT}, T_O^{PT}) - F_I h_I(P_O^{GG}, T_O^{GG}) - (F_{Nozzle\ Cooling} + F_{Nozzle\ Leak.}) h_O(P_O^{Comp.}, T_O^{Comp.})$$

Operating Compressor Poly Eff

$$\eta_{Poly} = \frac{\dot{W}_{Poly}}{\dot{W}_{Comp.ShaftPower}}$$

$$\dot{W}_{Poly} = \frac{n T_I \left(\left(\frac{P_O}{P_I} \right)^{\frac{1}{n}} - 1 \right)}{MW}$$

$$n = \frac{\left(\frac{P_O}{P_I} \right)}{\left(\frac{T_O}{T_I} \right)}$$

Operating Compressor Isen Efficiency

$$\eta_{Isen} = \frac{\dot{W}_{Isen}}{\dot{W}_{Comp.ShaftPower}}$$

$$\dot{W}_{Isen} = H_O(P_O^{Comp.}, S_I^{Comp.}) - H_I(P_I^{Comp.}, T_I^{Comp.})$$

Operating Compressor Shaft Power

$$\dot{W}_{Comp.ShaftPower} = H_O(P_O^{Comp.}, T_O^{Comp.}) - H_I(P_I^{Comp.}, T_I^{Comp.})$$

Operating GT Heat Rate

$$Heat\ Rate = \frac{F_{Fuel}H_{FuelLHV}}{W_{GTShaftPower}}$$

Operating GT Thermal Eff

$$\eta_{Thermal} = \frac{W_{GTShaftPower}}{F_{Fuel}H_{FuelLHV}}$$

18.2.7 Gas Turbine - Design KPI

18.2.7.1 Design GT Shaft Power

From exhaust flow vs power curve

$$\dot{W}_{ShaftPower} = f(W, F)$$

18.2.7.2 Design Exhaust Temperature

From exhaust flow vs exhaust temperature curve

$$T_{Ex} = f(T, F)$$

18.2.7.3 Design GT Heat Rate

From exhaust flow vs heat rate curve

$$Design\ Heat\ Rate = f(Heat\ Rate, F)$$

18.2.7.4 Design GT Thermal Efficiency

From exhaust flow vs efficiency curve

$$\eta_{Thermal} = f(\eta, F)$$

If efficiency curve not provided

$$\eta_{Thermal} = \frac{1}{Heat\ Rate}$$

18.2.8 Gas Turbine - Model Output

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	CombustorEff	Combustor Efficiency (Output)	%	Dynamic	Performance	Calculated
2	CombustorHeatOfCombustion	Combustor Heat Of Combustion (Output)	kJ/hr	Dynamic	Performance	Calculated
3	CompressorInletMassFlowRate	Compressor Inlet	kg/hr	Dynamic	Performance	Calculated

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
		Mass Flow Rate of Air (Output)				
4	EnergyIndex	Energy Index (Output)		Dynamic	Performance	Calculated
5	GTOutletVolumeFlowRate	Gas Turbine Outlet Volume Flow Rate (Output)	m ³ /hr	Dynamic	Performance	Calculated
6	HealthIndex	Health Index (Output)		Dynamic	Performance	Calculated
7	Operating GT Efficiency	Operating Gas Turbine Efficiency (Output)	%	Dynamic	Performance	Calculated
8	OperatingCompressorIsenEff	Operating Compressor Isentropic Efficiency (Output)	%	Dynamic	Performance	Calculated
9	OperatingCompressorPolyEff	Operating Compressor Polytropic Efficiency (Output)	%	Dynamic	Performance	Calculated
10	OperatingCompressorShaftPower	Operating Compressor Shaft Power (Output)	kW	Dynamic	Performance	Calculated
11	OperatingCycleHeatRatio	Operating Cycle Heat Ratio (Output)		Dynamic	Performance	Calculated
12	OperatingGTHeatRate	Operating Gas Turbine Heat Rate (Output)	kJ/kW-hr	Dynamic	Performance	Calculated
13	OperatingGTShaftPower	Operating Gas Turbine Shaft Power (Output)	kW	Dynamic	Performance	Calculated
14	OperatingGTThermalEff	Operating Gas Turbine Thermal Efficiency (Output)	%	Dynamic	Performance	Calculated
15	OperatingTurbineEff	Operating Turbine Efficiency (Output)	%	Dynamic	Performance	Calculated
16	OperatingTurbineInletTemp	Operating Turbine Inlet Temperature (Output)	°C	Dynamic	Performance	Calculated
17	OperatingTurbineOutletTemp	Operating Turbine Outlet Temperature	°C	Dynamic	Performance	Calculated

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
		(Output)				
18	OperatingTurbineShaftPower	Operating Turbine Shaft Power (Output)	kW	Dynamic	Performance	Calculated
19	PercentDeviationCompressorShaftPower	Percent Deviation Compressor Shaft Power between Operating and Design values (Output)	%	Dynamic	Performance	Calculated
20	PercentDeviationGTHeatRate	Percent Deviation Gas Turbine Heat Rate between Operating and Design values (Output)	%	Dynamic	Performance	Calculated
21	PercentDeviationGTShaftPower	Percent Deviation Gas Turbine Shaft Power between Operating and Design values (Output)	%	Dynamic	Performance	Calculated
22	PercentDeviationTurbineShaftPower	Percent Deviation Turbine Shaft Power between Operating and Design values (Output)	%	Dynamic	Performance	Calculated
23	PerformanceIndex	Performance Index (Output)		Dynamic	Performance	Calculated
24	TurbineInletMassFlowRate	Turbine Inlet Mass Flow Rate of Gas (Output)	kg/hr	Dynamic	Performance	Calculated
25	TurbineInletPressure	Turbine Inlet Pressure (Output)	barg	Dynamic	Performance	Calculated
26	TurbineOutletPressure	Turbine Outlet Pressure (Output)	barg	Dynamic	Performance	Calculated

18.2.9 Gas Turbine - Degradation Model

Every turbomachinery will degrade over time, this generally results in a decrease in efficiency, specifically thermal efficiency for Gas turbines. The typical lifespan of a Gas Turbine is 100,000 - 150,000 EOH (Equivalent Operating Hours). Typically, a Gas Turbine degrades 2 - 6% every 24000 EOH out of which 1 - 1.5% is permanent loss and the rest can be recovered by water washing or similar operations. Machines located in dry and humid environments degrade more.

To account for these losses in design calculation, the degradation model can be used to shift the baseline of the performance table i.e., calculate design KPIs based on the current age of equipment rather than when the equipment was commissioned.

To enable the degradation model, two additional configuration steps are required:

1. Enable **Metric** calculations for the specific asset. (Refer to “Honeywell\MES\AssetManager\Documents\Asset Performance Management Configuration Guide.pdf”)
2. Provide degradation matrix, this attribute contains Hours V/S Degradation information.

The below figure depicts a reference for Degradation Matrix,

Running Hours	Degradation %
12000	1
24000	2.5
48000	5

The Running Hours V/S Degradation can be obtained from the OEMs manual or observation made at the site. The model will calculate current degradation based on the current running hours and account for it in Design Thermal Efficiency and subsequently to Power, Exhaust Temperature, and Heat Rate.

NOTE

At least one entry is required in the degradation matrix.

18.3 Gas Turbine - Fault Model

18.3.1 Gas Turbine - APM Model

In Asset Performance Management, you can configure faults and symptoms models for gas turbine inputs and performance output to diagnose the problems early. The fault management can inform operators of any violations of the nominal turbine operating range and deviations from the expected characteristic. The fault symptoms or models are used to:

1. Warn against potential damage to gas turbine under unfavorable operating conditions.
2. Provide early warning of damages to the gas turbine.
3. The Fault Tree provides a graphical display of assets and associated faults and associated faults and symptoms.
4. The Fault Tree is used to locate the source of a fault in a single view. Hence, analyzing the fault is easier with navigation views.

18.3.1.1 Gas Turbine - Multi-Shaft Simple Cycle Performance Fault model

Model Type	APM Model
Category	Gas Turbine
Equipment Class	Turbine
Equipment Type	Gas Turbine - Multi-Shaft Simple Cycle

The following messages can be generated for process alarms or notifications to warn operating personnel of unfavorable operating conditions

1. **Low Compressor Efficiency:** It is determined by means of the deviation of the current compressor efficiency from the efficiency characteristic.
2. **Tip Rubs:** Blade-to-case rub can degrade the performance of Gas Turbines through the introduction of high amplitude shaft vibration and severe blade/seal wear. It can even lead to catastrophic failure in the worst occurrence.

3. **Fouling:** It is an important factor in Gas Turbines, caused due to deposition of material on the surfaces. The gas turbine's effective performance decreases when fouling occurs. Fouling predictions are used to predict the maintenance schedules for cleaning Gas Turbines.

The below table illustrates the list of faults and symptoms configured for a gas turbine to identify possible anomalies that affect the performance or health of a Gas Turbine:

Fault Name	Associated Symptoms	Recommendations
GG Compressor Efficiency degradation	Compressor Discharge Temperature RH (MCED)	1. Short term; Wash if possible. 2. Inspect the Air Filtration system. 3. Gas Carry out boroscope examination (IGV, blade, and/or seal). 4. Monitor efficiency, and determine if the degradation of efficiency is still progressing or if it has stopped. 5. If still progressing, determine the rate of degradation (auto). 6. Ensure compressor cooling pipework to GG nozzles and blades are not leaking, also ensure check valves in pipework are working correctly 7. Ensure bleed valves are not leaking.
	Compressor efficiency deviation RL (MCED)	
GT Thermal Efficiency degradation (high heat rate)	Heat Rate Deviation HIGH (MTHED)	1. Update LHV Inspect Air Filtration system Monitor efficiency, and determine if degradation of efficiency is still progressing or if it has stopped. 2. If still progressing, determine the rate of degradation (auto). 3. Carry out crank wash and boroscope inspection at the first opportunity (IGV, blade, and/or seal).
Combustors fault	Combustor in lean-lean mode ON (MCF)	1. If T spread high: For heavy-duty DLN: ensure the GT is not in extended lean-lean mode. 2. Ensure there is no fuel gas condensation taking place. 3. Determine which fuel nozzle it comes from (consult swirl angle diagram) or carry out borescope inspection on the combustor. 4. Inspect/replace defective fuel nozzle 5. High combustor dynamics: 1. Check Fuel gas skid functionality Perform a gas analysis, and determine if combustor re-tuning is required Carry out flow tests on fuel nozzles
	Temperature spread (T48 SPREAD) RH (MCF)	
Train component underperformance	Compressor efficiency deviation LOW (MTCUP)	1. Identify which components are responsible for the underperformance. 2. Perform the actions described in the dedicated component underperformance fault.
	Power Turbine Efficiency LOW (MTCUP)	
	Heat Rate Deviation LOW (MTCUP)	
Turbine inner cooling anomaly	Compressor Bleed Pressure(s) RL (MTICA)	1. Contact the gas turbine vendor to verify if an anomaly is genuine and discuss options.

Fault Name	Associated Symptoms	Recommendations
	Compressor Bleed Pressure(s) RH (MTICA)	
PT Efficiency degradation	GT Outlet Temperature RH (MPTED)	1. Check NGV actuator (where applicable) 2. Carry out a boroscope examination, particularly when high vibrations are also present.
	NGV Angle RH (MPTED)	
	Power Turbine Efficiency Deviation LOW (MPTED)	
NGV malfunction	NGV Angle RL (MNGVM)	1. Check NGV actuator and feed Check all linkage connections 2. Carry out a boroscope examination to identify the presence of internal damage.
	NGV Angle RH (MNGVM)	
Power Turbine Outside Allowable Speed Range	Power Turbine Speed HIGH (MPTOASR)	1. If speed below min: Check GT compressor turbine efficiency 2. Check process compressor efficiency 3. Check MW of process gas Change compressor operating point to verify control system is functional 4. Check bearing parameters for compressor and turbine Reduce load on the compressor side until speed recovers 5. If speed is above max: Consider shutting down the gas turbine and investigating options with the equipment vendor.
	Power Turbine Speed LOW (MPTOASR)	

18.3.1.2 Gas Turbine - Single Shaft Simple Cycle Performance Fault model

Model Type	APM Model
Category	Gas Turbine
Equipment Class	Turbine
Equipment Type	Gas Turbine - Single Shaft Simple Cycle

Fault Name	Associated Symptoms	Recommendations
GG Compressor Efficiency degradation	Compressor Discharge Temperature RH (CED)	1. Short term; Wash if possible. 2. Inspect the Air Filtration system. 3. Gas Carry out boroscope examination (IGV, blade, and/or seal). 4. Monitor efficiency, and determine if the degradation of efficiency is still progressing or if it has stopped. 5. If still progressing, determine the rate of degradation (auto). 6. Ensure compressor cooling pipework to GG nozzles and blades are not leaking, also ensure check valves in pipework are working correctly,

Fault Name	Associated Symptoms	Recommendations
	Compressor efficiency deviation RL (CED)	7. Ensure bleed valves are not leaking.
GT Thermal Efficiency degradation (high heat rate)	Heat Rate Deviation HIGH (SSTHED)	1. Update LHV Inspect Air Filtration system Monitor efficiency, and determine if degradation of efficiency is still progressing or if it has stopped. 2. If still progressing, determine the rate of degradation (auto). 3. Carry out crank wash and boroscope inspection at the first opportunity (IGV, blade, and/or seal).
Combustors fault	Combustor in lean-lean mode ON (CF)	1. If T spread high: For heavy-duty DLN: ensure the GT is not in extended lean-lean mode. 2. Ensure there is no fuel gas condensation taking place. 3. Determine which fuel nozzle it comes from (consult swirl angle diagram) or carry out borescope inspection on combustor Inspect/replace defective fuel nozzle 4. High combustor dynamics: Check Fuel gas skid functionality Perform a gas analysis, and determine if combustor re-tuning is required 5. Carry out flow tests on fuel nozzles
	Temperature spread (T48 SPREAD) RH (CF)	
Train component underperformance	Compressor efficiency deviation LOW (TCUP)	1. Identify which are the components responsible for the underperformance. 2. Perform the actions described in the dedicated component underperformance fault.
	Power Turbine Efficiency LOW (TCUP)	
	Heat Rate Deviation LOW (TCUP)	
Turbine inner cooling anomaly	Compressor Bleed Pressure(s) RL (TICA)	1. Contact the gas turbine vendor to verify if an anomaly is genuine and discuss options.
	Compressor Bleed Pressure(s) RH (TICA)	

18.3.2 Gas Turbine - APR Model

18.3.2.1 Gas Turbine- Bearing System Fault Model

Model Type	APR Model
Category	Gas Turbine
Equipment Class	Turbine
Equipment Type	Gas Turbine Bearing System

The below table illustrates the list of faults and symptoms configured for Gas Turbine Conditioning based monitoring.

Fault Name	Associated Symptoms	Recommendations
GT Thrust/Radial Bearing Anomaly due to Lube Oil Temp	GT-DE Radial vibration overall (X) RH (GLOT)	1. Check cooling water supply pressure. 2. Check the bypass control valve. 3. pen both coolers and stabilize temperature. 4. Check oil heater is not on (even if instruments indicate it is off). 5. Line walk the LO cooling pipework to check for leaks. 6. Inspect strainers on pumps. 7. Check instrumentation is working correctly. 8. If malfunction of one cooler is suspected, isolate and inspect it.
	GT-DE Radial vibration overall (Y) RH (GLOT)	
	GT-DE Radial bearing A temperature RH (GLOT)	
	GT-DE Radial bearing B temperature RH (GLOT)	
	GT-NDE Radial vibration overall (X) RH (GLOT)	
	GT-NDE Radial vibration overall (Y) RH (GLOT)	
	GT-NDE Radial bearing A temperature RH (GLOT)	
	GT-NDE Radial bearing B temperature RH (GLOT)	
	GT-Thrust bearing active side displacement Z1 RH (GLOT)	
	GT-Thrust bearing active side displacement Z2 RH (GLOT)	
	GT-Thrust bearing inactive side displacement Z1 RH (GLOT)	
	GT-Thrust bearing inactive side displacement Z2 RH (GLOT)	
	GT-OB Thrust Bearing A Temperature RH (GLOT)	
	GT-OB Thrust Bearing B Temperature RH (GLOT)	
	GT-OB Thrust Bearing C Temperature RH (GLOT)	
	GT-OB Thrust Bearing D	

Fault Name	Associated Symptoms	Recommendations
	Temperature RH (GLOT)	
	GT-IB Thrust Bearing A Temperature RH (GLOT)	
	GT-IB Thrust Bearing B Temperature RH (GLOT)	
	GT-IB Thrust Bearing C Temperature RH (GLOT)	
	GT-IB Thrust Bearing D Temperature RH (GLOT)	
	LO header temperature RH (GLOT)	
GT Thrust/Radial Bearing Anomaly due to Lube Oil Press	GT-DE Radial vibration overall (X) RH (GLOP)	1. Switch on both oil pumps (if possible) and stabilize the oil pressure. 2. Assess malfunction. 3. Line walk oil pipework if accessible and try to identify leaks in flanges/pipework. 4. Inspect filters and strainers for blockages. 5. Ensure the LO pressure transmitter/gauge is reading correct. 6. Inspect/borescope oil drains for oil coking and blockages.
	GT-DE Radial vibration overall (Y) RH (GLOP)	
	GT-DE Radial bearing A temperature RH (GLOP)	
	GT-DE Radial bearing B temperature RH (GLOP)	
	GT-NDE Radial vibration overall (X) RH (GLOP)	
	GT-NDE Radial vibration overall (Y) RH (GLOP)	
	GT-NDE Radial bearing A temperature RH (GLOP)	
	GT-NDE Radial bearing B temperature RH (GLOP)	
	GT-Thrust bearing active side displacement Z1 RH (GLOP)	
	GT-Thrust bearing active side displacement Z2 RH (GLOP)	
	GT-Thrust bearing	

Fault Name	Associated Symptoms	Recommendations
	inactive side displacement Z1 RH (GLOP)	
	GT-Thrust bearing inactive side displacement Z2 RH (GLOP)	
	GT-OB Thrust Bearing A Temperature RH (GLOP)	
	GT-OB Thrust Bearing B Temperature RH (GLOP)	
	GT-OB Thrust Bearing C Temperature RH (GLOP)	
	GT-OB Thrust Bearing D Temperature RH (GLOP)	
	GT-IB Thrust Bearing A Temperature RH (GLOP)	
	GT-IB Thrust Bearing B Temperature RH (GLOP)	
	GT-IB Thrust Bearing C Temperature RH (GLOP)	
	GT-IB Thrust Bearing D Temperature RH (GLOP)	
	LO header pressure RL (GLOP)	
GT Thrust/Radial Bearing Anomaly due to Lube Oil Quality	GT-DE Radial vibration overall (X) RH (GLOQ)	1. Check the oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, and TAN. 4. Ensure all man covers on tanks are closed. 5. Recondition or replace oil.
	GT-DE Radial vibration overall (Y) RH (GLOQ)	
	GT-DE Radial bearing A temperature RH (GLOQ)	
	GT-DE Radial bearing B temperature RH (GLOQ)	
	GT-NDE Radial vibration overall (X) RH (GLOQ)	

Fault Name	Associated Symptoms	Recommendations
	GT-NDE Radial vibration overall (Y) RH (GLOQ)	
	GT-NDE Radial bearing A temperature RH (GLOQ)	
	GT-NDE Radial bearing B temperature RH (GLOQ)	
	GT-Thrust bearing active side displacement Z1 RH (GLOQ)	
	GT-Thrust bearing active side displacement Z2 RH (GLOQ)	
	GT-Thrust bearing inactive side displacement Z1 RH (GLOQ)	
	GT-Thrust bearing inactive side displacement Z2 RH (GLOQ)	
	GT-OB Thrust Bearing A Temperature RH (GLOQ)	
	GT-OB Thrust Bearing B Temperature RH (GLOQ)	
	GT-OB Thrust Bearing C Temperature RH (GLOQ)	
	GT-OB Thrust Bearing D Temperature RH (GLOQ)	
	GT-IB Thrust Bearing A Temperature RH (GLOQ)	
	GT-IB Thrust Bearing B Temperature RH (GLOQ)	
	GT-IB Thrust Bearing C Temperature RH (GLOQ)	

Fault Name	Associated Symptoms	Recommendations
	GT-IB Thrust Bearing D Temperature RH (GLOQ)	
	LO Viscosity SH (GLOQ)	
	LO Viscosity SL (GLOQ)	
	LO TAN SH (GLOQ)	
	LO TAN SL (GLOQ)	
GT Thrust Bearing Anomaly due to Mechanical Issue	GT-Thrust bearing active side displacement Z1 RH (GTMI)	1. Inspect vibration instrumentation to check integrity. 2. If the unit has tripped borescope the internals of the turbine to identify any potential damage and inspect the thrust bearing (if this is accessible). 3. If damage has been seen the turbine will require overhauling. 4. If the unit hasn't tripped yet and has past the alarm but approaching the trip shut the unit down and carry out a borescope inspection and thrust bearing inspection (if the bearing is accessible).
	GT-Thrust bearing active side displacement Z2 RH (GTMI)	
	GT-Thrust bearing inactive side displacement Z1 RH (GTMI)	
	GT-Thrust bearing inactive side displacement Z2 RH (GTMI)	
	GT-OB Thrust Bearing A Temperature RH (GTMI)	
	GT-OB Thrust Bearing B Temperature RH (GTMI)	
	GT-OB Thrust Bearing C Temperature RH (GTMI)	
	GT-OB Thrust Bearing D Temperature RH (GTMI)	
	GT-IB Thrust Bearing A Temperature RH (GTMI)	
	GT-IB Thrust Bearing B Temperature RH (GTMI)	
	GT-IB Thrust Bearing C Temperature RH (GTMI)	
	GT-IB Thrust Bearing D Temperature RH (GTMI)	
GT DE RB Anomaly due to Mechanical Issue	GT-DE Radial vibration overall (X) RH (GDERBMI)	1. Inspect vibration instrumentation to check integrity. 2. If the unit has tripped borescope the internals of the turbine to identify any potential damage and inspect the thrust bearing (if this is accessible), if damage has been seen the turbine will require overhauling.

Fault Name	Associated Symptoms	Recommendations
	GT-DE Radial vibration overall (Y) RH (GDERBMI)	- If the unit hasn't tripped yet and has passed the alarm but approaching the trip shut the unit down and carry out a borescope inspection and thrust bearing inspection (if the bearing is accessible). - Check the gland seal performance.
	GT-DE Radial bearing A temperature RH (GDERBMI)	
	GT-DE Radial bearing B temperature RH (GDERBMI)	
GT NDE RB Anomaly due to Mechanical Issue	GT-NDE Radial vibration overall (X) RH (GNDERBMI)	- Inspect vibration instrumentation to check integrity. - If the unit has tripped borescope the internals of the turbine to identify any potential damage and inspect the thrust bearing (if this is accessible), if damage has been seen the turbine will require overhauling. - If the unit hasn't tripped yet and has passed the alarm but approaching the trip shut the unit down and carry out a borescope inspection and thrust bearing inspection (if the bearing is accessible). - Check the gland seal performance.
	GT-NDE Radial vibration overall (Y) RH (GNDERBMI)	
	GT-NDE Radial bearing A temperature RH (GNDERBMI)	
	GT-NDE Radial bearing B temperature RH (GNDERBMI)	
GT Rotor Anomaly	GT-DE Radial vibration overall (X) RH (GTRTR)	- Investigate with high-frequency vibration analysis. - Check Carbon brush integrity. - Check the magnetism of the rotor. - Check rotor unbalance misalignment looseness, and check coupling integrity. - Check foundation bolt integrity. - Check both bearings for clearance. - Check the gland seal performance.
	GT-DE Radial vibration overall (Y) RH (GTRTR)	
	GT-NDE Radial vibration overall (X) RH (GTRTR)	
	GT-NDE Radial vibration overall (Y) RH (GTRTR)	
PT Thrust/Radial Bearing Anomaly due to Lube Oil Temp	PT-Thrust bearing active side displacement Z1 RH (PLOT)	- Check cooling water supply pressure. - Check the bypass control valve. - Open both coolers and stabilize the temperature. - Check oil heater is not on (even if instruments indicate it is off). - Line walk the LO cooling pipework to check for leaks. - Inspect strainers on pumps. - Check instrumentation is working correctly. - If a malfunction of one cooler is suspected, isolate and inspect it.
	PT-Thrust bearing active side displacement Z2 RH (PLOT)	
	PT-Thrust bearing inactive side displacement Z1 RH (PLOT)	
	PT-Thrust bearing inactive side displacement Z2 RH	

Fault Name	Associated Symptoms	Recommendations
	(PLOT)	

Fault Name	Associated Symptoms	Recommendations
	PT-OB Thrust Bearing A Temperature RH (PLOT)	
	PT-OB Thrust Bearing B Temperature RH (PLOT)	
	PT-OB Thrust Bearing C Temperature RH (PLOT)	
	PT-OB Thrust Bearing D Temperature RH (PLOT)	
	PT-IB Thrust Bearing A Temperature RH (PLOT)	
	PT-IB Thrust Bearing B Temperature RH (PLOT)	
	PT-IB Thrust Bearing C Temperature RH (PLOT)	
	PT-IB Thrust Bearing D Temperature RH (PLOT)	
	PT-DE Bearing A Temperature RH (PLOT)	
	PT-DE Bearing B Temperature RH (PLOT)	
	PT-NDE Bearing A Temperature RH (PLOT)	
	PT-NDE Bearing B Temperature RH (PLOT)	
	PT-DE Radial X Vibrations RH (PLOT)	
	PT-DE Radial Y Vibrations RH (PLOT)	
	PT-NDE Radial X Vibrations RH (PLOT)	
	PT-NDE Radial Y Vibrations RH (PLOT)	

Fault Name	Associated Symptoms	Recommendations
	LO header temperature RH (PLOT)	
PT Thrust/Radial Bearing Anomaly due to Lube Oil Press	PT-Thrust bearing active side displacement Z1 RH (PLOP)	1. Switch on both oil pumps (if possible) and stabilize the oil pressure. 2. Assess malfunction. 3. Line walk oil pipework if accessible and try to identify leaks in flanges/pipework. 4. Inspect filters and strainers for blockages. 5. Ensure the LO pressure transmitter/gauge is reading correctly. 6. Inspect/borescope oil drains for oil coking and blockages.
	PT-Thrust bearing active side displacement Z2 RH (PLOP)	
	PT-Thrust bearing inactive side displacement Z1 RH (PLOP)	
	PT-Thrust bearing inactive side displacement Z2 RH (PLOP)	
	PT-OB Thrust Bearing A Temperature RH (PLOP)	
	PT-OB Thrust Bearing B Temperature RH (PLOP)	
	PT-OB Thrust Bearing C Temperature RH (PLOP)	
	PT-OB Thrust Bearing D Temperature RH (PLOP)	
	PT-IB Thrust Bearing A Temperature RH (PLOP)	
	PT-IB Thrust Bearing B Temperature RH (PLOP)	
	PT-IB Thrust Bearing C Temperature RH (PLOP)	
	PT-IB Thrust Bearing D Temperature RH (PLOP)	
	PT-DE Bearing A Temperature RH (PLOP)	
	PT-DE Bearing B	

Fault Name	Associated Symptoms	Recommendations
	Temperature RH (PLOP)	
	PT-NDE Bearing A Temperature RH (PLOP)	
	PT-NDE Bearing B Temperature RH (PLOP)	
	PT-DE Radial X Vibrations RH (PLOP)	
	PT-DE Radial Y Vibrations RH (PLOP)	
	PT-NDE Radial X Vibrations RH (PLOP)	
	PT-NDE Radial Y Vibrations RH (PLOP)	
	LO header pressure RL (PLOP)	
PT Thrust/Radial Bearing Anomaly due to Lube Oil Quality	PT-Thrust bearing active side displacement Z1 RH (PLOQ)	1. Check the oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, and TAN. 4. Ensure all man covers on tanks are closed. 5. Recondition or replace oil.
	PT-Thrust bearing active side displacement Z2 RH (PLOQ)	
	PT-Thrust bearing inactive side displacement Z1 RH (PLOQ)	
	PT-Thrust bearing inactive side displacement Z2 RH (PLOQ)	
	PT-OB Thrust Bearing A Temperature RH (PLOQ)	
	PT-OB Thrust Bearing B Temperature RH (PLOQ)	
	PT-OB Thrust Bearing C Temperature RH (PLOQ)	
	PT-OB Thrust Bearing D Temperature RH (PLOQ)	

Fault Name	Associated Symptoms	Recommendations
	PT-IB Thrust Bearing A Temperature RH (PLOQ)	
	PT-IB Thrust Bearing B Temperature RH (PLOQ)	
	PT-IB Thrust Bearing C Temperature RH (PLOQ)	
	PT-IB Thrust Bearing D Temperature RH (PLOQ)	
	PT-DE Bearing A Temperature RH (PLOQ)	
	PT-DE Bearing B Temperature RH (PLOQ)	
	PT-NDE Bearing A Temperature RH (PLOQ)	
	PT-NDE Bearing B Temperature RH (PLOQ)	
	PT-DE Radial X Vibrations RH (PLOQ)	
	PT-DE Radial Y Vibrations RH (PLOQ)	
	PT-NDE Radial X Vibrations RH (PLOQ)	
	PT-NDE Radial Y Vibrations RH (PLOQ)	
	LO Viscosity SH (PLOQ)	
	LO Viscosity SL (PLOQ)	
	LO TAN SH (PLOQ)	
	LO TAN SL (PLOQ)	
PT Thrust Bearing Anomaly due to Mechanical Issue	PT-Thrust bearing active side displacement Z1 RH (PTMI)	1. Inspect vibration instrumentation to check integrity. 2. If the unit has tripped borescope the internals of the turbine to identify any potential damage and inspect the thrust bearing (if this is accessible). 3. If damage has been seen the turbine will require overhauling. 4. If the unit hasn't tripped yet and has passed the alarm but

Fault Name	Associated Symptoms	Recommendations
	PT-Thrust bearing active side displacement Z2 RH (PTMI)	approaching the trip shut the unit down and carry out a borescope inspection and thrust bearing inspection (if the bearing is accessible).
	PT-Thrust bearing inactive side displacement Z1 RH (PTMI)	
	PT-Thrust bearing inactive side displacement Z2 RH (PTMI)	
	PT-OB Thrust Bearing A Temperature RH (PTMI)	
	PT-OB Thrust Bearing B Temperature RH (PTMI)	
	PT-OB Thrust Bearing C Temperature RH (PTMI)	
	PT-OB Thrust Bearing D Temperature RH (PTMI)	
	PT-IB Thrust Bearing A Temperature RH (PTMI)	
	PT-IB Thrust Bearing B Temperature RH (PTMI)	
	PT-IB Thrust Bearing C Temperature RH (PTMI)	
	PT-IB Thrust Bearing D Temperature RH (PTMI)	
PT DE RB Anomaly due to Mechanical Issue	PT-DE Bearing A Temperature RH (PDERBMI)	1. Inspect vibration instrumentation to check integrity. 2. If the unit has tripped borescope the internals of the turbine to identify any potential damage and inspect the thrust bearing (if this is accessible), if damage has been seen the turbine will require overhauling. 3. If the unit hasn't tripped yet and has passed the alarm but approaching the trip shut the unit down and carry out a borescope inspection and thrust bearing inspection (if the bearing is accessible). 4. Check the gland seal performance.
	PT-DE Bearing B Temperature RH (PDERBMI)	
	PT-DE Radial X Vibrations RH (PDERBMI)	
	PT-DE Radial Y Vibrations RH (PDERBMI)	
PT NDE RB Anomaly due to Mechanical Issue	PT-NDE Bearing A Temperature RH (PNDERBMI)	1. Inspect vibration instrumentation to check integrity. 2. If the unit has tripped borescope the internals of the turbine to identify any potential damage and inspect the thrust bearing (if this is accessible), if damage has been seen the turbine will

Fault Name	Associated Symptoms	Recommendations
	PT-NDE Bearing B Temperature RH (PNDERBMI)	require overhauling. 3. If the unit hasn't tripped yet and has passed the alarm but approaching the trip shut the unit down and carry out a borescope inspection and thrust bearing inspection (if the bearing is accessible). 4. Check the gland seal performance.
	PT-NDE Radial X Vibrations RH (PNDERBMI)	
	PT-NDE Radial Y Vibrations RH (PNDERBMI)	
	PT-NDE Radial Y Vibrations RH (PNDERBMI)	
PT Rotor Anomaly	PT-DE Radial X Vibrations RH (PTRTR)	1. Investigate with high-frequency vibration analysis. 2. Check Carbon brush integrity. 3. Check the magnetism of the rotor. 4. Check rotor unbalance misalignment looseness, and check coupling integrity. 5. Check foundation bolt integrity. 6. Check both bearings for clearance. 7. Check the gland seal performance.
	PT-DE Radial Y Vibrations RH (PTRTR)	
	PT-NDE Radial X Vibrations RH (PTRTR)	
	PT-NDE Radial Y Vibrations RH (PTRTR)	
PT rotor damage or fouling	PT-DE Radial X Vibrations RH (PTRD)	1. Carry out borescope inspection. 2. Check Lube Oil System for internal oil leaks Prepare for machine overhaul.
	PT-DE Radial Y Vibrations RH (PTRD)	
	PT-NDE Radial X Vibrations RH (PTRD)	
	PT-NDE Radial Y Vibrations RH (PTRD)	
	Power Turbine Outlet Temperature RH (PTRD)	
	Power Turbine Efficiency Deviation LOW (PTRD)	
High Axial thrust	PT-OB Thrust Bearing A Temperature RH (PTHAT)	1. Check thrust control valve integrity and functionality (if active thrust control) or that correct orifice is installed (if passive thrust control) 2. Check to balance chamber pressure.
	PT-OB Thrust Bearing B Temperature RH (PTHAT)	
	PT-OB Thrust Bearing C Temperature RH (PTHAT)	
	PT-OB Thrust Bearing D Temperature RH	

Fault Name	Associated Symptoms	Recommendations
	(PTHAT)	
	PT-IB Thrust Bearing A Temperature RH (PTHAT)	
	PT-IB Thrust Bearing B Temperature RH (PTHAT)	
	PT-IB Thrust Bearing C Temperature RH (PTHAT)	
	PT-IB Thrust Bearing D Temperature RH (PTHAT)	

18.3.2.2 Gas Turbine- Air Intake

Model Type	APR Model
Category	Gas Turbine
Equipment Class	Turbine
Equipment Type	Gas Turbine- Air Intake

The below table illustrates the list of faults and symptoms configured for the Gas Turbine- Air Intake

Fault Name	Associated Symptoms	Recommendations
Filter collapsed or bypassed	GT Inlet Filter Pressure Drop RL (AFCBP)	1. Short term; 1. Inspect the Air Filtration system. 2. Filter backwash. 2. Check below indoor is closed, 3. Check the integrity of all expansions bellows. 4. Check filter sealing is adequate, inspect ducting for holes, 5. Inspect the FOD screen, and wash it if possible. 6. Borescope inspection at the opportunity.
Filter obstruction	Compressor Inlet Pressure RL (AFO)	1. Short term; Wash if possible. 2. Borescope inspection at the opportunity. 3. Inspect the Air Filtration system. 4. Filter backwash.
	GT Inlet Filter Pressure Drop RH (AFO)	
FOD screen obstruction	Compressor Inlet Pressure RL (AFODSO)	1. Inspect the FOD screen.
	GT Inlet Filter Pressure Drop RH (AFODSO)	

18.3.2.3 Gas Turbine- Exhaust System

Model Type	APR Model
Category	Gas Turbine
Equipment Class	Turbine
Equipment Type	Gas Turbine- Exhaust System

The below table illustrates the list of faults and symptoms configured for the Gas Turbine- Exhaust System

Fault Name	Associated Symptoms	Recommendations
High counterpressure	GT Enclosure delta pressure RH (VOP)	1. Inspect at the first opportunity.
	GT Enclosure delta pressure RL (VUP)	
	GT Enclosure Temperature RH (VHT)	

18.3.2.4 Gas Turbine- Ventilation System

Model Type	APR Model
Category	Gas Turbine
Equipment Class	Turbine
Equipment Type	Gas Turbine- Ventilation System

The below table illustrates the list of faults and symptoms configured for a Ventilation System based on Advanced Predictive Analysis

Fault Name	Associated Symptoms	Recommendations
Overpressure	GT Enclosure delta pressure RH (VOP)	1. Inspect ventilation fans, damper, and ducting, Confirm pressure transmitter is reading correctly 2. Inspect turbine and pipework for leaks.
Underpressure	GT Enclosure delta pressure RL (VUP)	1. Inspect ventilation fans, damper, and ducting, 2. Confirm pressure transmitter is reading correctly, 3. Inspect the enclosure for leaks, and check that enclosure doors are air-tight.
High Temperature	GT Enclosure Temperature RH (VHT)	1. Confirm Thermocouple is reading correctly, 2. Verify that high temperature is not just a local hot spot, at the first opportunity: Inspect the ventilation fan, damper, and ducting, 3. Inspect turbine and pipework for leaks, and clean up any oil leaks.

18.3.2.5 Gas Turbine- FuelGas System

Model Type	APR Model
Category	Gas Turbine
Equipment Class	Turbine
Equipment Type	Gas Turbine- FuelGas System

The below table illustrates the list of faults and symptoms configured for a Gas Turbine- FuelGas System

Fault Name	Associated Symptoms	Recommendations
Loss of gas quality	Fuel LHV HIGH (FGLGQ)	1. Analyze the supplied gas, Borescope fuel gas system and inspect for sulfur deposits or water, Inspect gas Y strainer. 2. Check filter.
	Fuel LHV LOW (FGLGQ)	
	Fuel MWI HIGH (FGLGQ)	
	Fuel MWI LOW (FGLGQ)	
	Fuel metering valve position RH (FGLGQ)	
Insufficient FG supply	Fuel Inlet Pressure LOW (FGIFGS)	1. Check FG system inlet pressure Inspect FCV, PCV, SOV Borescope FGMV and inspect strainer Identify possible leakage Borescope fuel line.
	Fuel metering valve position RH (FGIFGS)	
	Driven equipment speed deviation LOW (FGIFGS)	
FG overpressure	Fuel Inlet Pressure HIGH (FGFGOP)	1. Check the PCV/FCV valve Check the Gas to Liquid Fuel switchover.
	Temperature Spread RH (FGFGOP)	
Loss of superheat	Fuel Inlet Temperature RL (FGLOS)	1. Perform a gas analysis, Check Fuel gas skid functionality, Check Superheater and heat tracing integrity, and Identify faulty injectors by swirl angle chart.
	Temperature Spread RH (FGLOS)	

18.3.2.6 Gas Turbine Lube Oil System APR Fault Model

Model Type	APR Model
Category	Gas Turbine
Equipment Class	Turbine
Equipment Type	Gas Turbine - Lube Oil System

The below table illustrates the list of faults and symptoms configured for a Gas Turbine Lube Oil system

Fault Name	Associated Symptoms	Recommendations
Loss of lube oil quality	LO Viscosity HIGH (GLOQ)	1. Check the oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, and TAN. 4. Check no breather caps or man covers have been removed and left open to allow debris into the tank. 5. Do filter cleaning Recondition or replace oil.
	LO Viscosity LOW (GLOQ)	
	LO Tan HIGH (GLOQ)	
	LO Tan LOW (GLOQ)	
Oil circuit leak	Oil Tank Level RH (GLOCL)	1. Line walk oil pipework if accessible and try to identify leaks in flanges /pipework. 2. Inspect filters and strainers for leaks ensure. 3. Check Oil pressure transmitter/gauge is reading correct. 4. Switch the oil cooler. 5. If leakage stops inspect the offline cooler and plan its repair.
	LO PCV Position RL (GLOCL)	
	LO PCV Return Flow RL (GLOCL)	
Lube Oil Filter Anomaly	Oil Filter Dp RH (GLFA)	1. Check filter DP. 2. Changeover and clean /replace the stand-by filter. 3. Check the Filter DP transmitter.
Loss of pressure due to Pump anomaly	LO Header Pressure RL (GLLPA)	1. Check the Discharge PT integrity. 2. Switch on both oil pumps and stabilize the oil pressure. 3. Assess malfunction. 4. Line walk the system and check for leaks. 5. Ensure PSV's are seated. 6. Check spillback/PCV valves are set correctly and not returning oil back to the tank. 7. Ensure no hand valves have been left shut for the standby pump. 8. Check the oil level in the tank, Open tank overhead vents for the release of pressure. 9. Check PCV function for Integrity. 10. Assess malfunction. 11. Check Auto Logic Function tests, and check the functioning of the auto change-over switch. 12. If maintenance has recently been performed check the correct rotation of the pump. 13. Ensure NRV's are working correctly.
	LO Pumps Discharge Pressure RL (GLLPA)	
	LO PCV Position RL (GLLPA)	
	Main Lube Oil Pump Current (Run Signal) ON (GLLPA)	
	Aux Lube Oil Pump Current (Run Signal) ON (GLLPA)	
	LO PCV Return Flow RL (GLLPA)	
Loss of cooling due to TCV Anomaly	LO Header Temperature RH (GLLTCV)	1. Check the bypass control valve for integrity. 2. Open both coolers and stabilize the temperature. 3. Line walk the system to check for leaks.

Fault Name	Associated Symptoms	Recommendations
	LO Cooler Bypass Valve (TCV Position) RH (GLLTCV)	4. Check the Isolation/ bypass valve of TCV.
Loss of cooling due to Cooler Anomaly	LO header temperature RH (LCCA)	1. Check the cooling water supply pressure. 2. Open both coolers and stabilize the temperature. 3. If a malfunction of one cooler is suspected, isolate and inspect it.
	Delta Temperature LO tank vs. LO cooler discharge RH (LCCA)	4. Line walk the system to check for leaks. 5. Check cooling water pumps for blockages, and ensure the strainer is clean.
Loss of heating due lube oil tank heater OFF	LO Header Temperature RL (GLHLTH)	1. Check the Heater ON/OFF.
	Oil Tank Temperature RL (GLHLTH)	2. Switch integrity and Change the switch as per the requirement.
	Lube Oil Tank Heater Run Signal OFF (GLHLTH)	3. Check the heater circuit for Voltage and current.
Loss of cooling due lube oil tank heater ON	LO Header Temperature RH (GLCLTH)	1. Check the Heater ON/OFF.
	Oil Tank Temperature RH (GLCLTH)	2. Switch integrity and Change the switch as per the requirement.
	Lube Oil Tank Heater Run Signal ON (GLCLTH)	3. Offline Check the heater circuit for Voltage and current.
Loss of cooling Due to Fan-Motor Issue	LO Header Temperature RH (GLLCFMI)	1. Check the Fan and Motor /coupling/belt Integrity and rectify.
	Air Cooler Fan-1 (Run Status) OFF (GLLCFMI)	
	Air Cooler Fan-2 (Run Status) OFF (GLLCFMI)	
Loss of Control Oil pressure	Control Oil Header Pressure RL (GLLCOP)	1. Switch on both oil pumps (if possible) and stabilize the oil pressure. 2. Assess malfunction. 3. Line walk oil pipework if accessible and try to identify leaks in flanges /pipework. 4. Inspect filters and strainers for blockages. 5. Ensure the Control Oil pressure transmitter /gauge is reading correctly.
Control Oil Filter Malfunction	Control Oil Filter Dp RH (GLCOFM)	1. Check the control oil filter and clean it.
LO overpressure failure	Tank Over Pressure RH (GLLOVPF)	1. Check the tank relief valve.

MULTI INJECTION STEAM TURBINE

Thermal Power unit has rotating components in which the enthalpy drop of steam is converted into mechanical energy in one or several stages. The steam turbine is used in industrial processes where there is a need for low- or medium-pressure steam. Turbine stages consist of a stationary stator row (guide vanes or nozzle ring) and a rotating rotor row. In guide vanes, high-pressure, high-temperature steam is expanded, resulting in high velocity. Guide vanes direct the flow to the rotor blades at an appropriate angle. In rotor, the flow direction is changed, and the kinetic energy of the working fluid is absorbed by the rotor shaft, producing mechanical energy. This mechanical energy is used to run an electric generator or mechanical equipment such as pumps, fans, compressors, etc.

As steam turbines are energy-intensive equipment, using performance monitoring tools can enable steam turbines to operate in desired conditions, saving energy eventually. Therefore, the availability of steam turbines is an extremely important factor. A special monitoring system is required for improving turbine availability and decreasing energy cost.

19.1 Multi Injection Steam Turbine - Fault Model

19.1.1 Multi Injection Steam Turbine - APR Model

19.1.1.1 Multi Injection Steam Turbine Seal System

Model Type	APR Model
Category	Steam Turbine
Equipment Class	Steam Turbine Component
Equipment Type	Multi Injection ST Seal System

Below table illustrates the list of faults and symptoms configured for a Multi Injection Steam Turbine Seal System

Fault Name	Associated Symptoms	Recommendations
Shaft Seal Failure	Seal Steam Flow R High	Causes: 1. Vibrations 2. Misalignment 3. Bad thermal expansion 4. Wrong Start-up 5. Condensation of bad-quality steam 6. Incorrect seal clearance 7. Excessive sealing steam pressure Effects: 1. Low Efficiency 2. Increased vibrations

Fault Name	Associated Symptoms	Recommendations
		3. Worn shaft on seal location
	Gland Steam Condenser Outlet Temperature R High	Recommendations: 1. Verify steam quality 2. Check turbine thermal expansion gauges vs manual 3. Ensure steam seals are fitted correctly with the correct clearance
Loss of Seal Steam	Seal Steam Header Pressure R Low	Causes: 1. PCV malfunction 2. Miss of supply seal steam Line failure/leakage 3. Valve Carbon rings failed and spring failed Seal steam vent valve (if any) stuck closed 4. Improper cooling/vacuum of gland steam condenser and backpressure
	Seal Steam Pressure DE R Low	Effects: 1. Low efficiency 2. Air sucking into the condenser 3. Increased vibrations 4. Leakage of steam from the turbine resulting in water entering the bearing - a potential safety hazard
	Seal Steam Pressure NDE R Low	Recommendations: 1. Verify the seal steam supply, the PCV, and the connection to the steam header 2. Ensure seal steam valve is free and open 3. Inspect valve carbon rings 4. Check performance/condition of gland steam condenser 5. As a last resort, use lead strips to ensure correct laby seals clearance
Gland steam condenser malfunction	Gland Steam Condenser Outlet Pressure R High	Causes: 1. Ejector failure 2. Cooling Water tube failure 3. Cooling Water supply failure Effects: 1. Loss of vacuum pressure 2. Steam leaks from main rotor shaft labys into atmosphere or bearing housing Recommendations: 1. Check cooling water flow

Fault Name	Associated Symptoms	Recommendations
		2. Check ejector function
		3. Check vacuum slot of condenser for liquid content
	Gland Steam Condenser Outlet Temperature R High	4. Check condenser temperature vs. pressure
		5. If all of the above are ok, either lines from shaft seal to condenser are restricted/blocked or there is an excessive leakage through worn turbine shaft labyrinths, which may cause system malfunction

19.1.1.2 Multi Injection Steam Turbine Bearing System APR Fault Model

Model Type	APR Model
Category	Steam Turbine
Equipment Class	Steam turbine Component
Equipment Type	Multi Injection ST Bearing System

Below table illustrates the list of faults and symptoms configured for a Multi Injection ST Bearing System

Fault Name	Associated Symptoms	Recommendations
Thrust Bearing Failure RH	IB Thrust Bearing Temperature A R High	Causes: 1. Reverse rotation 2. Coupling lock 3. Pad overload due to misalignment 4. Piping stress on Turbine Casing 5. Turbine Thermal Expansion 6. Loss of lubrication 7. Oil contamination 8. Electrical discharge 9. Pad pivot wear 10. LO degradation 11. LO temperature 12. Oil foaming 13. Low Oil Flow 14. Thrust Piston wear 15. Incorrect DBSE between couplings causing overloaded spacer, rotor gauss level too high, earth brush failure 16. Loose proximity probe 17. Damaged thrust collar 18. Incorrect float within bearing Effects:

Fault Name	Associated Symptoms	Recommendations
		1. Overload 2. Babbitt wear/ failure 3. Oil film degradation 4. Bearing damage
	IB Thrust Bearing Temperature B R High	5. Rotor damage 6. Displacement alarm/trip Seal damage 7. Increased bearing temperature 8. Rotor/diaphragm blade potential contact 9. Increased lube oil temperature
	OB Thrust Bearing Temperature A R High	10. Damaged/bent J strip seals 11. Damaged carbon seals Recommendations: 1. If unit has tripped borescope the internals of the turbine to identify any potential damage and inspect the thrust bearing (if this is accessible)
	OB Thrust Bearing Temperature B R High	2. In case of damage, the turbine will require overhauling. 3. If the unit hasn't tripped yet and has past the alarm and is approaching the trip, shut down the unit and carry out borescope inspection and thrust bearing inspection (if the bearing is accessible). 4. Inspect vibration instrumentation to check integrity
Thrust Bearing Failure ROC	IB Thrust Bearing Temperature A R High	Causes: 1. Reverse rotation. Coupling lock 2. Pad overload due to misalignment 3. Piping stress on Turbine Casing 4. Turbine Thermal Expansion 5. Loss of lubrication 6. Oil contamination 7. Electrical discharge 8. Pad pivot wear 9. LO degradation 10. LO temperature 11. Oil foaming 12. Low oil flow 13. Thrust piston wear 14. Incorrect DBSE between couplings causing overloaded spacer 15. Rotor gauss level too high 16. Earth brush failure

Fault Name	Associated Symptoms	Recommendations
		17. Loose proximity probe 18. Damaged thrust collar 19. Incorrect float within bearing Effects: 1. Overload 2. Babbitt wear/failure 3. Oil film degradation 4. Bearing damage 5. Rotor damage 6. Displacement alarm/trip Seal damage 7. Increased bearing temperature 8. Rotor/diaphragm blade potential contact 9. Increased lube oil temperature 10. Damaged/bent J strip seals 11. Damaged carbon seals Recommendations: 1. If unit has tripped borescope the internals of the turbine to identify any potential damage and inspect the thrust bearing (if this is accessible) 2. In case of damage, the turbine will require overhauling. 3. If the unit hasn't tripped yet and has past the alarm but approaching the trip shut the unit down and carry out a borescope inspection and thrust bearing inspection (if the bearing is accessible). Inspect vibration instrumentation to check integrity
	IB Thrust Bearing Temperature B R High	
	OB Thrust Bearing Temperature A R High	
	OB Thrust Bearing Temperature B R High	
	IB Thrust Bearing Temperature A ROC Long Term	
	IB Thrust Bearing Temperature B R ROC Long Term	
	OB Thrust Bearing Temperature A ROC Long Term	
	OB Thrust Bearing Temperature B R ROC Long Term	
DE Radial Bearing Failure RH	Radial Bearing Temperature A DE R High	Causes: 1. Unbalance 2. Misalignment 3. Loss of lubrication 4. Oil Contamination 5. Pad pivot wear 6. Electrical discharge 7. Babbitt failure 8. LO degradation 9. LO temperature 10. Oil foaming Low 11. Oil Flow Seal Steam leakage into Oil Piping stress on Turbine Casing 12. Turbine Thermal Expansion Rotor gauss level too high

Fault Name	Associated Symptoms	Recommendations
		13. Bearing clearance incorrect 14. damage journal on rotor 15. Loose proximity probe/accelerometer Effects:
	Radial Bearing Temperature B DE R High	1. Overload 2. Babbitt wear/ failure 3. Oil film degradation 4. Bearing damage 5. Rotor damage 6. Displacement alarm/trip Seal damage 7. Increased bearing temperature 8. Rotor/diaphragm blade potential contact
	Radial Bearing Vibration X DE R High	9. Increased lube oil temperature 10. Damaged/bent J strip seals 11. Damaged carbon seals Recommendations:
	Radial Bearing Vibration Y DE R High	1. If unit has tripped borescope the internals of the turbine to identify any potential damage and inspect the thrust bearing (if this is accessible), 2. In case of damage, the turbine will require overhauling. 3. If the unit hasn't tripped yet and has past the alarm but approaching the trip shut the unit down and carry out a borescope inspection and radial bearing inspection (if the bearing is accessible) 4. Inspect vibration instrumentation to check integrity
DE Radial Bearing Failure Vibration ROC	Radial Bearing Temperature A DE R High	Causes: 1. Unbalance 2. Misalignment 3. Loss of lubrication 4. Oil Contamination 5. Pad pivot wear 6. Electrical discharge 7. Babbitt failure 8. LO degradation 9. LO temperature 10. Oil foaming Low 11. Oil Flow Seal Steam leakage into Oil Piping stress on Turbine Casing

Fault Name	Associated Symptoms	Recommendations
		12. Turbine Thermal Expansion Rotor gauss level too high 13. Bearing clearance incorrect 14. Damage journal on rotor 15. Loose proximity probe/accelerometer
	Radial Bearing Temperature B DE R High	Effects: 1. Overload 2. Babbitt wear/ failure
	Radial Bearing Vibration X DE R High	3. Oil film degradation 4. Bearing damage 5. Rotor damage 6. Displacement alarm/trip Seal damage
	Radial Bearing Vibration Y DE R High	7. Increased bearing temperature 8. rotor/diaphragm blade potential contact 9. Increased lube oil temperature 10. Damaged/bent J strip seals 11. Damaged carbon seals
	Radial Bearing Vibration X DE ROC Long Term	Recommendations: 1. If unit has tripped borescope the internals of the turbine to identify any potential damage and inspect the thrust bearing (if this is accessible) 2. In case of damage, the turbine will require overhauling
	Radial Bearing Vibration Y DE ROC Long Term	3. If the unit hasn't tripped yet and has past the alarm but approaching the trip shut the unit down and carry out a borescope inspection and radial bearing inspection (if the bearing is accessible) 4. Inspect vibration instrumentation to check integrity
DE Radial Bearing Failure Temperature ROC	Radial Bearing Temperature A DE R High	Causes: 1. Unbalance 2. Misalignment 3. Loss of lubrication 4. Contamination 5. Pad pivot wear 6. Electrical discharge 7. Babbitt failure 8. LO degradation 9. LO temperature 10. Oil foaming Low 11. Oil Flow

Fault Name	Associated Symptoms	Recommendations
		12. Lube Oil cooler failure 13. TCV failure 14. Bearing clearance too tight Effects: 1. High vibration 2. Overload 3. Babbitt failure 4. Bearing/rotor damage 5. Oil film instability 6. High bearing temperature Recommendations: 1. If unit has tripped, inspect bearings and check its internal clearances; ensure oil pipework is clear of blockages. 2. Thoroughly inspect the lube oil system and associated hardware. 3. If the unit is still running and has passed the alarm point but hasn't tripped yet and is still increasing shut the unit down and inspect as above.
	Radial Bearing Temperature B DE R High	
	Radial Bearing Vibration X DE R High	
	Radial Bearing Vibration Y DE R High	
	Radial Bearing Temperature A DE ROC Long Term	
	Radial Bearing Temperature B DE ROC Long Term	
NDE Radial Bearing Failure Vibration ROC	Radial Bearing Temperature A NDE R High	Causes: 1. Unbalance 2. Misalignment 3. Loss of lubrication 4. Oil Contamination 5. Pad pivot wear 6. Electrical discharge 7. Babbitt failure 8. LO degradation 9. LO temperature 10. Oil foaming 11. Low Oil Flow Seal Steam leakage into Oil Piping stress on Turbine Casing 12. Turbine Thermal Expansion Rotor gauss level too high 13. Bearing clearance incorrect 14. Damage journal on rotor 15. Loose proximity probe/accelerometer Effects: 1. Overload 2. Babbitt wear/ failure

Fault Name	Associated Symptoms	Recommendations
		3. Oil film degradation 4. Bearing damage 5. Rotor damage 6. Displacement alarm/trip Seal damage 7. Increased bearing temperature 8. Rotor/diaphragm blade potential contact 9. Increased lube oil temperature 10. Damaged/bent J strip seals 11. Damaged carbon seals
	Radial Bearing Temperature B NDE R High	
	Radial Bearing Vibration X NDE R High	
	Radial Bearing Vibration Y NDE R High	
	Radial Bearing Vibration X NDE ROC Long Term	
	Radial Bearing Vibration Y NDE ROC Long Term	
		Recommendations: 1. If unit has tripped borescope the internals of the turbine to identify any potential damage and inspect the thrust bearing (if this is accessible), 2. In case of damage, the turbine will require overhauling. 3. If the unit hasn't tripped yet and has past the alarm but approaching the trip shut the unit down and carry out a borescope inspection and radial bearing inspection (if the bearing is accessible). 4. Inspect vibration instrumentation to check integrity
NDE Radial Bearing Failure Temperature ROC	Radial Bearing Temperature A NDE R High	Causes: 1. Unbalance 2. Misalignment 3. Loss of lubrication 4. Contamination 5. Pad pivot wear 6. Electrical discharge 7. Babbitt failure 8. LO degradation 9. LO temperature 10. Oil foaming Low 11. Oil Flow 12. Lube oil cooler failure 13. TCV failure 14. Bearing clearance too tight Effects: 1. High vibration 2. Overload 3. Babbitt failure

Fault Name	Associated Symptoms	Recommendations
	Radial Bearing Temperature B NDE R High	4. Bearing/rotor damage 5. Oil film instability. 6. High bearing temperature
	Radial Bearing Vibration X NDE R High	Recommendations: - If unit has tripped, inspect bearings and check its internal clearances, ensure oil pipework is clear of blockages. - Thoroughly inspect the lube oil system and associated hardware. - If the unit is still running and has passed the alarm point but hasn't tripped yet and is still increasing shut the unit down and inspect as above.
	Radial Bearing Vibration Y NDE R High	
	Radial Bearing Temperature A NDE ROC Long Term	
	Radial Bearing Temperature B NDE ROC Long Term	
NDE Radial Bearing Failure RH	Radial Bearing Temperature A NDE R High	Causes: - Unbalance - Misalignment - Loss of lubrication - Oil Contamination - Pad pivot wear - Electrical discharge - Babbitt failure - LO degradation - LO temperature - Oil foaming - Low Oil Flow Seal Steam leakage into Oil Piping stress on Turbine Casing - Turbine Thermal Expansion - Rotor gauss level too high - Bearing clearance incorrect - Damage journal on rotor - Loose proximity probe/accelerometer Effects: - Overload - Babbitt wear/ failure - Oil film degradation - Bearing damage - Rotor damage

Fault Name	Associated Symptoms	Recommendations
		6. Displacement alarm/trip Seal damage 7. Increased bearing temperature 8. Rotor/diaphragm blade potential contact 9. Increased lube oil temperature 10. Damaged/bent J strip seals 11. Damaged carbon seals
	Radial Bearing Temperature B NDE R High	
	Radial Bearing Vibration X NDE R High	Recommendations: 1. If the unit has tripped borescope the internals of the turbine to identify any potential damage and inspect the thrust bearing (if this is accessible) 2. In case of damage, the turbine will require overhauling 3. If the unit hasn't tripped yet and has past the alarm but approaching the trip shut the unit down and carry out a borescope inspection and radial bearing inspection (if the bearing is accessible) 4. Inspect vibration instrumentation to check integrity
	Radial Bearing Vibration Y NDE R High	
MIST_AXL_DISP_WARNING	Thrust Bearing Axial Displacement Z1 R High	Causes: 1. Unbalance 2. Misalignment 3. Loss of lubrication 4. Oil Contamination 5. Pad pivot wear 6. Electrical discharge 7. Babbitt failure 8. LO degradation 9. LO temperature 10. Oil foaming Low 11. Oil Flow Seal Steam leakage into Oil Piping stress on Turbine Casing 12. Turbine Thermal Expansion Rotor gauss level too high 13. Bearing clearance incorrect 14. Damage journal on rotor 15. Loose proximity probe/accelerometer Effects: 1. Overload 2. Babbitt wear/ failure 3. Oil film degradation 4. Bearing damage

Fault Name	Associated Symptoms	Recommendations
		5. Rotor damage 6. Displacement alarm/trip Seal damage 7. Increased bearing temperature 8. Rotor/diaphragm blade potential contact 9. Increased lube oil temperature 10. Damaged/bent J strip seals 11. Damaged carbon seals
	Thrust Bearing Axial Displacement Z2 R High	Recommendations: 1. If unit has tripped borescope the internals of the turbine to identify any potential damage and inspect the thrust bearing (if this is accessible), 2. If damage has been seen the turbine will require overhauling. 3. If the unit hasn't tripped yet and has past the alarm but approaching the trip shut the unit down and carry out a borescope inspection and radial bearing inspection (if the bearing is accessible) 4. Inspect vibration instrumentation to check integrity

19.1.1.3 Multi Injection Steam Turbine Condenser System APR Fault Model

Model Type	APR Model
Category	Steam turbine
Equipment Class	Steam turbine component
Equipment Type	Multi injection ST condenser system

Below table illustrates the list of faults and symptoms configured for Multi Injection ST Condenser System

Fault Name	Associated Symptoms	Recommendations
Tube Side CrackShell Side Outlet Flow Rate R Low	Tube Side Outlet Temp R Low	Causes: Tube integrity/thinning, Baffle, and its support integrity, etc. Effects: 1. Tube crack will lead to high-pressure steam/water leaking into the low-pressure tube (with process gas) leading to process integrity issues etc.. will lead to further corrosion further downstream of the assets. Recommendations: 1. Inspect tube integrity 2. Clean or plug the tube if required
	Tube Side Temperature Difference (O/L-I/L) R High	
	Shell Side Outlet Temp R High	
	Shell Pressure R Low	
Tube Side Inlet	Tube Side Inlet	Causes:

Fault Name	Associated Symptoms	Recommendations
Anomaly	Pressure R Low	Anomaly in the upstream Asset of the process gas train, process upsets, Pipeline integrity (leak/fouling), Pipeline CUI, Isolation / Control/Bypass/Non-Return valves stuck/leak Effects: The anomaly in the input will give rise to operating parameter changes in subsequent downstream assets, thus increasing the process upsets further. Recommendations: 1. Monitor the upstream assets of process gas. 2. Inspect integrity of upstream assets, pipelines, insulations, valves, etc. 3. Inspect for Pipelines CUI. Check the valves' internals.
	Tube Side Inlet Flow Rate R Low	
	Tube Side Inlet Temp R High	
	Tube Side Inlet Pressure R High	
	Tube Side Inlet Flow Rate R High	
Shell or Tube Side Fouling	Tube Side Outlet Temp R High	Causes: 1. Anomaly in the upstream Asset of the process gas train, process upsets, Pipeline integrity (leak/fouling), Pipeline CUI, Isolation/ "Scaling/fouling at either side of the tubes, Tubes clogging, etc. Effects: The fouling will lead to lesser heat transfer and hence effectiveness, duty, performance, etc. will be affected. Recommendations: 1. Inspect the tubes for fouling at inner/outer side, and clean the exchanger if necessary 2. Monitor the steam conductivity, pH of water, Sulfur content of the gas, etc.
	Tube Side Temperature Difference (O/L-I/L) R Low	
	Shell Side Outlet Temp R Low	
Corrosion And Fouling Risk	Shell Side Inlet Pressure R Low	Causes: 1. Process gas & water/condensate mixing 2. Bacterial/micro-organism growth in the fluid Effects: Any change in property will lead to either corrosion or fouling in the asset and its complete train. Recommendations: 1. Monitor the pH. Monitor the Steam conductivity. 2. Monitor the H₂S, SO₂ level in the process gas
	Shell Side Outlet Pressure R High	
Shell Side Inlet Anomaly	Shell Side Inlet Temp R High	Causes Anomaly in the upstream Asset of the Steam / Water train, process upsets, Pipeline integrity (leak/fouling), Pipeline CUI, Isolation / Control/Bypass/Non-Return valves stuck/leak, pressure controller issue Effects:

Fault Name	Associated Symptoms	Recommendations
	Shell Side Inlet Pressure R High	The anomaly in the input will give rise to operating parameter changes in subsequent downstream assets, thus increasing the process upsets further. Recommendations: 1. Monitor the upstream assets of Steam / Water/Condensate. 2. Inspect integrity of upstream assets, pipelines, insulations, valves, etc. Inspect for Pipelines CUI. 3. Check the valves' internals. 4. Carry out loop proof test of the steam pressure control loop
	Shell Side Inlet Temp R Low	
	Shell Side Inlet Flow Rate R Low	
	Shell Side Inlet Flow Rate R High	
	Shell Side Inlet Pressure R Low	
Shell Side Crack	Tube Side Temperature Difference (O/L-I/L) R Low	Causes 1. Shell integrity/thinning 2. Nozzle crack 3. Support integrity 4. Manway gasket / Bolt integrity 5. PSV Leak/passing etc Effects: Shell crack will lead to high-pressure steam/water leaking outside leading to safety issues, sustainability issues, etc. Recommendations: 1. Inspect shell integrity 2. Inspect insulation damage with hot spot or Shell CUI 3. Inspect Nozzle integrity. Also, inspect shell and piping support integrity 4. Check Manway gasket condition, Manway bolt tightening torques, and Manway integrity 5. Check PSV leakage
	Shell Pressure R Low	
	Shell Condensate Level -1 R Low	
	Shell Condensate Level -2 R Low	

19.1.1.4 Multi Injection Steam Turbine Lube Oil System APR Fault Model

Model Type	APR Model
Category	Steam Turbine
Equipment Class	Steam turbine Component
Equipment Type	Multi Injection ST Lube Oil System

Below table illustrates the list of faults and symptoms configured for Multi Injection ST Lube Oil System

Fault Name	Associated Symptoms	Recommendations
Loss of cooling	LO header temperature R High	Causes: 1. TCV failure 2. Cooler failure 3. Cooler fouling 4. CW temperature anomaly 5. LO pump performance degradation (low flow) 6. LO cooling water pipework leaking 7. LO cooling water 8. PCV passing 9. Malfunction of oil heater remaining on when not required Effects: 1. Oil instability, 2. Bearing overheating 3. LO varnishing Recommendations: 1. Check cooling water supply pressure. 2. Check bypass control valve. 3. Open both coolers and stabilize temperature. 4. Check oil heater is not on (even if instruments indicate it is off) 5. If malfunction of one cooler is suspected, isolate and inspect it. 6. Line walk the LO cooling pipework to check for leaks 7. Inspect strainers on pumps 8. check instrumentation is working correctly
	Oil tank temperature R High	
	LO Cooler bypass valve (TCV Position) R High	
	LO cooler outlet temperature R High	
Oil circuit leak	Oil tank level ROC Long Term	Causes: 1. Oil pipe crack 2. Gasket failure 3. Flange leakage 4. Bearing seal failure Effects: 1. Oil tank level decreases 2. PCVs opens to compensate for lost oil flow 3. Oil accumulates on the floor, close to the leakage source. Recommendations: 1. Line walk oil pipework if accessible and try to identify leaks in flanges/pipework 2. Inspect filters and strainers for leaks ensure.

Fault Name	Associated Symptoms	Recommendations
	LO PCV position/return flow Long Term	3. Check Oil pressure transmitter/gauge is reading correctly. Switch oil cooler: if leakage stops inspect offline cooler and plan its repair.
Loss of Control Oil pressure	Control Oil header pressure R Low	Causes: 1. PCV malfunction 2. Line failure/leakage 3. LO pump performance degradation 4. LO tank level low 5. Oil pipework leaking 6. Spillback valve on lube oil tank set wrong 7. PCV passing 8. Solenoid Malfunction Effects: 1. Turbine Trip 2. Process impacts Recommendations: 1. Check cooling water supply pressure. 2. Check bypass control valve. 3. Open both coolers and stabilize temperature. 4. Switch on both oil pumps (if possible) and stabilize oil pressure. 5. Assess malfunction. 6. Line walk oil pipework if accessible and try to identify leaks in flanges/pipework 7. Inspect filters and strainers for blockages 8. Ensure Control Oil pressure transmitter/gauge is reading correctly
Loss of pressure	LO header pressure R Low	Causes: 1. PCV malfunction 2. Line failure/leakage 3. LO pump performance degradation 4. LO tank level low 5. Oil pipework leaking, 6. Spillback valve on lube oil tank set wrong 7. PCV passing 8. Oil drain from bearing coked. Effects: 1. Bearing overheating

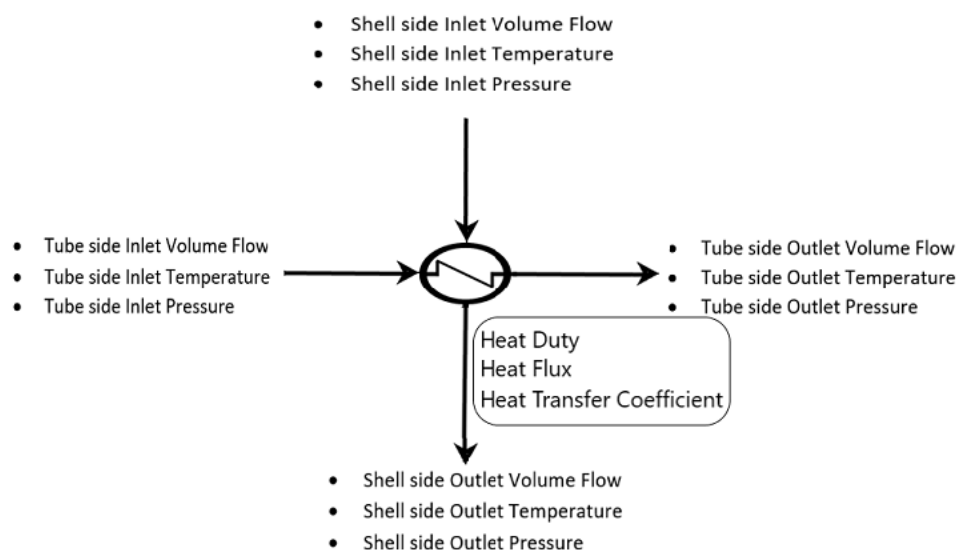
Fault Name	Associated Symptoms	Recommendations
	Oil tank level ROC Long Term	2. Bearing seizure 3. Babbitt failure 4. Journal damage 5. Low lube oil pressure. 6. Control oil low pressure, 7. ESV and throttle valves potentially close due to oil pressure
	LO Pumps discharge pressure R Low	
	LO PCV position/return flow R High	
	LO PCV position/return flow R High	
	Main Lube oil pump current (run signal) ON	Recommendations: 1. Switch on both oil pumps (if possible) and stabilize oil pressure. 2. Assess malfunction. 3. Line walk oil pipework if accessible and try to identify leaks in flanges/pipework 4. Inspect filters and strainers for blockages 5. Inspect/borescope oil drains for oil coking and blockages 6. Ensure LO pressure transmitter/gauge is reading correctly
	Aux Lube oil pump current (run signal) ON	
	LO PCV position/return flow R Low	

SHELL AND TUBE HEAT EXCHANGER

A heat exchanger is a device used to transfer thermal energy (enthalpy) between two or more fluids or between a solid surface and a fluid or between solid particulates and a fluid, at different temperatures and in thermal contact.

- Shell and tube heat exchanger is a tubular and indirect contact type heat exchanger. The tube bundle is arranged inside the shell. Baffles are placed in the shell to make shell-side fluid flow at higher turbulence and to induce higher heat transfer. To increase the heat exchange rate between two streams, industries generally use multi-shell heat exchangers. If more than one shell is used in a heat exchanger it is called a multi-shell heat exchanger. Shell fluid enters the shell, flows across the tube bundle, and leaves the shell.
- In the multi-shell and tube exchangers, the shells are connected either in series or in parallel. Fluid enters the shell flows across it and enters into the next shell. The flow conditions are neither parallel flow nor counter flow type. In these exchangers, flow is a combination of both parallel and counter flow types. Hence a correction factor 'F' must be introduced in the general heat equation.
- Proper design, operation, and maintenance of heat exchangers will make the process energy-efficient, reducing energy losses. Heat exchanger performance can deteriorate with time, off-design operations, and other interferences such as fouling, scaling, etc. It is necessary to assess periodically the heat exchanger's performance in order to maintain them at an optimum level.
- In addition to operating conditions, monitoring basic machine health data will help maintain reliable operations of a heat exchanger.

Category	EQUIPMENT
Equipment Class	HEAT EXCHANGER
Equipment Type	SHELL AND TUBE HEAT EXCHANGER



20.1 Shell and Tube Heat Exchanger - Performance Monitoring

The following measured parameters are monitored as they can negatively impact the performance and health of shell and tube exchanger if not maintained in the optimum operating range.

Shell/Tube Inlet Temperature: Increase or decrease of the inlet temperature will affect heat transfer within the heat exchanger and thereby varies the outlet temperature.

Shell/tube Mass/Volumetric flow rate: Increase in the flow rate of fluids results in an increase in the film heat transfer coefficient and thereby increase in the expected overall heat transfer coefficient.

Shell/Tube Inlet Pressure: Increase or decrease in pressure can vary parameters like temperature, velocity of flow etc., which ultimately may affect convective heat transfer coefficient.

Process Performance Overview: Process performance monitoring view in APM provides a list of the asset attributes and their details, including the current value. This list can be used to review and update asset attribute information and open an Asset Trend to view more details. It also highlights (in red color) the attributes if deviates from the high or low limits.

20.1.1 Notations

F	Mass Flow rate
P	Pressure
T	Temperature
Q	Heat duty
q	Heat flux
A	Area
U	Heat transfer Coefficient
h	Enthalpy
p	Temperature Effectiveness
R	Heat capacity ratio
HEx_{load}	Heat exchanger load
ρ	Density
η	Heat transfer Efficiency
ϕ_v	Phase fraction - vapor
ΔT_{lnCor}	Corrected log mean temperature difference
ΔT_{LMTD}^{Design}	Design log mean temperature difference
R_{hf}	Hot side fouling
R_{cf}	Cold side fouling
R_F	Overall fouling
ε	Effectiveness

20.1.2 Subscripts

<i>h</i>	Hot side
<i>c</i>	Cold side
<i>s</i>	Shell side
<i>t</i>	Tube side
<i>i</i>	Inlet
<i>o</i>	Outlet
<i>Design</i>	Design specification
<i>sat</i>	Saturated conditions
<i>Ref</i>	Reference value

20.1.3 Superscript

<i>Ex</i>	Expected
<i>Op</i>	Operating
<i>Dev</i>	Deviation
<i>Design</i>	Design specification

20.1.4 Shell and Tube Heat Exchanger - Model Assumptions

1. There is no internal heat generation.
2. The LMTD Correction factor is valid for any number of shell passes and only for an even number of tube passes per shell.
3. There is no leakage of fluid between the shell and tube side (Inlet flow = outlet flow).
4. For the condensing steam system, incoming steam is assumed to be completely vapor either saturated or superheated. If not, it is corrected to the saturated vapor condition by doing ${}^P\phi_v (= 1)$ flash.

$$h_{Sat}, \rho_{Sat}, T_{Sat} = f(P, \phi_{v=1})$$

5. For the evaporating water system, incoming feed is assumed to be completely liquid either saturated or subcooled. If not, it has to correct to the saturated liquid condition by doing ${}^P\phi_v (= 0)$ flash.

$$h_{Sat}, \rho_{Sat}, T_{Sat} = f(P, \phi_{v=0})$$

20.1.5 Shell and Tube Heat Exchanger - Model Input

No.	Attribute	Units	Attribute Category	Monitoring Category	Source
MEASURED PARAMETERS					
1	Tube Inlet Temperature	°C	Dynamic	Process	Field input
2	Tube Outlet Temperature	°C	Dynamic	Process	Field input
3	Tube Inlet Pressure	bar g	Dynamic	Process	Field input

No.	Attribute	Units	Attribute Category	Monitoring Category	Source
4	Tube Outlet Pressure	bar g	Dynamic	Process	Field input
5	Tube Inlet Mass flow	Kg hr ⁻¹	Dynamic	Process	Field input
6	Tube Inlet Volume flow	m ³ hr ⁻¹	Dynamic	Process	Field input
7	Shell Inlet Temperature	°C	Dynamic	Process	Field input
8	Shell Outlet Temperature	°C	Dynamic	Process	Field input
9	Shell Inlet Pressure	bar g	Dynamic	Process	Field input
10	Shell Outlet Pressure	bar g	Dynamic	Process	Field input
11	Shell Inlet Mass Flow	Kg hr ⁻¹	Dynamic	Process	Field input
12	Shell Inlet Volume flow	m ³ hr ⁻¹	Dynamic	Process	Field input
MODEL CONSTANTS					
1	Tube Design Mass Flow	Kg hr ⁻¹	Constant	Design	Datasheet
2	Tube Design Inlet Temperature	°C	Constant	Design	Datasheet
3	Tube Design Outlet Temperature	°C	Constant	Design	Datasheet
4	Shell Design Mass Flow	Kg hr ⁻¹	Constant	Design	Datasheet
5	Shell Design Inlet Temperature	°C	Constant	Design	Datasheet
6	Shell Design Outlet Temperature	°C	Constant	Design	Datasheet
7	Number of shell passes		Constant	Design	Datasheet
8	Heat Exchanger Area		Constant	Design	Datasheet
9	Heat Transfer Coefficient - Reference In Service State	kW m ⁻² C ⁻¹	Constant	Design	Datasheet
10	Design Duty	kW	Constant	Design	Datasheet
11	Shell side model type Type 1: None of the below Type 2: Steam Condensing Type 3: Water Evaporating		Constant	Design	User Input
12	Tube Side model type Type 1: None of the below Type 2: Steam Condensing Type 3: Water Evaporating		Constant	Design	User Input
13	Tube Outside diameter (optional)		Constant	Design	Datasheet
14	Tube Inside diameter (optional)		Constant	Design	Datasheet
15	Tube wall thickness (optional)		Constant	Design	Datasheet
16	Tube wall Thermal conductivity (optional)		Constant	Design	Datasheet

No.	Attribute	Units	Attribute Category	Monitoring Category	Source
17	Rated shell side fouling resistance (optional)		Constant	Design	Datasheet
18	Rated tube side fouling resistance (optional)		Constant	Design	Datasheet
19	Design fouling resistance		Constant	Design	Datasheet
20	Rated Shell Pressure Drop	bar	Constant	Design	Datasheet
21	Rated Tube Pressure Drop	bar	Constant	Design	Datasheet

20.1.6 Shell and Tube Heat Exchanger - KPIs

20.1.6.1 Operating KPIs

1. Operating Heat Duty
2. Operating Heat Flux
3. Operating Heat Transfer Coefficient

20.1.6.2 Expected KPIs

1. Expected Heat Duty
2. Expected Heat Flux
3. Expected Heat Transfer Coefficient
4. Expected Heat Transfer Coefficients
5. Expected Shell Pressure Drop
6. Expected Tube Pressure Drop

20.1.6.3 Deviation KPIs

1. Deviation Heat Duty
2. Deviation Heat Flux
3. Deviation Heat Transfer Coefficient
4. Deviation Shell Pressure Drop
5. Deviation Tube Pressure Drop

20.1.6.4 Heat Exchanger Efficiency

1. Duty Efficiency %
2. Heat Transfer Efficiency %

20.1.6.5 Energy Loss Parameters

1. Duty Loss due to design
2. Duty Loss due to design %
3. Duty Loss due to operation
4. Duty Loss due to operation %
5. Total Duty Loss
6. Total Duty Loss %

20.1.6.6 Heat Exchanger Health Parameters

1. Shell Fouling %
2. Tube Fouling %
3. Overall Heat Exchanger Fouling %

20.1.6.7 Pressure drop information

1. Shell side Pressure drop
2. Tube side pressure drop

20.1.6.8 Shell and Tube Heat Exchanger - Performance Modeling***Overall Mass Balance***

Shell side:

$$F_{si} = F_{so}$$

Tube side:

$$F_{ti} = F_{to}$$

Overall Force Balance

Shell side:

$$P_{si} = P_{so} + \Delta P_s$$

Tube side:

$$P_{ti} = P_{to} + \Delta P_t$$

Overall Energy Balance

Hot side:

$$F_h h_{hi}(P_{hi}, T_{hi}, z_{hij}) = F_h h_{ho}(P_{ho}, T_{ho}, z_{hoj}) + Q_h$$

Cold side:

$$F_c h_{ci}(P_{ci}, T_{ci}, z_{cij}) + Q_c = F_c h_{co}(P_{co}, T_{co}, z_{coj})$$

20.1.6.9 Shell and Tube Heat Exchanger - Expected KPIs***Expected Heat Duty***

Expected Heat Duty is the product of design duty (taken from the datasheet) and heat exchanger load. The Heat Exchanger load is the ratio of Design LMTD (calculated based on design temperatures) and Operating LMTD (calculated based on operating temperatures).

$$\Delta T_1^{Design} = T_{hi}^{Design} - T_{co}^{Design}$$

$$\Delta T_2^{Design} = T_{h_o}^{Design} - T_{c_i}^{Design}$$

$$\Delta T_{LMTD}^{Design} = \frac{\Delta T_1^{Design} - \Delta T_2^{Design}}{\ln \left(\frac{\Delta T_1^{Design}}{\Delta T_2^{Design}} \right)}$$

$$T_{load} = \frac{\Delta T_{LMTD}^{Design}}{\Delta T_{lnCor}}$$

$$Heat\ Exchanger\ Load = HEx_{load} = T_{load}$$

$$Expected\ Heat\ Duty = Q^{Ex} = Q_{Design} HEx_{load}$$

Expected Heat Flux

The Expected Heat flux is the ratio of Expected heat duty and the heat exchanger area. It gives us an idea of the expected heat transfer rate irrespective of area.

$$Expected\ Heat\ Flux = q^{Ex} = \frac{Q^{Ex}}{A}$$

Expected Heat Transfer Coefficient

The Expected heat transfer coefficient is the product of the heat transfer coefficient - Reference in service (taken from the datasheet) and the correction factor calculated with the ratios of operating and rated mass flowrates.

$$f1 = \left(\frac{shellMassFlow}{RatedShellMassFlow} \right)^{0.8}, \quad f2 = \left(\frac{TubeMassFlow}{RatedTubeMassFlow} \right)^{0.8}$$

$$F = \frac{2 \times f1 \times f2}{(f1 + f2)}$$

$$Expected\ Heat\ Transfer\ Coefficient = U_{Exp} = U_{Rated} \times F$$

Expected Design Effectiveness

Design effectiveness is calculated based on design temperatures,

$$\epsilon^{design} = Max(p^{design}, p^{design} R^{design})$$

Expected Design Fouling Resistance

Design fouling resistance from the datasheet needs to be provided.

$$R_F^{Design}$$

Expected Shell Pressure Drop

The Expected Shell Pressure Drop is the ratio operating shell mass flow and rated shell mass flow times the rated shell pressure drop.

$$\text{Expected Shell Pressure Drop} = \left(\frac{\text{Operating Shell Massflow}}{\text{Rated Shell Massflow}} \right)^2 \times \text{Rated Shell Pressure Drop}$$

Expected Tube Pressure Drop

The Expected Tube Pressure Drop is the ratio operating tube mass flow and rated tube mass flow times the rated tube pressure drop.

$$\text{Expected Tube Pressure Drop} = \left(\frac{\text{Operating Tube Massflow}}{\text{Rated Tube Massflow}} \right)^2 \times \text{Rated Tube Pressure Drop}$$

20.1.6.10 Shell and Tube Heat Exchanger - Operating KPIs

Operating Heat Duty

Heat Duty is the amount of energy transferred from the hot fluid to the cold fluid within the exchanger. It is calculated as the product of mass flow and the mass enthalpy difference of inlet and outlet for both sides of the exchanger and minimum duty is taken as the operating heat duty of the heat exchanger. Since we are calculating duty using enthalpy, it is applicable for phase change exchangers such as condensers or evaporators.

$$F_h h_{hi}(P_{hi}, T_{hi}, z_{hij}) = F_h h_{ho}(P_{ho}, T_{ho}, z_{hoj}) + Q_h$$

$$F_c h_{ci}(P_{ci}, T_{ci}, z_{cij}) + Q_c = F_c h_{co}(P_{co}, T_{co}, z_{coj})$$

$$\text{Operating Heat duty} = Q^{Op} = \text{Min}(Q_h, Q_c)$$

Operating Heat Flux

The Operating Heat flux is the ratio of Operating heat duty and the heat exchanger area. It gives us an idea of operating heat transfer rate irrespective of area.

$$\text{Operating Heat flux} = q^{Op} = \frac{Q^{Op}}{A}$$

Operating Heat Transfer Coefficient

The heat transfer coefficient or film coefficient is the proportionality constant between the heat flux and the thermodynamic driving force for the flow of heat (i.e., the temperature difference, ΔT). The overall heat transfer rate for combined modes is usually expressed in terms of an overall conductance or heat transfer coefficient, U. The Logarithmic Mean Temperature (LMTD) is the function of inlet and outlet temperatures of both sides and the number of Shell Passes.

$$\text{Temperature Effectiveness} = p = \frac{T_{co} - T_{ci}}{T_{hi} - T_{ci}}$$

$$\text{Heat Capacity Ratio} = R = \frac{T_{hi} - T_{ho}}{T_{co} - T_{ci}}$$

$$\alpha = \left(\frac{Rp - 1}{p - 1} \right)^{\left(\frac{1}{N} \right)}$$

N = Number of shell passes

$$X = \frac{1 - \alpha}{R - \alpha}$$

$$\text{Correction Factor} = F_t = \frac{(R^2 + 1)^{0.5} \ln \frac{1-X}{1-RX}}{(R-1) \ln \frac{2-X(R+1)-(R^2+1)^{0.5}}{2-X(R+1)+(R^2+1)^{0.5}}}$$

$$\Delta T_1 = T_{hi} - T_{co}$$

$$\Delta T_2 = T_{ho} - T_{ci}$$

$$\text{Operating LMTD} = \Delta T_{\ln Cor} = F_t \frac{\Delta T_1 - \Delta T_2}{\ln \frac{\Delta T_1}{\Delta T_2}}$$

$$\text{Operating Heat Transfer Coefficient} = U^{Op} = \frac{\text{Min}(Q_h, Q_c)}{A \Delta T_{\ln Cor}}$$

Heat Exchanger Approach Temperature

Higher the heat exchanger approach temperature we achieve, higher will be the heat transfer for the constant heat exchanger area.

$$\text{Approach Temperature} = T_{hi} - T_{co}$$

Heat Exchanger Efficiency

The Heat transfer percent is the ratio of actual heat transfer to the maximum possible heat transfer. One more indicator which will say about the heat exchanger efficiency is the duty efficiency percentage. It is the ratio of minimum heat duty to maximum heat duty and here, we are focusing on the actual heat energy received by the cold fluid apart from the heat losses.

$$\text{Duty efficiency Percent} = Q \% = \frac{\text{Min}(Q_h, Q_c)}{\text{Max}(Q_h, Q_c)} 100$$

$$\text{Heat Transfer Percent} = \eta = \text{Max}(100p, 100pR)$$

Energy Loss parameters

The total energy loss is the sum of energy losses due to design and the energy losses due to operation or fouling. There are some heat losses to surrounding which are considered as the design heat losses and heat loss due to fouling is considered as the operational heat losses.

Energy losses due to design:

$$Q_{Loss}^{Design} \% = 100 - Q\%$$

$$Q_{Loss}^{Design}(kW) = \max(Q_h, Q_c) \frac{Q_{Loss}^{Design}\%}{100}$$

Energy losses due to operation or fouling:

$$Q_{Loss}^{Fouling}\% = 100 \left(1 - \frac{\min(U_h^{Op}, U_c^{Op})}{U^{Ex}} \right)$$

$$Q_{Loss}^{Fouling}(kW) = \max(Q_h, Q_c) \frac{Q_{Loss}^{Fouling}\%}{100}$$

Total Energy Loss:

$$Q_{Loss}^{Total}(kW) = Q_{Loss}^{Design} + Q_{Loss}^{Fouling}$$

$$Q_{Loss}^{Total}\% = Q_{Loss}^{Design}\% + Q_{Loss}^{Fouling}\%$$

Heat exchanger health parameters

Heat exchanger performance can deteriorate with time, off design operations, and other interferences such as fouling, scaling, etc. It is necessary to assess periodically the heat exchanger's health in order to maintain them at an optimum level. Fouling causes a decrease in surface area reducing the heat transfer coefficient.

$$\text{Shell side Heat transfer coefficient} = U_s = \frac{Q_s}{A \Delta T_{lnCor}}$$

$$\text{Tube side Heat transfer coefficient} = U_t = \frac{Q_t}{A \Delta T_{lnCor}}$$

Shell side Fouling %:

If,

$$U_s = U^{Op}$$

$$R_{sf}\% = 100 \left(1 - \frac{U_s}{U^{Ex}} \right)$$

Otherwise

$$R_{sf}\% = 100 \left(1 - \frac{U_s Q\%}{U^{Ex} 100} \right)$$

Tube side Fouling %:

If,

$$U_t = U^{Op}$$

$$R_{tf}\% = 100 \left(1 - \frac{U_t}{U^{Ex}} \right)$$

Otherwise

$$R_{tf}\% = 100 \left(1 - \frac{U_t Q \%}{U^{Ex} 100} \right)$$

Heat Exchanger Overall Fouling %:

$$R_F\% = 100 \frac{U^{Ex} - U^{op}}{U^{Ex}}$$

Pressure drop

The pressure drop is the difference between inlet and outlet pressures of the individual sides. The increase in pressure drop on a side indicates the fouling at that particular side of the heat exchanger.

$$\text{Shell side Pressure drop} = \Delta P_s = P_{si} - P_{so}$$

$$\text{Tube side Pressure drop} = \Delta P_t = P_{ti} - P_{to}$$

Shell side and Tube side Temperature Difference

Hot side identification,

$$T_{hi} = \max(T_{si}, T_{ti})$$

$$\text{If } T_{hi} = T_{si},$$

then

$$\Delta T_{shell} = T_{so} - T_{si}$$

$$\Delta T_{tube} = T_{ti} - T_{to}$$

Else,

$$\Delta T_{shell} = T_{si} - T_{so}$$

$$\Delta T_{tube} = T_{to} - T_{ti})$$

Average Heat Duty

$$Q_{avg} = \frac{Q_h + Q_c}{2}$$

Operating Effectiveness

$$\varepsilon^{op} = \text{Max}(p, p R)$$

Operating Fouling Factor

If required design parameters such as tube out diameter, tube In diameter, tube wall thermal conductivity, tube wall thickness, and rated tube side fouling resistance are provided, and rated shell side fouling resistance is provided, the operating fouling resistance can be calculated using those inputs. Otherwise, it will be the calculated using heat transfer coefficient.

$$R_F^{op} = \left(\frac{1}{U_s} \right) + \left(\frac{1}{U_t} \right) \left(\frac{d_{t_o}}{d_{t_i}} \right) + \left(\frac{t_w}{K_w} \right) \left(\frac{d_{t_o}}{d_{t_i}} \right) + R_s^{design} + R_t^{design} \left(\frac{d_{t_o}}{d_{t_i}} \right)$$

OR

$$R_F^{op} = \frac{1}{U^{Ex}} - \frac{1}{U^{op}}$$

20.1.6.11 Shell and Tube Heat Exchanger - Deviations KPIs

The deviation KPIs such as Deviation Duty, flux, and Heat transfer coefficient represents the deviation of operating KPI with expected KPI at current operating conditions.

$$\text{Deviation Heat Duty} = Q^{Dev} = \frac{Q^{Op} - Q^{Ex}}{Q^{Ex}} 100$$

$$\text{Deviation Heat Flux} = q^{Dev} = \frac{q^{Op} - q^{Ex}}{q^{Ex}} 100$$

$$\text{Deviation Heat Transfer Coefficient} = U^{Dev} = \frac{U^{Op} - U^{Ex}}{U^{Ex}} 100$$

$$\Delta T_{LMTD}^{Dev} = \frac{(\Delta T_{InCor} - \Delta T_{LMTD}^{Design})}{\Delta T_{LMTD}^{Design}} 100$$

$$R_F^{Dev} = \frac{(R_F^{op} - R_F^{Design})}{R_F^{Design}} 100$$

$$\varepsilon^{Dev} = \frac{(\varepsilon^{op} - \varepsilon^{design})}{\varepsilon^{design}} 100$$

Deviation Shell Pressure Drop

$$= 100 \times \left(\frac{\text{Shell Pressure Drop} - \text{Expected Shell Pressure Drop}}{\text{Expected Shell Pressure Drop}} \right)$$

Deviation Tube Pressure Drop

$$= 100 \times \left(\frac{\text{Tube Pressure Drop} - \text{Expected Tube Pressure Drop}}{\text{Expected Tube Pressure Drop}} \right)$$

20.1.7 Shell and Tube Heat Exchanger - Model Output

No.	Attribute	Units	Attribute Category	Monitoring Category	Source
Expected KPIs					
1	Expected Heat Duty	kW	Dynamic	Performance	Calculated
2	Expected Heat Flux	kW m ⁻²	Dynamic	Performance	Calculated
3	Expected Heat Transfer Coefficient	kW m ⁻² °C ⁻¹	Dynamic	Performance	Calculated
4	Expected Shell Flow Load	Kg hr ⁻¹	Dynamic	Performance	Calculated
5	Expected Tube Flow Load	Kg hr ⁻¹	Dynamic	Performance	Calculated
6	Design Effectiveness		Constant	Design	Calculated
7	Design Fouling Resistance		Constant	Design	Datasheet
8	Expected Heat Transfer Coefficient	kW m ⁻² °C ⁻¹	Dynamic	Performance	Calculated
9	Expected Shell Pressure Drop	bar	Dynamic	Performance	Calculated
10	Expected Tube Pressure Drop - bar - Dynamic - Performance - Calculated	bar	Dynamic	Performance	Calculated
Operating KPIs					
1	Operating Heat Duty	kW	Dynamic	Performance	Calculated
2	Operating Heat Flux	kW m ⁻²	Dynamic	Performance	Calculated
3	Operating Heat Transfer Coefficient	kW m ⁻² °C ⁻¹	Dynamic	Performance	Calculated
4	Shell Relative Flow Load	%	Dynamic	Performance	Calculated
5	Tube Relative Flow Load	%	Dynamic	Performance	Calculated
6	Heat Exchanger Load	%	Dynamic	Performance	Calculated
7	Log Mean Temperature Difference	°C	Dynamic	Performance	Calculated
8	Heat Exchanger Approach Temperature	°C	Dynamic	Performance	Calculated
9	Shell Side Heat Duty	kW	Dynamic	Performance	Calculated
10	Tube Side Heat Duty	kW	Dynamic	Performance	Calculated
11	Shell side temperature difference		Dynamic	Performance	Calculated
12	Tube side temperature difference		Dynamic	Performance	Calculated
13	Average heat duty		Dynamic	Performance	Calculated
14	Operating Fouling resistance		Dynamic	Performance	Calculated

No.	Attribute	Units	Attribute Category	Monitoring Category	Source
15	Operating Effectiveness		Dynamic	Performance	Calculated
Deviation KPIs					
1	Deviation Heat Duty	%	Dynamic	Performance	Calculated
2	Deviation Heat Flux	%	Dynamic	Performance	Calculated
3	Deviation Heat Transfer Coefficient	%	Dynamic	Performance	Calculated
4	Deviation LMTD		Dynamic	Performance	Calculated
5	Deviation Fouling Factor		Dynamic	Performance	Calculated
6	Deviation Effectiveness		Dynamic	Performance	Calculated
7	Deviation Shell Pressure Drop	bar	Dynamic	Performance	Calculated
8	Deviation Tube Pressure Drop	bar	Dynamic	Performance	Calculated
Heat Exchanger Efficiency					
1	Duty Efficiency Percent	%	Dynamic	Performance	Calculated
2	Heat Transfer Percent	%	Dynamic	Performance	Calculated
Energy Loss Information					
1	Design Heat Loss	kW	Dynamic	Performance	Calculated
2	Design Heat Loss Percent	%	Dynamic	Performance	Calculated
3	Operating Heat Loss	kW	Dynamic	Performance	Calculated
4	Operating Heat Loss Percent	%	Dynamic	Performance	Calculated
5	Total Duty Loss	kW	Dynamic	Performance	Calculated
6	Total Duty Loss Percent	%	Dynamic	Performance	Calculated
Fouling Information					
1	Shell Fouling Percent	%	Dynamic	Performance	Calculated
2	Tube Fouling Percent	%	Dynamic	Performance	Calculated
3	Heat Exchanger Overall Fouling Percent	%	Dynamic	Performance	Calculated
Pressure Drop Information					
1	Shell side Pressure Drop	bar	Dynamic	Performance	Calculated
2	Tube side Pressure Drop	bar	Dynamic	Performance	Calculated

20.2 Shell and Tube Heat Exchanger - Fault Model

20.2.1 Shell and Tube Heat Exchanger - APM Model

Below table illustrates the list of faults and symptoms configured for a Shell and Tube Heat Exchanger.

Fault Name	Fault Symptoms	Recommendations
COND PERF DEGRADATION ANOMALY	TubeSideHeatDuty SL (PDA)	1. Check sensor calibration. 2. Check all the previous fault models and act as per the action plan for that fault model
	ShellSideHeatDuty SL (PDA)	
	Avg HeatDuty SL (PDA)	
	DeviationHeatDuty LOW (PDA)	
	LMTD SL (PDA)	
	DeviationLMTD LOW (PDA)	
	OperatingEffectiveness SL (PDA)	
	DeviationEffectiveness LOW (PDA)	
	OperatingHTC SL (PDA)	
	DeviationHTC LOW (PDA)	
	OperatingFoulingResistance SH (PDA)	
	DeviationFoulingResistance HIGH (PDA)	
	HeatExchangerFoulingPercent SH (PDA)	
	OverallHTC NTU SL (PDA)	
	OverallHTC FR SL (PDA)	
	DeviationShellDP HIGH (PDA)	
	DeviationTubeDP HIGH (PDA)	

20.2.2 Shell and Tube Heat Exchanger - APR Model

20.2.2.1 Condenser Type

Below table illustrates the list of faults and symptoms configured for a Shell and Tube Heat Exchanger

Fault Name	Fault Symptoms	Recommendations
COND SHELL SIDE CRACK ANOMALY	Outlet2 Temperature RH (SSCA)	1. Check sensor calibration. 2. Inspect the insulation damage with hot spot or Shell CUI. 3. Inspect Nozzle integrity. 4. Inspect shell and piping support integrity. 5. Check the Manway gasket condition. Manway bolt tightening torques. Manway integrity. 6. Check PSV leakage
	TubeSideTemperatureDifference RH (SSCA)	
	Inlet1 MassFlow RL (SSCA)	
	ShellCondesateLevel1 RL (SSCA)	
	ShellCondesateLevel2 RL (SSCA)	
	ShellPressure RL (SSCA)	
COND SHELL TUBE FOULING ANOMALY	Outlet2 Temperature RH (STSFA)	1. Check sensor calibration. 2. Inspect the tubes for fouling at the inner/outer side and clean the exchanger if necessary. 3. Monitor the BFW Quality: conductivity, pH, oxygen level, phosphate injection to waste heat boiler steam drum
	TubePressureDrop RH (STSFA)	
	TubeSideTemperatureDifference RH (STSFA)	
	DeviationTubeDP HIGH (STSFA)	
COND TUBE SIDE CRACK ANOMALY	Outlet2 Temperature RL (TSCA)	1. Check sensor calibration. 2. Inspect the Tube integrity. 3. Monitor the BFW quality: pH, oxygen level, phosphate injection 4. Carryout RFT (NDT) test of tubes to detect flaw /thickness loss from inside of tube bore during shutdown 5. Clean or plug the tube if required
	TubePressureDrop RL (TSCA)	
	TubeSideTemperatureDifference RL (TSCA)	
	Inlet1 MassFlow RL (TSCA)	
	ShellPressure RL (TSCA)	
COND FOULING RISK ANOMALY	ShellSideFlowConductivity RH (CFRA)	1. Check sensor calibration. 2. Monitor the pH. 3. Monitor the Steam conductivity. 4. Monitor the H2S and SO2 levels in the process of gas 5. Monitor the lab reports of the condensate 6. Assess if the risk has contributed to any process upsets in any upstream or downstream assets in the line
	ShellSideFlowpH RL (CFRA)	
	ShellSideFlowpH RH (CFRA)	
	TubeWallThickness SL (CFRA)	
	BFWAnalysis PH SH (CFRA)	
	BFWAnalysis PH SL (CFRA)	
	BFWAnalysis Conductivity SH (CFRA)	
	BFWAnalysis Oxygen SL (CFRA)	
	WaterAnalysis Chloride SH (CFRA)	
	WaterAnalysis Fe SH (CFRA)	
	WaterAnalysis CO2 SH (CFRA)	

Fault Name	Fault Symptoms	Recommendations
	WaterAnalysis TDS SH (CFRA)	
	WaterAnalysis ResidualBiocides SH (CFRA)	
	WaterAnalysis Hardness SH (CFRA)	
	WaterAnalysis Alkalinity SH (CFRA)	
	CorrosionRate ShellSide RH (CFRA)	
	CorrosionRate TubeSide RH (CFRA)	
	CorrosionRate RBI SH (CFRA)	
	HotSpotTemperature SH (CFRA)	
	DeviationShellIDP HIGH (CFRA)	
COND SHELL SIDE INLET ANOMALY	Inlet1 Temperature RL (SSIA)	1. Check sensor calibration. 2. Assess if there is air entrainment during start-up and vent it if necessary. 3. Monitor the upstream assets of Steam / Water/Condensate 4. Inspect the integrity of upstream assets. pipelines. insulations. valves etc. 5. Check the upstream valves' internals. 6. Carry out loop proof test of the steam pressure control loop
	Inlet1 Temperature RH (SSIA)	
	Inlet1 Pressure RL (SSIA)	
	Inlet1 Pressure RH (SSIA)	
	Inlet1 MassFlow RL (SSIA)	
	Inlet1 MassFlow RH (SSIA)	
COND TUBE SIDE INLET ANOMALY	Inlet2 Temperature RL (TSIA)	1. Check sensor calibration. 2. Monitor the upstream assets of the process gas 3. Inspect integrity of upstream assets, pipelines, insulations, valves etc. 4. Inspect for Pipelines' CUI. 5. Check the valves' internals
	Inlet2 Temperature RH (TSIA)	
	Inlet2 Pressure RL (TSIA)	
	Inlet2 Pressure RH (TSIA)	
	Inlet2 MassFlow RL (TSIA)	
	Inlet2 MassFlow RH (TSIA)	
COND TUBE SIDE CORROSION RISK	TubeSideFlowConductivity RH (TSCR)	1. Check sensor calibration. 2. Monitor the pH 3. Monitor the conductivity 4. Monitor the H₂S and SO₂ level in the process gas. 5. Assess if the risk has contributed to any process upsets in any upstream or downstream assets in the line
	TubeSideFlowpH RH (TSCR)	
	ProcessH ₂ SDetection RH (TSCR)	
	ProcessSO ₂ Detection RH (TSCR)	
COND THERMAL	-	1. Check sensor calibration. 2. Reduce severity of cyclic operation via process changes.

Fault Name	Fault Symptoms	Recommendations
FATIGUE RISK TUBE		3. Inspection / replacement of damaged parts.
COND THERMAL FATIGUE RISK SHELL	-	1. Check sensor calibration. 2. Reduce severity of cyclic operation via process changes. 3. Inspection / replacement of damaged parts.
COND HIGH VELOCITY EROSION TUBE	-	1. Maintain fluid velocity within design limit and monitor during operation regularly
COND HIGH VELOCITY EROSION SHELL	-	1. Maintain fluid velocity within design limit and monitor during operation regularly
COND SULFIDATION RISK SHELL	CorrosionRateGeneral RH (SRS)	1. Process conditions should be monitored for any increase in temperature beyond the acceptable limit and/or changing H2S levels beyond the specified. 2. In opportunity shutdown carryout external ultrasonic thickness measurement and profile radiography to identify the metal loss /thinning of reheater tubes. 3. In opportunity shutdown carryout PMI of tubes and shell materials for alloy verification. This is recommended where sulfidic corrosion is involved. Si level in Carbon steel >0.1% 4. Carryout repair work in case of abnormality.
	CorrosionRateTLC RH (SRS)	
	SandErosionRate RH (SRS)	
COND H2 SULFIDATION RISK SHELL	DeviationinCorrosionRate HIGH (HSRS)	1. Evaluate if actual operating temperatures can be decreased for control of the corrosion rate, UT, VT and RT thickness readings are used to monitor loss in thickness of shell and tube.
COND ENV CRACK RISK TUBE	-	1. Evaluate the process conditions Process engineers and Corrosion/ materials specialists 2. Identify when the conditions are most likely to promote wet H2S damage and address it to prevent Wet H2S damage mechanism. 3. Avoid process cooling below dew point (condensation of the water vapors). 4. During shutdown opportunity, carryout visual inspection followed by applicable NDT tech techniques e.g WFMT, RT, UT, PT for crack detection
COND CREEP RISK TUBE	Inlet2 Temperature Static HIGH (CRT)	1. Visual inspection of elongated tubes, if any. 2. Reduce severity of high temperature operation
	Inlet2 Temperature Dynamic HIGH (CRT)	
	Outlet2 Temperature Static HIGH (CRT)	
	Outlet2 Temperature Dynamic	

Fault Name	Fault Symptoms	Recommendations
	HIGH (CRT)	
COND CREEP RISK SHELL	Inlet1 Temperature Static HIGH (CRS)	1. Visual inspection of elongated tubes, if any. 2. Reduce severity of high temperature operation
	Inlet1 Temperature Dynamic HIGH (CRS)	
	Outlet1 Temperature Static HIGH (CRS)	
	Outlet1 Temperature Dynamic HIGH (CRS)	
COND MECH FATIGUE SHELL	-	1. Check sensor calibration. 2. Maintain fluid velocity within design limit and monitor during operation regularly. 3. Inspect and measure the clearance between baffle hole and tube OD in opportunity shutdown /turnaround and repair if required.
COND MECH FATIGUE TUBE	-	1. Check sensor calibration. 2. NDE techniques such as visual and PT, should be carried out to detect surface cracks at all weld joints of tube-to-tube sheet during opportunity shutdown. 3. Repair if flaw identified.
COND TIME TO CLEAN	Time to Clean Estimation LOW (TTC)	1. Check the sensors' calibration. 2. Investigate the reason of fouling or rise in pressure difference across the Hex. 3. Plan to clean the HEx

20.2.2.2 Evaporator Type

Below table illustrates the list of faults and symptoms configured for a Shell and Tube Heat Exchanger

Fault Name	Fault Symptoms	Recommendations
EVAP SHELL SIDE CRACK ANOMALY	Outlet1 Temperature RH (SSCA)	1. Check sensor calibration. 2. Inspect the Shell integrity. 3. Inspect the insulation damage with hot spot or Shell CUI. 4. Inspect Nozzle integrity. 5. Inspect shell and piping support integrity. 6. Check Manway gasket condition, Manway bolt tightening torques, Manway integrity
	Outlet1 Pressure RL (SSCA)	
	Inlet1 MassFlow RL (SSCA)	
	ShellSideTemperatureDifference RH (SSCA)	
	ShellPressure RL (SSCA)	

Fault Name	Fault Symptoms	Recommendations
EVAP SHELL TUBE FOULING ANOMALY	TubePressureDrop RH (STSFA)	1. Check sensor calibration, 2. Inspect the tubes for fouling, and clean the exchanger, if necessary, 3. Inspect the tubes for fouling at inner/outer side, and clean the exchanger, if necessary, 4. Monitor the BFW quality: conductivity, pH, oxygen level, phosphate injection to waste heat boiler steam drum
	Outlet1 Temperature RL (STSFA)	
	ShellSideTemperatureDifference RL (STSFA)	
	DeviationTubeDP HIGH (STSFA)	
EVAP TUBE SIDE CRACK ANOMALY	Inlet2 MassFlow RL (TSCA)	1. Check sensor calibration. 2. Inspect the Tube integrity. 3. Clean or plug the tube if required
	Outlet1 Temperature RH (TSCA)	
	Outlet1 Pressure RH (TSCA)	
	Inlet1 MassFlow RH (TSCA)	
	ShellSideTemperatureDifference RH (TSCA)	
EVAP FOULING RISK ANOMALY	ShellSideFlowConductivity RH (CFRA)	1. Check sensor calibration. 2. Assess if the risk has contributed to any process upsets in any upstream or downstream assets in the line. 3. Monitor temperature during shutdowns or other event leading to temperature dropdown. 4. Carryout RFT (NDT) test of tubes to detect flaw /thickness loss from inside of tube bore during shutdown
	ShellSideFlowpH RL (CFRA)	
	ShellSideFlowpH RH (CFRA)	
	ProcessH2SDetection RH (CFRA)	
	ProcessSO2Detection RH (CFRA)	
	TubeWallThickness SL (CFRA)	
	CorrosionRate ShellSide RH (CFRA)	
	CorrosionRate RBI SH (CFRA)	
	HotSpotTemperature SH (CFRA)	
	DeviationShellIDP HIGH (CFRA)	

Fault Name	Fault Symptoms	Recommendations
EVAP SHELL SIDE INLET ANOMALY	Inlet1 Temperature RL (SSIA)	1. Check sensor calibration. 2. Monitor the upstream assets of process gas. 3. Inspect integrity of upstream assets, pipelines, insulations, valves etc. 4. Check the valves' internals. 5. Carry out loop proof test of the steam pressure control loop
	Inlet1 Temperature RH (SSIA)	
	Inlet1 Pressure RL (SSIA)	
	Inlet1 Pressure RH (SSIA)	
	Inlet1 MassFlow RL (SSIA)	
	Inlet1 MassFlow RH (SSIA)	
	ShelleSideControlValvePosition RL (SSIA)	
	ShelleSideControlValvePosition RH (SSIA)	
EVAP TUBE SIDE INLET ANOMALY	Inlet2 Temperature RL (TSIA)	1. Check sensor calibration. 2. Monitor the upstream assets of process gas. 3. Inspect integrity of upstream assets, pipelines, insulations, valves etc. 4. Check the valves' internals. 5. Carry out loop proof test of the steam pressure control loop
	Inlet2 Temperature RH (TSIA)	
	Inlet2 Pressure RL (TSIA)	
	Inlet2 Pressure RH (TSIA)	
	Inlet2 MassFlow RL (TSIA)	
	Inlet2 MassFlow RH (TSIA)	
	TubeSideControlValvePosition RL (TSIA)	
	TubeSideControlValvePosition RH (TSIA)	
EVAP TUBE CORROSION FOULING RISK	TubeSideFlowConductivity RH (TSCFR)	1. Check sensor calibration. 2. Monitor the pH. 3. Monitor the Steam conductivity. 4. Monitor the H2S and SO2 level in the process gas. 5. Monitor the lab reports of the condensate. 6. Assess if the risk has contributed to any process upsets in any upstream or downstream assets in the line
	TubeSideFlowpH RL (TSCFR)	
	TubeSideFlowpH RH (TSCFR)	
	TubeWallThickness SL (TSCFR)	
	BFWAnalysis PH SH (TSCFR)	
	BFWAnalysis PH SL (TSCFR)	
	BFWAnalysis Conductivity SH (TSCFR)	
	BFWAnalysis Oxygen SL (TSCFR)	
	WaterAnalysis Chloride SH (TSCFR)	
	WaterAnalysis Fe SH (TSCFR)	
	WaterAnalysis CO2 SH (TSCFR)	
	WaterAnalysis TDS SH	

Fault Name	Fault Symptoms	Recommendations
	(TSCFR)	
	WaterAnalysis ResidualBiocides SH (TSCFR)	
	WaterAnalysis Hardness SH (TSCFR)	
	WaterAnalysis Alkalinity SH (TSCFR)	
	CorrosionRate TubeSide RH (TSCFR)	
	CorrosionRate RBI SH (TSCFR)	
	HotSpotTemperature SH (TSCFR)	
EVAP THERMAL FATIGUE RISK TUBE	-	1. Reduce the severity of cyclic operation via process changes. 2. Inspection/replacement of damaged parts
EVAP THERMAL FATIGUE RISK SHELL	-	1. Reduce the severity of cyclic operation via process changes. 2. Inspection/replacement of damaged parts
EVAP HIGH VELOCITY EROSION TUBE	-	Maintain fluid velocity within design limit and monitor during operation regularly
EVAP HIGH VELOCITY EROSION SHELL	-	Maintain fluid velocity within design limit and monitor during operation regularly
EVAP SULFIDATION RISK SHELL	CorrosionRateGeneral RH (SRS)	1. Process conditions should be monitored for any increase in temperature beyond the acceptable limit and/or changing H2S levels beyond the specified. 2. In opportunity shutdown carry out external ultrasonic thickness measurement and profile radiography to identify the metal loss /thinning of reheater tubes. 3. In opportunity shutdown carryout PMI of tubes and shell materials for alloy verification. 4. This is recommended where sulfidic corrosion is involved. 5. Si level in Carbon steel >0.1%. 6. Carry out repair work in case of abnormality.
	CorrosionRateTLC RH (SRS)	
	SandErosionRate RH (SRS)	
EVAP H2 SULFIDATION RISK SHELL	DeviationinCorrosionRate HIGH (HSRS)	1. Evaluate if actual operating temperatures can be decreased for control of the corrosion rate. 2. UT, VT and RT thickness readings are used to monitor loss in thickness of shell and tube
EVAP ENV CRACK RISK	-	1. Evaluate the process conditions Process engineers and Corrosion/ materials specialists.

Fault Name	Fault Symptoms	Recommendations
SHELL		- Identify when the conditions are most likely to promote wet H₂S damage and address it to prevent Wet H₂S damage mechanism. - Avoid process cooling below dew point (condensation of the water vapors). - During shutdown opportunity, carryout visual inspection followed by applicable NDT tech techniques e.g WFMT, RT, UT, PT for crack detection
EVAP CREEP RISK TUBE	Inlet2 Temperature Static HIGH (CRT)	- Bias / drift in the sensor, - Visual inspection of elongated tubes, if any. - Reduce severity of high temperature operation
	Inlet2 Temperature Dynamic HIGH (CRT)	
	Outlet2 Temperature Static HIGH (CRT)	
	Outlet2 Temperature Dynamic HIGH (CRT)	
EVAP CREEP RISK SHELL	Inlet1 Temperature Static HIGH (CRS)	- Bias / drift in the sensor, - Visual inspection of elongated tubes, if any. - Reduce severity of high temperature operation
	Inlet1 Temperature Dynamic HIGH (CRS)	
	Outlet1 Temperature Static HIGH (CRS)	
	Outlet1 Temperature Dynamic HIGH (CRS)	
EVAP MECH FATIGUE SHELL	-	- Bias / drift in the sensor - Maintain fluid velocity within design limit and monitor during operation regularly. - Inspect and measure the clearance between baffle hole and tube OD in opportunity shutdown /turnaround and repair if required
EVAP MECH FATIGUE TUBE	-	- Bias / drift in the sensor. - NDE techniques such as visual and PT, should be carried out to detect surface cracks at all weld joints of tube-to-tube sheet during opportunity shutdown. - Repair if flaw identified
EVAP TIME TO CLEAN	Time to Clean Estimation LOW (TTC)	- Check the sensors' calibration. - Investigate the reason of fouling or rise in pressure difference across the Hex. - Plan to clean the Hex
EVAP TUBE SIDE OUT ANOMALY	DrumCondesateLevel1 RL (TSOA)	- Check sensor calibration. - Inspect Drum internals for any issue. - Inspect the drum level control valve.

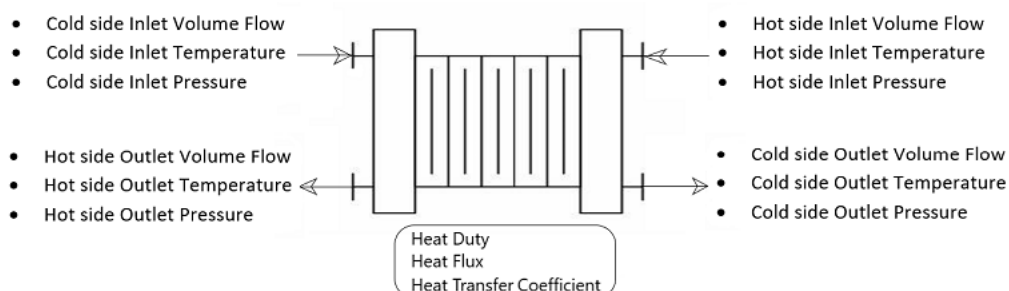
Fault Name	Fault Symptoms	Recommendations
		4. Conduct loop proof test for drum level control loop
	DrumCondesateLevel1 RH (TSOA)	
	DrumCondesateLevel2 RL (TSOA)	
	DrumCondesateLevel2 RH (TSOA)	

PLATE AND FRAME HEAT EXCHANGER

A heat exchanger is a device used to transfer thermal energy (enthalpy) between two or more fluids, between a solid surface and a fluid, or between solid particulates and a fluid, at different temperatures and in thermal contact.

- The proper design, operation, and maintenance of heat exchangers will make the process energy efficient and minimize energy losses. Heat exchanger performance can deteriorate with time, off design operations, and other interferences such as fouling, scaling, etc. It is necessary to assess periodically the heat exchanger's performance in order to maintain them at an optimum level.
- Monitoring basic machine health data in addition to operating conditions helps to maintain reliable operations of a heat exchanger.

Category	EQUIPMENT
Equipment Class	HEAT EXCHANGER
Equipment Type	PLATE AND FRAME HEAT EXCHANGER



21.1 Plate and Frame Heat Exchanger - Performance Monitoring

The following measured parameters are monitored as they have impact on performance and health of plate and frame heat exchanger, if not maintained in operating range.

21.1.1 Hot/Cold Side Inlet Temperature:

The inlet temperatures in the Heat Exchangers are considered as the load variable since it depends on the process or upstream equipment. The increase or decrease of inlet temperature will affect the heat transfer approach temperature and thereby affects the heat transfer effectiveness/percent.

21.1.2 Hot/Cold Side Mass/Volumetric flow rate:

The increase in the inlet flow rate of fluids results in increase in the heat duty and heat transfer coefficient but also increases the pressure drop of that particular stream.

21.1.3 Hot/Cold Side Inlet pressure:

The inlet Pressure, at the given inlet temperature, affects the phase fraction of the fluid and also affects the saturation conditions of the fluid.

21.1.4 Process Performance Overview:

Process performance monitoring view in APM provides a list of the asset attributes and their details, including the current value. The list can be used to review and update asset attributes information, as well as on the asset trend to view more details. It also highlights (in red color) the attributes from the high or low limits.

21.1.5 Notations

F	Mass Flow rate
P	Pressure
T	Temperature
Q	Heat duty
q	Heat flux
A	Heat Transfer Area
U	Heat transfer Coefficient
h	Enthalpy
p	Temperature Effectiveness
R	Heat capacity ratio
HEx_{load}	Heat exchanger load
ρ	Density
η	Heat transfer Efficiency
ϕ_v	Phase fraction - vapor
ΔT_{lnCor}	Corrected log mean temperature difference
ΔT_{LMTD}^{Design}	Design log mean temperature difference
R_{hf}	Hot side fouling
R_{cf}	Cold side fouling
R_F	Overall fouling

21.1.6 Subscripts

h	Hot side
c	Cold side
s	Shell side
t	Tube side
i	Inlet
o	Outlet

<i>Design</i>	Design specification
<i>sat</i>	Saturated conditions
<i>Ref</i>	Reference value

21.1.7 Superscript

<i>Ex</i>	Expected
<i>Op</i>	Operating
<i>Dev</i>	Deviation
<i>Design</i>	Design specification

21.1.8 Plate and Frame Heat Exchanger - Model Assumptions

1. There is no internal heat generation.
2. There is no leakage of fluid between the hot and cold sides (Inlet flow = outlet flow).
3. For the condensing steam system, incoming steam is assumed to be complete vapor either saturated or superheated. If not, it is corrected to the saturated vapor condition by doing $^P\phi_v(=1)$ flash.

$$h_{Sat}, \rho_{Sat}, T_{Sat} = f(P, \phi_{v=1})$$

4. For the evaporating water system, incoming feed is assumed to be completely liquid either saturated or subcooled. If not, it has to correct to the saturated liquid condition by doing $^P\phi_v(=0)$ flash.

$$h_{Sat}, \rho_{Sat}, T_{Sat} = f(P, \phi_{v=0})$$

21.1.9 Plate and Frame Heat Exchanger - Model Input

No.	Attribute	Units	Attribute Category	Monitoring Category	Source
MEASURED PARAMETERS					
1	Cold side Inlet Temperature	°C	Dynamic	Process	Field input
2	Cold side Outlet Temperature	°C	Dynamic	Process	Field input
3	Cold side Inlet Pressure	bar g	Dynamic	Process	Field input
4	Cold side Outlet Pressure	bar g	Dynamic	Process	Field input
5	Cold side Inlet Mass flow	Kg hr ⁻¹	Dynamic	Process	Field input
6	Cold side Inlet Volume flow	m ³ hr ⁻¹	Dynamic	Process	Field input
7	Hot side Inlet Temperature	°C	Dynamic	Process	Field input
8	Hot side Outlet Temperature	°C	Dynamic	Process	Field input
9	Hot side Inlet Pressure	bar g	Dynamic	Process	Field input
10	Hot side Outlet Pressure	bar g	Dynamic	Process	Field input
11	Hot side Inlet Mass flow	Kg hr ⁻¹	Dynamic	Process	Field input

No.	Attribute	Units	Attribute Category	Monitoring Category	Source
12	Hot side Inlet Volume flow	m ³ hr ⁻¹	Dynamic	Process	Field input
MODEL CONSTANTS					
1	Hot side Design Mass Flow	Kg hr ⁻¹	Constant	Design	Datasheet
2	Hot side Design Inlet Temperature	°C	Constant	Design	Datasheet
3	Hot side Design Outlet Temperature	°C	Constant	Design	Datasheet
4	Cold side Design Mass Flow	Kg hr ⁻¹	Constant	Design	Datasheet
5	Cold side Design Inlet Temperature	°C	Constant	Design	Datasheet
6	Cold side Design Outlet Temperature	°C	Constant	Design	Datasheet
7	Number of Passes		Constant	Design	Datasheet
8	Heat Exchanger Area		Constant	Design	Datasheet
9	Heat Transfer Coefficient - Reference In Service State	kW m ⁻² C ⁻¹	Constant	Design	Datasheet
10	Design Heat Duty	kW	Constant	Design	Datasheet
11	Design fouling resistance		Constant	Design	Datasheet

21.1.10 Plate and Frame Heat Exchanger - KPI

21.1.10.1 Operating KPIs

1. Operating Heat Duty
2. Operating Heat Flux
3. Operating Heat Transfer Coefficient

21.1.10.2 Expected KPIs

1. Expected Heat Duty
2. Expected Heat Flux
3. Expected Heat Transfer Coefficient

21.1.10.3 Deviation KPIs

1. Deviation Heat Duty
2. Deviation Heat Flux
3. Deviation Heat Transfer Coefficient

21.1.10.4 Heat Exchanger Efficiency

1. Duty Efficiency %
2. Heat Transfer Efficiency %

21.1.10.5 Energy Loss Parameters

1. Duty Loss due to design
2. Duty Loss due to design %
3. Duty Loss due to operation
4. Duty Loss due to operation %
5. Total Duty Loss
6. Total Duty Loss %

21.1.10.6 Heat Exchanger Health Parameters

1. Hot side Fouling %
2. Cold side Fouling %
3. Overall Heat Exchanger Fouling %

21.1.10.7 Pressure drop information

1. Hot side Pressure drop
2. Cold side Pressure drop

21.1.10.8 Plate and Frame Heat Exchanger - Performance Modeling***Overall Mass Balance***

For Hot side,

$$F_{s_i} = F_{s_o}$$

For Cold side,

$$F_{t_i} = F_{t_o}$$

Overall Force Balance

For Hot side,

$$P_{s_i} = P_{s_o} + \Delta P_s$$

For Cold side,

$$P_{t_i} = P_{t_o} + \Delta P_t$$

Overall Energy Balance

For hot side,

$$F_h h_{hi}(P_{hi}, T_{hi}, z_{hij}) = F_h h_{ho}(P_{ho}, T_{ho}, z_{hoj}) + Q_h$$

For cold side,

$$F_c h_{ci}(P_{ci}, T_{ci}, z_{cij}) + Q_c = F_c h_{co}(P_{co}, T_{co}, z_{coj})$$

21.1.10.9 Plate and Frame Heat Exchanger - Expected KPIs

Expected Heat Duty

Expected Heat Duty is the product of design duty (taken from the datasheet) and heat exchanger load. The Heat Exchanger load is the ratio of Design LMTD (calculated based on design temperatures) and Operating LMTD (calculated based on operating temperatures).

$$\Delta T_1^{Design} = T_{h_i}^{Design} - T_{c_o}^{Design}$$

$$\Delta T_2^{Design} = T_{h_o}^{Design} - T_{c_i}^{Design}$$

$$SQ_{Design} = \sqrt{\left(T_{h_i}^{Design} - T_{h_o}^{Design}\right)^2 - \left(T_{c_o}^{Design} - T_{c_i}^{Design}\right)^2}$$

$$LN_{design} = \ln \left(\left| \frac{\Delta T_1^{Design} + \Delta T_2^{Design} + SQ_{Design}}{\Delta T_1^{Design} + \Delta T_2^{Design} - SQ_{Design}} \right| \right)$$

$$\Delta T_{LMTD}^{Design} = \frac{SQ_{Design}}{LN_{design}}$$

$$T_{load} = \frac{\Delta T_{LMTD}^{Design}}{\Delta T_{lnCor}}$$

$$Heat\ Exchanger\ Load = HEx_{load} = T_{load}$$

$$Expected\ Heat\ Duty = Q^{Ex} = Q_{Design} HEx_{load}$$

Expected Heat Flux

The Expected Heat flux is the ratio of Expected heat duty and the heat exchanger area. It gives us an idea of the expected heat transfer rate irrespective of area.

$$Expected\ Heat\ Flux = q^{Ex} = \frac{Q^{Ex}}{A}$$

Expected Heat Transfer Coefficient

The Expected heat transfer coefficient is the product of the heat transfer coefficient - Reference in service (taken from the datasheet) and the heat exchanger load.

$$Expected\ Heat\ Transfer\ Coefficient = U^{Ex} = U_{Ref} HEx_{load}$$

Expected Design Effectiveness

Design effectiveness is calculated based on design temperatures,

$$\varepsilon^{design} = \text{Max}(p^{design}, p^{design} R^{design})$$

Expected Design Fouling Resistance

Design fouling resistance from the datasheet needs to be provided.

$$R_F^{Design}$$

21.1.10.10 Plate and Frame Heat Exchanger - Operating KPIs**Operating Heat Duty**

Heat Duty is the amount of energy transferred from hot fluid to cold fluid within the exchanger. It is calculated as the product of mass flow and the mass enthalpy difference of inlet and outlet for both sides of the exchanger and minimum duty is taken as the operating heat duty of the heat exchanger. Since we are calculating duty using enthalpy, it is applicable for phase change exchangers such as condensers or evaporators.

$$F_h h_{hi}(P_{hi}, T_{hi}, z_{hi}) = F_h h_{ho}(P_{ho}, T_{ho}, z_{ho}) + Q_h$$

$$F_c h_{ci}(P_{ci}, T_{ci}, z_{ci}) + Q_c = F_c h_{co}(P_{co}, T_{co}, z_{co})$$

$$\text{Operating Heat duty} = Q^{Op} = \text{Min}(Q_h, Q_c)$$

Operating Heat Flux

The Operating Heat flux is the ratio of Operating heat duty and the heat exchanger area. It gives us an idea of the operating heat transfer rate irrespective of area.

$$\text{Operating Heat flux} = q^{Op} = \frac{Q^{Op}}{A}$$

Operating Heat Transfer Coefficient

The heat transfer coefficient or film coefficient is the proportionality constant between the heat flux and the thermodynamic driving force for the flow of heat (i.e., the temperature difference, ΔT). The overall heat transfer rate for combined modes is usually expressed in terms of an overall conductance or heat transfer coefficient, U. The Logarithmic Mean Temperature (LMTD) is the function of inlet and outlet temperatures of both sides and the number of Passes.

$$\text{Temperature Effectiveness} = p = \frac{T_{co} - T_{ci}}{T_{hi} - T_{ci}}$$

$$\text{Heat Capacity Ratio} = R = \frac{T_{hi} - T_{ho}}{T_{co} - T_{ci}}$$

$$\alpha = \left(\frac{Rp - 1}{p - 1} \right)^{\left(\frac{1}{N} \right)}$$

N = Number of passes

$$X = \frac{1 - \alpha}{R - \alpha}$$

$$\text{Correction Factor} = F_t = \frac{(R^2 + 1)^{0.5} \ln \frac{1-X}{1-RX}}{(R-1) \ln \frac{2-X(R+1)-(R^2+1)^{0.5}}{2-X(R+1)+(R^2+1)^{0.5}}}$$

$$\Delta T_1 = T_{hi} - T_{co}$$

$$\Delta T_2 = T_{ho} - T_{ci}$$

$$SQ = \sqrt{|(T_{hi} - T_{ho})^2 - (T_{co} - T_{ci})^2|}$$

$$LN = \ln \left(\left| \frac{\Delta T_1 + \Delta T_2 + SQ}{\Delta T_1 + \Delta T_2 - SQ} \right| \right)$$

$$\text{Operating LMTD} = \Delta T_{lnCor} = Ft \frac{SQ}{LN}$$

$$\text{Operating Heat Transfer Coefficient} = U^{Op} = \frac{\text{Min}(Q_h, Q_c)}{A \Delta T_{lnCor}}$$

Heat Exchanger Approach Temperature

The higher the heat exchanger approach temperature we achieve, the higher will be the heat transfer for the constant heat exchanger area.

$$\text{Approach temperature} = T_{hi} - T_{co}$$

Heat exchanger efficiency

The Heat transfer efficiency is the ratio of actual heat transfer to the maximum possible heat transfer. One more indicator which will say about the heat exchanger efficiency is the duty efficiency percentage. It is the ratio of minimum heat duty to maximum heat duty and here, we are focusing on the actual heat energy received by the cold fluid apart from the heat losses.

$$\text{Duty efficiency Percent} = Q \% = \frac{\text{Min}(Q_h, Q_c)}{\text{Max}(Q_h, Q_c)} 100$$

$$\text{Heat Transfer Percent} = \eta = \text{Max}(100p, 100pR)$$

Energy Loss parameters

The total energy loss is the sum of energy losses due to design and the energy losses due to operation or fouling. There are some heat losses to the surrounding which are considered as the design heat losses and heat loss due to fouling is considered as the operational heat losses.

Energy losses due to design:

$$Q_{Loss}^{Design} \% = 100 - Q \%$$

$$Q_{Loss}^{Design}(kW) = \max(Q_h, Q_c) \frac{Q_{Loss}^{Design}\%}{100}$$

Energy losses due to operation or fouling:

$$Q_{Loss}^{Fouling}\% = 100 \left(1 - \frac{\min(U_h^{Op}, U_c^{Op})}{U^{Ex}} \right)$$

$$Q_{Loss}^{Fouling}(kW) = \max(Q_h, Q_c) \frac{Q_{Loss}^{Fouling}\%}{100}$$

Total Energy Loss:

$$Q_{Loss}^{Total}(kW) = Q_{Loss}^{Design} + Q_{Loss}^{Fouling}$$

$$Q_{Loss}^{Total}\% = Q_{Loss}^{Design}\% + Q_{Loss}^{Fouling}\%$$

Heat exchanger health parameters

Heat exchanger performance can deteriorate with time, off-design operations, and other interferences such as fouling, scaling, etc. It is necessary to assess periodically the heat exchanger's health in order to maintain them at an optimum level. Fouling causes a decrease in surface area reducing the heat transfer coefficient.

$$\text{Hot side Heat transfer coefficient} = U_h = \frac{Q_h}{A \Delta T_{lnCor}}$$

$$\text{Cold side Heat transfer coefficient} = U_c = \frac{Q_c}{A \Delta T_{lnCor}}$$

Hot side Fouling %:

If,

$$U_h = U^{Op}$$

$$R_{hf}\% = 100 \left(1 - \frac{U_h}{U^{Ex}} \right)$$

Otherwise

$$R_{hf}\% = 100 \left(1 - \frac{U_h Q\%}{U^{Ex} 100} \right)$$

Cold side Fouling %:

If,

$$U_c = U^{Op}$$

$$R_{cf}\% = 100 \left(1 - \frac{U_t}{U^{Ex}} \right)$$

Otherwise

$$R_{tf}\% = 100 \left(1 - \frac{U_c Q \%}{U^{Ex} 100} \right)$$

Heat Exchanger Overall Fouling %:

$$R_F\% = 100 \frac{U^{Ex} - U^{op}}{U^{Ex}}$$

Pressure drop

The pressure drop is the difference between the inlet and outlet pressures of the individual sides. The increase in pressure drop on a side indicates the fouling at that particular side of the heat exchanger.

$$\text{Hot side Pressure drop} = \Delta P_s = P_{si} - P_{so}$$

$$\text{Cold side Pressure drop} = \Delta P_t = P_{ti} - P_{to}$$

Hot side and Cold side Temperature Difference

Hot side identification

$$\Delta T_{hot} = T_{hi} - T_{ho}$$

Cold side identification

$$\Delta T_{cold} = T_{co} - T_{ci}$$

Average Heat Duty

$$Q_{avg} = \frac{Q_h + Q_c}{2}$$

Operating Fouling Factor

Fouling factor as an inverse of HTC

$$R_F^{op} = \frac{1}{U^{Ex}} - \frac{1}{U^{op}}$$

Operating Effectiveness

$$\varepsilon^{op} = \text{Max}(p, p R)$$

21.1.10.11 Plate and Frame Heat Exchanger - Deviation KPIs

The deviation KPIs such as Deviation Duty, flux, and Heat transfer coefficient represent the deviation of operating KPI with expected KPI at current operating conditions.

$$\text{Deviation Heat Duty} = Q^{Dev} = \frac{Q^{Op} - Q^{Ex}}{Q^{Ex}} 100$$

$$\text{Deviation Heat Flux} = q^{Dev} = \frac{q^{Op} - q^{Ex}}{q^{Ex}} 100$$

$$\text{Deviation Heat Transfer Coefficient} = U^{Dev} = \frac{U^{Op} - U^{Ex}}{U^{Ex}} 100$$

$$\Delta T_{LMTD}^{Dev} = \frac{(\Delta T_{InCor} - \Delta T_{LMTD}^{Design})}{\Delta T_{LMTD}^{Design}} 100$$

$$R_F^{Dev} = \frac{(R_F^{Op} - R_F^{Design})}{R_F^{Design}} 100$$

$$\varepsilon^{Dev} = \frac{(\varepsilon^{Op} - \varepsilon^{Design})}{\varepsilon^{Design}} 100$$

21.1.11 Plate and Frame Heat Exchanger - Model Output

No.	Attribute	Units	Attribute Category	Monitoring Category	Source
Expected KPIs					
1	Expected Heat Duty	kW	Dynamic	Performance	Calculated
2	Expected Heat Flux	kW m ⁻²	Dynamic	Performance	Calculated
3	Expected Heat Transfer Coefficient	kW m ⁻² C ⁻¹	Dynamic	Performance	Calculated
4	Expected hot Flow Load	Kg hr ⁻¹	Dynamic	Performance	Calculated
5	Expected cold Flow Load	Kg hr ⁻¹	Dynamic	Performance	Calculated
6	Design Effectiveness		Constant	Design	Calculated
7	Design Fouling Resistance		Constant	Design	Datasheet
Operating KPIs					
1	Operating Heat Duty	kW	Dynamic	Performance	Calculated
2	Operating Heat Flux	kW m ⁻²	Dynamic	Performance	Calculated
3	Operating Heat Transfer Coefficient	kW m ⁻² C ⁻¹	Dynamic	Performance	Calculated
4	Hot side Relative Flow Load	%	Dynamic	Performance	Calculated
5	Cold side Relative Flow Load	%	Dynamic	Performance	Calculated

No.	Attribute	Units	Attribute Category	Monitoring Category	Source
6	Heat Exchanger Load	%	Dynamic	Performance	Calculated
7	Log Mean Temperature Difference	°C	Dynamic	Performance	Calculated
8	Duty Efficiency Percent	%	Dynamic	Performance	Calculated
9	Heat Transfer Percent	%	Dynamic	Performance	Calculated
10	Heat Exchange Approach Temperature	°C	Dynamic	Performance	Calculated
11	Hot Side Heat Duty	kW	Dynamic	Performance	Calculated
12	Cold Side Heat Duty	kW	Dynamic	Performance	Calculated
13	Hot side temperature difference		Dynamic	Performance	Calculated
14	Cold side temperature difference		Dynamic	Performance	Calculated
15	Average heat duty		Dynamic	Performance	Calculated
16	Operating Fouling resistance		Dynamic	Performance	Calculated
17	Operating Effectiveness		Dynamic	Performance	Calculated
Deviation KPIs					
1	Deviation Heat Duty	%	Dynamic	Performance	Calculated
2	Deviation Heat Flux	%	Dynamic	Performance	Calculated
3	Deviation Heat Transfer Coefficient	%	Dynamic	Performance	Calculated
4	Deviation LMTD		Dynamic	Performance	Calculated
5	Deviation Fouling Factor		Dynamic	Performance	Calculated
6	Deviation Effectiveness		Dynamic	Performance	Calculated
Energy Loss Information					
1	Design Heat Loss	kW	Dynamic	Performance	Calculated
2	Design Heat Loss Percent	%	Dynamic	Performance	Calculated
3	Operating Heat Loss	kW	Dynamic	Performance	Calculated
4	Operating Heat Loss Percent	%	Dynamic	Performance	Calculated
5	Total Duty Loss	kW	Dynamic	Performance	Calculated
6	Total Duty Loss Percent	%	Dynamic	Performance	Calculated
Fouling Information					
1	Hot side Fouling Percent	%	Dynamic	Performance	Calculated
2	Cold side Fouling Percent	%	Dynamic	Performance	Calculated
3	Heat Exchanger Overall Fouling Percent	%	Dynamic	Performance	Calculated
Pressure Drop Information					
1	Hot side Pressure Drop	bar	Dynamic	Performance	Calculated
2	Cold side Pressure Drop	bar	Dynamic	Performance	Calculated

21.2 Plate and Frame Heat Exchanger - Fault Model

21.2.1 Plate and Frame Heat Exchanger - APM Model

Below table illustrates the list of faults and symptoms configured for a Plate and Frame Heat Exchanger.

Fault Name	Associated Symptoms	Recommendations
Performance Low Warning	Cold Side Outlet Temperature Low	1. Check inlet and outlet temperatures of shell and Tube side. 2. Refer Operational Dashboard. 3. Check for Tube Outlet Temperature indications. 4. Refer Exchanger Homepage Graphic.
	Hot Side Outlet Temperature High	
	Operating Duty Low	
	Operating HTC Low	

21.2.2 Plate and Frame Heat Exchanger - APR Model

Below table illustrates the list of faults and symptoms configured for a Plate and Frame Heat Exchanger.

Fault Name	Associated Symptoms	Recommendations
PHE Fouling Warning	Cold Side Pressure Drop High	1. Check for Pressure indications on Shell Side. 2. Exchanger may be severely fouled and Check Cleaning Cycle. 3. Check for Pressure indications on Tube Side.
	Fouling Percentage High	
	Hot Side Pressure Drop High	

ELECTRIC ROPE SHOVEL

More efficient than diesel powered machines, electric rope shovels are entirely electric and therefore they experience less heat loss. And, they have available dipper payloads of up to 109 tonnes (120 tons).

Electric Rope Shovels use regenerative braking technology to convert kinetic energy from shovel motions into electrical energy when braking. The electrical energy that is generated is then fed back to the grid. Without regenerative braking, the kinetic energy would be burned off as heat.

Electric Rope Shovel have fewer component change-outs that result in less waste. Component change-out intervals for electric rope shovels are generally longer than those for similar-sized hydraulic machines. This saves you money and reducing waste in the environment, electric rope shovel motors and gear cases can be rebuilt.

22.1 Electric Rope Shovel - Fault Model

22.1.1 Electric Rope Shovel - APR Model

22.1.1.1 Electric Rope Shovel-Cooling System

Model Type	APR Model
Category	Mining Asset
Equipment Class	Mining Asset
Equipment Type	Electric Rope Shovel-Cooling System

Below table illustrates the list of faults and symptoms configured for Electric Rope Shovel-Cooling System

Fault Name	Associated Symptoms	Recommendations
ERS_Abnormal_Operations_Planetary_GB	ERS_Load_Capacity_Bucket_High	Cause: 1. Excess speed of prime mover 2. Improper installation 3. Misalignment between prime mover and gearbox creates noise 4. Damaged bearing 5. High input speed 6. Low viscosity oil 7. Overload 8. Improper backlash Effects:

Fault Name	Associated Symptoms	Recommendations
		1. Improper installation of gearbox which results in loose bolts of mounting
	ERS_Torsional_Force_High	2. Increase noise
	ERS_Viscosity_Oil_Low	3. Brakes the output shaft
	ERS_Oil_Pump_Inlet_Low	4. Worn out gear and pinion 5. Damages sun and star gears Recommendation: 1. Check the rate of heat dissipation 2. Lubricants circulated through a heat exchanger 3. Temperature of Gear Box
ERS_Weak_performance_Stick_Crowding	ERS_Oil_Level_Low	Cause: 1. Either stalls the equipment (heavy penetration) or jacks the boom backwards which is detrimental to both boom and boom suspension ropes (cables) due to heavy vibration caused 2. Hard rock digging ability 3. Weak electric motor power 4. Low electricity voltage 5. Excess temperature of Planetary Gearbox Effects: 1. Rock digging ability 2. Rock swell factor 3. Dipper fill factor 4. Swing angle 5. Position of ERS with respect to bank 6. Boom jacking generates severe vibration Recommendation: 1. Check the alignment of Planetary Gear Box and Prime Mover 2. Foundation bolts of Gearbox 3. Voltage of electricity 4. Output speed of Motor and Gearbox 5. Temperature 6. Oil level 7. Pressure of Oil
	ERS_Speed_Prime_Mover_High	
	ERS_Heat_Dissipation_Rate_Low	
	ERS_PitchBrace_Lengths_Low	
	ERS_Rake_AndTooth_Angle_Dipper_Low	
	ERS_SwingAngle_Low	

Fault Name	Associated Symptoms	Recommendations
ERS_Abnormal_Transmission_Swing_Operation	ERS_Oil_Level_Low	Cause: 1. Low level of Oil 2. The compact footprint of planetary gears makes heat dissipation more difficult 3. Planetary gearbox, the motor and the driven equipment must be in line with each other, although manufacturers offer right-angle designs that incorporate other gearsets Effects: 1. Helical or Spur gear damages 2. Bearing seizes 3. Low torque transmission 4. Low output of axial load 5. Damages of internal components of pumps Recommendation: 1. Check oil level and temperature of oil 2. Alignment of prime mover and gearbox 3. Proper bearing mounting and loads
	ERS_Oil_Temperature_High	
	ERS_Axial_and_Radial_Load_High	
	ERS_Lubricating_Oil_Pressure_Low	
	ERS_Electric_MTR_Speed_Low	
	ERS_OilInSpiiterBoxOverheat_Power_Pack_1_Alarm_ON	
	ERS_OilInSpiiterBoxOverheat_Power_Pack_2_Alarm_ON	

22.1.1.2 Electric Rope Shovel-Electric System

Model Type	APR Model
Category	Mining Asset
Equipment Class	Mining Asset
Equipment Type	Electric Rope Shovel-Electric System

Below table illustrates the list of faults and symptoms configured for Electric Rope Shovel-Electric System

Fault Name	Associated Symptoms	Recommendations
Transmission_ Modules_Working_ anomaly	ERS_voltage_flow_Low	Cause: 1. The existence of harmonics indicates that electrical distribution is disturbed and power quality is not optimum. 2. The presence of harmonics in a system results in excess heat being produced in equipment which can often result in premature component failures. 3. An arc flash is the result of a breakdown of the air between two or more conductors Effect: 1. Large amounts of energy being released in the form of an explosion. 2. Harmonics indicates that electrical distribution is disturbed and power quality is not optimum Recommendation: 1. Check the large amounts of energy being released in the form of an explosion. 2. With software package enable the user to perform an Arc Flash Hazard Assessment of particular equipment in accordance with the IEEE 1584 or NFPA 70E standards. Along with determining the incident energy
	ERS_Fluctn_rate_current_Low	
	ERS_Bearing_Temperature_High	
	ERS_crowd_armut_volt_Low	
	ERS_sys_lube_OFF	
AC_IGBT_Acutrol_ drive_system_ Anomaly	ERS_voltage_flow_IGBT_Low	Cause: 1. High power drives in mining equipment cause degradation of power quality in the power network due to harmonics, long-distance cables, continuously changing power distribution topology and high power pulsating loads 2. Voltage drop affects the shovel's operation triggering drives protections affecting severely mining production Effect: 1. Affects the shovel's operation triggering drives protections affecting severely mining production Recommendation: 1. To perform the harmonic analysis of the Hail Creek Mine power system with accurate harmonic models for the DC and AC shovels. 2. Identify the magnitude and phase angle of the voltage at each bus and determines the active and reactive power flowing in each line with SKM power tool
	ERS_Specf_power_consumn_Low	
	ERS_Fatigue_Resist_High	
	ERS_Bearing_Temperature_High	
	ERS_sys_lube_OFF	
Electric_Motors_ Performance_ Anomaly	ERS_payload_dipper_High	Cause: Voltage fluctuations and voltage drop. It was determined that the voltage drop under normal operating conditions should be no more than 7% and under starting conditions should be no

Fault Name	Associated Symptoms	Recommendations
	ERS_Peck_power_supply_Low	more than 35%. Perform load flow studies of the proposed arrangement of the shovels as well as the option of connecting both shovels to the transportable substation of 4100XPC AC Shovel. Effect: 1. High power drives in mining equipment cause degradation of power quality in the power network due to harmonics Load flow and the average and peak loadings could be affected Recommendation: 1. Check voltage fluctuations and connections of Shovel and Substations 2. Identify the magnitude and phase angle of the voltage at each bus and determine the active and reactive power flowing in each line 3. Check Automatic lubrication system covers all regular lube points, pressure, contamination, and pump condition
	ERS_loading_rate_High	
	ERS_mucking_rate_High	
	ERS_equivnt_digg_energy_High	

22.1.1.3 Electric Rope Shovel- Lower Work System Fault Model

Model Type	APR Model
Category	Mining Asset
Equipment Class	Mining Asset
Equipment Type	Electric Rope Shovel- Lower Work

Below table illustrates the list of faults and symptoms configured for an Electric Rope Shovel- Lower Work System

Fault Name	Associated Symptoms	Recommendations
ERS_Turn_Table_Movement_Anomaly	ERS_Swing_Armatr_Volt_Low	Cause: 1. Swing Motors not working properly 2. Rack and Pinion worn out 3. Main Shaft Loose 4. Straddle-mounted rollers damages 5. Planetary drive worn out 6. Roller Bearing damages Effects: 1. Weak swinging 2. Load stalling and Poor movement of turn table with load 3. Noisy Movement 4. Roller Bearing Temperature increases

Fault Name	Associated Symptoms	Recommendations
	ERS_Swin_MTR_Speed_Low	Recommendation: 1. Check the rack and pinion, Main Shaft, Rollers and bearings, Swing Motors and Propel Motors, Voltage of Power supply
	ERS_Voltage_Electric_Supply_Low	
	ERS_Digging_Force_High	
	ERS_Payload_High	
	ERS_Vibration_Energy_High	
ERS_Swing_Hoist_Motor_Anomaly	ERS_Bearin_Temp_High	Cause: 1. Imbalance of Rotor 2. Shaft Misalignment 3. Looseness 4. Bearing Faults 5. Stator Issues 6. Broken Rotor Bars 7. Improper Assembly
	ERS_Payload_High	
	ERS_Transmission_Modulan_Supply_Pressure_High	
	ERS_Hoist_Armatur_Volt_Low	
	ERS_Main_Power_Supply_Rate_Low	
		Effects: 1. Damages rotor 2. Shaft 3. Weak performance of Swing and Hoist motors 4. Low productivity 5. Bearing seizes Recommendation: Check: 1. Balance of Rotor Shaft alignment 2. Any Looseness 3. Bearing Faults 4. Stator Issues 5. Broken Rotor Bars
ERS_One_side_Non_working_Crawler	ERS_Torque_Output_Low	Cause: 1. Broken plate in clutch pack 2. Transmission Pressure and Response time uneven 3. Low Hydraulic fluid level 4. Failure of one side electric motor, weak recoil spring, failure of rear-drive, loose triple crawler shoes Effects:

Fault Name	Associated Symptoms	Recommendations
		1. Loose the track and possibility of coming out, sprockets splice worn out,
	ERS_Transmission_ Output_Low	2. Failure of one side hydraulic pumps, weak engine working leads to providing less power to pump, Flexible drive coupling rubber segments worn out.
	ERS_Hydrostatic_ TransMTR_Dispt_ Solenoid_OFF	Recommendation: Check:
	ERSTransmission_ Lock_Switch_OFF	1. The plates in clutch pack 2. Transmission and response pressure 3. internal leakages at torque convertor or transmission clutch packs 4. hydraulic fluid level 5. Working condition of both side hydraulic pumps, Motors, Sprockets and Recoil spring, Drive couplings
ERS_Crawler_ Movement_Anomaly	ERS_Bearin_Temp_ High	Cause: 1. Low power supply to Motors, 2. Propel Motors not working properly, 3. Sprocket and Recoil spring weak, 4. Uneven pit floor, 5. Damages the Track, shoes of links, damages of Track Rollers,
	ERS_Swing_Armatr_ Volt_Low	Effects: 1. Slow movement of Shovel 2. Sprocket and Recoil spring damages 3. Track and shoes of link worn out or loose
	ERS_Swin_MTR_ Speed_Low	Recommendation: Check:
	ERS_Voltage_ Electric_Supply_Low	1. The voltage of Power supply 2. Propel Motors 3. Sprocket and Recoil Springs 4. Track shoes and Rollers 5. Rack and Pinion
	ERS_Voltage_Propel_ MTR_Low	

22.1.1.4 Electric Rope Shovel- Steering System Fault Model

Model Type	APR Model
Category	Mining Asset
Equipment Class	Mining Asset

Model Type	APR Model
Equipment Type	Electric Rope Shovel- Steering System

Below table illustrates the list of faults and symptoms configured for an Electric Rope Shovel- Steering System

Fault Name	Associated Symptoms	Recommendations
ERS_Loss_control_over_the_steering_Mechanism	ERS_Hydraulic_Pump_Pressure_Low	Cause: 1. Problems with Electric Cylinder 2. Double acting cylinder 3. Linear Magnetic Motors 4. Various electric circuits 5. Two steering members must be in clockwise direction and same must be retracted in the same manner 6. Proper pressure in working line Effects: 1. Low pressure in working line 2. Four-way valve and hydraulic pump not supply hydraulic to reservoir 3. relief valve 4. loss of control of four-way valve so it will not supply at certain time hydraulic pressure to four cylinders Recommendation: Check: 1. The pressure of hydraulic 2. Leakage of pressure line 3. Air contamination of oil 4. Contaminated oil 5. Pressure relief valve 6. Four-way valve, reservoir 7. Actuating valve 8. Retracting piston working in proper condition.
	ERS_Left_Steering_Solenoid_OFF	
	ERS_Left_Track_Steering_Lever_Position_Sensor_OFF	
	ERS_Right_Track_Steering_Lever_Position_Sensor_OFF	
	ERS_Left_Steering_Clutch_Solenoid_OFF	
	ERS_Right_Steering_Clutch_Solenoid_OFF	
ERS_Weak_Response_Steering_or_Difficult_to_Steer	ERS_Steering_Mode_Switch_OFF	Cause: 1. Loose or disconnected steering or loader connections, 2. Faulty entire joystick control circuit board, 3. Safety switches malfunction, 4. Bad circuits can also harm control system Effects: 1. Leakages of fittings and seals on the control valves.

Fault Name	Associated Symptoms	Recommendations
	ERS_Hydraulic_Pump_Pressure_Low	2. Slow response of Steering 3. Joystick controls for the skid steer's steering and loader aren't responding to their directions Recommendation: Check: 1. The wires connecting the actuators and hydraulic pump to the joystick to ensure they're attached appropriately 2. Entire joystick control circuit board 3. Safety switches
	ERS_Rear_Steering_Manual_Switch_OFF	
	ERS_Left_Track_Steering_Lever_Position_Sensor_OFF	
	ERS_Steering_Pump_Oil_Pressure_Low	
	ERS_Steering_Oil_Filter_Differential_Pressure_Switch_OFF	
ERS_Anomaly_operatio_Right_Turn_or_Left_Turn	ERS_Hydraulic_Pump_Pressure_Low	Cause: 1. Break down of hydraulic pumps 2. Malfunction of transfer valve Malfunctioning of hydraulic by pass valve, 3. In the pump Oil leakages 4. Problem in tele control system Effects: 1. Linkage between the tiller and rudder angle transmitter detached. 2. The steering gear continued to operate normally, but the transmitter lost its input signal and, as a result, the bridge mounted rudder angle indicator stopped working. Recommendation: Check 1. Hydraulic pump 2. Oil pressure 3. Relief valve 4. Four-way valve 5. Hydraulic oil level
	ERS_Right_Steering_Solenoid_OFF	
	ERS_Left_Steering_Solenoid_OFF	
	ERS_Right_Side_Steering_Manifold_Solenoid_OFF	
	ERS_Left_Side_Steering_Manifold_Solenoid	
	ERS_Steering_Pump_Oil_Pressure_Low	
	ERS_Steering_Oil_Level_Switch_OFF	
	ERS_Left_Steering_Clutch_Solenoid_OFF	

22.1.1.5 Electric Rope Shovel- Transmission System Fault Model

Model Type	APR Model
Category	Mining Asset
Equipment Class	Mining Asset
Equipment Type	Electric Rope Shovel- Transmisson System

Below table illustrates the list of faults and symptoms configured for an Electric Rope Shovel - Transmission System

Fault Name	Associated Symptoms	Recommendations
ERS_Failure_Clutch_Operation	ERS_Hydraulic_Oil_Level_Low	Cause: 1. Fuel filter choke ups 2. Fuel Supply not properly coming 3. Fuel pump not working properly 4. Steiner dirty 5. Sludge in Fuel tank Effects: 1. Fuel Filter choke ups 2. Engine not taking load 3. Power loss of Engine 4. Efficiency reduces Recommendation: 1. Fuel Filter Clean, and change it periodically 2. Fuel Tank Clean 3. Drain water separator
	ERS_Hydrostatic_Transmission_Control_On	
	ERS_Rate_Dipper_Fill_Load_high	
	ERS_Transmission_Oil_Level_Low	
	ERS_Transmission_Oil_Pressure_high	
	ERS_Hydrostatic_Trans_MTR_Displacement_Solenoid_OFF	
	ERS_Torque_Converter_Impeller_Clutch_Solenoid_OFF	
	ERS_Torque_Converter_Lockup_Clutch_Solenoid_OFF	
ERS_Abnormal_Operation_Gear_Box	ERS_Transmiss_Output_Speed__R_L	Cause: 1. Faulty Torque Converter, 2. Problems in Hydrostatic Transmission, 3. Low pressure of Hydraulic Oil, High temperature, 4. Damaged High-pressure relief valve, 5. Low electric power supply, 6. Weak electric motors Effects: 1. Loss of Power due to slippage of torque, 2. Weak performance of stick and crowding, 3. Stalling and jacking of Boom, 4. Low productivity Recommendation: Check: 1. The Oil pressure 2. Oil level 3. Electrical power supply

Fault Name	Associated Symptoms	Recommendations
	ERS_Hydraulic_Pressure_R_L	4. Pressure pipeline and valves 5. Electric short circuits
	ERS_Transmission_Oil_Temperature_Sensor_OFF	
	ERS_DataLink_Translation_Module_Switch_OFF	
	ERS_Pay_Load_HIGH	
	ERS_Hydrostatic_Trans_MTR_Displacement_Solenoid_OFF	
	ERS_Transmission_Lock_Switch_ON	

22.1.1.6 Electric Rope Shovel-Upper Work System Fault Model

Model Type	APR Model
Category	Mining Asset
Equipment Class	Mining Asset
Equipment Type	Electric Rope Shovel-Upper Work System

Below table illustrates the list of faults and symptoms configured for an Electric Rope Shovel-Upper Work

Fault Name	Associated Symptoms	Recommendations
ERS_Weak_Performance_Shovel	ERS_Power_Output_Low	Cause: 1. Poor Electric supply 2. Severe stress loading of the shovel front-end assembly 3. Material flows 4. High stresses 5. Harsh excavation conditions can initiate cracks in the dipper-teeth assembly 6. Gear scuffing and wear Effects: 1. Cracks in the dipper -teeth assembly, 2. Rack and Pinion Teeth broken, these cracks resulting in fatigue failure of front end assembly. Recommendation:

Fault Name	Associated Symptoms	Recommendations
	ERS_Swing_Speed_Low	To check the electric supply, Motor performance, bucket, teeth and shackle block, shipper shaft, Pulley and Rope conditions, surface of the mining area and height of the loosened material.
	ERS_Cutting_Angle_Low	
	ERS_Tool_Velocity_Low	
	ERS_Fill_Factor_Low	
	ERS_Digging_Rate_Low	
	ERS_Excavation_Forces_High	
	ERS_Specific_Energy_Consumed_High	
	ERS_PLC_system_Lubricant_OFF	
ERS_Swinging_Hoisting_anomaly	ERS_Excavation_Forces_High	Cause: 1. Problems in swing gears 2. Pinion and electric circuit 3. Low level of lubricant oil 4. Low voltage 5. Faulty Trail cable 6. Improper lubricant and grease system 7. Planetary gearbox teeth worn out 8. Low hoist torque transfer from the electric motor to the hoist drum gear Effects: 1. Reduces machine speed 2. Damages gearbox teeth 3. Low torque output of electric motors Recommendation: Check the required voltage of power supply, IGBT Acutrol drive system, Proper Dual-output pinion shafts from each gearbox engage the swing rack and pumps of lubrication and greasing
	ERS_Specific_Energy_Consumed_High	
	ERS_Swing_Speed_Low	
	ERS_Swing_Armature_Voltage_Low	
	ERS_Swing_Motor_Speed_Low	
	ERS_PLC_system_Lubricant_OFF	
	ERS_Swing_Angle_Low	

Fault Name	Associated Symptoms	Recommendations
ERS_Boom_Jacking_Stall_Equipment	ERS_Swing_Angle_Low	Cause: 1. Excessive Crowd force and Breakout (Bail Pull) Force 2. Poor Crowd force 3. Heavy penetrations 4. Heavy vibrations of boom and boom suspension ropes (Cables) 5. Improper rake and tooth angle 6. Swing angle 7. Position of ERS with respect to bank Effects: 1. Decrease the digging force of the Dipper 2. Stall the equipment and jacks the boom backward 3. Poor dip fill 4. Shallow penetrations 5. Low productivity Recommendation: 1. Check the shackle block teeth, Shipper shaft 2. Check crowd force, Hoist force, boom and boom suspensions rope, rake and tooth angle of dipper, swing angle
	ERS_Crowd_outForce_Low	
	ERS_BreakOut_Force_Low	
	ERS_Crowd_Angle_Low	
	ERS_Cutting_Angle_High	
ERS_Weak_performance_Dipper	ERS_Crowd_outForce_Low	Cause: 1. Improper dipper rake and tooth angle, Pitch brace length, machine parameters (bail pull, crowd force) operator skill and experience, swing angle, position of the shovel relative to the bank, dual or single-sided loading etc. rock density, rock swell factor, dipper fill factor, 2. degree of bench blasting and fragmentation, Effects: Heavy and shallow penetration on the dipper, Excessive hoist motion; stalling in the bank and Boom Jacking Recommendation: 1. Check delta drive and Planetary Transmissions on swing and Propel 2. Adjust the dipper rake and tooth angle 3. Pitch brace length, 4. Check the power supply, and Electric motor output
	ERS_ElectricMotor_Output_per_m3_Low	
	ERS_Dipper_Rake_and_Tooth_angle_Low	
	ERS_Crowd_Angle_Low	
	ERS_Rock_Density_High	

Fault Name	Associated Symptoms	Recommendations
ERS_Abnormal_operation_Lower_Work	ERS_Fill_Factor_High	Cause: Problems with slewing mechanism, loose main shaft in holder, bearing damages, wear of cylindrical cones, Track damages, damages undercarriage frame, slewing ring Effects: Low productivity of digging and dumping, bearing damages, Main shaft loose, damages undercarriage frame, slewing ring and track. Recommendation: Check the slewing mechanism, Main shaft of lower works, Track and bearings, lubrication of track,
	ERS_Rotation_speed_of_Lower_works_Low	
	ERS_Swing_Armature_Voltage_Low	
	ERS_Swing_Motor_Speed_Low	

HYDRAULIC SHOVEL

Hydraulic shovel is used to fill compacted, unballasted material. Hydraulic shovel takes less time than other equipment like wheel loader. It requires small loading area and cycle time is faster than the other equipment. Its bucket capacity is 9.2 yd³ and maximum production is 1468 to 1660 Tones/ hr .

Hydraulic shovels are used on the surface of metal and non-metal mines for overburden and over loading. They are becoming increasingly popular because of their mobility, selective mining capability, and electric power independence. Hydraulic shovels can also work on steeper slopes than electric shovels. They can be fitted with a backhoe attachment, which permits them to dig below ground level.

The ropes on electric shovels have been replaced with hydraulics, and this has led to hydraulic excavators that are often called hydraulic shovels. This additional degree of freedom achieved by the machine results in better performance.

23.1 Hydraulic Shove - Fault Model

23.1.1 Hydraulic Shove - APR Model

23.1.1.1 Hydraulic Shovel- Battery System

Model Type	APR Model
Category	Mining Asset
Equipment Class	Mining Asset
Equipment Type	Hydraulic Shovel - Battery System

Below table illustrates the list of faults and symptoms configured for Hydraulic Shovel- Battery System

Fault Name	Associated Symptoms	Recommendations
HS_Battery_Discharge_Anomaly	HS_Level_of_Distill_Water_High	Cause: 1. Battery terminal is loose 2. Wiring connections from Alternator to battery are disconnected 3. Defective charging system or Alternator - any short circuit may cause excessive current draw Effect: 1. Clicking sound at the time of starting 2. Slow cranking 3. Backfiring 4. Leakage of acid from battery 5. Needing to press the accelerator pedal

Fault Name	Associated Symptoms	Recommendations
		Recommendation:
	HS_Battery_Load_High	1. Check wire connections, battery terminal, battery cells.
	HS_Battery_Capacity_Low	2. Maintain the electrolyte level to the "FULL" mark on the battery
		3. Check the condition of the electrolyte with the 245-5829 Coolant Battery Tester Refractometer.
HS_Battery_Recharging_Anomaly	HS_Rate_Of_Charging_Low	Cause:
		1. Battery unable to take load
		2. Alternator connections are either loose or disconnected
		3. Low electrolyte liquid level in batteries
	HS_Battery_Load_High	4. Alternator's fan belt is either slipping or loose
		Effect:
		1. Battery discharge
		2. Low voltage supply to battery
		3. Engine starting trouble
		Recommendation:
		1. Check for distilled water level
		2. Check terminal wire connections
		3. Check battery terminal
		4. Check Alternator output voltage
HS_Alternator_Anomaly	HS_Alternator_Power_Supply_Rate_Low	Cause:
		1. Malfunction of diode in Alternator
		2. A slipping belt reducing the developed voltage
		3. Slow or malfunctioning accessories in the battery system
	HS_Alternator_RPM_Low	Effect:
		1. Battery not charging
		2. Low voltage supply to battery
		Recommendation:
		1. Check
		1. wire connections
		2. Alternator terminal
		3. Alternator charging rate
		4. Battery cells
		2. Maintain the electrolyte level to the "FULL" mark on the battery,
HS_Starter_Pinion_	HS_Ignition_Key_Switch_Status_ON	Cause:

Fault Name	Associated Symptoms	Recommendations
Anomaly		1. Starter Splice worn out 2. Wire connections loose or sparking 3. A spark causes combustible gases to ignite
	HS_Battery_Capacity_Low	Effect: 1. Grinding noise and motor will not spin 2. Pinion splice worn out 3. Pinion stuck up
	HS_Open_Circuit_Voltage_Low	Recommendation: 1. Check Starter splince 2. Turn the key start switch to the OFF position 3. Check open circuit voltage 4. Check if the pinion stuck up or not any connections loose or disconnected
HS_Starting_Anomaly	HS_Ignition_Key_Switch_Status_ON	Cause: 1. Starter rotor is stuck 2. Battery unable to take load 3. Alternator connections are either loose or disconnected 4. Low electrolyte liquid level in batteries
	HS_Level_of_Distill_Water_Low	Effect: 1. Engine starting trouble 2. Not charging the battery 3. Low voltage supply to battery
	HS_Battery_Load_High	Recommendation: 1. Check Battery Gravity & distilled water level 2. Check terminal wire connections, Battery terminals & Battery load.
	HS_Battery_Capacity_Low	

23.1.1.2 Hydraulic Shovel- Brake System

Model Type	APR Model
Category	Mining Asset
Equipment Class	Mining Asset
Equipment Type	Hydraulic Shovel- Break System

Below table illustrates the list of faults and symptoms configured for Hydraulic Shovel- Break System

Fault Name	Associated Symptoms	Recommendations
Brake Failure	Brake_Fluid_Pressure_RL	Cause: 1. Loss of brake fluid 2. Worn-out brake cylinder 3. Booster diaphragm 4. Black smoke in oil circuit, oil pump & engine's three filters, hydraulic pump, or distribution valve leak 5. Hot car brakes due to an increase in internal pressure of the high-pressure oil pump 6. Oil detreated or dirty Effect: 1. Brake fluid level reduced 2. hot car brakes 3. hydraulic pump or distribution valve leaks MC & SC failure 4. Pump power will be greater than the engine power 5. Cold car brakes Recommendation: 1. Check 1. The brake booster diaphragm 2. Hot car, cold car brakes 3. Oil circuit 4. Engine intake and engine filters 5. Pump regulator 6. Engine valve clearance 7. PRV Solenoid valve voltage
	Travel_Brake_Solenoid_Status_OFF	
	Brake_Air_Pressure_RL	
	Brake_Fluid_Level_RL	
Brake sticking	Brake_Fluid_Pressure_RL	Cause: 1. Caliper stuck due to dust, moisture 2. Piston stuck up 3. Low fluid pressure 4. Failure of oil pump, oil circuit, distribution valves 5. Weak charge pump 6. Air in brake system Effect: 1. Weak brake 2. Piston gets stuck due to low pressure 3. Brake pad wears out faster 4. High-pitched noise of brakes 5. Vehicles pull one side

Fault Name	Associated Symptoms	Recommendations
		Recommendation:
	Parking_Brake_Sensor_ON	1. Check 1. Brake fluid level 2. Oil brake circuit 3. Oil pump pressure 4. Distribution valve pressure 5. Oil quality 6. Brake pack disc
	Brake_Fluid_Level_RL	
Parking Brake Failure	HST_Powered_status_OFF	Cause: 1. Worn or broken parking brake cable 2. Seized parking brake 3. Premature wearing of brake pads or shoes 4. Drive with parking brake 5. Faulty rear caliper motor Effect: 1. Parking brake will not hold the weight of vehicle 2. Due to engaged brake damage the brake drum 3. Excessive wear of brake pads and disc Recommendation: 1. Check 1. Hydraulic pressure 2. Check EPB failure 3. Rear caliper motor 4. Ignition switch and parking brake position 5. The brake drum
	Auto_Park_Brake_Status_ON	
	Ignition_Switch_Status_OFF	
	Parking_Brake_Sensor_ON	

23.1.1.3 Hydraulic Shovel - Cooling System Fault Model

Model Type	APR Model
Category	Mining Asset
Equipment Class	Mining Asset
Equipment Type	Hydraulic Shovel- Cooling System

Table illustrates the list of faults and symptoms configured for Hydraulic Shovel - Cooling System

Fault Name	Associated Symptoms	Recommendations
HS_Hydraulic_Fluid_Anomaly	HS_Engine_Coolant_Outlet_Pressure_High	Cause: 1. Improper working of heat exchanger

Fault Name	Associated Symptoms	Recommendations
		2. Internal leakages of pump, valve, and actuator create overheating 3. Foaming of oil causes cavitation in pump
	HS_Thermostat_Operating_Temperature_Low	Effect: - Heat will reduce viscosity of circulating fluid - Reducing its capacity to remove heat - Damages components and restricts flow and normal operation
	HS_Heat_Rejection_Per_Minute_Low	Recommendation: - Check - Heat exchanger - Oil foaming - Water contamination - Back pressure of valve - Viscosity - Quality of oil
	HS_Hydrauli_Fluid_Temperature_High	
HS_Engine_OverHeating	HS_Engine_Coolant_Level_Low	Cause: - Low coolant level - Improper coolant ratio - Thermostat failure - Radiator fins clogged or dirty
	HS_Thermostat_Operating_Temperature_Low	Effect: - Coolant will boil - Engine efficiency reduces - Weak power strokes - Piston and rings will wear out
	HS_Coolant_Ratio_Low	
	HS_Engine_Oil_Level_Low	Recommendation: - Check - The coolant level and its quality - Thermostat valve function - Fan belt tension - Radiator clogged fins
	HS_Amount_Air_Flow_Through_Radiator_Low	
HS_Water_Pump_Failure	HS_Thermostat_Operating_Temperature_Low	Cause: - Excessive wear in water pump - Bearing seized - Damaged impeller, transmission shaft alignment mismatch Effect: - Over temperature of coolant

Fault Name	Associated Symptoms	Recommendations
		2. Engine efficiency reduces
	HS_Water_Pump_Pressure_Low	3. Weak power strokes
		4. Piston and rings will wear out
	HS_FanBelt_Tension_Low	Recommendation:
	ERSTransmission_Lock_Switch_OFF	1. Check 1. water pump shaft play 2. Fan belt tension 3. Water pump bearing
HS_Malfunction_Arm_or_Boom_Operation	HS_Hydraulic_Pressure_Low	Cause: 1. Oil pump cavitates and develops excessive clearance 2. Air pockets in suction line 3. Less oil level 4. Water contamination in hydraulic fluid Effect: 1. Damages internal components of pump, valve, actuators 2. Oil contaminations will result in an increase in circulating oil temperature Recommendation: 1. Check 1. Oil level 2. Temperature 3. Oil pressure 4. Any leakages in suction line 5. Water contamination
	HS_Hydraulic_Fluid_Temperature_High	
HS_Thermostate_Failure	HS_Engine_Coolant_Outlet_Pressure_High	Cause: 1. Thermostat sensor damaged or malfunctioning 2. Restriction in thermostat valve opening Effect: 1. Coolant will not circulate properly, 2. Engine will overheat leading to engine block damage or crack Recommendation: 1. Check 1. Coolant temperature 2. Thermostat opening function 3. Suction of water pump
	HS_Thermostat_Operating_Temperature_Low	
	HS_Water_Pump_Pressure_Low	
	HS_Coolant_Temperature_High	

23.1.1.4 Hydraulic Shovel - Engine System Fault Model

Model Type	APR Model
Category	Mining Asset
Equipment Class	Mining Asset
Equipment Type	Hydraulic Shovel- Engine System

Below table illustrates the list of faults and symptoms configured for Hydraulic Shovel- Engine System

Fault Name	Associated Symptoms	Recommendations
Engine_Oil_Consumption_Anomaly	Engine_Coolant_Temperature_RH	Cause: 1. Engine oil entering combustion chamber 2. Broken piston rings 3. Excessive engine oil pressure 4. Aging of engine sealing 5. Valve jam Effect: 1. Engine cylinder bore damage 2. Black smoke 3. Weak combustion and power 4. Overheating of engine 5. Push rod bend Recommendation: 1. check 1. Piston rings 2. Oil pump pressure 3. Oil level 4. Engine seal 5. Any leakages of engine oil
	Engine_Fuel_Rate_RL	
	Engine_Load_Capacity_RH	
	EO_Level_RL	
	Engine_oil_Pressure_RL	
	Engine_Speed_RL	
slow_operation_Bucket_Showel	Engine_Load_Capacity_RL	Cause: 1. Poor fuel quality 2. Incorrect valve clearance and injection timing 3. Flow rate of hydraulic pump does not match the output power of engine Effect: 1. Engine will not take load 2. Bucket operation will be slow or jerky Recommendation:

Fault Name	Associated Symptoms	Recommendations
		1. Check 1. Hydraulic system 2. Mechanical transmission system 3. Flow rate 4. Output pressure of the pump 5. Overhaul the internal hydraulic components of the pump 6. Overload 7. Overflow valve setting mechanism
	Engine_oil_Pressure_RL	
	Engine_Speed_RL	
	HydraPump_Pressure_RL	
Engine_OverHeat	EC_Level_RL	Cause: 1. Long-term use of engine oil 2. Engine oil quality and its viscosity 3. Low oil level 4. Improper lubrication of engine parts
	Engine_Coolant_Flow_RL	
	Engine_Coolant_Temperature_RH	
	Engine_Load_Capacity_RH	
	EO_Level_RH	Effect: 1. Coolant level decreases and engine oil level increases 2. Engine gets overheated and experiences power loss
	Engine_Speed_RL	
	Aftercooler_Coolant_Level_RL	Recommendation: 1. Routine change of engine oil and oil filter 2. Tune up engine periodically 3. Check the timing of engine 4. Change air filter and oil periodically at the time of preventive maintenance
	Ambient_Air_Temperature_RH	
	Engine_Fan_Speed_RL	
	Front_Aftercooler_Temperature_Coolant_RH	
	Left_Exhaust_Temperature_RH	
	Rear_Aftercooler_Temperature_Coolant_RH	
	Right_Exhaust_Temperature_RH	
	Right_Left_Exhaust_Temp_RH	
	Engine_Oil_Temperature_RH	
	Inlet_Manifold_Temperature_RH	

Fault Name	Associated Symptoms	Recommendations
Engine_Air_Filter_Restiction	Engine_Speed_RH	Cause: 1. Air filter chock up 2. Dirty oil 3. Sludge in oil or abnormal viscosity of oil Effect: 1. Air filter chock up 2. Dirty oil 3. Engine power loss Recommendation: 1. Check 1. Air filter indicator 2. Oil level 3. Air dryer 4. Oil viscosity 5. Contamination of oil 2. Clean air filter
	Crankcase_Air_Pressure_Gauge_RH	
	Left_Exhaust_Temperature_RH	
	Peak_Air_Filter_Restriction_RH	
	Right_Exhaust_Temperature_RH	
	Right_Left_Exhaust_Temp_RH	
	Engine_Fan_Speed_RH	
Weak_Performance_Engine	Engine_Load_Capacity_RH	Cause: 1. Air in fuel line 2. Fuel filter chock up 3. Inertial oil bath air filter oil level low 4. Oil blockage or plunger of FI Pump will loose speed control bracket Effect: 1. Rack and governor lever are separated 2. Safety device of Governor will loosen 3. Governor spring will break and thrust bearing or governor bearing will be damaged Recommendation: 1. Check 1. Fuel suction line 2. Air filter 3. Oil blockage 4. Oil pump load rejection situation 5. Governor drive sleeve is stuck 6. Positioning screw is too long 7. Throttle valve position
	EO_Level_RL	
	Engine_oil_Pressure_RL	
	Engine_Speed_RL	
	Engine_Fan_Speed_RH	
	Fuel_Filter_Bypass_OFF	
	Left_Exhaust_Temperature_RH	
	Right_Exhaust_Temperature_RH	
	Right_Left_Exhaust_Temp_RH	
	RightTurbocharger_Inlet_Pressure_RL	
	AutoRetarder_Link_mode_ON	
	TransmissionLink_Active_OFF	

Fault Name	Associated Symptoms	Recommendations
Coolant_temp_Diff_Anomaly	EC_Level_RL	Cause: 1. Oil pump faulty or lube filter chock up 2. Engine overheat 3. Engine block crack 4. Head Gasket leak 5. Power loss Effect: 1. High-temperature rise 2. Engine overheat 3. Piston rings and 4. Piston expands 5. Push rod bends Recommendation: 1. Check 1. Oil pump function 2. Oil pressure 3. Lube filter 4. Oil level 5. Viscosity
	Engine_Coolant_Flow_RL	
	Engine_Coolant_Temperature_RH	
	Aftercooler_Coolant_Level_RL	
	Ambient_Air_Temperature_RH	
	Engine_Fan_Speed_RL	
	Front_Aftercooler_Temperature_Coolant_RH	
	Fuel_Filter_Bypass_OFF	
	Left_Exhaust_Temperature_RH	
	Peak_Air_Filter_Restriction_RH	
	Right_Exhaust_Temperature_RH	
	Right_Left_Exhaust_Temp_RH	

23.1.1.5 Hydraulic Shovel- Fuel System Fault Model

Model Type	APR Model
Category	Mining Asset
Equipment Class	Mining Asset
Equipment Type	Hydraulic Shovel- Fuel System

Below table illustrates the list of faults and symptoms configured Hydraulic Shovel- Fuel System

Fault Name	Associated Symptoms	Recommendations
Engine Racing anomaly	Fuel_Pump_Suction_Rate_RL	Cause: 1. Fuel filter chock ups 2. Fuel is not properly supplied 3. Fuel pump is not working properly 4. Strainer is plugged

Fault Name	Associated Symptoms	Recommendations
		5. Presence of sludge in fuel tank
	Fuel_Quantity_Supplied_RL	Effect:
	Fuel_Filter_Differential_Pressure_RH	1. Power loss of Engine 2. Reduced efficiency
	Injector_Pressure_RL	Recommendation: 1. Clean fuel filter, change it if necessary
	Solenoid_Switch_Status_ON	2. Fuel tank cleaning 3. Drain the water separator
Engine load Anomaly	Fuel_Pump_Suction_Rate_RL	Cause: 1. Inaccurate engine timing 2. Fuel pump is not working properly 3. Piston rings broken 4. Presence of sludge in fuel tank
	Fuel_Quantity_Supplied_RL	Effect: 1. Fuel Filter chock ups 2. Crank shaft damages 3. Piston rings broken 4. Power loss of Engine 5. Efficiency reduces
	FuelFlow_asper_load_condition_RL	
	Fuel_Filter_Differential_Pressure_RH	Recommendation: 1. Adjust the timing of engine 2. Clean the strainer and fuel filters 3. Clean air cleaner and change air cleaner oil regularly
Engine Exhaust Anomaly	Fuel_Pump_Suction_Rate_RL	Cause: 1. Clogged air filter 2. Faulty injector or dribbling of Injector 3. Incomplete combustion 4. Blocked exhaust manifold Effect: 1. Fuel knocking 2. Incomplete fuel combustion 3. Malfunction of fuel injector 4. Loss of engine power and efficiency Recommendation: 1. Adjust fuel injection pump timing

Fault Name	Associated Symptoms	Recommendations
	Fuel_Quantity_Supplied_RL	2. Check whether oil and fuel are mixing 3. Over fueling and non-availability of sufficient oxidant
	Injector_Pressure_RL	
Engine Starting Anomaly	Fuel_Pump_Suction_Rate_RL	Cause: 1. Fuel filter is choked 2. Fuel is not properly supplied 3. Fuel pump is not working properly 4. Strainer dirty 5. Sludge in fuel tank Effect: 1. Fuel filter choke ups 2. Engine not taking load 3. Power loss of engine and efficiency reduces 4. Engine taking long self Recommendation: 1. Adjust the timing of engine fuel injection 2. Clean the strainer and fuel filters 3. Clean the air cleaner 4. Check the solenoid switch 5. Change air cleaner oil if necessary
	Fuel_Filter_Differential_Pressure_RL	
	Solenoid_Switch_Status_ON	
Engine Motion Fluctuation	Centrifugal_Force_FI_Pump_RL	Cause: 1. Air in fuel line 2. Fuel filter choke ups 3. Fuel is not properly supplied 4. Fuel pump not working properly 5. Strainer dirty 6. Presence of sludge in Fuel tank Effect: 1. Fuel Filter choke ups 2. Engine not taking load 3. Power loss of engine and efficiency reduces Recommendation: 1. Check air in fuel suction line 2. Clean the strainer and fuel filters

Fault Name	Associated Symptoms	Recommendations
	Injector_Pressure_RL	3. Clean air cleaner and change air cleaner oil regularly
	Fuel_Injection_Rate_RL	

23.1.1.6 Hydraulic Shovel - Hydraulic System Fault Model

Model Type	APR Model
Category	Mining Asset
Equipment Class	Mining Asset
Equipment Type	Hydraulic Shovel- Hydraulic System

Below table illustrates the list of faults and symptoms configured for Hydraulic Shovel- Hydraulic System

Fault Name	Associated Symptoms	Recommendations
Arm Operation Anomaly	Hydraulic_Oil_Level_RL	Cause: 1. Air and water contamination 2. No oil in system 3. Insufficient oil 4. Oil line restriction or leakage in suction pump 5. Badly worn components 6. Oil filter dirty or clogged
	HydraulicPump_Pressure_RL	
	Flow_Rate_Of_Hydraulic_Pump_RL	
	Swing_Brake_Solenoid_Switch_ON	
	HO_Filter_DP_Switch_ON	
	Electronic_Governor_Relief_valve_pressure_RL	
	EHC_VSP_Boom_Raise_Ope_RH	Effect: 1. Increased noise and heat 2. Erratic operation of cylinders 3. Inability to develop full output 4. Decreased speed of cylinders and pump failure Recommendation: 1. Check 1. Oil in system 2. Wrong oil or any restrictions in oil line 3. Dirty or clogged filter 4. Any leakages 5. Performance of hydraulic pumps
	EHC_VSHP_Bucket_Tilt_In_Ope_RL	
	EHC_VSP_Right_Swing_Ope_RL	
	EHC_VSHP_R_Travel_Forward_RL	
	EHC_VSLP_R_Travel_Forward_RL	
	Lift_level_Sensor_ON	

Fault Name	Associated Symptoms	Recommendations
Failure of Hydraulic Pump	Hydraulic_Oil_Level_RL	Cause: 1. Air trapped in hydraulic system 2. Water contamination 3. High temperature 4. Clogged oil filter Effect: 1. Increased noise 2. Increased heat 3. Erratic operation of cylinders 4. Difficulty or inability to develop full output 5. Decreased speed of cylinder Recommendation: 1. Check 1. Oil filter dirty or clogged 2. Check air trapped in system via pump inlets 3. Water contamination 2. Maintain fluid level and quality of fluid
	Swing_Hydraulic_Inhibit_Solenoid_Status_ON	
	HydraulicPump_Pressure_RL	
	Pump_1Contamination_Alarm_Status_ON	
	Pump_2Contamination_Alarm_Status_ON	
	Pump_3Contamination_Alarm_Status_ON	
	Hydraulic_Oil_Pressure_RL	
Inability to build pressure	Hydraulic_Oil_Level_RL	Cause: 1. Bad relief valve, or pump compensator 2. Tank line temperature of the relief valve 3. Increase in viscosity of oil 4. Debris or water in oil 5. Secondary flows within the fittings and valves Effect: 1. Difficulty to develop full output 2. Decreased the speed of cylinder 3. Inability to lift load efficiently Recommendation: 1. Check 1. Directional control setting 2. Pump is receiving fluid 3. Drive motor is not reversed 4. Pump relief valve or mechanical damage
	HydraulicPump_Pressure_RL	
	Air_Inlet_Pressure_RL	
	Flow_Rate_Of_Hydraulic_Pump_RL	
	Electronic_Governor_Relief_valve_pressure_RL	
	EHC_VSP_Left_Swing_Ope_RL	
	Lift_level_Sensor_ON	
Abnormal Temperature of Fluid	Swing_Pump_2Contamination_Alarm_Status_ON	
	Hydraulic_Oil_Temperature_RH	Cause: 1. Blocked or damaged heat exchanger 2. Air and Water contamination

Fault Name	Associated Symptoms	Recommendations
		- No oil in system - Insufficient oil - Oil line restriction - Air leakage in suction pump - Badly worn components - Oil filter dirty or clogged
	Hydraulic_Oil_Pressure_RL	Effect: - Cold temperature affects viscosity of the fluid - High-temperature damage the system components - Seal breaks and oil degrades faster
	Air_Inlet_Pressure_RH	Recommendation: - Check - Fluid quality and level - Blocked or damaged heat exchanger - Oil filters - Debris or dirt in oil - Oil pump - Pressure relief valve - Flow control valve
Jerky operation of Boom	Hydraulic_Oil_Level_RL	Cause: - Plunger pump failure - Main overflow valve improper adjustment - Oil pump cavitates and develops excessive clearance - Air pockets in suction line - Less oil level - Water contamination Effect: - Damages oil pump internal components of pump, valve, actuators - Plunger pump servo piston failure - Overflow valve damaged - Oil contamination creates temperature Recommendation: - Check and repair - LS Valve - OverHaul PC Valve - OverHaul Hydraulic components

Fault Name	Associated Symptoms	Recommendations
		2. Check
	Air_Inlet_Pressure_RL	1. Oil level
		2. Temperature
		3. Oil Pressure
	Flow_Rate_Of_Hydraulic_Pump_RL	4. Any leakages in suction line
		5. Water contamination

23.1.1.7 Hydraulic Shovel- Steering System Fault Model

Model Type	APR Model
Category	Mining Asset
Equipment Class	Mining Asset
Equipment Type	Hydraulic Shovel- Steering System

Below table illustrates the list of faults and symptoms configured for Hydraulic Shovel- Steering System

Fault Name	Associated Symptoms	Recommendations
Hard_Steering	Steering_Oil_Level_RL	Cause: - Steering Oil Pump Failure - Oil Level Less - Steering Pump Hose Leakage - Contaminated oil Effect: - Steering Oil Pump seizes - Steering Recker arm bend - Tie rod bend - Drop Arm Bend - Ball Joints Come out and Steering Hard Recommendation: - Check - Steering Oil Level - Oil pressure, and quality of Steering oil - Oil Pump Pressure - Wheel Bearing play and Kingpin free play
	Oil_Pump_Suction_Rate_RL	
	Port_Flow_Buffer_Valve_Pressure_RL	
	Regulating_Valve_Sensor_OFF	
Steering_Oil_Pump_Damage	Steering_Oil_Level_RL	Cause: - Contaminated Oil - Less Oil Level - Steering Pump Hose Leak

Fault Name	Associated Symptoms	Recommendations
		4. High oil pressure Effect:
	Steering_Oil_Pressure_RH	1. Steering Oil Pump seizes 2. Steering Recker arm bend 3. Tie rod bend
	Steering_OilFil_DP_Switch_status_OFF	4. Drop Arm Bend 5. Ball Joints Come out and Steering Hard Recommendation:
	Regulating_Valve_Sensor_OFF	1. Check 1. Steering Oil Level 2. Oil pressure and quality of Steering oil 3. Oil Pump Pressure
Weak_Steering_Performance	Steering_Oil_Level_RL	Cause: 1. Water leakages from the outside of the software of hydraulic press 2. External oil leakages 3. Poor pipe joints 4. Sealing 5. Cracked pipes 6. Embrittlement of seals 7. Hydraulic oil level Effect: 1. Piston rod 2. Cylinder and seal damages 3. internal leakages of steering pump 4. Inner hole of baffle plate is damaged or too large 5. Sealing of Rotary Joints damages 6. High-pressure oil leaks Recommendation: 1. Check 1. Oil level 2. Oil Pressure 3. Sealing and oil pump joints 4. Central rotary joint 5. Steering gear motor 6. Single channel steady flow valve 7. Steering cylinder
	Hydraulic_Power_Amplification_Sensor_OFF	
	Steering_Oil_Pressure_RL	

Fault Name	Associated Symptoms	Recommendations
Steering_Lock	Steering_Oil_Level_RL	Cause: 1. Steering Oil Pump Fails 2. Oil Level Less 3. Steering Pump Hose Leak 4. Oil Dirty
	Steering_Oil_Pressure_RL	Effect: 1. Steering Oil Pump seizes 2. Steering Recker arm bend 3. Tie rod bend 4. Drop Arm Bend 5. Ball Joints Come out and Steering Hard
	Steering_Lock_Switch_status_OFF	Recommendation: 1. Check 1. Steering Oil Level 2. Oil pressure and quality of Steering oil 3. Oil Pump Pressure

23.1.1.8 Hydraulic Shovel- Transmission System Fault Model

Model Type	APR Model
Category	Mining Asset
Equipment Class	Mining Asset
Equipment Type	Hydraulic Shovel- Transmission System

Below table illustrates the list of faults and symptoms configured for Hydraulic Shovel- Transmission System

Fault Name	Associated Symptoms	Recommendations
Clutch Operation Abnormal	Final_drive_Bypass_Status_OFF	Cause: 1. Low level of Transmission Fluid 2. Broken or loose clutch cable 3. Broken linkage 4. Faulty hydraulic master or slave cylinder 5. Improper fit of throwout bearing with pressure plate 6. Seized clutch plate Effect: 1. Clutch Plate seizes 2. Clutch pedal chatters when accelerating

Fault Name	Associated Symptoms	Recommendations
		3. Clutch disc torsion springs losses or broken 4. Thrust bearing failure 5. Loss of Transmission Power Recommendation: 1. Check 1. Dirty oil or seized clutch plate 2. Pressure plate 3. Transmission fluid level 4. Linkages at the time of preventive maintenance 5. Smooth clutch operation
	Torque_Output_RL	
	Transmission_Charge_Filter_Bypass_OFF	
	Transmission_Output_Speed_RL	
	Transmission_Link_Status_OFF	
	Power_Transmission_RL	
	Neutralizer_Pressure_Switch_OFF	
Low power of Arm	DP_LO_RL	Cause: 1. Low flow of oil to both hydraulic motors 2. Failure of safety valves 3. Air trapped in suction line 4. Cavitation of oil 5. Low flow rate and pressure to directional valves 6. Failure of hydraulic circuit. Effect: 1. Damages hydraulic pump 2. Pressure relief valves 3. Piston gets stuck 4. Transmission Power loss and Power reductions Recommendation: 1. Maintain Oil level and its quality, 2. Check 1. Hydraulic pump 2. Clogged oil filter 3. Cavitation 4. Corrosion and erratic oil 5. Temperature of oil 6. Vibration 7. Installation of pump
	Transmission_Output_Speed_RL	
	Transmission_Link_Status_OFF	
	Hydraulic_Pump_Pressure_RL	

Fault Name	Associated Symptoms	Recommendations
Jump Transmission	DP_LO_RL	Cause: 1. Sliding gear sleeve and Fixed gear sleeve due to frequent shifting damages 2. Self-locking mechanism degrades, or disappears 3. Improper adjustment of shifting device 4. Sudden change of external load 5. Improper operation method Effect: 1. Sliding gear and fixed gear sleeves ground into cone shape 2. Spring of self-locking weakens or broken 3. Steel ball or groove on the fork shaft worn out and transmission jumps Recommendation: 1. Check 1. Proper adjustment of shifting device 2. Steel balls and spring weakness 3. Operate proper shifting process
	Torque_Output_RL	
	Transmission_Output_Speed_RL	
	GB_Oil_Level_RL	
	Left_Drive_Motor_Speed_Sensor_OFF	
	Right_Drive_Motor_Speed_Sensor_OFF	
One side Non-Working of Crawler	Torque_Output_RL	Cause: 1. Low Hydraulic fluid level 2. Failure of one side electric motor 3. Weak recoil spring, 4. Failure of rear-drive 5. Loose triple crawler shoes Effect: 1. Loose the track and possibility of coming out 2. Sprockets splice worn out 3. Failure of one side hydraulic pumps 4. Weak engine working leads to providing less power to pump 5. Flexible drive coupling rubber segments worn out Recommendation: 1. check 1. Hydraulic fluid level, 2. Working condition of both side hydraulic pumps, Motors, Sprockets, and Recoil spring 3. Drive couplings
	Transmission_Output_Speed_RL	
	GB_Oil_Level_RL	
	Left_Drive_Motor_Speed_Sensor_OFF	
	Right_Drive_Motor_Speed_Sensor_OFF	
	Transmission_Link_Status_OFF	
	Transmission_Pressure_Supply_RL	
	Power_Transmission_RL	

Haul trucks are off-highway, rigid dump trucks specifically engineered for use in high-production mining and heavy-duty construction environments. Haul trucks are also used for transporting construction equipment from job site to job site. Some of these trucks are multi-axle designed for supporting equipment being hauled.

Most haul trucks have a two-axle design. Haul truck capacities range from 40 short tons to 496 short tons. Large quarry-sized trucks range from 40 to 100 short tons. Quarry operations are typically smaller than, say, a gold/copper mining, and require smaller trucks.

The largest, highest-payload-capacity haul trucks are referred to as ultra class trucks. The ultra-class includes all haul trucks with a payload capacity of 300 short tons.

A rear-eject configuration is an alternative haul truck body style. Instead of lifting the bed vertically, the hydraulic cylinder pushes a ram-face horizontally through the body to eject the hauled load.

24.1 Haul Truck - Fault Model

24.1.1 Haul Truck - APR Model

24.1.1.1 Haul Truck Unit Performance and Health Fault Model

Model Type	APR Model
Category	Mining Asset
Equipment Class	MovingMiningAsset
Equipment Type	HaulTruck_Battery_System

Below table illustrates the list of faults and symptoms configured for Haul Truck Battery System

Fault Name	Associated Symptoms	Recommendations
Battery_ Discharge_ Anomaly	HT_BS_FB_ Tension_BDA_L	Causes: 1. Loose battery terminal 2. Wiring connections from Alternator to battery loose or disconnected 3. Defective charging system or Alternator 4. Short circuit may cause excessive current draw Effect: 1. Clicking sound at the time of starting 2. Slow cranking 3. Backfiring

Fault Name	Associated Symptoms	Recommendations
		4. Needing to press the accelerator pedal
	HT_BS_DistillWater_L_BDA_L	Recommendations: 1. Check wire condition Battery terminal, Battery cell
	HT_BS_Cooling_Capacity_BDA_L	2. Maintain the electrolyte level to the "FULL" mark on the battery 3. Check the condition of the electrolyte with Coolant Battery Tester Refractometer.
Alternator Anomaly	HT_BS_FB_Tension_AA_L	Causes: 1. Bad diode of Alternator 2. A slipping belt reduced the voltage 3. Slow and malfunctioning accessories Effect: 1. Not charging the battery 2. Low voltage supply to battery
	HT_BS_Alternator_RPM_AA_L	Recommendations: 1. Check wire connections, battery terminal, and battery cells 2. Maintain the electrolyte level to the "FULL" mark on the battery 3. Check the condition of the electrolyte with Coolant Battery Tester Refractometer
Starting Anomaly	HT_BS_DistillWater_L_SA_L	Causes: 1. Starter rotor stuck up 2. Battery not taking load 3. Alternator connections loose or disconnected 4. Low electrolyte level
	HT_BS_Cooling_Capacity_SA_L	Effect: 1. Not charging the battery 2. Low voltage supply to battery
	HT_BS_Cooling_Load_SA_H	Recommendations: 1. Check battery terminal looseness or distilled water level 2. Check wire connections, battery terminal, and battery cells 3. Maintain the electrolyte
	HT_BS_IgnitionKey_S_SA_OFF	
Battery Recharging Anomaly	HT_BS_FB_Tension_BRA_L	Causes: 1. Battery not taking load 2. Alternator connections loose or disconnected 3. Low electrolyte level, Alternator fan belt loose Effect:

Fault Name	Associated Symptoms	Recommendations
		1. Not charging the battery
	HT_BS_Alternator_RPM_BRA_L	2. low voltage supply to battery
	HT_BS_Charging_Rate_BRA_L	Recommendations:
	HT_BS_Cooling_Load_BRA_H	1. Check Battery terminal loose or distilled water level 2. Check wire connections, Battery terminal, Battery cells 3. Maintain the electrolyte
Starting_Anomaly	HT_BS_IgnitionKey_S_SPA_ON	Causes: 1. Starter Splice worn out 2. Wire connections loose or Sparking 3. A spark can cause the combustible gases to ignite. This can result in severe performance. Effect:
	HT_BS_Cooling_Capacity_SPA_L	1. Battery can spoil and blast 2. Grinding noise & does not spin Motor Recommendations: 1. Batteries give OFF combustible gases which can explode. 2. A spark can cause the combustible gases to ignite. This can result in severe personal injury.

24.1.1.2 Haul Truck Power Transmission

Model Type	APR Model
Category	Mining Asset
Equipment Class	MovingMiningAsset
Equipment Type	HaulTruck_Brake_System

Below table illustrates the list of faults and symptoms configured for Haul Truck Brake System

Fault Name	Associated Symptoms	Recommendations
Air Compressor performance Anomaly	Left Rear Brake Temp High	Causes: 1. Air leakage from Air Dryer 2. Air Pressure Tank Drain plug leak 3. Unloader valve leak 4. Air leakage from Air Haul Truck adaptor.
		Effects:

Fault Name	Associated Symptoms	Recommendations
		1. Taking long time to fill air pressure
	Left Rear Wheel Speed High	2. Weak Brake
	Unloader Valve Timing Low	3. Possibility of Brake Stuck
	AP Tank Capacity	Recommendations:
	Load Switch Status ON	Check 1. Air dryer leakage 2. Unloader Valve Faulty or leak 3. Air Pressure Tank Drain plug leak and air pipeline leakages
Brake Valve Leak	Brake Air Pr Low	Causes: 1. Air Pressure leakages 2. Air Tank Leakages 3. Air Haul Truck Pressure Building Time Increases
	Left From Brake Temp high	Effects: 1. Weak Brake 2. Possibility of Brake drum stuck and vehicle stop, 3. Wheel Drum scoring
	Air Haul Truck Performance High	Recommendations:
	AP TanknCapacity Low	Check 1. Overhaul of Air Brake Valve 2. Air Pressure Tank leakages 3. Air Haul Truck Pressure building Time
Brake Weak & Spongy	Brake Air Pressure Low	Causes: 1. Brake Drum Polished or Scoring 2. S cam sticky 3. Brake shoe sticking and not releasing 4. Unloader Valve leak 5. Water in Air Pressure Tank 6. Faulty Air Dryer
	Air Dryer Performance Low	Effects: Dangerous to drive vehicle, not possible to stop vehicle during emergency Recommendations: Overhaul of Air Brake Valve Check Air Pressure Tank leakages Air Haul Truck Pressure building Time

Fault Name	Associated Symptoms	Recommendations
Driving with Parking Brake Set	Brake Air Pressure Low	Causes: 1. Brake Drum Polished or Scoring 2. S cam sticky, M C & SC leakages 3. Brake Fluid level less 4. Brake Fluid Dirty 5. Brake Booster diaphragm leak Effects: 1. Brake Weak 2. Brake Drum Polished or Scoring 3. S cam sticky 4. Brake shoe sticking and not releasing Recommendations: Clean 1. Brake Drum 2. Lubricate S cam and Brake Shoe Arm pins 3. Overhaul of MC and Sc
	Brake Temp High	
	Unloader Valve Timings Low	
	Parking Brake Status ON	
Master Cylinder Slave Cylinder, Booster Abnormal Operation	Brake Stroke Low	Causes: 1. M C & SC leakages 2. Brake Fluid level less 3. Brake Fluid Dirty 4. Brake Booster diaphragm leak Effects: 1. Brake Weak or Slip and brake pedal goes down hard Recommendations: Routine check of 1. Brake fluid level 2. Brake Booster Pressure 3. Overhauling of Master Cylinder 4. In Air brake actuator high tensile loaded spring can cause injury
	Brake Fluid Level High	

24.1.1.3 Haul Truck Cooler System Fault Model

Model Type	APR Model
Category	Mining Asset
Equipment Class	MovingMiningAsset

Model Type	APR Model
Equipment Type	HaulTruck_Engine_System

Below table illustrates the list of faults and symptoms configured for a Haul Truck Engine System

Fault Name	Associated Symptoms	Recommendations
Engine Efficiency Anomaly	Coolant Level Low	Potential Causes: 1. Wrong Timing 2. Wrong firing order and Tappet clearance increased Effects: 1. High vibration 2. Increase of brg clearance 3. Seal leakage 4. Power loss of Engine 5. Engine noisy Recommended Action: 1. Tune up Engine periodically 2. Check the Timing of Engine 3. Change Air filter & Oil, Coolant periodically at the time of preventive maintenance
	Load High	
	Oil level Low	
	Oil Pressure Low	
	Speed Low	
	Fan Speed High	
	Fuel Rate Low	
	Fuel Filter Bypass Status Off	
	Left Exhaust Temp High	
	R Exhaust Temp High	
	Right Left Exhaust Temp High	
	Turbocharger Pr Low	
	Transmission Link Off	
Engine vibration or misfiring of engine	Coolant Temp High	Potential Causes: 1. Engine Foundation broken 2. Tappet clearance increased 3. Improper Timing Effects: 1. High vibration 2. Increase of brg clearance 3. Seal leakage 4. Power loss of Engine Recommended Action: 1. Routine Checking of Engine foundation & Alignment
	Fuel Rate Low	
	Engine Load High	
	EngineOil Level LOW	
	EngineOil Pressure LOW	
	Engine Speed LOW	

Fault Name	Associated Symptoms	Recommendations
Engine Head Gasket leak	Coolant Level Low	Potential Causes: 1. Engine Head gasket leak 2. Piston rings broken 3. Water jackets corradated and circulation stops Effects: 1. Coolant level decreases and Engine oil level increases 2. Engine gets overheated and power loss Recommended Action: Engine head gasket change and check Oil level periodically
	Engine Load High	
	EngineOil Pressure Low	
	Engine Speed Low	
Engine OverHeat	Coolant Level Low	Potential Causes: 1. Long time of Engine oil uses 2. Engine oil quality and its viscosity 3. Low Oil level 4. Improper lubrication of Engine parts Effects: 1. Coolant level decreases and Engine oil level increases 2. Engine gets overheated and power loss Recommended Action: 1. Routine change of Engine oil, Oil Filter 2. Tune up Engine periodically 3. Check the Timing of Engine 4. Change Air filter & Oil periodically at the time of preventive maintenance
	Coolant FlowRate Low	
	Coolant Temp High	
	Engine Load High	
	EngineOil Level High	
	Engine Speed Low	
	AC Coolant Level Low	
	Air Temp High	
	Engine FanSpeed Low	
	AC Front Temp Low	
	L Exhaust Temp Low	
	AC Rear Temp Low	
	R Exhaust Temp Low	
	RL Exhaust Temp High	

Fault Name	Associated Symptoms	Recommendations
Coolant Temp Diff Anomaly	Coolant Level Low	Potential Causes: 1. Oil pump faulty, or lube filter chock up 2. Oil Pump faulty, or lube filter chock up 3. Engine Overheat 4. Engine block crack 5. Head Gasket leak 6. Power loss Effects: 1. High-Temperature Rise 2. Engine Overheat 3. Piston Rings and Piston expands 4. Push Rod bends Recommended Action: 1. Check Oil pump function and Oil pressure 2. Check Oil pump function and Oil pressure 3. Check Lube Filter, Oil Level, and Viscosity
	Coolant FlowRate Low	
	Coolant Temp High	
	AC Coolant Level Low	
	Air Temp High	
	Engine FanSpeed Low	
	Fuel Rate High	
	AC Front Temp High	
	FuelFilter Bypass OFF	
	L Exhaust Temp High	
	AirFilter DP High	
	R Exhaust Temp High	
	RL Exhaust Temp High	
Engine Air Filter Restriction	Engine Load High	Potential Causes: 1. Air filter chock up 2. Dirty Oil Effects: 1. Air filter chock up 2. Dirty Oil 3. Engine power loss Recommended Action: Check Air filter Indicator, Oil level, and Air Dryer.
	Engine Speed High	
	CC Air P High	
	Engine FanSpeed High	
	L Exhaust Temp High	
	AirFilter DP High	
	R Exhaust Temp High	
	RL Exhaust Temp High	

Fault Name	Associated Symptoms	Recommendations
Oil Differential Pressure Alert	Fuel Rate High	Potential Causes: 1. Differential Oil Level less 2. Axle packing worn out 3. Difference in breather chock up
	Engine Load High	
	OilFilter DP Low	
	EngineOil Level Low	Effects: 1. Differential teeth worn out and axle splice damages 2. Power loss
	Engine Speed High	
	AC Coolant_Level High	Recommended Action: 1. Check the breather 2. Difference in Oil level and axle nuts
	Engine FanSpeed High	

24.1.1.4 Haul Truck Cooling System Fault Model

Model Type	APR Model
Category	Mining Asset
Equipment Class	MovingMiningAsset
Equipment Type	HaulTruck_Cooling_System

Illustrates the list of faults and symptoms configured for a Haul Truck Cooling System

Fault Name	Associated Symptoms	Recommendations
Water pump failure & Engine overHeat	Coolant Level Low	Potential Causes: 1. The engine overheats frequently 2. Foaming is observed 3. Oil has entered the cooling system and the coolant is contaminated 4. Water pump bearing ceased Effects: 1. Engine Overheat and compression leak 2. Engine efficiency will be reduced and power loss Recommended Action: 1. Check water pump pulley 2. Check coolant Temperature alarm 3. Check Thermostat Valve, Coolant level & Temperature
	ThermostatOp Level Low	
	Coolant Ratio Low	
	Coolant Temp High	
Air Bubble in	ThermostatOp Level	Potential Causes:

Fault Name	Associated Symptoms	Recommendations
Radiator / Radiator hose Leakage	Low	1. Gasket leak 2. Radiator hose broken or Hose clip damaged Effects: 1. Coolant level will drop and coolant will not circulate properly Recommended Action: 1. Check Radiator Coolant level 2. Check hose clips 3. Check spongy hoses
	Pump Pressure Low	
	Coolant Flowrate High	
Thermostat Failure	Engine Temp High	Potential Causes: 1. Thermostat sensor damaged or not detecting pressure 2. Not opening Thermostat valve at a certain temperature Effects: 1. Coolant will not circulate properly, due to Engine overheat possibility of engine block crack Recommended Action: 1. Check the coolant temperature 2. Check suction of water pump
	Thermostat Op Temp Low	
	Pump P Low	
	Coolant Temp Low	
Water Boiling and pressure come out from Radiator	Engine Temp High	Potential Causes: 1. Fan belt corradiated or hard 2. Fan Belt Cut due to long run or burs ON pulley Effects: 1. Coolant will not circulate properly, due to Engine overheat possibility of engine block crack Recommended Action: 1. Check Fan belt, Water pump circulation, and Thermostat working 2. Check Radiator Coolant level 3. Check Coolant Temperature
	Thermostat Op Temp High	
	Coolant Ratio Lo	
	FB Tension Low	

Fault Name	Associated Symptoms	Recommendations
Low Radiator Coolant level / Radiator cap missing	Coolant level Low	Potential Causes: 1. Coolant leak and enters into crank case 2. Fan Belt broken 3. Thermostat not working 4. Water pump faulty 5. Any Leakages of coolant from hoses 6. Clamp loose 7. Cap lock broken
	Pump Pressure Low	Effects: 1. Engine Temperature will not maintain properly 2. Engine Power Loss 3. Compression Leak Recommended Action: 1. Check Fan belt, Water pump circulation, and Thermostat working 2. Check Radiator Coolant level, Coolant Ratio
Old coolant or rusty coolant	Additive Elements High	Potential Causes: 1. Engine oil mixing in coolant 2. Reservoir cap missing / Radiator cap missing Effects: 1. Engine Temperature will not maintain properly / Engine Overheat due to rusty coolant Recommended Action:: Periodically change coolant at the time of preventive Maintenance

24.1.1.5 Haul Truck Fuel System Fault Model

Model Type	APR Model
Category	Mining Asset
Equipment Class	MovingMiningAsset
Equipment Type	HaulTruck_Fuel_System

Illustrates the list of faults and symptoms configured for a Haul Truck Fuel System

Fault Name	Associated Symptoms	Recommendations
Engine Racing anomaly	Fuel Supply Rate Low	Potential Causes: 1. Fuel filter choke ups 2. Fuel Supply not properly coming 3. Fuel pump not working properly 4. Steiner dirty 5. Sludge in Fuel tank Effects: 1. Fuel Filter choke ups, Engine not taking load 2. Power loss of Engine and efficiency reduces Recommended Action: 1. Fuel Filter Clean and change it periodically 2. Clean Fuel Tank 3. Drain water separator
	Suction Rate Low	
	GV_Flow Low	
Engine load Anomaly	Fuel Supply Rate Low	Potential Causes: 1. Fuel filter choke ups 2. Fuel Supply not properly coming 3. Fuel pump not working properly 4. Steiner dirty 5. Sludge in Fuel tank Effects: 1. Fuel Filter choke ups 2. Engine not taking load 3. Power loss of Engine and efficiency reduces Recommended Action: 1. Adjust the timing of Engine and clean the strainer and fuel filters 2. Clean air cleaner & change air cleaner oil regularly
	FP_Performance Low	
	Injector Pressure Low	

Fault Name	Associated Symptoms	Recommendations
Engine Motion Fluctuation	Fuel Supply Rate Low	Potential Causes: 1. Fuel filter choke ups 2. Fuel Supply not properly coming 3. Fuel pump not working properly 4. Steiner dirty 5. Sludge in Fuel tank Effects: 1. Fuel Filter choke ups 2. Engine not taking load 3. Power loss of Engine and efficiency reduces Recommended Action: 1. Adjust the timing of Engine and clean the strainer and fuel filters 2. Clean air cleaner & change air cleaner oil regularly
	Suction Rate Low	
	FP_Centrifugal Force High	
	Governor Control Low	
Engine Exhaust Anomaly	Fuel Supply Rate Low	Potential Causes: 1. Clogged Air Filter 2. Faulty Injector or Injector dribbling 3. Incomplete combustion 4. Blocked exhaust manifold Effects: 1. Incomplete Fuel Consumption, 2. Leaking fuel injector 3. Loss of engine power and efficiency Recommended Action: 1. Engine exhausts black smoke 2. Engine power loss 3. Fuel consumption 4. Malfunction of fuel injection system
	Injector Low	

Fault Name	Associated Symptoms	Recommendations
Engine Starting Anomaly	Fuel Supply Rate Low	Potential Causes: 1. Fuel filter choke ups 2. Fuel Supply not properly coming 3. Fuel pump not working properly 4. Steiner dirty 5. Sluge in Fuel tank Effects: 1. Fuel Filter choke ups 2. Engine not taking load 3. Power loss of Engine and efficiency reduces 4. Engine taking long self Recommended Action: 1. Adjust the timing of Engine and clean the strainer and fuel filters 2. Clean air cleaner & change air cleaner oil regularly
	Suction Rate Low	
	Fuel To Air Ratio Low	

24.1.1.6 Haul Truck Steering System Fault Model

Model Type	APR Model
Category	Mining Asset
Equipment Class	MovingMiningAsset
Equipment Type	HaulTruck_Steering_System

Illustrates the list of faults and symptoms configured for a Haul Truck Steering System

Fault Name	Associated Symptoms	Recommendations
Hard Steering	Steering Oil Level Low	Potential Causes: 1. Steering Oil Pump Fails 2. Oil Level Less 3. Steering Pump Hose Leak 4. Oil Dirty Effects: 1. Steering Oil Pump ceases 2. Steering Racker arm bend 3. Tie rod bend 4. Drop Arm Bend 5. Ball Joints Come out 6. Steering Hard Recommended Action: Check 1. Steering Oil Level 2. Oil pressure 3. Quality of Steering oil 4. Oil Pump Pressure
	SO Pump Suction Rate Low	
Steering Oil Pump Damage	Steering Oil Level Low	Potential Causes: 1. Steering Oil Pump Fails 2. Oil Level Less 3. Steering Pump Hose Leak 4. Oil Dirty Effects: 1. Steering Oil Pump ceases, 2. Steering Racker arm bend, 3. Tie rod bend 4. Drop Arm Bend 5. Ball Joints Come out and Steering Hard Recommended Action: Check 1. Steering Oil Level 2. Oil pressure 3. Quality of Steering oil 4. Oil Pump Pressure
	Steering Oil Pressure High	
	SO Pump Suction Rate Low	

Fault Name	Associated Symptoms	Recommendations
Steering Lock	Steering Oil Level Low	Potential Causes: 1. Steering Oil Pump Fails 2. Oil Level Less 3. Steering Pump Hose Leak 4. Oil Dirty Effects: 1. Steering Oil Pump ceases 2. Steering Racker arm bend 3. Tie rod bend 4. Drop Arm Bend 5. Ball Joints Come out and Steering Hard Recommended Action: Check 1. Steering Oil Level 2. Oil pressure 3. Quality of Steering oil 4. Oil Pump Pressure
	Steering Oil Pressure Low	

24.1.1.7 Haul Truck Suspension System Fault Model

Model Type	APR Model
Category	Mining Asset
Equipment Class	MovingMiningAsset
Equipment Type	HaulTruck_ Suspension_System

Illustrates the list of faults and symptoms configured for a Haul Truck Suspension System

Fault Name	Associated Symptoms	Recommendations
Suspension Spring Weak Anomaly	Hoist Lever Position High	Potential Causes: 1. Heavy Load or Weak spring Leaf 2. Main leaf broken or other leafs shifted 3. Shock Absorber weak or leak Effects: 1. Suspension Springs break 2. Shock Absorber leak or sitting down 3. Rubber Buffer Pad breaks 4. Air balloon leak or tear

Fault Name	Associated Symptoms	Recommendations
	Payload High	Recommended Action: 1. Routine checking of Suspension System and 2. Check oil of Shock Absorber
Vehicle Tilted Anomaly	Body position Angle High	Potential Causes: 1. One side shock absorber weak or leak and spring weak. Effects: 1. Suspension Springs break 2. Shock Absorber leak or sitting down 3. Rubber Buffer Pad break 4. Air balloon leak or tear Recommended Action: 1. Routine checking of Suspension System 2. Check oil of Shock Absorber
	Desired Hoist Status Off	
	Tyre Pressure low	
	Payload Low	
Body Up Anomaly Alarm	Body Position Angle Low	Potential Causes: 1. Improper Body Lever 2. Body Position angle & Body 3. Hoist Lever Position Effects: 1. Body Lever 2. Body Position 3. Body Position Angle 4. Desired Hoist Status 5. Haul Distance 6. Heartbeat tag 7. Hoist Lever Position Recommended Action: 1. Check Position of body lever, angle, and desired hoist Position
	Hoist Lever Position Low	
	Body Up Alarm On	
Axle Position Anomaly	Body Position Angle Low	Potential Causes: 1. Suspension Spring Center bolt broken and damage Effects: 1. Suspension Springs breaks 2. Shock Absorber leak or sitting down 3. Rubber Buffer Pad breaks 4. Air balloon leak or tear Recommended Action:

Fault Name	Associated Symptoms	Recommendations
	Tyre Pressure low	1. Routine checking of Suspension System 2. Check oil of Shock Absorber
	Payload High	

24.1.1.8 Haul Truck Transmission System Fault Model

Model Type	APR Model
Category	Mining Asset
Equipment Class	MovingMiningAsset
Equipment Type	HaulTruck_Transmission_System

Illustrates the list of faults and symptoms configured for a Haul Truck Transmission System

Fault Name	Associated Symptoms	Recommendations
Vehicle Pick up Anomaly	HT_TS_VEHiclePickUpAnomaly	Potential Causes: 1. Low Oil Level 2. Clutch riding by Drivers 3. Dirty Oil or Viscosity is not according to recommendations Effects: 1. Clutch Plate ceases 2. Gear Box Teeth worn out 3. Differential Crown & Pinion Teeth worn out 4. Transmission Power loss and Power reductions Recommended Actions: Check 1. Dirty oil or ceased clutch plate 2. Pressure plate 3. Maintain Oil level at the time of preventive maintenance
	HT_TS_VEHiclePickUpAnomaly	
	HT_TS_VEHiclePickUpAnomaly	
	HT_TS_VEHiclePickUpAnomaly	
In manual Transmission system-worn gear synchronizer	HT_TS_ManualTransistionAnomaly	Potential Causes: 1. Oil Leakage 2. Bad Driving Habits 3. Not pressing properly clutch pedal Effects: 1. Clutch Plate ceases 2. Gear Box Teeth worn out 3. Differential Crown & Pinion Teeth worn out 4. Transmission Power loss and Power reductions

Fault Name	Associated Symptoms	Recommendations
	HT_TS_ ManualTransistionAnomaly	Recommended Actions: 1. Maintain Oil level and its quality 2. Check worn or loose shift fork, levers, and shaft
	HT_TS_ ManualTransistionAnomaly	
Failed needle roller bearings in automatic Transmission	HT_TS_ RollerBearingAnomaly	Potential Causes: 1. Oil Leakage 2. Bad Driving Habits 3. Not pressing properly clutch pedal Effects: 1. Clutch Plate ceases 2. Gear Box Teeth worn out 3. Differential Crown & Pinion Teeth worn out 4. Transmission Power loss and Power reductions Recommended Actions: 1. Maintain Oil level at the time of preventive maintenance
	HT_TS_ RollerBearingAnomaly	
	HT_TS_ RollerBearingAnomaly	
	HT_TS_ RollerBearingAnomaly	
Abnormal Clutch Operation and Gear Box	HT_TS_ AbnormalClutchOperation	Potential Causes: 1. Oil Leakage 2. Bad Driving Habits 3. Not pressing properly clutch pedal Effects: 1. Clutch Plate ceases 2. Gear Box Teeth worn out 3. Differential Crown & Pinion Teeth worn out 4. Transmission Power loss and Power reductions Recommended Actions: 1. Inspect incorrect lubrication and Maintain Oil level at the time of preventive maintenance
	HT_TS_ AbnormalClutchOperation	
	HT_TS_ AbnormalClutchOperation	
	HT_TS_ AbnormalClutchOperation	
Low Torque Transmission	HT_TS_ LowTorqueTransistion	Potential Causes: 1. Oil Leakage 2. Bad Driving Habits 3. Not pressing clutch pedal properly Effects: 1. Clutch Plate ceases 2. Gear Box Teeth worn out 3. Differential Crown & Pinion Teeth worn out 4. Transmission Power loss and Power reductions

Fault Name	Associated Symptoms	Recommendations
	HT_TS_ LowTorqueTransistion	Recommended Actions: 1. Inspect incorrect lubrication and Maintain Oil level at the time of preventive maintenance.
	HT_TS_ LowTorqueTransistion	
	HT_TS_ LowTorqueTransistion	
	HT_TS_ LowTorqueTransistion	

FURNACE MONITORING

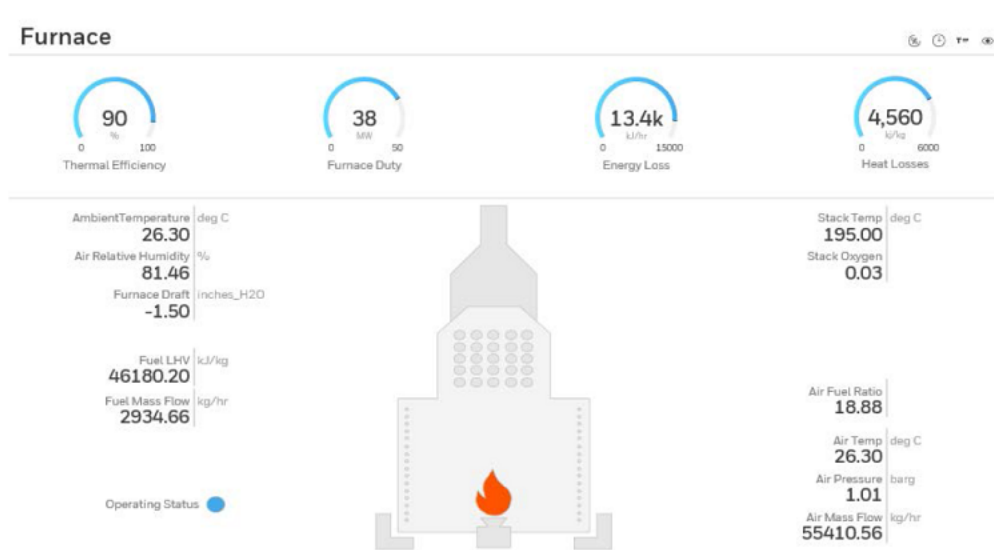
Furnace is an exchanger that transfers heat from the combustion of fuel to fluids contained in tubular coils within an internally insulated enclosure. It is used to provide heat for a process or can serve as a reactor that provides heat for reaction. Furnace designs vary as to its function, heating duty, type of fuel, and method of introducing combustion air. Heat is generated by an Furnace by mixing fuel with air or oxygen, or from electrical energy. The residual heat will exit the furnace as flue gas.

An industrial Furnace comprises below segments,

Radiant section: The radiant section is where the tubes receive almost all its heat by radiation from the flame. In a vertical, cylindrical Furnace, the tubes are vertical. Tubes can be vertical or horizontal, placed along the refractory wall, in the middle, etc., or arranged in cells. Studs are used to hold the insulation together and on the wall of the furnace. The tubes are a distance away from the insulation so radiation can be reflected to the back of the tubes to maintain a uniform tube wall temperature. Tube guides at the top, middle, and bottom hold the tubes in place.

Convection section: The convection section is located above the radiant section where it is cooler to recover additional heat. Heat transfer takes place by convection here, and the tubes are finned to increase heat transfer. The first two tube rows in the bottom of the convection section and at the top of the radiant section is an area of bare tubes (without fins) and are known as the shield section ("shock tubes"), so named because they are still exposed to plenty of radiation from the firebox and they also act to shield the convection section tubes, which are normally of less resistant material from the high temperatures in the firebox. The area of the radiant section just before flue gas enters the shield section and into the convection section is called the bridge zone. A crossover is a tube that connects from the convection section outlet to the radiant section inlet. The crossover piping is normally located outside so that the temperature can be monitored and the efficiency of the convection section can be calculated. The sight glass at the top allows personnel to see the flame shape and pattern from above and visually inspect if flame impingement is occurring. Flame impingement happens when the flame touches the tubes and causes small isolated spots of very high temperature.

Radiant coil: This is a series of tubes horizontal/ vertical hairpin type connected at ends (with 180° bends) or helical in construction. The radiant coil absorbs heat through radiation. They can be single-pass or multi-pass depending upon the process-side pressure drop allowed. The radiant coils and bends are housed in the radiant box. Radiant coil materials vary from carbon steel for low-temperature services to high alloy steels for high-temperature services. These are supported by the radiant side walls or hanging from the radiant roof. Material of these supports is generally high alloy steel. While designing the radiant coil, care is taken so that provision for expansion (in hot conditions) is kept.



Burner: The burner in the vertical, cylindrical Furnace is located on the floor and fires upward. Some Furnace has side-fired burners, such as in train locomotives. The burner tile is made of high-temperature refractory and is where the flame is contained. Air registers located below the burner and at the outlet of the air blower are devices with movable flaps or vanes that control the shape and pattern of the flame, whether it spreads out or even swirls around. Flames should not spread out too much, as this will cause flame impingement. Air registers can be classified as primary, secondary, and if applicable, tertiary, depending on when their air is introduced. The primary air register supplies primary air, which is the first to be introduced in the burner. Secondary air is added to supplement primary air. Burners may include a pre-mixer to mix the air and fuel for better combustion before introducing it into the burner. Some burners even use steam as pre-mix to preheat the air and create better mixing of the fuel and heated air. The floor of the furnace is mostly made of a different material from that of the wall, typically hard castable refractory to allow technicians to walk on its floor during maintenance. A furnace can be lit by a small pilot flame or in some older models, by hand. Most pilot flames nowadays are lit by an ignition transformer (much like a car's spark plugs). The pilot flame in turn lights up the main flame. The pilot flame uses natural gas while the main flame can use both diesel and natural gas. When using liquid fuels, an atomizer is used, otherwise, the liquid fuel will simply pour onto the furnace floor and become a hazard. Using a pilot flame for lighting the furnace increases safety and ease compared to using a manual ignition method (like a match).

Sootblower: Sootblowers are found in the convection section. As this section is above the radiant section and air movement is slower because of the fins, soot tends to accumulate here. Sootblowing is normally done when the efficiency of the convection section is decreased. This can be calculated by looking at the temperature change from the crossover piping and at the convection section exit.

Sootblowers utilize flowing media such as water, air, or steam to remove deposits from the tubes. This is typically done during maintenance with the air blower turned on. There are several different types of sootblowers used. Wall blowers of the rotary type are mounted on furnace walls protruding between the convection tubes. The lances are connected to a steam source with holes drilled into it at intervals

along its length. When it is turned on, it rotates and blows the soot off the tubes and out through the stack.

Stack: The flue gas stack is a cylindrical structure at the top of all the heat transfer chambers. The breeching directly below it collects the flue gas and brings it up high into the atmosphere where it will not endanger personnel. The stack damper contained within works like a butterfly valve and regulates draft (pressure difference between air intake and air exit) in the Furnace, which is what pulls the flue gas through the convection section. The stack damper also regulates the heat lost through the stack. As the damper closes, the amount of heat escaping the Furnace through the stack decreases, but the pressure or draft in the furnace increases which poses risks to those working around it if there are air leakages in the furnace, the flames can then escape out of the firebox or even explode if the pressure is too great.

Individually, each of these important indicators collectively provides a complete picture of the actual condition of the Furnace. This document explains Furnace's monitoring philosophy for process, performance, energy, and health parameters and their importance.

Monitoring Basic machine health data in addition to Furnace operating conditions helps to maintain reliable Furnace operations. There are three areas that help to monitor pump performance to increase availability and decrease energy costs.

- Process and condition monitoring
- Energy monitoring
- Performance and Health monitoring

Item	Type
Category	Equipment
Equipment Class	Heaters
Equipment Type	Furnace

25.1 Furnace Monitoring - Performance Monitoring

The following measured parameters are monitored as it has following impact on performance and health of the furnace if not operated in operating range.

Stack O₂ % : This indicates Air or O₂ percent in the exit gases of the furnace. More O₂ indicates that the furnace is operating at more AF (air to fuel) ratio. More O₂ content than desired indicates losses to preheat excess air supply to combustion temperature. Less O₂ indicates that the furnace is operating at lower AF (air to fuel) ratio. This can cause incomplete combustion of fuel and reduce thermal efficiency of the furnace.

NO_x Emissions : This indicates current NO_x ppm levels in the exit gases of the furnace. More of NO₂ emissions is hazardous for the environment.

CO Emission : This indicates current CO ppm levels in the exit gases of the furnace. More the level indicates incomplete combustion of fuel and lower efficiency of the furnace. More CO levels adds risk for plant and environment safety as well.

Flue Gas Temp/ Stack Temp: This indicates exit gas/stack gas temperature of the furnace. More the temperature indicates in-efficient use of combustion energy.

Furnace Draft : It indicates furnace draft pressure.

Process Performance Overview : Process Performance monitoring view in APM provides a list of the asset attributes and their details, including the current value. This list can be used to review and update asset attribute information, as well as to open an Asset Trend to view more details. It also highlights (in red color) the attributes if deviates from the high or low limits.

25.1.1 Furnace Monitoring - Model Inputs

S.No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Running Status	Operational Status of Asset (Input)		Dynamic	Process	Field Input
2	Mean Sp Heat of Air In & Gas Out	Mean Specific Heat of Air In & Gas Out (Required Input)	kJ/kg- C	Dynamic	Thermo Properties	Field Input
3	Sp Heat of Diluted Gas Out	Specific heat of Diluted Gas Out (Required Input)	kJ/kg- C	Dynamic	Thermo Properties	Field Input
4	Gas Temperature Leaving (Undiluted)	Gas Temperature Leaving (Undiluted)	°C	Dynamic	Process	Field Input
5	Air Inlet Temperature	Air Heater Air Inlet Temperature (Required Input)	°C	Dynamic	Process	Field Input
6	Air Exit Temperature	Air Heater Air Exit Temperature (Required Input)	°C	Dynamic	Process	Field Input
7	Gas Inlet Temperature	Air Heater Gas Inlet Temperature (Required Input)	°C	Dynamic	Process	Field Input
8	Gas Exit Temperature	Air Heater Gas Exit Temperature (Required Input)	°C	Dynamic	Process	Field Input
9	Flue Gas Oxygen Before AH	Measured Flue Gas Oxygen Before AH (Required Input)	weight percent	Dynamic	Process	Field Input
10	Flue Gas Oxygen After AH	Measured Flue Gas Oxygen After AH (Required Input)	weight percent	Dynamic	Process	Field Input

25.1.2 Furnace Monitoring - KPIs

The following indicators help to know the overall performance of the furnace Thermal Efficiency: Thermal efficiency is defined as the percentage of heat available to heat process fluid with respect to heat supplied. It accounts losses due to air and fuel preheating, air relative humidity, stack, and radiation losses. Process side heat balance is not considered in furnace efficiency calculations.

Total moisture in combustion air

$$\text{Moisture in air} = \frac{\text{Vapour pr. of water}}{101.325} + \frac{\text{Relative humidity}}{100} + \frac{18}{28.85}$$

Total wet air per Kg of fuel required

$$\text{Kg of wet air per Kg of fuel} = \frac{\text{Kg of air required}}{1 - \text{moisture in air}}$$

Total moisture per Kg of fuel

$$\text{Kg of moisture per Kg of fuel} = \text{Kg of wet air per Kg of fuel} - \text{Kg of air required}$$

Total excess air per Kg of fuel

$$\text{Kg of excess air per Kg of fuel} = \frac{(28.85 * \% O_2) \left(\frac{N_2 \text{ formed}}{28} + \frac{Co_2 \text{ formed}}{44} + \frac{H_2O \text{ formed}}{18} \right)}{20.95 - \% O_2 \left[\left(1.6028 * \frac{\text{Kg of } H_2O}{\text{Kg of air required}} \right) + 1 \right]}$$

Percentage excess air

$$\text{Percentage excess air} = \frac{\text{Kg of excess air per kg of fuel}}{\text{Kg of air required}} * 100$$

Net thermal efficiency

$$\text{Net thermal efficiency} = \frac{\text{Total heat absorbed}}{\text{Total heat input}} * 100$$

$$E = \frac{(hL + \Delta ha + \Delta hf + \Delta hm) - (hr + hs)}{(hL + \Delta ha + \Delta hf + \Delta hm)} * 100$$

Where,

Stack heat loss is calculated based on Flue gas enthalpy difference at stack temperature and at datum temperature (15 Deg. C). The Flue gas quantity is calculated based on theoretical air requirement for complete combustion of fuel and excess O₂ content in the stack. Radiation losses are expressed as fixed percentage of Fuel LHV as per norms defined in API 560. Radiation Loss = Radiation Loss % × LHV / 100 Radiation loss% is defined in APM template as below,

1.5% - No air preheater means air enters the Furnace at ambient temperature

2.5% - Air is preheated to improve the furnace efficiency. In this air temperature value should be taken based on measurement after air preheater.

Mean specific heat of process fluid	Kcal/Kg °C	Cp
Lower heating value of the fuel burned	Kcal /Kg	hL

Mean specific heat of process fluid	Kcal/Kg °C	Cp
Higher heating value of the fuel burned	Kcal /Kg	hH
Specific heat capacity of the air	Kcal /Kg °C	Cp a
Specific heat capacity of the fuel	Kcal /Kg °C	Cp f
Specific heat capacity of the atomizing medium	Kcal/Kg °C	Cp m
Ambient air sensible heat correction due to SPH	Kcal/Kg	Δha
Fuel sensible heat correction	Kcal /Kg	Δhf
Atomizing medium sensible heat correction	Kcal /Kg	Δhm
Radiation heat loss	Kcal /Kg	hr
Stack heat loss	Kcal /Kg	hs

Fuel efficiency

$$\text{Fuel efficiency} = \frac{\text{Total heat absorbed}}{\text{Heat input due to combustion of fuel only}} * 100$$

$$Ef = \frac{(hL + \Delta ha + \Delta hf + \Delta hm) - (hr + hs)}{hL} * 100$$

Projected Energy Loss

$$\text{Energy Loss} = \text{Heat Losses} * Mf / 1000$$

$$\text{Energy Loss PerHr} = (\text{Energy Loss} * (\text{Fuel Cost}/1000)) / (3600/1000)$$

$$\text{Projected Energy Loss} = \text{Energy Loss PerHr} * 8400$$

25.1.3 Furnace Monitoring - Model Outputs

S.No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Air Side Effectiveness	Air Side Effectiveness (Output)	%	Dynamic	Performance	Calculated
2	Gas Side Efficiency	Gas Side Efficiency (Output)	%	Dynamic	Performance	Calculated
3	Gas Side Effectiveness	Gas Side Effectiveness (Output)	%	Dynamic	Performance	Calculated
4	Air Heater Leakage (Approximate)	Air Heater Leakage (Approximate) (Output)	%	Dynamic	Performance	Calculated
5	Thermal Performance (X-ratio)	Thermal Performance (X-ratio) (Output)		Dynamic	Performance	Calculated

25.2 Furnace Monitoring - Fault Model

In APM, a fault model is built to identify possible anomalies that affects the performance or health of a furnace. The fault management can inform operators of any violations of the nominal furnace operating range and deviations from the expected characteristic and make this data available for further processing through the historian. The fault symptoms or models are used to:

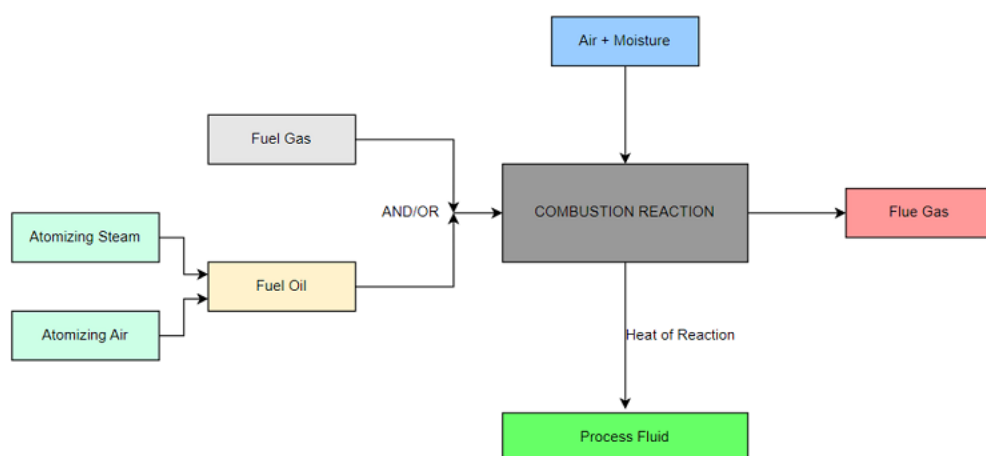
25.2.1 Furnace Monitoring - APM Model

Fault Name	Recommendations
Performance Degradation	Tube leakage High
	Air Outlet Temp Low
	Air Side Effectiveness Low
	Gas Side Effectiveness Low
	Inlet Flue Gas Temp Low
Performance Low Warning	Fuel Efficiency Low Thermal Efficiency Low
	Stack Temperature High
	Stack O2 High with High Draft
	Stack O2 Low with High Draft
	Stack O2 High with Low Draft
	Stack O2 Low with Low Draft
	Stack O2 Low at Target Draft
	Stack O2 High at Target Draft
Poor Operations	Positive Ach Pressure
	Fuel BackPressure High
Incomplete Combustion	CO Emissions High
	NOx Emission High
	Air to Fuel Ratio Low
	Air to Fuel Ratio High
Performance Low Warning	Fuel Efficiency Low
	Stack Oxygen High
	Stack Oxygen Low
	Stack Oxygen Low
	Stack Temperature High
Poor Operations	Positive Ach Pressure
	Fuel BackPressure High

Fault Name	Recommendations
Incomplete Combustion	CO Emissions High
	NOx Emission High
	Air to Fuel Ratio Low
	Air to Fuel Ratio High

25.2.2 Furnace Monitoring - APR Model

Fault Name	Associated Symptoms
Potential SuperHeater Tubeleak	Furnace Pressure Rapid Change
	Damper Opening Rapid Change
	SH Temp Diff Rapid Change
Potential Convection WaterWall Tubeleak	Flue Gas Temp Rapid Change
	Furnace Pressure Rapid Change
	Damper Opening Long Term Change



- Combustors, also commonly called furnaces/ fired heaters, are a primary source of thermal energy for process heating operations in petroleum refining and chemical plants.
- Furnaces and fired heaters essentially consist of tubes, burners, a refractory-lined firebox, and a stack. The firebox and the tubes within it are termed the “radiant section,” which is characterized by radiant heat transfer. The convection section is characterized by convective heat transfer and may be used to preheat process fluid before it enters the radiant tubes or to generate or superheat steam.
- Burners provide the means of heating, and the fuel can be fuel gas, fuel oil or it can be dual fired.
- Liquid fuels may require an atomizing medium of either air or steam. Pilots may use a different fuel than the main burners which may require additional piping to the burners.

The ultimate analysis provides a convenient method for reporting the major organic elemental composition of the fuel. For this analysis, a fuel sample is combusted in an ultimate analyzer, which measures the weight percent of carbon, hydrogen, nitrogen, sulfur, and ash from a fuel sample.

26.1 Combustor - Model Input

26.1.1 Ultimate Analyzer - Model Data

No.	Attribute	Attribute Category	Monitoring Category	Source
1	Datum Temperature	Design	Process	Constant
2	Inlet1	Port	Process	Port
3	Mass Flow	Dynamic	Process	Field input
4	Inlet1_Temperature	Dynamic	Process	Constant
5	Specification • Spec=0 Manually enter LHV and HHV; (Default)	Design	Process	Constant

No.	Attribute	Attribute Category	Monitoring Category	Source
	- Spec=1 Calc LHV and HHV from Composition; - Spec=2 Calc Hydrogen Carbon Ratio;			
6	LowerHeatingValue (Optional, Required if Spec=0)	Design	Process	Constant
7	HigherHeatingValue (Optional, Required if Spec=0)	Design	Process	Constant

26.1.2 Combustor - Model Data

No.	Attribute	Attribute Category	Monitoring Category	Source
1	Air Temperature	Dynamic	Process	Field input
2	Air Pressure	Dynamic	Process	Field input
3	Ambient Temperature	Dynamic	Process	Field input
4	Air Relative Humidity	Design	Process	Constant
5	Stack O2	Dynamic	Process	Field input
6	Percent Radiation Loss	Design	Process	Datasheet
7	Flue Gas Temperature	Dynamic	Process	Field input
8	Atomising Air to Fuel Ratio (Optional, required if Fuel Oil used = 1)	Dynamic	Process	Constant
9	Atomising Steam to Fuel Ratio (Optional, required if Fuel Oil used = 1)	Dynamic	Process	Constant
10	Datum Temperature (Optional, default = 15.6 C)	Design	Process	Constant
11	Enthalpy of Atomising Steam Actual (Optional, required if Fuel Oil used = 1)	Dynamic	Process	Constant
12	Enthalpy of Atomising Steam Datum (Optional, required if Fuel Oil used = 1)	Dynamic	Process	Constant
13	Fuel Oil Used (0=Fuel Gas (Default) & 1 = Fuel Oil/Dual fired)	Design	Process	Constant
14	Specific Heat of Atomising Air (Optional, required if Fuel Oil used = 1)	Dynamic	Process	Constant
15	Temperature of Atomising Air (Optional, required if Fuel Oil used = 1)	Dynamic	Process	Constant
16	Inlet-1 (Ultimate Analyser - 1)	Port	Process	Port
17	Inlet-2 (Ultimate Analyser - 2) Optional	Port	Process	Port
18	Inlet-3 (Ultimate Analyser - 3) Optional	Port	Process	Port

26.2 Combustor - Variables

26.2.1 Excess Air (%)

$$F_{Air}^{excess}(\%) = \frac{F_{Air} - F_{Air}^{min}}{F_{Air}^{min}} 100$$

26.2.2 Sensible Heat of Flue Gas/ Exhaust (KJ/Kg)

$$q_{ex}^{sensible} = C_{p_{ex}}(T_{ex} - T_{Datum})$$

26.2.3 Specific Sensible Heat Air (KJ/Kg)

$$q_{Air}^{sensible} = \frac{F_{Air}}{F_{Fuel}^{total}} C_{p_{Air}}(T_{Air} - T_{Datum})$$

$$F_{Fuel}^{total} = F_{Fuel1} + F_{Fuel2} + F_{Fuel3}$$

26.2.4 Specific Sensible Heat Moisture (KJ/Kg)

$$q_{Moisture}^{sensible} = \frac{MW_{H_2O}}{MW_{Air}} H_{Abs}(T_{Moisture} - T_{Datum})$$

$$q_{Moisture}^{sensible} = \frac{18}{28.97} H_{Abs}(T_{Moisture} - T_{Datum})$$

26.2.5 Specific Sensible Heat Fuel (KJ/Kg)

$$q_{Fuel}^{sensible} = \frac{(q_{Credit1}F_{Fuel1} + q_{Credit2}F_{Fuel2} + q_{Credit3}F_{Fuel3})}{F_{Fuel}^{total}}$$

$$q_{Credit1} = C_{p_{Fuel1}}(T_i^{Fuel1} - T_{Datum})$$

$$q_{Credit2} = C_{p_{Fuel2}}(T_i^{Fuel2} - T_{Datum})$$

$$q_{Credit3} = C_{p_{Fuel3}}(T_i^{Fuel3} - T_{Datum})$$

26.2.6 Heat Credits (KJ/Kg)

$$q_{credits} = q_{Air}^{sensible} + q_{Moisture}^{sensible} + q_{Fuel}^{sensible}$$

26.2.7 Stack Heat Loss (KJ/Kg)

$$q_{stack}^{Loss} = h_{ex}(T_{ex}, F_{Air}^{excess})$$

26.2.8 Radiation Heat Loss (KJ/Kg)

$$q_{rad}^{Loss} = \frac{q_{rad}^{Loss} \% LHV_{fuel}}{100}$$

$$LHV_{fuel} = \frac{(LHV_{fuel1} F_{Fuel1} + LHV_{fuel2} F_{Fuel2} + LHV_{fuel3} F_{Fuel3})}{F_{Fuel}^{total}}$$

26.2.9 Total Heat Losses (KJ/Kg)

$$q_{loss}^{total} = q_{stack}^{Loss} + q_{rad}^{Loss}$$

26.2.10 Atomising Steam Heat Correction (KJ/Kg)

$$AS_{Hcorr} = SF_{ratio} (h_{atomSteam_{act}} - h_{atomSteam_{datum}})$$

26.2.11 Atomising Air Heat Correction (KJ/Kg)

$$AA_{Hcorr} = C_{pAir} AF_{ratio} (T_{atomAir} - T_{Datum})$$

26.2.12 Thermal Efficiency (%)

$$\eta_{thermal}(\%) = \frac{(LHV_{fuel} + q_{credits} + AS_{Hcorr} + AA_{Hcorr} - q_{loss}^{total})}{LHV_{fuel} + q_{credits} + AS_{Hcorr} + AA_{Hcorr}} 100$$

26.2.13 Fuel Efficiency (%)

$$\eta_{Fuel}(\%) = \frac{(LHV_{fuel} + q_{credits} + AS_{Hcorr} + AA_{Hcorr} - q_{loss}^{total})}{LHV_{fuel}} 100$$

26.2.14 Furnace Duty (MW)

$$Q_{furnace} = \frac{(F_{Fuel1} LHV_{fuel1} + F_{Fuel2} LHV_{fuel2} + F_{Fuel3} LHV_{fuel3})}{3600 * 1000}$$

26.2.15 Energy Loss (MW)

$$Q_{loss} = \frac{q_{loss}^{total} F_{Fuel}^{total}}{3600 * 1000}$$

26.2.16 Absorbed Duty (MW)

$$Q_{Absorbed} = Q_{furnace} - Q_{loss}$$

26.2.17 Flue Gas or Exhaust Oxygen (mole fraction)

$$y_{O_2}^{ex} = \frac{y_{O_2}^{stack}(\%)}{100}$$

26.2.18 Absolute Humidity (kg of water/kg of air)

$$H_{Abs} = \frac{RH}{100} \left(\frac{P_{air}^{sat}}{P_{air}} \right) \left(\frac{MW_{water}}{MW_{air}} \right)$$

26.3 Combustor - Model Output

26.3.1 Ultimate Analyzer - Model Data

No.	Model Outputs	Attribute Category	Monitoring Category	Source
1	Ash	Dynamic	Performance	Calculated
2	Carbon	Dynamic	Performance	Calculated
3	CarbonHydrogenRatio	Dynamic	Performance	Calculated
4	HeatCredit	Dynamic	Performance	Calculated
5	HHV	Dynamic	Performance	Calculated
6	Hydrogen	Dynamic	Performance	Calculated
7	Inlet1_CompoundData	Dynamic	Performance	Calculated
8	Inlet1_Cp	Dynamic	Performance	Calculated
9	Inlet1_Density	Dynamic	Performance	Calculated
10	LHV	Dynamic	Performance	Calculated
11	Moisture	Dynamic	Performance	Calculated
12	Nitrogen	Dynamic	Performance	Calculated
13	Oxygen	Dynamic	Performance	Calculated
14	Sulfur	Dynamic	Performance	Calculated
15	Temperature	Dynamic	Performance	Calculated
16	Ash	Dynamic	Performance	Calculated

26.3.2 Combustor - Model Data

No.	Model Outputs	Attribute Category	Monitoring Category	Source
1	Absolute Humidity	Dynamic	Performance	Calculated
2	Absorbed Duty	Dynamic	Performance	Calculated
3	Air Density	Dynamic	Performance	Calculated
4	Air Enthalpy	Dynamic	Performance	Calculated
5	Air Mass Flow	Dynamic	Performance	Calculated
6	Air Min Flow	Dynamic	Performance	Calculated
7	Air Specific Heat	Dynamic	Performance	Calculated
8	Air To Fuel Ratio	Dynamic	Performance	Calculated

No.	Model Outputs	Attribute Category	Monitoring Category	Source
9	Atomising Air Heat Correction	Dynamic	Performance	Calculated
10	Atomising Steam Heat Correction	Dynamic	Performance	Calculated
11	Carbon	Dynamic	Performance	Calculated
12	Compound Data	Dynamic	Performance	Calculated
13	Energy Loss	Dynamic	Performance	Calculated
14	Excess Air Percentage	Dynamic	Performance	Calculated
15	Flue Gas Ash	Dynamic	Performance	Calculated
16	Flue Gas Carbon Dioxide	Dynamic	Performance	Calculated
17	Flue Gas Carbon Monoxide	Dynamic	Performance	Calculated
18	Flue Gas Cp	Dynamic	Performance	Calculated
19	Flue Gas Density	Dynamic	Performance	Calculated
20	Flue Gas Enthalpy	Dynamic	Performance	Calculated
21	Flue Gas Mass Flow	Dynamic	Performance	Calculated
22	Flue Gas Nitrogen	Dynamic	Performance	Calculated
23	Flue Gas Oxygen	Dynamic	Performance	Calculated
24	Flue Gas Sulphur Dioxide	Dynamic	Performance	Calculated
25	Flue Gas Water	Dynamic	Performance	Calculated
26	Fuel Ash	Dynamic	Performance	Calculated
27	Fuel Density	Dynamic	Performance	Calculated
28	Fuel Efficiency	Dynamic	Performance	Calculated
29	Fuel Hydrogen	Dynamic	Performance	Calculated
30	Fuel Inlet Mass Flow	Dynamic	Performance	Calculated
31	Fuel LHV	Dynamic	Performance	Calculated
32	Fuel Mass Flow	Dynamic	Performance	Calculated
33	Fuel Nitrogen	Dynamic	Performance	Calculated
34	Fuel Oxygen	Dynamic	Performance	Calculated
35	Fuel Sensible Heat	Dynamic	Performance	Calculated
36	Fuel Sulphur	Dynamic	Performance	Calculated
37	Fuel Compound Data	Dynamic	Performance	Calculated
38	Furnace Duty	Dynamic	Performance	Calculated
39	Heat Credits	Dynamic	Performance	Calculated
40	Heat Losses	Dynamic	Performance	Calculated
41	Inlet1_Energy	Dynamic	Performance	Calculated

No.	Model Outputs	Attribute Category	Monitoring Category	Source
42	Inlet1_FlowRate	Dynamic	Performance	Calculated
43	Inlet1_FuelFlow	Dynamic	Performance	Calculated
44	Inlet1_HeatCredit	Dynamic	Performance	Calculated
45	Inlet2_CompoundData	Dynamic	Performance	Calculated
46	Inlet2_Energy	Dynamic	Performance	Calculated
47	Inlet2_FlowRate	Dynamic	Performance	Calculated
48	Inlet2_FuelFlow	Dynamic	Performance	Calculated
49	Inlet2_HeatCredit	Dynamic	Performance	Calculated
50	Inlet3_CompoundData	Dynamic	Performance	Calculated
51	Inlet3_Energy	Dynamic	Performance	Calculated
52	Inlet3_FuelFlow	Dynamic	Performance	Calculated
53	Inlet3_HeatCredit	Dynamic	Performance	Calculated
54	Inlet3_MassFlow	Dynamic	Performance	Calculated
55	Radiation Loss	Dynamic	Performance	Calculated
56	Sensible Heat Air	Dynamic	Performance	Calculated
57	Sensible Heat Fuel	Dynamic	Performance	Calculated
58	Stack Heat Loss	Dynamic	Performance	Calculated
59	Thermal Efficiency	Dynamic	Performance	Calculated

AIR PREHEATER MONITORING

Air Preheaters are used for preheating the air which is fed to the boiler or furnace for combustion of fuels. The primary objective of an APH is to extract the waste heat from the flue gases leaving the boiler. It also allows control over the temperature of gases leaving the stack (to meet emissions regulations, for example). It is installed between the economizer and chimney.

Air preheater performance monitoring plays an important role to predict the cleaning and maintenance schedules with temperature gradient, leakage. Monitoring Basic machine health data in addition to Air preheater operating conditions helps to maintain reliable operations of an Air preheater. There are two areas that help to monitor Air preheater performance to increase availability and decrease energy costs.

- Process and condition monitoring
- Performance and Health monitoring

Air PreHeater



This document explains Air Preheater monitoring philosophy for process, performance, energy, and health parameters and their importance.

27.1 Air Preheater Monitoring - Performance Monitoring

The following measured parameters are monitored as they have impact on performance and health of Air preheater, if not maintained in operating range.

Air Inlet Temperature: It is ambient air temperature going to air preheater, higher the ambient temperature higher the outlet temperature of air preheater results in less fuel consumption.

Flue Gas Inlet Temperature: Higher the Flue gas inlet temperature to air preheater higher heat transfer to air result in higher air outlet temperature.

Process Performance Overview: Process Performance monitoring view in APM provides a list of the asset attributes and their details, including the current value. This list can be used to review and update asset attribute information, as well as to open an Asset Trend to view more details. It also highlights (in red color) the attributes if deviates from the high or low limits.

27.1.1 Air Preheater Monitoring - Model Inputs

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Running Status	Operational Status of Asset (Input)		Dynamic	Process	Field Input
2	Mean Sp Heat of Air In & Gas Out	Mean Specific Heat of Air In & Gas Out (Required Input)	kJ/kg- C	Dynamic	Thermo Properties	Field Input
3	Sp Heat of Diluted Gas Out	Specific heat of Diluted Gas Out (Required Input)	kJ/kg- C	Dynamic	Thermo Properties	Field Input
4	Gas Temperature Leaving (Undiluted)	Gas Temperature Leaving (Undiluted)	°C	Dynamic	Process	Field Input
5	Air Inlet Temperature	Air Heater Air Inlet Temperature (Required Input)	°C	Dynamic	Process	Field Input
6	Air Exit Temperature	Air Heater Air Exit Temperature (Required Input)	°C	Dynamic	Process	Field Input
7	Gas Inlet Temperature	Air Heater Gas Inlet Temperature (Required Input)	°C	Dynamic	Process	Field Input
8	Gas Exit Temperature	Air Heater Gas Exit Temperature (Required Input)	°C	Dynamic	Process	Field Input
9	Flue Gas Oxygen Before AH	Measured Flue Gas Oxygen Before AH (Required Input)	weight_percent	Dynamic	Process	Field Input
10	Flue Gas Oxygen After AH	Measured Flue Gas Oxygen After AH (Required Input)	weight_percent	Dynamic	Process	Field Input

27.1.2 Air Preheater Monitoring - KPIs

The following indicators help us to know overall performance of the Air preheater.

Air Side Effectiveness

It indicates how effectively heat is transferred to Air from flue gases based on air temperature difference. Higher effectiveness indicates more effective heat transfer.

Gas Side Effectiveness

It indicates how effectively heat is transferred from flue gases to Air based on flue gas temperature difference. Higher effectiveness indicates more effective heat transfer.

X- Ratio (Thermal Performance)

A low X-ratio indicates excessive gas weight through the air heater or that airflow is bypassing the air heater. A lower than design X-ratio leads to higher than design gas outlet temperature & can be used as an indication of excessive tempering air to the mills or excessive boiler infiltration.

Leakage Percentage

It indicates the outside air passing in flue gases, which dilutes the flue gases causing decrease in temperature.

27.1.3 Air Preheater Monitoring - Model Outputs

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Air Side Effectiveness	Air Side Effectiveness(Output)	%	Dynamic	Performance	Calculated
2	Gas Side Efficiency	Gas Side Efficiency (Output)	%	Dynamic	Performance	Calculated
3	Gas Side Effectiveness	Gas Side Effectiveness (Output)	%	Dynamic	Performance	Calculated
4	Air Heater Leakage (Approximate)	Air Heater Leakage (Approximate) (Output)	%	Dynamic	Performance	Calculated
5	Thermal Performance (X-ratio)	Thermal Performance(X-ratio) (Output)		Dynamic	Performance	Calculated

27.2 Air Preheater Monitoring - Fault Model

In Asset Performance Management, faults, symptoms, and fault models are configured based on Air preheater performance model outputs and process parameters measured from field to diagnose the problems early. The fault management can inform operators of any violations of the nominal Air preheater operating range and deviations from the expected characteristic and make this data available for further processing through the historian. The fault symptoms or models are used to:

1. Warn against potential damage to Air preheater under unfavorable operating conditions.
2. Provide early warning of damages to the Air preheater

The following messages can be generated for process alarms or notifications to warn operating personnel of unfavorable operating conditions

1. **Heat transfer degradation:** It occurs mainly due to fouling and leakage in air preheater.
2. **Low effectiveness:** It occurs due to variation in temperatures at inlet and outlet. It can be avoided by operating the air preheater at optimal conditions.
3. **Leakage:** It can be detected by checking the material balance.
4. **Fouling:** It is an important factor in air preheater, caused due to deposition of material on the tubes. The air preheater effective heat duty and heat transfer coefficient decreases when fouling occurs. Fouling predictions are used to predict the maintenance schedules for cleaning heat exchanger.
5. **Air outlet temperature:** It determines the efficiency of other sources like furnace, boilers since higher air outlet temperature lower heat duty required to achieve desired temperature.

Below table illustrates the list of faults and symptoms configured for an Air preheater.

Fault Name	Associated Symptoms
Performance Degradation	Tube leakage High
	Air Outlet Temp Low
	Air Side Effectiveness Low
	Gas Side Effectiveness Low
	Inlet Flue Gas Temp Low

BOILER MONITORING

A boiler is an enclosed vessel that provides a means for combustion heat to be transferred into water until it becomes steam. Depending on industry requirements boiler generated steam can be utilized in various applications like power generation units, as a source of energy for unit operations, etc.

A survey shows that scope of savings in boiler house ranges from 28% to 46% for different industries. Also, it indicated steam consumption increased by 16% when boiler efficiency is dropped by 10%.

The boiler system comprises of Combustor, BFW system, Steam drum, Blowdown, and fuel system.

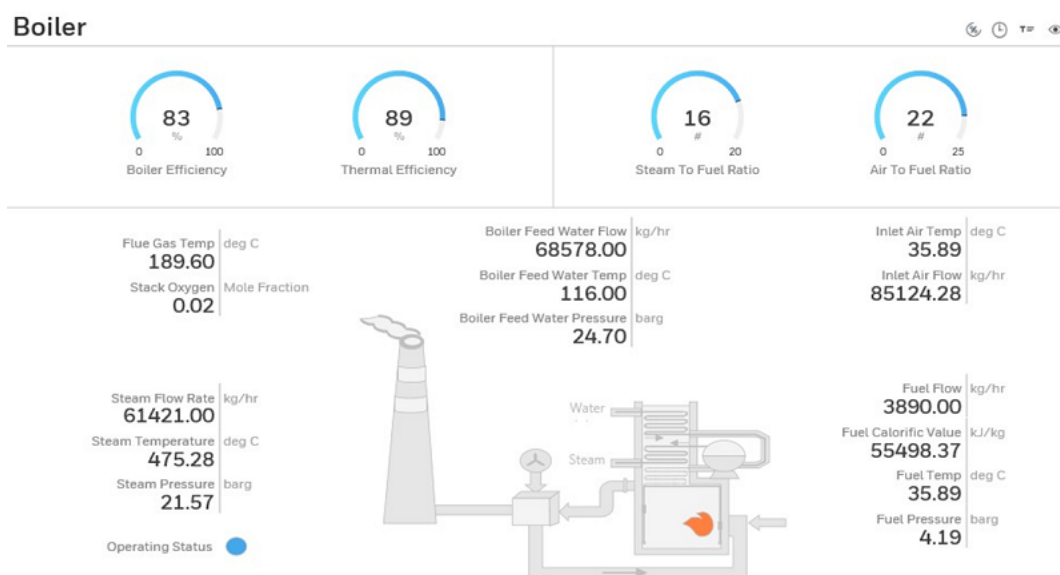
Combustor: The combustor is used to burn fuel with air to generate heat which will be used to heat boiler feed water to generate steam or superheated steam.

BFW system: The boiler feed water system provides water to the boiler and regulates it automatically to meet the steam demand.

Steam Drum: The steam drum collects and controls the steam produced in the boiler. Steam is directed through a piping system to the point of use. Throughout the system, steam pressure is regulated using valves and checked with steam pressure gauges.

Blowdown: Boiler blowdown is the removal of water from a boiler. Its purpose is to control boiler water parameters within prescribed limits to minimize scale, corrosion, carryover, and other specific problems. Blowdown is also used to remove suspended solids present in the system. These solids are caused by feedwater contamination, by internal chemical treatment precipitates, or by exceeding the solubility limits of otherwise soluble salts.

The fuel system includes all equipment used to provide fuel to generate the necessary heat.



Monitoring Basic machine health data in addition to boiler operating conditions helps to maintain reliable boiler operations. There are three areas that help to monitor boiler performance to increase availability and decrease energy costs.

1. Process and condition monitoring
2. Energy monitoring
3. Performance and Health monitoring

28.1 Boiler Monitoring - Performance Monitoring

The following measured parameters are monitored as it has following impact on performance and health of the boiler if not operated in operating range.

Feed water flow: A sudden increase or decrease in Feed water flow at constant fuel firing changes generation of steam flow which changes the pressure in the steam drum and boiler circuit. The change in pressure will cause a change in both the boiling point and density of the water and steam.

Feed water Temperature: Improper heating of feed water results in lower boiler efficiency, incomplete mechanical deaeration, and can cause cracking and catastrophic failure to the feedwater lines, boiler shell, and boiler tubes through thermal shocking. Low temperatures can affect the boiler feed water quality as well, as it will result in higher levels of dissolved gases, oxygen and carbon dioxide, that can corrode the internal metal surfaces the water comes in contact with. Thermal shock also occurs any time there is a sudden change in temperature of a metal or uneven temperatures throughout the boiler. Pre- heating the feed water to achieve an ideal boiler feed water temperature will allow the boiler to run most efficiently as the heat input required to raise the temperature of the water up is reduced.

Fuel Flow: Fuel flow directly impact generation of steam in boiler. Higher the fuel flow in furnace increase steam pressure and vice-versa.

Flue Temperature: Flue gas temperature measures the temperature of the combustion gases when they leave the boiler. If the flue gas temperature is high, it suggests the heat created by the boiler isn't being effectively used to generate steam. In other words, a high flue gas temperature suggests heat is being lost.

Air Temperature at boiler inlet: The temperature of the combustion air entering the boiler can have a relatively noticeable effect on industrial boiler efficiency. It can also have an impact on industrial boiler efficiency calculations due to it affecting the net stack temperature. Net stack temperature is the difference between the flue gas temp and the boiler inlet air temperature. Although it's tempting to minimize ambient temperature with hopes of lowering flue gas temp, a 40-degree alteration in ambient temperature can affect industrial boiler efficiency by one percent or more. Achieving the optimum balance is a delicate situation that involves strategic consideration.

Steam Pressure: The pressure of generated steam directly impact generated steam temperature. To achieve more steam temperature, pressure of steam drum need to increase to achieve desired steam temperature.

Stack O2 %: The stack O2 content reflects the amount of excess air supplied to the burner. The higher the excess air, the lower the boiler efficiency. The O2 analyzer allows the boiler operator to optimize boiler fuel consumption.

Deaerator drum level: If the water in deaerator storage tank level increases then the pressure in the deaerator increases which effects the inlet steam flow. This affects heating of inlet water which reduces the temperature of the water. So the removal of dissolved gasses will be disturbed. The low level may not have significance affect on the deaeration process but due to this low level the outlet flow may reduce which is delivered to steam generator. There will be a trip signal to BFP on deaerator low level. So, the deaerator level should be maintained.

Conductivity: Conductivity measurement is key parameter for boiler control, as it directly relates to energy savings. Lack of conductivity monitoring leads to operational problems like scaling inside the boiler, necessitating increased fuel consumption. Conductivity monitoring is important for minimizing the mineral level in the boiler by replacing process water with fresh water. This removal of water is referred to as blowdown. Conductivity monitoring also helps improve condensate recovery.

Blowdown Flow: Blowdown is an important process, excessive blowdown should be prevented to avoid wasting water and energy. Increase in blowdown flow decreases the boiler efficiency as it contributes to heat losses.

Process Performance Overview: Process Performance monitoring view in APM provides a list of the asset attributes and their details, including the current value. This list can be used to review and update asset attribute information, as well as to

open an Asset Trend to view more details. It also highlights (in red color) the attributes if deviates from the high or low limits.

28.1.1 Boiler Monitoring - Model Inputs

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Air Flow Rate	Air Flow Rate (Input)	kg/hr	Dynamic	Process	Field Input
2	Air Inlet Temperature	Air Inlet Temperature (Input)	°C	Dynamic	Process	Field Input
3	BFW Flow Rate	BFW Flow Rate (Input)	kg/hr	Dynamic	Process	Field Input
4	BFW Pressure	BFW Pressure (Input)	barg	Measured	Process	Field Input
5	BFW Temperature	BFW Temperature (Input)	°C	Calculated	Process	Field Input
6	BFW Stream_In	Asset Connection BFW Stream Inputs		Static	Connection	
7	Blowdown Rate	Blowdown Rate (Input)	kg/hr	Dynamic	Process	Field Input
8	BlowdownStream_In	Asset Connection Blowdown Stream Inputs		Static	Connection	
9	Boiler Drum Level	Boiler Drum Level (Input)	%	Dynamic	Process	Field Input
10	Combustor_In	Asset Connection Combustor Inputs		Static	Connection	
11	Conductivity	Conductivity (Input)		Dynamic	Process	Field Input
12	Deaerator Drum Level	Deaerator Drum Level (Input)	%	Dynamic	Process	Field Input
13	Dissolved Oxygen	Dissolved Oxygen (Input)		Dynamic	Process	Field Input
14	Enthalpy of Feed water	Enthalpy of Feed water (Input)	kJ/kg	Dynamic	Process	Field Input
15	Flue Gas Temp	Flue Gas Temp (Input)	°C	Dynamic	Process	Field Input
16	Fuel Flow Rate	Fuel Flow Rate (Input)	kg/hr	Dynamic	Process	Field Input
17	Furnace Pressure	Furnace Pressure (Input)	barg	Dynamic	Process	Field Input
18	Stack Oxygen	Stack Oxygen (Input)	Mole Fraction	Dynamic	Process	Field Input
19	Enthalpy of Steam	Enthalpy of Steam (Input)	kJ/kg	Dynamic	Process	Field Input
20	Steam Flow	Steam Flow (Input)	kg/hr	Dynamic	Process	Field Input
21	Steam Pressure	Steam Pressure (Input)	barg	Dynamic	Process	Field Input
22	Steam Temperature	Steam Temperature (Input)	°C	Dynamic	Process	Field Input

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
23	Steam Stream_In	Asset Connection Steam Stream Inputs		Static	Connection	
24	UAnalyzer_In	Asset Connection Ultimate Analyzer Inputs		Static	Connection	
25	Fuel Calorific Value	Fuel Calorific Value (Input)	kJ/kg	Dynamic	Internal	Internally calculated
26	Fuel Moisture	Fuel Moisture (Input)	%	Calculated	Internal	Internally calculated
27	Fuel Pressure	Fuel Pressure (Input)	barg	Dynamic	Internal	Internally calculated
28	FuelStream_In	Asset Connection FuelStream Inputs		Static	Internal	Constant
29	Enthalpy at saturated Pressure	Enthalpy at saturated Pressure(Input)	kJ/kg	Dynamic	Internal	Internally calculated

28.1.2 Boiler Monitoring - KPIs

The following indicators help to know the overall performance of the boiler

Boiler Efficiency by direct method

Boiler Efficiency is defined as percentage of heat input that is effectively utilized to generate steam.

$$\eta = \frac{\text{Energy output}}{\text{Energy input}} \times 100$$

$$\eta = \frac{Q(H-h)}{q \cdot \text{GCV}} \times 100$$

Where,

Q= Quantity of steam generated (kg/hr)

H= Enthalpy of steam (Kcal/kg)

h= Enthalpy of water (Kcal/kg)

GCV= Gross calorific value for the fuel (Kcal/kg)

q= Mass of fuel burnt (kg/hr)

Thermal Efficiency of Fuel

Thermal Efficiency is the percentage of heat available to heat process fluid with respect to heat supplied. It accounts for losses due to air and fuel preheating, air relative humidity, stack, and radiation losses. Process side heat balance is not considered in furnace efficiency calculations. Further detailed calculations are available in the Furnace document under [Key performance indicator](#) section.

$$\text{Thermal Efficiency} = \frac{\text{Total Heat Input} - \text{Stack Heat Loss} - \text{Radiation Heat Loss}}{\text{Total Heat Input}} * 100$$

$$e = \frac{(\text{LHV} + \text{Ha} + \text{Hf}) - \text{Qs} - \text{Qr}}{(\text{LHV} + \text{Ha} + \text{Hf})} * 100$$

Where,

e= Net thermal efficiency (%)

LHV= Heat input or lower heating value of the fuel (BTU/lb of fuel)

Ha= Heat input in the form of sensible heat of air (BTU/lb of fuel)

Hf= Heat input in the form of sensible heat of air (BTU/lb of fuel)

Qs= Calculated stack heat losses (BTU/lb of fuel)

Qr= Assumed radiation heat loss (BTU/lb of fuel)

Steam to Fuel Ratio

Higher steam to fuel ratio indicates good performance of the boiler and vice versa.

Steam to fuel ratio= Steam Flow(kg/hr)/Fuel Flow(kg/hr)

Air to Fuel Ratio

It is measured in range and this range varies with fuel type. High Air to fuel ratio contributes to heat losses and low Air to fuel ratio contributes to incomplete combustion.

Air to Fuel Ratio = Air Flow(kg/hr)/Fuel Flow(kg/hr)

Water Balance

Water balance shows the quantity of water flow in and out of system with steam generation.

Water Balance =(BFW_Flow-Steam_Flow-BlowdownRate)

28.1.3 Boiler Monitoring - Model Outputs

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Air to Fuel Ratio	Air to Fuel Ratio (Output)		Dynamic	Performance	Calculated
2	BlowDown Loss	Blow Down Loss (Output)	kJ/hr	Dynamic	Energy	Calculated
3	Boiler Efficiency	Boiler Efficiency (Output)	%	Dynamic	Performance	Calculated
4	Steam to Fuel Ratio	Steam to Fuel Ratio (Output)		Dynamic	Performance	Calculated
5	Thermal Efficiency	Thermal Efficiency Indirect Method (Output)	%	Dynamic	Performance	Calculated
6	Water Balance	Water Balance(Output)	kg/hr	Dynamic	Performance	Calculated

28.2 Boiler Monitoring - Fault Model

In Asset Performance Management, you can configure faults, symptoms, and fault models on boiler performance output to diagnose the problems early. The fault management can inform operators of any violations of the nominal boiler operating range and deviations from the expected characteristic and make this data available for further processing through the historian. The fault symptoms or models are used to:

1. Warn against potential damage to boiler under unfavorable operating conditions.
2. Provide early warning of damages to the boiler.

The following messages can be generated for process alarms or notifications to warn operating personnel of unfavorable operating conditions

1. **Steam to Fuel Ratio Low:** It indicates lower steam generation for higher fuel consumption. This can happen due to low BFW temperature or heat losses from flue gases released at stack.
2. **Flue Gas Exit Temp High:** It indicates heat losses to flue gas which result in lower boiler efficiency. It causes due to excess air flow.
3. **Boiler Efficiency Low:** It indicates lower steam generation at higher energy input.
4. **Thermal Efficiency Low:** It indicate supplied energy in form of fuel is not fully utilized to generate steam.
5. **Air to Fuel Ratio low:** Lower air to fuel ratio indicates incomplete combustion of fuel.
6. **NOx Emission High:** It indicate higher concentration of nitrogenous compound in fuel gas. This indication can also help to monitor environmental norms and limit NOx emission.
7. **CO Emission High:** Higher CO Emission indicate incomplete combustion of fuel result in lower boiler efficiency.
8. **Boiler drum level low:** Sudden increaser or decrease in boiler drum impact generation of steam flow which Impact pressure of steam drum. This can be happened due to fluctuation in BFW flow.
9. **Steam Drum Priming:** It causes due to sudden change in boiler load or high level in steam drum. It can nullify by decreasing boiler load.
10. **Steam Drum Foaming:** It causes when Boiler feed water contaminated with oil. It can nullify by removing contaminated water vie blowdown.
11. **Carry over from steam drum:** It impact quality of steam generated, it causes due to sudden change in boiler load.
12. **Steam Drum Abnormal Level:** Sudden increase or decrease in steam drum level affect pressure of steam drum.
13. **Abnormal Combustion:** Abnormal combustion impact heat input to BFW which Impact generation of steam.
14. **Deaerator Level Low:** Low water level in deaerator drum decreased the pressure in the deaerator drum which effects the inlet steam flow. This affects heating of inlet water which increases the temperature of the water.
15. **Dissolved Oxygen High:** Dissolved oxygen is the major cause of boiler corrosion and carbon dioxide makes the water acidic which harm boiler.

Below table illustrates the list of faults and symptoms configured for a boiler to identify possible anomalies that affects the performance or health of a boiler.

Fault Name	Associated Symptoms
Energy Consumption High	Blowdown Losses High
	Steam Temperature High
Steam Drum Instability	Steam Drum Level Low
	Water Balance High
Health Degradation Warning	Conductivity Low
	Dissolved Oxygen High

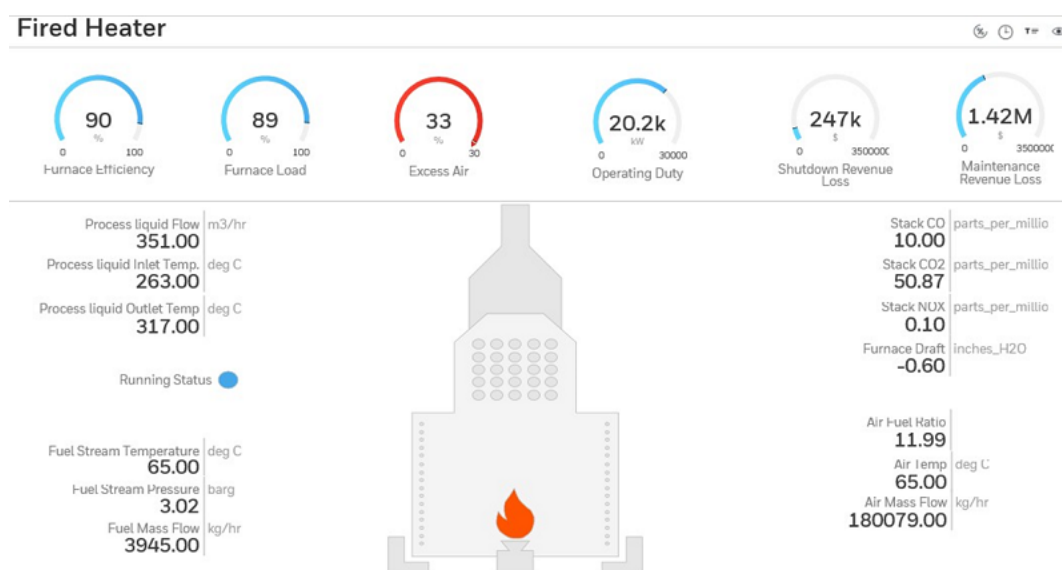
Fault Name	Associated Symptoms
Performance Degradation Warning	Boiler Efficiency Low
	Steam to Fuel Ratio Low
	Potential Water Carry-over
	Steam Drum Pressure Low

FIRED HEATER MONITORING

Fired heater heats the process fluid to a high temperature from the heat generated by the combustion of fuel. An industrial fired heater comprises below segments:

Radiant section: In this section of a heater, the portion of heat is transferred to the tubes by radiation from the flame. The Tubes can be vertical or horizontal which are placed a distance away from the insulation so radiation can be reflected to the back of the tubes to maintain a uniform tube wall temperature.

Convection section: In this portion of a heater, the portion of heat is transferred to the tubes by radiation from the flame. Heat transfer takes place by convection here, and the tubes are finned to increase heat transfer.



Burner: Introduces fuel and air into a heater at the desired velocities and concentration to establish and maintain proper ignition and combustion.

Stack: The flue gas stack is a cylindrical structure at the top of all the heat transfer chambers, which is used to discharge flue gas into the atmosphere.

Monitoring Basic machine health data in addition to fired heater operating conditions helps to maintain reliable fired heater operations. There are three areas that help to monitor fired heater performance to increase the availability and decrease the energy costs.

1. Process and condition monitoring
2. Energy monitoring
3. Performance and Health monitoring

29.1 Fired Heater Monitoring - Performance Monitoring

The following measured parameters are monitored as it has following impact on performance and health of the fired heater if not operated in operating range.

Air Outlet Temperature: It depends upon the flue gas inlet temperature to air preheater, higher the flue gas temperature increases inlet temperature of air to fired heater which helps in thermal efficiency of fired heater.

Flue Gas Temp/ Stack Temp: This indicates exit gas/stack gas temperature of the fired heater. More the temperature indicates in-efficient use of combustion energy.

Fuel gas composition: The change of the composition of hydrocarbons and inert gases like CO₂ and N₂ influences the ignitability and the combustion behavior of the fuel gas mixture. The lower quality fuels results in poor combustion, misfire or engine knocking. Changing combustion parameters influence the exhaust emissions, the cylinder peak pressure and the knock margin. High frequency fluctuations have an impact on the fired heater load controller and can result in unstable operation and varying emissions levels.

Draft pressure: Optimum operation requires that excess air in the flue gas entering the convection section be minimized and there should be a very small negative pressure at the convection section inlet. Excessive draft, either positive pressure or negative pressure, can lead to severe problems.

NO_x Emission: Stringent emission limits require greater control of NO_x and other stack components. Operating the heater at optimum efficiency, with low excess air firing is the simplest and least expensive way to reduce NO_x emissions. The high temperature in the flame and radiant section, together with combustion turbulence at the burners, causes reaction of oxygen with nitrogen forming NO (nitric oxide) and NO₂ (nitrogen dioxide). Low excess air firing (LEA) is the simplest way to reduce NO_x formation and improve efficiency. The more excess air, the more oxygen is available to produce NO_x. Low NO_x burners stage the combustion reducing the available oxygen, temperature or residence time to limit the formation of NO_x.

CO Emission: This indicates current CO ppm levels in the exit gases of the fired heater. More the level indicates incomplete combustion of fuel and lower efficiency of the fired heater. More CO levels adds risk for plant and environment safety as well.

Stack O₂ %: This indicates Air or O₂ percent in the exit gases of the fired heater. More O₂ indicates that the fired heater is operating at more AF (air to fuel) ratio. More O₂ content than desired indicates losses to preheat excess air supply to combustion temperature. Less O₂ indicates that the fire heater is operating at lower AF (air to fuel) ratio. This can cause incomplete combustion of fuel and reduce thermal efficiency of the fired heater.

29.1.1 Fired Heater Monitoring - Model Inputs

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Running Status	Running Status of Asset (Input)		Dynamic	Process	Field Input
2	APH Flue Gas Pressure	Air Preheater Section Flue Gas Pressure (Input)	barg	Dynamic	Process	Field Input
3	APH Flue Gas Inlet Temp.	Air Preheater Section Flue Gas Inlet Temp (Input)	°C	Dynamic	Process	Field Input
4	APH Flue Gas Outlet Temp.	Air Preheater Section Flue Gas Outlet Temp(Input)	°C	Dynamic	Process	Field Input
5	Air through APH Flow	Air through Air Preheater Flow (Input)	m ³ /hr	Dynamic	Process	Field Input
6	APH Tube Inlet Pressure	Air Preheater Tube Inlet Pressure(Input)	barg	Dynamic	Process	Field Input
7	APH Tube Outlet Pressure	Air Preheater Tube Outlet Pressure (Input)	barg	Dynamic	Process	Field Input
8	APH Tube Inlet Temp	Air Preheater Tube Inlet Temp (Input)	°C	Dynamic	Process	Field Input

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
9	APH Tube Outlet Temp	Air Preheater Tube Outlet Temp(Input)	°C	Dynamic	Process	Field Input
10	Air through APH Flow Diverter Valve Opening	Air through Air Preheater Flow Diverter Valve Opening(Input)		Dynamic	Process	Field Input
11	Air N2	Percentage Volume Air N2 (Input)	%	Dynamic	Process	Field Input
12	Air O2	Percentage Volume Air O2 (Input)	%	Dynamic	Process	Field Input
13	Cost of HEX Cleaning Operation	Cost of HEX Cleaning Operation (Input)	\$	Dynamic	Process	Field Input
14	Duration of Cleaning Operation	Duration of Cleaning Operation(Input)	hour	Dynamic	Process	Field Input
15	Combustion Air Flow	Combustion Air Flow (Input)	m ³ /hr	Dynamic	Process	Field Input
16	Combustion Air Temperature	Combustion Air Temperature (Input)	°C	Dynamic	Process	Field Input
17	CS Flue Gas Pressure	Convection Section Flue Gas Pressure (Input)	barg	Dynamic	Process	Field Input
18	CS Inlet Temp.	Convection Section Inlet Temp.(Input)	°C	Dynamic	Process	Field Input
19	CS Outlet Temp.	Convection Section Outlet Temp.(Input)	°C	Dynamic	Process	Field Input
20	CS Tube Stream Flow	Convection Tube Stream Flow (Input)	m ³ /hr	Dynamic	Process	Field Input
21	CS Tube Inlet Pressure	Convection Tube Inlet Pressure(Input)	barg	Dynamic	Process	Field Input
22	CS Tube Outlet Pressure	Convection Tube Outlet Pressure (Input)	barg	Dynamic	Process	Field Input
23	CS Tube Inlet Temp	Convection Tube Inlet Temp (Input)	°C	Dynamic	Process	Field Input
24	CS Tube Outlet Temp	Convection Tube Outlet Temp (Input)	°C	Dynamic	Process	Field Input
25	Flue Gas OUT Analysis CO	Flue Gas OUT Analysis CO (Input)	%	Dynamic	Process	Field Input
26	Flue Gas OUT Analysis CO2	Flue Gas OUT Analysis CO2 (Input)	%	Dynamic	Process	Field Input
27	Flue Gas OUT Analysis N2 (atm)	Flue Gas OUT Analysis N2 (atm) (Input)	%	Dynamic	Process	Field Input
28	Flue Gas OUT Analysis NOX	Flue Gas OUT Analysis NOX (Input)	%	Dynamic	Process	Field Input

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
29	Flue Gas OUT Analysis O2	Flue Gas OUT Analysis O2 (Input)	%	Dynamic	Process	Field Input
30	Gas Fuel Heating Value (LHV or HHV)	Gas Fuel Heating Value (LHV or HHV) (Input)	kJ/kg	Dynamic	Process	Field Input
31	Gas Fuel Flow	Gas Fuel Flow(Input)	m ³ /hr	Dynamic	Process	Field Input
32	Cost of Fuel	Cost of Fuel(Input)		Dynamic	Process	Field Input
33	Fuel Higher Heating Value	Fuel Higher Heating Value (Input)	MJ/kg	Dynamic	Process	Field Input
34	Draft Pressure	Furnace Pressure (Input)	inches_H ₂ O	Dynamic	Process	Field Input
35	RS Flue Gas Pressure	Radiation Section Flue Gas Pressure (Input)	barg	Dynamic	Process	Field Input
36	RS Flue Gas Inlet Temp	Radiation Section Flue Gas Inlet Temp (Input)	°C	Dynamic	Process	Field Input
37	RS Flue Gas Outlet Temp	Radiation Section Flue Gas Outlet Temp(Input)	°C	Dynamic	Process	Field Input
38	RS Tube Stream Flow	Radiation Tube Stream Flow (Input)		Dynamic	Process	Field Input
39	RS Tube Inlet Pressure	Radiation Tube Inlet Pressure (Input)	barg	Dynamic	Process	Field Input
40	RS Tube Outlet Pressure	Radiation Tube Outlet Pressure (Input)	barg	Dynamic	Process	Field Input
41	RS Tube Inlet Temp.	Radiation Tube Inlet Temp. (Input)	°C	Dynamic	Process	Field Input
42	RS Tube Outlet Temp.	Radiation Tube Outlet Temp. (Input)	°C	Dynamic	Process	Field Input
43	Stack Dampner Opening	Stack Dampner Opening (Input)	%	Dynamic	Process	Field Input
44	Furnace Skin Temperature s	Furnace Skin Temperatures (Input)	°C	Dynamic	Process	Field Input
45	APH Tube Area	Air Preheater Tube Area (Input)	m ²	Dynamic	Design	Constant from Manufacturer datasheet
46	APH Tube Product Density	Air Preheater Tube Product Density (Input)	kg/m ³	Dynamic	Design	Constant from Manufacturer datasheet
47	APH Heat Transfer Coeff.	Air Preheater Section Heat Transfer Coeff. (Input)	W/m ² * °C	Dynamic	Design	Constant from Manufacturer datasheet
48	CS Design Tube FLOW	Convection Section Design	kg/hr	Dynamic	Design	Constant from

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
		Tube Flow (Input)				Manufacturer datasheet
49	CS Tube Area	Convection Tube Area (Input)	m ²	Dynamic	Design	Constant from Manufacturer datasheet
50	CS Tube Product Density	Convection Tube Product Density (Input)	kg/m ³	Dynamic	Design	Constant from Manufacturer datasheet
51	CS Heat Transfer Coeff	Convection Section Heat Transfer Coeff (Input)	W/m ² * °C	Dynamic	Design	Constant from Manufacturer datasheet
52	HEX Design Duty	HEX Design Duty (Input)	kW	Dynamic	Design	Constant from Manufacturer datasheet
53	Fuel Gas to Product Ratio	Furnace Fuel Gas to Product Ratio (Input)		Dynamic	Design	Constant from Manufacturer datasheet
54	HEX Load Constant A	HEX Load Constant A (Input)		Dynamic	Design	Constant from Manufacturer datasheet
55	HEX Load Constant B	HEX Load Constant B (Input)		Dynamic	Design	Constant from Manufacturer datasheet
56	HEX Load Constant C	HEX Load Constant C (Input)		Dynamic	Design	Constant from Manufacturer datasheet
57	RS Design Tube Flow	Radiant Section Design Tube Flow (Input)	kg/hr	Dynamic	Design	Constant from Manufacturer datasheet
58	RS Tube Area	Radiation Tube Area (Input)	m ²	Dynamic	Design	Constant from Manufacturer datasheet
59	RS Tube Product Density	Radiation Tube Product Density (Input)	kg/m ³	Dynamic	Design	Constant from Manufacturer datasheet
60	RS Heat Transfer Coeff.	Radiation Section Heat Transfer Coeff (Input)	W/m ² * °C	Dynamic	Design	Constant from Manufacturer datasheet
61	Gas Fuel C ₂ H ₆	Gas Fuel C ₂ H ₆ (Input)	%	Dynamic	Fuel Composition	Lab Analysis
62	Gas Fuel C ₃ H ₈	Gas Fuel C ₃ H ₈ (Input)	%	Dynamic	Fuel Composition	Lab Analysis
63	Gas Fuel C ₄ H ₁₀ (n/i-	Gas Fuel C ₄ H ₁₀ (n/i- C ₄ H ₁₀)	%	Dynamic	Fuel	Lab Analysis

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
	C4H10)	(Input)			Composition	
64	Gas Fuel C4H8	Gas Fuel C4H8 (Input)	%	Dynamic	Fuel Composition	Lab Analysis
65	Gas Fuel C5H12	Gas Fuel C5H12 (Input)	%	Dynamic	Fuel Composition	Lab Analysis
66	Gas Fuel C6H14	Gas Fuel C6H14 (Input)	%	Dynamic	Fuel Composition	Lab Analysis
67	Gas Fuel C6H6	Gas Fuel C6H6 (Input)	%	Dynamic	Fuel Composition	Lab Analysis
68	Gas Fuel C7H8	Gas Fuel C7H8 (Input)	%	Dynamic	Fuel Composition	Lab Analysis
69	Gas Fuel CH4	Gas Fuel CH4(Input)	%	Dynamic	Fuel Composition	Lab Analysis
70	Gas Fuel CO	Gas Fuel CO(Input)	%	Dynamic	Fuel Composition	Lab Analysis
71	Gas Fuel CO2	Gas Fuel CO2(Input)	%	Dynamic	Fuel Composition	Lab Analysis
72	Gas Fuel H2	Gas Fuel H2(Input)	%	Dynamic	Fuel Composition	Lab Analysis
73	Gas Fuel H2Ov	Gas Fuel H2Ov (Input)	%	Dynamic	Fuel Composition	Lab Analysis
74	Gas Fuel H2S	Gas Fuel H2S(Input)	%	Dynamic	Fuel Composition	Lab Analysis
75	Gas Fuel N2	Gas Fuel N2(Input)	%	Dynamic	Fuel Composition	Lab Analysis
76	Gas Fuel O2	Gas Fuel O2(Input)	%	Dynamic	Fuel Composition	Lab Analysis
78	Gas Fuel SO2	Gas Fuel SO2(Input)	%	Dynamic	Fuel Composition	Lab Analysis
79	APH Tube Inlet Specific Heat	Air Preheater Tube Inlet Specific Heat (Input)	J/kg*°C	Dynamic	Thermo Properties	Constant
80	APH Tube Outlet Specific Heat	Air Preheater Tube Outlet Specific Heat (Input)	J/kg*°C	Dynamic	Thermo Properties	Constant
81	CS Tube Inlet Specific Heat	Convection Tube Inlet Specific Heat (Input)	J/kg*°C	Dynamic	Thermo Properties	Constant
82	CS Tube Outlet Specific Heat	Convection Tube Outlet Specific Heat (Input)	J/kg*°C	Dynamic	Thermo Properties	Constant
83	RS Tube Inlet Specific Heat	Radiation Tube Inlet Specific Heat (Input)	J/kg*°C	Dynamic	Thermo Properties	Constant
84	RS Tube Outlet	Radiation Tube Outlet Specific	J/kg*°C	Dynamic	Thermo	Constant

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
	Specific Heat	Heat (Input)			Properties	
85	Air Inlet Specific Heat	Air Inlet Specific Heat (Internal calculation)	kJ/kg-C	Dynamic	Thermo Properties	Calculated
86	Air Outlet Specific Heat	Air Outlet Specific Heat (Internal calculation)	kJ/kg-C	Dynamic	Thermo Properties	Calculated
87	Combustion Air Density	Combustion Air Density (Internal calculation)	kg/m ³	Dynamic	Thermo Properties	Calculated

The Asset Performance Management process and equipment monitoring models require thermodynamic properties such as enthalpy, entropy, density, and specific heat to be calculated.

There are two options available to compute thermodynamic properties:

1. Native Asset Performance Management Thermodynamic models uses Be
2. Linked UniSim Thermodynamic models

29.1.2 Fired Heater Monitoring - KPIs

The following indicators help to know the overall performance of the Fired Heater

Furnace Efficiency

The efficiency of a furnace is the ratio of useful output to heat input. The furnace efficiency can be determined by both direct and indirect method.

Excess Air

For fired heater safety and maintenance needs, additional air beyond the theoretical "air to fuel ratio" needs to be added this is called as "excess air." With fired heater combustion, if some excess air is not added to the combustion process, unburned fuel, soot, smoke, and carbon monoxide exhaust will create additional emissions and surface fouling.

Air to fuel ratio

The ratio of air to fuel is the most efficient for combustion in fired heater, the quantity of fuel and air would be in a perfect ratio to provide perfect combustion with no unused fuel or air.

Convection section heat transfer efficiency

The convection section heat transfer efficiency calculated using terminal temperature difference.

Radiation section heat transfer efficiency

The radiation section heat transfer efficiency calculated using terminal temperature difference.

Shutdown Revenue loss

Shutdown revenue loss is calculated based on Fuel Compensation for Furnace duty loss in various section multiplied by the unit rate of fired fuel.

Maintenance revenue loss

Maintenance revenue loss is calculated based on revenue loosed for cleaning assets.

Performance Overview

Equipment Key Performance Indicators (KPIs) view in APM provides a list of KPIs and their details, including the current value. This list can be used to review and update asset attribute information, as well as to open an Asset Trend to view more details. It also highlights (in red color) the attributes if deviates from the high or low limits.

29.1.3 Fired Heater Monitoring - Model Outputs

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Air to Fuel Ratio	Air to Fuel Gas Stoichiometric Ratio (Output)		Dynamic	Performance	Calculated
2	Furnace Load	Furnace Load (Output)	%	Dynamic	Performance	Calculated
3	Shutdown Revenue Loss	Shutdown Revenue Loss (Output)	kg/hr	Dynamic	Performance	Calculated
4	CS Heat Transfer Efficiency	Convection Section Heat Transfer Efficiency (Output)	%	Dynamic	Performance	Calculated
5	Furnace Efficiency	Furnace Input/Output Efficiency (Output)	%	Dynamic	Performance	Calculated
6	Furnace Revenue Loss	Furnace Revenue Loss(Output)	USD	Dynamic	Performance	Calculated
7	Maintenance Revenue Loss	Furnace Maintenance Revenue Loss (Output)	USD	Dynamic	Performance	Calculated
8	Excess Air	Excess Air(Output)	%	Dynamic	Performance	Calculated
9	RS Heat Transfer Efficiency	Radiation Section Heat Transfer Efficiency(Output)	%	Dynamic	Performance	Calculated
10	Furnace Expected Duty	Furnace Expected Duty (Output)	kW	Dynamic	Energy	Calculated
11	Total Heat Input	Total Heat Input (Output)	kW	Dynamic	Energy	Calculated
12	Total Heat Output	Total Heat Output (Output)	kW	Dynamic	Energy	Calculated
13	Duty Loss	Duty Loss (Output)	kW	Dynamic	Energy	Calculated
14	Fuel Compensation for Duty Loss	Fuel Compensation for Duty Loss (Output)	kg/hr	Dynamic	Energy	Calculated
15	CS Fouling Percentage	Convection Section Fouling Percentage (Output)	%	Dynamic	Energy	Calculated
16	CS Heat Flux	Convection Section Heat Flux (Output)	kilowatts/m ²	Dynamic	Energy	Calculated
17	Furnace Heat Flux	Furnace Heat Flux (Output)	kilowatts/m ²	Dynamic	Energy	Calculated
18	RS Fouling Percentage	Radiation Section Fouling Percentage (Output)	%	Dynamic	Energy	Calculated
19	RS Heat Flux	Radiant Section Heat Flux (Output)	kilowatts/m ²	Dynamic	Energy	Calculated

29.2 Fired Heater Monitoring - Fault Model

In APM, a fault model is built to identify possible anomalies that affects the performance or health of a Fired Heater. The fault management can inform operators of any violations of the nominal fired heater operating range and deviations from the expected characteristic and make this data available for further processing through the historian. The fault symptoms or models are used to:

1. Warn against potential damage to fired heater under unfavorable operating conditions.
2. Provide early warning of damages to the fired heater.

The following messages can be generated for process alarms or notifications to warn operating personnel of unfavorable operating conditions

1. **Low fired heater efficiency:** It is determined by means of the deviation of the current fired heater efficiency from the design efficiency.
2. **Incomplete combustion:** It indicate fuel is not completely utilized to generate heat in form of energy. It can be determining by stack CO, NOX and O2 concentration.
3. **Backfire:** It indicate pressure inside fired heater is higher which resist burning of fuel. This can be happed due to high draft pressure and low stack damper opening.
4. **Hotspot on tubes:** This determines by high skin temperature and can damage tubes of fired heater.

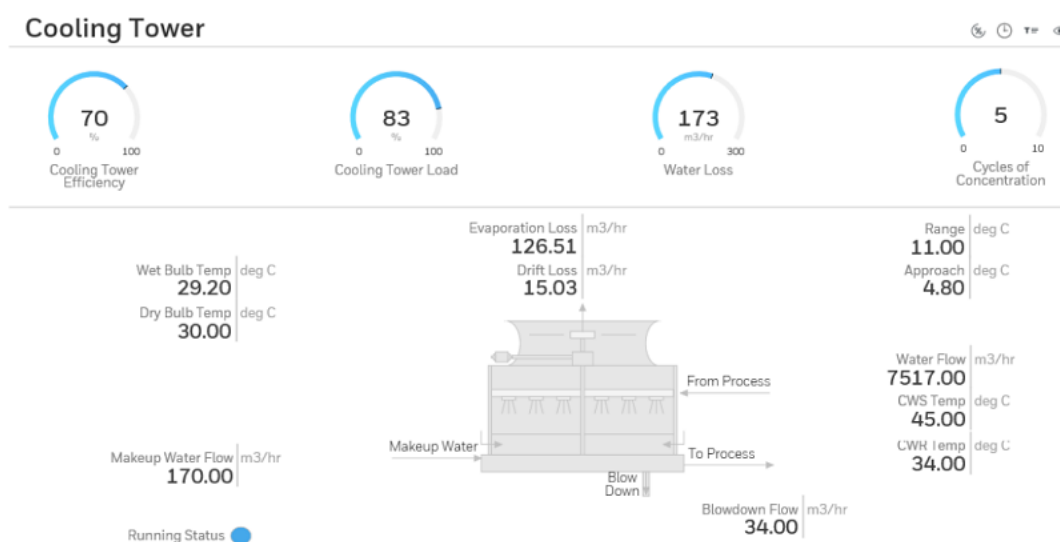
Below table illustrates the list of faults and symptoms configured for a fired heater.

Fault Name	Associated Symptom
Performance Degradation	Fired Heater Efficiency Low
	Radiation Section Efficiency Low
	Convection Section Efficiency Low
	Radiation Section Fouling High
	Convection Section Fouling High
	Oxygen Level High
Health Degradation	High Fuel Gas to Product Ratio
	Flame impingement
	HotSpot Warning
Backfire Warning	Furnace Internal Pressure High
	Stack Damper Closed
Cyclic Combustion	Flue Gas NOX Level High
	Flame Detection failure
Incomplete Combustion	Combustion Air Low
	O2 in Flue Gas Low
	CO in Flue Gas High

COOLING TOWER MONITORING

A cooling tower is a specialized heat exchanger in which, air and water are brought into direct contact with each other in order to reduce the water's temperature. During this process, a small volume of water is evaporated by reducing temperature of the water being circulated through the tower. Water heated by an industrial process or in an air-conditioning condenser is pumped to the cooling tower through pipes. The water sprays through nozzles and exposes as much water surface area as possible for maximum air-water contact. As the water flows through the cooling tower, it is exposed to air, which is pulled through the tower by the electric motor-driven fan. When the water and air meet, a small amount of water is evaporated, creating a cooling action. The cooled water is then pumped back to the condenser or process equipment where it absorbs heat. It will then be pumped back to the cooling tower to be cooled once again.

Cooling towers are a very important part of many chemical plants. Cooling tower performance monitoring plays an important role to predict the cleaning and maintenance schedules with fouling, scaling, leakage, etc. Monitoring Basic machine health data in addition to Cooling tower operating conditions helps to maintain reliable operations of a Cooling tower. There are two areas that help to monitor Cooling tower performance to increase the availability and decrease energy costs.



30.1 Cooling Tower Monitoring - Performance Monitoring

The following measured parameters are monitored as they have impact on performance and health of Cooling tower, if not maintained in operating range.

Air Inlet Temperature (dry bulb) : The Dry Bulb temperature, usually referred to as air temperature. The Dry Bulb Temperature refers basically to the ambient air temperature. It is called "Dry Bulb" because the air temperature is indicated by a thermometer not affected by the moisture of the air.

Wet Bulb Temperature : Wet bulb temperature is an important factor in performance of evaporative water-cooling equipment. It is a controlling factor from the aspect of minimum cold-water temperature to which water can be cooled by the evaporative method. Thus, the wet bulb temperature of the air entering the cooling tower determines operating temperature levels throughout the plant, process, or system. Theoretically, a cooling tower will cool water to the entering wet bulb temperature, when operating without a heat load.

Water Makeup Flow : As water evaporates in a cooling tower system, the water vapor enters the atmosphere while any dissolved solids remain behind, building concentration in the remaining water. Most water treatment systems control this concentration by bleeding off the high concentration water, which is made up by fresh city water with a lower solid concentration.

Process Performance Overview : Process Performance monitoring view in APM provides a list of the asset attributes and their details, including the current value. This list can be used to review and update asset attribute information, as well as to open an Asset Trend to view more details. It also highlights (in red color) the attributes if deviates from the high or low limits.

30.1.1 Cooling Tower Monitoring - Model Inputs

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Running Status	Operational Status of Asset (Input)		Dynamic	Process	Field Input
2	No of Running Fans	Actual No of Running Fans (Input)		Dynamic	Process	Field Input
3	Air Volume Flow Rate	Air Volume Flow Rate (Required Input)	m ³ /hr	Dynamic	Process	Field Input
4	Air Inlet Pressure	Air Inlet Pressure (atmospheric)	barg	Dynamic	Process	Field Input
5	Air Inlet Temperature (dry bulb)	Air Inlet Temperature (Required Input)	°C	Dynamic	Process	Field Input
6	Cooling Water Pressure	Cooling Water Pressure(Input)	barg	Dynamic	Process	Field Input
7	Water Blow down Flow Rate	Water Blow down Flow Rate (Required Input)	kg/hr	Dynamic	Process	Field Input
8	Wet Bulb Temperature	Wet Bulb Temperature (Required Input)	°C	Dynamic	Process	Field Input
9	Water Circulation Flow Rate	Water Circulation Flow Rate (Required Input)	kg/hr	Dynamic	Process	Field Input
10	Water Makeup Flow Rate	Water Makeup Flow Rate (Required Input)	kg/hr	Dynamic	Process	Field Input
11	Cold Water Outlet pH	Cold Water Outlet pH(Input)		Dynamic	Process	Field Input
12	Hot Water Inlet Temperature	Hot Water Inlet Temperature (Required Input)	°C	Dynamic	Process	Field Input
13	Cold Water Outlet Temperature	Cold Water Outlet Temperature (Required Input)	°C	Dynamic	Process	Field Input
14	Air Inlet Sp Heat	Air Inlet Specific Heat(Required Input, if fluid property data is not given)	kJ/kg-C	Dynamic	Thermo Properties	Field Input
15	Latent Heat	Latent Heat (Required Input, if fluid property data is not given)		Dynamic	Thermo Properties	Field Input
16	Actual Air Density	Actual Air Density (Required Input, if fluid property data is not given)	kg/m ³	Dynamic	Thermo Properties	Field Input

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
17	Liquid to Gas Ratio	Liquid to Gas Ratio(Required Input)		Static	Design	Constant
18	Design Air Inlet Temperature	Design/Rated Air Inlet Temperature (Required Input)	°C	Static	Design	Constant
19	Design Rated Air Outlet Temperature	Design/Rated Air Outlet Temperature (Required Input)	°C	Static	Design	Constant
20	Rated Duty PerFan	Rated Duty Per Fan(Required Input)	kW	Static	Design	Constant
21	Rated Heat Duty	Rated Heat Duty (Required Input)	kW	Static	Design	Constant
22	Design Hot Water Inlet Temperature	Design/Rated Hot Water Inlet Temperature (Required Input)	°C	Static	Design	Constant
23	Design Cold Water Outlet Temperature	Design/Rated Cold Water Outlet Temperature (Required Input)	°C	Static	Design	Constant
24	Heat Transfer Area per volume unit	Heat Transfer Area per volume unit (Required Input)	m ² / m ³	Static	Design	Constant
25	Heat Transfer Volume	Heat Transfer Volume (Required Input)	m ³	Static	Design	Constant
26	Hot Water Inlet Sp Heat	Hot Water Inlet Specific Heat (Required Input)	kJ/kg-C	Static	Design	Constant
27	Water in Air Loss Factor	Water in Air Loss Factor(Required Input)	%	Static	Design	Constant
28	Water Density	Water Density (Required Input)	kg/m ³	Static	Design	Constant

30.1.2 Cooling Tower Monitoring - KPIs

Approach

Approach is the difference between the cooling tower outlet cold water temperature and ambient wet bulb temperature.

Cycles of Concentration

Cycles of concentration is the ratio of dissolved solids in circulating water to the dissolved solids in makeup water.

Range

It is the difference between the cooling tower water inlet and outlet temperature.

Evaporation Loss

Evaporation loss is the water quantity evaporated for cooling duty and, theoretically, for every 10,00,000-kCal heat rejected, evaporation quantity works out to 1.8 m³. An empirical relation used often is:

Evaporation Loss (m³/hr) = 0.00085 x 1.8 x circulation rate (m³/hr) x (T₁-T₂) T₁-T₂ = Temperature difference between inlet and outlet water.

Cooling Tower Efficiency

Cooling tower effectiveness (in percentage) is the ratio of range, to the ideal range,

i.e., difference between cooling water inlet temperature and ambient wet bulb temperature, or in other words it is = Range / (Range + Approach).

Blow down loss

Blow down losses depend upon cycles of concentration and the evaporation losses and is given by relation:

$$\text{Blow Down} = \text{Evaporation Loss} / (\text{Cycles of Concentration} - 1)$$

Cooling Tower Duty

Cooling Tower Duty is the heat rejected in kCal/hr or TR, given as product of mass flow rate of water, specific heat and temperature difference.

30.1.3 Cooling Tower Monitoring - Model Outputs

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Cooling Tower Efficiency	Cooling Tower Efficiency (Output)	%	Dynamic	Performance	Calculated
2	Cooling Tower Load	Cooling Tower Load(Output)	%	Dynamic	Performance	Calculated
3	Cooling Tower Duty	Cooling Tower Duty(Output)	kW	Dynamic	Performance	Calculated
4	Range	Range (Output)	°C	Dynamic	Performance	Calculated
5	Approach	Approach (Output)	°C	Dynamic	Performance	Calculated
6	Cycles of Concentration	Cycles of Concentration (Output)		Dynamic	Performance	Calculated
7	Water Loss	Water Loss (Output)	m ³ /hr	Dynamic	Performance	Calculated
8	Actual HTC	Actual Heat Transfer Coefficient (Output)	kJ/m ² -hr-C	Dynamic	Performance	Calculated
9	Expected No of Running Fans	Expected No of Running Fans (Output)		Dynamic	Performance	Calculated
10	Air Exhaust Temperature	Air Exhaust Temperature (Output)	°C	Dynamic	Energy	Calculated
11	Operating LMTD	Operating Log Mean Temperature Difference (Output)	°C	Dynamic	Energy	Calculated
12	Design LMTD	Design Log Mean Temperature Difference (Output)	°C	Dynamic	Design	Calculated
13	Humidity Ratio	Humidity Ratio (Output)		Dynamic	Energy	Calculated
14	NTU	Number of Transfer Units (Output)		Dynamic	Energy	Calculated
15	Blow down Loss	Blow down Loss(Output)	m ³ /hr	Dynamic	Energy	Calculated
16	Drift Loss	Drift Loss (Output)	m ³ /hr	Dynamic	Energy	Calculated
17	Evaporation Loss	Evaporation Loss(Output)	m ³ /hr	Dynamic	Energy	Calculated
18	Expected HTC	Expected Heat Transfer Coefficient (Output)	kJ/m ² -hr-C	Dynamic	Energy	Calculated

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
19	Heat Transfer Efficiency	Heat Transfer Efficiency[%] (Output)	%	Dynamic	Energy	Calculated

30.2 Cooling Tower Monitoring - Fault Model

In Asset Performance Management, faults, symptoms, and fault models are configured based on Cooling tower performance model outputs and process Parameters measured from field to diagnose the problems early. The fault management can inform operators of any violations of the nominal Cooling tower operating range and deviations from the expected characteristic and make this data available for further processing through the historian. The fault symptoms or models are used to:

Fault Name	Associated Symptoms
Performance Degradation	Cooling Water Outlet Temp High
	Water Loss High
	Makeup Water Flow Low
	Heat Transfer Efficiency Low
	Cooling Tower Efficiency Low
Health Degradation	Cooling Tower Fouling

DIESEL GENERATOR MONITORING

A Generator is a device that converts mechanical energy obtained from an external source into electrical energy as the output. Generator uses the mechanical energy supplied to it to force the movement of electric charges present in the wire of its windings through an external electric circuit. This flow of electric charges constitutes the output electric current supplied by the generator. The main components of a generator are as below.

Engine: The engine is the source of the input mechanical energy to the generator. The size of the engine is directly proportional to the maximum power output the generator can supply.

The Alternator: This is the part that turns the mechanical energy (the rotation of the shaft) into electrical power through induction. When a diesel engine turns the central shaft, the rotor constantly moves the north and south poles of its magnetic fields across the bundles of wire that surround them. This causes a great deal of electrical current to flow back and forth through the wires.

The Fuel System: This is typically the diesel fuel supply for the engine. This tank may be inside the generator housing for smaller, or portable units, or it may be a separate external structure for larger, permanently installed units.

The Voltage Regulator: It regulates the AC current provided according to the speed of the engine.

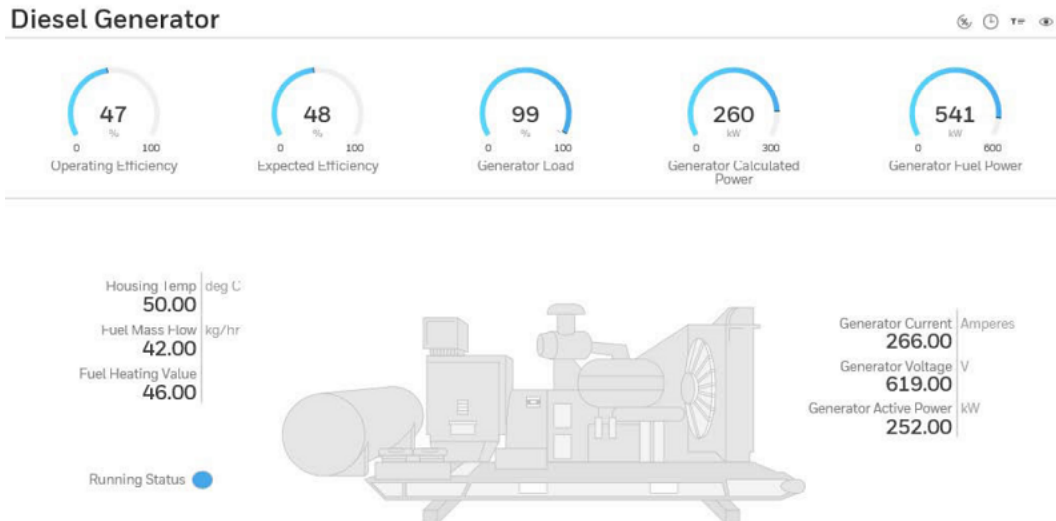
The Cooling System: As the generator produces a great deal of waste heat in addition to mechanical energy, this heat should be soaked up by a cooling system.

The Lubrication (oil) System: Any engine requires lubrication, and this is handled by an oil pump and reservoir attached to the engine itself.

There are two areas that help to monitor Diesel Generator performance to increase the availability and decrease energy costs.

- Process and condition monitoring
- Performance and Health monitoring

Diesel Generator



31.1 Diesel Generator Monitoring - Performance Monitoring

The following measured parameters are monitored as they have impact on performance and health of diesel generator, if not maintained in operating range.

Measured Voltage (V): When a motor is operated at reduced voltage, even within the allowable 10% limit, the motor will draw increased current to produce the torque requirements imposed by the load. This causes an increase in both stator and rotor resistance losses with subsequent increased heating at rated horsepower load or service factor operation. Operating a motor at the outer limits of its voltage requirements reduces its efficiency and causes premature failure. Whereas increase in voltage, magnetizing current increases by an exponential function. Depending upon design of the motor, saturation of the core iron will occur, and overheating will result. At about 10% to 15% above the motor's design voltage (overvoltage), both efficiency and power factor significantly decrease for standard efficiency motors, while the full-load slip decreases.

Measured Current (Amps): An increase in current is a danger to the motor only if that current exceeds the motor's nameplate current rating. When amps go above the nameplate rating, heat begins to build up in the motor. The more heat and the longer the exposure to it, the more damage to the motor. Whereas decreasing in current than require result in decrease efficiency of equipment.

Measured Power (KW): To drive a fixed mechanical load connected to the shaft, a motor must draw a fixed amount of power from the line. Power is directly impact rotational speed and torque. As more load is applied the motor draws more current, which increases torque. However, as power increase the speed of motor increase. Fluctuation in power impact efficiency and operation of equipment.

Process Performance Overview: Process Performance monitoring view in APM provides a list of the asset attributes and their details, including the current value. This list can be used to review and update asset attribute information, as well as to open an Asset Trend to view more details. It also highlights (in red color) the attributes if deviates from the high or low limits.

31.1.1 Diesel Generator Monitoring - Model Inputs

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Constant A (Expected efficiency)	Constant A (Expected efficiency) (Input)		Static	Design	Constant From Manufacturer Data Sheet
2	Constant B (Expected Efficiency)	Constant B (Expected Efficiency) (Input)		Static	Design	Constant From Manufacturer Data Sheet
3	Constant C (Expected Efficiency)	Constant C (Expected Efficiency) (Input)		Static	Design	Constant From Manufacturer Data Sheet
4	Rated Power	Rated Power (Input)		Static	Design	Constant
5	Active Power	Active Power (Input)	kW	Dynamic	Process	Constant
6	Fuel Heating Value	Fuel Heating Value (Input)		Dynamic	Process	Constant
7	Fuel Mass Flow	Fuel Mass Flow (Input)	kg/hr	Dynamic	Process	Constant
8	Current	Current (Input)	Amperes	Dynamic	Process	Constant
9	Power Factor	Power Factor (Input)		Dynamic	Process	Constant
10	Housing Temperature	Housing Temperature (Input)	°C	Dynamic	Process	Constant

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
11	Voltage	Voltage (Input)	V	Dynamic	Process	Constant
12	Running Status	Running Status		Dynamic	Process	Constant

31.1.2 Diesel Generator Monitoring - KPIs

The following indicators help us to know overall performance of the diesel generator.

Motor Efficiency

Motor Efficiency of diesel generator is calculated as "Percentage of Actual Power Output / Actual Power Input".

Motor efficiency = (Actual Power Output / Actual Power Input) * 100

Higher the efficiency, better the motor. For ideal motor, the efficiency is 100%. However, efficiency cannot be more than 100%.

Actual Power Input is calculated as, $P_i = (V \times I \times PF \times \sqrt{3})/1000$

Where:

P_i = Three-phase power in kW

V = RMS voltage, mean line-to-line of 3 phases I = RMS current, mean of 3 phases

PF = Power factor as a decimal

Actual power output = Shaft power of equipment (Power transmitted by motor to shaft)

Expected Efficiency

Expected Efficiency of DC motors is calculated using following empirical equation. Expected efficiency = $A \cdot ML^2 + B \cdot ML + ML$

Where,

ML= Motor Load A = -0.00125

B=0.175

Motor Load

This is the ratio of Actual Input Power to Rated Power (Nameplate Power). Actual Input Power is calculated as,

$P_i = (V \times I \times PF \times \sqrt{3})/1000$ Where:

P_i = Three-phase power in kW

V = RMS voltage, mean line-to-line of 3 phases I = RMS current, mean of 3 phases

PF = Power factor as a decimal

Rated power is calculated as, $P_{ir} = (hp \times 0.7457)/\eta_{fl}$

Where:

P_{ir} = Input power at full-rated load in kW hp = Nameplate rated horsepower

η_{fl} = Efficiency at full-rated load

Electric motor datasheet usually provides the efficiency and power factor at full load and at various part loads (for example 50%,75%).

31.1.3 Diesel Generator Monitoring - Model Outputs

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Operating Efficiency	Operating Efficiency (Output)	%	Double	Performance	Calculated
2	Operating Power	Operating Power (Output)	kW	Double	Performance	Calculated
3	Fuel Power	Fuel Power (Output)	kW	Double	Performance	Calculated
4	Expected Efficiency	Expected Efficiency (Output)	%	Double	Performance	Calculated
5	Generator Load	Generator Load (Output)	%	Double	Performance	Calculated

31.2 Diesel Generator Monitoring - Fault Model

In APM, a fault model is built to identify possible anomalies that affects the performance or health of a Diesel Generator. The fault management can inform operators of any violations of the nominal diesel generator operating range and deviations from the expected characteristic and make this data available for further processing through the historian.

The following messages can be generated for process alarms or notifications to warn operating personnel of unfavorable operating conditions.

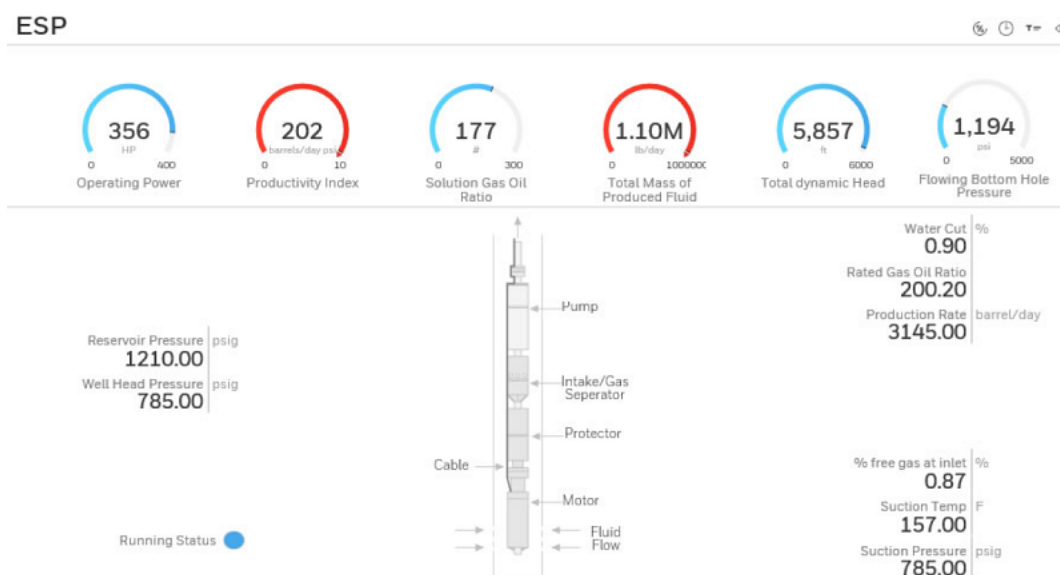
1. **Low Efficiency:** Occurs mainly due to improper load conditions. At low load, the frictional losses cause low efficiency of the motor. At high load, as the load torque increases the speed of the motor decreases which will impact the motor efficiency. Hence, it is required to optimize the load on motor to achieve maximum efficiency.
2. **Overheating:** Occurs due to excessive load on motor which will cause damage to the motor. It is important to monitor the housing and winding temperatures to maintain the health of the motor.

Below table illustrates the list of faults and symptoms configured for a Diesel Generator.

Fault Name	Associated Symptoms
Performance Degradation	Load High
	Load Low
	Efficiency Low
Health Degradation	Temperature High

ELECTRICAL SUBMERSIBLE PUMP MONITORING

Electrical submersible pumps are used to pump liquid in process industries. An electric motor drives the pump to increase the fluid's kinetic energy which converts into pressure. This pressure lifts the fluid through the pump. ESPs are basically centrifugal pumps with vertical shafts. The main advantage of this type of pump is that it prevents pump cavitation, a problem associated with a high elevation difference between pump and the fluid surface.



This document explains ESP monitoring philosophy for process, performance, energy, and health parameters and their importance. Monitoring Basic machine health data in addition to ESP operating conditions helps to maintain reliable ESP operations. There are three areas that help to monitor ESP performance to increase availability and decrease energy costs.

- Process and condition monitoring
- Energy monitoring
- Performance and Health monitoring

32.1 Electrical Submersible Pump Monitoring - Performance Monitoring

The following measured parameters are monitored as it has following impact on performance and health of the ESP, if not operated in operating range.

Reservoir Pressure: Reservoir pressure is important factors that control the deliverability of gas wells and the only one that decreases continuously with production going on; in fact, the decline of reservoir pressure dominates the reduction of deliverability. Reservoir pressure is a basic control on gas capacity and reservoir behavior and consists of two principal components, one is lithostatic pressure and other one is hydrostatic pressure.

Wellhead Pressure: Wellhead pressure helps to control fluid flow. When wellhead pressure low the reservoir is able to produce more fluids at same drawdown.

Pump suction Temp: The ESP suction temperature used in the design is based primarily on the dynamic well temperature and the effect of produced fluid flowing between the ESP unit and the casing. Suction temperature exceeding the ESP's design rating can reduce the expected life.

Producing Gas Oil Ratio: Presence of free gas at pump intake affects the pump performance negatively, reducing the liquid rates and the pressure added by the pump. This performance reduction resulting in unstable pump operation, significant production losses, lower reliability and increased operating costs due to premature equipment failure. If the bore fluid contains a relatively high oil-gas ratio, If the gas-oil ratio is low the gas separator is not required at the pump inlet.

Process Performance Overview: Process Performance monitoring view in APM provides a list of the asset attributes and their details, including the current value. This list can be used to review and update asset attribute information, as well as to open an Asset Trend to view more details. It also highlights (in red color) the attributes if deviates from the high or low limits.

32.1.1 Electrical Submersible Pump Monitoring - Model Inputs

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Correction Factor	Correction Factor (Input)		Dynamic	Design	Constant
2	Specific gravity of gas	Specific gravity of gas (Input)		Dynamic	Design	Constant
3	Inlet Tube Diameter	Inlet Tube Diameter (Input)	inch	Dynamic	Design	Constant
4	Oil gravity API	oil gravity API (Input)	API_gravity	Dynamic	Design	Constant
5	Pump Intake Depth	Pump Intake Depth (Input)	m	Dynamic	Design	Constant
6	Specific gravity of Water	Specific gravity of Water (Input)		Dynamic	Design	Constant
7	Top Perforation	Top Perforation (Input)	m	Dynamic	Design	Constant
8	Z factor	Z factor (Input)		Dynamic	Design	Constant
9	Producing Gas Oil Ratio	Producing Gas Oil Ratio (Input)		Dynamic	Process	Field Input
10	Head/Stage	Head/Stage (Required Input)	foot	Dynamic	Process	Field Input
11	Pump inlet pressure	Pump inlet pressure (Required Input)	psig	Dynamic	Process	Field Input
12	Pump Depth	Pump Depth (Required Input)	foot	Dynamic	Process	Field Input
13	Power Per Stage	Power Per Stage (Required Input)	horsepower	Dynamic	Process	Field Input
14	Reservoir Pressure	Reservoir Pressure (Required Input)	psig	Dynamic	Process	Field Input
15	Production Rate	Production Rate (Required Input)	barrel/day	Dynamic	Process	Field Input
16	Suction Temperature	Suction Temp (Required Input)	F	Dynamic	Process	Field Input
17	Water Cut	Water Cut (Required Input)	%	Dynamic	Process	Field Input
18	Wellhead Pressure	Wellhead Pressure (Input)	psig	Dynamic	Process	Field Input

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
19	Performance Data	Performance Data		Static	Design	Constant
20	Running Status	Running Status		Dynamic	Process	Field Input

32.1.2 Electrical Submersible Pump Monitoring - KPIs

The following indicators help to know the overall performance of the ESP

Percentage free gas to total volume of fluid at inlet

It is ratio of volume of gas at pump inlet to total capacity of pump intake.

Free gas to total volume of fluid at inlet= (Volume of gas at the pump intake / Total fluid volume at the pump intake) *100

Total number of stages required

It is ratio of total dynamic head required by pump to head given by one stage.

Total number of stages required= Total Dynamic Head / (Head/Stage)

Productivity Index

The productivity index (PI) is equal to the liquid flow rate divided by the well drawdown (Pr-Pwf).

Productivity Index=Production Rate /(Reservoir Pressure -Flowing bottom hole pressure)

Pump Power required

It is power per stage to total no of stages required.

Pump Power required: Power Per Stage * Total number of stages required *Specific Gravity of Fluid

Total Dynamic Head(ft)

It is net head of electrical submersible pump.

Total Dynamic Head= Net Well Lift+ Tubing Friction Loss + Wellhead Pressure Head

32.1.3 Electrical Submersible Pump Monitoring - Model Outputs

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Percentage free gas to total volume of fluid at inlet	% free gas to total volume fluid at inlet (Output)	%	Dynamic	Performance	Calculated
2	Total number of stages required	Total number of stages required (Output)		Dynamic	Performance	Calculated
3	Productivity Index	Productivity Index (Output)	barrel_per_day_per_psi	Dynamic	Performance	Calculated
4	Pump Power required	Pump Power required (Output)	horsepower	Dynamic	Performance	Calculated
5	Flowing bottom hole pressure	Flowing bottom hole pressure (Output)	psig	Dynamic	Performance	Calculated

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
6	Solution gas- oil ratio	Solution gas-oil ratio (Output)		Dynamic	Performance	Calculated
7	Total Dynamic Head	Total Dynamic Head (Output)	foot	Dynamic	Performance	Calculated
8	Total Mass of Produced Fluid	Total Mass of Produced Fluid (Output)	pound_mas s_per_day	Dynamic	Performance	Calculated

32.2 Electrical Submersible Pump Monitoring - Fault Model

In Asset Performance Management, you can configure faults, symptoms, and fault models on ESP performance output to diagnose the problems early. The fault management can inform operators of any violations of the nominal ESP operating range and deviations from the expected characteristic and make this data available for further processing through the historian. The fault symptoms or models are used to:

1. Warn against potential damage to ESP under unfavorable operating conditions.
2. Provide early warning of damages to the ESP.

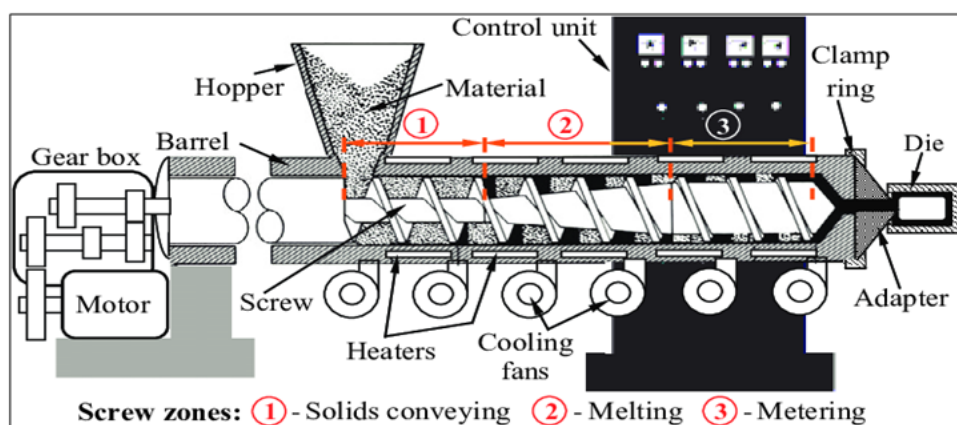
The following messages can be generated for process alarms or notifications to warn operating personnel of unfavorable operating conditions

1. **Low pump efficiency:** It is determined by means of the deviation of the current pump efficiency from the efficiency characteristic.
2. **High gas content:** It is determined by means of the reduction in delivery height or head.
3. **Blockage:** It is determined on the basis of a limit value for an electrical power being undershot.
4. **Incorrect direction of rotation (the motor was connected incorrectly and rotates in the wrong direction):** It is determined when the delivery height or head falls significantly ($> 40\%$) but with only a slight deviation ($< 20\%$) of the power characteristic.
Limit value violation for power values (e.g. electrical overload).
5. **Deviation of the operating point from the flow characteristic (i.e. reduction in delivery height or head:** Indication of high gas content, cavitation, blockage, or dry running). This is configured by flow loss or capacity loss of pump.
6. **Deviation of the operating point from the power characteristic:** It is configured by percent deviation in power between operating and design values.

Below table illustrates the list of faults and symptoms configured for a pump to identify possible anomalies that affects the performance or health of a pump.

Fault Name	Associated Symptoms
Low ESP Performance	High Power Consumption
	Low Productivity Index
ESP Health Degradation	High Suction Temp

Extrusion is a process used to create objects of a fixed cross-sectional profile by pushing material through a die of the desired cross-section. The screw extruder is a key component of the polymer processing industry, which is concerned with the conversion of polymer in raw material form (i.e., solid polymer pellets, granules, or powder) into finished products. An extruder is used to convert the solid polymer granules into melt and pump the highly viscous melt through a die at high pressure to give it a shape for example a rod or a tube. The solid polymer is gravity/vacuum fed through the feed hopper onto the screw and then is compressed and driven forward by the rotating screw. As the polymer moves forward it is gradually melted. The heat required for melting is supplied from two sources: frictional heat developed through the shearing action of the screw and barrel on the polymer, and heat conducted from the barrel wall.



- Inlet Pressure (Kg/cm²)
- Inlet Temperature (C)
- Inlet Volume flow (m³/hr) or mass flow (kg/hr)
- Screw Speed (rpm)

- Outlet Pressure (Kg/cm²)
- Outlet Temperature (C)
- Outlet Volume flow (m³/hr) or mass flow (kg/hr)



- Motor power (kW)
- Specific power (kW)
- Heating power (kW)
- Cooling duty (kW)
- Screw Speed

33.1 Extruder - Performance Monitoring

33.1.1 Notations

F	Flow
h	Specific Enthalpy
Q	Volume Flow
P	Pressure
W	Power

N	Screw Speed
T	Temperature
V	Electrical Voltage
I	Electrical Current
P_f	Electrical Power Factor

33.1.2 Subscripts

i	Inlet
o	Outlet

33.1.3 Extruder - Model Input

No.	Attribute	Description	Units	Attribute Category	Source
1	Total Blower Power	Total Blower Power (Input)	kW	Dynamic	Measurement
2	Inlet Flowrate for Cooler 1	Inlet Flowrate for Cooler 1 (Input)	kg/hr	Dynamic	Measurement
3	Inlet Enthalpy for Cooler 1	Inlet Enthalpy for Cooler 1 (Input)	kJ/kg	Dynamic	Measurement
4	Outlet Enthalpy for Cooler 1	Outlet Enthalpy for Cooler 1 (Input)	kJ/kg	Dynamic	Measurement
5	Inlet Flowrate for Cooler 2	Inlet Flowrate for Cooler 2 (Input)	kg/hr	Dynamic	Measurement
6	Inlet Enthalpy for Cooler 2	Inlet Enthalpy for Cooler 2 Input)	kJ/kg	Dynamic	Measurement
7	Outlet Enthalpy for Cooler 2	Outlet Enthalpy for Cooler 2 (Input)	kJ/kg	Dynamic	Measurement
8	Inlet Flowrate for Cooler 3	Inlet Flowrate for Cooler 3 (Input)	kg/hr	Dynamic	Measurement
9	Inlet Enthalpy for Cooler 3	Inlet Enthalpy for Cooler 3 (Input)	kJ/kg	Dynamic	Measurement
10	Outlet Enthalpy for Cooler 3	Outlet Enthalpy for Cooler 3 (Input)	kJ/kg	Dynamic	Measurement
11	Inlet Flowrate for Cooler 4	Inlet Flowrate for Cooler 4 (Input)	kg/hr	Dynamic	Measurement
12	Inlet Enthalpy for Cooler 4	Inlet Enthalpy for Cooler 4 (Input)	kJ/kg	Dynamic	Measurement
13	Outlet Enthalpy for Cooler 4	Outlet Enthalpy for Cooler 4 (Input)	kJ/kg	Dynamic	Measurement
14	Inlet Flowrate for Cooler 5	Inlet Flowrate for Cooler 5 (Input)	kg/hr	Dynamic	Measurement
15	Inlet Enthalpy for Cooler 5	Inlet Enthalpy for Cooler 5 Input)	kJ/kg	Dynamic	Measurement
16	Outlet Enthalpy for Cooler 5	Outlet Enthalpy for Cooler 5 (Input)	kJ/kg	Dynamic	Measurement
17	Inlet Flowrate for Cooler 6	Inlet Flowrate for Cooler 6 (Input)	kg/hr	Dynamic	Measurement
18	Inlet Enthalpy for Cooler 6	Inlet Enthalpy for Cooler 6 (Input)	kJ/kg	Dynamic	Measurement
19	Outlet Enthalpy for Cooler 6	Outlet Enthalpy for Cooler 6 (Input)	kJ/kg	Dynamic	Measurement
20	Inlet Flowrate for Cooler 7	Inlet Flowrate for Cooler 7 (Input)	kg/hr	Dynamic	Measurement
21	Inlet Enthalpy for Cooler 7	Inlet Enthalpy for Cooler 7 (Input)	kJ/kg	Dynamic	Measurement
22	Outlet Enthalpy for Cooler 7	Outlet Enthalpy for Cooler 7 (Input)	kJ/kg	Dynamic	Measurement
23	Inlet Flowrate for Cooler 8	Inlet Flowrate for Cooler 8 (Input)	kg/hr	Dynamic	Measurement

No.	Attribute	Description	Units	Attribute Category	Source
24	Inlet Enthalpy for Cooler 8	Inlet Enthalpy for Cooler 8 (Input)	kJ/kg	Dynamic	Measurement
25	Outlet Enthalpy for Cooler 8	Outlet Enthalpy for Cooler 8 (Input)	kJ/kg	Dynamic	Measurement
26	Inlet Flowrate for Cooler 9	Inlet Flowrate for Cooler 9 (Input)	kg/hr	Dynamic	Measurement
27	Inlet Enthalpy for Cooler 9	Inlet Enthalpy for Cooler 9 (Input)	kJ/kg	Dynamic	Measurement
28	Outlet Enthalpy for Cooler 9	Outlet Enthalpy for Cooler 9 (Input)	kJ/kg	Dynamic	Measurement
29	Degasser Temperature	Degasser Temperature (Input)	°C	Dynamic	Measurement
30	Die Temperature	Die Temperature (Input)	°C	Dynamic	Measurement
31	Power of Electrical Heater 1	Power of Electrical Heater 1 (Input)	kW	Dynamic	Measurement
32	Power of Electrical Heater 2	Power of Electrical Heater 2 (Input)	kW	Dynamic	Measurement
33	Power of Electrical Heater 3	Power of Electrical Heater 3 (Input)	kW	Dynamic	Measurement
34	Mass Throughput	Mass Throughput (Required Input)	kg/hr	Dynamic	Measurement
35	Melt Pressure	Melt Pressure (Input)	kg/cm ² (a)	Dynamic	Measurement
36	Melt Temperature	Melt Temperature (Input)	°C	Dynamic	Measurement
37	Motor Power	Motor Power (Required Input)	kW	Dynamic	Measurement
38	Inlet Enthalpy for Oil Heater 1	Inlet Enthalpy for Oil Heater 1 (Input)	kJ/kg	Dynamic	Measurement
39	Outlet Enthalpy for Oil Heater 1	Outlet Enthalpy for Oil Heater 1 (Input)	kJ/kg	Dynamic	Measurement
40	Inlet Flowrate for Oil Heater 1	Inlet Flowrate for Oil Heater 1 (Input)	kg/hr	Dynamic	Measurement
41	Inlet Enthalpy for Oil Heater 2	Inlet Enthalpy for Oil Heater 2 (Input)	kJ/kg	Dynamic	Measurement
42	Outlet Enthalpy for Oil Heater 2	Outlet Enthalpy for Oil Heater 2 (Input)	kJ/kg	Dynamic	Measurement
43	Inlet Flowrate for Oil Heater 2	Inlet Flowrate for Oil Heater 2 (Input)	kg/hr	Dynamic	Measurement
44	Inlet Enthalpy for Oil Heater 3	Inlet Enthalpy for Oil Heater 3 (Input)	kJ/kg	Dynamic	Measurement
45	Outlet Enthalpy for Oil Heater 3	Outlet Enthalpy for Oil Heater 3 (Input)	kJ/kg	Dynamic	Measurement
46	Inlet Flowrate for Oil Heater 3	Inlet Flowrate for Oil Heater 3 (Input)	kg/hr	Dynamic	Measurement
47	Inlet Enthalpy for Oil Heater 4	Inlet Enthalpy for Oil Heater 4 (Input)	kJ/kg	Dynamic	Measurement
48	Outlet Enthalpy for Oil Heater 4	Outlet Enthalpy for Oil Heater 4 (Input)	kJ/kg	Dynamic	Measurement

No.	Attribute	Description	Units	Attribute Category	Source
49	Inlet Flowrate for Oil Heater 4	Inlet Flowrate for Oil Heater 4 (Input)	kg/hr	Dynamic	Measurement
50	Inlet Enthalpy for Oil Heater 5	Inlet Enthalpy for Oil Heater 5 Input)	kJ/kg	Dynamic	Measurement
51	Outlet Enthalpy for Oil Heater 5	Outlet Enthalpy for Oil Heater 5 (Input)	kJ/kg	Dynamic	Measurement
52	Inlet Flowrate for Oil Heater 5	Inlet Flowrate for Oil Heater 5 (Input)	kg/hr	Dynamic	Measurement
53	Inlet Enthalpy for Oil Heater 6	Inlet Enthalpy for Oil Heater 6 (Input)	kJ/kg	Dynamic	Measurement
54	Outlet Enthalpy for Oil Heater 6	Outlet Enthalpy for Oil Heater 6 (Input)	kJ/kg	Dynamic	Measurement
55	Inlet Flowrate for Oil Heater 6	Inlet Flowrate for Oil Heater 6 (Input)	kg/hr	Dynamic	Measurement
56	Inlet Enthalpy for Oil Heater 7	Inlet Enthalpy for Oil Heater 7 (Input)	kJ/kg	Dynamic	Measurement
57	Outlet Enthalpy for Oil Heater 7	Outlet Enthalpy for Oil Heater 7 (Input)	kJ/kg	Dynamic	Measurement
58	Inlet Flowrate for Oil Heater 7	Inlet Flowrate for Oil Heater 7 (Input)	kg/hr	Dynamic	Measurement
59	Inlet Enthalpy for Oil Heater 8	Inlet Enthalpy for Oil Heater 8 Input)	kJ/kg	Dynamic	Measurement
60	Outlet Enthalpy for Oil Heater 8	Outlet Enthalpy for Oil Heater 8 (Input)	kJ/kg	Dynamic	Measurement
61	Inlet Flowrate for Oil Heater 8	Inlet Flowrate for Oil Heater 8 (Input)	kg/hr	Dynamic	Measurement
62	Inlet Enthalpy for Oil Heater 9	Inlet Enthalpy for Oil Heater 9 (Input)	kJ/kg	Dynamic	Measurement
63	Outlet Enthalpy for Oil Heater 9	Outlet Enthalpy for Oil Heater 9 (Input)	kJ/kg	Dynamic	Measurement
64	Inlet Flowrate for Oil Heater 9	Inlet Flowrate for Oil Heater 9 (Input)	kg/hr	Dynamic	Measurement
65	Screw Speed	Screw Speed (Required Input)	rpm	Dynamic	Measurement
66	First Zone Temperature	First Zone Temperature (Input)	°C	Dynamic	Measurement
67	Second Zone Temperature	Second Zone Temperature (Input)	°C	Dynamic	Measurement
68	Third Zone Temperature	Third Zone Temperature (Input)	°C	Dynamic	Measurement
69	Fourth Zone Temperature	Fourth Zone Temperature (Input)	°C	Dynamic	Measurement
70	Fifth Zone Temperature	Fifth Zone Temperature (Input)	°C	Dynamic	Measurement

No.	Attribute	Description	Units	Attribute Category	Source
71	Sixth Zone Temperature	Sixth Zone Temperature (Input)	°C	Dynamic	Measurement
72	Seventh Zone Temperature	Seventh Zone Temperature (Input)	°C	Dynamic	Measurement
73	Hot Oil Temp in Zone1	Hot Oil Temp in Zone1 (Input)	°C	Dynamic	Measurement
74	Hot Oil Temp in Zone2	Hot Oil Temp in Zone2 (Input)	°C	Dynamic	Measurement
75	Hot Oil Temp in Zone3	Hot Oil Temp in Zone3 (Input)	°C	Dynamic	Measurement
76	Hot Oil Temp in Zone4	Hot Oil Temp in Zone4 (Input)	°C	Dynamic	Measurement
77	Hot Oil Temp in Zone5	Hot Oil Temp in Zone5 (Input)	°C	Dynamic	Measurement
78	Hot Oil Temp in Zone6	Hot Oil Temp in Zone6 (Input)	°C	Dynamic	Measurement
79	Hot Oil Temp in Zone7	Hot Oil Temp in Zone7 (Input)	°C	Dynamic	Measurement
80	Hot Oil Temp in Zone8	Hot Oil Temp in Zone8 (Input)	°C	Dynamic	Measurement
81	Hot Oil Temp in Zone9	Hot Oil Temp in Zone9 (Input)	°C	Dynamic	Measurement

33.1.4 Extruder - KPIs

33.1.4.1 Input KPI

- Mass Throughput
- Melt Pressure
- Melt Temperature
- Degasser Temperature
- Heating Fluid Inlet Temperature
- Heating Fluid Outlet Temperature
- Heating Fluid Inlet Pressure
- Heating Fluid Outlet Pressure
- Heating Fluid Mass Flowrate
- Cooling Fluid Inlet Temperature
- Cooling Fluid Outlet Temperature
- Cooling Fluid Inlet Pressure
- Cooling Fluid Outlet Pressure
- Cooling Fluid Mass Flowrate
- Electric Current
- Electric Voltage

33.1.4.2 Output KPI

Expected KPI

- Expected Motor Power
- Expected Specific Power
- Expected Screw Speed

Operating KPI

- Operating Motor Power
- Operating Specific Power
- Operating Screw Speed
- Operating Heater Power
- Operating Cooling Power
- Operating Total Power

33.1.4.3 Extruder - Performance Modeling

The Performance modeling of the Extruder helps to determine the performance KPI. The following is the procedure,

Overall Mass Balance

Applying the continuity equation,

$$F_i = F_o$$

Overall Energy Balance

$$F_o h_o(P_o, T_o) = F_i h_i(P_i, T_i) + W_{Total}$$

$$\dot{W}_{Total} = \dot{W}_{Motor} + \dot{W}_{Heating} - \dot{W}_{Cooling}$$

33.1.4.4 Extruder - Expected KPIs

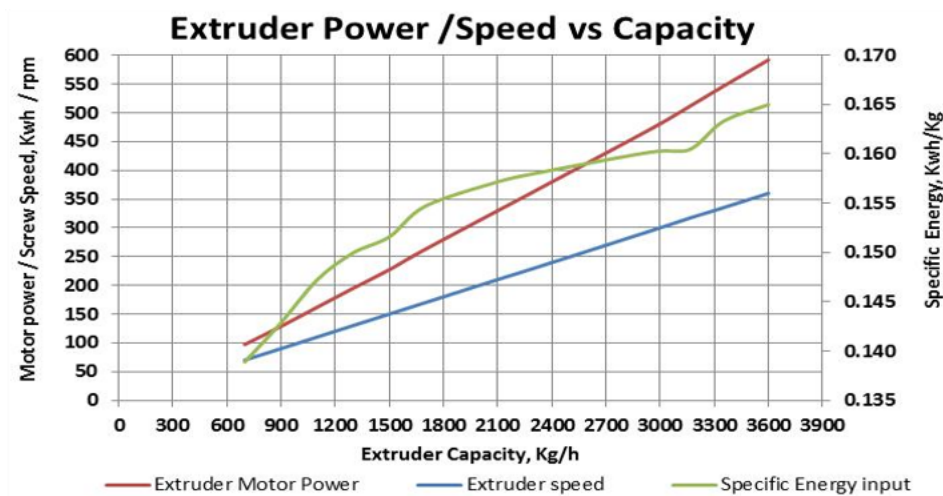
The following indicators help us to know the expected overall performance of the extruder.

Expected Motor Power (kW)

The expected motor power will be calculated from the chart below as a function of the operating extruder capacity (kg/hr).

$$\dot{W}_{Motor}^{Expected} = f(F_i)$$

This plot will be different for a polymer based on the mold flow index of a polymer.



Expected Specific Power (kW)

The expected Specific Power is calculated using the above chart supplied by the user,

$$\dot{W}_{Specific}^{Expected} = f(F_i)$$

Expected Screw Speed (rpm)

The expected Specific Power is calculated using the above chart supplied by the user,

$$N^{Expected} = f(F_i)$$

33.1.4.5 Extruder - Operating KPIs**Operating Motor Power (kW)**

An attribute for Motor power will be created and values for the same will be fetched from motor assets through PHD tag mapping.

Operating Heating Duty (kW)

The operating heating duty will be calculated based on the type of heating system used. In an extruder two types of heating systems can be used, the first is done by an electrical band heater, and the second is done by fluid heaters.

1. Electrical Heaters:

An attribute for electrical heater power will be created and values for the same will be fetched from motor assets through PHD tag mapping.

NOTE

- The model is capable of handling a maximum of three electrical heaters.
- For more than three electrical heaters, a basic calculation can be written to sum all heaters power and mapped to the electrical heater power attribute.

2. Fluid Heater:

$$\dot{W}_{Heating}^{Operating} = \text{Change in Enthalpy of Hot Fluid}$$

$$\dot{W}_{Heating}^{Operating} = F_i h_i(P_i, T_i) - F_o h_o(P_o, T_o)$$

NOTE

- The model is capable of handling a maximum of nine oil/fluid heaters.
- If inlet and outlet conditions (composition, temperature, and pressure) are known, then the streams need to be configured and enthalpies need to map to respective heater inlet and outlet enthalpy attributes.
- If inlet and outlet conditions (composition, temperature, and pressure) are unknown, enthalpy values need to map directly.
- For more than nine heaters, a basic calculation can be written, and the power consumption will be mapped.

Operating Cooling Duty (kW)

Here cooling may be done by Fan/Blower or Fluid Cooler.

1. Blower:

An attribute for Blower power will be created and values for the same will be fetched from the blower asset through PHD tag mapping.

NOTE

- If the blower is motor-driven then, the motor will be mapped as blower power.
- For multiple blowers, the total blower power will be the sum of all blower power.

2. Fluid Cooler:

$$\dot{W}_{Cooling}^{Operating} = \text{Change in Enthalpy of Cooling Fluid}$$

$$\dot{W}_{Cooling}^{Operating} = F_o h_o(P_o, T_o) - F_i h_i(P_i, T_i)$$

NOTE

- The model is capable of handling a maximum of nine coolers.
- If inlet and outlet conations (composition, temperature, and pressure) are known, then the streams need to be configured and enthalpies need to map to respective cooler inlet and outlet enthalpy attributes.
- If inlet and outlet conations (composition, temperature, and pressure) are unknown, enthalpy values need to map directly.
- For more than nine coolers, a basic calculation can be written, and the power consumption will be mapped.

Operating Total Power (kW)

The operating total power will be calculated as the sum of all energy input and output as follows:

$$\dot{W}_{Total}^{Operating} = \dot{W}_{Motor}^{Operating} + \dot{W}_{Heating}^{Operating} + \dot{W}_{Cooling}^{Operating}$$

33.1.5 Extruder - Model Output

No.	Attribute	Description	Units	Attribute Category	Source
1	Deviation Motor Power	Deviation Motor Power (Output)	%	Dynamic	Calculated
2	Deviation Screw Speed	Deviation Screw Speed (Output)	%	Dynamic	Calculated
3	Deviation Specific Power	Deviation Specific Power (Output)	%	Dynamic	Calculated
4	Expected Motor Power	Expected Motor Power (Output)	kW	Dynamic	Calculated

No.	Attribute	Description	Units	Attribute Category	Source
5	Expected Screw Speed	Expected Screw Speed (Output)	rpm	Dynamic	Calculated
6	Expected Specific Power	Expected Specific Power (Output)	kilowatt_ hours/kilogram	Dynamic	Calculated
7	Operating Specific Power	Operating Specific Power (Output)	kilowatt_ hours/kilogram	Dynamic	Calculated
8	Total Power use from all Sources	Operating Total Power (Output)	kW	Dynamic	Calculated
9	Total Cooler Power	Total Cooler Power (Output)	kW	Dynamic	Calculated
10	Total Heater Power	Total Heater Power (Output)	kW	Dynamic	Calculated

33.2 Extruder - Fault Model

33.2.1 Extruder - APM Model

Model Type	APM Model
Category	Extruder
Equipment Class	Extruder
Equipment Type	AT_Extruder

The below table illustrates the list of faults and symptoms configured for the Extruder Advanced Model:

Fault Name	Fault Symptoms	Recommendations
Hot Oil Supply Anomaly	Hot Oil flow in Zone1 RL (EHS)	1. Check Hot oil heaters and clean them if required. 2. Take the stand-by pump online if the running pump is not able to develop pressure. 3. Check the hot oil circuit for any leaks. 4. Check the Hot oil quality.
	Hot Oil flow in Zone2 RL (EHS)	
	Hot Oil flow in Zone3 RL (EHS)	
	Hot Oil flow in Zone4 RL (EHS)	
	Hot Oil flow in Zone5 RL (EHS)	
	Hot Oil flow in Zone6 RL (EHS)	
	Hot Oil flow in Zone7 RL (EHS)	
	Hot Oil flow in Zone8 RL (EHS)	
	Hot Oil flow in Zone9 RL (EHS)	
	Hot Oil Temp in Zone1 RL (EHS)	
	Hot Oil Temp in Zone2 RL (EHS)	
	Hot Oil Temp in Zone3 RL (EHS)	
	Hot Oil Temp in Zone4 RL (EHS)	
	Hot Oil Temp in Zone5 RL (EHS)	
	Hot Oil Temp in Zone6 RL (EHS)	
	Hot Oil Temp in Zone7 RL (EHS)	
	Hot Oil Temp in Zone8 RL (EHS)	
	Hot Oil Temp in Zone9 RL (EHS)	
Individual Zone Temperature Low	First Zone Temperature RL (UETD)	1. Increase the feed zone temperature to feed the material to the transition zone properly. 2. Check the temperature resistance and cooling system. 3. Check the screen to catch the degraded, burned particles. 4. Increase the barrel temperatures gradually equally for each zone.
	Second Zone Temperature RL (UETD)	
	Third Zone Temperature RL (UETD)	
	Fourth Zone Temperature RL (UETD)	
	Fifth Zone Temperature RL (UETD)	
	Sixth Zone Temperature RL (UETD)	
	Seventh Zone Temperature RL (UETD)	
Individual Zone Temperature High	First Zone Temperature RH (IZTH)	1. Check the screen and clean it if it is blocked. 2. Check for blockages in the hopper. 3. Check the screw configuration. 4. Clean the screw before production.
	Second Zone Temperature RH (IZTH)	
	Third Zone Temperature RH	

Fault Name	Fault Symptoms	Recommendations
	(IZTH)	
	Fourth Zone Temperature RH (IZTH)	
	Fifth Zone Temperature RH (IZTH)	
	Sixth Zone Temperature RH (IZTH)	
	Seventh Zone Temperature RH (IZTH)	
Melting Temperature or Pressure High	Melting Temperature RH (MTPH)	1. Check the Vent to the atmosphere and clear if any choking exists. 2. Check the Cooling circuit for any leaks. 3. Check Cooling pumps for any issue
	Motor Current RH (MTPH)	
Shear Pin Fault	Motor Current ROCH ST (SPF)	1. Control the flow of material from the Hopper. 2. There might be a blockage inside the hopper. 3. Increase the die temperature to dissolve the blockage.
Cooling System Anomaly	Cooling water pump discharge pressure RL (CSA)	1. Check the Cooling Water Pump. 2. Check the radiator of water cooler and clean the filters.

33.2.2 Extruder - APR Model

33.2.2.1 Extruder Bearing System

Model Type	APR Model
Category	Extruder
Equipment Class	Extruder
Equipment Type	Extruder - Bearing

The below table illustrates the list of faults and symptoms configured for the Extruder Bearing System.

Fault Name	Associated Symptoms	Recommendations
Bearing Anomaly due to Lube Oil Temperature	Radial Bearing Vibration X DE RH (LOT)	1. Check the calibration of the sensor. 2. Check cooling water supply pressure. 3. Check the bypass control valve. 4. Check cooling water pumps for blockages, and ensure the strainer is clean. 5. If the cooler is an air cooler check fan and motor integrity. 6. If a malfunction of one cooler is suspected isolate and inspect it. 7. Line walk the system to check for leaks.
	Radial Bearing Vibration Y DE RH (LOT)	
	Radial Bearing Temperature A DE RH (LOT)	

Fault Name	Associated Symptoms	Recommendations
	Radial Bearing Temperature B DE RH (LOT)	
	Radial Bearing Vibration X NDE RH (LOT)	
	Radial Bearing Vibration Y NDE RH (LOT)	
	Radial Bearing Temperature A NDE RH (LOT)	
	Radial Bearing Temperature B NDE RH (LOT)	
	Lube Oil Header Temperature RH (LOT)	
Bearing Anomaly due to Lube Oil Pressure	Radial Bearing Vibration X DE RH (LOP)	1. Check the calibration of the sensor. 2. Check the oil level in the tank. 3. Check filter DP. 4. Line walk the system and check for leaks. 5. Ensure PSVs are seated, and check spillback/PCV valves are set correctly. 6. If maintenance has recently been performed check the correct rotation of the pump. 7. Ensure no hand valves have been left shut, and ensure NRV's are working correctly. 8. Check the oil supply or return line for any blockages.
	Radial Bearing Vibration Y DE RH (LOP)	
	Radial Bearing Temperature A DE RH (LOP)	
	Radial Bearing Temperature B DE RH (LOP)	
	Radial Bearing Vibration X NDE RH (LOP)	
	Radial Bearing Vibration Y NDE RH (LOP)	
	Radial Bearing Temperature A NDE RH (LOP)	
	Radial Bearing Temperature B NDE RH (LOP)	
	Lube Oil Header Pressure RL (LOP)	

Fault Name	Associated Symptoms	Recommendations
Bearing Anomaly due to Lube Oil Quality	Radial Bearing Vibration X DE RH (LOQ)	1. Take an oil sample and check water content, viscosity, and TAN. 2. Check no breather caps or man covers have been removed and left open to allow debris into the tank. 3. Recondition or replace oil.
	Radial Bearing Vibration Y DE RH (LOQ)	
	Radial Bearing Temperature A DE RH (LOQ)	
	Radial Bearing Temperature B DE RH (LOQ)	
	Radial Bearing Vibration X NDE RH (LOQ)	
	Radial Bearing Vibration Y NDE RH (LOQ)	
	Radial Bearing Temperature A NDE RH (LOQ)	
	Radial Bearing Temperature B NDE RH (LOQ)	
	Lube Oil Viscosity SH (LOQ)	
	Lube Oil Viscosity SL (LOQ)	
DE Radial Bearing Failure due to Mechanical Issue	Radial Bearing Vibration X DE RH (RBDM)	1. Check the calibration of the sensor. 2. Investigate with high-frequency vibration analysis. 3. Assess the likely cause and time to fail(trip) if intervention is not possible. 4. Intervention options, reduce speed, check parts availability for replacement Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 5. Change bearing if required.
	Radial Bearing Vibration Y DE RH (RBDM)	
	Radial Bearing Temperature A DE RH (RBDM)	
	Radial Bearing Temperature B DE RH (RBDM)	
NDE Radial Bearing Failure due to Mechanical Issue	Radial Bearing Vibration X NDE RH (RBNDM)	1. Check the calibration of the sensor. 2. Investigate with high-frequency vibration analysis. 3. Assess the likely cause and time to fail(trip) if intervention is not

Fault Name	Associated Symptoms	Recommendations
	Radial Bearing Vibration Y NDE RH (RBNDM)	possible. 4. Intervention options, reduce speed, check parts availability for replacement Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 5. Change bearing if required.
	Radial Bearing Temperature A NDE RH (RBNDM)	
	Radial Bearing Temperature B NDE RH (RBNDM)	

33.2.2.2 Extruder - Lube Oil System

Model Type	APR Model
Category	Extruder
Equipment Class	Extruder
Equipment Type	Extruder - Lube Oil System

The below table illustrates the list of faults and symptoms configured for an Extruder Lube oil system

Fault Name	Associated Symptoms	Recommendations
Loss of Lube Oil Quality	Lube Oil Viscosity HIGH (LOLQ)	1. Take an oil sample and check water content, viscosity, and TAN. 2. Check no breather caps or man covers have been removed and left open to allow debris into the tank. 3. Do filter cleaning. 4. Recondition or replace oil.
	Lube Oil Viscosity LOW (LOLQ)	
	Lube Oil TAN HIGH (LOLQ)	
Lube Oil Filter Anomaly	Lube Oil Filter DP RH (LFDF)	1. Check filter DP. 2. Changeover and clean/replace the stand-by filter. 3. Check the Filter DP transmitter
Loss of Pressure due to Pump Anomaly	Lube Oil Header Pressure RL (LOPPF)	1. Check the Discharge PT integrity. 2. Line walk the system and check for leaks, and ensure PSVs are seated. 3. Check spillback/PCV valves are set correctly. 4. Check the oil level in the tank, Open tank overhead vents for the release of pressure. 5. Check the PCV function for Integrity. 6. If maintenance has recently been performed check the correct rotation of the pump. 7. Ensure NRVs are working correctly.
	Lube Oil Pump Discharge Pressure RL (LOPPF)	
	Lube Oil Main Pump Current (Run Status) ON (LOPPF)	
Loss of Cooling due to TCV Anomaly	Lube Oil Header Temperature (Cooler Out Temp) RH (LOCTF)	1. Check the calibration of the sensor. 2. Check the bypass control valve for integrity.

Fault Name	Associated Symptoms	Recommendations
Loss of Cooling due to Cooler Anomaly	LO Cooler Bypass Valve (TCV) Position RH (LOCTF)	3. Line walk the system to check for leaks. 4. Check the Isolation/bypass valve of TCV.
	Lube Oil Header Temperature (Cooler Out Temp) RH (LOCCF)	1. Check the calibration of the sensor. 2. Check cooling water supply pressure. 3. If a malfunction of one cooler is suspected, isolate and inspect it.
	LO Tank and Cooler Outlet Temperature Delta RH (LOCCF)	4. Line walk the system to check for leaks. 5. Check cooling water pumps for blockages, and ensure the strainer is clean. 6. Check supply and return headers for blockages.

33.2.2.3 Extruder Transmission Fault Model

Model Type	APR Model
Category	Extruder
Equipment Class	Extruder
Equipment Type	Extruder - Transmission

The below table illustrates the list of faults and symptoms configured for an Extruder Transmission System

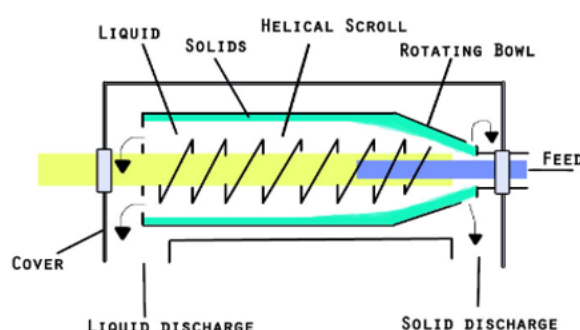
Fault Name	Associated Symptoms	Recommendations
Gearbox Radial Bearings malfunction /Failure	Gearbox IPShaftBearingVibration X DE RH (GPRB)	1. Assess likely cause and time to trip if intervention is not possible. 2. Intervention options: reduce load. 3. Check Oil temperature. 4. Restore oil properties. 5. Ensure parts availability. 6. Take an oil sample and have it tested. 7. At first opportunity Inspect journal and thrust bearing. 8. Check clearances if temperature and vibrations do not stabilize plan for bearing replacement.
	Gearbox IPShaftBearingVibration Y DE RH (GPRB)	
	Gearbox IPShaftBearingVibration X NDE RH (GPRB)	
	Gearbox IPShaftBearingVibration Y NDE RH (GPRB)	
	Gearbox OPShaftBearingVibration X DE RH (GPRB)	
	Gearbox OPShaftBearingVibration Y DE RH (GPRB)	
	Gearbox OPShaftBearingVibration X NDE RH (GPRB)	
	Gearbox	

Fault Name	Associated Symptoms	Recommendations
	OPShaftBearingVibration Y NDE RH (GPRB)	
	Gearbox Oil Temperature RH (GPRB)	
	Gearbox Oil TAN HIGH (GPRB)	
	Gearbox Oil TAN LOW (GPRB)	
	Gearbox Oil Viscosity HIGH (GPRB)	
	Gearbox Oil Viscosity LOW (GPRB)	

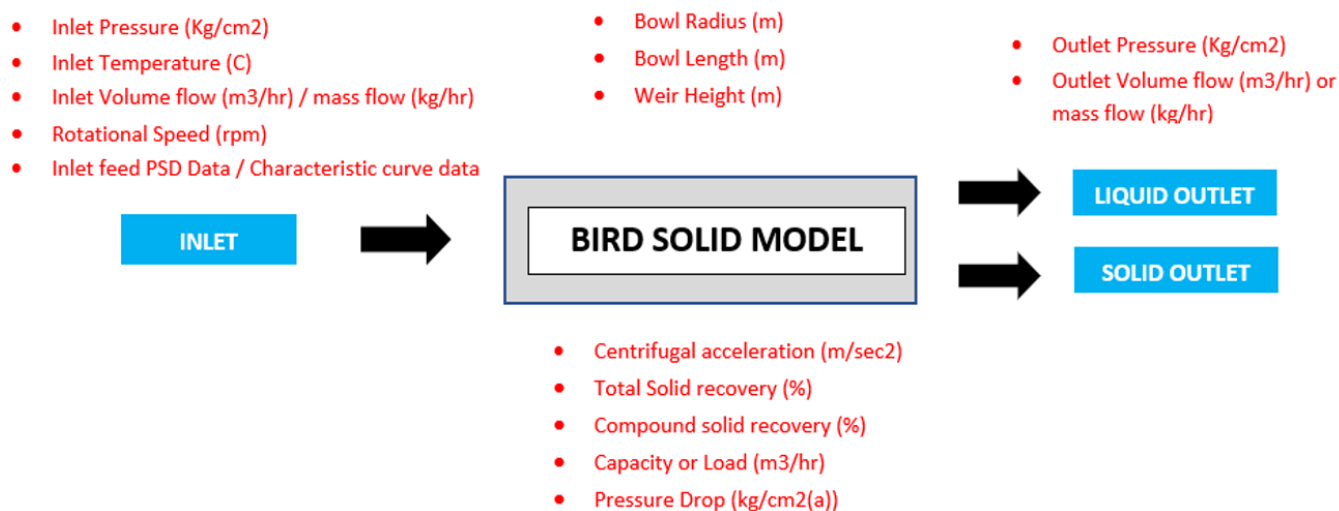
The two typical variants of a centrifuge asset that operate on a continuous mode are,

- Bird Solid Bowl
- Laval disc centrifuge

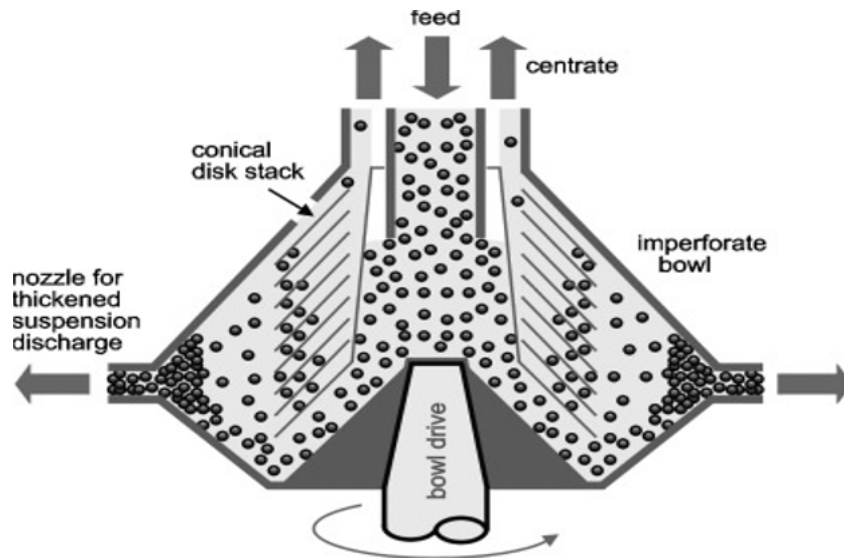
Bird Solid Bowl



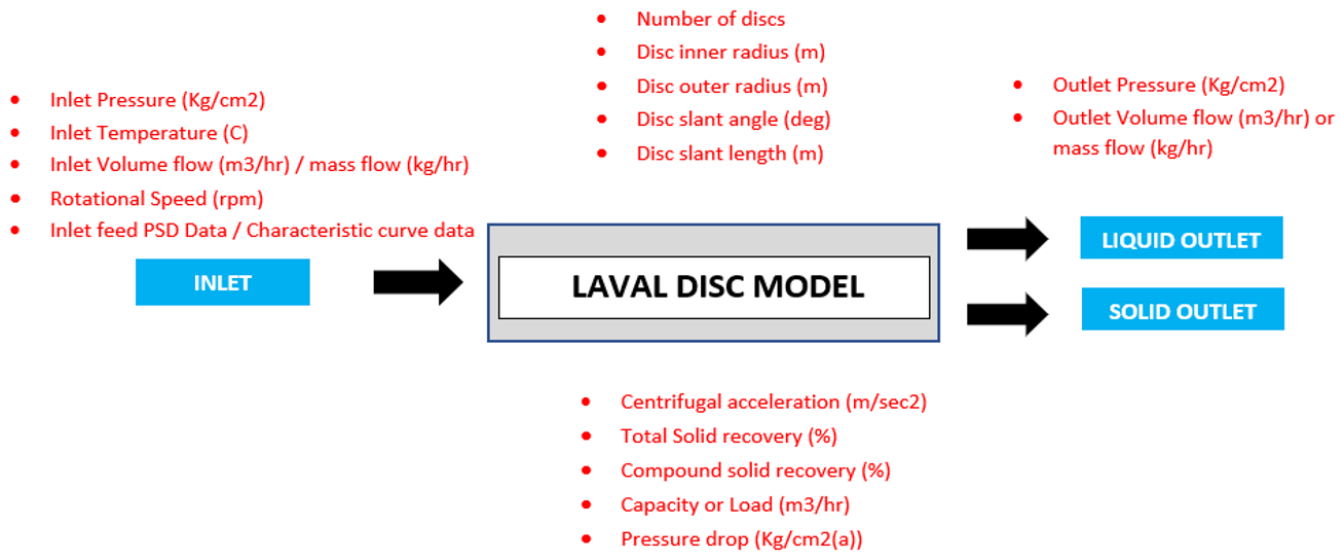
For bird solid bowls with the help of a helical screw conveyor, solid bowl centrifuges separate two substances with different densities by the centrifugal force formed under fast rotation. Feed slurry enters the conveyor and is delivered into the rotating bowl through discharge ports. There is a slight speed difference between the rotation of the conveyor and the bowl, causing the solids to convey from the stationary zone where the feed is introduced to the bowl wall. By centrifugal force, the collected solids move along the bowl wall, out of the pool, and up the dewatering beach located at the tapered end of the bowl. The clarified liquid flows through the conveyor in the opposite direction through adjustable overflow parts.



Laval Disc Centrifuge



In case of Laval disc centrifuge the feed slurry enters from the top through a stationary pipe and flows into different discharge spouts. Inside the bowl are closely spaced disks, which are actual cones of sheet metal set one above the other. Matching holes in the disks about halfway between the axis and the wall of the bowl form channels through which liquid passes. Feed slurry enters the bowl at the bottom, flows into the channels, and upwards past the disks. The liquid is carried above, whereas solid particle accumulates inside and forms a thickened suspension which is further removed through side nozzles.



34.1 Centrifuge - Performance Monitoring

34.1.1 Notations

Below are the notations used in the equations of Performance Modeling, Expected KPI, and Operating KPI

F	Feed Mass flow
L	Liquid Product Mass Flow
S	Solid Product Mass Flow
Q	Feed volume flow
P	Pressure
T	Temperature

H	Mass enthalpy
η	Efficiency
r_o	Outer disc radius
r_i	Inner disc radius
L	Bowl length
r	Bowl radius
h	Weir height
G	Centrifugal acceleration
ω	Rotational Speed

34.1.2 Subscripts

i	Inlet
o	Outlet

34.1.3 Centrifuge - Model Input

34.1.3.1 Bird Solid Bowl Model Input

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Feed Inlet Pressure	Feed Inlet Pressure (Input)	Kg/cm ² (a)	Dynamic	Measurement	Field Input
2	Feed Inlet Temperature	Feed Inlet Temperature (Input)	°C	Dynamic	Measurement	Field Input
3	Feed Mass Flow	Feed Mass Flow (Required Input if actual volumetric flow not provided)	kg/hr	Dynamic	Measurement	Field Input
4	Feed Actual Volumetric flow	Feed Actual Volumetric flow (Required Input if the mass flow not provided)	m ³ /hr	Dynamic	Measurement	Field Input
5	Feed Solid Percent	Feed Solid Percent (Required Input)	%	Dynamic	Measurement	Field Input
6	Feed Density	Feed Density (Required Input)	kg/m ³	Dynamic	Measurement	Laboratory measurement / Thermo-stream properties
7	Liquid Product Pressure	Liquid Product Pressure (Input)	Kg/cm ² (a)	Dynamic	Measurement	Field Input
8	Ideal Feed Liquid Volume Flow	Ideal Feed Liquid Volume Flow (Input)	m ³ /hr	Dynamic	Measurement	Field Input

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
9	Liquid Phase Density	Liquid Phase Density (Input)	kg/m ³	Dynamic	Measurement	Laboratory measurement / Thermo-stream property
10	Liquid Phase Viscosity	Liquid Phase Viscosity (Input)	centipoise	Dynamic	Measurement	Laboratory based measurement / Thermo-stream property
11	Solid Product Mass Flow	Solid Product Mass Flow (Required Input)	kg/hr	Dynamic	Measurement	Field Input
12	Average Solid Density	Average Solid Density (Input used if PSD Data is provided)	kg/m ³	Dynamic	Measurement	Laboratory measurement / Thermo-stream property
13	Solid Component 1	Solid Component 1 (Required Input if PSD Data is provided)	Mass Fraction	Dynamic	Measurement	Laboratory Measurement/Field Input
14	Solid Component 2	Solid Component 2 (Required Input if PSD Data is provided)	Mass Fraction	Dynamic	Measurement	Laboratory Measurement/Field Input
15	Solid Component 3	Solid Component 3 (Required Input if PSD Data is provided)	Mass Fraction	Dynamic	Measurement	Laboratory Measurement/Field Input
16	Solid Component 4	Solid Component 4 (Required Input if PSD Data is provided)	Mass Fraction	Dynamic	Measurement	Laboratory Measurement/Field Input
17	Solid Component 5	Solid Component 5 (Required Input if PSD Data is provided)	Mass Fraction	Dynamic	Measurement	Laboratory Measurement/Field Input
18	Bowl Length	Bowl Length (Input)	m	Static	Design	Equipment Data sheet
19	Bowl Radius	Bowl Radius (Input)	m	Static	Design	Equipment Data sheet
20	Weir Height	Weir Height (Input)	m	Static	Design	Equipment Data sheet
21	Rotational Speed	Rotational Speed (Input)	rpm	Dynamic	Measurement	Field Input
22	Characteristic Curve Data	Characteristic Curve Data (Input - Solid Product Rate Vs Feed Solid Percent Vs Feed Volume flow)		Static	Design	Performance curve mapping
23	PSD Property	PSD Property Data (Input - Feed Solid size Vs Mass Percent)		Static	Design	Performance curve mapping

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
	Data					
24	Solid Compound Density Table	Solid Compound Density Table (Input - Solid Info Vs Compound Density)		Static	Design	Performance curve mapping
25	Custom Performance Data Input	Custom Performance Data Input (Optional Input)		Static	Design	Performance curve mapping
26	Interpolation Type	Interpolation Type (Required input if custom performance data is provided)		Static	Design	Static Input

34.1.3.2 Laval Disc Model Input

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Feed Inlet Pressure	Feed Inlet Pressure (Input)	Kg/cm ² (a)	Dynamic	Measurement	Field Input
2	Feed Inlet Temperature	Feed Inlet Temperature (Input)	°C	Dynamic	Measurement	Field Input
3	Feed Mass Flow	Feed Mass Flow (Required Input if actual volumetric flow not provided)	kg/hr	Dynamic	Measurement	Field Input
4	Feed Actual Volumetric flow	Feed Actual Volumetric flow (Required Input if mass flow not provided)	m ³ /hr	Dynamic	Measurement	Field Input
5	Feed Solid Percent	Feed Solid Percent (Required Input)	%	Dynamic	Measurement	Field Input
6	Feed Density	Feed Density (Required Input)	kg/m ³	Dynamic	Measurement	Laboratory measurement / Thermo-stream properties
7	Liquid Product Pressure	Liquid Product Pressure (Input)	Kg/cm ² (a)	Dynamic	Measurement	Field Input
8	Ideal Feed Liquid Volume Flow	Ideal Feed Liquid Volume Flow (Input)	m ³ /hr	Dynamic	Measurement	Field Input
9	Liquid Phase Density	Liquid Phase Density (Input)	kg/m ³	Dynamic	Measurement	Laboratory measurement / Thermo-stream property
10	Liquid Phase Viscosity	Liquid Phase Viscosity (Input)	centipoise	Dynamic	Measurement	Laboratory based measurement / Thermo-stream property

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
11	Solid Product Mass Flow	Solid Product Mass Flow (Required Input)	kg/hr	Dynamic	Measurement	Field Input
12	Average Solid Density	Average Solid Density (Input used if PSD Data is provided)	kg/m ³	Dynamic	Measurement	Laboratory measurement / Thermo-stream property
13	Solid Component 1	Solid Component 1 (Required Input if PSD Data is provided)	Mass Fraction	Dynamic	Measurement	Laboratory measurement /Field Input
14	Solid Component 2	Solid Component 2 (Required Input if PSD Data is provided)	Mass Fraction	Dynamic	Measurement	Laboratory measurement /Field Input
15	Solid Component 3	Solid Component 3 (Required Input if PSD Data is provided)	Mass Fraction	Dynamic	Measurement	Laboratory measurement /Field Input
16	Solid Component 4	Solid Component 4 (Required Input if PSD Data is provided)	Mass Fraction	Dynamic	Measurement	Laboratory measurement /Field Input
17	Solid Component 5	Solid Component 5 (Required Input if PSD Data is provided)	Mass Fraction	Dynamic	Measurement	Laboratory measurement /Field Input
18	Disc Inner Radius	Disc Inner Radius (Input)	m	Static	Design	Equipment Data sheet
19	Disc Outer Radius	Disc Outer Radius (Input)	m	Static	Design	Equipment Data sheet
20	Disc Slant Angle	Disc Slant Angle (Input)	Degrees	Static	Design	Equipment Data sheet
21	Number of Discs	Number of Discs (Input)		Static	Design	Equipment Data sheet
22	Rotational Speed	Rotational Speed (Input)	rpm	Dynamic	Measurement	Field Input
23	Characteristic Curve Data	Characteristic Curve Data (Input - Solid Product Rate Vs Feed Solid Percent Vs Feed Volume flow)		Static	Design	Performance curve mapping
24	PSD Property Data	PSD Property Data (Input - Feed Solid size Vs Mass Percent)		Static	Design	Performance curve mapping
25	Solid Compound Density Table	Solid Compound Density Table (Input - Solid Info Vs Compound Density)		Static	Design	Performance curve mapping
26	Custom	Custom Performance Data Input		Static	Design	Performance curve

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
	Performance Data Input	(Optional Input)				mapping
27	Interpolation Type	Interpolation Type (Required input if custom performance data is provided)		Static	Design	Static Input

34.1.4 Centrifuge - Model Output

34.1.4.1 Bird Solid Bowl Model Output

No	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Centrifugal Acceleration	Centrifugal Acceleration (Output)	m/s ²	Dynamic	Performance	Calculated
2	Operating Pressure Drop	Operating Pressure Drop (Output)	Kg/cm ² (a)	Dynamic	Performance	Calculated
3	Operating Total Solid Recovery	Operating Total Solid Recovery (Output)	%	Dynamic	Performance	Calculated
4	Expected Compound Solid Recovery	Expected Compound Solid Recovery (Output)		Dynamic	Performance	Calculated
5	Expected Feed Flow	Expected Feed Flow (Output)	m ³ /hr	Dynamic	Performance	Calculated
6	Expected Pressure Drop	Expected Pressure Drop (Output)	Kg/cm ² (a)	Dynamic	Performance	Calculated
7	Expected Solid Percent	Expected Solid Percent (Output)	%	Dynamic	Performance	Calculated
8	Expected Total Solid Recovery	Expected Total Solid Recovery (Output)	%	Dynamic	Performance	Calculated
9	Deviation Capacity Loss	Deviation Capacity Loss (Output)	%	Dynamic	Deviation	Calculated
10	Deviation Feed Solid Percent	Deviation Feed Solid Percent (Output)	%	Dynamic	Deviation	Calculated
11	Deviation Pressure Drop	Deviation Pressure Drop (Output)	%	Dynamic	Deviation	Calculated
12	Deviation Total Solid Recovery	Deviation Total Solid Recovery (Output)	%	Dynamic	Deviation	Calculated
13	Custom Performance Data Output	Custom Performance Data Output (Output - ID = 0 Input data, ID = 1 Operating Curve, ID = 2 Expected point and ID = 3 Operating point)		Dynamic	Performance	Calculated

34.1.4.2 Laval Disc Model Output

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Centrifugal Acceleration	Centrifugal Acceleration (Output)	m/s ²	Dynamic	Performance	Calculated
2	Operating Pressure Drop	Operating Pressure Drop (Output)	Kg/cm ² (a)	Dynamic	Performance	Calculated
3	Operating Total Solid Recovery	Operating Total Solid Recovery (Output)	%	Dynamic	Performance	Calculated
4	Expected Compound Solid Recovery	Expected Compound Solid Recovery (Output)		Dynamic	Performance	Calculated
5	Expected Feed Flow	Expected Feed Flow (Output)	m ³ /hr	Dynamic	Performance	Calculated
6	Expected Pressure Drop	Expected Pressure Drop (Output)	Kg/cm ² (a)	Dynamic	Performance	Calculated
7	Expected Solid Percent	Expected Solid Percent (Output)	%	Dynamic	Performance	Calculated
8	Expected Total Solid Recovery	Expected Total Solid Recovery (Output)	%	Dynamic	Performance	Calculated
9	Deviation Capacity Loss	Deviation Capacity Loss (Output)	%	Dynamic	Deviation	Calculated
10	Deviation Feed Solid Percent	Deviation Feed Solid Percent (Output)	%	Dynamic	Deviation	Calculated
11	Deviation Pressure Drop	Deviation Pressure Drop (Output)	%	Dynamic	Deviation	Calculated
12	Deviation Total Solid Recovery	Deviation Total Solid Recovery (Output)	%	Dynamic	Deviation	Calculated
13	Custom Performance Data Output	Custom Performance Data Output (Output - ID = 0 Input data, ID = 1 Operating Curve, ID = 2 Expected point and ID = 3 Operating point)		Dynamic	Performance	Calculated

34.1.5 Centrifuge - KPIs

The following indicators help us to know overall performance of the asset.

34.1.5.1 Model Inputs

- Feed Inlet Pressure
- Liquid Outlet Pressure
- Feed Inlet Temperature
- Feed Inlet Volume Flow
- Rotational Speed

- Bowl radius
- Bowl length
- Weir height
- Number of discs
- Inner disc radius
- Outer disc radius
- Disc slant angle
- Feed PSD Data (Solid Info, Solid feed size, and Solid mass percent)
- Characteristic curve Data (Solid feed percent, Solid production rate, and Feed Inlet Volume Flow)

34.1.5.2 Model Output KPI's

Operating KPI's

- Centrifugal acceleration.
- Operating Total solid recovery
- Operating Pressure Drop.

Expected KPI's

- Expected Total solid recovery
- Expected Compound solid recovery
- Expected Feed Flow
- Expected Feed Solid percent
- Expected Pressure Drop.

34.1.5.3 Centrifuge - Expected KPI's

Below are the expected KPI formulations of centrifuge assets.

Expected Total Solid Recovery (%)

The expected total solid recovery is formulated based on two approaches, they are PSD Approach and the Characteristic curve approach.

PSD Information Approach

First Cut Size prediction needs to be conducted, kindly refer Appendix section for the derivation of cut size diameter.

$$D_{pc} = \frac{3\sqrt{(Q\mu)/L}}{\sqrt{\pi} \omega r_{eq} \sqrt{(\rho_p - \rho)}}$$

$$D_{pc} = \frac{3\sqrt{(Q\mu)/L}}{\sqrt{\pi} \omega r_{eq} \sqrt{(\rho_p - \rho)}}$$

Here D_{pc} is the pivotable variable which can get updated or changed based on all dependent variables on RHS of the above equation

$$Q = Q_{Feed\ Solid} + Q_{Ideal\ Liquid}$$

$$Q_{Ideal\ Liquid} = \frac{FM_{w_{pure\ liquid}}}{M_{w_{feed}}\rho_{liquid}}$$

For Bird Model

$$r_{eq} = r - r_h$$

For Laval disc Model

$$r_{eq} = \frac{r_o^2 - r_i^2}{\ln\left(\frac{2r_o}{r_i + r_o}\right)}$$

Cut Size or diameter is also a minimum particle diameter, that the settling chamber can separate from the solid-liquid mixture. Further using PSD of feed, we can the cumulative mass percent of feed solid that pass towards solid product streams

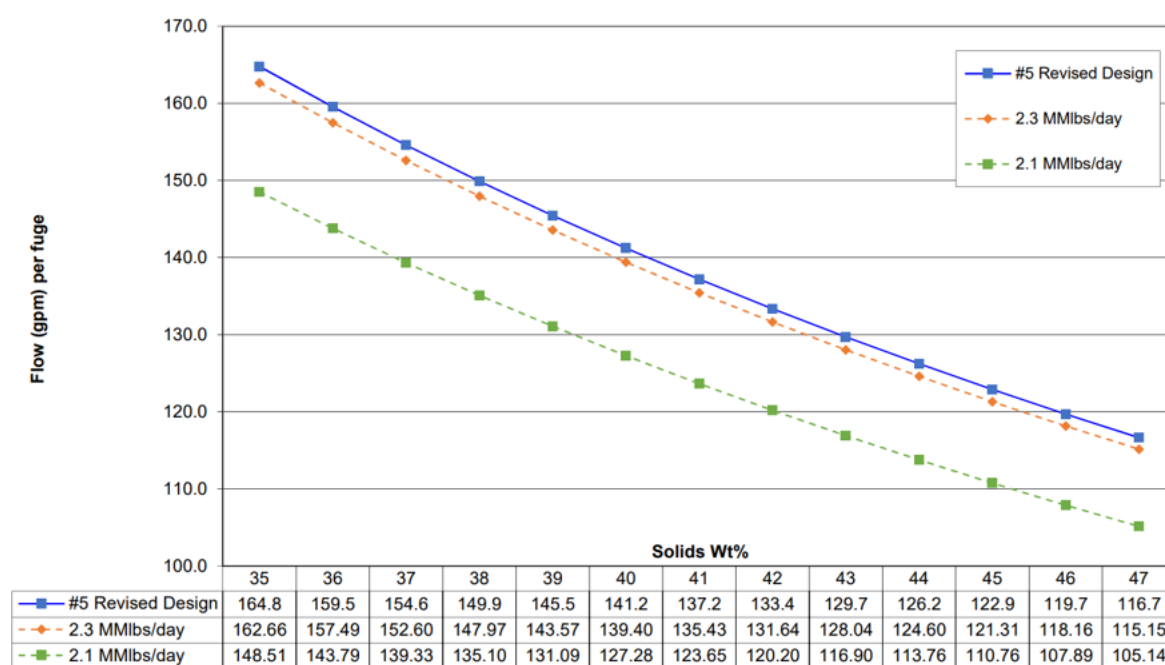
- If $D_p < D_{pc}$ then the following mass percent of feed solid component will pass towards liquid product.
- If $D_p \geq D_{pc}$ then the following mass percent of feed solid component will pass towards solid product.

Using the compositional details against cut size we can predict the total solid recovery as,

$$\eta_{TSR}^{Expected} = \frac{SZ_s}{FZ_f}$$

Characteristic Curve Approach

From the below characteristic curve, we can provide the current solid production rate against the feed solid weight percent and back calculate the Centrifuge flow (capacity). Which later can be used to back predict the total solid recovery of the centrifuge asset



$$Q_{Expected} = f(z_f, Sz_s)$$

$$\eta_{TSR}^{Expected} = \frac{Sz_s}{Q_{expected} \rho z_f}$$

Expected Feed Flow (m³/hr)

From interpolation of the characteristic curve (Figure 3) we get,

$$Q_{Expected} = f(z_f, Sz_s)$$

NOTE

Expected feed flow can be predicted only if Characteristic curve data is provided.

Expected Feed Solid Percent (%)

From interpolation of the characteristic curve (Figure 3) we get,

$$z_{f_{Expected}} = f(Q_i, Sz_s)$$

NOTE

Expected feed solid percent can be predicted only if Characteristic curve data is provided.

Expected Compound Solid Recovery (%)

Here i = Solid component:

$$\eta_{TSR}^{Expected} = \frac{Sz_{s_i}}{Fz_{f_i}}$$

NOTE

Expected compound recovery can be predicted only if PSD data is provided.

Expected Pressure Drop (%)

In case of the Laval disc model

$$\Delta P_{Expected} = \frac{1}{2} \omega^2 \rho (r - h)^2$$

In case of the bird solid bowl model

$$\Delta P_{Expected} = \frac{1}{2} \omega^2 \rho (r_o^2 - r_i^2)$$

34.1.5.4 Centrifuge - Operating KPI's

Below are operating KPI formulations of centrifuge asset

Centrifugal acceleration (m/sec^2)

Centrifugal force at operational condition,

$$G = \omega^2 r_o$$

NOTE

In the case of Laval disc model outer radius will be the outer disc radius and whereas in the bird model outer radius will be the bowl radius

Operating Total Solid Recovery (%)

$$\eta_{TSR}^{operating} = \frac{\text{Solid production rate}}{\text{Solid feed rate}} = \frac{S z_s}{F z_f}$$

Operating Pressure Drop (%)

Difference of pressure between feed inlet pressure and liquid product outlet pressure.

$$\Delta P_{Operating} = P_{f_i} - P_{l_o}$$

34.1.5.5 Centrifuge - Performance Modeling

Performance modeling of the Centrifuge helps to determine the performance KPI. The following is the procedure,

Overall Mass Balance

Feed slurry will be separated as liquid product and solid product

$$F = L + S$$

Liquid component Balance, here x denotes the liquid mass fraction

For Liquid Side:

$i = 1, 2, 3, \dots, n$ --- For n number of liquid components

$$F x_{i_f} = L x_{i_l} + S x_{i_s}$$

For Solid Side:

$i = 1, 2, 3, \dots, n$ --- For n number of solid components

$$F z_{i_f} = L z_{i_l} + S z_{i_s}$$

Overall Force Balance

Applying the force balance equation, the same balance follows for both the solid and liquid product sides.

$$\Delta P = P_o - P_i$$

Overall Energy Balance

Applying the energy balance equation,

$$FH_f = L H_i + SH_s$$

34.2 Centrifuge - Fault Model

34.2.1 Centrifuge - APM Model

Bird Solid Bowl Centrifuge/ Laval Disc Centrifuge

Item	Type
Category	Equipment
Equipment Class	Centrifuge
Equipment Type	Bird Solid Bowl Centrifuge/ Laval Disc Centrifuge

Below table illustrates the list of faults and symptoms configured for a Bird Solid Bowl Centrifuge/ Laval Disc Centrifuge.

Fault Name	Fault Symptoms	Recommendations
Centrifugal Force Low	Rotational Speed RL (BSBCFFL)	1. Check motor performance.
Poor Performance	Deviation Total Solid Recovery LOW (BSBPPER)	1. Check the temperature of the inlet feed. 2. Check the motor performance for low rotational speed. 3. Check the flow rate of the feed.
	Feed Solid Percent LOW (BSBPPER)	
	Feed Actual Volumetric flow RH (BSBPPER)	
	Feed Mass Flow RH (BSBPPER)	
	Feed Inlet Pressure RL (BSBPPER)	
	Liquid Product Mass Flow RL (BSBPPER)	
	Solid Product Mass Flow LOW (BSBPPER)	
	Operating Total Solid Recovery LOW (BSBPPER)	
	Feed Inlet Temperature RH (BSBPPER)	

34.2.2 Centrifuge - APR Model

34.2.2.1 Centrifuge - Bearing System

Item	Type
Category	Equipment
Equipment Class	Centrifuge
Equipment Type	Centrifuge - Bearing System

Below table illustrates the list of faults and symptoms configured for a Centrifuge-Bearing System.

Fault Name	Fault Symptoms	Recommendations
Bearing Failure Due to Mechanical Issue	Feed End Bearing Temperature RH (BFMI)	1. Check calibration of sensor. 2. Investigate with high-frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if intervention is not possible. 4. Intervention options, reduce speed, Parts availability. 5. Check the trends to establish if it's either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 7. Change bearing if required.
	Feed End Bearing Vibration - Horizontal RH (BFMI)	
	Feed End Bearing Vibration - Vertical RH (BFMI)	
Bearing Failure Due to Lube Oil Pressure	Feed End Bearing Temperature RH (BFLOP)	1. Check the calibration of sensor. 2. Switch on both oil pumps and stabilize oil pressure. 3. Check oil level in tank. 4. Check filter DP. 5. Assess malfunction. 6. Line walks the system and check for leaks. 7. Ensure PSV's are seated, check spillback/PCV valves are set correctly and not returning oil back to the tank. 8. If maintenance has recently been performed check correct rotation of pump. 9. Ensure no hand valves have been left shut, and ensure NRV's are working correctly. 10. Check functioning of the auto change-over switch.
	Feed End Bearing Vibration - Horizontal RH (BFLOP)	
	Feed End Bearing Vibration - Vertical RH (BFLOP)	
	Lube Oil Pressure RL (BFLOP)	
	Lube Oil Flow RL (BFLOP)	
Bearing Failure Due to Lube Oil Temperature	Feed End Bearing Temperature RH (BFLOT)	1. Check the calibration of sensor. 2. Check cooling water supply pressure. 3. Check bypass control valve. 4. Open both coolers and stabilize temperature. 5. Check cooling water pumps for blockages, and ensure strainer is clean. 6. If the cooler is an air cooler check fan and motor integrity. 7. If a malfunction of one cooler is suspected. 8. isolate and inspect it. 9. Line walks the system to check for leaks.
	Feed End Bearing Vibration - Horizontal RH (BFLOT)	
	Feed End Bearing Vibration - Vertical RH (BFLOT)	
	Lube Oil Temperature RH (BFLOT)	
Spindle Main Bearing Failure Due to Mechanical Issue	Spindle Top Main Bearing Temperature RH (SMBFMI)	1. Check calibration of sensor. 2. Investigate with high-frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if

Fault Name	Fault Symptoms	Recommendations
	Spindle Bottom Main Bearing Temperature RH (SMBFMI)	intervention is not possible. - Intervention options, reduce speed, Parts availability. - Check the trends to establish if it's either both X and Y probes or one. - Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. - Change bearing if required.
	Spindle Radial Vibration RH (SMBFMI)	
	Spindle Axial Vibration RH (SMBFMI)	
	Axial Frame Vibration RH (SMBFMI)	
Spindle Main Bearing Failure Due to Lube Oil Pressure	Spindle Top Main Bearing Temperature RH (SMBFLOP)	- Check the calibration of sensor. - Switch on both oil pumps and stabilize oil pressure. - Check oil level in tank. - Check filter DP. - Assess malfunction. - Line walks the system and check for leaks. - Ensure PSV's are seated, check spillback/PCV valves are set correctly and not returning oil back to the tank. - If maintenance has recently been performed check correct rotation of pump. - Ensure no hand valves have been left shut, and ensure NRV's are working correctly. - Check functioning of the auto change-over switch.
	Spindle Bottom Main Bearing Temperature RH (SMBFLOP)	
	Spindle Radial Vibration RH (SMBFLOP)	
	Spindle Axial Vibration RH (SMBFLOP)	
	Spindle Oil Supply Pressure RL (SMBFLOP)	
	Spindle Bearing Flow RL (SMBFLOP)	
Spindle Main Bearing Failure Due to Lube Oil Temperature	Spindle Top Main Bearing Temperature RH (SMBFLOT)	- Check the calibration of sensor. - Check cooling water supply pressure. - Check bypass control valve. - Open both coolers and stabilize temperature. - Check cooling water pumps for blockages, and ensure strainer is clean. - If the cooler is an air cooler check fan and motor integrity. - If malfunction of one cooler is suspected. - Isolate and inspect it. - Line walks the system to check for leaks.
	Spindle Bottom Main Bearing Temperature RH (SMBFLOT)	
	Spindle Radial Vibration RH (SMBFLOT)	
	Spindle Axial Vibration RH (SMBFLOT)	
	Spindle Supply Temperature RH (SMBFLOT)	

34.2.2.2 Centrifuge - Gearbox System Fault Model

Item	Type
Category	Equipment
Equipment Class	Centrifuge

Item	Type
Equipment Type	Centrifuge - Gearbox System

Below table illustrates the list of faults and symptoms configured for a Centrifuge - Gearbox System.

Fault Name	Fault Symptoms	Recommendations
End Bearing Failure Due to Mechanical Issue	Gearbox End Bearing Vibration - Horizontal RH (BFMI)	1. Check calibration of sensor. 2. Investigate with high-frequency vibration analysis. 3. Assess likely cause and time to fail(trip) if intervention is not possible. 4. Intervention options, reduce speed, Parts availability. 5. Check the trends to establish if it's either both X and Y probes or one. 6. Inspect bearing for the following, wiping, internal clearances, oil coking / hot spots. 7. Change bearing if required.
	Gearbox End Bearing Vibration - Vertical RH (BFMI)	
	Gearbox Temperature RH (BFMI)	
	Torque Overload Pressure RH (BFMI)	
End Bearing Failure Due to Gearbox Oil Pressure	Gearbox End Bearing Vibration - Horizontal RH (BFGBOP)	1. Check the calibration of sensor. 2. Switch on both oil pumps and stabilize oil pressure. 3. Check oil level in tank. 4. Check filter DP. 5. Assess malfunction. 6. Line walks the system and check for leaks. 7. Ensure PSV's are seated, check spillback/PCV valves are set correctly and not returning oil back to the tank. 8. If maintenance has recently been performed check correct rotation of pump. 9. Ensure no hand valves have been left shut, and ensure NRV's are working correctly. 10. Check functioning of the auto change-over switch.
	Gearbox End Bearing Vibration - Vertical RH (BFGBOP)	
	Gearbox Temperature RH (BFGBOP)	
	Gearbox Oil Supply Pressure RL (BFGBOP)	
	Gearbox Flow RL (BFGBOP)	
End Bearing Failure Due to Gearbox Oil Temperature	Gearbox End Bearing Vibration - Horizontal RH (BFGBOT)	1. Check the calibration of sensor. 2. Check cooling water supply pressure. 3. Check bypass control valve. 4. Open both coolers and stabilize temperature. 5. Check cooling water pumps for blockages, and ensure strainer is clean. 6. If the cooler is an air cooler check fan and motor integrity. 7. If malfunction of one cooler is suspected. 8. Isolate and inspect it. 9. Line walks the system to check for leaks.
	Gearbox End Bearing Vibration - Vertical RH (BFGBOT)	
	Gearbox Temperature RH (BFGBOT)	
	Gearbox Supply Temperature RH (BFGBOT)	

34.2.2.3 Centrifuge - Lube Oil System

Item	Type
Category	Equipment
Equipment Class	Centrifuge
Equipment Type	Centrifuge - Lube Oil System

Below table illustrates the list of faults and symptoms configured for a Centrifuge - Lube Oil System.

Fault Name	Fault Symptoms	Recommendations
Lube Oil Pressure Low	Spindle Reservoir Level RL (LOPL)	1. Check oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, TAN. 4. Check no breather caps or man covers have been removed and left open to allow debris into the tank. 5. Do filter cleaning. 6. Recondition or replace oil.
	Gearbox Reservoir Level RL (LOPL)	
	Lube Oil Pressure RL (LOPL)	
	Lube Oil Flow RL (LOPL)	
Lube Oil Temperature High	Spindle Supply Temperature RH (LOTH)	1. Check the calibration of sensor. 2. Check bypass control valve for integrity. 3. Open both coolers and stabilize temperature. 4. Line walks the system to check for leaks. 5. Check the Isolation/bypass valve of TCV. 6. Check cooling water pumps for blockages, and ensure strainer is clean.
	Spindle Return Temperature RH (LOTH)	
	Gearbox Supply Temperature RH (LOTH)	
	Gearbox Return Temperature RH (LOTH)	
	Spindle Reservoir Temperature RH (LOTH)	
	Gearbox Reservoir Temperature RH (LOTH)	
	Lube Oil Temperature RH (LOTH)	
Spindle Supply Oil Pressure Low	Spindle Supply RL (SSOPL)	1. Check oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination. 3. Take an oil sample and check water content, viscosity, TAN. 4. Check no breather caps or man covers have been removed and left open to allow debris into the tank. 5. Do filter cleaning. 6. Recondition or replace oil.
	Spindle Pressure After Filter RL (SSOPL)	
	Spindle Oil Supply Pressure RL (SSOPL)	
	Spindle Bearing Flow RL (SSOPL)	
Spindle Oil Filter	Spindle Oil Supply Differential	1. Check filter DP.

Fault Name	Fault Symptoms	Recommendations
Anomaly	Pressure RH (SOFA)	2. Changeover and clean/replace the standby filter. 3. Check the Filter DP transmitter.
Gearbox Oil Pressure Low	Gearbox Pump Pressure RL (GBOPL)	1. Check oil tank for free water by draining 1L into a sample bottle. 2. Switch cooler if water contamination.
	Gearbox Oil Supply Pressure RL (GBOPL)	3. Take an oil sample and check water content, viscosity, TAN. 4. Check no breather caps or man covers have been removed and left open to allow debris into the tank.
	Gearbox Flow RL (GBOPL)	5. Do filter cleaning. 6. Recondition or replace oil.
Gearbox Oil Filter Anomaly	Gearbox Pressure After Filter RH (GBOFA)	1. Check filter DP. 2. Changeover and clean/replace the standby filter. 3. Check the Filter DP transmitter.

CONVEYOR BELT

35.1 Conveyor Belt - APR Model

35.1.1 Conveyor Belt - Belt System Fault Model

Item	Type
Category	Equipment
Equipment Class	MiningAsset
Equipment Type	Conveyor Belt- Belt System

Below table illustrates the list of faults and symptoms configured for a Conveyor Belt - Belt System

Fault Name	Fault Symptoms	Recommendations
Damage of Conveyor Belt	The number Of Textile Plies in The Belt Core LOW (DCB)	1. Adjust the alignment of belt, use heat resistant and Vulcan prime, basalt or glass layer high tensile, cold resistant belt. 2. Check the tension and drive motor speed, pulley bearing and edge of the pulley, idler.
	Belt Temperature RH (DCB)	
	Belt Speed RH (DCB)	
	BELT LOAD RH (DCB)	
Conveyor Belt Slippage/ Swaying	Conveyor VIB X RH (CBSS)	1. Check the belt tension, head pulley, rollers, alignment of frames, pulley size, bearing, load of material, whether belt is scored or lose, slow motor speed, and proper installation of pulley lagging.
	Conveyor VIB Y RH (CBSS)	
	Steep Angle of Inclination RL (CBSS)	
	Misalignment Transmitter Alert/ Swaying Alert ON (CBSS)	
	Belt Damage Monitor Sensor ON (CBSS)	
	Conveyor Material Tracking System Status OFF (CBSS)	
	Motion Monitoring and Protection Sensor Status OFF (CBSS)	

35.1.2 Conveyor Belt - Brake System Fault Model

Item	Type
Category	Equipment
Equipment Class	MiningAsset
Equipment Type	Conveyor Belt- Brake System

Below table illustrates the list of faults and symptoms configured for a Conveyor Belt - Brake System.

Fault Name	Fault Symptoms	Recommendations
Unable to activate the brakes	Controllable Moving Device Sensor Status OFF (UAB)	1. Check whether the spring-applied, hydraulically released BSFH 520 brakes act on a 1,6 m (5,2 ft) diameter disc mounted on the low-speed side of the drive shaft. 2. The brake system provides up to 134,354 ft-lbs (182,160 Nm) nominal braking torque to stop the conveyor in a controlled manner over a 50-second time period.
	Oil Temperature RH (UAB)	
	Braking Force RL (UAB)	
	Retraction Time Indicator Status OFF (UAB)	
	VFD Motor Controller Status OFF (UAB)	
Not applying sufficient braking force	Oil Level RH (NASBF)	1. Check the hydraulic oil level, temperature of oil, brake disc, brake caliper, brake drum and oil pressure, any leakages of oil and oily drums, proper working of the three SOBO controllers.
	Oil Temperature RH (NASBF)	
	Braking Force RL (NASBF)	
	Brake ON/OFF Indicator Status OFF (NASBF)	
	Brake Disc Sensor status ON (NASBF)	
Delayed Braking Anomaly	Overspeed Alarm Status ON (DBA)	1. To check the take-up system like gravity, screw adjusted, or winch is working properly and maintain the pressure and tension of belt, working of motor VFD proper setting distance of idlers, friction between the pulley and the idlers.
	Brake Ramp Status OFF (DBA)	
	VFD Motor Controller Status OFF (DBA)	
	Brake Disc Sensor	

Fault Name	Fault Symptoms	Recommendations
	status OFF (DBA)	

35.1.3 Conveyor Belt - Drive System Fault Model

Item	Type
Category	Equipment
Equipment Class	MiningAsset
Equipment Type	Conveyor Belt- Drive System

Below table illustrates the list of faults and symptoms configured for a Conveyor Belt - Drive System.

Fault Name	Fault Symptoms	Recommendations
Radial Bearing Vib Malfunction rH	Motor DE Radial VIB X RH (RBVM)	1. Inspect bearings and internal clearances. 2. Check Alignment. 3. Plan for bearing replacement in Upcoming opportunities.
	Motor DE Radial VIB Y RH (RBVM)	
	Motor DE Radial VIB X ROCH ST (RBVM)	
	Motor DE Radial VIB Y ROCH ST (RBVM)	
	Motor NDE Radial VIB X RH (RBVM)	
	Motor NDE Radial VIB Y RH (RBVM)	
	Motor NDE Radial VIB X ROCH ST (RBVM)	
	Motor NDE Radial VIB Y ROCH ST (RBVM)	
Radial Bearing Temp Malfunction rH	Motor DE RB TEMP RH (RBTM)	1. Check Lubrication. 2. Check Alignment. 3. Check cooling fan for any damage.
	Motor DE RB TEMP ROCH ST (RBTM)	
	Motor NDE RB TEMP RH (RBTM)	
	Motor NDE RB TEMP ROCH ST (RBTM)	
Radial Bearing BDF Malfunction rH	DE RB BDF RH (RBBDFM)	1. Replacement of Lubrication. 2. Check Bearing Clearances.
	NDE RB BDF RH (RBBDFM)	

Fault Name	Fault Symptoms	Recommendations
Radial Bearing SF/Kur Malfunction rH	DE RB Shock Finder RH (RBSFKM)	1. Replacement of Lubrication. 2. Check Bearing Clearances. 3. Plan for bearing replacement in coming opportunities.
	DE RB Kurtosis MVX RH (RBSFKM)	
	NDE RB Shock Finder RH (RBSFKM)	
	NDE RB Kurtosis MVX RH (RBSFKM)	
Motor overheating and/or winding overheating due to overload	Winding Temperature R1 RH (MTROVHT)	1. The cooling fan condition needs to be checked. 2. Test Insulation resistance of the winding. 3. Cleaning of Dust from cooling fins.
	Winding Temperature R2 RH (MTROVHT)	
	Winding Temperature Y1 RH (MTROVHT)	
	Winding Temperature Y2 RH (MTROVHT)	
	Winding Temperature B1 RH (MTROVHT)	
	Winding Temperature B2 RH (MTROVHT)	
	Current 3-Phase RH (MTROVHT)	
Motor speed low	Motor Speed RL (MTRSL)	1. Check Motor RPM with Techometer in No load condition. 2. Checking of Bearing. 3. Check Greasing condition.
Weak Electric Motor Performance	Belt Speed RH (WEMP)	1. Check the power supply of electricity, bearing of pulley, proper alignment of frame, output shaft bearing, coupling fittings.
	Belt Force RH (WEMP)	
	Belt Tension RH (WEMP)	
	Load Capacity of Belt RH (WEMP)	

35.1.4 Conveyor Belt - Pully System Fault Model

Item	Type
Category	Equipment
Equipment Class	MiningAsset
Equipment Type	Conveyor Belt- Pully System

Below table illustrates the list of faults and symptoms configured for a Conveyor Belt - Pulley System.

Fault Name	Fault Symptoms	Recommendations
Pulley Bearing Vib Malfunction rH	Head Pulley DE RB VIB RH (PBVM)	1. Inspect bearings and internal clearances. 2. Check Alignment. 3. Plan for bearing replacement in Upcoming opportunities.
	Tail Pulley DE RB VIB RH (PBVM)	
	Head Pulley NDE RB VIB RH (PBVM)	
	Tail Pulley NDE RB VIB RH (PBVM)	
	Head Pulley DE RB VIB ROCH ST (PBVM)	
	Tail Pulley DE RB VIB ROCH ST (PBVM)	
	Head Pulley NDE RB VIB ROCH ST (PBVM)	
	Tail Pulley NDE RB VIB ROCH ST (PBVM)	
	Pressure Pulley RB VIB RH (PBVM)	
Pulley Bearing Temp Malfunction rH	Head Pulley DE RB TEMP RH (PBTM)	1. Check Lubrication. 2. Check Alignment. 3. Check cooling fan for any damage.
	Tail Pulley DE RB TEMP RH (PBTM)	
	Head Pulley NDE RB TEMP RH (PBTM)	
	Tail Pulley NDE RB TEMP RH (PBTM)	
	Head Pulley DE RB TEMP ROCH ST (PBTM)	
	Tail Pulley DE RB TEMP ROCH ST (PBTM)	
	Head Pulley NDE RB TEMP ROCH ST (PBTM)	
	Tail Pulley NDE RB TEMP ROCH ST (PBTM)	
	Pressure Pulley RB TEMP RH (PBTM)	
Pulley Bearing BDF Malfunction rH	Head Pulley DE RB DEF RH (PBBDFM)	1. Replacement of Lubrication. 2. Check Bearing Clearances.
	Head Pulley NDE RB DEF RH (PBBDFM)	
	Tail Pulley DE RB DEF RH (PBBDFM)	
	Tail Pulley NDE RB DEF RH (PBBDFM)	
	Pressure Pulley RB DEF RH (PBBDFM)	
Pulley Bearing SF/Kur Malfunction rH	Head Pulley DE RB Shock Finder RH (PBSFKM)	1. Replacement of Lubrication. 2. Check Bearing Clearances. 3. Check Bearing Roller/Race condition. 4. Plan for bearing replacement in coming opportunities.

Fault Name	Fault Symptoms	Recommendations
	Head Pulley DE RB Kurtosis MVX RH (PBSFKM)	
	Head Pulley NDE RB Shock Finder RH (PBSFKM)	
	Head Pulley NDE RB Kurtosis MVX RH (PBSFKM)	
	Tail Pulley DE RB Shock Finder RH (PBSFKM)	
	Tail Pulley DE RB Kurtosis MVX RH (PBSFKM)	
	Tail Pulley NDE RB Shock Finder RH (PBSFKM)	
	Tail Pulley NDE RB Kurtosis MVX RH (PBSFKM)	
	Pressure Pulley RB Shock Finder RH (PBSFKM)	
	Pressure Pulley RB Kurtosis MVX RH (PBSFKM)	

35.1.5 Conveyor Belt - Transmission System Fault Model

Item	Type
Category	Equipment
Equipment Class	MiningAsset
Equipment Type	Conveyor Belt- Transmission System

Below table illustrates the list of faults and symptoms configured for a Conveyor Belt - Transmission System.

Fault Name	Fault Symptoms	Recommendations
Gear box Bearing Vib Malfunction rH	GEARBOX IP DE VIB X RH (BVM)	1. Correct Alignment. 2. Bearing clearances need to be checked. 3. Plan for bearing replacement in coming opportunities.
	GEARBOX IP DE VIB Y RH (BVM)	
	GEARBOX OP NDE VIB X RH (BVM)	
	GEARBOX OP NDE VIB Y RH (BVM)	
	GEARBOX IP DE VIB X ROCH ST (BVM)	
	GEARBOX IP DE VIB Y ROCH ST (BVM)	
	GEARBOX OP NDE VIB X ROCH ST (BVM)	
	GEARBOX OP NDE VIB Y ROCH ST (BVM)	
	GEARBOX IP NDE VIB X RH (BVM)	
	GEARBOX IP NDE VIB Y RH (BVM)	
	GEARBOX IP NDE VIB X ROCH ST (BVM)	
	GEARBOX IP NDE VIB Y ROCH ST (BVM)	
	GEARBOX OP DE VIB X RH (BVM)	
	GEARBOX OP DE VIB Y RH (BVM)	
	GEARBOX OP DE VIB X ROCH ST (BVM)	
	GEARBOX OP DE VIB Y ROCH ST (BVM)	
	GEARBOX INT VIB X RH (BVM)	
	GEARBOX INT VIB Y RH (BVM)	
	GEARBOX INT VIB X ROCH ST (BVM)	
	GEARBOX INT VIB Y ROCH ST (BVM)	
	GEARBOX OIL SUPPLY PRESSURE RH (BVM)	
	GEARBOX OIL FLOW RH (BVM)	

Fault Name	Fault Symptoms	Recommendations
Gear box Bearing Temp Malfunction rH	GEARBOX IP DE TEMP RH (BTM)	1. Replace Lubrication. 2. Check Alignment.
	GEARBOX OP NDE TEMP RH (BTM)	
	GEARBOX IP DE TEMP ROCH ST (BTM)	
	GEARBOX OP NDE TEMP ROCH ST (BTM)	
	GEARBOX IP NDE TEMP RH (BTM)	
	GEARBOX IP NDE TEMP ROCH ST (BTM)	
	GEARBOX OP DE TEMP RH (BTM)	
	GEARBOX OP DE TEMP ROCH ST (BTM)	
	GEARBOX INT TEMP RH (BTM)	
	GEARBOX INT TEMP ROCH ST (BTM)	
	GEARBOX OIL TEMP RH (BTM)	
Gear box Bearing BDF Malfunction rH	GEARBOX IP DEF RH (BBDFM)	1. Replacement of Lubrication. 2. Check Bearing Clearances. 3. Plan for bearing replacement in coming opportunities.
	GEARBOX INT DEF RH (BBDFM)	
	GEARBOX OP DEF RH (BBDFM)	
Gear box Bearing SF/Kur Malfunction rH	GEARBOX IP Shock Finder RH (BSFKM)	1. Replacement of Lubrication. 2. Check Bearing Clearances. 3. Check Bearing Roller/Race condition. 4. Plan for bearing replacement in coming opportunities.
	GEARBOX IP Radial Bearing Kurtosis MVX RH (BSFKM)	
	GEARBOX INT Shock Finder RH (BSFKM)	
	GEARBOX INT Radial Bearing Kurtosis MVX RH (BSFKM)	
	GEARBOX OP Shock Finder RH (BSFKM)	
	GEARBOX OP Radial Bearing Kurtosis MVX RH (BSFKM)	

ROTARY PRESSURE FILTER

36.1 Rotary Pressure Filter - Fault Model

36.1.1 Rotary Pressure Filter - APM Model

Model Type	APR Model
Category	Filter
Equipment Class	Filter
Equipment Type	Rotary Pressure Filter

Below table illustrates the list of faults and symptoms configured for the Rotary Pressure Filter System Fault Model

Fault Name	Fault Symptoms	Recommendations
High Moisture Content	M1-1423 PTA Dryer Moisture HIGH (RMH)	1. Check speed is not high. 2. Check for additional water input. 3. Check the Filtrate separators outlet PCV's integrity. 4. Regulate Steam flow. 5. Check Dryer Motor Performance.
	M1-1423 PTA Dryer Amps RH (RMH)	
	Main Filtrate Separator PCV A OP RH (RMH)	
	Main Filtrate Separator PCV B OP RH (RMH)	
	Second Wash Filtrate Separator PCV OP RH (RMH)	
	Steam Flow to dryer RH (RMH)	
Low Throughput Feed Line issue	Delta P (RPF & Main Filtrate Separator) HIGH (RTLF)	1. Take Pump changeover. 2. Check for feed line choke. 3. Check Filtrate separators level and PCV functions. 4. Do Caustic Wash if required.
	Delta P (RPF & Secondary Wash Filtrate Separator) HIGH (RTLF)	
	Feed Drum Level RL (RTLF)	
	Main Filtrate Separator Level RL (RTLF)	
	Main Filtrate Separator Level 2 RL (RTLF)	
	RPF Feed Flow RL (RTLF)	
	Second Wash Filtrate Separator Level RL (RTLF)	

Fault Name	Fault Symptoms	Recommendations
Low Throughput Operational Issue	Feed Drum Level RL (RTLO)	1. Increase DP between PDCV of Feed drum and Main and Secondary. 2. Filtrate the drum to a high limit. 3. Increase Water Wash.
	First Wash Flow RL (RTLO)	
	Main Filtrate Separator Level RL (RTLO)	
	Main Filtrate Separator Level 2 RL (RTLO)	
	RPF Feed Flow RL (RTLO)	
	Second Wash Filtrate Separator Level RL (RTLO)	
	Second Wash Flow RL (RTLO)	
RPF drum filtrate pipe choked	Inert Gas Flow to RPF RL (RDFPC)	1. Check the agitator for proper mixing in the feed tank. 2. Check the slurry strength and take the necessary action as required. 3. Increase Water Wash. 4. RPF water wash, if remains the same will be a caustic wash.
	Main Filtrate Separator Level RL (RDFPC)	
	Main Filtrate Separator Level 2 RL (RDFPC)	
Low washing efficiency	First Wash Flow RL (RLE)	1. Increase wash water. 2. Check the water distribution spray test. 3. Regulate the speed of RPF to increase washing time. 4. Check water line pressure for any leaks.
	Second Wash Flow RL (RLE)	
	Wash Water Temperature RH (RLE)	
High RPF drum torque	RPF Main Drive Torque RH (RTH)	1. Check for cake removal. 2. Water wash the RPF, if still not improved caustic wash will be done. 3. Check motor merger. 4. Check gearbox vibration. 5. Check RPF bearing for clearances and other anomalies.
RPF Drum Seal leakage	Seal Water Pressure RL (RDSL)	1. Check leakages in Seal System and plan for replacement. 2. Check Seal Water Supply.
	Seal Water Pressure RH (RDSL)	
	Seal Water Return Flow from RPF Main Shaft RL (RDSL)	
	Seal Water Supply Flow to RPF Main Shaft RL (RDSL)	
	Seal Water Supply Flow to RPF Main Shaft RH (RDSL)	
	Seal DP for Main Shaft (Seal Pr -RPF Pr) RL (RDSL)	

Fault Name	Fault Symptoms	Recommendations
RPF screw seal leakage	Seal Water Pressure RL (RSSL)	1. Check leakages in Seal System and plan for replacement. 2. Check Seal Water Supply.
	Seal Water Pressure RH (RSSL)	
	Seal Water Return Flow from RPF Screw RL (RSSL)	
	Seal Water Supply Flow to RPF Screw RL (RSSL)	
	Seal Water Supply Flow to RPF Screw RH (RSSL)	
	Seal DP for Main Shaft (Seal Pr -RPF Pr) RL (RSSL)	
RPF Drum Level Anomaly	RPF Drum Level LT15111 RL (RDLA)	1. Check Feed pumps. 2. Check the feed line for any choke. 3. Check the solution for slurry content in the feed tank if the level is high. 4. Check the agitator for proper mixing. 5. Adjust the overflow weir plate for adjusting the level in the drum.
	RPF Drum Level LT15111 RH (RDLA)	
RPF Feed Pump Discharge Anomaly	RPF Feed Pump Discharge Pressure RL (RFDP)	1. Check the filtrate line for choke. 2. Check the agitator for integrity. 3. Check pump internals and bearing clearances.
	RPF Feed Pump Motor Current RH (RFDP)	

36.1.2 Rotary Pressure Filter - APR Model

36.1.2.1 Rotary Valve System

Model Type	APR Model
Category	Filter
Equipment Class	Filter
Equipment Type	Rotary Valve

Below table illustrates the list of faults and symptoms configured for Rotary Valve System

Fault Name	Fault Symptoms	Recommendations
Rotary Valve Torque Anomaly	Rotary Valve Torque (XI15138) RH (RRA)	1. Check Water wash RPF Try to increase speed for some time. 2. Check chain. 3. Check motor merger. 4. Check valve and motor bearings for clearances and other defects.
Rotary Valve Speed	Rotary Valve Speed (SIA15138A.PV) RL	1. Check the valve motor for integrity.

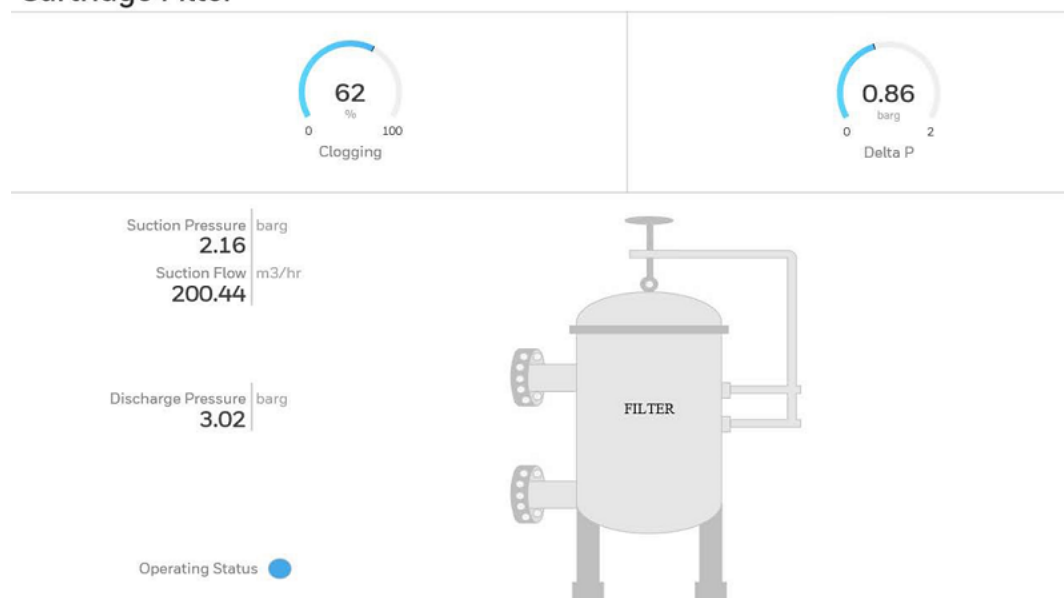
Fault Name	Fault Symptoms	Recommendations
Anomaly	(RRS)	2. Check the VFD drive of the motor for malfunction.
	Rotary Valve Speed (SIA15138A.PV) RH (RRS)	
Rotary Differential Temperature Anomaly	Differential Temp between Screw feeder and Inner casing LOW (RRD)	1. Increase the LP steam flow to maintain temperature. 2. Regulate water wash as per requirement. 3. Check gas flow for drying and cake blowback.
	Differential Temp between Inner and Outer casing LOW (RRD)	

CARTRIDGE FILTER MONITORING

Cartridge filters are used for filtration purpose which uses filtration technology of removing solid matter and suspended impurities from a fluid stream by passing it through a variety micro porous filters, ultrafiltration units, sand cartridge filters, activated carbon cartridge filters etc.

A cartridge filter consists of one or more filter cartridges, which slot into a housing. Cartridges can be made of a rigid or flexible material. They have to be thrown away when they become blocked.

Cartridge Filter



This document explains cartridge filter monitoring philosophy for process, performance, energy and health parameters and their importance.

37.1 Cartridge Filter Monitoring - Performance Monitoring

The following measured parameters are monitored as they have impact on performance and health of Cartridge filters, if not maintained in operating range.

Suction Pressure: The suction pressure of cartridge filter affects yarn hairiness and compactness. As the suction pressure increases, the fibers are tightly pressed over filter with less slippage. Therefore, yarn diameter decreases due to more transverse force on surface fibers. Therefore, more compact structure of the yarn has been formed with less diameter, Result in higher flow rate.

Filter Flow: It is useful to keep records of the flow that has passed through a filter to get a feel for how long a cartridge will last and when it may need replacing.

Discharge Pressure: Higher discharge pressure restricts flow of fluid through filter at same time it causes solid unloading from filter and fluid migration. Therefore, pressure drop across filter is closely monitor by operator for deciding changeover of filter.

Process Performance Overview: Process Performance monitoring view in APM provides a list of the asset attributes and their details, including the current value. This list can be used to review and update asset attribute information, as well as to open an Asset Trend to view more details. It also highlights (in red color) the attributes if deviates from the high or low limits.

37.1.1 Cartridge Filter Monitoring - Model Inputs

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Rated Air Flow	Rated Air Flow (Input)		Static	Design	Field Input
2	Reference Density	Reference Density (Input)	barg	Static	Thermo Properties	Field Input
3	Suction Pressure	Suction Pressure (Input)	°C	Dynamic	Process	Field Input
4	Discharge Pressure	Discharge Pressure (Input)	°C	Dynamic	Process	Field Input
5	Volumetric Flow Rate	Volumetric Flow Rate (Input)	m ³ /hr	Dynamic	Process	Field Input
6	Fluid Density	Fluid Density (Input)	barg	Dynamic	Thermo Properties	Field Input

37.1.2 Cartridge Filter Monitoring - KPIs

The following indicators help us to know overall performance of the Cartridge filter.

Differential Pressure

This indicates the pressure drop across the filter. The rise in differential pressure indicates deterioration of filter performance.

$$dP = P1 - P2$$

Where,

P1= Pressure at filter Inlet P2= Pressure at filter Outlet

Percentage clogging

This indicates current clogging in the filter. Above 80% clogging, the filter elements are recommended to clean or replace. Clogging is calculated by comparing design flow, pressure drop with current operating conditions.

$$\% \text{Clogging} = \frac{\text{Rated Flowrate} * (\text{Actual Pressure drop} - \text{Housing Pressure drop})}{\text{Actual Flowrate} * \text{Max allowable pressure drop}} * 100$$

Housing Pressure drop = Fluid Constant*Fluid density*Flow rate

37.1.3 Cartridge Filter Monitoring - Model Outputs

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	Filter Relative Clogging	Filter Relative Clogging (Output)		Dynamic	Performance	Calculated
2	Filter Actual DP	Filter Actual DP (Output)		Dynamic	Performance	Calculated

37.2 Cartridge Filter Monitoring - Fault Model

In Asset Performance Management, you can configure faults, symptoms, and fault models on Cartridge Filter performance output to diagnose the problems early. The fault management can inform operators of any violations of the nominal Cartridge Filter operating range and deviations from the expected characteristic and make this data available for further processing through the historian. The fault symptoms or models are used to:

1. Warn against potential damage to Cartridge Filter under unfavorable operating conditions.
2. Provide early warning of damages to the Cartridge Filter.

The following messages can be generated for process alarms or notifications to warn operating personnel of unfavorable operating conditions

1. Higher differential pressure
2. High filter clogging
3. Incoming fluid too dirty
4. Sudden changes in flow rate to a filter
5. Changing cartridges too late
6. Cartridge pore size is too large
7. Cartridge not seated or installed correctly
8. Cartridge is starting to unload
9. Cartridge has ruptured (sudden loss in differential pressure)
10. By concentrating particles in one area

Below table illustrates the list of faults and symptoms configured for a Cartridge Filter to identify possible anomalies that affects the performance or health of a Cartridge Filter.

Fault Name	Associated Symptom
Performance Degradation Warning	Filter Differential Pressure High
Health Degradation Warning	Filter Clogging High

ROTARY VACUUM FILTER

38.1 Rotary Vacuum Filter - APR Model

38.1.1 Rotary Vacuum Filter - RVF Drum

Item	Type
Category	Equipment
Equipment Class	Filter
Equipment Type	Rotary Vacuum Filter - RVF Drum

Below table illustrates the list of faults and symptoms configured for a Rotary Vacuum Filter - RVF Drum.

Fault Name	Fault Symptoms	Recommendations
Bridging of Outlet Chute	RVF Vacuum Pressure RH (BROTC)	1. Check for blockage of chute. 2. Adjust feed flow and speed of drum to optimize cake formation.
	Outlet screw Amps RVF A RL (BROTC)	
	Outlet screw Amps RVF A RH (BROTC)	
	Outlet screw Amps RVF B RL (BROTC)	
	Outlet screw Amps RVF B RH (BROTC)	
	Outlet screw Amps RVF C RL (BROTC)	
	Outlet screw Amps RVF C RH (BROTC)	
	Motor Current RVF A RH (BROTC)	
	Motor Current RVF B RH (BROTC)	
	Motor Current RVF C RH (BROTC)	
	Case Pressure RVF A RH (BROTC)	
	Case Pressure RVF B RH (BROTC)	

Fault Name	Fault Symptoms	Recommendations
	Case Pressure RVF C RH (BROTC)	
Cylinder Speed and Submergence Malfunction	Filter Speed RVF A RL (CYCSSBMF)	1. Adjust the speed, level, and inlet slurry flow to the drum.
	Filter Speed RVF A RH (CYCSSBMF)	
	Filter Speed RVF B RL (CYCSSBMF)	
	Filter Speed RVF B RH (CYCSSBMF)	
	Filter Speed RVF C RL (CYCSSBMF)	
	Filter Speed RVF C RH (CYCSSBMF)	
	Feed RVF A RH (CYCSSBMF)	
	Feed RVF B RH (CYCSSBMF)	
	Feed RVF C RH (CYCSSBMF)	
	VAT Level RVF A RL (CYCSSBMF)	
	VAT Level RVF A RH (CYCSSBMF)	
	VAT Level RVF B RL (CYCSSBMF)	
	VAT Level RVF B RH (CYCSSBMF)	
	VAT Level RVF C RL (CYCSSBMF)	
	VAT Level RVF C RH (CYCSSBMF)	
Cloth Rinse Malfunction	Cloth Rinse Flow RVF A RL (CLRMF)	1. Adjust the control valve of recycle gas and valve on the blow back.
	Cloth Rinse Flow RVF B RL (CLRMF)	
	Cloth Rinse Flow RVF C RL (CLRMF)	
	Cloth Rinse Temperature RL (CLRMF)	
	Case Pressure RVF A RL	

Fault Name	Fault Symptoms	Recommendations
	(CLRMF)	
	Case Pressure RVF A RH (CLRMF)	
	Case Pressure RVF B RL (CLRMF)	
	Case Pressure RVF B RH (CLRMF)	
	Case Pressure RVF C RL (CLRMF)	
	Case Pressure RVF C RH (CLRMF)	
High Load on RVF	Feed RVF A RH (HLRVF)	1. Adjust the flow rate of feed. 2. Check level of feed in RVF drum.
	Feed RVF B RH (HLRVF)	
	Feed RVF C RH (HLRVF)	
	Motor Current RVF A RH (HLRVF)	
	Motor Current RVF B RH (HLRVF)	
	Motor Current RVF C RH (HLRVF)	
Pressure Buildup in Casing	RVF Vacuum Pressure RH (PRBUC)	1. Check vacuum pump condition. 2. Check flow of feed to RVF. 3. Check for bridging of chute.
	Case Pressure RVF A RH (PRBUC)	
	Case Pressure RVF B RH (PRBUC)	
	Case Pressure RVF C RH (PRBUC)	
	Filtrate Flow RL (PRBUC)	
RVF at risk of overflow	RVF Vacuum Pressure RH (ROFRVF)	1. Check if RVF is not overflowing (loss of submergence) & adjust feed, as required Check the RVF-speed and reduce it if required. 2. Check Filtrate outlet chain valve operation.
	VAT Level RVF A RH (ROFRVF)	
	VAT Level RVF B RH (ROFRVF)	
	VAT Level RVF C RH (ROFRVF)	
	DP control OP RVF A RH (ROFRVF)	
	DP control OP RVF B RH	

Fault Name	Fault Symptoms	Recommendations
	(ROFRVF)	
	DP control OP RVF C RH (ROFRVF)	
VAT Level Low	VAT Level RVF A LOW (VLVL)	1. Check & confirm physical VAT level, at local. 2. Increase RVF Feed flow. 3. Check feed pump motor.
	VAT Level RVF B LOW (VLVL)	
	VAT Level RVF C LOW (VLVL)	
	Vacuum pump discharge pressure controller OP HIGH (VLVL)	
	RVF Vacuum Pressure Controller OP LOW (VLVL)	
Low mean time between wash	Cloth DP RVF A RH (LMTW)	1. Check the solvent conditions. 2. Visually check the physical condition of cloth.
	Cloth DP RVF B RH (LMTW)	
	Cloth DP RVF C RH (LMTW)	
	Dryer- Chromium HIGH (LMTW)	
	Dryer - Cobalt HIGH (LMTW)	
High vapor in RVF-VAT drum	Dryer - Manganese HIGH (LMTW)	1. Check the temperature of feed. 2. Lower recycle seal flow.
	RVF Vacuum Pressure RL (HVRVD)	
	3rd Crystallizer Temperature RH (HVRVD)	
	Seal Recycle Flow RH (HVRVD)	
Low Throughput (dry basis)	Filtrate Flow RL (LTH)	1. Check RVF Cloth (visually) - for any choking and RVF service life. 2. Check if 4-CBA is not higher than 0.37% and if high, bring it down. 3. Change over RVF and do caustic wash.
High solids slippage in the filtrate	Mother Liquor Solid Content HIGH (HSSFRVF)	1. Check physical condition of cloth. 2. Check rinse-bar falls down or not. 3. Check filter cloth service life. 4. Change-over RVF (if one RVF fail). 5. Shutdown (if both the running RVF have same problem).
	Filtrate Drum Agitator Current RH (HSSFRVF)	

Fault Name	Fault Symptoms	Recommendations
Low solvent flow	Solvent Flow RL (LSFRVF)	1. Check for blockage in solvent line. 2. Check for solvent pump motor.
	Recycle Solvent Flow RL (LSFRVF)	
	Sp Consumption of Solvent Flow RL (LSFRVF)	
	Sp Consumption of Recycle Solvent Flow RL (LSFRVF)	

38.1.2 Rotary Vacuum Filter - Screw Conveyor and Dryer

Item	Type
Category	Equipment
Equipment Class	Rotary Vacuum Filter
Equipment Type	Rotary Vacuum Filter - Screw Conveyor and Dryer

Below table illustrates the list of faults and symptoms configured for a Rotary Vacuum Filter - Screw Conveyor and Dryer.

Fault Name	Fault Symptoms	Recommendations
Outlet Screws Load Variability	Outlet screw Amps RVF A RL (OTSCLV)	1. Check for High fines in the solids. 2. Check for high organics.
	Outlet screw Amps RVF A RH (OTSCLV)	
	Outlet screw Amps RVF B RL (OTSCLV)	
	Outlet screw Amps RVF B RH (OTSCLV)	
	Outlet screw Amps RVF C RL (OTSCLV)	
	Outlet screw Amps RVF C RH (OTSCLV)	
	Moisture based on SM Steam HIGH (OTSCLV)	
	Moisture based on SL Steam HIGH (OTSCLV)	
	Cake Moisture HIGH (OTSCLV)	
Outlet screw overload	Current of Discharge Feed Screw Conveyor RH (OTSCOL)	1. Check for the integrity of switches and solenoid. 2. Check for the state and quantity of material, ensure material is in proper condition as per recommendations, remove extra material dumped in the conveyor.
	Feed RVF A RH (OTSCOL)	
	Feed RVF B RH (OTSCOL)	

Fault Name	Fault Symptoms	Recommendations
	Feed RVF C RH (OTSCOL)	
Outlet screw shaft fatigue	Outlet screw Amps RVF A RH (OTSCSF)	1. Periodic elimination of accumulated hardened material or add trough liners or mount the screw on larger diameter casing with a heavier wall thickness.
	Outlet screw Amps RVF B RH (OTSCSF)	
	Outlet screw Amps RVF C RH (OTSCSF)	

38.1.3 Rotary Vacuum Filter - Vacuum System

Item	Type
Category	Equipment
Equipment Class	Rotary Vacuum Filter
Equipment Type	Rotary Vacuum Filter - Vacuum System

Below table illustrates the list of faults and symptoms configured for a Rotary Vacuum Filter - Vacuum System.

Fault Name	Fault Symptoms	Recommendations
Vacuum pump not performing due to Seal Issue	RVF Vacuum Pressure RH (VPNSI)	1. Check vacuum condenser for fouling and coolant temperature. 2. Check seal flow.
	Motor Current Vacuum pump A RL (VPNSI)	
	Motor Current Vacuum Pump B RL (VPNSI)	
	Motor Current Vacuum Pump C RL (VPNSI)	
	Recycle (seal) Flow RL (VPNSI)	
	Seal fluid cooler OL Temperature RH (VPNSI)	
High load on vacuum pump	RVF Vacuum Pressure RH (HLVP)	1. Check cloth tear - high solids slippage in filtrate. 2. Check for Low cloth binding/load. 3. Check mechanical condition/healthiness of the pump.
	Motor Current Vacuum pump A RH (HLVP)	
	Motor Current Vacuum Pump B RH (HLVP)	
	Motor Current Vacuum Pump C RH (HLVP)	
Leakage or no cake on cloth	RVF Vacuum Pressure RH (LNCC)	1. Check for PVRV pick-up / passing from LP absorber side Check for no overflow condition and/or no cake on cloth.

Fault Name	Fault Symptoms	Recommendations
	DP control OP RVF A RH (LNCC)	
	DP control OP RVF B RH (LNCC)	
	DP control OP RVF C RH (LNCC)	
	Oxygen in recycle gas RH (LNCC)	

HRSG MONITORING

A heat recovery steam generator (HRSG) is one of the major pieces of equipment in a gas turbine combined cycle power plant that boasts a high thermal efficiency and produces minimal CO₂ emissions. An HRSG is a kind of heat exchanger that recovers heat from the exhaust gases of a gas turbine to an extreme degree. The heat is recovered in the form of steam which is served as the power source of a power-generating steam turbine.

For the heat-transfer tubes of an HRSG, finned tubes with excellent heat-transfer performance are employed. By adopting a compact design, the installation footprint of the equipment is reduced.

HRSGs consist economizer, evaporator and superheater. The different components are put together to meet the operating requirements of the unit.

Economiser: The first major HRSG circuit component is the economizer. An economizers function is to increase water temperature to an "approach" temperature near the saturation temperature. Approach temperature is carefully chosen to ensure maximum heat energy absorption efficiency and operational flexibility.

Evaporator: Evaporator sections are where the boiling process or steam generation occurs. As heat energy is absorbed by water from the gas stream, the water temperature increases. When water reaches the boiling point or saturation temperature, some of the water evaporates or vaporizes to steam.

Superheater: The last major component of a steam generator circuit is the Superheater. The major function of a Superheater is to increase steam temperature above saturation. High steam temperature minimizes the introduction of water as liquid to the steam turbine and improves steam cycle efficiency. The Superheater absorbs heat energy from the CT exhaust gas and transfers this energy to the steam. The steam superheat energy level is measured as an increase in steam temperature beyond the steam temperature achieved in the evaporator section. The Superheater sections typically have the highest metal temperatures in the HRSG. Superheater sections are composed of extended or finned tube surface modules. The high-pressure Superheater modules are one-pass, two-row modules and are arranged in a series/parallel configuration to reach the desired final steam temperature and capacity.



Monitoring Basic machine health data in addition to HRSG operating conditions helps to maintain reliable boiler operations. There are three areas that helps to monitor HRSG performance to increase the availability and decrease the energy costs.

1. Process and condition monitoring
2. Energy monitoring
3. Performance and Health monitoring

39.1 HRSG Monitoring - Performance Monitoring

The following measured parameters are monitored as it has following impact on performance and health of the boiler if not operated in operating range.

BlowDown Flow Rate: Blowdown is used to reduced dissolved solids, ph, conductivity and silica contain of HRSG water. Reducing blowdown rate can substantially reduce energy losses, as the temperature of the blown-down liquid is the same as that of the steam generated in the boiler.

Minimizing blowdown will also reduce makeup water and chemical treatment costs. Insufficient blowdown may lead to carryover of boiler water into the steam, or the formation of deposits. Excessive blowdown will waste energy, water, and chemicals.

Conductivity: It is important monitoring parameter for HRSG since its used for detecting impurity in leakage. If a condenser tube fails or impurities enter from another source like makeup water the sodium, calcium, and magnesium salts in the water are converted to their respective acids, primarily dilute sulfuric and hydrochloric acids, by the cation exchange column. The dilute acids that emerge (HCl and H₂SO₄, primarily) are more conductive than their respective salts and have an immediate influence on conductivity, thus providing a quick indication of upsets. In general, the cation conductivity of a clean condensate sample should be less than 0.2 micromhos (microsiemens) per centimeter (μS). This limit is mandatory to monitor to minimize corrosion in economizers and evaporator tubes.

Deaerator Drum Level & Dissolved Oxygen: The deaerator drum should maintain dissolved oxygen concentrations below 10 ppb. Deaerator outlet sampling for dissolved oxygen (grab sampling is most common) helps the plant operator or chemist monitor deaerator performance. An increase in dissolved oxygen levels over normal concentrations could indicate problems with the internal deaerator components. Trays may become misaligned. Or, the deaerator vents may not be adjusted properly.

Steam Drum Level: The steam drums are steam/water separators, storage tanks, and water treatment sites for steam purity control. Drum boilers operate in the area on the saturation curve. The steam/water mixture entering the drum from the riser tubes usually is 5-10% steam depending on the boiler load and pressure. Staying at low-quality levels protects the tubes from overheat failures due to the nature of the boiling process. In the steam drum, saturated steam is separated from the steam/water mixture.

Feed Water Mass Flow Rate: Feed water flow influence generation of steam however quality of feed water also plays important role on generated steam quality. Excessive feedwater contamination will reduce the boiler cycles of concentration and require increased blowdown.

HRSG Inlet Temperature: Inlet temperature of HRSG directly impact generated steam temperature, higher inlet temperature of HRSG produce high temperature of steam and increase efficiency of HRSG.

Flue Gas Volume Flow Rate: It indicate load to HRSG, higher flue gas flowrate higher load to HRSG and produce high steam rate.

Auxiliary Fuel flowrate: Auxiliary fuel is used to utilized oxygen in turbine exhaust gases that can be utilized to fire additional fuel to the exhaust gas and thus increase its temperature. Increase in inlet gas temperature generates additional steam in the evaporator, which increases the capacity of the heat sink, which is the economizer. The auxiliary fuel firing is an excellent way of generating additional steam in HRSGs and fuel utilization is nearly 100% or even more. That is, one can generate additional steam with nearly the same amount or lesser amount of fuel.

Process Performance Overview: Process Performance monitoring view in APM provides a list of the asset attributes and their details, including the current value. This list can be used to review and update asset attribute information, as well as to open an Asset Trend to view more details. It also highlights (in red color) the attributes if deviates from the high or low limits.

39.1.1 HRSG Monitoring - Model Inputs

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	BlowDown Flow Rate	BlowDown Flow Rate (Input)	kg/hr	Dynamic	Process	Field Input
2	Conductivity	Conductivity (Input)		Dynamic	Process	Field Input
3	Deaerator Drum Level	Deaerator Drum Level (Input)	%	Dynamic	Process	Field Input
4	Dissolved Oxygen	Dissolved Oxygen (Input)		Dynamic	Process	Field Input
5	Steam Drum Level	Steam Drum Level (Input)	%	Dynamic	Process	Field Input
6	Feed Water Mass Flow Rate	Feed Water Mass Flow Rate (Input)	kg/hr	Dynamic	Process	Field Input
7	Feed Water Pressure	Feed Water Pressure (Input)	barg	Dynamic	Process	Field Input
8	Feed Water Temperature	Feed Water Temperature (Input)	°C	Dynamic	Process	Field Input
9	Flue Gas Mass Flow Rate	Flue Gas Mass Flow Rate (Input)	kg/hr	Dynamic	Process	Field Input
10	HRSG Inlet Temperature	Flue Gas Inlet Temperature (Output)	°C	Dynamic	Process	Field Input
11	HRSG Outlet Temperature	Flue Gas Outlet Temperature (Output)	°C	Dynamic	Process	Field Input
12	Flue Gas Volume Flow Rate	Flue Gas Volume Flow Rate (Input)	m ³ /hr	Calculated	Process	Field Input
13	HRSG Fuel Mass Flow Rate	HRSG Fuel Mass Flow Rate (Input)	kg/hr	Dynamic	Process	Field Input
14	Auxiliary Fuel Pressure	Auxiliary Fuel Pressure (Input)	barg	Dynamic	Process	Field Input
15	Auxiliary Fuel Temperature	Auxiliary Fuel Temperature (Input)	°C	Dynamic	Process	Field Input
16	GT Fuel Mass Flow Rate	GT Fuel Mass Flow Rate (Input)	kg/hr	Dynamic	Process	Field Input
17	GT Power in MW	GT Power in MW (Input)	megawatts	Dynamic	Process	Field Input
18	Steam Mass Flow Rate	Steam Mass Flow Rate (Input)	kg/hr	Dynamic	Process	Field Input

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
19	Steam Pressure	Steam Pressure (Input)	barg	Dynamic	Process	Field Input
20	Steam Temperature	Steam Temperature (Input)	°C	Dynamic	Process	Field Input
21	Steam Drum Pressure	Steam Drum Pressure (Input)	barg	Dynamic	Process	Field Input

39.1.2 HRSG Monitoring - KPIs

The following indicators help to know the overall performance of the HRSG

HRSG Efficiency

It is expressed as heat generated in form of steam by heat input as fuel. This efficiency can be evaluated using the formula

$$\text{HRSG Efficiency} = \text{Heat Output} / \text{Heat Input} \times 100$$

$$\text{HRSG Efficiency} = Q \times (h_g - h_f) / q \times \text{GVC} \times 100$$

Where,

Q= Quantity of steam generated per hour in kg/hr

q = Quantity of fuel used per hour in kg/hr

hg= Enthalpy of saturated steam in kCal/kg of steam

hf= Enthalpy of feed water in kCal/kg of water

GVC- gross calorific value of the fuel in kCal/kg of fuel

CoGeneration Efficiency

The cogeneration efficiency is explained, as power generated by gas turbine calculations using steam generation to heat provided to HRSG and it is expressed as,

$$\text{CoGeneration Efficiency} = \frac{\text{Power output} + \text{HRSG output}}{\text{Gas turbine fuel input} \times \text{LHV of Fuel}}$$

Steam to Fuel Ratio

It is ratio of steam flow to fuel flow to HRSG. Higher steam to fuel ratio indicates good performance of the boiler and vice versa.

$$\text{Steam to fuel ratio} = \text{Steam Flow (kg/hr)} / \text{Fuel Flow (kg/hr)}$$

Water Balance

Water balance show the quantity of water flow in and out of system with steam generation.

$$\text{Water Balance} = (\text{BFW_Flow} - \text{Steam_Flow} - \text{BlowdownRate})$$

39.1.3 HRSG Monitoring - Model Outputs

No.	Attribute	Description	Units	Attribute Category	Monitoring Category	Source
1	BlowDown Loss	BlowDown Loss (Output)	kJ/hr	Dynamic	Internal	Calculated
2	CoGeneration Efficiency	CoGeneration Efficiency (Output)	%	Dynamic	Performance	Calculated
3	Feed Water Enthalpy	Feed Water Enthalpy (Input)	kJ/kg	Dynamic	Internal	Calculated
4	Flue Gas Specific Heat	Flue Gas Specific Heat (Input)	kJ/kg-C	Dynamic	Internal	Calculated
5	Fuel Calorific Value	Fuel Calorific Value (Input)	kJ/kg	Dynamic	Internal	Calculated
6	GT Total Power	GT Total Power (Input)	kJ/hr	Dynamic	Internal	Calculated
7	HRSG Efficiency	HRSG Efficiency (Output)	%	Dynamic	Performance	Calculated
8	Steam Enthalpy	Steam Enthalpy (Input)	kJ/kg	Dynamic	Internal	Calculated
9	Steam to Fuel Ratio	Steam to Fuel Ratio (Output)		Dynamic	Performance	Calculated
10	Water Balance	Water Balance (Output)	kg/hr	Dynamic	Performance	Calculated

39.2 HRSG Monitoring - Fault Model

In Asset Performance Management, you can configure faults, symptoms, and fault models on HRSG performance output to diagnose the problems early. The fault management can inform operators of any violations of the nominal HRSG operating range and deviations from the expected characteristic and make this data available for further processing through the historian. The fault symptoms or models are used to:

1. Warn against potential damage to HRSG under unfavorable operating conditions.
2. Provide early warning of damages to the HRSG.

The following messages can be generated for process alarms or notifications to warn operating personnel of unfavorable operating conditions

1. **Steam to Fuel Ratio Low:** It indicates lower steam generation for higher fuel consumption. This can happen due to low BFW temperature or heat losses from flue gases released at stack.

Below table illustrates the list of faults and symptoms configured for a HRSG to identify possible anomalies that affects the performance or health of a HRSG.

Fault Name	Associated Symptoms
HRSG Performance Low	HRSG Efficiency Low
	Co-Gen Efficiency Low
	Steam to Fuel Ratio Low
	Supplementary Firing Stopped
	Potential Water Carry Over in Steam
	Steam Drum Pressure Low
Health Degradation Warning	Conductivity Low
	Dissolved Oxygen High
Steam Drum Instability	Steam Drum Level Low
	Water Balance High
Energy Consumption High	Blowdown Losses High
	Steam Temperature High
	HRSG Outlet Temp High

UNISIM DESIGN (USD) INTEGRATION

The key performance parameters of assets that are modeled using USD can be calculated and the results of calculation sent back to Asset Performance Management. The input variables and output variables of these assets must be configured in USD and mapped to the relevant tags in Asset Performance Management (see "UniSim Design Integration" in the Asset Performance Management Configuration Guide).

The plant data is pre-processed and initialized UniSim model. The results from UniSim are historized and analyzed for faults.

NOTE

Models that are in "Steady" state can only be used for analysis in USD.

USD data sources in Asset Performance Management are configured to execute modeling in two different modes:

1. **Simulation Calculation:** the calculation of equipment/process performance using the configured UniSim design model.
2. **Optimization:** identifying the optimal process condition or reconciling measurements, equipment parameters using the configured UniSim design model.

40.1 USD Snapshots

When a USD model executes, a snapshot file is saved to the location configured on the USD data source (see "Configure a UniSim Design (USD) Data source" in the Asset Performance Management Configuration Guide). Depending on the execution result, the snapshot will either be saved to the Success or Fail folders within the folder location.

If the model needs to be adjusted (either the change the calculations of KPIs, or to address errors), the snapshot files can be loaded in UniSim Design, when the model can be reviewed and re-tuned.

Once the model has been adjusted in UniSim Design, a new case file can be exported, and loaded into the USD data source.

40.2 USD Parameters

The following parameters are used with the Asset Performance Management UniSim Interface, and the UniSim Optimization Status.

40.2.1 Asset Performance Management-UniSim Interface Parameters

1. **Execution Mode:** The current mode of the APM interface to UniSim. The Execution Mode changes while the UniSim model is loaded, running, completed, and so on.
2. **Execution Flag:** The UniSim calculation model can be scheduled based on the interval configured or executed whenever this particular attribute value changes to one. The APM Engine resets the value to zero once it recognizes the request to execute so that subsequent triggering can happen to re-execute the UniSim model.
3. **Error Status:** The status of the last UniSim model execution.
4. **Remaining Time:** The UniSim calculation is configured with a time-out. Whenever the execution starts, the remaining time is set to timeout and decreases until the execution is completed.

40.2.2 UniSim Optimization Status Parameters

1. **Phase:** Displays the current phase of the Optimizer algorithm. Values include Initialize, Setup, OPT Deriv, OPT Search, and Results.
2. **Evaluation Count:** Reports the number of model evaluations performed during the course of the optimization.
3. **FP Iterations:** Displays the number of minor iterations since the last major iteration.
4. **Gradient Iterations:** Reports the number of gradient evaluations performed during the course of the optimization.

40.3 Simulator Interface

Asset Performance Management interface with UniSim Design to perform KPI calculation by running simulator in calculation mode.

40.3.1 Simulator Execution Mode

The following are the possible values for the execution mode attribute in UniSim Interface asset.

1. CALCULATION Run In Progress = 9
2. CALCULATION Run returned SUCCESS = 12
3. CALCULATION Run ERROR or not converged= 11
4. CALCULATION Run Timedout = 10

40.3.2 Simulator Error Status

The follow are the possible values for the Error Status attribute in UniSim Interface asset.

1. Not Converged =13
2. Success =1
3. Timed out = 20

40.4 Optimizer Interface

NOTE

Refer to Section 7: Derivative Utility & Optimizer in the RTO Reference Guide for how to setup optimizer in UniSim Design. This document should be available in your UniSim Design installation folder.

In Asset Performance Management configure the USD interface with execution mode as optimizer to execute optimizer and generate snapshots periodically with plant preprocess data.

40.4.1 Optimization Execution Mode

The following are the possible values for the execution mode attribute in UniSim Interface asset:

1. OPTIMIZATION Run In Progress = 14
2. OPTIMIZATION Run returned SUCCESS = 17
3. OPTIMIZATION Run returned ERROR = 16
4. OPTIMIZATION Run Timedout = 15

40.4.2 Optimizer Error Status

NOVA Optimization Error Code	Description	Success
0	Successful solution	Yes
1	Infeasible problem	Yes
-2	Unbounded solution	Yes
2	Running	NA
3	Not Run	No
-3	Bounds crossed for some constraint	No
-4	Invalid input	No
-5	User model abort command received	No
-6	Insufficient work space for nlp	No
-7	Insufficient work space for reduced Hessian	No
-8	Insufficient work space for LU factors	No
-9	Maximum iterations	No
-10	Bad derivative information	No
-11	Too much accuracy requested	No
-12	External abort command received	No
-13	System error	No

Legacy SQP Optimization Error Code	Description	Success
1	Success	Yes
4	Step convergence	Yes
5	Cost convergence	Yes
7	Gradient convergence	Yes
11	Timeout feasible	Yes
13	Not converged	Yes
16	Relaxed OK	Yes
18	Cost convergence - at feasible point limit	Yes
2	Impossible	No
8	Globally infeasible	No
9	Infeasible	No
6	Flat convergence	No
10	Unbounded	No
12	Timeout infeasible	No
14	Not Run	No

Legacy SQP Optimization Error Code	Description	Success
3	No variables	No
15	Stopped	No

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